

Structural origin of the Barbero–Immirzi parameter and CKM mixing angles; derivation of the Planck, Boltzmann, and universal gravitational constants

Relationship between action, area, information, and entanglement entropy

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Abstract

The Planck constants in their action sections, the Barbero–Immirzi parameter, and the Cabibbo–Kobayashi–Maskawa (CKM) mixing angles are obtained, and Boltzmann transduction and the gravitational section are determined in their explicit dimensional realizations. The local Hadamard lattice quotient has sixteen classes; its multiplicative norm and the self-scale fix $\text{ord}_{10}(\varphi\alpha^{16}) = -34$; regional characters determine the action sections before and after return, related by $h_a = 2\pi\hbar_a$. Hexadic–octadic incidence and the oriented turn produce the Barbero–Immirzi functional, whose evaluation is $\gamma_{\text{HMT}} = 0.2740076726944592747 \dots$. The sector rules yield three CKM angles through an invertible rational transformation, together with their complex phase and the Jarlskog invariant. Conversion between degrees and radians transports the angular arguments and coefficients exactly.

Action and period determine the energy per tritic decision: $k_B\Theta_{\text{clk},a} \log 3 = h_a/T_{108}$. For a fixed positive thermal section, this identity determines a unique Boltzmann transducer; its individual coordinate retains the energy and temperature bases and their normalization. Composing the Hadamard return with incidence inscription defines the length $L = (5759/23040)\alpha_{\text{HMT}}^{16}L_*$. The positive radial reader, energy and conjugate length preserve the same character; their two normalized products determine a unique coefficient $G_a = c_{\text{int}}^3 L^2/\hbar_a$ in the reduced-radius realization. The compatible clock satisfies $R_{108} = L$. The proof makes explicit the incorporated longitudinal and metric rules, their antecedents and their dimensional sections. Reciprocity between the action and gravitational sections preserves $G_a\hbar_a$ and hence the area scale $\ell_P^2 = G_a\hbar_a/c_{\text{int}}^3$. In the horizon realization, energy, temperature, entropy, and half-action satisfy the identities of a single cell.

The thermal construction determines return capacity, the carrier of multiplicities 1, 380, 649, its positive energy-bearing memory, and the marked spectrum of intensity 1.380649×10^{-23} . An explicit rule of marked entropy and conditioned energy turns this intensity into a thermometric coefficient and determines a unique free-energy minimum. Degree assignment and the reading rule retain their explicit status within this realization.

These determinations relate action, area, and boundary information. The Barbero–Immirzi parameter weights the elementary area increment; area and the modular Hamiltonian jointly recover the sector occupations of the reduced state when $\Delta_{\text{mod}} \neq 0$ and the normalization is known. The gravitational realization retains a five-component tensor carrier and its explicit transverse projection. The action ellipse distinguishes symplectic and antisymplectic transports. Its Gaussian realization identifies the semiaxes with two standard deviations, saturates the Robertson–Schrödinger inequality, and preserves area h under reciprocity.

The common origin is the APP–TRIT–TPK composition of Triadic Modular Holography and Dimensional Mechanics. APP, *All Possible Paths*, uses the discrete support $(\mathbb{Z}/9\mathbb{Z})^2$ and the additive Cayley graph generated by $\pm(1, 0), \pm(0, 1)$, with sum and product evaluations. TRIT assigns the $\{-1, 0, +1\}$ signature of orientation and regime; TPK, *Topological Phase Kernel*, composes selection, transport, and carry and memory updates. The construction of numerical regions and the dodecaphase record precedes action characters and angular coordinates. The correlated analytic representation of alpha prolongs, at every depth, the coordinate generated by dodecaphase closure. Incidence evaluation produces twenty-five coordinates from the specified ledger of visits and traces. The memory update admits a finite identity with a terminal state, a series prolongation with tail control, and an exact reduction jointly transforming operator, source, and reader.

Dodecaphase–Witt transport realizes the angular reader isometrically, preserves its induced metric, and recovers its initial coordinates. CKM and Pontecorvo–Maki–Nakagawa–Sakata (PMNS) retain their respective spectral frames and realizations. Exceptional incidence extends to Moonshine, and the reciprocal radius is realized through momentum–winding exchange in the domain of T-duality. Numerical values and quantitative comparisons retain both action sections, the units, the individual normalizations, and the status of each physical reference. The explanation of each constant thus combines its defining operation, its value, and the structural information preserved by its transformations.

Keywords: Triadic Modular Holography; reduced Planck constant; Barbero–Immirzi parameter; hexadic–octadic incidence; entanglement entropy; gravitation; T-duality; Moonshine; CKM mixing; Boltzmann constant; Planck length.

Contents

1	Introduction	7
1.1	Determination of constants and relations between action, area, and information	7
1.2	Arithmetic support, orientation, and generating dynamics	7
1.3	Action, incidence, and boundary information	8
1.4	Capacity, energy-bearing memory, and thermometric realization	8
1.5	Longitudinal composition and determination of gravitational coupling	9
1.6	Reciprocity, angular propagation, and exceptional continuity	9
1.7	Identities for area, entropy, and gravitational scale	10
	Notation and domains of the constructions	11
2	Formal kernel of Triadic Modular Holography	13
2.1	APP: positional support and arithmetic evaluations	13
2.2	TRIT: orientation and quadratic regimes	16
2.3	TPK: Topological Phase Kernel	22
2.4	Accumulated memory and transport resolvents	28
2.5	Correspondence between positional support and the oriented calendar	32
2.6	Operator architecture of transport, memory, and prolongation	32
3	Topological Phase Kernel and generation of the seed catalogue	44
3.1	Phase selection, transport and enriched state	44
3.2	Initial conditions and reversible enumeration	45
3.3	Sheet evaluation and arithmetic record	46
3.4	Six windows and decimal block emission	47
3.5	Complete calculation of an emission beginning with 501	48
3.6	Factorisation of the finite census	49
3.7	Oriented ternary reading and preservation of fibres	51
3.8	Dihedral action and the forty-three orbits	52
3.9	Self-contained enumeration and verification algorithm	53
3.10	Preservation of the record and the function of the seeds	53
4	Development of the formal kernel: initial conditions, geometry, and refinement	53
4.1	Initial states and the emission recurrence	54
4.2	Calendar closure and memory advance	55
4.3	Observable compression and orthogonal conservation	56
4.4	Local development: commutation, Lorentzian signature, and discrete curvature	57
4.5	Cubic completion and two notions of boundary	58
4.6	Cubic incidence and Lorentzian realization	60
4.7	Compatibility of modular and positional reductions	61
4.8	Number as a compatible section and value as a reading	62
4.9	Composition of histories and memory multiplier	62
4.10	Double variance of refinement and conservation of unity	63
4.11	Phase return, Weyl representation, and two readings	64
5	Coinductive generation of π, φ, e to arbitrary depth	65
5.1	Orbits, regions, and characters of closure, propagation, and self-scaling	65
5.2	Finite transport and the prolongation boundary	66
5.3	Five closure regions and angular compensation	66
5.4	Propagation reader and normalization of the unit	67
5.5	Modal incidence and generation of self-scaling	68
5.6	Exponential realizations of the generated coordinates	69
5.7	Compatible cylinders and arbitrary depth	70
5.8	Specular involution and synchronization of the regions	71
5.9	Sturmian helical suspension	72
5.10	Nonadic connection, holonomy, and monodromy representation	75
5.11	Block clock and capacity clock	76
5.12	Symbolic suspension and aperiodic time crystal	77
5.13	Specular symmetry of the regions and dual realization	78
5.14	Temporal transport and tritic exponential composition	78
5.15	Multiplicative irreducibility and return periods	79
6	Regional generation and refinement	82
6.1	Positional emission through rational separation of cells	82
6.2	Regional blocks of six trits at arbitrary depth	83

6.3	Transition registers and reconstruction of linear lifts	84
6.4	Simultaneous refinement in ternary and decimal bases	85
6.5	Selection of the next block by the regional reader	86
6.6	Ternary signature, carry, Witt incidence, and reading clock	87
6.7	Composition theorem for the block, cylinder, and signature	88
6.8	Calendar census and simultaneous compatibility of the regional biographies	89
7	Regional determination of the dodecaphasic descriptor and selection of the coupling register	90
7.1	Regional bands, encoding, and integer signatures	90
7.2	Conversion clock and first zero advance	91
7.3	Oriented construction of the twenty-four-trit descriptor	92
7.4	Functional panel and terminal repertoire	93
7.5	Comparative readers, affine reader, and temporal incidence	97
7.6	Enumeration of cylinders and decimal candidates	98
7.7	Exceptional incidence and heterotypy condition	100
7.8	Regional composition and continuity of the realizations	102
7.9	Executable certification and traceability of the operations	103
7.10	Complete census of cylinders and selection of the terminal boundary	103
8	Dodecaphasic record and rational phase constant	108
8.1	Cyclic extension and memory grading	108
8.2	The oriented record and the integral $1 + 5 + 6$ chart	109
8.3	Incidence evaluation preceding the dodecaphasic record	110
8.4	Conjugation of counts and boundary terms	111
8.5	Two readings of the terminal state	112
8.6	From the channel record to the signed vector	113
8.7	Hadamard transformation by orbits and integer reconstruction	114
8.8	Cyclic differences and the uniform component	115
8.9	Integral compatibility and determination of the transverse record	116
8.10	Two rational readings: κ_{36} and κ_{per}	119
8.11	Reversibility of the rational constant and phase change	120
8.12	Periodicity and cokernel of the transport operator	120
9	Derivation of the fine-structure constant	122
9.1	Positional determination through integer dodecaphase closure	122
9.2	Analytic determination through the vacancy structure	128
9.3	Identification criterion for compatible readouts	130
9.4	Compatibility between the dodecaphasic and analytic representations of alpha	131
10	Dodecaphasic update and observable memory	137
10.1	Quadratic balance	137
10.2	Transport resolvents and preservation of memory	138
11	Dodecaphasic seal and reversible rational reading	139
12	Cyclic decomposition and ratios of determinants	140
12.1	Oriented information and the complex sector	140
13	Planck constants and construction of the action scale	141
13.1	Hadamard quotient, multiplicative norm, and decimal order	141
13.2	Regional persistence and uniqueness of the action selector	142
13.3	Word length and the determinant-based return character	143
13.4	Action sections and dimensionless return quotient	144
13.5	Structural definition and role of the Planck constants	147
13.6	Angular coordinates, normalization of the turn, and units of action	147
13.7	Covariance of the channels and the incidence functional	148
14	Hexadic–octadic incidence and the Barbero–Immirzi functional	149
14.1	The weights precede the pole coefficient	149
14.2	Parity and structural sensitivity	150
14.3	Structural dependence of the functional and area normalization	150
14.4	Structural definition and role of the Barbero–Immirzi parameter	151
15	Action ellipse and modular reciprocity	152
15.1	Semixes, foci, and interior radii	152
15.2	Parametric phase, polar angle, and action area	153
15.3	Shape reciprocity and interior distance	154
15.4	Bohr radius: a consequence of the action scale	154

15.5	Inductive–capacitive realization and constitutive continuity	155
15.6	Canonical covariance and reciprocity of the action ellipse	155
16	Symplectic transport of the action ellipse	158
16.1	Quadratic form and inductive–capacitive realization	158
16.2	Transport invariant and Hamiltonian connection	158
16.3	Genealogical compatibility and conformally symplectic transport	159
17	CKM mixing angles, CP phase, and leptonic transport	160
17.1	Common antecedent and separation of the two transports	160
17.2	Sectorial incidence and construction of the angular reader	160
17.3	Rational CKM chart and angular units	164
17.4	Geometric decomposition under sheet inversion	165
17.5	Transport of sheet inversion	165
17.6	Record parity and CP realization	166
17.7	Unitary amplitudes, CP phase, and the Jarlskog invariant	166
17.8	Conjugation of amplitudes and oriented invariant	167
17.9	Leptonic record, operators, and spectral frames	168
17.10	Record kernel and rank sufficiency	169
17.11	Check of the rank-one phase	170
17.12	Leptonic phase and the Dirac–Majorana criterion	170
17.13	Closure of angular transport	171
18	Equivariant realization of the angular reader in Witt incidence	172
18.1	Regional triplet and orientation line	172
18.2	Dodecaphase sector and variational selection of orientation	173
18.3	Reynolds intertwiner and polar normalization	175
18.4	Point–hexad incidence and joint gauge change	175
18.5	Isometric composition and coordinate recovery	176
18.6	Faithful realization of the angular sectorial system	177
18.7	Exact certification and finite unitary composition	178
18.8	Structural result and scope of the composition	178
19	Area operator and reconstruction of sectorial information	179
19.1	Tritic regrouping and boundary incidence	179
19.2	Transport of the collective incidence modes	179
19.3	Angular character and area operator	180
19.4	State normalization and the meaning of the modular Hamiltonian	181
19.5	Reduced state, Schmidt decomposition, and entanglement entropy	181
19.6	Genealogical memory and informational content of a reading	182
19.7	Areal and volumetric sheets: incidence, growth, and measure	182
19.8	Admissible continuations and survival information	183
19.9	Cyclic correlation and preservation of path order	183
19.10	Equation of state for sectorial information	184
19.11	Occupation inversion and entropic invariants	186
19.12	First modular law and finite area variations	186
20	Simultaneous rigidity of the longitudinal reader and radial coupling	188
20.1	Incidence record and constitutive restrictions	188
20.2	Simultaneous determination theorem	190
20.3	Reciprocity and phase rigidity	191
20.4	Planck scales and the unit-character specialization	191
20.5	Archive covariance and change of units	192
20.6	Exact elimination of coupled memory	192
20.7	Joint reduction of radius, energy and conjugate length	193
20.8	Dynamic memory and induced temporal norm	193
20.9	Refinement, spectral domains and scope	194
21	Action, gravity, and thermal information	195
21.1	Dodecaphasic period and circular realization of action	195
21.2	Thermal normalization and energy per cycle	196
21.3	Barbero–Immirzi in the Cartan–Holst realization	196
21.4	Phase return and representation of accumulated memory	197
21.5	Cubic incidence and boundary soldering	198
21.6	Geometric realization of transport and soldering	199
21.7	Oriented transport of the connection and area	201
21.8	Icosahedral incidence and five-component gravitational coupling	201

21.9	Intertwiner with the space of symmetric traceless tensors	208
21.10	Transverse traceless projection	208
21.11	Decomposition into 5 polarization modes	209
21.12	Physical interpretation of the representational chain	210
21.13	Structural definition and function of the gravitational constant	210
22	Boltzmann constant, information, and thermal transduction	212
22.1	Dimensional structure of transduction	212
22.2	Genealogical memory, information reduction, and thermal transduction	212
23	Spectral capacity, energy-bearing memory and thermal transport	217
23.1	Capacity, vacancy and joint renewal	217
23.2	Capacity returns and cylinder valuation	218
23.3	Relation carriers and their interchange realization	218
23.4	Multiplicative seeds and exact fibre compensation	219
23.5	Double grading and recovery from its marginals	220
23.6	Positive energy dilation of memory	220
23.7	Memory loop and decimal spectrum	221
23.8	Marked spectral section and complete intensity	222
23.9	Boundary currents, compression and offset rigidity	222
23.10	Thermal equivalence with transported reference and scale	224
23.11	Thermal response of the marker and preparation cost	225
23.12	Entropy and heat capacity of the graded carrier	225
24	Composition of action, area, and marked thermal reading	227
24.1	Return energy and areal resolution	227
24.2	Provenance information and horizon response	228
24.3	Spectral reference and preserved multiplicities	228
24.4	Entropy increment of the tensor record	229
24.5	Normalized realization and conditioned energy reading	229
24.6	Transducer fixed by the joint reading rule	230
24.7	Composition with the chronological section and its transports	230
24.8	Energy origin, reciprocity, and the electronic section	231
25	Additive selection and the thermal reading instrument	232
25.1	Spectral events and relative measure	232
25.2	Unitary apparatus, recovery, and energy domain	233
25.3	Tritic transport with remainder and carry	234
25.4	Equivalent embeddings and memory enlargement	234
25.5	Complete preparation, entropy, and conditioning	235
25.6	Characterization by additivity and refinement	236
25.7	Realization by visits between returns	236
25.8	Common reference and correlations between systems	237
25.9	Thermometric selection and composition with action and horizon	238
25.10	Structural definition and function of the Boltzmann constant	239
26	Confluence of action, area, information, and gravitation	240
26.1	Energy and entropy of the half-action cell	240
26.2	Conversion between decision energy and area increment	240
26.3	Physical meaning of area confluence	241
26.4	Spectral entropy and preservation under transport	242
26.5	Effective multiplicity and hyperbolic realization	242
27	Evaluation of the constants and quantitative comparison	243
27.1	Reduced Planck constant and action per turn: two orientations	243
27.2	Barbero–Immirzi: internal value and comparison between constructions	244
27.3	CKM angles, amplitudes, and CP violation	245
27.4	Boltzmann and universal gravitation: quantity, chart, and reference	245
27.5	Reproduction of the evaluations	247
28	Exceptional prolongation: from dodecaphasic incidence to Moonshine	248
28.1	Structural function of K and α in exceptional incidence	248
28.2	Compatibility of the Archimedean and incidence realizations	251
28.3	The finite realization and the ternary code	252
28.4	Dodecaphasic flag associated with K and α	253
28.5	Isometric intertwining of the incidence representations	254
28.6	The Witt star and the generation of M_{12}	255
28.7	The binary residue and the five regions of π	256

28.8	From the ternary code to the Niemeier lattice	257
28.9	Incidence selection of the rootless unimodular neighbor	258
28.10	The vertex operator algebra and the Monster	259
28.11	Graded Moonshine traces and representation of the Monster	259
29	From the action ellipse to T-duality	261
29.1	The action shape determines the normalized radius	261
29.2	Lattice exchange and equivalence of operators	261
29.3	Nonadic sections and carry transport	262
29.4	Momentum, winding, and realization scale	263
29.5	Modular evaluation and the Moonshine character	264
30	Conclusions	266
30.1	Planck constants, Barbero–Immirzi, and CKM mixing	266
30.2	The Boltzmann constant and energy per tritic decision	267
30.3	Gravitation and preservation of the area scale	267
30.4	Boundary information and entanglement entropy	268
30.5	Reciprocity and exceptional continuation	268
30.6	Evaluation of the record and preservation of observation	269
30.7	Structural definitions and quantitative results	269
30.8	Conservation of the areal scale and recovery of boundary information	270
A	Electronic antecedent and dimensional realization	273
A.1	Center, orientation, and vacancy on a preserved fiber	274
A.2	Commutator norm of the arithmetic projections	275
A.3	Transfer between sheets and coefficients of the basal stage	277
A.4	Global torsion and the basal coordinate	278
A.5	Two readers of the electronic return	279
A.6	Complete electronic composition and normalization	281
A.7	Dimensional realization: energy, mass, and chronological scale	281
A.8	Lorentzian transport and electromagnetic representation	283
A.9	Comparison of the reduced Planck constant	284
B	Conditions of the realizations	285
B.1	Alpha coordinate and mixing frames	285
B.2	Thermal bases and gravitational normalization	285
B.3	Provenance of the longitudinal and radial composition	285
	References	286

1 Introduction

1.1 Determination of constants and relations between action, area, and information

This article determines the Planck action sections and their return, the Barbero–Immirzi functional, and the CKM angular coordinates, and establishes Boltzmann transduction and gravitational reciprocity in the specified dimensional realizations. These results are connected by effective operations: action converts frequency into energy; Barbero–Immirzi weights the incidence that resolves area; Boltzmann converts dimensionless information into entropy; and gravitation transforms action into a common geometric scale. The physical question is how these functions and their values are determined from a single discrete structure instead of entering as measured parameters.

The determination of Planck separates the decimal order of action from its return coordinates. The local quotient of the record has sixteen sheets; its multiplicative norm, followed by autoscaling, gives $\text{ord}_{10}(\varphi\alpha^{16}) = -34$. Regional characters then fix $\hbar_{\text{pre}} = H_5 10^{-34} \mathcal{U}_S$ and $\hbar_{\text{ret}} = \eta_{\text{ret}} 10^{-34} \mathcal{U}_S$, with $0 < \eta_{\text{ret}} < H_5$. The ratio $R_{\text{act}} = H_5/\eta_{\text{ret}} > 1$ preserves the effect of return under a change of the common positive basis; in each section, $h_a = 2\pi\hbar_a$. The local reader, the character, and the regional continuation rule are part of this determination, developed in 13.

The Barbero–Immirzi functional evaluates an oriented chain of twelve sectors and a return seam on incidences of cardinalities 90 and 120. Evaluation yields $\gamma_{\text{HMT}} = 0.2740076726944592747 \dots$, odd under angular reversal. The CKM sector rules, in turn, produce a rational matrix of determinant $-3857/6$, three mixing angles, and the phase $\delta_{\text{deg}} = 9A_{\text{deg}}$. Their complex realization is unitary and determines the Jarlskog invariant. Both results receive the angular coordinates already generated: the Barbero functional does not select the mixing angles, nor do those angles retrospectively select the sector rules.

Longitudinal composition precedes gravitational evaluation. Incidence inscription and the complementary archive yield $r_N = 5759/23040$. Longitudinal rule R4 applies this coefficient to $\alpha_{\text{HMT}}^{16} L_*$ and determines L ; metric rule R5 determines the positive radial operator of the same character used by the energy. Section 20 states both compositions and proves uniqueness of their joint realization with phase action, the conjugate clock, and reduced-radius identification R8. The reference L_* , declared equal to one metre in the SI evaluation, retains its dimensional role.

The relation between Boltzmann and gravitation then takes the explicit form

$$k_B \Theta_{\text{clk},a} \log 3 = \frac{h_a}{T_{108}}, \quad G_a = \frac{c_{\text{int}}^3 R_{108}^2}{\hbar_a}, \quad \frac{G_a \hbar_a}{c_{\text{int}}^3} = \ell_P^2.$$

Here $R_{108} = L$ and $t_0 = \pi_{\text{HMT}} L / (54c_{\text{int}})$. The first identity determines the energy per tritic decision and a unique thermal transducer once the positive temperature section is fixed. The second is the circular specialization of the common coefficient determined by the radial proof. The third shows that reciprocity between action and gravitation preserves the area unit. These thermal and geometric conditions relate the results without identifying determination of a dimensional section with the choice of its numerical coordinate.

The distinction requires attention to the nature of each object. A dimensionless ratio expresses a relation between quantities. A dimensional constant occupies a line of quantities and acquires a numerical coordinate when a unit is chosen. A boundary condition fixes a realization within a collection of solutions. All three determinations may enter the same formula and preserve different roles in its proof. Duff, Okun, and Veneziano’s debate on dimensional constants illustrates the importance of making these choices explicit [11]. The current formulation of the International System in turn specifies the role of defining constants in the realization of units [10]; [13].

1.2 Arithmetic support, orientation, and generating dynamics

In Triadic Modular Holography, HMT, arithmetic support, ternary orientation, and dynamics with memory precede scalar readings. APP, *All Possible Paths*, uses the discrete support $X_9 = (\mathbb{Z}/9\mathbb{Z})^2$, with the additive Cayley graph generated by $\pm(1, 0)$ and $\pm(0, 1)$. At each position of its 1, ..., 9 chart, the two evaluations, sum and product, act while retaining their residues and quotients. TRIT assigns a signature $\{-1, 0, +1\}$ distinguishing orientation and local regime. TPK, *Topological Phase Kernel*, composes selection, transport, carry and memory updates, and compatible state prolongation. Phase return preserves memory advance. The operations and regional generation in 2–10 construct the numerical regions, the dodecaphase record, and alpha before their use in action, incidence, and information readers. The correlated analytic representation of alpha prolongs, at every depth, the coordinate produced by dodecaphase closure. The Archimedean realization determines values of compatible cylinders; the incidence realization preserves other information

from the same origin. This common dependency allows each constant to be studied as value and structure, with measurement subsequently serving as a comparison.

This distinction already takes material form in the constitution of the record. The evaluation of visits and oriented traces in 8.3 produces three integer counts per window, preserves their residues and quotients, and determines the twenty-five coordinates of (8.5). The domain includes the incidence collection, its multiplicities, and its orientation. Proposition 8.3 characterizes exactly the residual information determining that image; the transformations in 8.6 and 8.7 subsequently allow the signed record to be recovered. Evaluation of the ledger and reconstruction of its output are thus linked, with a distinct demonstrative role for each operation.

This structure acts effectively in the equations. Incidence cardinalities enter a return functional; action sections enter a gravitational reciprocity; boundary occupations enter area and the modular spectrum; inversion of a quadratic form enters the exchange of momentum and winding. Each connection is expressed through a map, an identity, and a domain of validity.

1.3 Action, incidence, and boundary information

The specific argument of Article II focuses on the relation between action, area, entropy, and gravity. The Barbero–Immirzi parameter occupies a central position because it links the incidence functional to an areal normalization and its realization in gravitational action. Holst’s formulation provides a language of subsequent realization for this coefficient [12]. In the construction studied here, the coefficient is evaluated from oriented incidence and the angular channels already obtained.

The inclusion of a marked hexad in an octad preserves the incidence pair determining cardinalities 90 and 120. The dodecaphase turn distinguishes twelve sectors from an oriented return boundary. The linear reader of that chain produces the functional’s multiplicities. The pole coefficient and decimal evaluation are computed afterward. The sequence reveals why the Barbero–Immirzi value belongs to a structure of orientation and area, in addition to occupying a position in a table of constants.

Area and the modular Hamiltonian describe complementary aspects of the boundary. The first reading aggregates occupations. The second weights their sectorial provenance. For $\Delta_{\text{mod}} \neq 0$ and known normalization, the Hamiltonian yields the weighted observable D ; together with the total N , it reconstructs the two occupations originating the aggregate. Inversion of the pair (N, D) does not require that additional condition. This recoverability specifies the content of the genealogical thesis: two configurations can share a scalar reading and preserve differences accessible to another reading of the same state. The argument is developed first in the corresponding finite space and then in the declared realization.

Memory preservation also specifies how to reuse dynamics when auxiliary variables are eliminated. Identity (10.5) retains the terminal state of each truncation; Proposition 2.18 provides the prolongation and its tail bound. Proposition 2.19 proves that the complete observation is preserved by jointly transforming the reduced operator, source, and reader. This rule identifies the operative content that must be transmitted between two descriptions of the same transport.

Entanglement entropy additionally requires a bipartition and a state. The exposition specifies both through a purification and its reduced state. The spectrum’s dimensionless information is distinguished from dimensional entropy, whose normalization incorporates the Boltzmann constant. This precision allows the energy of tritic periodicity to be related to action without confusing a state count, a probability distribution, and a temperature.

1.4 Capacity, energy-bearing memory, and thermometric realization

Section 23 develops the finite mechanism connecting renewal of the two APP sheets, capacity and the first vacancy, double grading of the carrier, and memory energy. The bilateral loop determines a decimal spectrum; tritic grouping retains the remainder. The capacity marker and incidence sectors then produce a complete intensity with registered energy origin. Connected currents fix the intersector gaps, and centered transport preserves the reader between realizations with different scales.

Section 24 composes these results with action, area, Barbero, and horizon. It adds a thermometric realization retaining the spectral reference in a tensor record: the entropy channel reads marked information, and the energy channel reads the conditioned mean. The proof obtains the coefficient from their differential ratio and establishes the corresponding variational principle. The fixed reference and marked–conditioned pair of functionals are the constitutive conditions of the realization. The chronological capacity Hamiltonian retains degrees 23, 26, 29 and determines its normalized state; thermal application of the intensity is thereby related to clock and horizon, preserving their scales and effective transports.

1.5 Longitudinal composition and determination of gravitational coupling

Gravitation forms part of the central argument. The composition of inscription, length, and metric first fixes L and the radial response. The conjugate clock satisfies $\omega_{108}L = c_{\text{int}}$; its period is distributed over the 108 steps of the dodecaphasic calendar. This gives $T_{108} = 2\pi_{\text{HMT}}L/c_{\text{int}}$ and $t_0 = T_{108}/108$. Reduced-radius condition R8 determines G on the same character already expressing the energy. Circular coincidence retains the equivalent form $\vartheta_{108}^\circ = (54/\pi_{\text{HMT}})^2$. In the resulting action sections, the gravitational coefficient transforms reciprocally, and the product of action and gravitation remains invariant. The area unit is L^2 in both orientations.

The preserved equality has a geometric consequence: passage between the sections affects certain coordinates while preserving a common geometric quantity. The meaning of the value is obtained from the transport that produces it. The relation to the horizon then introduces energy, temperature, and areal entropy under explicit normalizations. The confluence of these quantities in an action cell is proved by substitution of the preceding constructions.

Thermal dependency is presented with equal precision. An identity for the product of the Boltzmann constant and a temperature determines an energy scale. Determining each individual coordinate requires its corresponding chart. The normalization of each realization specifies the bases and the dimensional transport of that coordinate. Physical comparison is performed on these outputs and their units.

1.6 Reciprocity, angular propagation, and exceptional continuity

Angular coordinates enter two distinct realizations of the same construction. The oriented-incidence functional determines the Barbero–Immirzi parameter; the species reader determines the CKM mixing angles and their phase. Section 17 presents the rational matrix, its inverse, and the complex unitary realization. The determinant of the transformation and the Jarlskog invariant provide complementary controls: the former examines recovery of the coordinates; the latter examines the phase of the amplitudes. The numerical values are compared after those constructions.

Theorem 18.5 realizes the angular coordinates in the Witt incidence module through an explicit isometry. Corollary 18.7 preserves the metric of the sectorial system and provides an inverse on its image. Thus passage to exceptional incidence transmits the coordinates produced by the angular rules, together with their metric relations. Sectorial evaluation and isometric transport compose in that order, according to (18.15).

The action ellipse introduces a fixed-determinant form and an anisotropy parameter. Exchanging its semiaxes inverts the modular radius. Two transports can preserve the same energy form and act with different signs on the symplectic form. The orientation of action then distinguishes symplectic and antisymplectic transports with the same scalar energy. This property underlies the attention to transport and memory in the model's continuation.

The action–covariance relation specifies the quantum content of this geometry. Proposition 15.6 constructs, in the canonical representation and with the Gaussian state specified there, a covariance of determinant $\hbar^2/4$ attaining equality in Robertson–Schrödinger. Proposition 15.7 identifies the action contour with the two-standard-deviation ellipse: its semiaxes are $2\Delta Q$ and $2\Delta P$, and its area is $h = 2\pi\hbar$. Reciprocity exchanges dispersions and preserves that determinant. Passage to position and momentum keeps the dimensional chart explicit, according to (15.14). The previously constructed action, its angular distribution, and the fluctuations of a defined operator realization are thereby related.

Section 16 proves exact conservation of quadratic action under transport between ellipses, derives its Hamiltonian connection term, and establishes the positivity condition of the generator. The conformally symplectic version preserves normalized action and specifies the transported form.

Momentum–winding exchange prolongs this reciprocity in a lattice space with radius. Its proof preserves the carry quotient, operator domain, and unitary transformation. The corresponding modular function is evaluated on the radius thus transported. Exceptional incidence is developed through to Moonshine as part of the second realization of the common structure. The connections between these realizations are presented through their concrete maps, allowing what changes and what is preserved to be distinguished.

Comparison of the quark and lepton sectors requires separately following the antecedents that the angular development and species operators transmit to CKM and PMNS. Each transport preserves its own frame; this distinction makes it possible to specify where amplitude conjugation, route orientation, and mixing invariants act. The manuscript maintains construction differences in passing from structural coordinates to the respective realizations.

1.7 Identities for area, entropy, and gravitational scale

The relation between information and gravitation is expressed through three articulated determinations. The areal unit satisfies $\ell_P^2 = G\hbar/c^3$. Oriented incidence organizes its elementary resolution through $\gamma_{\text{HMT}}a_0$. The reduced state and bipartition determine accessible information, whose dimensional expression uses k_B . In the sectorial collection developed in 19.10, these dependencies take the form

$$\langle \hat{\mathcal{A}} \rangle = \gamma_{\text{HMT}} a_0 \frac{G\hbar}{c^3} \langle N \rangle, \quad S = k_B (\log Z + \Delta_{\text{mod}} \langle D \rangle). \quad (1.1)$$

The quantities N and D are the two readings of boundary occupations; Z normalizes the state. Their joint map recovers those occupations through (19.8). The adjacent proofs also show how their fluctuations govern the responses of area and entropy.

The next step preserves its explicit geometric data. In the horizon realization of Section 26, $S_H/k_B = A_H/(4\ell_P^2)$ and $T_H S_H = E_\Lambda$. Microscopic area, reduced-state entropy, and horizon entropy retain their definitions and realization maps. This organization allows incidence, action, information, and gravitation to be read continuously, specifying at each transition the object preserved and the observable that changes.

The scalar gravitational section also has a tensor realization. Incidence descent and the rank-five projector in 21.8 determine a five-component carrier; its intertwiner realizes it as symmetric trace-free tensors before the rank-two transverse reduction. Component orientation and reduced observation thus remain distinct from the common coupling. Microscopic area and sector information, constructed in 19 and 19.10, retain their definitions when these transports are composed. Exceptional continuity in 28.2 and 28.9 preserves incidence up to Moonshine; the duality in 29 preserves the unitary action and operator domains under momentum–winding exchange.

Notation and domains of the constructions

The table identifies the main symbols by their domain and the definition introducing them. Subscripts, typography, and operator context are part of the notation. References point to the effective definitions; anticipatory occurrences retain that same meaning.

Symbol	Object and meaning	Definition and location
$\mathcal{U}_t$	Transition on the state with memory: selection, transport, and record update, in that order.	(2.12).
$\mathcal{R}_{12}, R_{\text{sgn}}$	Twelve-window record with route, orientation, carry, and incidences; signed reader with codomain $\mathbb{Z}^{12}$.	§8.2; composition (8.7).
A_m, C_m, V_m	Three integer counts per window: residual visits, difference of oriented traces, and signed boundary incidence.	§8.3, definition of the counts.
z_m	Decimal incidence block in $\{0, \dots, 999\}$: $100[A_m]_{10} + 10[C_m]_{10} + [V_m]_{10}$.	(8.4).
$U, U^{(r)}$	Signed integer vector of twelve coordinates and its three four-coordinate harmonic blocks.	(8.9) and (8.10).
K, K_j	Integral record $K = \mathcal{H}_{12}U/4$, with fixed phase and orientation; coordinate prolongation by $K_{j+12} = K_j$.	§8.7, definition of the integral transformation; §9.1.1.
$\mathcal{K}_{\text{ph}}$	Transfer kernel on $\mathcal{U}_6 \times C_9$, also called K_{ph} when describing the reduced-memory representation.	§2.6, “Reduced transfer”; return (2.60).
A_j, c_j	Base-1000 block of α and integer quotient of Euclidean division; index j runs over the positional expansion.	(9.1).
$\mathbf{z}_m, y_m$	Accumulated vector memory in $\mathbb{R}^{12}$ and coordinate in the moving frame: $y_m = S^{-m}\mathbf{z}_m$.	§10, increment proposition; update (10.2).
$S, R = S^{-1}$	Cyclic permutation of the memory basis, $S\mathbf{e}_m = \mathbf{e}_{m+1}$; R denotes its inverse within the resolvent development.	§10; §10.2, opening paragraph.
η_m	Vector increment produced by the oriented event: $\eta_m = a_m S^{-1}\mathbf{e}_0$.	(10.1) and (10.2).
$U : X \rightarrow X, A$	In memory reduction, U is the state update; $A : V_0 \rightarrow V_0$ is the retained block of its linear extension $T\delta_x = \delta_{U(x)}$.	§2.4, “Memory elimination”; Proposition 2.19.
$U : \mathcal{H} \rightarrow \mathcal{H}$	Isometric realization of one-turn transport, $U^*U = I$, in observable compression.	§4, “Observable compression”; (4.7).
$R = \mathbb{Z}/9\mathbb{Z}$	APP residue ring, with maximal ideal $\mathfrak{m} = (3)$.	§2, “Ternary reduction and multiplicative radical”.
Local R, η	In the central block $B_c = 7I + 2R$, R exchanges the two coordinates and $R^2 = I$. The rapidity of its normalization is $\eta = \frac{1}{2} \log(9/5)$.	§4, “Local development”: $B_c/\sqrt{45} = \exp(\eta R)$.
Σ_H	Dimensionless norm-and-self-scale reader $\Sigma_H = \varphi\alpha^{16}$; its decimal valuation is $\text{ord}_{10} \Sigma_H = -34$.	(13.2) and (13.3).

Symbol	Object and meaning	Definition and location
$\mathcal{L}_S, \mathcal{U}_S$	Oriented action line and positive basis in which its sections are expressed.	§13, “Action sections and dimensionless return quotient”.
H_5, η_{ret}	Dimensionless action coefficients before and after return; $\eta_{\text{ret}} = H_5 - (90/\pi)\mathcal{C}_\pi(\alpha)$.	(13.4)–(13.7).
$\hbar_{\text{pre}}, \hbar_{\text{ret}}, h_a$	Sections $H_5 10^{-34}\mathcal{U}_S$ and $\eta_{\text{ret}} 10^{-34}\mathcal{U}_S$. For $a \in \{\text{pre}, \text{ret}\}$, $h_a = 2\pi\hbar_a$ expresses action per turn.	(13.9); §13, passage to action per turn.
$R_{\text{act}}, \delta_{\text{ret}}$	Multiplicative ratio H_5/η_{ret} and additive difference $H_5 - \eta_{\text{ret}}$ of the return coordinates.	(13.10).
$A_{\text{deg}}, C_{\text{deg}}^*$	Angular coordinates 1000α and $2(\eta_{\text{ret}} + \alpha)$. Their radian coordinates are $x = \pi A_{\text{deg}}/180$, $y = \pi C_{\text{deg}}^*/180$.	(13.14); §13.6, definition of the charts.
$\gamma_{\text{HMT}}, \gamma$	Conjugate evaluation of the oriented chain c_{12}^+ at $q_\pm$, with angular component $\pi AC^*/180$, for $0 < q_\pm < 1$. The positive area increment uses $ \gamma_{\text{HMT}} $.	§14.4; evaluation (14.2).
$\eta_{\text{el}}, a_S, b_S$	Shape parameter $\text{artanh}(C^*/A)$ and balanced canonical semiaxes of the ellipse, with $a_S b_S = 2\hbar$.	§15, initial definition.
$R_{\text{el}}, \tau_{\text{el}}$	Dimensionless modular radius $\sqrt{b_S/a_S}$ and modulus $\tau_{\text{el}} = iR_{\text{el}}^2$.	§15, reciprocity of the ellipse.
K_a^i	Extrinsic curvature in the normal convention of the geometric realization; enters $\mathcal{A}_a^i = \Gamma_a^i(E) + \gamma K_a^i$.	§21.7, definition of the connection.
L_*, L, ℓ_P	Positive longitudinal reference and length $L = (5759/23040)\alpha_{\text{HMT}}^{16}L_*$. Realization R1–R8 determines $\ell_P^2 = L^2$; the SI evaluation declares $L_* = 1$ m.	(20.6) and (20.16).
R_X, Λ_X, E_L	Positive radial reading, operator-valued conjugate length, and energy scale $E_L = \hbar_{\text{ret}}c_{\text{int}}/L$. They satisfy $R_X\Lambda_X = L^2I$.	(20.10) and (20.15).
R_{108} , horizon R	Circular radius $c_{\text{int}}/\omega_{108}$ of the dodecaphasic period, equal to L for the conjugate clock; $R > 0$ is the radial variable of the horizon realization.	§21, initial definition; (26.2).
G_a	Gravitational section reciprocal to an action $\hbar_a > 0$: $G_a = c_{\text{int}}^3 L^2/\hbar_a$, with $R_{108} = L$, $c_{\text{int}} > 0$ and radius coincidence in the convention $r_g = Gm/c_{\text{int}}^2$.	§21.13; (21.56) and (21.57).
$\mathbb{G}_{5,a}$	Operator $G_a P_5^{\text{ico}}$ on $V_{12} = \mathbb{R}[A_5/C_5]$. The orthogonal projector selects the summand V_5 ; section G_a is common to its five components.	(21.37); Theorem 21.16.
$k_B, \Theta_{\text{clk},a}$	Positive linear transducer $k_B : \mathcal{L}_\Theta \rightarrow \mathcal{L}_E$ and positive thermal section, related by $k_B \Theta_{\text{clk},a} \log 3 = h_a/(108t_0)$. The thermal section and declared bases determine its individual coordinate.	§25.10; (22.8) and (22.9).

In the incidence record, windows carry indices $1, \dots, 12$. In the memory update they are numbered from 0 to 11, and their prolongation uses $m \geq 0$. The index shift preserves the same partition into nonadic windows. In angular formulas, A, C^* retain the chart declared in each expression; the quotient C^*/A is invariant under the common conversion between degrees and radians.

2 Formal kernel of Triadic Modular Holography

Triadic Modular Holography, abbreviated HMT, begins with a finite structure of positions and operations. The order of construction matters: positions receive arithmetic evaluations, transitions acquire a ternary signature, and the dynamics preserves the information needed to prolong them. Numerical realizations are formed only afterwards. A constant recognized at the end of this process does not replace the rule that produced its region or the history that determines its evaluation.

The name *All Possible Paths*, APP, expresses the role of paths on the support. It is adopted as the name of the construction, with a conceptual reference to the path formulation of quantum mechanics; it does not imply that its arithmetic law is Feynman's path integral. The *Topological Phase Kernel*, TPK,¹ denotes the composition of selection, transport, memory update, and prolongation. The acronyms are retained only after the objects they represent have been defined.

2.1 APP: positional support and arithmetic evaluations

Consider the nine interior marks on a line graduated from zero to ten,

$$D_9 = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}.$$

Their horizontal and vertical arrangement produces the eighty-one positions of D_9^2 . The interior marks must be distinguished from the ten unit intervals of the reference line. This arrangement fixes an alphabet and two directions; it does not yet use a real expansion to determine a trajectory.

Cyclic identification of the positions yields the support

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2.$$

The chart of positive representatives retains the symbol nine for the zero class. For every positive integer, define

$$\rho_9(n) = 1 + ((n - 1) \bmod 9), \quad q_9(n) = \frac{n - \rho_9(n)}{9}, \quad n = \rho_9(n) + 9q_9(n). \quad (2.1)$$

The positive digital root is therefore a choice of representative of the residue class. By itself, it does not preserve the integer before reduction; the coordinate q_9 supplies precisely the missing information.

At each position (i, j) , accumulating i , repeated j times, produces ij . The two integer evaluations and their projections are

$$S(i, j) = i + j, \quad P(i, j) = ij, \quad \Sigma(i, j) = \rho_9(i + j), \quad \Pi(i, j) = \rho_9(ij).$$

The symmetry $(i, j) \mapsto (j, i)$ interchanges the directions of accumulation and preserves both evaluations. This is a property of a single support, not a subsequent identification between two independent matrices.

Definition 2.1 (Lifted arithmetic evaluation). The APP evaluation that preserves both sheets is

$$\mathcal{A}(i, j) = ((R_{ij}^+, q_{ij}^+), (R_{ij}^\times, q_{ij}^\times)) = ((\rho_9(i + j), q_9(i + j)), (\rho_9(ij), q_9(ij))). \quad (2.2)$$

Here the term *sheet* denotes an evaluation on the same support: additive or multiplicative. The two quotient coordinates belong to the arithmetic state and are not discarded when the residues are reduced.

Proposition 2.2 (Reconstruction of the evaluations). *The two pairs in (2.2) determine $i + j$ and ij exactly. The residues alone do not determine these integer evaluations or their quotients.*

Proof. The first statement is (2.1) applied to each sheet. For the second, $10 = 1 + 9$ and $19 = 1 + 2 \cdot 9$ have the same residue and different quotients. On the two-dimensional support, positions $(5, 5)$ and $(8, 2)$ give the same sum and the same product residue, but products $25 = 7 + 18$ and $16 = 7 + 9$. The residue projection identifies data that the lifted evaluation distinguishes. $\square$

¹The authors also dedicate the initials TPK to Tan Pengkiang. This dedication is independent of the mathematical definition.

	1	2	3	4	5	6	7	8	9	
1	1	2	3	4	5	6	7	8	9	One support: 81 positions.
2	2	4	6	8	1	3	5	7	9	Two evaluations: $S(i, j) = i + j, P(i, j) = ij.$
3	3	6	9	3	6	9	3	6	9	Each row is generated by accumulation: $P(i, j + 1) - P(i, j) = i.$
4	4	8	3	7	2	6	1	5	9	
5	5	1	6	2	7	3	8	4	9	
6	6	3	9	6	3	9	6	3	9	The chart represents 0 (mod 9) by 9.
7	7	5	3	1	8	6	4	2	9	The highlighted block is $B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix},$
8	8	7	6	5	4	3	2	1	9	with eigenvalues 9 and 5.
9	9	9	9	9	9	9	9	9	9	

Figure 1: APP multiplication table in positive representatives. Non-unit rows and columns are distinguished by gray shading; the frame identifies the adjacent diagonal block with equal diagonals. The displayed table is an observable of the arithmetic torus; its representation edges do not constitute an intrinsic topological boundary of the torus.

2.1.1 Cayley graph, toroidal topology, and winding number

The cardinal translations are given, in row and column coordinates, by

$$\nu_N = (-1, 0), \quad \nu_E = (0, 1), \quad \nu_S = (1, 0), \quad \nu_O = (0, -1), \quad T_d(x) = x + \nu_d.$$

The first coordinate increases southwards and the second eastwards. The graph

$$\mathcal{G}_9 = \text{Cay}(X_9, \{\nu_N, \nu_E, \nu_S, \nu_O\})$$

is the two-dimensional cyclic lattice: it has eighty-one vertices and four outgoing oriented edges per vertex. It is a Cayley graph for the *additive* structure of the support. Multiplication of residues has zero divisors and does not turn all APP rows into a multiplicative group.

A cardinal word $w = d_1 \cdots d_r$ and an origin x_0 determine

$$x_k = x_0 + \sum_{j=1}^k \nu_{d_j} \pmod{9}.$$

There are $81 \cdot 4^r$ oriented paths of length r , including returns and backtracking. When the path closes, its integer displacement before reduction defines

$$\text{wind}(w) = \frac{1}{9}(\#S(w) - \#N(w), \#E(w) - \#O(w)) \in \mathbb{Z}^2. \quad (2.3)$$

Path reversal changes the sign of this vector. Returning to the same residue position therefore does not require the winding to vanish. This is the first example of a distinction that will reappear in phase return: one coordinate may close while another records a nontrivial evolution.

2.1.2 Transport quotient and cocycle identity

Let $s_9 : \mathbb{Z}/9\mathbb{Z} \rightarrow D_9$ be the section of positive representatives. The transport quotient associated with additive composition, also called the *carry* in positional arithmetic, is

$$c_9(a, b) = \frac{s_9(a) + s_9(b) - s_9(a + b)}{9}.$$

The compositional character of this memory is expressed by an exact identity.

Lemma 2.3 (Cocycle identity). *For $a, b, c \in \mathbb{Z}/9\mathbb{Z}$,*

$$c_9(a, b) + c_9(a + b, c) = c_9(b, c) + c_9(a, b + c).$$

Proof. Both sums equal

$$\frac{s_9(a) + s_9(b) + s_9(c) - s_9(a + b + c)}{9}.$$

Associativity of the sum of lifted representatives requires precisely this additional term when composing in the quotient. $\square$

With positive representatives, this cocycle is not normalized at the identity: $c_9(0, a) = 1$. If s_0 takes representatives in $\{0, \dots, 8\}$, its normalized carry c_0 satisfies

$$c_9(a, b) = c_0(a, b) + \mathbf{1}_{a=0} + \mathbf{1}_{b=0} - \mathbf{1}_{a+b=0}.$$

The difference is the change of section; it does not alter the cocycle identity. The subsequent decomposition of the calendar into phase and window will use representatives from zero to eight.

The totals of both evaluations distinguish the visible contribution from that retained in the quotient:

$$\begin{aligned} \sum S &= 810, & \sum P &= 2025, & \sum \Sigma &= 405, & \sum \Pi &= 459, \\ \sum q^+ &= 45, & \sum q^\times &= 174. \end{aligned}$$

It follows that

$$2025 - 810 = (459 - 405) + 9(174 - 45) = 54 + 9 \cdot 129. \quad (2.4)$$

The sum–product difference is not exhausted by the visible defect 54. The identity also preserves the quotient defect 129. Removing it would change the arithmetic being transported, even if the residue table remained identical.

2.1.3 Ternary reduction and multiplicative radical

The ring $R = \mathbb{Z}/9\mathbb{Z}$ has unit group $\{1, 2, 4, 5, 7, 8\}$ and maximal ideal

$$\mathfrak{m} = (3) = \{0, 3, 6\}, \quad \mathfrak{m}^2 = 0.$$

Every additive row is a permutation of the nine representatives. A multiplicative row is such a permutation exactly when its index is a unit. Rows three and six contain three repeated values; row nine contains only the representative of the zero class. Thus multiplicative folding already distinguishes the units from the nilpotent sector before any continuous geometry is introduced.

The map $x \mapsto 3x$ has kernel and image both equal to $\mathfrak{m}$. It induces an isomorphism of vector spaces over $\mathbb{F}_3$,

$$R/\mathfrak{m} \longrightarrow \mathfrak{m}, \quad [x] \longmapsto 3x,$$

but not a ring isomorphism. In particular, the visible sequence 3, 6, 9 admits two different readings: its direct reduction modulo three is 0, 0, 0, whereas extracting the factor three and then reducing its internal coordinate gives 1, 2, 0. This second operation preserves a coordinate of the ideal; it is not the same projection as directly reducing the original number.

2.1.4 Latin realization of the additive evaluation

The difference between the two evaluations can also be expressed through operators. Let $(e_r)_{r \in \mathbb{Z}/9\mathbb{Z}}$ be the orthonormal basis of $\mathbb{C}^9$. The assignment $v_{ab} = e_{a+b}$ produces an orthonormal basis in every row and column. The projectors $P_{ab} = v_{ab}v_{ab}^*$ satisfy

$$\sum_b P_{ab} = I_9 = \sum_a P_{ab}, \quad P_{ab}^2 = P_{ab}.$$

Indeed, each translation $b \mapsto a + b$ permutes the nine residues. This yields a realization of a Latin square by rank-one projectors, in the terminology of quantum Latin squares [3]; [5]. This construction starts from the additive sheet already defined and makes explicit the map to its operator representation.

The multiplicative evaluation preserves different content: for $a = 3$, the vectors e_{3b} repeat the residues 0, 3, 6, and

$$\sum_{b \in \mathbb{Z}/9\mathbb{Z}} |e_{3b}\rangle\langle e_{3b}| = 3(|e_0\rangle\langle e_0| + |e_3\rangle\langle e_3| + |e_6\rangle\langle e_6|).$$

The multiplicity is recorded in the operator and characterizes the multiplicative radical. Thus the toroidal support, the Latin property, and the operator resolution of the identity are properties requiring different definitions. The quantum magic squares studied by De las Cuevas, Drescher, and Netzer require positive entries and row and column sums equal to the identity; their theory also distinguishes dilations with commuting entries[4]. This reference provides a precise language for comparing realizations of APP.

2.2 TRIT: orientation and quadratic regimes

Definition 2.4 (Ternary signature and local lift). The TRIT signature uses the balanced identification

$$\mathbb{F}_3 \longrightarrow \{-1, 0, +1\}, \quad 0 \mapsto 0, \quad 1 \mapsto +1, \quad 2 \mapsto -1.$$

For $n \in \mathbb{Z}$, its lift preserves $n = r + 3c$, where $r \in \{-1, 0, +1\}$ and $c \in \mathbb{Z}$. In a transition of bounded local amplitude, the three states $(0, -1)$, $(0, 0)$, $(0, +1)$ may exhibit the same zero residue and different carries.

The signature does not replace the trajectory. A zero residue does not mean an absence of orientation, phase, or quotient information. Nor does it introduce a physical quantity by itself. Its function is to distinguish algebraically the transport classes that will be realized next.

2.2.1 Three quadratic regimes and a common exponential

Once the ternary signature has been obtained, its quadratic realization is defined by

$$\mathbb{A}_\tau = \mathbb{R}[X]/(X^2 + \tau), \quad \tau \in \{-1, 0, +1\}. \quad (2.5)$$

If J_τ is the class of X , then $J_\tau^2 = -\tau I$. The structure distinguishes the elliptic type $\tau = +1$, the parabolic type $\tau = 0$, and the hyperbolic type $\tau = -1$. The real realization does not generate the signature: it expresses the types already produced by oriented reduction.

Proposition 2.5 (Exponentiation of the three types). *In each algebra (2.5),*

$$\exp(tJ_\tau) = C_\tau(t)I + S_\tau(t)J_\tau, \quad (2.6)$$

where

$$C_\tau(t) = \sum_{n \geq 0} \frac{(-\tau)^n t^{2n}}{(2n)!}, \quad S_\tau(t) = \sum_{n \geq 0} \frac{(-\tau)^n t^{2n+1}}{(2n+1)!}.$$

The three realizations are

τ	J_τ^2	$\exp(tJ_\tau)$
+1	$-I$	$\cos t I + \sin t J_\tau$
0	0	$I + t J_\tau$
-1	I	$\cosh t I + \sinh t J_\tau$

Proof. Separating the even and odd terms of the exponential series and substituting $J_\tau^{2n} = (-\tau)^n I$ and $J_\tau^{2n+1} = (-\tau)^n J_\tau$ gives the formulas. For $\tau = 0$, all terms of degree at least two vanish. In the other two cases, the trigonometric or hyperbolic series are recovered. $\square$

The exponential is common to all three regimes: it is not assigned exclusively to the parabolic branch. The invariants

$$C_\tau(t)^2 + \tau S_\tau(t)^2 = 1$$

follow from $\exp(tJ_\tau)\exp(-tJ_\tau) = I$. The same algebraic principle preserves unity through realizations that differ in their quadratic type.

In the hyperbolic regime, $P_\pm = (I \pm J_{-1})/2$ are complementary projectors and

$$\exp(tJ_{-1}) = e^t P_+ + e^{-t} P_-.$$

Growth and contraction appear jointly in two eigendirections. In the elliptic regime, the parameter describes a rotation; in the parabolic regime, a nilpotent translation. The temporal interpretation adopted by HMT associates +1 with return with memory, 0 with the threshold, and -1 with bidirectional opening. This interpretation requires an ordering of the parametrization; it does not add a fourth algebraic type.

2.2.2 Balanced division and composition of lifts

Ternary reduction expresses an integer evaluation of APP through a visible coordinate and a prolongation coordinate. Its function is not merely to abbreviate the notation for an integer: it provides a composition rule determining how the quotient changes when evaluations are added or multiplied. The balanced choice also makes the compatibility of this rule with sign reversal explicit.

Definition 2.6 (Balanced ternary coordinates). For $n \in \mathbb{Z}$, let

$$q_3(n) = \left\lfloor \frac{n+1}{3} \right\rfloor, \quad r_3(n) = n - 3q_3(n).$$

The map

$$\mathcal{B} : \mathbb{Z} \longrightarrow \{-1, 0, +1\} \times \mathbb{Z}, \quad n \longmapsto (r_3(n), q_3(n))$$

is called the balanced ternary decomposition. Its inverse is $\mathcal{L}(r, c) = r + 3c$.

Indeed, $r_3(n) \in \{-1, 0, +1\}$. If two pairs represent the same integer, $r + 3c = s + 3d$, then $r - s$ is a multiple of 3 and belongs to the interval $[-2, 2]$; necessarily $r = s$ and $c = d$. The representation is therefore unique. Division can be repeated on $q_3(n)$, so that each level preserves the data needed to reconstruct the preceding one. The quotient is not a perturbation of the residue: it is its complementary coordinate.

The signature obtained from an evaluation F , with $F = S$ or $F = P$ in APP, is $r_3 \circ F$. When reduction acts on the phase cursor, the same balanced section produces the classes $\{1, 4, 7\}$, $\{2, 5, 8\}$, and $\{3, 6, 9\}$, with values +1, -1, 0. The TPK selector defined below uses these classes to determine the operating mode. Reading an evaluation and selecting a mode thus share the same ternary reduction, but have different domains and functions.

Proposition 2.7 (Arithmetic of balanced pairs). Let $n = r + 3c$ and $m = s + 3d$, with $r, s \in \{-1, 0, +1\}$. Define

$$\chi(r, s) = \frac{r + s - r_3(r + s)}{3}.$$

Integer operations are transported to the pairs by

$$(r, c) \boxplus (s, d) = (r_3(r + s), c + d + \chi(r, s)), \quad (2.7)$$

$$(r, c) \boxtimes (s, d) = (rs, rd + sc + 3cd). \quad (2.8)$$

The map $\mathcal{L}$ preserves both operations.

Proof. The equality $r + s = r_3(r + s) + 3\chi(r, s)$ gives

$$n + m = r_3(r + s) + 3(c + d + \chi(r, s)).$$

On the other hand,

$$nm = rs + 3(rd + sc + 3cd).$$

Since the product of two balanced representatives again belongs to $\{-1, 0, +1\}$, it requires no additional residue reduction. Uniqueness of the decomposition proves both formulas. $\square$

For example, $2 = (-1) + 3$ and $4 = 1 + 3$. Their sum and product are reconstructed as

$$(-1, 1) \boxplus (1, 1) = (0, 2), \quad (-1, 1) \boxtimes (1, 1) = (-1, 3).$$

The resulting evaluations are 6 and 8. Adding the quotients suffices in this additive example; the product involves the cross terms rd , sc , and $3cd$. The discrepancy between the two operations also resides in the prolongation coordinate, not only in the residue.

The term χ is a cocycle of the balanced section. If $r \oplus s = r_3(r + s)$, it satisfies

$$\chi(r, s) + \chi(r \oplus s, t) = \chi(s, t) + \chi(r, s \oplus t).$$

Both sides represent the quotient of the same sum of three integers. This identity ensures that regrouping the operations does not change the reconstructed evaluation. At the same time, projection onto the first component suppresses this accounting: $1 + 1 = -1 + 3$, whereas the residue reading retains only -1 . The information removed is determined, in this case, by $\chi(1, 1) = 1$.

The relation with modulus 9 preserves a second discrete scale. The map

$$\beta : \{-1, 0, +1\} \times \mathbb{F}_3 \longrightarrow \mathbb{Z}/9\mathbb{Z}, \quad (r, \bar{c}) \longmapsto r + 3c \pmod{9}$$

is well defined and bijective. Changing the representative of $\bar{c}$ adds a multiple of 9; conversely, equality of images fixes r first, by reduction modulo 3, and then $\bar{c}$. Addition transported by this bijection contains the term $\chi(r, s)$ from (2.7), now reduced modulo 3. Thus the two ternary levels do not compose by independent coordinatewise addition: normalization of the first level modifies the second.

In particular, the classes 3, 6, and 0 of $\mathbb{Z}/9\mathbb{Z}$ have first component 0, but second components 1, 2, and 0, respectively. The distinction already observed between direct reduction and extraction of the ideal coordinate appears as a property of β . Preserving this coordinate prevents the zero residue sector from being artificially collapsed into a single position. Modulus 1000, used later to organize decimal blocks, serves a different positional-representation function; its quotient is preserved through the corresponding division. A change of base preserves the integer when it retains residue and quotient, whereas coincidence of residues in different bases does not by itself identify their transport states.

2.2.3 Inversion, residue class, and non-visible coordinates

Additive inversion lifts unambiguously:

$$\mathcal{B}(-n) = (-r_3(n), -q_3(n)).$$

In particular, 3, 0, and -3 have coordinates $(0, 1)$, $(0, 0)$, and $(0, -1)$. They share a zero signature but represent different evaluations. In a transition restricted to a quotient of maximum amplitude 1, these are the three possibilities of the visible fiber over zero; in a general integer lift, that fiber contains all pairs $(0, c)$, with $c \in \mathbb{Z}$.

The quotient coordinate recovers the evaluated integer, but does not by itself recover the trajectory that produced it. Two APP paths may share evaluation, residue, and quotient while differing in orientation, sequence of modes, or winding. Thus decomposition $\mathcal{B}$ is integrated into the transport state rather than replacing it. The arithmetic memory of a division and the memory of a trajectory are related coordinates, not synonyms.

For a signature sequence of length n , oriented reversal is

$$C_{\text{rev}}(\tau_1, \dots, \tau_n) = (-\tau_n, \dots, -\tau_1).$$

Applying it twice restores the sequence. The operation reverses the order of positions and the sign of each signature; it preserves the length and the zero positions in reversed order. In the adopted parametrization, it interchanges return with memory, associated with $+1$, and bidirectional opening, associated with -1 , while retaining the threshold 0. Transport of the remaining trajectory data is specified by the corresponding TPK involution.

This signature reversal must be distinguished from conjugation within a quadratic algebra. The former may change the type τ ; the latter, constructed below, acts within a type already fixed. This distinction prevents attributing an algebraic equivalence between different regimes to a sign reversal.

2.2.4 Quadratic product, conjugation, and algebraic norm

An element of $\mathbb{A}_\tau$ has a unique expression $aI + bJ_\tau$. The relation $J_\tau^2 = -\tau I$ determines its product:

$$(aI + bJ_\tau)(cI + dJ_\tau) = (ac - \tau bd)I + (ad + bc)J_\tau. \quad (2.9)$$

This is a real realization of the type already selected. The coordinates a, b are not APP residues, and the product (2.9) does not replace (2.8); it expresses another operation in the declared quadratic codomain.

Definition 2.8 (Conjugation and algebraic norm). Conjugation of type τ and its algebraic norm are

$$\overline{aI + bJ_\tau} = aI - bJ_\tau, \quad N_\tau(aI + bJ_\tau) = a^2 + \tau b^2.$$

Proposition 2.9 (Multiplicativity and invertibility). For $z, w \in \mathbb{A}_\tau$,

$$z\bar{z} = N_\tau(z)I, \quad N_\tau(zw) = N_\tau(z)N_\tau(w).$$

Moreover, z is invertible if and only if $N_\tau(z) \neq 0$, in which case

$$z^{-1} = \frac{\bar{z}}{N_\tau(z)}.$$

Proof. The product of $aI + bJ_\tau$ and $aI - bJ_\tau$ is $(a^2 + \tau b^2)I$. Conjugation is multiplicative and the algebra is commutative, so that $(zw)\bar{z}\bar{w} = (z\bar{z})(w\bar{w})$. If the norm does not vanish, the stated formula supplies the inverse. Conversely, if $zw = I$, multiplicativity implies $N_\tau(z)N_\tau(w) = 1$. $\square$

The quadratic form N_τ is positive definite only in the elliptic type. For $\tau = 0$, it equals a^2 , and the element J_0 is nonzero although its square and norm are zero. For $\tau = -1$, it equals $a^2 - b^2$; the nonzero elements $I + J_{-1}$ and $I - J_{-1}$ have zero norm and their product vanishes. Here the algebraic norm is a multiplicative quadratic form: it does not presuppose a positive vector-space norm.

The three types admit faithful realizations by

$$aI + bJ_\tau \mapsto \begin{pmatrix} a & -\tau b \\ b & a \end{pmatrix}.$$

The determinant of this matrix is $N_\tau(aI + bJ_\tau)$. The complex structure, the dual-number structure, and the split real structure are thus represented by a single matrix expression. Each preserves its own law and zero divisors. The form of signature $(1, 1)$ in the last case admits a Lorentzian interpretation in dimension $1 + 1$; identification of a physical metric additionally requires a specific space, field, and transport rule.

2.2.5 Multiplicative conservation and composition of evolutions

The exponential already obtained belongs to the set of norm-one elements:

$$N_\tau(\exp(tJ_\tau)) = 1.$$

This conservation persists under composition of parameters. Indeed, all factors are functions of the same generator, and

$$\exp(tJ_\tau)\exp(sJ_\tau) = \exp((t + s)J_\tau).$$

Comparing the two components gives the addition laws

$$C_\tau(t + s) = C_\tau(t)C_\tau(s) - \tau S_\tau(t)S_\tau(s), \quad (2.10)$$

$$S_\tau(t + s) = S_\tau(t)C_\tau(s) + C_\tau(t)S_\tau(s). \quad (2.11)$$

Conjugation changes t to $-t$ and coincides with inversion within this subgroup. It does not change τ : it reverses an evolution of the chosen regime.

In the elliptic type, the preserved form is $a^2 + b^2$, and the exponential trajectory traverses the unit-norm circle. In the hyperbolic type, the eigencoordinates are $u = a + b$, $v = a - b$, with $uv = N_{-1}(z)$. The exponential gives

$$(u, v) = (e^t, e^{-t}), \quad uv = 1.$$

Growth of one coordinate is accompanied by contraction of the other. Conservation expresses a relation between the two coordinates; it does not assert that each remains constant.

In the parabolic type,

$$(I + tJ_0)(I + sJ_0) = I + (t + s)J_0, \quad (I + tJ_0)^{-1} = I - tJ_0.$$

The evolution does not stop because $J_0^2 = 0$. Nilpotency removes terms of order at least two, but preserves the linear term that transports the parameter. The visible symbol 0, the zero element of the algebra, and the nonzero generator J_0 are distinct. This distinction is necessary to describe the threshold without identifying it with an absence of state or dynamics.

Proposition 2.10 (Composition of affine coordinates). *In the chart where the scalar component does not vanish, represent a direction by $I + uJ_\tau$. If $1 - \tau uv \neq 0$, the direction of its product with $I + vJ_\tau$ has coordinate*

$$u \star_\tau v = \frac{u + v}{1 - \tau uv}.$$

Proof. We have

$$(I + uJ_\tau)(I + vJ_\tau) = (1 - \tau uv)I + (u + v)J_\tau.$$

Dividing by the scalar component gives the stated expression. $\square$

The affine chart preserves the ratio of components, not the scalar factor divided out. When $1 - \tau uv = 0$, the product must be examined before changing charts: it may have a well-defined direction with zero scalar component, or be the zero element in the presence of zero divisors. The composition law still belongs to the full algebra; a coordinate singularity does not automatically imply a singularity of the object. This formula will reappear in the elliptic regional reading. Its use requires retaining the denominator and, where applicable, the normalization factor.

2.2.6 Transport of type and subsequent realizations

The algebras $\mathbb{A}_{+1}$, $\mathbb{A}_0$, and $\mathbb{A}_{-1}$ are not isomorphic as unital real algebras. The first is a field; the second contains a nonzero nilpotent; the third has nontrivial idempotents and is isomorphic to $\mathbb{R} \times \mathbb{R}$. Thus a change of signature organizing a route cannot be interpreted as an undifferentiated isomorphism between the three realizations. The TPK preserves the regime index, the domains, and the effective operator acting at each transition.

In the electron chapter, the pair of projections of the APP sheets produces a normalized commutator whose square is $-I$. The same block contains an involution of square I and two nilpotent directions. The joint realization will be proved for that concrete pair of projections; the preceding classification explains why the three quadratic relations can appear within one matrix algebra without being identified with one another.

In the exceptional prolongation, the Paley matrix reduced modulo 3 satisfies $A_W^2 = -I_6$. This relation endows the ternary space with a structure of dimension 3 over $\mathbb{F}_9$. Its connection to a real operator root is expressed through the integral presentation

$$\mathbb{Z}[u]/(u^2 + 1).$$

Base changes to $\mathbb{F}_3$ and $\mathbb{R}$ produce representations of the same relation. Each retains its characteristic, dimension, and incidence information. Developing the respective operators establishes the connection, not equality of their cardinalities.

Norm-one conservation consequently describes the quadratic realizations of evolutions. Conservation of quotients, orientation, and trajectory belongs to the transported arithmetic state. Both properties participate in the construction, with different functions: one determines the identities of the representations; the other makes it possible to compose, reconstruct, and prolong the operations that give rise to them.

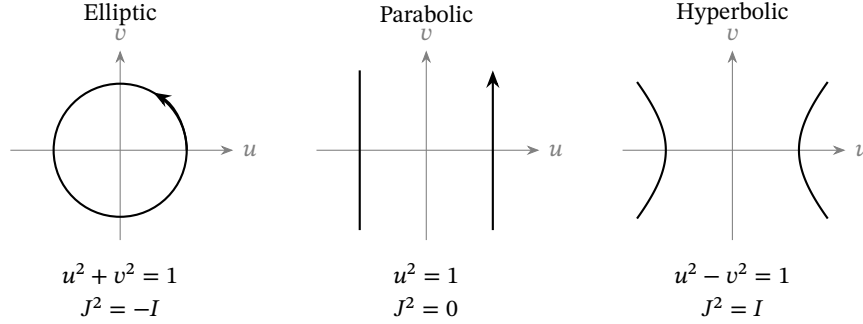


Figure 2: Quadratic realizations of TRIT. For $J_\tau = \begin{pmatrix} 0 & -\tau \\ 1 & 0 \end{pmatrix}$, the exponential preserves $u^2 + \tau v^2$. Parabolic degeneration preserves u and transports v linearly. The three diagrams represent orbits of quadratic forms; their geometric type corresponds to the algebraic relation of the generator.

2.3 TPK: Topological Phase Kernel

The dynamics uses the phase selector

$$M_{\text{ph}}(r) = \begin{cases} +1, & r \in \{1, 4, 7\}, \\ -1, & r \in \{2, 5, 8\}, \\ 0, & r \in \{3, 6, 9\}. \end{cases}$$

Its value determines which evaluation operates. Cardinal transport moves the corresponding cursor; the record preserves the previous value, the new quotient, the orientation, and the transition. Selection and transport are therefore different functions: a ternary sign is not a displacement on the torus.

We shall denote by $\hat{X}$ the space of arithmetic states with trajectory memory. An element contains the positions and directions of both cursors, the active mode, the TRIT signature, the phase, the residues and quotients of both evaluations, the carries, the constructed prefix, and the boundary data. When a refinement cylinder is introduced, it forms part of the state. This notation avoids replacing the object by a reduced version that retains only the visible symbol.

Definition 2.11 (TPK update). Let Sel_t be mode selection through M_{ph} , Tra_t the lifted cardinal transport, and Mem_t the record update. The transition is

$$\mathcal{U}_t = \text{Mem}_t \circ \text{Tra}_t \circ \text{Sel}_t. \quad (2.12)$$

Each application preserves the fields it does not modify and updates the remaining fields by Euclidean division and trajectory concatenation.

The notation (2.12) expresses the order of composition; it does not replace the effective rules. The following section specifies its finite two-cursor realization completely. Prolongation will then be obtained not by erasing its memory and repeating the first emission, but by transporting the prefixes and their compatible boundaries. Thus APP supplies the evaluations, TRIT distinguishes their regimes, and TPK composes their evolution.

2.3.1 Dynamic state, observables, and operational dependencies

The Topological Phase Kernel organizes the evolution of the two APP sheets with the orientation determined by TRIT. Its definition comprises the state on which it acts, the transition rules, and the relations that make prolongations compatible. Equation (2.12) acquires operational content when the coordinates received and modified by each factor are specified. Position belongs to the toroidal support; phase belongs to the calendar; orientation determines transport; quotients record the arithmetic evaluation before reduction. These coordinates are related through the transition and retain different mathematical functions.

The observable projection of a state is

$$q = (x_+, d_+, x_-, d_-, \theta) \in \mathcal{Q}_{\text{obs}} := (X_9 \times \mathcal{D})^2 \times \Theta_{108}, \quad \mathcal{D} = \{N, E, S, O\}, \quad \Theta_{108} = \mathbb{Z}/108\mathbb{Z}. \quad (2.13)$$

The subscripts distinguish the additive and multiplicative cursors. The fibers of complete states over q preserve the ordered trajectory, the integer evaluations of both sheets, their residues and quotients, and the orientation and boundary records. A precision prolongation adds the prefix and the compatible cylinder. Thus two states with the same projection q may retain different records.

Operational dependencies have three levels. At the local level, the selector activates a sheet, transport changes its position, and the update computes the new arithmetic coordinates. At the trajectory level, these transitions compose and retain the records of preceding steps. At the refinement level, extension relations determine which continuation preserves the predecessor's prefix, signature, and boundary. The finite calendar describes the phase of these operations; holonomy describes their composition on the history.

2.3.2 Tritic selection and lifted cardinal transport

Let $\bar{\theta} \in \{0, \dots, 107\}$ be the calendar representative. The operating phase and the dodecaphase sector are

$$r(\theta) = 1 + (\bar{\theta} \bmod 9), \quad \beta(\theta) = \left[\left[\frac{\bar{\theta}}{9} \right] \right]_{12}. \quad (2.14)$$

The previously defined selector may be written $\varepsilon_9 = M_{\text{ph}}$. Its value vector in the ordered chart of D_9 is

$$m_{\text{ph}} = (1, -1, 0, 1, -1, 0, 1, -1, 0).$$

The function $M_{\text{ph}} : D_9 \rightarrow \{-1, 0, +1\}$ and this vector are two presentations of the same selector, with their respective domains.

Define the activation indicators

$$a_+(\theta) = \mathbf{1}_{\{1,4,7\}}(r(\theta)), \quad a_\times(\theta) = \mathbf{1}_{\{2,5,8\}}(r(\theta)).$$

Cursor transport is then

$$x'_+ = x_+ + a_+(\theta)v_{d_+}, \quad x'_\times = x_\times + a_\times(\theta)v_{d_\times} \quad \text{in } X_9. \quad (2.15)$$

In phases 3, 6, 9, both positions remain unchanged; the neutral event forms part of the chronology and its record. Cardinal directions determine the vectors v_d , whereas the phase signature determines which vector is applied.

The integer lift preserves boundary crossing. If $s : X_9 \rightarrow D_9^2$ chooses positive representatives and $x' = x + av_d$, with $a \in \{0, 1\}$, define

$$b_d(x, a) = \frac{s(x) + av_d - s(x')}{9} \in \mathbb{Z}^2. \quad (2.16)$$

Divisibility by 9 follows from equality in X_9 . If $z \in \mathbb{Z}^2$ records previous crossings, the update is $z' = z + b_d(x, a)$. Thus

$$s(x') + 9z' = s(x) + 9z + av_d.$$

The integer transport data remain reconstructible after the position has been reduced on the torus.

Proposition 2.12 (Conservation of lifted displacement). *For a cardinal trajectory of n steps, with activation indicators a_t ,*

$$s(x_n) + 9z_n = s(x_0) + 9z_0 + \sum_{t=0}^{n-1} a_t v_{d_t}.$$

The same equality is preserved under concatenation of two compatible trajectories.

Proof. The one-step identity is (2.16). Summing it over the n steps cancels the intermediate coordinates. Splitting the summation interval into two segments produces exactly the concatenation law. For a closed trajectory, the difference $z_n - z_0$ is the previously defined winding number. $\square$

The same principle governs APP evaluations. If a sheet takes the integer value N_t , its record simultaneously preserves $N_t = R_t + 9q_t$. The difference between two steps satisfies

$$N_{t+1} - N_t = (R_{t+1} - R_t) + 9(q_{t+1} - q_t).$$

Spatial-crossing memory z_t and arithmetic quotient q_t have different definitions; preserving them jointly maintains the geometric and arithmetic provenance of every transition.

2.3.3 Two-cursor chronology and composite returns

The cardinal involution j interchanges N with S and E with O . After applying (2.15), both directions are reversed at the end of a block of 27 steps. Writing

$$h(\theta) = \mathbf{1}_{\{0,27,54,81\}}((\bar{\theta} + 1) \bmod 108),$$

the complete rule on the observable projection is

$$F_{\text{obs}}(q) = (x'_+, j^{h(\theta)}d_+, x'_\times, j^{h(\theta)}d_\times, \theta + 1). \quad (2.17)$$

The reading that forms the emission uses the positions before transport, according to the explicit recurrence in Section 4. The observable update and emission therefore compose in a fixed order.

Proposition 2.13 (Reversibility and period of the observable calendar). *The map F_{obs} is a permutation of $\mathcal{Q}_{\text{obs}}$ and satisfies $F_{\text{obs}}^{108} = I$. Within a block of 54 steps, positions and directions are restored; the coordinate in Θ_{108} advances by 54.*

Proof. Given the final state, the preceding phase is first recovered by subtracting one in Θ_{108} . That phase determines whether directions were reversed and which cursor moved. Applying the corresponding involution and subtracting the cardinal vector uniquely recovers the initial state.

Each interval of 27 steps contains nine activations of each cursor. The following block preserves the activations and reverses the directions, so the integer displacements cancel over 54 steps. The rule has period 54 in its position and orientation increments; the same cancellation holds for any calendar origin. After 108 steps, θ is also restored. $\square$

We reserve

$$\mathcal{K}_{27} = F_{\text{obs}}^{27} : \mathcal{Q}_{\text{obs}} \longrightarrow \mathcal{Q}_{\text{obs}}$$

for the iterate of 27 transitions. Its fourth iterate is the identity on the observable state defined in (2.13). Projecting history onto the calendar makes this finite check possible. The trajectory record, its evaluations, and the new prefixes remain in the dynamic state on which $\mathcal{U}_t$ acts; observable equality does not reinitialize them.

2.3.4 Affine memory transport and composition of states

The observable projection admits fibers of arithmetic and geometric data. Over a configuration q , we write a state in the form

$$x = (q, o, h, c, \sigma, C, b, \lambda).$$

Here o preserves orientation and sheet; h , residues and quotients; c , additive transport memory; σ , the boundary signature; C , the precision cylinder; b , its boundary label; and λ , the recorded trajectory. The signature is determined by a map

$$\text{Sig}_q : H_q \times C_q \times \text{Hist}_q \longrightarrow S_q, \quad \sigma = \text{Sig}_q(h, c, \lambda).$$

The symbols H_q , C_q , Hist_q , and S_q denote, respectively, the sets of residue–quotient data, the abelian transport-memory group, the recorded trajectories, and the boundary signatures. Realizations of this signature are specified component by component in the regional and terminal constructions. This dependence prevents treating the signature as a free parameter once the data producing it are fixed.

The elementary residue–quotient chart is explicit. For $n \in \mathbb{Z}$, let

$$r = 1 + ((n - 1) \bmod 9), \quad u = \frac{n - r}{9}.$$

Then $n = r + 9u$, with $r \in D_9$ and $u \in \mathbb{Z}$. Uniqueness follows because two representatives in D_9 congruent modulo 9 are equal. This chart supplies local integer reconstruction. The precision towers additionally incorporate the truncation maps and cylinders developed in 4 and 5.

Let $\gamma : q \rightarrow q'$ be an observable trajectory. Fiber transport is expressed through maps

$$H(\gamma) : H_q \rightarrow H_{q'}, \quad C(\gamma) : C_q \rightarrow C_{q'}, \quad \text{Hist}(\gamma) : \text{Hist}_q \rightarrow \text{Hist}_{q'}.$$

The maps respect identities and composition, and $C(\gamma)$ is an abelian-group homomorphism. The increment $k_{\text{tr}}(\gamma) \in C_{q'}$ satisfies

$$\begin{aligned} k_{\text{tr}}(\text{id}_q) &= 0, \\ k_{\text{tr}}(\gamma_2 \circ \gamma_1) &= k_{\text{tr}}(\gamma_2) + C(\gamma_2)k_{\text{tr}}(\gamma_1). \end{aligned} \tag{2.18}$$

Here k_{tr} denotes the transport cocycle; the rational coordinate κ of record K retains its notation in the dodecaphase-closure chapter.

Transportable coordinates are updated by

$$\begin{aligned} h' &= H(\gamma)h, & c' &= C(\gamma)c + k_{\text{tr}}(\gamma), \\ \lambda' &= \gamma \circ \lambda, & \sigma' &= \text{Sig}_{q'}(h', c', \lambda'). \end{aligned} \tag{2.19}$$

The second equation is an affine action: the linear term transports previous memory and the cocycle adds the increment produced by the trajectory. In the spatial-crossing realization for each cursor separately, $C(\gamma) = I$ and $k_{\text{tr}}(\gamma)$ is the sum of the vectors in (2.16). This concrete instance connects the abstract law with the cardinal operations already defined.

Proposition 2.14 (Compatibility of the update with concatenation). *Updating by (2.19) along γ_1 , followed by updating along γ_2 , coincides with updating along $\gamma_2 \circ \gamma_1$.*

Proof. For the affine coordinate, two updates give

$$c'' = C(\gamma_2)C(\gamma_1)c + C(\gamma_2)k_{\text{tr}}(\gamma_1) + k_{\text{tr}}(\gamma_2).$$

Functoriality of C and (2.18) turn this expression into $C(\gamma_2 \circ \gamma_1)c + k_{\text{tr}}(\gamma_2 \circ \gamma_1)$. The statement for h follows from composition of H , and that for λ from associativity of trajectories. The final signature agrees because it is evaluated on the same three reconstructed coordinates. $\square$

The cylinder and its boundary complete this law through the relations Ext_γ , defined in 2.3.7. Their elements determine which of the preceding updates preserve an admissible prolongation. States, together with the triples (γ, x, x') admitted by these relations, form a category over observable trajectories:

$$(\gamma_2, x', x'') \circ (\gamma_1, x, x') = (\gamma_2 \circ \gamma_1, x, x'').$$

The identity of x is (id_q, x, x) . Composition preserves arithmetic transport and boundary prolongability simultaneously. This is the structure in which returns with memory and successive refinements take place.

2.3.5 Dodecaphase record, phase, and accumulated memory

The 108 transitions are distributed over the ordered windows

$$E_m = \{9m - 8, \dots, 9m\}, \quad 1 \leq m \leq 12.$$

The record preserves the phase origin, orientation, and data produced during each window. Its map is

$$\mathcal{R}_{12}(x) = \left((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m) \right)_{m=1}^{12},$$

where τ_m is the window's subroute, σ_m its orientation, κ_m the integer boundary increment, and Λ_m the local incidence record. We adopt the chart developed in 8.2; its window indices correspond to $m = \beta + 1$. The domain comprises compatible TPK histories, and the codomain their oriented temporal records. Composition of segments and their increments is governed by (2.19). By contrast, the group

C_{12} describes translation of the window index; Θ_{108} preserves the step calendar. This distinction specifies what is transformed and what is observed when a cycle is completed.

In coordinates $\theta = 9b + r$, with $0 \leq r < 9$, advancing 9 steps preserves r and increases b by one. Addition of phases introduces the quotient

$$c_0(r, r') = \left\lfloor \frac{r + r'}{9} \right\rfloor,$$

whose cocycle identity ensures associativity of composition. The finite calendar preserves $[b]_{12}$; a record of turns preserves $b \in \mathbb{Z}$, or $b \in \mathbb{N}_0$ for a history with fixed origin. Section 4 proves the nonsplit exact sequence (4.5). Its function is to describe the relation between these coordinates before they are used to construct the signed state.

The signed record receives phase and memory data through the composition developed in Section 8:

$$R_{\text{sgn}} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12}. \quad (2.20)$$

Its domain is the terminal state with memory; its output belongs to $\mathbb{Z}^{12}$. The reversible transformation between this signed state and K , together with the rational reading of K , is proved in the corresponding chapter. Record operations precede these transformations. Reconstructing a record from its cyclic differences verifies its recoverability; its dynamic provenance must be preserved through the events that produced it.

For the encoded trace (d_t) retained by the record, the event reader counts codes $\{2, 4, 6, 8\}$ in each window with orientation $(+, +, +, -, -, -, +, +, +, -, -, -)$. These codes and the cardinal symbols of $\mathcal{D}$ are different record coordinates. Chapter 8 develops this reader, the integral decomposition $1 + 5 + 6$, the congruences characterizing its image, and its exact inverse. The inversion proof establishes which data this chart preserves; the definition of $\mathcal{R}_{12}$ specifies the oriented history on which it acts.

2.3.6 Phase incidence and internal channel construction

The selector determines three subsets of D_9 :

$$H_+ = \{1, 4, 7\}, \quad H_- = \{2, 5, 8\}, \quad H_0 = \{3, 6, 9\}.$$

Let $H_{\text{act}} = H_+ \sqcup H_-$ be the set of active phases and $O_{\text{ph}} = D_9 \setminus \{9\}$ the chart preceding the marked return. Denote by $P_2(H_{\text{act}})$ the set of unordered pairs of active phases. Internal incidence is given by

$$\mathcal{F}_6^{\text{ph}} = H_{\text{act}} \times P_2(H_{\text{act}}), \quad \mathcal{F}_8^{\text{ph}} = O_{\text{ph}} \times P_2(H_{\text{act}}), \quad \Theta_{\text{ph}} = D_9 \times P_2(H_{\text{act}}). \quad (2.21)$$

The indices describe the number of positions in the first component. The construction uses the phases and calendar origin already fixed; subsequent exceptional realizations will transport these incidences to their own domains.

Proposition 2.15 (Cardinalities and oriented phase incidence). *The sets in (2.21) have cardinalities 90, 120, 135, respectively. If*

$$\partial_{\text{ph}} = \Theta_{\text{ph}} \times \{-1, +1\}, \quad E_{\text{ph}} = \partial_{\text{ph}} \sqcup \{\infty\}, \quad L_{\text{ph}} = H_0 \times P_2(H_{\text{act}}),$$

then

$$|\partial_{\text{ph}}| = 270, \quad |E_{\text{ph}}| = 271, \quad |L_{\text{ph}}| = 45, \quad |\partial_{\text{ph}} \sqcup L_{\text{ph}}| = 315.$$

Simultaneously translating the phase origin and marked return preserves all these cardinalities and inclusions.

Proof. There are six active phases, so $|P_2(H_{\text{act}})| = \binom{6}{2} = 15$. Multiplying by 6, 8, 9 gives 90, 120, 135. Orientation doubles 135; the return mark adds one element; the three neutral phases give $3 \cdot 15 = 45$. A calendar translation transports each factor by a bijection and carries the marked point to its image. Products and disjoint unions thus preserve the incidence. $\square$

Here the cardinality 271 has a phase-incidence realization. The cubic completion in Section 4 has its own construction of the corona 1000 – 729. The correspondence between the two realizations must preserve the maps and marks identifying them, in addition to their cardinality. Likewise, the selector ε_9 , the incidence (90, 120), and the extractor $\mathcal{E}_{108}^{90,120}$ retain the domains and operations just distinguished.

2.3.7 Trajectory reversal and conjugate prolongations

Bidirectionality begins with an operation on trajectories before scalar values are assigned to them. For $\gamma = (x_0; d_1, \dots, d_n)$, with endpoint x_n , define

$$\text{rev } \gamma = (x_n; jd_n, \dots, jd_1).$$

The reversed trajectory starts at the endpoint of the original. Composition satisfies

$$\text{rev}^2 \gamma = \gamma, \quad \text{rev}(\gamma_2 \circ \gamma_1) = \text{rev } \gamma_1 \circ \text{rev } \gamma_2,$$

and integer displacement changes sign. These identities follow by reversing the order of steps and replacing each cardinal vector with its opposite. For a loop, it follows that $\text{wind}(\text{rev } \gamma) = -\text{wind}(\gamma)$.

On records, reversal also requires transporting the order of components and the orientation of events. Route reversal and signature conjugation are distinguishable operations: their composition is specified when applied to a concrete record. In particular, the regional involution interchanging closure and propagation components preserves the self-scaling axis; its formula is given in Section 5.13. This regional operation acts on a different domain from cardinal trajectory reversal.

Extensions over a route $r : q \rightarrow q'$ are expressed by relations between fibers of compatible states. If $\mathcal{F}_q$ denotes the fiber of complete states over q , we require

$$\text{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'}, \quad \Delta_{\mathcal{F}_q} \subseteq \text{Ext}_{\text{id}_q},$$

$$\text{Ext}_{r_2} \circ \text{Ext}_{r_1} \subseteq \text{Ext}_{r_2 \circ r_1}.$$

Their composition preserves the evaluations of both sheets, orientation, quotients, prefix, and boundary. Route reversal does not by itself imply that every extension relation is a bijection. Conjugate prolongation is defined on the domain where transport of these data satisfies the corresponding compatibilities.

2.3.8 Nonadic holonomy and continuity of refinement

A phase turn composes transitions on the state with memory:

$$\Gamma_{9,k} = \mathcal{U}_{k+8} \circ \dots \circ \mathcal{U}_k. \quad (2.22)$$

On an individualized history, this functional composition realizes the holonomy relation; on the full domain we retain $\Gamma_{9,k} \subseteq \widehat{X}_k \times \widehat{X}_{k+9}$, with the admissible prolongations developed later. The cyclic base has nine phases, while the fiber preserves trajectory, quotients, and boundary. Monodromy is the action of one turn of this base on the fibers; holonomy $\Gamma_{9,k}$ expresses its realization through the effective TPK transitions.

If g observes phase and M the number of retained turns,

$$g(\Gamma_{9,k}x) = g(x), \quad M(\Gamma_{9,k}x) = M(x) + 1.$$

The first identity expresses phase closure and the second identifies the new temporal depth. Coinductive prolongation also preserves its cylinders and prefixes. Restrictions between depths compose an inverse system; a compatible section gathers its antecedents without replacing them by the latest observation. This arithmetic of histories and its unity are developed in 4.8–4.10; the regions and their readers are constructed in Section 5.

Phase transfer on emissions already constructed has a different domain. Let $\mathcal{A}_+$ and $\mathcal{A}_\times$ be the additive and multiplicative signature alphabets, with cardinalities 18 and 26. Their product identifies the catalog $\mathcal{U}_6$. For $f : \mathcal{A}_+ \times \mathcal{A}_\times \rightarrow \mathbb{R}$, the sheet averages are

$$(E_+f)(s, e) = \frac{1}{18} \sum_{s' \in \mathcal{A}_+} f(s', e), \quad (E_\times f)(s, e) = \frac{1}{26} \sum_{e' \in \mathcal{A}_\times} f(s, e').$$

These emissions and their cardinalities are constructed through the recurrence in Section 4. Once they have been obtained, the transfer operator is defined on $\mathcal{U}_6 \times C_9$ by

$$P_+ = E_+, \quad P_- = E_\times, \quad P_0 = I, \quad \mathcal{K}_{\text{ph}}((u, r), (v, r+1)) = P_{M_{\text{ph}}(r)}(u, v),$$

with zero weight for all other phase transitions. The index r uses the positive representative of the cyclic class. The phase order $(+1, -1, 0)^3$ then produces the renewal

$$\mathcal{K}_{\text{ph}}^9|_r = E_+E_\times.$$

Indeed, both averages are idempotent, commute, and average different factors; their composition assigns weight $1/(18 \cdot 26) = 1/468$ to each emission. This equality characterizes a subsequent transfer observable. History transport remains given by (2.22), with its memory and boundary.

The resulting hierarchy establishes the function of the following chapters. Finite emission constructs the domain and regions; extension relations preserve their prefixes; the nonadic connection prolongs them; the dodecaphase record determines the data entering K and the closure of α . Numerical and incidence realizations receive this preceding structure. This order allows every consequence to be presented in its own chapter while preserving, from the formal kernel onwards, the objects that make it possible.

2.4 Accumulated memory and transport resolvents

Calendar return does not remove increments produced during the journey. To express this distinction in operator form, an accumulated trajectory-event record is constructed first, followed by its generating series. Reduction of memory variables will be performed on this transport, also preserving the source and output reader.

2.4.1 Oriented events and record update

Let (d_t) be the sequence of integer direction codes retained by a TPK trajectory. The code d_t and the cursor's cardinal direction are different coordinates; no identification of their alphabets is assumed. We number the windows starting from zero:

$$I_m = \{9m + 1, \dots, 9m + 9\}, \quad m \geq 0.$$

In the first cycle, I_m is window E_{m+1} of the dodecaphase record. Its orientation is

$$\sigma = (1, 1, 1, -1, -1, -1, 1, 1, 1, -1, -1, -1).$$

For a prolonged history, $\sigma_m \in \{-1, 1\}$ is the sign retained in the corresponding window. The signed event is

$$a_m = \sigma_m \sum_{t \in I_m} \mathbf{1}_{\{2,4,6,8\}}(d_t), \quad |a_m| \leq 9. \quad (2.23)$$

Each summand is incorporated when its transition occurs. This definition counts trace events; it does not select directions through a value that is to be obtained later.

Let $\mathbf{e}_0, \dots, \mathbf{e}_{11}$ be the basis of $\mathbb{Z}^{12}$ and $\mathbf{S}\mathbf{e}_j = \mathbf{e}_{j+1 \bmod 12}$. Write $R = S^{-1}$. The accumulated record in fixed coordinates satisfies

$$z_{m+1} = z_m + a_m \mathbf{e}_{m \bmod 12}.$$

A prefix with no accumulated history starts at $z_0 = 0$; at a different origin, z_0 retains the preceding record. The frame accompanying the calendar is $y_m = S^{-m}z_m$.

Proposition 2.16 (Update and return with memory). *The record in the moving frame satisfies*

$$y_{m+1} = Ry_m + \eta_m, \quad \eta_m = a_m R \mathbf{e}_0. \quad (2.24)$$

For every $n \geq 1$,

$$y_n = R^n y_0 + \sum_{j=0}^{n-1} R^{n-1-j} \eta_j. \quad (2.25)$$

In particular,

$$y_{12} = y_0 + \sum_{j=0}^{11} a_j \mathbf{e}_j. \quad (2.26)$$

Proof. Multiplying the update of z_m by $S^{-(m+1)}$ gives $y_{m+1} = S^{-1}y_m + a_m S^{-1} \mathbf{e}_0$, since $S^{-m} \mathbf{e}_{m \bmod 12} = \mathbf{e}_0$. Formula (2.25) follows by induction. For $n = 12$, $R^{12} = I$ and $R^{11-j} \eta_j = a_j R^{12-j} \mathbf{e}_0 = a_j \mathbf{e}_j$, which proves (2.26). $\square$

Thus twelve windows restore the frame and preserve the event vector. The count alone does not recover the order of all steps within each window; that order remains in the recorded trajectory.

To specify the balance, define

$$D_3 = I - S^3, \quad D_4 = I - S^4, \quad B = D_3^\top D_3 + D_4^\top D_4,$$

$$\mathcal{M}(y) = \frac{1}{2} y^\top B y, \quad q(y) = \sum_{j=0}^{11} y_j.$$

The first is a quadratic form of the record; the second records the sum of its coordinates. They are not identified here with physical energy or charge.

Proposition 2.17 (Event-induced balance). *Update (2.24) satisfies*

$$\begin{aligned} \mathcal{M}(y_{m+1}) - \mathcal{M}(y_m) &= a_m (B y_m)_0 + 2a_m^2, \\ q(y_{m+1}) - q(y_m) &= a_m. \end{aligned} \quad (2.27)$$

Proof. Expanding the products gives $B = 4I - S^3 - S^{-3} - S^4 - S^{-4}$; hence $R^\top B R = B$ and $B_{00} = 4$. Since $y_{m+1} = R(y_m + a_m \mathbf{e}_0)$, the quadratic difference is $a_m \mathbf{e}_0^\top B y_m + \frac{1}{2} a_m^2 B_{00}$. The coordinate sum is invariant under permutation R , giving the second equality. $\square$

2.4.2 Generating series and uniform memory control

The series preserves all increments, not only their sum at the end of a cycle. With a formal variable u , let

$$Y(u) = \sum_{m \geq 0} y_m u^m, \quad H_\eta(u) = \sum_{m \geq 0} \eta_m u^m.$$

Proposition 2.18 (Resolvent and tail bound). *The formal identity*

$$Y(u) = (I - uR)^{-1}(y_0 + uH_\eta(u)) \quad (2.28)$$

holds. Both series converge for $|u| < 1$. If $\ell : \mathbb{R}^{12} \rightarrow \mathbb{R}$ is linear, $L = \|\ell\|_{1 \rightarrow \mathbb{R}}$, $M_0 = \|y_0\|_1$, and $0 < r < 1$, then, for every integer $N \geq 0$ and $|u| \leq r$,

$$\left| \sum_{m > N} \ell(y_m) u^m \right| \leq L r^{N+1} \left(\frac{M_0 + 9(N+1)}{1-r} + \frac{9r}{(1-r)^2} \right). \quad (2.29)$$

The derivative series converges uniformly on every disk $|u| \leq r < 1$.

Proof. Multiplying (2.24) by u^{m+1} and summing gives $Y - y_0 = uRY + uH_\eta$. The constant term of $I - uR$ is invertible, so its formal inverse is $\sum_{k \geq 0} u^k R^k$. Moreover, R preserves the ℓ^1 norm and $\|\eta_m\|_1 \leq 9$, whence $\|y_m\|_1 \leq M_0 + 9m$. For the tail, sum the majorants $L(M_0 + 9m)r^m$, using

$$\sum_{m > N} r^m = \frac{r^{N+1}}{1-r}, \quad \sum_{m > N} m r^m = r^{N+1} \left(\frac{N+1}{1-r} + \frac{r}{(1-r)^2} \right).$$

Finally, $\sum_{m \geq 1} Lm(M_0 + 9m)r^{m-1} < \infty$ majorizes the derivative series and proves its uniform convergence. $\square$

The variable u counts windows of nine transitions. Identifying it with another reader's variable requires preserving that clock and the map relating the two readings.

2.4.3 Memory elimination: operator, source, and reader

On a domain where the update is fixed as $U : X \rightarrow X$, the free vector space $V = \mathbb{Q}^{(X)}$ on the states admits the operator $T\delta_x = \delta_{U(x)}$. The clock is included in x if the transition depends on it. This linear extension preserves distinct states even when they share an observable phase.

Consider a decomposition $V = V_0 \oplus V_1$, where V_1 is the auxiliary sector to be eliminated, and write

$$T = \begin{pmatrix} A & B_1 \\ C & D \end{pmatrix}, \quad v = \begin{pmatrix} v_0 \\ v_1 \end{pmatrix}, \quad \ell = (\ell_0, \ell_1).$$

Here $A : V_0 \rightarrow V_0$, $B_1 : V_1 \rightarrow V_0$, $C : V_0 \rightarrow V_1$, and $D : V_1 \rightarrow V_1$. The following identities are formal; they also hold wherever the resolvents converge.

Proposition 2.19 (Schur reduction with source and reader). *Let*

$$D_u = (I - uD)^{-1}, \quad \mathcal{F}(u) = I - uA - u^2 B_1 D_u C.$$

The retained component of $(I - uT)^{-1}v$ is

$$w_0 = \mathcal{F}(u)^{-1}(v_0 + uB_1 D_u v_1), \quad w_1 = D_u(v_1 + uCw_0). \quad (2.30)$$

Its complete reading satisfies

$$\begin{aligned} \ell(I - uT)^{-1}v &= (\ell_0 + u\ell_1 D_u C) \mathcal{F}(u)^{-1}(v_0 + uB_1 D_u v_1) \\ &\quad + \ell_1 D_u v_1. \end{aligned} \quad (2.31)$$

Proof. The two equations of $(I - uT)w = v$ are $w_0 = v_0 + uAw_0 + uB_1w_1$ and $w_1 = v_1 + uCw_0 + uDw_1$. Solving the second and substituting it into the first gives (2.30); applying ℓ_0 and ℓ_1 to the two components gives (2.31). All formal inverses exist because their constant terms are identities. $\square$

The term $u^2B_1D_uC = \sum_{k \geq 0} u^{k+2}B_1D^kC$ describes departure into memory, k steps within it, and a return. By contrast, $u\ell_1D_uC$ reads memory before that return. Thus reducing only the operator does not necessarily preserve the observation: the source and reader must also be transformed.

The distinction is visible in the cycle already constructed. If P retains $\mathbf{e}_0$, then $PRP = 0$, but

$$\langle \mathbf{e}_0, (I - uR)^{-1}\mathbf{e}_0 \rangle = \frac{1}{1 - u^{12}}, \quad \langle \mathbf{e}_{11}, (I - uR)^{-1}\mathbf{e}_0 \rangle = \frac{u}{1 - u^{12}}. \quad (2.32)$$

Indeed, $R^n\mathbf{e}_0 = \mathbf{e}_{-n \bmod 12}$: the two series collect, respectively, $n \equiv 0$ and $n \equiv 1 \pmod{12}$. The resolvent of the compression is I , whereas compression of the resolvent preserves the omitted returns. This example concerns the twelve-window record, not the entire TPK state.

2.4.4 Segment composition and coherence of reduction

Let three composable affine transports be $F_i(y) = R_iy + b_i$, with compatible domains and codomains. The memory produced by the complete journey is

$$F_3F_2F_1(y) = R_3R_2R_1y + b_3 + R_3b_2 + R_3R_2b_1. \quad (2.33)$$

The proof is successive substitution of the three affine formulas. Grouping the first two segments first, or the last two, produces the same expression; the factors transport each increment from its point of origin. In (2.24), $R_i = R$ and b_i is the event of its window.

Eliminating several layers preserves a similar coherence. To make it explicit, now decompose $V = V_0 \oplus V_1 \oplus V_2$ and write

$$T = \begin{pmatrix} A & B_1 & B_2 \\ C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{pmatrix}, \quad Q_2 = (I - uD_{22})^{-1}.$$

Eliminating the third component leaves the four blocks

$$\begin{aligned} A' &= A + uB_2Q_2C_2, & B'_1 &= B_1 + uB_2Q_2D_{21}, \\ C'_1 &= C_1 + uD_{12}Q_2C_2, & D'_{11} &= D_{11} + uD_{12}Q_2D_{21}. \end{aligned} \quad (2.34)$$

Proposition 2.20 (Transitivity of elimination). *If $D_* = (D_{ij})_{i,j=1}^2$, the complement obtained in two stages coincides with that obtained by eliminating both memories together:*

$$\begin{aligned} I - uA' - u^2B'_1(I - uD'_{11})^{-1}C'_1 &= I - uA \\ &\quad - u^2(B_1 \ B_2)(I - uD_*)^{-1} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}. \end{aligned} \quad (2.35)$$

Proof. Solving the third equation of $(I - uT)w = v$ and substituting it into the first two produces (2.34). Eliminating the second component then gives the left-hand side of (2.35). Solving the two memory equations together gives the right-hand side, by Proposition 2.19. The formal solution is unique because $I - uT$ has constant term I . Both procedures also transform the source and reader according to (2.30)–(2.31). $\square$

Balance (2.27) describes record increments. The preceding identities allow them to be transported when memory is reused in dodecaphase closure; they do not by themselves determine additional analytic coefficients.

2.5 Correspondence between positional support and the oriented calendar

The calendar representation distinguishes the positions of APP from the oriented transitions of its evolution. We revisit this correspondence to make explicit the reading of the segment containing the junction between turns.

Consider the alphabet of nine indices

$$D_9 = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}.$$

Its horizontal and vertical arrangement produces the eighty-one positions of D_9^2 . The oriented cycle C_9 is also used to describe motion. Its nine edges are assigned, in order, the indices of D_9 . The index of an edge specifies a transition between states, whereas a vertex specifies its boundary. The map assigning each edge its index allows the same alphabet to be used in the positional support and in the calendar; it preserves the distinction between position, oriented interval, and transported state.

In the integer lift of the calendar, the first turn comprises the nine consecutive intervals $[0, 1], \dots, [8, 9]$. The continuation $[9, 10]$ starts the next turn with index one and updated memory. The segment drawn up to ten shows both turns at their junction. This representation specifies the nine transitions of one turn and the following transition; the holonomy of the state is constructed subsequently through the effective composition of the TPK.

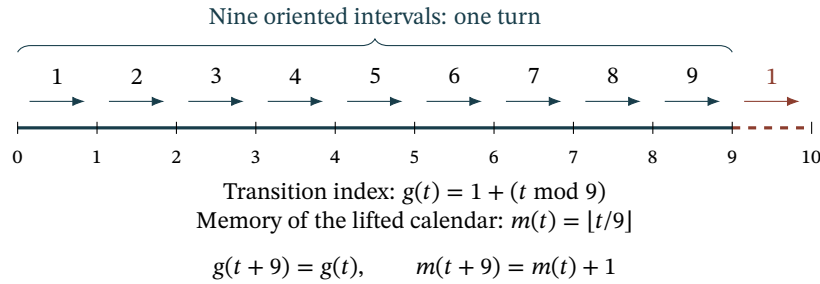


Figure 3: Lift of the cycle of nine transitions. The upper numbers label oriented intervals; the lower numbers locate their boundaries. The dashed interval shows the first transition of the next turn. The phase index returns and the calendar quotient increases. This diagram represents the calendar; the action on residues, quotients, orientation, boundary, and memory is defined through the operations of the TPK.

2.6 Operator architecture of transport, memory, and prolongation

The topological phase kernel organizes the evolution of the two APP evaluations on oriented histories. Its structure brings together a path support, local update rules, prolongation relations, temporal records, and reading maps. Each component has its own domain. This hierarchy allows the observable position, the transport quotient, and the memory of the crossed boundaries to be preserved within a single construction. The composition in equation (2.12) thus acquires a precise realization at each of these levels.

2.6.1 Bivariate paths and lifted transport

Let P_{APP} be the free category of the cardinal graph on $X_9 = (\mathbb{Z}/9\mathbb{Z})^2$. Its objects are the APP positions; its morphisms are cardinal words, with composition by concatenation. The category

$$P_{APP}^{(2)} = P_{APP} \times P_{APP}$$

preserves, in order, the path of the additive sheet and that of the multiplicative sheet. Both paths traverse the same arithmetic support and keep their evaluations distinct.

In the chart $I_9 = \{0, \dots, 8\}$, the generators are

$$\nu_N = (-1, 0), \quad \nu_E = (0, 1), \quad \nu_S = (1, 0), \quad \nu_O = (0, -1), \quad T_d(x) = [x + \nu_d]_9.$$

For $w = d_1 \cdots d_r$, the lift to the integer covering space preserves

$$\tilde{x}_k = \tilde{x}_0 + \sum_{j=1}^k \nu_{d_j}, \quad x_k = [\tilde{x}_k]_9.$$

When $x_r = x_0$, the winding class is

$$\text{wind}(w) = \frac{1}{9}(\#S - \#N, \#E - \#O) \in \mathbb{Z}^2. \quad (2.36)$$

The final position and winding are two different readings of the same route. The second records the displacement of the lift even when the first has returned.

Path reversal has a determined domain and codomain:

$$\text{rev} : \text{Hom}(x, y) \longrightarrow \text{Hom}(y, x), \quad d_1 \cdots d_r \longmapsto \bar{d}_r \cdots \bar{d}_1,$$

where $\bar{N} = S, \bar{E} = O$. It satisfies $\text{rev}^2 = I$, reverses the order of composition, and changes the sign of (2.36). This is the precise operation that must be transported when comparing conjugate orientations of a route.

2.6.2 Full state and preservation through updates

The description of $\hat{X}$ is organized over the positions and directions of the two cursors. Over each observable configuration q , the arithmetic fiber is written

$$\mathcal{F}_q = O_q \times H_q \times C_q \times S_q \times B_q \times \Lambda_q. \quad (2.37)$$

The coordinates O_q preserve orientation and sheet; H_q , the remainders and quotients of the evaluations; C_q , compatible carries; S_q , the boundary signature; B_q , the prefix and its cylinders; and Λ_q , the ordered transport record. The fiber is restricted by the corresponding arithmetic, incidence, and truncation identities. The full state also preserves the route that produced it and the origin from which its phases are read.

Mode selection has type

$$M_{\text{ph}} : D_9 \longrightarrow \{-1, 0, +1\}, \quad \text{Sel}_t : \hat{X} \longrightarrow \hat{X}_t^{\text{sel}}.$$

The second map incorporates the value of the first into the state and activates the corresponding sheet. Lifted transport

$$\text{Tra}_t : \hat{X}_t^{\text{sel}} \longrightarrow \hat{X}_t^{\text{tra}}$$

applies T_d to the active cursor, concatenates its route letter, and preserves the selected sheet. Finally,

$$\text{Upd}_t : \hat{X}_t^{\text{tra}} \longrightarrow \hat{X}$$

updates the evaluations and records. In this notation, Upd_t is the update Mem_t of (2.12), made explicit in its components.

Each integer evaluation a is preserved through $a = \rho_9(a) + 9q_9(a)$; each ternary evaluation, through its oriented remainder and carry. For a concatenation $r_2 r_1$, carry transport satisfies the cocycle law

$$k_{\text{tr}}(r_2 r_1) = k_{\text{tr}}(r_2) + C(r_2)k_{\text{tr}}(r_1), \quad (2.38)$$

where $C(r_2)$ is the transport action of the second route on the carry coordinate of the first. In the ordered record, updating is performed by adjoining the event produced:

$$\Lambda_{t+1} = \Lambda_t \| (\text{sheet}, d_t, \text{remainder}, \text{quotient}, \text{carry}, \text{orientation}, \text{boundary})_t.$$

Preservation has a dynamic meaning here: earlier contributions remain recoverable within the record being prolonged. Their coordinates may change when the state is transported.

2.6.3 Affine action on transport memory

For a trajectory $r : q \rightarrow q'$, the maps

$$H(r) : H_q \rightarrow H_{q'}, \quad C(r) : C_q \rightarrow C_{q'}$$

respect identity and concatenation; C_q is the abelian group of transport memory, and $C(r)$ its homomorphism. The increment $k_{\text{tr}}(r) \in C_{q'}$, with $k_{\text{tr}}(\text{id}_q) = 0$, is the cocycle of (2.38). This symbol distinguishes the transport increment from the rational coordinates of the dodecaphasic record.

In a chart with data (h, c, λ) , the update has the explicit form

$$h' = H(r)h, \quad c' = C(r)c + k_{\text{tr}}(r), \quad \lambda' = r \circ \lambda. \quad (2.39)$$

The signature is evaluated through $\text{Sig}_{q'}(h', c', \lambda')$. Its residence in the fiber $S_{q'}$ preserves its dependence on those data.

Proposition 2.21 (Composition of affine memory). *Updating by r_1 , followed by updating by r_2 , agrees with updating by $r_2 r_1$.*

Proof. Two memory updates give

$$c'' = C(r_2)C(r_1)c + C(r_2)k_{\text{tr}}(r_1) + k_{\text{tr}}(r_2).$$

Compatibility of C with concatenation and identity (2.38) turn this expression into $C(r_2 r_1)c + k_{\text{tr}}(r_2 r_1)$. The coordinate h satisfies the same compatibility by functoriality of H ; the trajectory does so by associativity. The signature agrees because it is evaluated on the same data. $\square$

The spatial chart of each cursor provides an elementary realization of this law. For $x \in I_9^2$ and a cardinal direction d ,

$$a_d(x) = \frac{x + \nu_d - T_d(x)}{9} \in \mathbb{Z}^2, \quad x' = T_d(x), \quad c' = c + a_d(x).$$

Here $C(r) = I$, and $k_{\text{tr}}(r)$ sums the integer boundary crossings. For a route of length n , the telescoping sum gives

$$x_n = x_0 + \sum_{j=1}^n \nu_{d_j} - 9 \sum_{j=1}^n a_{d_j}(x_{j-1}).$$

On a loop, the last sum equals $\text{wind}(r)$. The residual position returns while the lift preserves the number and orientation of its crossings. The remaining memory components retain their own transport maps and prolongation relations.

2.6.4 Chronology of the two cursors and observable return

The finite calendar lives in $\Theta_{108} = \mathbb{Z}/108\mathbb{Z}$. For $0 \leq \tilde{\theta} < 108$, set

$$r(\theta) = 1 + (\tilde{\theta} \bmod 9), \quad \beta(\theta) = \left[\left[\frac{\tilde{\theta}}{9} \right] \right]_{12}, \quad \varepsilon_9(r) = M_{\text{ph}}(r).$$

Presentation (2.13) groups position and direction by cursor. This extension groups the two positions first and then the two directions. The two charts are identified through the factor permutation

$$\begin{aligned} \mathfrak{p}_{\text{obs}} : (X_9 \times \mathcal{D})^2 \times \Theta_{108} &\longrightarrow X_9^2 \times \mathcal{D}^2 \times \Theta_{108}, \\ (x_+, d_+, x_-, d_-, \theta) &\longmapsto (x_+, x_-, d_+, d_-, \theta), \end{aligned} \quad (2.40)$$

with $\mathcal{D} = \{N, E, S, O\}$. The inverse restores each direction alongside its cursor. If superscripts agr and sep distinguish these two expressions of the observable successor, then

$$\mathfrak{p}_{\text{obs}} F_{\text{obs}}^{\text{agr}} = F_{\text{obs}}^{\text{sep}} \mathfrak{p}_{\text{obs}}.$$

Indeed, the selector depends on the same phase, the cardinal vector is applied to the same cursor, and direction reversal occurs at the same instant; only the coordinate order changes. We retain $\mathcal{Q}_{\text{obs}}$ and F_{obs} for the space and map so identified. This presentation permutation leaves the events and memory coordinates of the history unchanged.

The observable space is

$$\mathcal{Q}_{\text{obs}} = X_9^2 \times \{N, E, S, O\}^2 \times \Theta_{108}.$$

Write a state as $q = (x_+, x_-, d_+, d_-, \theta)$. The active sheet is first evaluated at the current position. The positions are then updated by

$$(x'_+, x'_-) = \begin{cases} (T_{d_+} x_+, x_-), & \varepsilon_9(r) = +1, \\ (x_+, T_{d_-} x_-), & \varepsilon_9(r) = -1, \\ (x_+, x_-), & \varepsilon_9(r) = 0. \end{cases} \quad (2.41)$$

If $\tilde{\theta} + 1$ is a multiple of 27, (d_+, d_-) is replaced by $(\bar{d}_+, \bar{d}_-)$. In all cases θ is incremented modulo 108. These rules define the successor $F_{\text{obs}} : \mathcal{Q}_{\text{obs}} \rightarrow \mathcal{Q}_{\text{obs}}$. The emission realization in section 4 applies exactly this order of evaluation, transport, and reversal.

Proposition 2.22 (Observable return and preservation of history). *The map F_{obs} is a permutation. The operator $\mathcal{K}_{27} = F_{\text{obs}}^{27}$ satisfies $\mathcal{K}_{27}^4 = I$. A complete history of one hundred eight updates also preserves the events and carry accumulated during that turn.*

Proof. Knowing the later phase recovers the preceding phase. The rule indicates whether a direction reversal occurred; undoing it and applying inverse cardinal transport to the active cursor recovers the previous state. Each segment of twenty-seven transitions contains nine activations of each sheet. From an endpoint of a segment, each cursor makes nine displacements in the same direction, whose sum is zero in X_9 ; it then reverses its orientation. Two segments restore the orientations, and four also restore the calendar. The identity is transported to any phase by invertibility of F_{obs} . The full record preserves the concatenation of the one hundred eight events and their counter, which have been updated at every transition. $\square$

The periodicity of this permutation belongs to the observable quotient. In the full history, the integer window counter increases by twelve over the same interval. Exact sequence (4.5) expresses the reduction of the counter to its dodecaphasic coordinate and preserves its defect of additivity.

2.6.5 Bidirectional electronic transport and arithmetic record

The central representation uses a cell shared by the two APP evaluations. Its coordinates belong to $D_9 = \{1, \dots, 9\}$, and the inversion $(r, c) \mapsto (10 - r, 10 - c)$ has the unique fixed point $(5, 5)$. The visible state is written (r, c, h, v, n) , with $h, v \in \{-1, 1\}$ and integer chronological index $n \geq 0$. The tritic selector repeats $(+1, -1, 0)$. On the additive sheet, $r + c$ is read and the column is translated in direction h ; on the multiplicative sheet, rc is read and the row is translated in direction v . The threshold preserves position. Both orientations are reversed upon completing each interval of twenty-seven updates.

To express reading and carry with the same positive convention, define

$$[z]_9^+ = 1 + ((z - 1) \bmod 9), \quad q_9(z) = \frac{z - [z]_9^+}{9}.$$

Evaluation precedes displacement. If a_n is the integer read, the event preserves $(a_n, [a_n]_9^+, q_9(a_n))$, the sheet, previous position, orientation, and boundary crossing. The tritic symbol is $[a_n]_9^+ \bmod 3$. The spatial lift retains, for example, the integer $q_9(c + h)$ when c is replaced by $[c + h]_9^+$.

The shared-cell representation and the two-cursor representation of (2.41) have different visible states. Their records are related to the TPK state through their respective representation maps. For the central trajectory, the full update on the shared cell producing the electronic word is used here.

Proposition 2.23 (Invertible update with preserved quotients). *Add to the visible state two integer records Q_+ , $Q_\times$ and two winding numbers W_r , W_c . Each active reading increments its sheet's record by $q_9(a_n)$; each displacement increments the corresponding winding by its boundary crossing. This update has an explicit inverse on its image.*

Proof. The later index determines the preceding mode and whether an orientation reversal has just occurred. This reversal is undone first. If the mode was additive, the previous column is recovered as $[c - h]_9^+$; if multiplicative, the row is recovered as $[r - v]_9^+$. The neutral mode preserves both coordinates. The recovered cell again determines the integer read, its remainder, quotient, and boundary crossing. Subtracting these increments from Q_+ , $Q_\times$, W_r , W_c and reversing the chronological index recovers the preceding state exactly. Composing these inverses recovers an entire finite history. Event concatenation also preserves the order of readings. $\square$

The records Q_+ , $Q_\times$ are an explicit representation by sums of the quotient history. The construction is integrated into the memory fiber alongside its other coordinates. Its composition law, for concatenable histories u, v , is

$$Q_s(uv) = Q_s(u) + Q_s(v), \quad s \in \{+, \times\},$$

where the second history begins at the first one's final state. Algebraic inversion of an update subtracts its increments; subsequently traversing a spatial return path records new evaluations. This distinction simultaneously preserves transport reversibility and history duration.

Proposition 2.24 (Central return and exact accumulation). *Starting from $(r, c, h, v, n) = (5, 5, 1, 1, 0)$ and zero initial records, the spatial projection and its orientations return after fifty-four updates. Over that interval,*

$$(\Delta W_r, \Delta W_c) = (0, 0), \quad (\Delta Q_+, \Delta Q_\times) = (10, 46).$$

After $54N$ updates, for every integer $N \geq 0$, one has $(Q_+, Q_\times) = (10N, 46N)$, while the spatial coordinates and their orientations return. The integer index takes the value $54N$; the dodecaphasic calendar modulo 108 returns for even N .

Proof. Each pair of active modes displaces each coordinate once. In twenty-seven updates, both complete a turn of nine positions and return to the central diagonal; h, v are then reversed. The next block traverses the opposite turn. Spatial crossings cancel between the two blocks.

In each block, the additive readings run through $2k$, $1 \leq k \leq 9$, whose positive quotient sums to five. The multiplicative readings of the positive block run through $k[k + 1]_9^+$. Their integer sum and remainder sum are

$$\sum_{k=1}^9 k[k + 1]_9^+ = 249, \quad \sum_{k=1}^9 [k[k + 1]_9^+]_9^+ = 42.$$

The second identity follows from the list $(2, 6, 3, 2, 3, 6, 2, 9, 9)$. By Euclidean division, the quotient sum is $(249 - 42)/9 = 23$. Substitution $k \mapsto [k - 1]_9^+$ shows that the negative block contains the same multiset of products. The two blocks therefore produce ten and forty-six units in the respective records. The spatial projection and its orientations return to their initial values. The activation and reversal rules repeat upon shifting the index by 54, and the increments do not depend on the accumulated records. The preceding additive law proves the formula for every N by induction. $\square$

The tritic reading and window orientation provide, on the same trajectory,

$$\gamma_e = ((100000220)^3(120200000)^3)^2, \quad \tau_e = (+, +, +, -, -, -, +, +, +, -, -, -).$$

Powers represent concatenations. The discrepancy and window readers developed in (A.13) and (A.14) both compute $(80, 54, 6)$. The evaluation is obtained from the trajectory before incorporating α , the action section, or a mass. Quotient memory preserves information additional to that triple: in one hundred eight updates it records $(20, 92)$, with zero net spatial winding. Each representation thus retains the type of the observable producing it.

2.6.6 Bidirectionality of the product reader and quotient memory

The product reading preserves four arithmetic coordinates before determining a signature. For $x = (i, j) \in D_9^2$, $D_9 = \{1, \dots, 9\}$, define

$$N(x) = ij, \quad r(x) = 1 + ((ij - 1) \bmod 9), \quad q(x) = \frac{N(x) - r(x)}{9},$$

and the discrete reader

$$\ell(1, 2, 4, 8, 7, 5) = (0, 1, 2, 3, 4, 5), \quad \ell(3) = \ell(6) = \ell(9) = 0.$$

Thus $N = r + 9q$ simultaneously preserves the integer product, positive residue, and arithmetic quotient. The function $f_\ell(x) = \ell(r(x))$ is the coordinate used by the regional signature. The functions N, r, q , and f_ℓ have a positional domain; the transport direction remains in the oriented seed.

Advance, half-turn, and reading before transport. Let $\mathcal{X}_\times = D_9^2 \times \{N, E, S, O\}$. The cardinal vectors are $\nu_N = (-1, 0)$, $\nu_E = (0, 1)$, $\nu_S = (1, 0)$, $\nu_O = (0, -1)$. Positional addition is reduced to positive representatives modulo nine. Define

$$P(x, d) = (x + \nu_d, d), \quad H(x, d) = (x, \bar{d}), \quad \nu_{\bar{d}} = -\nu_d.$$

These operators satisfy $P^9 = I$, $H^2 = I$, and $HPH = P^{-1}$.

In the first fifty-four TPK transitions, the product channel is activated in phases 2, 5, 8 of each nonadic window. Each activation reads the position before advance. After transitions 27 and 54, H is applied. Thus the eighteen positions read are, for a seed s ,

$$P^0s, P^1s, \dots, P^8s, \quad HP^0s, PHP^0s, \dots, P^8HP^0s,$$

with the understanding that the positional reading omits only direction. For any positional function f , put $f_n = f(P^n s)$, with indices modulo nine. The word of raw readings is

$$\mathcal{R}_f(s) = (f_0, f_1, \dots, f_8, f_0, f_8, f_7, \dots, f_1). \quad (2.42)$$

The six window sums, each comprising three activations, are

$$A_j^f(s) = f_{3j} + f_{3j+1} + f_{3j+2}, \quad 0 \leq j < 3, \quad (2.43)$$

$$A_{3+j}^f(s) = f_{-3j} + f_{-3j-1} + f_{-3j-2}, \quad 0 \leq j < 3. \quad (2.44)$$

For $f = f_\ell$, the signature is $E_{\text{sig}}(s) = ([A_0^f]_{10}, \dots, [A_5^f]_{10})$. The decimal quotient of each accumulator is $c_j^{10} = [A_j^f / 10]$, and it preserves $A_j^f = [A_j^f]_{10} + 10c_j^{10}$.

Proposition 2.25 (Half-turn and permutation of half-cycles). *For every oriented seed and every positional function f ,*

$$\mathcal{R}_f(Hs) = \text{rot}_9 \mathcal{R}_f(s), \quad A_j^f(Hs) = A_{j+3 \bmod 6}^f(s).$$

The same permutation transports the decimal residues and quotients of the accumulators. It applies, in particular, to N, r, q , and f_ℓ .

Proof. The relation $P^n H = H P^{-n}$ and equality $f(Hs) = f(s)$ give $f(P^n Hs) = f(P^{-n} s) = f_{-n}$. Substitution into (2.42) exchanges its two groups of nine entries. Summation in groups of three produces the window identity. Euclidean division by ten is performed componentwise and respects the permutation. $\square$

Advance also has an exact law on the accumulators:

$$A_j^f(Ps) - A_j^f(s) = f_{3j+3} - f_{3j}, \quad 0 \leq j < 3, \quad (2.45)$$

$$A_{3+j}^f(Ps) - A_{3+j}^f(s) = f_{1-3j} - f_{-2-3j}, \quad 0 \leq j < 3. \quad (2.46)$$

The proof consists in canceling the two common summands of each window. Transport of a signature therefore requires the boundary evaluations appearing in these differences, in addition to the sums already expressed.

Chronological inversion and boundary terms. Let $\gamma = (x_0, \dots, x_n)$ be a recorded trajectory, and let $\gamma^\dagger = (x_n, \dots, x_0)$ be its chronological inversion, with each edge oriented in the opposite direction. For a positional function f , the reading before each advance is

$$C_f(\gamma) = \sum_{t=0}^{n-1} f(x_t).$$

Then

$$C_f(\gamma^\dagger) = C_f(\gamma) + f(x_n) - f(x_0). \quad (2.47)$$

If $n = 3m$ and A_j^f groups three consecutive edges,

$$A_j^f(\gamma^\dagger) = A_{m-1-j}^f(\gamma) + f(x_{3(m-j)}) - f(x_{3(m-j-1)}). \quad (2.48)$$

Both formulas follow because the reverse trajectory reads the arrival endpoints of the original edges. The interior summands coincide; the difference preserves the endpoints.

For an antisymmetric edge observable $a(x, y) = -a(y, x)$, its word does transform according to

$$\mathcal{R}_a(\gamma^\dagger) = -\text{rev } \mathcal{R}_a(\gamma).$$

This identity acts on oriented edges. Identities (2.47)–(2.48) act on positional readings. The half-turn H changes the initial direction and preserves calendar order; chronological inversion reverses the complete record. Their domains and reading rules are specified separately.

Occupation cocycle and memory preserved on outward and return paths. For a cardinal edge $x \rightarrow y$, with positive representatives $\tilde{x}, \tilde{y}$, the covering crossing is

$$b(x, y) = \frac{\tilde{x} + v_d - \tilde{y}}{9} \in \mathbb{Z}^2.$$

The identity $b(y, x) = -b(x, y)$ holds. In the free category of recorded paths, define the additive reader

$$\mathbf{m}(\gamma) = \left(\sum_{t=0}^{n-1} b(x_t, x_{t+1}), C_q(\gamma), n \right). \quad (2.49)$$

Concatenation satisfies $\mathbf{m}(\gamma_2 \circ \gamma_1) = \mathbf{m}(\gamma_1) + \mathbf{m}(\gamma_2)$. It is an additive cocycle on recorded trajectories: the record preserves outward and return edges as different events.

By (2.47),

$$\mathbf{m}(\gamma^\dagger \circ \gamma) = (0, 2C_q(\gamma) + q(x_n) - q(x_0), 2n), \quad (2.50)$$

where $\gamma^\dagger \circ \gamma$ denotes the temporal concatenation of the outward path followed by the return. The oriented sum of crossings cancels; arithmetic occupation and length remain. This reader does not descend to the quotient that identifies a path followed by its inverse with the empty trajectory, since that quotient eliminates the events preserved by the second and third coordinates.

There is also a purely boundary-based even reading:

$$V_b(\gamma) = \sum_{t=0}^{n-1} b(x_t, x_{t+1})b(x_t, x_{t+1})^\top, \quad V_b(\gamma^\dagger \circ \gamma) = 2V_b(\gamma).$$

The odd and even readings record distinct properties of the same edges; both are computed before the path is forgotten.

Exact application to the signature fiber 555555. The finite enumeration of $\mathcal{X}_\times$ contains 324 seeds, 26 signatures, and 24 seeds with signature 555555. The stable partition under P, H refines this fiber into three classes of eight elements. They can be labeled by their signature after one advance:

$$\mathcal{C}_{177}, \quad \mathcal{C}_{717}, \quad \mathcal{C}_{771}.$$

The half-turn action is

$$HC_{177} = C_{177}, \quad HC_{717} = C_{771}, \quad HC_{771} = C_{717}.$$

The attached finite certificate verifies these identities on all representatives. In the twenty-four seeds, the six raw accumulators of f_ℓ are exactly five. Hence the decimal quotients c_j^{10} vanish on this fiber. The additional information is observed in the positions, integer evaluations, and arithmetic quotients, which have readings different from that signature.

The fixed transverse coordinate a of the cardinal path is 4 or 5 in these seeds. During one turn of nine advances, each value of the moving coordinate $u \in D_9$ is traversed once. Since $\gcd(a, 9) = 1$, the positive residues $r(au)$ permute D_9 . This gives, without using a physical constant,

$$\sum_{u=1}^9 au = 45a, \quad \sum_{u=1}^9 r(au) = 45, \quad \sum_{u=1}^9 q(au) = 5(a-1).$$

The subsequent return traverses the same positions in the opposite direction. Over the eighteen activations,

$$\sum N = 90a, \quad \sum r = 90, \quad \sum q = 10(a-1) = \begin{cases} 30, & a = 4, \\ 40, & a = 5. \end{cases} \quad (2.51)$$

The accumulated integer product is, respectively, 360 or 450. The net winding is zero; the sum of quotients and the eighteen incidences preserve the history of the reading. Each of the three signature classes contains representatives of both transverse values. The same signature and the same transport class of that finite quotient can therefore coexist with distinct arithmetic occupations.

These results connect preserved memory with concrete operations: permutation of half-cycles, boundary differences, arithmetic quotients, and incidence records. The physical-mathematical identification of any subsequent reader preserves its own maps; the identities proved here belong to the internal dynamics of the product channel.

2.6.7 Arithmetic memory of the multiplicative channel

Preservation of the regional signature extends to the full arithmetic evaluation of APP. For an oriented position $x = (i, j, d) \in \{1, \dots, 9\}^2 \times \{N, E, S, O\}$, let $\chi(x) = (a, b, \epsilon)$, where a is the fixed coordinate, b the moving coordinate, and ϵ its orientation:

$$\chi(i, j, N) = (j, i, -1), \quad \chi(i, j, S) = (j, i, +1), \quad \chi(i, j, E) = (i, j, +1), \quad \chi(i, j, O) = (i, j, -1).$$

The axis label $\iota \in \{i, j\}$ preserves which of the two original coordinates moves. Writing $\langle b \rangle_9 = 1 + ((b-1) \bmod 9)$, the operators and evaluation are

$$\bar{P}(a, b, \epsilon) = (a, \langle b + \epsilon \rangle_9, \epsilon), \quad \bar{H}(a, b, \epsilon) = (a, b, -\epsilon), \quad (2.52)$$

$$n(a, b) = ab, \quad r(a, b) = \langle ab \rangle_9, \quad q(a, b) = \frac{ab - r(a, b)}{9}. \quad (2.53)$$

Proposition 2.26 (Exact representation of arithmetic transport). *The identities $\chi P = \bar{P}\chi$ and $\chi H = \bar{H}\chi$ hold. The representation χ has 162 states and is the coarsest of the representations stable under P, H that preserve the integer product or, equivalently, the pair (r, q) . Additionally preserving ι recovers the 324 original oriented positions.*

Proof. Advance modifies only b , and the half-turn only ϵ , proving the two transport identities. The nine observations $n_k = n(\bar{P}^k \chi(x))$, $0 \leq k < 9$, run through $a, 2a, \dots, 9a$. Thus one recovers $a = \min_k n_k$, $b = n_0/a$, and the sign through $(n_1/a - n_0/a) \bmod 9 \in \{1, 8\}$. Every stable equivalence that preserves n preserves these nine observations and therefore distinguishes all states (a, b, ϵ) . The identity $n = r + 9q$ proves equivalence of the two evaluations. Each fiber of χ contains the two transposed routes: $(i, j, d) \mapsto (j, i, d^\top)$, with $N^\top = O, O^\top = N, E^\top = S, S^\top = E$. The axis label distinguishes these two preimages and allows position and direction to be reconstructed. $\square$

The TRIT selector activates the product at $t \equiv 2, 5, 8 \pmod{9}$, before each advance. The half-turn acts at the end of every 27 instants. If y_{t-1} denotes the preceding arithmetic state, the increment and its accumulated memory are determined by

$$\eta_q(t) = \mathbf{1}_{\{t \equiv 2 \pmod{3}\}} q(y_{t-1}), \quad \lambda_q(t) = \lambda_q(t-1) + \eta_q(t), \quad \lambda_q(0) = 0. \quad (2.54)$$

The bound $0 \leq q(a, b) \leq a-1$ holds. To invert an update, one first undoes the corresponding half-turn, then the advance, and finally subtracts the quotient recovered from the preceding state. The record retains phase, axis, event order, and memory; λ_q is an additive cocycle of this transport.

Each block of 27 contains nine advances and traverses the moving coordinate once. Hence the return of position and orientation at 54, and of phase at 108, coexist with

$$\lambda_q(108N) = NQ_X^+(a), \quad Q_X^+(a) = \frac{180a - 4 \sum_{b=1}^9 r(a, b)}{9}, \quad N \geq 0. \quad (2.55)$$

Indeed, each block sums the products $a, 2a, \dots, 9a$; summing $n = r + 9q$ over the four blocks gives the formula. If a is coprime to 9, the residues sum to 45, and $Q_X^+(a) = 20(a-1)$. In window j , counted from $j = 0$, the relative calendar sign is $\sigma_j = (-1)^{\lfloor j/3 \rfloor}$, and the effective orientation is $\epsilon_j = \epsilon_0 \sigma_j$. The charge signed by σ_j vanishes; the charge weighted by the effective orientation likewise vanishes after multiplication by ϵ_0 . The ledger preserves the initial orientation, the accumulated charge, and its increments. These quantities belong to the multiplicative cursor; the readers of a trajectory that alternates between both sheets act on the record of that trajectory.

The finite emission, its coupling of signatures, and ternary reduction have successive types:

$$S^2 \xrightarrow{s \times e} \text{im } s \times \text{im } e \xrightarrow{G} \mathcal{U}_6 \xrightarrow{\Psi_6} W_6 \hookrightarrow \mathbb{F}_3^6.$$

Section 4 proves the factorization of the 104 976 seeds into $18 \cdot 26 = 468$ emissions and the 243 visible words in the ambient space of 729 elements. The maps G and Ψ_6 are those in (4.3) and (4.4); the record preserves the arithmetic preimages together with this finite image.

2.6.8 Compatibility of quotients and refinement cylinders

Prolongation compares states that simultaneously preserve remainders, quotients, and boundaries. In the six-coordinate linear chart of the record, the matrix

$$A_H = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \quad \det A_H = 1,$$

gives the division map

$$\mathcal{H} : \mathbb{Z}^6 \longrightarrow \{0, 1, 2\}^6 \times \mathbb{Z}^6, \quad x \longmapsto (u, x^+), \quad y = xA_H, \quad u = [y]_3, \quad x^+ = (y - u)/3. \quad (2.56)$$

Rows are interpreted as vectors. The pair (u, x^+) preserves $y = u + 3x^+$ and permits recovery of $x = (u + 3x^+)A_H^{-1}$. The ternary orientation of u is then obtained through ρ_3^{or} , preserving the preceding quotient. This makes the reversible content of the local chart explicit.

The cylindrical chart relates a ternary prefix of length L , with integer index N , to K decimal triads, with index D . The corresponding half-open cylinders have rational endpoints

$$I_3(N, L) = \left[\frac{N}{3^L}, \frac{N+1}{3^L} \right), \quad I_{1000}(D, K) = \left[\frac{D}{1000^K}, \frac{D+1}{1000^K} \right).$$

Define the boundary integers

$$a = N1000^K - D3^L, \quad b = (N+1)1000^K - D3^L.$$

Lemma 2.27 (Integer update of compatible cylinders). *The preceding cylinders intersect if and only if $a < 3^L$ and $b > 0$. Adjoining a ternary block $u \in \{0, \dots, 728\}$ gives $N' = 729N + u$ and $L' = L + 6$. For $\Delta K \in \{0, 1\}$, the boundaries are updated by*

$$\begin{aligned} \Delta K = 0 : \quad a' &= 729a + u1000^K, \\ \Delta K = 1 : \quad D' &= 1000D + \delta, \quad a' = 729000a + u1000^{K+1} - \delta 729 3^L, \end{aligned}$$

where $\delta \in \{0, \dots, 999\}$ is the triad of the compatible prolongation. In both cases, $b' = a' + 1000^{K+\Delta K}$.

Proof. Two half-open intervals intersect exactly when each left endpoint is less than the other's right endpoint. Multiplying those two inequalities by $3^L 1000^K$ gives $a < 3^L$, $b > 0$. Substituting N' , L' , and D' into $a' = N' 1000^{K+\Delta K} - D' 3^{L'}$ produces the two recurrences. Subtracting the definitions of b' and a' gives the last identity. $\square$

These operations act on integer prefixes and their compatible restrictions. The Archimedean reading appears when the resulting cylinder system is evaluated. The prolongation domain also incorporates orientation, carry, and the joint boundary signature of (2.37); interval comparison is one component of this joint compatibility.

2.6.9 Extension relations and joint nonadic connection

For an observable route $r : q \rightarrow q'$, prolongation has type

$$\text{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'}, \quad \text{Ext}_{r_2} \circ \text{Ext}_{r_1} \subseteq \text{Ext}_{r_2 r_1}.$$

The identities satisfy $\Delta \mathcal{F}_q \subseteq \text{Ext}_{\text{id}_q}$. It preserves the truncations, sheets, orientations, carries, and boundary incidences of the two histories. The identities and admissible concatenations form the category of TPK states and prolongations. This relational structure admits local fibers with more than one continuation and allows their persistence to be imposed jointly.

Let $\mathcal{H}_m$ be the set of admissible histories of depth m , with truncations $p_{n,m}$. Survival is expressed by

$$\text{Surv}_m^{(N)} = p_{N,m}(\mathcal{H}_N), \quad \text{Surv}_m = \bigcap_{N \geq m} \text{Surv}_m^{(N)}, \quad X_\chi = \varprojlim_m \text{Surv}_m^{(\chi)}.$$

The compatible section simultaneously preserves all its prefixes. The local connection relations $\mathcal{R}_k$ compose into

$$\Gamma_{9,k} = \mathcal{R}_{k+8} \circ \dots \circ \mathcal{R}_k, \quad \Gamma_{9,k} : \hat{X}_k \rightarrow \hat{X}_{k+9}. \quad (2.57)$$

The indicated domain is that of compatible histories. On an individualized history, the nine relations represent its effective transport; the system retains all admissible prolongations before this individualization.

The return law is written

$$g(\Gamma_{9,k}x) = g(x), \quad q_{\text{mem}}(\Gamma_{9,k}x) = q_{\text{mem}}(x) + 1.$$

The phase returns and the history is prolonged. The prefixes $w_6, w_{12}, w_{18}, w_{24}, w_{30}$, the boundary R_{36} , and the charts G_9 are states, extensions, or presentations of this construction, with types distinct from the holonomy Γ_9 . Their regional realization is developed in section 5; the composition of turns and their representations are proved in 5.11–5.13.

2.6.10 Phase incidence and arithmetic origin of the channels

The selector M_{ph} induces the classes

$$H_+ = \{1, 4, 7\}, \quad H_- = \{2, 5, 8\}, \quad H_0 = \{3, 6, 9\}, \quad H_{\text{act}} = H_+ \sqcup H_-.$$

The phase origin fixes the terminal representative $\star_{\text{ph}} = 9$ and the set $O_{\text{ph}} = D_9 \setminus \{\star_{\text{ph}}\}$. Write $P_2(H_{\text{act}})$ for the unordered pairs of active phases and define

$$\mathcal{F}_6^{\text{ph}} = H_{\text{act}} \times P_2(H_{\text{act}}), \quad \mathcal{F}_8^{\text{ph}} = O_{\text{ph}} \times P_2(H_{\text{act}}), \quad \Theta_{\text{ph}} = D_9 \times P_2(H_{\text{act}}).$$

Theorem 2.28 (Incidence of the phase semicorona). *At the declared window origin,*

$$|\mathcal{F}_6^{\text{ph}}| = 90, \quad |\mathcal{F}_8^{\text{ph}}| = 120, \quad |\Theta_{\text{ph}}| = 135.$$

The oriented boundary, pointed return, and lateral closure,

$$\partial_{\text{ph}} = \Theta_{\text{ph}} \times \{-1, +1\}, \quad E_{\text{ph}} = \partial_{\text{ph}} \sqcup \{\infty\}, \quad L_{\text{ph}} = H_0 \times P_2(H_{\text{act}}), \quad P_{\text{ph}} = \partial_{\text{ph}} \sqcup L_{\text{ph}},$$

satisfy

$$|\partial_{\text{ph}}| = 270, \quad |E_{\text{ph}}| = 271, \quad |L_{\text{ph}}| = 45, \quad |P_{\text{ph}}| = 315.$$

Translation of the origin transports all these sets by bijections and preserves their incidences.

Proof. There are six active phases and $\binom{6}{2} = 15$ unordered pairs. Products by 6, 8, 9 give 90, 120, 135. Orientation doubles the last set, the pointed return adds one element, and the three neutral phases contribute $3 \cdot 15 = 45$ lateral incidences. The C_9 action simultaneously transports the selector, the terminal representative, and each factor of the Cartesian products. $\square$

The difference of active cardinalities is $120 - 90 = 30$. The normalized defect therefore has the exact expression

$$\frac{30}{120} - \frac{30}{90} = -\frac{1}{12}. \quad (2.58)$$

Here 90 and 120 are cardinalities of incidences produced by the phase rule. The subsequent hexadic-octadic realization transports this incidence to its combinatorial objects. The angular coordinates of these objects are established through their realization maps.

2.6.11 Temporal record, signed extraction, and reversible coordinates

Let $E_m = \{9(m-1) + 1, \dots, 9m\}$. Restricting a history to twelve windows preserves in each its subroute τ_m , orientation σ_m , boundary carry κ_m , and local record Λ_m :

$$\mathcal{R}_{12}(x) = ((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m))_{m=1}^{12}.$$

The domain of $\mathcal{R}_{12}$ is histories; the elements of the group C_{12} are sector classes, and Θ_{108} preserves their calendar. This distinction keeps the data preserved by each realization visible.

The signed extraction of section 8 factors through

$$\mathcal{X}_{\text{TPK}}^{\text{term, mem}} \xrightarrow{\mathcal{R}_{12}} \mathcal{R}_{12}^{\text{mem}} \xrightarrow{\mathcal{E}_{108}^{90, 120}} \mathbb{Z}^{12} \times \mathbb{Z}^{12} \times \mathbb{Z} \xrightarrow{\Pi_H} \mathbb{Z}^{12}. \quad (2.59)$$

The second map extracts the channels $(b^{90}, b^{120}, Q_{\text{TPK}})$; the third constructs the signed vector U . The rule $\varepsilon_9 = M_{\text{ph}}$ belongs to phase selection, whereas $\mathcal{E}_{108}^{90, 120}$ acts on the full channel record. Their domains and outputs distinguish them.

Equations (8.8) and (8.9) give the arithmetic action of Π_H . The transformation $\mathcal{H}_{12}$, constructed from the three step-three orbits, then reversibly relates U to the record K . The cyclic differences and total charge provide subsequent readings of the same record. This preserves the common provenance of the regional coordinates and the signed coordinate before the integer closure determining α .

2.6.12 Reduced transfer and memory preserved by the dynamics

Once the emission factorization has been constructed, $\mathcal{U}_6 \simeq \mathcal{A}_+ \times \mathcal{A}_\times$, with $|\mathcal{A}_+| = 18$ and $|\mathcal{A}_\times| = 26$, there are two projections on functions of the catalog:

$$(E_+ f)(s, e) = \frac{1}{18} \sum_{s' \in \mathcal{A}_+} f(s', e), \quad (E_\times f)(s, e) = \frac{1}{26} \sum_{e' \in \mathcal{A}_\times} f(s, e').$$

The signature alphabets $\mathcal{A}_+$, $\mathcal{A}_\times$ are distinct from the seed sets of the two cursors. Each projection averages the active coordinate and preserves the other. With $P_{+1} = E_+$, $P_{-1} = E_\times$, $P_0 = I$, the transfer kernel on $\mathcal{U}_6 \times C_9$ is

$$\mathcal{K}_{\text{ph}}((u, r), (v, r + 1)) = P_{M_{\text{ph}}(r)}(u, v).$$

The two projections commute and are idempotent. Thus their composition during one phase turn satisfies

$$\mathcal{K}_{\text{ph}}^9|_r = E_+ E_\times, \quad (E_+ E_\times)(u, v) = \frac{1}{468}. \quad (2.60)$$

This operator is a reduced-memory representation on emissions already constructed. The holonomy Γ_9 also preserves the coordinates of the history on which that representation is obtained.

In an isometric realization U_{Γ_9} of transport, and for an isometric inclusion J of the observed coordinate, its two components are

$$T = J^* U_{\Gamma_9} J, \quad \eta = (I - J J^*) U_{\Gamma_9} J.$$

One obtains directly

$$U_{\Gamma_9} J = J T + \eta, \quad J^* \eta = 0, \quad I - T^* T = \eta^* \eta.$$

The component η is computed from the same realized update and the declared projection. The telescoping identity (4.8) preserves both the contributions expressed in the orthogonal complement and the terminal term. This separation is the operator basis for the subsequent distinction between recorded information, entropy of an observation, and dimensional realization of action.

3 Topological Phase Kernel and generation of the seed catalogue

The two APP evaluations acquire generative efficacy once their activation, evaluation position, positional transport and the information preserved by each transition have been specified. The *Topological Phase Kernel*, abbreviated TPK, combines these operations into a dynamics on enriched states. Its definition comprises selection, transport, arithmetic updating and preservation of the record. The finite construction developed in this section provides the seeds on which the regional lifts and coinductive prolongation subsequently act.

The principal result of this stage is a generated and classified catalogue: the initial conditions of the two cursors produce integer emissions; their ternary reading determines oriented words; and the dihedral action organises these words into orbits. Each step preserves its preimages and the relations that permit recovery of the preceding description. The chain of cardinalities

$$104\,976 \longrightarrow 468 \longrightarrow 243 \subset 729, \quad 243 \longrightarrow 43,$$

abbreviates distinct objects. Each object is constructed and each count justified below, before any of them is used as data for regional selection.

3.1 Phase selection, transport and enriched state

3.1.1 Tritic selector of operative phase

The nonadic phase takes its values in $D_9 = \{1, \dots, 9\}$. The tritic selector is the function

$$M_{\text{ph}} : D_9 \longrightarrow \{-1, 0, +1\}, \quad M_{\text{ph}}(g) = \begin{cases} +1, & g \in \{1, 4, 7\}, \\ -1, & g \in \{2, 5, 8\}, \\ 0, & g \in \{3, 6, 9\}. \end{cases} \quad (3.1)$$

In the emitter realisation, the value $+1$ activates the additive evaluation, -1 activates the multiplicative evaluation, and 0 records the neutral event. The cardinal orientation of the cursor is another state coordinate. This distinction specifies both the operation being read and the direction in which its evaluation point is transported.

For transitions indexed by $t \geq 1$, the phase preceding the reading is

$$g_t = 1 + [(t-1)]_9, \quad \epsilon_t = M_{\text{ph}}(g_t), \quad (3.2)$$

where $[n]_b$ denotes the representative of n modulo b in $\{0, \dots, b-1\}$. A window of nine transitions contains the sequence $(+1, -1, 0, +1, -1, 0, +1, -1, 0)$.

3.1.2 Oriented cursors and cardinal transport

The indices $I_9 = \{0, \dots, 8\}$ are used for positions, and the positive labels $(a, b) = (i+1, j+1)$ for APP evaluations. A finite cursor is

$$c = (i, j; d) \in \mathcal{S} := I_9^2 \times \mathcal{D}, \quad \mathcal{D} = \{N, E, S, O\}.$$

The four directions act by

$$v_N = (-1, 0), \quad v_E = (0, 1), \quad v_S = (1, 0), \quad v_O = (0, -1).$$

The cardinal map corresponding to d is

$$T_d(i, j) = ([i + (v_d)_1]_9, [j + (v_d)_2]_9).$$

Thus M_{ph} and T_d have distinct domains, codomains and actions. Tritic activation selects one of the two cursors; cardinal transport changes its position in the direction already retained by that cursor.

The integer lift of the cursor is $\tilde{c} = (x, y; d) \in \mathbb{Z}^2 \times \mathcal{D}$, with finite position $([x]_9, [y]_9)$. Its winding counts are recovered from

$$x = 9[x/9] + [x]_9, \quad y = 9[y/9] + [y]_9. \quad (3.3)$$

These two divisions concern spatial transport. The division of an additive or multiplicative evaluation constitutes a distinct arithmetic record, which is preserved together with them.

3.1.3 Composition of the transition

The state of the finite realisation includes the two lifted cursors, their directions, the phase, the accumulators of the active window, the ordered list of blocks already emitted and the transition record. Each local record contains the preceding positions, the integer evaluations of both sheets, their residues and quotients, the active regime and the orientations. Prefix and boundary fields arising in a prolongation are incorporated into the same state with their compatibility rules.

The transition is ordered as

$$\mathcal{U}_t = \text{Mem}_t \circ \text{Tra}_t \circ \text{Sel}_t. \quad (3.4)$$

Selection determines the regime and preserves the sample taken before any displacement. Transport applies the active movement and the cardinal reversal prescribed by the schedule. Updating incorporates that sample into the accumulators and the record; at the end of a window, it emits the corresponding block. The composition therefore uses a sample taken before transport, although its definitive storage occurs at the end of the transition.

3.2 Initial conditions and reversible enumeration

The additive and multiplicative sheets have independent cursors. The domain of initial conditions is the ordered product

$$\Omega_{\text{seed}} = \mathcal{S}_+ \times \mathcal{S}_\times, \quad |\mathcal{S}| = 9^2 \cdot 4 = 324.$$

Consequently,

$$|\Omega_{\text{seed}}| = 324^2 = 104\,976. \quad (3.5)$$

The number 81 counts positions; 324 counts the oriented positions of one cursor; 104 976 counts ordered pairs of cursors. Choosing a position in APP retains only part of an initial condition of the emitter.

Fix the numbering

$$\text{ind}(N) = 0, \quad \text{ind}(E) = 1, \quad \text{ind}(S) = 2, \quad \text{ind}(O) = 3,$$

and define

$$\nu(i, j; d) = 4(9i + j) + \text{ind}(d). \quad (3.6)$$

Proposition 3.1 (Exact enumeration of the initial conditions). *The map ν is a bijection from $\mathcal{S}$ onto $\{0, \dots, 323\}$. The pair of indices is reversibly encoded by $324\nu_+ + \nu_\times \in \{0, \dots, 104\,975\}$.*

Proof. Division by four recovers

$$\text{ind}(d) = [\nu]_4, \quad j = [\lfloor \nu/4 \rfloor]_9, \quad i = \lfloor \nu/36 \rfloor.$$

The bounds on the domain ensure that these three quantities belong to their respective index sets and reconstruct ν through (3.6). For the pair, Euclidean division by 324 recovers the quotient ν_+ and the residue $\nu_\times$. $\square$

The enumeration traverses the complete domain before any regional grouping. The indices locate the preimages of an emission while keeping explicit the condition that produced each trajectory.

3.3 Sheet evaluation and arithmetic record

At a positive position (a, b) , the integer evaluations are $A_+(a, b) = a + b$ and $A_\times(a, b) = ab$. For $n > 0$, retain

$$\rho_9(n) = 1 + [n - 1]_9, \quad q_9(n) = \lfloor (n - 1)/9 \rfloor, \quad n = \rho_9(n) + 9q_9(n). \quad (3.7)$$

Additive activation incorporates $\rho_9(A_+)$ into the corresponding accumulator. Multiplicative activation uses an observable of the residue $\rho_9(A_\times)$, defined next.

3.3.1 Discrete exponent in the units modulo nine

The powers of 2 modulo 9 traverse

$$2^0, 2^1, \dots, 2^5 \equiv 1, 2, 4, 8, 7, 5 \pmod{9}, \quad 2^6 \equiv 1 \pmod{9}.$$

The discrete logarithm in this group of six units is represented by

$$\ell_9(1) = 0, \quad \ell_9(2) = 1, \quad \ell_9(4) = 2, \quad \ell_9(8) = 3, \quad \ell_9(7) = 4, \quad \ell_9(5) = 5. \quad (3.8)$$

The emission law extends the observable by $\ell_9(3) = \ell_9(6) = \ell_9(9) = 0$. To preserve its provenance, the recorded sample includes the original residue and the indicator of membership in the group of units. An observed value of zero can therefore originate from the unit 1 or from one of the three non-invertible residues, with its origin remaining recoverable.

3.3.2 Order of reading and movement

Evaluations are performed at the positions preceding the transition. If $\epsilon_t = +1$, the additive reading is accumulated and its cursor is displaced. If $\epsilon_t = -1$, the multiplicative discrete exponent is accumulated and the second cursor is displaced. If $\epsilon_t = 0$, the neutral counter is incremented and both positions are preserved.

At the end of steps 27 and 54, both directions are replaced by their opposites. The windows are concatenated with positions and orientations preserved. At the start of each window, only its three local accumulators are reset to zero; the preceding record remains available.

Proposition 3.2 (Inversion of the lifted transport). *Let $a, b \in \{0, 1\}$ be the activation and reversal indicators of a cursor. If opp interchanges $N \leftrightarrow S$ and $E \leftrightarrow O$, the transformation*

$$F_{a,b}(z, d) = (z + av_d, \text{opp}^b(d))$$

has inverse

$$F_{a,b}^{-1}(z', d') = (z' - av_{\text{opp}^b(d')}, \text{opp}^b(d')).$$

Spatial reduction modulo nine intertwines this transformation with the finite transport.

Proof. Applying opposition twice recovers the initial direction. Once that direction has been recovered, subtracting av_d reverses the displacement. For the spatial projection, it suffices to use $[z + av_d]_9 = [[z]_9 + av_d]_9$ componentwise. Directional reversal preserves this equality. $\square$

The phase determines the indicators used by the inverse. The spatial and arithmetic records can therefore be preserved jointly, including when the boundary of the representative I_9^2 is crossed.

3.4 Six windows and decimal block emission

3.4.1 Accumulation within a nonadic window

Window m , with $1 \leq m \leq 6$, comprises steps $9m - 8, \dots, 9m$. Let S_m be the sum of its active positive additive residues, E_m the sum of its active discrete exponents, and Z_m its number of neutral events.

Lemma 3.3 (Activation multiplicities). *Each window contains three additive activations, three multiplicative activations and three neutral activations. In particular, $Z_m = 3$.*

Proof. The phases of a window traverse exactly $1, \dots, 9$. Each of the three sets in (3.1) has cardinality three. $\square$

The six windows comprise 54 transitions and produce six blocks. This length belongs to the initial emitter defined here. Continuation in depth subsequently acts on the words and their compatible records.

3.4.2 Decimal encoding rule

The emission law specifies the digits

$$\begin{aligned} d_{m,1} &= [S_m]_{10}, \\ d_{m,2} &= [S_m + E_m]_{10}, \\ d_{m,3} &= [3S_m + 5E_m + 7Z_m]_{10}, \\ U_m &= 100d_{m,1} + 10d_{m,2} + d_{m,3}. \end{aligned} \quad (3.9)$$

The weights 100, 10, 1 are the hundreds, tens and units positions. The subscript 10 denotes the decimal residue, and each block satisfies $0 \leq U_m \leq 999$. The emitted word is the ordered sequence $\mathbf{U} = (U_1, \dots, U_6)$, with a fixed width of three digits per block.

The coefficients 3, 5, 7 constitute the explicit weights of this emission law. Once the rule has been fixed, each digit is calculated from the accumulators, and the results of this section follow from that rule. The number 1 appearing in its reduced form has a further specific provenance: $7Z_m = 21 \equiv 1 \pmod{10}$.

If $s_m = [S_m]_{10}$ and $e_m = [E_m]_{10}$, one obtains

$$G(s_m, e_m) = 100s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10} = U_m. \quad (3.10)$$

Here the letter e_m denotes a component of the discrete-exponent signature. Its definition belongs to the finite emitter.

Proposition 3.4 (Recovery of the two signatures). *The map $G : \{0, \dots, 9\}^2 \rightarrow \{0, \dots, 999\}$ is injective. For a block $U = 100a + 10b + c$ in its image, the inverse is*

$$s = a, \quad e = [b - a]_{10}, \quad c = [3s + 5e + 1]_{10}.$$

Consequently, the word $\mathbf{U}$ determines the two six-component signatures $\mathbf{s}$ and $\mathbf{e}$.

Proof. The last two summands of (3.10) belong to $\{0, \dots, 99\}$, so the hundreds digit recovers s . The tens digit is $[s + e]_{10}$; subtracting s and reducing modulo ten recovers e . The units digit verifies the third congruence. The reconstruction applies independently to the six positions of the word. $\square$

The integer totals S_m, E_m are recovered from the recorded readings; the hundreds and tens digits recover their residues. For example, a total $S_m = 15$ produces $s_m = 5$. This distinction identifies precisely which information each level of encoding preserves.

3.4.3 Positional encoding of the six-block word

The six blocks admit a fixed-width integer encoding,

$$\mathcal{N}(\mathbf{U}) = \sum_{m=1}^6 U_m 1000^{6-m}, \quad 0 \leq \mathcal{N}(\mathbf{U}) < 1000^6. \quad (3.11)$$

The first block occupies the position of greatest weight, and the sixth the base-thousand units position. A block 058 occupies a complete position with integer value 58; its three-digit width belongs to the decimal presentation of that position.

Proposition 3.5 (Positional recovery of the emissions). *The encoding (3.11) determines the complete word by*

$$U_m = \left\lfloor \left\lfloor \frac{\mathcal{N}(\mathbf{U})}{1000^{6-m}} \right\rfloor \right\rfloor_{1000}, \quad 1 \leq m \leq 6.$$

Proof. On division by 1000^{6-m} , the preceding blocks yield integer multiples of 1000, block m yields U_m , and the contribution of the subsequent blocks belongs to $[0, 1)$. Taking the integer part eliminates the latter contribution, and the residue modulo 1000 eliminates the former. $\square$

This encoding preserves the six emissions. Its domain has ordered base-thousand positions, whereas the APP domain has spatial positions modulo nine. The composition retains both indices: m identifies an emission window, and (i, j) identifies the cell visited during a transition in that window.

3.5 Complete calculation of an emission beginning with 501

Fix the initial conditions

$$c_{+,0} = (0, 4; N), \quad c_{\times,0} = (2, 3; N), \quad \nu_+ = 16, \quad \nu_{\times} = 84.$$

The initial positive positions are $(1, 5)$ and $(3, 4)$. The preceding chronological index is $t = 0$, the first phase read is $g_1 = 1$, and the record is empty. These coordinates are the inputs to the calculation; the blocks are obtained by executing the recurrence.

Table 2: First window: preceding readings and active accumulation.

t	ϵ_t	position +	position $\times$	ΔS	ΔE	(S, E, Z)
1	+1	(1, 5)	(3, 4)	6	0	(6, 0, 0)
2	-1	(9, 5)	(3, 4)	0	0	(6, 0, 0)
3	0	(9, 5)	(2, 4)	0	0	(6, 0, 1)
4	+1	(9, 5)	(2, 4)	5	0	(11, 0, 1)
5	-1	(8, 5)	(2, 4)	0	3	(11, 3, 1)
6	0	(8, 5)	(1, 4)	0	0	(11, 3, 2)
7	+1	(8, 5)	(1, 4)	4	0	(15, 3, 2)
8	-1	(7, 5)	(1, 4)	0	2	(15, 5, 2)
9	0	(7, 5)	(9, 4)	0	0	(15, 5, 3)

Note. Positions are positive labels of the cells preceding movement. Neutral events increment only Z .

The three active additive evaluations are

$$1 + 5 = 6, \quad 9 + 5 = 14 = 5 + 9, \quad 8 + 5 = 13 = 4 + 9,$$

so $S_1 = 6 + 5 + 4 = 15$. The three active multiplicative evaluations are $3 \cdot 4 = 12$, $2 \cdot 4 = 8$ and $1 \cdot 4 = 4$. Their positive residues are 3, 8, 4, with observables 0, 3, 2; hence $E_1 = 5$. Finally, $Z_1 = 3$. The decimal rule gives

$$d_{1,1} = [15]_{10} = 5, \quad d_{1,2} = [15 + 5]_{10} = 0, \quad d_{1,3} = [45 + 25 + 21]_{10} = 1,$$

and one obtains $U_1 = 501$.

Table 3: The six windows of initial condition (16, 84).

m	steps	S_m	E_m	Z_m	s_m	U_m	$[-U_m]_3$
1	1–9	15	5	3	5	501	0
2	10–18	6	5	3	6	614	1
3	19–27	24	5	3	4	498	0
4	28–36	21	5	3	1	169	2
5	37–45	12	5	3	2	272	1
6	46–54	12	5	3	2	272	1

The second additive reading above has arithmetic quotient 1, whereas its lifted cursor is at $(-1, 4; N)$, with vertical winding count -1 . These are distinct data in the same record. Both participate in the complete recovery of the evaluation and its position.

The resulting word and its signatures are

$$\mathbf{U} = (501, 614, 498, 169, 272, 272),$$

$$\mathbf{s} = (5, 6, 4, 1, 2, 2), \quad \mathbf{e} = (5, 5, 5, 5, 5, 5).$$

Cardinal reversal after step 27 changes the trajectories of the final three windows. The repeated block 272 corresponds to two successive windows with their own positions and chronological record.

As a second case, the minimal initial condition $(\nu_+, \nu_-) = (0, 0)$ produces

$$\mathbf{U} = (252, 109, 252, 903, 850, 850), \quad \mathbf{s} = (2, 1, 2, 9, 8, 8), \quad \mathbf{e} = (3, 9, 3, 1, 7, 7).$$

In this case, both cursors return to their initial positions and orientations after 54 transitions. The nonadic phase completes six cycles, the chronological counter advances by 54 units, and the record contains 54 samples. The positional reading of the return and the recorded history describe distinct properties of the execution.

3.6 Factorisation of the finite census

The signatures $f_+(\nu) = \mathbf{s}$ and $f_-(\nu) = \mathbf{e}$ are calculated by executing the corresponding cursor through the six windows. The independence of positions and directions between the two cursors gives

$$T_{\text{em}}(\nu_+, \nu_-) = G^{(6)}(f_+(\nu_+), f_-(\nu_-)), \quad (3.12)$$

where $G^{(6)}$ applies (3.10) at each position. This factorisation first permits enumeration of $324 + 324$ trajectories and then combination of the distinct signatures, preserving the multiplicities of their preimages.

3.6.1 Images of the two signatures

The following tables include every signature. Its multiplicity is the number of initial conditions producing it; the minimal index locates one such condition. The complete preimage is defined by

$$f_{\sigma}^{-1}(a) = \{\nu \in \{0, \dots, 323\} : f_{\sigma}(\nu) = a\}.$$

The recurrence of the preceding sections determines every element of this set by a finite integer calculation.

Table 4: The eighteen additive signatures.

signature	multiplicity	$\nu_{\min}$	signature	multiplicity	$\nu_{\min}$
122564	18	17	564122	18	16
122898	18	24	645212	18	4
212645	18	5	654889	18	33
212988	18	0	889221	18	13
221456	18	29	889654	18	32
221889	18	12	898122	18	25
456221	18	28	898465	18	20
465898	18	21	988212	18	1
546988	18	9	988546	18	8

Table 5: The twenty-six multiplicative signatures.

signature	multiplicity	$\nu_{\min}$	signature	multiplicity	$\nu_{\min}$
000000	108	8	555555	24	84
177393	8	1	555717	8	14
177555	8	54	555771	8	18
177771	8	11	555933	8	24
339555	8	6	717555	8	12
339771	8	15	717717	8	21
339933	8	59	717933	8	19
393177	8	0	771177	8	9
393393	8	45	771339	8	13
393555	8	40	771555	8	16
555177	8	52	933339	8	57
555339	8	4	933555	8	26
555393	8	41	933717	8	17

Proposition 3.6 (Image and multiplicities of the emitter). *The image $\mathcal{U}_6 = \text{im } T_{\text{em}}$ contains 468 distinct words. Their preimages are distributed into 432 fibres of cardinality 144, 18 of cardinality 432 and 18 of cardinality 1944.*

Proof. Exhaustive evaluation of the algorithm in Section 3.9 on the indices $0, \dots, 323$ gives the two tables above. The first partition has 18 classes of cardinality 18; the second has 24 classes of cardinality 8, one of cardinality 24 and one of cardinality 108. The sums $18 \cdot 18 = 324$ and $24 \cdot 8 + 24 + 108 = 324$ verify the total mass of both partitions. Proposition 3.4 makes the coupling of their images injective; there are therefore $18 \cdot 26 = 468$ emissions. By (3.12), the preimage of a pair of signatures is the product of their two preimages. This gives $18 \cdot 24 = 432$ products of cardinality $18 \cdot 8 = 144$, 18 of cardinality $18 \cdot 24 = 432$, and 18 of cardinality $18 \cdot 108 = 1944$. Finally,

$$432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976.$$

The enumerative part is an exact finite check of the recurrence; the factorisation explains why this count covers every pair. $\square$

The emission in Example 3.5 has a fibre of cardinality $18 \cdot 24 = 432$. The example with indices $(0, 0)$ has a fibre of cardinality $18 \cdot 8 = 144$. In both cases, the emitted word recovers the signatures, whereas identification of a particular initial condition uses its index or its trajectory record.

3.7 Oriented ternary reading and preservation of fibres

Oriented reduction acts separately on each block:

$$\Psi : \mathcal{U}_6 \longrightarrow \mathbb{F}_3^6, \quad \Psi(U_1, \dots, U_6) = ([-U_1]_3, \dots, [-U_6]_3). \quad (3.13)$$

Its sign fixes an orientation of the catalogue. The balanced identification $0 \mapsto 0, 1 \mapsto +1, 2 \mapsto -1$ is then applied componentwise.

Modulo nine organises APP evaluations and positions; residues modulo ten form each digit; base thousand combines three decimal positions; and modulo three determines the ternary signature of each block. These four uses have defined objects and functions. In particular, Ψ reads six blocks and produces six trits; a positional conversion of the integer formed by eighteen digits would be a different operation.

For the two conditions already calculated, one obtains

$$\begin{aligned} \Psi(501, 614, 498, 169, 272, 272) &= (0, 1, 0, 2, 1, 1), \\ \Psi(252, 109, 252, 903, 850, 850) &= (0, 2, 0, 0, 2, 2). \end{aligned}$$

The decimal emitter is defined before this reduction. The function of Ψ is to produce the ternary object on which orbital symmetry and regional selectors act. This function explains its position in the genealogy: it connects an already calculated emission with a structural classification.

The image $\mathcal{W}_6 := \Psi(\mathcal{U}_6)$ contains exactly 243 words within the ambient set of $3^6 = 729$ words. Enumeration by the final algorithm gives the histogram

$ \Psi^{-1}(w) $	1	2	3	4	5	6
number of words w	132	66	18	3	6	18.

(3.14)

The sums for the two readings are

$$132 + 66 + 18 + 3 + 6 + 18 = 243, \quad 132 + 2 \cdot 66 + 3 \cdot 18 + 4 \cdot 3 + 5 \cdot 6 + 6 \cdot 18 = 468.$$

The first counts ternary words; the second recovers the number of emissions distributed among their fibres.

Example 3.7 (Two emissions with the same ternary signature). The emissions

$$(498, 501, 614, 252, 252, 109), \quad (870, 810, 923, 252, 252, 109)$$

are distinct and both have image $(0, 0, 1, 0, 0, 2)$ under Ψ . Each has 144 initial conditions. The ternary word preserves a reading class; the emissions and their preimages distinguish the data that converge in that class.

Preservation of fibres is expressed concretely by

$$\mathcal{F}_w = \{\mathbf{U} \in \mathcal{U}_6 : \Psi(\mathbf{U}) = w\}, \quad \Omega_{\mathbf{U}} = T_{\text{em}}^{-1}(\mathbf{U}).$$

The complete preimage of w in the initial domain is therefore the disjoint union of the sets $\Omega_{\mathbf{U}}$, with $\mathbf{U} \in \mathcal{F}_w$. The enriched record can retain the emission, its signatures and the initial condition together with its ternary reading. The exact inverse then concerns the recorded datum with its type; the isolated six-trit word retains the precise scope of (3.13).

3.8 Dihedral action and the forty-three orbits

Define the following permutations on six coordinates:

$$r(u_1, u_2, u_3, u_4, u_5, u_6) = (u_3, u_1, u_2, u_5, u_6, u_4), \quad (3.15)$$

$$s(u_1, u_2, u_3, u_4, u_5, u_6) = (u_4, u_5, u_6, u_1, u_2, u_3). \quad (3.16)$$

The first rotates the two groups of three coordinates in conjugate directions; the second interchanges the groups. Their composition satisfies $r^3 = s^2 = 1$ and $srs = r^{-1}$. They therefore generate an action of the six-element dihedral group D_3 .

Proposition 3.8 (Equivariance and stability of the catalogue). *Reduction by Ψ commutes with the action of D_3 . The sets $\mathcal{U}_6$ and $\mathcal{W}_6$ are stable under its generators.*

Proof. The equality $\Psi(g\mathbf{U}) = g\Psi(\mathbf{U})$ holds at every coordinate because g permutes positions and Ψ applies the same reduction at each position. Stability of the catalogue is a further check: generating its 468 elements by (3.12), one verifies that $r\mathbf{U}$ and $s\mathbf{U}$ again belong to that set. The algorithm in Section 3.9 performs these membership checks. Equivariance then transfers stability to the ternary image. $\square$

Proposition 3.9 (Complete orbital decomposition). *The set $\mathcal{W}_6$ decomposes into 43 orbits: 38 of cardinality six and 5 of cardinality three.*

Proof. For each $w \in \mathcal{W}_6$, form

$$\mathcal{O}(w) = \{w, rw, r^2w, sw, srw, sr^2w\}.$$

Choose the lexicographically least word in this set as representative. Two words have the same representative exactly when they belong to the same orbit. The algorithm generates the list of representatives in Table 6; their cardinalities sum to $38 \cdot 6 + 5 \cdot 3 = 243$. Since every word in the catalogue participates in this operation and the catalogue is stable, the list covers the entire image. $\square$

Table 6: Minimal representatives and cardinalities of the forty-three orbits.

representative	cardinality	representative	cardinality	representative	cardinality
001002	6	002112	6	012222	6
001011	6	002211	6	021022	6
001012	6	002220	6	021121	6
001021	6	011021	6	021202	6
001022	6	011112	6	022022	3
001100	3	011120	6	022112	6
001101	6	011201	6	022121	6
001110	6	011211	6	022202	3
001112	6	011220	6	022211	6
001202	6	011222	6	022222	6
001210	6	012112	6	112112	3
001211	6	012121	6	112211	3
001220	6	012202	6	112222	6
001222	6	012210	6		
002012	6	012220	6		

The orbits inherit integer invariants from their fibres. For each word, $n(w) = |\mathcal{F}_w|$ counts emissions, whereas

$$M(w) = \sum_{\mathbf{U} \in \mathcal{F}_w} |\Omega_{\mathbf{U}}|$$

counts initial conditions. The symbol census $c(w) = (n_0, n_1, n_2)$ records the number of occurrences of each ternary class. Together with the orientation and the support of the trajectories, these data provide the predicates for subsequent regional selection. The classification thus preserves more information than the form of a representative word.

3.9 Self-contained enumeration and verification algorithm

The enumeration uses integers, finite lists and the operations already defined. The following procedure specifies its execution; σ takes the values $+$ and $\times$.

1. For each $v = 0, \dots, 323$, recover $(i, j; d)$ using the inverse of (3.6). Initialise an empty signature list.
2. For each window $m = 1, \dots, 6$, initialise the accumulator $A = 0$. Traverse $t = 9m - 8, \dots, 9m$, calculating ϵ_t by (3.2).
3. If $\sigma = +$ and $\epsilon_t = +1$, add $\rho_9(i + j + 2)$ to A and update $(i, j) \leftarrow T_d(i, j)$. If $\sigma = \times$ and $\epsilon_t = -1$, add $\ell_9(\rho_9((i + 1)(j + 1)))$ and perform the same transport.
4. At the end of $t = 27$ or $t = 54$, replace d by its opposite. At the end of the window, append $[A]_{10}$ to the signature list. Preserve the position and direction for the next window.
5. Group the 324 indices according to equality of their six-component lists. These two groupings give f_+ , $f_\times$ and all their preimages.
6. For each pair of distinct signatures, apply $G^{(6)}$. Assign to the emission the product multiplicity of the two preimages. Verify signature recovery in each block and the sum of multiplicities 104 976.
7. Apply Ψ to each emission and group by equality of words. Count the 243 values and the multiplicities in (3.14).
8. Apply r, s to each emission and verify membership in the catalogue. For each ternary word, form its six dihedral images, remove repetitions and record its lexicographic minimum. Count the 43 orbits and the two sizes in Table 6.

The execution terminates: the first stage evaluates 648 trajectories of 54 transitions; the second combines 468 pairs of signatures; the third groups a set of 243 words. An additional direct check can execute the 104 976 pairs and compare the result with the factorisation (3.12); its scope is the same finite domain.

3.10 Preservation of the record and the function of the seeds

Let $\text{Sample}(q_t)$ be the sample preceding the transition, and $\mathcal{M}_t$ the accumulated record. The memory update is

$$\mathcal{M}_{t+1} = \mathcal{M}_t \parallel \text{Sample}(q_t), \quad (3.17)$$

where $\parallel$ denotes ordered concatenation. Induction gives

$$\mathcal{M}_n = \mathcal{M}_0 \parallel (\text{Sample}(q_0), \dots, \text{Sample}(q_{n-1})).$$

The accumulators are reconstructed by summing the active contributions of each segment of nine records. Their quotients, the winding counts of the cursors and the orientations remain associated with the same transitions. Execution determines that record through the recurrence; storage preserves the specific provenance of the readings.

At the end of this stage, the complete domain of initial conditions, decimal emission, its two recoverable signatures, the ternary image, its fibres and its orbital classification have been constructed. The counts 104 976, 468, 243, 729 and 43 now have a domain and a mechanism of derivation. Regional selections receive these structures with their invariants, and the lifts act on oriented words with preserved provenance. This continuity makes the finite catalogue a constructive antecedent of the coinductive development.

4 Development of the formal kernel: initial conditions, geometry, and refinement

The eighty-one-position support does not coincide with the domain of initial conditions for the dynamics. An initial condition must also specify an orientation, and the additive and multiplicative sheets

have independent cursors. This distinction allows exhaustive reconstruction of the emission without selecting a numerical region in advance.

4.1 Initial states and the emission recurrence

For enumeration we use indices $I_9 = \{0, \dots, 8\}$, related to positive labels by $i \mapsto i + 1$. Let

$$\mathcal{S} = I_9^2 \times \{N, E, S, O\}, \quad \Omega_{\text{seed}} = \mathcal{S}_+ \times \mathcal{S}_\times.$$

Each initial condition $\omega = (s_+, s_\times)$ determines two positions and two directions. The domain comprises all ordered pairs, so

$$|\mathcal{S}| = 324, \quad |\Omega_{\text{seed}}| = 324^2 = 104\,976. \quad (4.1)$$

The independent annex enumerates all these conditions; the body presents the law transforming them. The rows of an emission table are fibers of this initial-condition domain.

The finite realization uses six windows of nine transitions. At each instant t , the phase signature is $M_{\text{ph}}(1 + ((t - 1) \bmod 9))$. If its value is $+1$, the positive residue $\Sigma(i + 1, j + 1) = \rho_9(i + j + 2)$ of the first cursor is read and that cursor is then moved; if its value is -1 , the positive residue $\Pi(i + 1, j + 1) = \rho_9((i + 1)(j + 1))$ of the second cursor is read and the second cursor is then moved. Zero increments the neutral counter. At the end of transitions twenty-seven and fifty-four, cardinal orientations are reversed. Integer evaluations and their quotients are retained in the record, but the emitter uses the residue readings specified here.

In the unit sector, the multiplicative reader uses the base-two discrete logarithm in $(\mathbb{Z}/9\mathbb{Z})^\times$:

$$\ell_9(1) = 0, \quad \ell_9(2) = 1, \quad \ell_9(4) = 2, \quad \ell_9(8) = 3, \quad \ell_9(7) = 4, \quad \ell_9(5) = 5.$$

For finite emission it is extended by zero to the non-unit sector. This extension is an observable, not an identification of the elements 3, 6, 9 with the multiplicative identity. The sheet and its membership in the radical sector remain in the trajectory record.

Let S_m be the sum of the three positive additive residues Σ in window m , E_m the sum of the three discrete exponents of the multiplicative residues Π , and $Z_m = 3$ the number of neutral events. Thus S_m does not sum the integer evaluations $S = i + j$, whose quotients remain in the record. With $[a]_{10} \in \{0, \dots, 9\}$, the decimal emission rule is

$$\begin{aligned} d_{m,1} &= [S_m]_{10}, \\ d_{m,2} &= [S_m + E_m]_{10}, \\ d_{m,3} &= [3S_m + 5E_m + 7Z_m]_{10}, \\ U_m &= 100d_{m,1} + 10d_{m,2} + d_{m,3}. \end{aligned} \quad (4.2)$$

The integer coefficients and calendar of this recurrence are explicit parts of the declared emission law. Values of π, φ, e or α are not received. Distinguishing an emission rule from a constant recognized from its output prevents the genealogy from being hidden in a decimal table.

If $s_m = [S_m]_{10}$ and $e_m = [E_m]_{10}$, the same rule is written

$$U_m = 100s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10}. \quad (4.3)$$

The final one is the residue of $7Z_m = 21$ modulo ten. The oriented ternary reduction of the catalog adopts the convention

$$\Psi(U_1, \dots, U_6) = ([-U_1]_3, \dots, [-U_6]_3). \quad (4.4)$$

The sign of this chart is explicit: changing it modifies the words and requires transport of subsequent orientations. The balanced identification of each component of $\mathbb{F}_3$ with TRIT is performed after (4.4).

Proposition 4.1 (Injective factorization of emission). *The word $U = (U_1, \dots, U_6)$ determines both signatures $s = (s_1, \dots, s_6)$ and $e = (e_1, \dots, e_6)$. The emission image has $18 \cdot 26 = 468$ elements. The ternary projection has 243 distinct values in the ambient set of 729 words.*

Proof. The hundreds digit of U_m recovers s_m . If b_m is its tens digit, then $e_m = [b_m - s_m]_{10}$. The units digit checks the third congruence of (4.3). Hence coupling is injective on the signature product.

Enumerating the 324 conditions for one cursor through the preceding recurrence produces eighteen additive signatures, each with eighteen preimages, and twenty-six multiplicative signatures, of which twenty-four have eight preimages, one has twenty-four, and one has one hundred and eight. The annex preserves these signatures and their initial conditions, allowing the finite enumeration to be repeated without a reference numerical catalog. Injectivity of the coupling gives $18 \cdot 26$ emissions and multiplicities

$$\begin{array}{ccc} 432 & 18 & 18 \\ 18 \cdot 24 & \text{fibers of } 144, & \text{fibers of } 432, \quad \text{fibers of } 1944. \end{array}$$

The sum is $432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976$. Applying (4.4) to the 468 emissions produces exactly 243 words. This last number is the cardinality of the image, not of the space $\mathbb{F}_3^6$. $\square$

This factorization explains the difference between finite exhaustiveness and unlimited depth. Domain (4.1) is complete for the declared support and emitter; it does not contain every prefix of every prolongation in advance. Depth belongs to the system of compatible extensions constructed over those conditions.

4.2 Calendar closure and memory advance

The calendar of one hundred and eight transitions consists of twelve nonadic windows. Its group structure is more precise than an informal product of two periods:

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0, \quad \iota([b]) = [9b], \quad p([t]) = [t]_9. \quad (4.5)$$

If $t = 9b + r$ with $0 \leq r < 9$, adding two phases produces the carry

$$(b, r) + (b', r') = \left(b + b' + \left\lfloor \frac{r + r'}{9} \right\rfloor, [r + r']_9 \right),$$

where the first coordinate is reduced modulo twelve. Composition preserves the term linking the two coordinates.

Proposition 4.2 (Nonsplitting of the calendar). *Sequence (4.5) is exact and admits no section that is a group homomorphism.*

Proof. The kernel of reduction modulo nine is the subgroup of multiples of nine, the image of ι . If the extension split, commutativity would give $C_{108} \cong C_{12} \times C_9$. The first group has exponent 108, whereas the second has exponent $\text{lcm}(12, 9) = 36$, a contradiction. $\square$

Nine transitions restore the nonadic phase and advance one window; one hundred and eight restore the finite calendar. Neither fact requires erasing the trajectory or restoring a prolongation prefix. Nonadic holonomy Γ_9 denotes transport through one turn on states with memory, so that

$$g(\Gamma_9 x) = g(x), \quad M(\Gamma_9 x) = M(x) + 1. \quad (4.6)$$

Phase equality and memory increment are compatible and constitute two observations of the same transport.

4.2.1 Temporal parametrization of compatibility and tritic orientation

The temporal interpretation of TRIT is established on transitions of the complete state. Let $x \xrightarrow{\text{adm}} y$ be the admissibility relation determined by TPK operations on $\hat{X}$. A trajectory parametrized by a discretely ordered set $\mathcal{T}$ is a map

$$\gamma : \mathcal{T} \longrightarrow \hat{X}, \quad \gamma(t) \xrightarrow{\text{adm}} \gamma(t^+),$$

where t^+ is the successor of t . The compatibility relation precedes the choice of parameter: the order allows its transitions to be expressed as an evolution. When the term section is used, the projection $p : E \rightarrow \mathcal{T}$ and condition $p \circ \gamma = \text{id}_{\mathcal{T}}$ are also declared. The state space, temporal order, and transported information are thereby specified.

The signature τ selects a quadratic regime and admits a temporal reading relative to this parametrization. Type +1, with $J_{+1}^2 = -I$, represents return and memory; type 0, with $J_0^2 = 0$, represents the transition threshold; type -1, with $J_{-1}^2 = I$, represents opening in two eigendirections. In the model's temporal terminology, these functions correspond, respectively, to past, present, and future. The correspondence concerns the operational position of a transition within a history: retained antecedent, update, and admissible prolongation.

This interpretation acquires content through three distinct operations. Restriction of a trajectory recovers the prefix already constructed. The update $\mathcal{U}_t$ composes selection, transport, and recording in the observed transition. Prolongation considers continuations that preserve the boundary conditions of that prefix. If $\mathcal{P}_n(x)$ denotes the set of admissible continuations of length n from x , deleting the last step defines

$$r_n : \mathcal{P}_{n+1}(x) \longrightarrow \mathcal{P}_n(x).$$

An unlimited continuation is a collection $(\gamma_n)_{n \geq 1}$ satisfying $r_n(\gamma_{n+1}) = \gamma_n$. The inverse system thus preserves the restrictions each depth imposes on preceding depths. Existence of a compatible collection is established in the corresponding prolongation domain; merely specifying the sets $\mathcal{P}_n(x)$ does not replace that proof.

Bidirectionality therefore combines reconstruction of antecedents with compatibility of continuations. On the signature sequence, the involution C_{rev} reverses order and changes the sign of each component. The trajectory is also subject to the transformations of position, sheet, and boundary specified by transport. The reversed signature is an observation of the conjugate trajectory, not a replacement for its other coordinates. The threshold remains distinguished: the visible value 0 retains its transition function even when its quotient or historical record is nonzero.

The relation with exponential geometries is equally precise. In each fixed regime, $\exp(tJ_\tau)$ composes parameters and preserves the algebraic norm. Elliptic closure belongs to that local representation. Phase return of Γ_0 , by contrast, acts on the trajectory with memory, according to (4.6). A trajectory can recover its phase while retaining a new antecedent: historical data have changed although the phase observation is the same. The parabolic threshold retains a linear displacement; hyperbolic opening separates two eigendirections of expansion and contraction. These three realizations provide composition laws, whereas the TPK determines their selection and effective concatenation.

The aperiodic suspension in Section 5 will make this distinction explicit through a finite phase factor, an irrational rotation, and a memory degree. The link between TRIT and temporality is thus expressed by operations on trajectories and their quadratic representations. Its content exceeds a classification of signs: it preserves the regime, the direction of transport, and the relation between antecedent, transition, and continuation.

The monodromic representation of this return is developed in Section 5.10. There, the holonomy of the state is distinguished from its representation on phase and counter. The symbolic suspension in Section 5.12 defines the aperiodic time-crystal model, and Section 5.14 composes this temporality with the three Euler realizations.

4.3 Observable compression and orthogonal conservation

An isometric realization provides a quantitative expression of what a reduced observation loses. Let $U : \mathcal{H} \rightarrow \mathcal{H}$ be an isometric realization of transport through one turn, $U^*U = I$, and $J : \mathcal{H}_{\text{vis}} \rightarrow \mathcal{H}$ an isometric inclusion. Define

$$T = J^*UJ, \quad D = (I - JJ^*)UJ.$$

Then

$$UJ = JT + D, \quad J^*D = 0, \quad I - T^*T = D^*D. \quad (4.7)$$

The last equality follows by taking the adjoint product of the first and using orthogonality. Contraction of the visible component does not by itself constitute destruction of information in the total state.

Proposition 4.3 (Finite-horizon conservation). *If $I - T^*T = D^*D$, then, for every $n \geq 1$,*

$$I = \sum_{r=0}^{n-1} T^{*r} D^* D T^r + T^{*n} T^n. \quad (4.8)$$

Proof. Multiplying $D^*D = I - T^*T$ by T^{*r} and T^r , then summing, produces a telescoping sum. The interior terms cancel, leaving $I - T^{*n} T^n$. $\square$

The final summand preserves the terminal state. Removing it before justifying a limit would alter the equality. This distinction is the operational content of conservation of unity under compression: the norm is distributed among the visible component, memory, and terminal, rather than being replaced by a single residue.

4.4 Local development: commutation, Lorentzian signature, and discrete curvature

The adjacent diagonal block with equal diagonals modulo nine is unique. Indeed,

$$k^2 \equiv (k+1)^2 \pmod{9} \iff 2k+1 \equiv 0 \pmod{9}$$

gives $k = 4$. Its visible matrix is

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 7I + 2R, \quad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad R^2 = I.$$

With $P_{\pm} = (I \pm R)/2$, one obtains $B_c = 9P_+ + 5P_-$. The unity of stochastic normalization and conservation of an indefinite form are two realizations of this block:

$$\frac{B_c}{9} = P_+ + \frac{5}{9}P_-, \quad \frac{B_c}{\sqrt{45}} = \exp(\eta R), \quad \eta = \frac{1}{2} \log \frac{9}{5}.$$

For $Q = \text{diag}(1, -1)$, the second matrix satisfies $B_c^T Q B_c = 45Q$. This yields a local realization in $SO^+(1, 1)$; no prior introduction of four-dimensional spacetime is required to prove it.

Uniform averaging over the classes $C_1 = \{1, 4, 7\}$, $C_2 = \{2, 5, 8\}$, and $C_0 = \{3, 6, 9\}$ gives

$$M_{\Pi} = \begin{pmatrix} 4 & 5 & 6 \\ 5 & 4 & 6 \\ 6 & 6 & 9 \end{pmatrix}, \quad M_{\Sigma} = \begin{pmatrix} 5 & 6 & 4 \\ 6 & 4 & 5 \\ 4 & 5 & 6 \end{pmatrix}.$$

The characteristic polynomial of M_{Π} is

$$(\lambda + 1)((\lambda - 9)^2 - 72),$$

so its quadratic form has signature $(2, 1)$. The local spectral gap $9 - 5 = 4$, the coefficient 2, the aggregate difference $51 - 45 = 6$, and the integrated defect $9 \cdot 6 = 54$ describe different levels of the same arithmetic comparison. They are not interchangeable indices of a single operator.

Discrete curvature can be described directly before this spectral realization. In residues, let $S(x, y) = x + y$, $P(x, y) = xy$, and let Δ_x, Δ_y be forward differences. Link variables are differences of potentials, $A_x^T = \Delta_x T$, $A_y^T = \Delta_y T$, and we set

$$F(T_x, T_y) = \Delta_x \Delta_y T_y - \Delta_y \Delta_x T_x.$$

Since $\Delta_x S = \Delta_y S = 1$, $\Delta_x P = y$, and $\Delta_y P = x$,

$$F(S, S) = F(P, P) = 0, \quad F(S, P) = +1, \quad F(P, S) = -1 \pmod{9}. \quad (4.9)$$

Sum and product, taken separately, induce flat connections; their ordered mixture changes the sign of plaquette curvature. Summing over a rectangle of sides a, b cancels interior edges, and the oriented boundary sum equals $\pm ab$ modulo nine. An exact relation thus appears between composition order and boundary orientation.

4.5 Cubic completion and two notions of boundary

Expanding the cubic support of side nine to that of side ten admits an exact positional stratification:

$$10^3 = 9^3 + 3 \cdot 9^2 + 3 \cdot 9 + 1 = 729 + 243 + 27 + 1. \quad (4.10)$$

The first term corresponds to three interior coordinates; the second to one new coordinate; the third to two; and the last to the vertex with three new coordinates. The expansion corona contains 271 positions. By contrast, the internal boundary of the discrete cube of side nine contains $9^3 - 7^3 = 386$. The difference between these counts is a difference of domain, not a conflict of values.

Dividing (4.10) by 1000 gives a normalized partition of unity. Cube incidence distinguishes six faces, twelve edges, and eight vertices. The subsequent connection to Steiner hexads and octads requires the corresponding combinatorial maps; cube cardinalities do not replace them. Here this completion supplies the relation between bases 729 and 1000, while exceptional prolongation will be developed after numerical generation and the electron.

4.5.1 Stratification of the completion and restoration of the apex

The completion admits a set-theoretic description making its operation explicit. Let E_9 be the set of nine residue classes, represented positively by $1, \dots, 9$, and let $*$ be an additional element. The symbol $*$ differs from the zero class, already represented by 9. Consequently, extending E_9 to $\hat{E} = E_9 \sqcup \{*\}$ increases the cardinality from nine to ten without changing the internal modular identification. The cubic completion is $\hat{C} = \hat{E}^3$, and the initial cube is $C = E_9^3$.

Definition 4.4 (Strata of combinatorial codimension). For $0 \leq r \leq 3$, define

$$S_r = \{(x_1, x_2, x_3) \in \hat{E}^3 : \#\{j : x_j = *\} = r\}.$$

The apex of the completion is $a_* = (*, *, *)$, and the punctured completion is $\hat{C}^\circ = \hat{C} \setminus \{a_*\}$.

Proposition 4.5 (Exact decomposition of the completion). *The strata are disjoint and satisfy*

$$\hat{C} = S_0 \sqcup S_1 \sqcup S_2 \sqcup S_3, \quad |S_r| = \binom{3}{r} 9^{3-r}, \quad (4.11)$$

$$|\hat{C}^\circ| = 729 + 243 + 27 = 999, \quad |\hat{C}| = 999 + 1 = 1000. \quad (4.12)$$

The punctured expansion contains 270 elements in addition to the initial cube; the complete expansion contains 271.

Proof. An element of S_r is obtained by choosing the r coordinates containing $*$, in $\binom{3}{r}$ ways, and assigning one of the nine classes of E_9 to each remaining coordinate. Classification by the number of additional coordinates is exhaustive and its classes are disjoint. The sole element of S_3 is a_* . Removing it subtracts one from the total cardinality; restoring it recovers the complete Cartesian product. $\square$

The expression *holographic extreme and mean ratio*, abbreviated EMRH in the corpus, here denotes this articulation of the initial body, expansion strata, and unit restoration. Its immediate content is the exact partition of a finite product and its normalization. The number 999 represents a definite object, the punctured completion; 1000 represents its closure through restoration of the apex. The added unit therefore has an explicit incidence location.

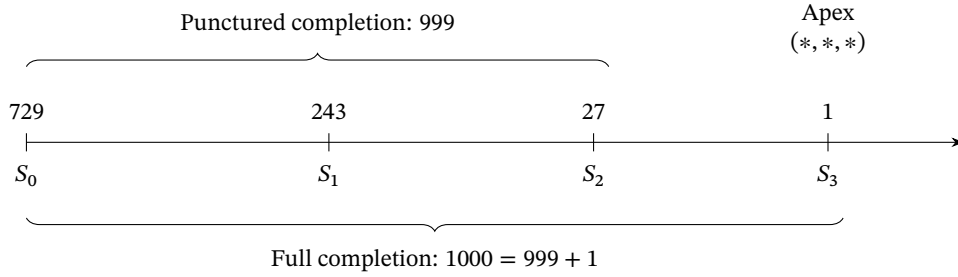


Figure 4: Strata of the cubic completion. The arrangement represents the set-theoretic decomposition, not a metric scale. The 243 elements of S_1 form three coordinate strata of 81 elements; the 27 elements of S_2 form three strata of 9. The apex completes the partition.

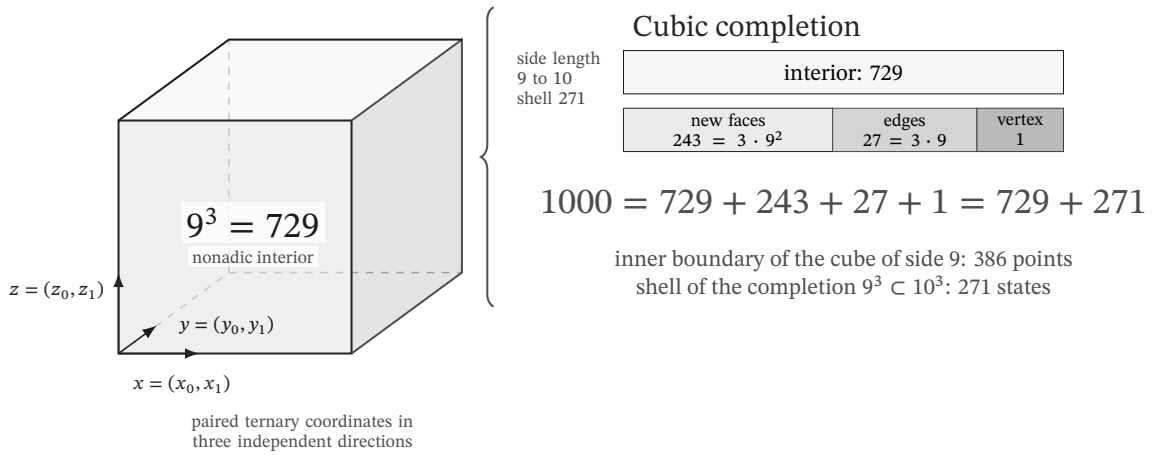


Figure 5: Cubic representation of the expansion from nine to ten classes per coordinate. Strata S_1, S_2, S_3 locate the added incidences on faces, edges, and the apex; their cardinalities form the 271-element shell. The 386-element boundary belongs to the initial cube. This geometric perspective complements the partition in Figure 4; graphical spacing is schematic.

4.5.2 Conservation of unity and dimensional compatibility

Normalization is not limited to dividing an isolated identity. In dimension d , the product $\widehat{E}^d$ carries the uniform measure $\mu_d(\{x\}) = 10^{-d}$. The coordinate projection $p_d : \widehat{E}^{d+1} \rightarrow \widehat{E}^d$ deletes the last coordinate.

Proposition 4.6 (Projective compatibility of normalized measures). *For every $d \geq 1$,*

$$(p_d)_* \mu_{d+1} = \mu_d, \quad \mu_d(\widehat{E}^d) = 1.$$

The mass of the stratum with r additional coordinates is $\binom{d}{r} 9^{d-r} / 10^d$, and the masses of all strata sum to one.

Proof. Each point of $\widehat{E}^d$ has ten preimages. Their total mass is $10 \cdot 10^{-(d+1)} = 10^{-d}$. Equality for any subset follows by summing over its points. The stratified formula follows from the same count as (4.11); the binomial $(9 + 1)^d$ gives total normalization. $\square$

In dimension three, removing the apex leaves mass 999/1000, and restoring it adds 1/1000. Renormalizing before restoration would define a different measure: the distinction is operationally relevant. In particular, conservation of probability mass, conservation of transport records, and thermodynamic dissipation concern different objects. The proposition proves the first and provides the compatible support for studying the second; the third requires a specified dynamic and energetic realization.

The internal geometric boundary also remains distinct. On the Cartesian representative $\{1, \dots, 9\}^3$, removing $\{2, \dots, 8\}^3$ leaves 386 points. This boundary belongs to the initial cube, whereas $\widehat{C} \setminus C$ belongs to its expansion. Cubic completion, base-thousand positional expansion, and exceptional-incidence reading share support data; their respective maps determine which information from these data each representation uses.

4.6 Cubic incidence and Lorentzian realization

Cubic completion supplies a relation between interior and boundary. Its geometry can be developed through incidence of the six faces with the eight vertices. Label faces by (i, ε) , $i \in \{1, 2, 3\}$, $\varepsilon \in \{+1, -1\}$, and vertices by $v \in \{+1, -1\}^3$. The incidence matrix is

$$M_{(i,\varepsilon),v} = \mathbf{1}_{\{v_i=\varepsilon\}}.$$

Let $u = (1, \dots, 1)^\top \in \mathbb{R}^6$, let R_F denote face opposition, and let $P_u = uu^\top/6$, $P_- = (I - R_F)/2$.

Proposition 4.7 (Four-dimensional incidence quotient). *The incidence Gram matrix satisfies*

$$MM^\top = 12P_u + 4P_-.$$

Its eigenvalues 12, 4, 0 have multiplicities 1, 3, 2. Consequently,

$$Q_\square = \mathbb{R}^6 / \ker(MM^\top) \simeq \mathbb{R}u \oplus E_-, \quad E_- = \text{im } P_-.$$

Proof. A face contains four vertices; two opposite faces have empty intersection, and two distinct non-opposite faces share two vertices. The matrix $12P_u + 4P_-$ has exactly these entries. The uniform subspace has dimension one; differences between opposite faces generate a three-dimensional space orthogonal to it. Its complement has dimension two and belongs to the kernel. $\square$

The decomposition 1 + 3 comes from incidence. The temporal choice is declared through the bilinear form

$$B_\square([x], [y]) = \langle x, (P_- - P_u)y \rangle.$$

It is well defined on the quotient because both projectors annihilate the kernel. In the basis

$$t = [u]/\sqrt{6}, \quad d_i = [e_{i,+} - e_{i,-}]/\sqrt{2},$$

its matrix is $\text{diag}(-1, 1, 1, 1)$. The domain, decomposition, and sign convention producing the Minkowski realization are thus explicit. The positive Gram matrix determines the quotient; the temporal assignment determines the indefinite form on it.

The operator

$$W_{\square} = \sqrt{6}P_u + \sqrt{2}P_v, \quad W_{\square}e_{i,\varepsilon} = t + \varepsilon d_i$$

sends the six face labels to null directions, since $B_{\square}(t + \varepsilon d_i, t + \varepsilon d_i) = -1 + 1 = 0$. Direction reversal and face opposition thus acquire a concrete geometric realization.

4.6.1 Soldering, reversal, and subdivision

The cellular realization uses an invertible transport $U_m(a)$, preserving the Lorentzian form, between the endpoint fibers of an oriented dual edge a , a face label $\lambda_m(a)$, and an orientation $\sigma_m(a)$. With $\ell_m = \ell_0/9^m$, the soldering map is

$$\beta_m(a) = \ell_m \sigma_m(a) W_{\square, s(a)} e_{\lambda_m(a)}.$$

Compatible reversal satisfies

$$\beta_m(\bar{a}) = -U_m(a) \beta_m(a).$$

Identify the source fibers at two consecutive levels through the declared refinement isometry. If the nine subsegments $a_1, \dots, a_9$, at level $m+1$, preserve label and orientation when transported to that common fiber, then

$$\beta_m(a) = \sum_{j=1}^9 U_{m+1}(a_1 \cdots a_{j-1})^{-1} \beta_{m+1}(a_j). \quad (4.13)$$

Each transported term equals $\ell_m/9$ times the same null direction, proving the identity. The compatibility hypothesis determines which subdivisions realize this refinement. The equation brings together direction, scale, and transport, rather than limiting geometric correspondence to a dimension count.

With orientation and the Lorentzian form also fixed, Hodge duality on $\Lambda^2 Q_{\square}^*$ satisfies $\star^2 = -I$. In dimension four, the sign is $(-1)^{2(4-2)+1} = -1$. This yields a real complex structure on two-forms and, after complexification, the eigenspaces with eigenvalues $+i$ and $-i$. The electromagnetic realization will use this domain after constructing the electronic section and declaring its coupling.

4.7 Compatibility of modular and positional reductions

The block-decimal base enters through Euclidean division preserving residue and quotient simultaneously:

$$n = 1000q + r, \quad 0 \leq r < 1000.$$

Since $1000 \equiv 1 \pmod{9}$ and $1000 \equiv 1 \pmod{3}$,

$$n \equiv q + r \pmod{9}, \quad n \equiv q + r \pmod{3}.$$

Reduction of a block therefore requires the quotient's contribution to recover the integer's class. The numbers 0 and 1000 have the same remainder modulo 1000, but different classes modulo 9 and 3. This is an operational reason for preserving memory through a change of resolution.

Ternary refinement and determination of a decimal prefix can be coordinated through a single integer remainder. If $N/3^L$ is a cylinder endpoint and $D/1000^r$ a decimal endpoint, define

$$A = 1000^r N - D 3^L.$$

On adding six ternary symbols, $N' = 729N + \nu$, $L' = L + 6$, with $0 \leq \nu < 729$. If the decimal prefix remains fixed, one obtains

$$A' = 729A + \nu 1000^r.$$

If a decimal block δ is also incorporated, then $D' = 1000D + \delta$, $r' = r + 1$, and

$$A' = 729000A + \nu 1000^{r+1} - 729\delta 3^L. \quad (4.14)$$

Both identities follow by substitution of the definitions. Comparing endpoints is thus reduced to integer operations retaining each prolongation's contribution. This compatibility makes it possible to refine a region before evaluating its real coordinate.

4.8 Number as a compatible section and value as a reading

Distinguishing state from observation specifies what object is prolonged as resolution increases. At depth n , let X_n be the set of admissible states produced by APP-TRIT-TPK. Their data include position, sum and product sheets, residues and quotients, regime, orientation, carry, phase, route, boundary, memory, and survival. These are not independent coordinates: earlier arithmetic evaluations and transports impose their relations. A prolongation preserves these relations in each antecedent, even while adding new steps and memory data. This is the meaning of the arithmetic of histories developed in [development and scope in Sections 8 and 9.4]; [Sections 4.8, 5.10, and 9.2.1].

Let $r_{m,n} : X_m \rightarrow X_n$, for $m \geq n$, be depth restrictions, with $r_{n,n} = \text{id}$ and $r_{\ell,n} = r_{m,n} \circ r_{\ell,m}$. Each restriction preserves the prefix and the information already determined in it.

Definition 4.8 (Compatible numerical section). An HMT number is a compatible history of the state system:

$$x = (x_n)_{n \geq 0} \in X_\infty := \varprojlim_n X_n, \quad r_{m,n}(x_m) = x_n. \quad (4.15)$$

A reader is an additional operation on a specified domain of these histories. A digit is a finite observation; a scalar value is an evaluation of the reader. The pair consisting of a section and its reader constitutes a read presentation, not a redefinition of the section's identity.

On the Archimedean domain X_∞^{Arch} , the reader associates a closed cylinder with each prefix, preserves nesting, and makes its diameter tend to zero. It thus determines

$$\text{ev}_{\mathbb{R}} : X_\infty^{\text{Arch}} \longrightarrow \mathbb{R}, \quad \{\text{ev}_{\mathbb{R}}(x)\} = \bigcap_n I_n(x). \quad (4.16)$$

The concrete construction of these cylinders and the determination of their boundaries are developed in Section 5. Definition (4.15) does not by itself assert that the image of (4.16) is the entire line: a coverage claim requires its own theorem. Nor does equality of values establish equality of histories without an injectivity result. This separation allows precision to be discussed as depth while preserving the distinction between the arithmetic antecedent and its scalar realization.

4.9 Composition of histories and memory multiplier

Fix a sector of states at depth n and a category $\mathcal{C}$ whose arrows are its admissible histories. Composition is written $vu = v \circ u$: first traverse u , then v . Elementary steps come from TPK selection, transport, and updating. A local tritic fiber admits the notation $\mathbb{R}[J_x]/(J_x^2 + \tau(x)1_x)$, where $\tau(x) \in \{+1, 0, -1\}$. Regime and orientation belong to the transported state; a transition between regimes does not amount to identifying their quadratic algebras.

The following formulation specifies an algebraic sector of this composition. For each object x , a unital associative real algebra A_x , possibly noncommutative, is given. For each $u : x \rightarrow y$, a unital homomorphism $\rho_u : A_x \rightarrow A_y$ is given. Assume a strict action:

$$\rho_{\text{id}_x} = \text{id}_{A_x}, \quad \rho_v \rho_u = \rho_{vu}. \quad (4.17)$$

For each composable pair $u : x \rightarrow y$, $v : y \rightarrow z$, fix an invertible central multiplier $\Omega(v, u) \in Z(A_z)^\times$. Require the normalization $\Omega(\text{id}_y, u) = \Omega(u, \text{id}_x) = 1_y$ and, for every composable triple, the identity

$$\rho_w(\Omega(v, u))\Omega(w, vu) = \Omega(w, v)\Omega(wv, u). \quad (4.18)$$

These are explicit hypotheses on transports and their coefficients, not a consequence of calling the dynamics holonomic. Applying this presentation to a TPK sector requires identifying its ρ_u and $\Omega(v, u)$. Transitions expressed through bimodules require preserving that structure and are not replaced by (4.17).

The space of finite sums

$$\mathcal{H}(\mathcal{C}, A, \rho, \Omega) = \bigoplus_{u \in \text{Mor}(\mathcal{C})} A_{t(u)} T_u$$

receives the bilinear product

$$(aT_v)(bT_u) = a\rho_v(b)\Omega(v, u)T_{vu}, \quad (4.19)$$

when the arrows are composable, and zero otherwise. Here $t(u)$ denotes the terminal state. Addition gathers histories with coefficients; multiplication concatenates histories while preserving the record multiplier.

Proposition 4.9 (Sectorial associativity with a central multiplier). *Under (4.17) and (4.18), product (4.19) is associative. If the object set is finite, its identity is $\sum_x 1_x T_{\text{id}_x}$.*

Proof. For $u : x \rightarrow y, v : y \rightarrow z, w : z \rightarrow t$, let $a \in A_y, b \in A_z, c \in A_t$. The two coefficients of T_{wvu} are

$$\begin{aligned} ((cT_w)(bT_v))(aT_u) &: c\rho_w(b)\Omega(w, v)\rho_{wv}(a)\Omega(wv, u), \\ (cT_w)((bT_v)(aT_u)) &: c\rho_w(b)\rho_w\rho_v(a)\rho_w(\Omega(v, u))\Omega(w, vu). \end{aligned}$$

The strict action identifies $\rho_w\rho_v(a)$ with $\rho_{wv}(a)$. Centrality of $\Omega(w, v)$ allows it to move to the right of that factor; (4.18) then equates the remaining products. The coefficients a, b, c are not permuted. If the triple is not composable, both parenthesizations are zero because of their initial and terminal states. Bilinearity completes the proof. Local identities and normalization of Ω verify the stated identity element. $\square$

Calendar carry provides an effective example. In the category with one object and arrows C_9 , take coefficients $A = \mathbb{R}[z, z^{-1}]$, the identity action, and

$$\Omega(b, a) = z^{c_0(a, b)}, \quad c_0(a, b) = \left\lfloor \frac{a+b}{9} \right\rfloor, \quad 0 \leq a, b < 9.$$

The two carry sums of a triple are $\lfloor (a+b+c)/9 \rfloor$; hence (4.18) holds. The formal variable z records the turn without assigning it a numerical value. Subsequently imposing $z^{12} = 1$ recovers the dodecaphase record of calendar (4.5); retaining z without that relation distinguishes successive turns. The example realizes this calendar's memory without identifying it with all TPK memory.

4.10 Double variance of refinement and conservation of unity

Restricting states prolongs observables in the opposite direction. Consider the function algebras, with pointwise product, $D_n = \text{Fun}(X_n, \mathbb{R})$. Precomposition defines

$$r_{m,n}^* : D_n \longrightarrow D_m, \quad r_{m,n}^* f = f \circ r_{m,n}. \quad (4.20)$$

These are unital morphisms; they are also injective when the corresponding restrictions are surjective. The direct limit $D_{\text{cyl}} = \varinjlim_n (D_n, r_{n+1,n}^*)$ gathers finite-depth observations. This algebra is not identified here with the whole route algebra: prolonging transport operators also requires their compatibility with refinement.

Lemma 4.10 (Identity and naturality of evaluation). *If e_x is the indicator of $x \in X_n$, then*

$$r_{m,n}^* e_x = \sum_{y: r_{m,n}(y)=x} e_y, \quad r_{m,n}^* 1_n = 1_m, \quad (4.21)$$

and, for every $f \in D_n$, $y \in X_m$,

$$\langle r_{m,n}^* f, y \rangle = \langle f, r_{m,n} y \rangle.$$

Each section $x \in X_\infty$ determines a well-defined evaluation $D_{\text{cyl}} \rightarrow \mathbb{R}$, given by $[f] \mapsto f(x_n)$ for $f \in D_n$.

Proof. The fibers of $r_{m,n}$ partition X_m . Their indicators are orthogonal and sum to 1_m , proving (4.21). Naturality is the definition of precomposition. Finally, $(r_{m,n}^* f)(x_m) = f(x_n)$, so two compatible representatives give the same evaluation in the direct limit. $\square$

The global identity is the constant function one on all possibilities; an individual section selects a compatible history. Preserving the former does not mean immobilizing the latter. Refinement may increase the number of extensions and the depth of each history while preserving the identity and all evaluations of its antecedents.

4.11 Phase return, Weyl representation, and two readings

Law (4.6) acquires an operator realization in the bilateral suspension constructed in Section 5. There, T is one-step advance, V_9 observes the nonadic phase, W_δ the rotation phase, and D_M the memory degree. Writing $\hat{\Gamma}_9 = T^9$ for the turn in this representation, the Weyl relations (5.13) imply, on finitely supported vectors,

$$\begin{aligned} V_9 \hat{\Gamma}_9 &= \hat{\Gamma}_9 V_9, & W_\delta \hat{\Gamma}_9 &= e^{2\pi i 9\delta} \hat{\Gamma}_9 W_\delta, \\ [D_M, \hat{\Gamma}_9] &= \hat{\Gamma}_9. \end{aligned} \tag{4.22}$$

The first equality follows from $\zeta_9^9 = 1$; the second from nine applications of the covariance of W_δ ; the third expresses the unit memory increment. The finite phase returns while irrational rotation and memory distinguish the next state. Characters and parameter δ are realized subsequently from the numerical construction specified there: they do not select the seeds retrospectively. This representation describes suspension observables and translations; it does not replace all TPK coordinates or operators.

The double reading extends the same distinction between history and observable. On its Archimedean domain, (4.16) applies; on the calibrated incidence domain, the maps Q_n of Section 28.2 preserve words and boundary frames. The equality proved there, $r_{n,m} Q_n = Q_m p_{n,m}$, determines their prolongation Q_∞ . On the common domain of both readings, $\text{ev}_\mathbb{R}(x)$ and $Q_\infty(x)$ can be considered jointly: the former retains the value; the latter, the transported incidence preserved by its codomain. The section precedes both. This articulation makes it possible to pass from the arithmetic of histories to constants and exceptional reading without turning their realizations into new initial data.

5 Coinductive generation of π, φ, e to arbitrary depth

The six-block emission is a finite stage of the construction, not a complete decimal expansion. Its function is to produce regions with multiplicity, orientation, and transport rules. Prolongation must preserve these data and determine a compatible collection of prefixes of arbitrary length. The word *generation* designates this operation; *genealogy* designates the reconstruction of its dependencies. Both notions enter the account, but serve different functions.

5.1 Orbits, regions, and characters of closure, propagation, and self-scaling

On a word $w = (w_1, \dots, w_6)$, define

$$r(w) = (w_3, w_1, w_2, w_5, w_6, w_4), \quad s(w) = (w_4, w_5, w_6, w_1, w_2, w_3).$$

We have $r^3 = s^2 = I$ and $srs = r^{-1}$; hence these permutations realize the dihedral group D_3 . The action preserves the catalog and multiplicities and commutes with Ψ . The quotient of the 243 visible words has 43 orbits: thirty-eight of cardinality six and five of cardinality three.

The closure class is recognized by a fiber predicate: its six orientations have five emissions per word, total multiplicity 1008, and distribution $432 + 4 \cdot 144$. There is a single orbit with these properties in the catalog. To specify the other two selections, write $f(w) = \#\Psi^{-1}(w)$, $m(w) = \sum_{U \in \Psi^{-1}(w)} \text{mult}(U)$, and $\nu(w) = (n_0, n_1, n_2)$. The additive diagonal class of an emission is $d(U) = [i + j]_9$ in indices from zero to eight; its additive signature preserves this class and the *ES* or *NO* orientation. Let $\mathcal{D}(\mathcal{O})$ be the set of these classes for all emissions over an orbit.

Among the size-six orbits, the joint predicate

$$f(w) = 1, \quad m(w) = 144 \quad (w \in \mathcal{O}), \quad \mathcal{D}(\mathcal{O}) = \{0, 3, 6\}$$

leaves exactly six orbits. In this sector, the census $\nu = (2, 3, 1)$, equal to that of closure, uniquely selects propagation; the census $\nu = (2, 2, 2)$ uniquely selects self-scaling. The diagonal support cannot be omitted: without it there are four balanced singleton-fiber orbits of multiplicity 144, not one. Finally, the existence of an emission with $d = 6$ and orientation *ES* selects one representative of each orbit and yields

$$w^{\text{cl}} = 010211, \quad w^{\text{pr}} = 201101, \quad w^{\text{au}} = 121200. \quad (5.1)$$

These symbols first designate regions of the catalog. The scalar identification will be proved through their readers; they are not chosen because they match digits of a known constant.

The closure word 010211 has the following five preimages:

Six-block emission	Multiplicative signature	Role	Multiplicity
501, 614, 498, 169, 272, 272	555555	transverse	432
810, 923, 870, 169, 272, 272	339555	positive orientation	144
810, 983, 810, 169, 272, 272	393555	positive orientation	144
870, 923, 810, 169, 272, 272	933555	positive orientation	144
870, 923, 810, 418, 674, 521	933717	oriented return	144

The multiplicity 432 and constant signature 555555 characterize the transverse region. They do not identify the central position (5, 5) of APP. Confusing a six-window signature with a position in the support would eliminate the distinction between numerical region and electronic structure.

The microfiber also has a precise bipartite morphology. Each emission is a product of an additive cursor class and a multiplicative class. The three positive regions share the same additive class of eighteen conditions; together they contain three multiplicative classes of eight conditions each. The transverse region has a product 18×24 ; the three positive regions together have another 18×24 ; and the return has a product 18×8 . The decomposition by emissions has five parts, whereas the decomposition by connected components of the bipartite graph has three. Both preserve the same 1008 seeds without reducing form to a cardinality.

5.2 Finite transport and the prolongation boundary

The endomorphisms L_0, L_1 of the ternary-word space act before the Archimedean realization. Their determination begins with the transitions of the arithmetic state, not with the prescription of two matrices. If U_t is the composite selection, transport, and update step, the observed block satisfies

$$b_j^\chi = \Pi_6(x_j^\chi), \quad x_{j+1}^\chi = U_{9j+9} \cdots U_{9j+1}(x_j^\chi), \quad \chi \in \{\text{cl}, \text{pr}, \text{au}\}.$$

Six transitions are formed for each regime: two consecutive transitions in each of the three regions. The pairs of source and target matrices obtained in the record are

X_0	Y_0	X_1	Y_1
010211	012222	010211	002111
012222	010211	002111	110221
201101	121221	102011	012222
121221	102011	012222	102011
121200	112202	121020	010210
112202	121020	010210	010200

Each six-digit string represents a row in $\mathbb{F}_3^6$. Since $\det X_0 = 1$ and $\det X_1 = -1$ in $\mathbb{F}_3$, the equations $X_a L_a = Y_a$ uniquely determine $L_a = X_a^{-1} Y_a$. In this row-vector chart, the resulting matrices are

$$L_0 = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \quad L_1 = \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix} \pmod{3}.$$

The calendar (L_0, L_0, L_1, L_1) produces four new six-component blocks. Their concatenations form

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30}.$$

The subscripts of w indicate accumulated length. The boundary R_{36} preserves the terminal structure and opens the prolongation relation; it is not renamed as a final periodic block that could be repeated indefinitely.

The identity $L = X^{-1}Y$ proves uniqueness once the transitions are fixed; it does not replace the production of those transitions by U_t . The provenance of their record and the examination of the sixteen binary calendars are documented in the technical supplement. The indicated calendar is the only one of those sixteen that jointly reproduces the three five-block sequences.

5.3 Five closure regions and angular compensation

The closure region contains a transverse component and four accompanying components. The symmetric reader acts on the space of its five positions through

$$P_5 = \frac{1}{5} \mathbf{1} \mathbf{1}^\top, \quad \mathbf{1} = (1, 1, 1, 1, 1)^\top.$$

The invariance $P_5 \lambda = \lambda$ and conservation $\mathbf{1}^\top \lambda = 1$ force $\lambda = \mathbf{1}/5$. This normalization distributes the unit among reader positions; it does not replace the seed multiplicities 432, 144, 144, 144, 144 with uniform frequencies. The reader's elliptic chart realizes each coordinate through $1 + \lambda_j J$; its slope is λ_j , so the primitive slope is $1/5$. This makes explicit the rule passing from a normalized coordinate to a slope, rather than identifying the two notions without a map. The reader composes the four accompanying positions through the law

$$a \oplus b = \frac{a + b}{1 - ab}. \quad (5.2)$$

This law is obtained by multiplying $(1 + aJ)(1 + bJ)$ in the regime $J^2 = -I$ and dividing by its scalar component. The composition of slopes is therefore a realization of the quadratic structure of TRIT, not an ordinary sum of four fractions.

Proposition 5.1 (Closure compensator). *The composition of four slopes $1/5$ is $120/119$. The condition of return to slope one determines a unique positive compensator of the form $1/q$, with $q > 1$, and gives $q = 239$.*

Proof. The first doubling gives $(1/5) \oplus (1/5) = 5/12$; the second gives $(5/12) \oplus (5/12) = 120/119$. The condition

$$\frac{120/119 - 1/q}{1 + (120/119)/q} = 1$$

is equivalent to $(120 - 119)q = 120 + 119$, so $q = 239$. $\square$

The resulting closure character has realization

$$\Pi_{\text{cl}} = 4 \left(4 \arctan \frac{1}{5} - \arctan \frac{1}{239} \right). \quad (5.3)$$

Machin's angular identity recognizes $\Pi_{\text{cl}} = \pi$. The meaning of the construction is precise: the slope and number of compositions are declared by the penta-fiber reader; the compensation equation forces 239. Once this character has been produced, its evaluation through alternating series does not select the seed orbit anew.

For $0 < x < 1$, the alternating sums

$$A_n(x) = \sum_{j=0}^n \frac{(-1)^j x^{2j+1}}{2j+1}$$

enclose $\arctan x$ between two consecutive truncations, with difference $x^{2n+3}/(2n+3)$. Applied to the two slopes in (5.3), they provide rational endpoints with explicit error. The numerical value is thus computed as a consequence of the exact character, without introducing a target prefix.

5.4 Propagation reader and normalization of the unit

The propagation region is realized through a formal collection $P_t \in \mathcal{A}[[t]]$, where $\mathcal{A}$ is a unital commutative algebra over $\mathbb{Q}$. Its composition satisfies $P_{t+s} = P_t P_s$, with unit $P_0 = 1$, and the oriented elementary transport fixes the scalar generator $P'_0 = +1$. The condition fixes its sign, not merely its magnitude. Writing

$$P_t = \sum_{n \geq 0} a_n t^n,$$

the normalization determines $a_0 = a_1 = 1$. Comparing the coefficient of $s^n t$ in $P_{s+t} = P_s P_t$ gives $(n+1)a_{n+1} = a_n a_1$; equivalently, $P'_t = P_t$. Thus,

$$(n+1)a_{n+1} = a_n, \quad a_n = \frac{1}{n!}. \quad (5.4)$$

The character at one parameter unit is $\Pi_{\text{pr}} = P_1$, subsequently recognized as e .

The normalization of the generator matters. Without this datum, the semigroup law would admit $P_t = e^{ct}$ for different c ; the condition fixing the parameter unit in the reader is therefore not omitted. This condition precedes the decimal comparison and belongs to the exposition of the operational rule, not to the choice of an experimental value.

Proposition 5.2 (Rational propagation bounds). *For $n \geq 1$, let $E_n = \sum_{j=0}^n 1/j!$. Then*

$$E_n < \Pi_{\text{pr}} < E_n + \frac{n+2}{(n+1)(n+1)!}.$$

Proof. The first omitted term is $1/(n+1)!$. Each subsequent ratio is at most $1/(n+2)$. The geometric majorant of the tail is

$$\frac{1}{(n+1)!} \frac{1}{1 - 1/(n+2)} = \frac{n+2}{(n+1)(n+1)!}.$$

The strict inequality follows from the decrease in successive ratios. $\square$

5.5 Modal incidence and generation of self-scaling

The tritic selector produces the cycle $(+1, -1, 0)$. In modes Σ, Π , the rule is $+1 : m \mapsto \Sigma, -1 : m \mapsto \Pi, 0 : m \mapsto m$. With cyclic closure, the observed mode is (Σ, Π, Π) . Its edges are $\Sigma \rightarrow \Pi, \Pi \rightarrow \Pi$, and $\Pi \rightarrow \Sigma$. Hence its incidence, in that order of modes, is

$$F_{av} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \quad (5.5)$$

The four entries are obtained from the calendar, not from an equation previously supplied with the value of φ . Repeating the calendar gives counts $3F_{av}, 9F_{av}, 36F_{av}$ at lengths nine, twenty-seven, and one hundred eight. These counts are not powers of the matrix.

Theorem 5.3 (Positive self-scaling character). *The incidence (5.5) has a unique strictly positive eigenray. Normalizing it as $(1, x)^T$, its coordinate x satisfies $x^2 - x - 1 = 0$, $x > 0$, and equals $\varphi = (1 + \sqrt{5})/2$.*

Proof. The eigenvalue equation gives $x = \lambda$ and $1 + x = \lambda x$. Eliminating λ produces the polynomial. Only one of its two roots is positive. Since F_{av}^2 has all entries positive, the positive ray is unique and simple. The radical expression recognizes the coordinate obtained; it does not participate in producing F_{av} . $\square$

With $F_0 = 0, F_1 = 1, F_{n+1} = F_n + F_{n-1}$, the incidence induces, for $n \geq 1$,

$$F_{av}^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}.$$

The consecutive ratios F_{n+1}/F_n alternate around φ . The determinant identity

$$F_{n+1}^2 - F_n F_{n+2} = (-1)^n$$

gives an exact width $1/(F_n F_{n+1})$ between two consecutive ratios. Their growth makes this rational bracket arbitrarily narrow.

The measure on all compatible histories does not coincide with the frequency of modes in the three-event calendar. The chronological frequency is $(1/3, 2/3)$. The maximal-entropy measure of the symbolic envelope defined by F_{av} is constructed from its eigenvectors and has weight ratio φ^{-2} . The latter enters the areal-volumetric reading; it is not obtained simply by replacing one counting frequency with another.

After the characters have been generated, the two orientations of the reader

$$R(x, y) = \frac{x}{y^2}$$

provide

$$\rho_A = \frac{\pi}{\varphi^2}, \quad \rho_E = \frac{\varphi}{\pi^2}, \quad \varphi^3 \rho_A^2 \rho_E = 1. \quad (5.6)$$

The second expression will appear in the electronic exponent. The first is also obtained as the limit of the marked reading $\pi F_{n-1}/F_{n+1}$. Exchanging the arguments relates the two, but does not turn a reparameterization of the electronic equation into a second independent derivation of all its coefficients.

5.6 Exponential realizations of the generated coordinates

The regional coordinates already determined allow recognition of concrete realizations of the TRIT quadratic collection. The order is substantive: construction of the regions precedes evaluation of an exponential at parameters π or $2 \log \varphi$. These expressions identify the role of the coordinates in operators constructed from APP and TPK; they do not participate in selecting the prefixes that produce them.

5.6.1 Ternary cycle, nilpotent transport, and sheet incidence

Multiplication $a(r) = 4r$ in $\mathbb{Z}/9\mathbb{Z}$ has order 3. On the units it determines the orbits $\{1, 4, 7\}$ and $\{2, 8, 5\}$. In the augmentation module of either orbit, consisting of integer combinations whose coefficients sum to zero, the basis $(e_r - e_{4r}, e_{4r} - e_{7r})$ gives

$$C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad C^2 + C + I = 0.$$

The restricted form has Gram matrix $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$. On residue evaluations modulo 9, the action distinguishes the sheets: $S(ax, ay) = aS(x, y)$, whereas $P(ax, ay) = a^{-1}P(x, y)$. The cyclic realization therefore preserves a contragredient orientation between sum and product. Quotients of the integer lift are retained in the corresponding arithmetic record.

Elementary parabolic transport is represented by

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

The areal–volumetric incidence already constructed in (5.5) gives

$$F_{av} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad A = F_{av}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

The three operators act on declared real spaces: the augmentation module tensored with $\mathbb{R}$, the nilpotent-transport plane, and the modal-incidence plane. Sharing a matrix dimension does not identify these spaces or their coordinates.

Proposition 5.4 (Concrete generators of the three types). *The operators*

$$J_e = \frac{2C + I}{\sqrt{3}}, \quad J_p = N, \quad J_h = \frac{2A - 3I}{\sqrt{5}}$$

satisfy, respectively,

$$J_e^2 = -I, \quad J_p^2 = 0, \quad J_h^2 = I.$$

Proof. The relation for C implies $(2C + I)^2 = -3I$. Direct multiplication gives $N^2 = 0$. Finally, A has trace 3 and determinant 1, so $A^2 - 3A + I = 0$ and $(2A - 3I)^2 = 5I$. $\square$

Proposition 5.5 (Three evaluations of the exponential identity). *In the preceding realizations,*

$$\boxed{\exp(\pi J_e) + I = 0, \quad \exp(J_p) = I + J_p, \quad \exp(2 \log \varphi J_h) = F_{av}^2.} \quad (5.7)$$

Proof. The elliptic specialization of the tritic exponential gives $\cos \pi I + \sin \pi J_e = -I$. Nilpotency removes exactly the terms of degree at least 2 in the second expression. For the third, $\varphi^2 = \varphi + 1$ implies

$$\cosh(2 \log \varphi) = \frac{3}{2}, \quad \sinh(2 \log \varphi) = \frac{\sqrt{5}}{2}.$$

Thus the exponential equals $\frac{3}{2}I + \frac{\sqrt{5}}{2}J_h = A$. $\square$

The first equality represents the elliptic half-turn; the complete turn satisfies $\exp(2\pi J_e) = I$. The second retains a nonzero linear term: the parabolic regime expresses transport, not absence of evolution. The third realizes a unimodular hyperbolic advance. Its eigenvalues are φ^2 and φ^{-2} ; one direction expands and the other contracts while preserving the determinant. The three evaluations use different parameters of one common exponential collection.

In this collection, e enters through the scalar evaluation $\exp(I) = eI$ and the law $\exp(sI)\exp(tI) = \exp((s+t)I)$. Its role is not restricted to the parabolic type. The coordinate e , previously obtained from its region, is recognized here as the unit evaluation of the exponential; this identity does not replace its coinductive genealogy.

5.6.2 Polynomial resolution of the regimes

The three realizations admit a compact presentation on the direct sum of their spaces. Define

$$\mathbb{J} = J_e \oplus J_p \oplus J_h, \quad Q = \mathbb{J}^2.$$

The preceding relations imply

$$\mathbb{J}^6 = \mathbb{J}^2, \quad Q^3 = Q.$$

Using Q keeps this operator square distinct from the dodecaphase record K .

Proposition 5.6 (Polynomial regime projectors). *The maps*

$$p_e = \frac{Q^2 - Q}{2}, \quad p_p = I - Q^2, \quad p_h = \frac{Q^2 + Q}{2}$$

are mutually orthogonal idempotents, sum to I , and recover the three summands of the representation.

Proof. The three polynomials take values $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ at the points $-1, 0, +1$, respectively. The differences between their squares and themselves, as well as their cross products, are multiples of $X^3 - X$. Substituting Q , which annihilates that polynomial, proves the identities. On the three blocks, Q acts as $-I, 0, I$. $\square$

One thus obtains

$$\begin{aligned} \exp(t\mathbb{J}) &= p_e(\cos tI + \sin t\mathbb{J}) + p_p(I + t\mathbb{J}) \\ &\quad + p_h(\cosh tI + \sinh t\mathbb{J}). \end{aligned} \tag{5.8}$$

Each summand preserves its quadratic relation, and conjugation reverses its evolution. The total representation gathers the three regimes, but does not by itself define a transition operator between them. Such a transition requires the TPK transport maps with their domains, orientation, and memory. The exponential articulation allows recognition of their types without replacing those dynamics by an identification of fibers.

5.7 Compatible cylinders and arbitrary depth

The realization of a ternary block $b = (b_1, \dots, b_6)$ is the integer

$$\nu(b) = \sum_{j=1}^6 b_j 3^{6-j} \in \{0, \dots, 728\}.$$

A block sequence defines integer prefixes through $N_{n+1} = 729N_n + \nu(b_{n+1})$, starting at $N_0 = m$, where m is the integer part determined by the reader. The preceding brackets place the closure, propagation, and self-scaling characters in $(3, 4)$, $(2, 3)$, and $(1, 2)$, respectively; hence their initial values are $m = 3, 2, 1$. The following blocks refine the fractional part. The corresponding cylinder is

$$I_n = \left[\frac{N_n}{729^n}, \frac{N_n + 1}{729^n} \right], \quad |I_n| = 729^{-n}. \tag{5.9}$$

The half-open convention is used to select a representative when a value lies on a boundary. For passage to the limit, the closures of these intervals are used, and the two equivalent boundary expansions are identified. This qualification prevents the assumption that every intersection of nested half-open intervals is automatically nonempty.

Theorem 5.7 (Determination of the scalar realization). *A compatible block history determines a unique limit of the left endpoints of (5.9). If an exact character has rational brackets of arbitrarily small width and is separated from the boundary of every decimal cylinder examined, each decimal prefix of finite length is determined after a finite computation. The prefixes thus obtained are mutually compatible.*

Proof. Extending by one block preserves the preceding prefix through Euclidean division. The closed cylinders are nested and their diameter tends to zero; their left endpoints are Cauchy and have a unique limit. For a fixed decimal length, a value separated from the partition boundaries has positive distance from them. A sufficiently narrow bracket lies within a unique cell and determines its prefix. Uniqueness of the cell and nesting of the partitions prove compatibility under truncation. At points with a terminating expansion, the exact boundary decision and its representation convention must be added; that decision does not follow from numerical width alone. $\square$

The closure, propagation, and self-scaling characters have the preceding brackets. After their recognition, the irrationality of π, e, φ guarantees their separation from the rational boundaries at any finite decimal precision. Arbitrary depth means *for every finite horizon there exists a computation determining its prefix*; it does not require a finite machine to store infinitely many digits at once. The mathematical object is the full compatible collection, and an execution produces a finite projection of it.

5.8 Specular involution and synchronization of the regions

The nonadic connection jointly transports the three characters. In the ninth-phase chart, the self-scaling axis and the ternary sum of closure and propagation remain fixed under the specular exchange. Helical inversion changes the direction of transport; the mirror of the two channels changes their relative orientation. These are different involutions, although both act on the same trajectory system.

The census coincidences illustrate this coordination:

Phase	N_π	N_e	N_φ	Equality of counts
4	207	207	122	$N_\pi = N_e$
5	39	151	151	$N_e = N_\varphi$
6	110	110	69	$N_\pi = N_e$
7	81	29	81	$N_\varphi = N_\pi$
8	59	59	59	triple synchronization
9	43	43	43	triple synchronization

These integers count compatible coordinates of the regions. They do not assert equality of π, e , and φ as real numbers. If

$$\partial(a, b, c) = (a - b, b - c, c - a),$$

its image belongs to the lattice $A_2 = \{(u, v, w) \in \mathbb{Z}^3 : u + v + w = 0\}$. In phases eight and nine, the relative component vanishes, but the diagonal component changes from (59, 59, 59) to (43, 43, 43). Relative synchronization coexists with a displacement of -16 per channel. The closure leading to α must preserve both pieces of information: relative agreement and the evolution of memory.

5.9 Sturmian helical suspension

The relation between bases nine and ten in the completion defines the dimensionless parameter

$$\delta = \frac{\log(10/9)}{\log 10}. \quad (5.10)$$

It is not obtained from a target physical constant. It is the relative logarithmic scale of two bases already present in the construction. Its use as a symbolic reading of refinement has an exact consequence.

Lemma 5.8 (Irrationality of the slope). *The number δ is irrational.*

Proof. If $\delta = p/q$, $q > 0$, then $(10/9)^q = 10^p$ and $9^q = 10^{q-p}$. The factorization of the left-hand side contains the prime three, whereas that of the right-hand side contains only the primes two and five. Equality is impossible. $\square$

The lower mechanical word, with intercept θ , is

$$z_n(\theta) = \lfloor (n+1)\delta + \theta \rfloor - \lfloor n\delta + \theta \rfloor \in \{0, 1\}.$$

It is obtained by observing the rotation $\theta \mapsto \theta + \delta \pmod{1}$ through the interval $[1 - \delta, 1)$. The event sequence has density δ , and its sum over any window of length L equals $\lfloor L\delta \rfloor$ or $\lceil L\delta \rceil$. Since $21 < 1/\delta < 22$, event separations are twenty-one or twenty-two. The word indicating which intervals are short is another mechanical word of slope $\sigma = 22 - 1/\delta$; its events are separated by six or seven primary intervals. The second scale is induced by the first, not a second free parameter.

5.9.1 Capacity index, vacancies, and induction of the tritic regime

The vacancy sequence is linked to refinement through an increment identity. To exhibit it, fix the phase origin $\theta = 0$. Let

$$\rho = \log_{10} 9 = 1 - \delta, \quad \mathcal{C}_t = \lfloor (t+1)\rho \rfloor \quad (t \geq 0).$$

The index $\mathcal{C}_t$ expresses the decimal capacity of refinement in the normalization comparing a ternary block of 6 positions, with 729 possibilities, and a decimal block of 3 positions, with 1000 possibilities. Indeed, $\log_{1000} 729 = \log_{10} 9 = \rho$. The letter K is reserved for the dodecaphase record constructed later.

Proposition 5.9 (Complementarity of capacity and vacancy). *For $t \geq 1$,*

$$\mathcal{C}_t - \mathcal{C}_{t-1} = 1 - z_t(0). \quad (5.11)$$

Consequently, if $0 \leq a < b$,

$$\mathcal{C}_b - \mathcal{C}_a + \sum_{t=a+1}^b z_t(0) = b - a.$$

Proof. Irrationality of δ gives

$$\mathcal{C}_t = t + 1 - \lfloor (t+1)\delta \rfloor = t - \lfloor (t+1)\delta \rfloor.$$

Subtracting consecutive terms produces (5.11). Summation over the interval telescopes. $\square$

Each transition increases the block counter by one. If $z_t = 0$, capacity also increases; if $z_t = 1$, capacity remains unchanged while a transition is incorporated into the history. Here a vacancy is a capacity-retention event, with a determined position and antecedent. The preceding balance preserves length exactly between acquired capacity and recorded vacancies. The observed phase $g_t^{\text{cap}} = \lfloor \mathcal{C}_t \rfloor_9$ satisfies

$$g_t^{\text{cap}} - g_{t-1}^{\text{cap}} \equiv 1 - z_t \pmod{9}.$$

This capacity phase and chronological phase $[t]_9$ are two different observations. Local capacity retention and a complete holonomy turn have distinct mechanisms, although both allow the phase observation to be preserved while the state changes.

The relation with TRIT can be followed in the vacancy positions themselves. For $j \geq 1$, define

$$p_j = \left\lfloor \frac{j}{\delta} \right\rfloor, \quad v_j = \mathcal{C}_{p_j}, \quad h_j = p_{j+1} - p_j.$$

The inequality $p_j \delta < j < (p_j + 1)\delta$ identifies p_j as the position of event j . Since $0 < \delta < 1$, it also gives $\lfloor (p_j + 1)\delta \rfloor = j$, and hence

$$v_j = p_j - j, \quad v_{j+1} - v_j = h_j - 1.$$

If $s_j = 22 - h_j$, then $s_j \in \{0, 1\}$ and

$$[v_{j+1}]_3 = [v_j - s_j]_3. \quad (5.12)$$

An interval of length 22 preserves the class; one of length 21 shifts it by one in the negative direction. The canonical selector is applied to the positive phase $\bar{r}_j = 1 + [v_j]_9$:

$$\tau_j = M_{\text{ph}}(\bar{r}_j).$$

A tritic regime is thus determined by the capacity index retained at each event, in addition to the vacancy's binary encoding.

The positions of the short intervals form the induced mechanical coding of slope $22 - 1/\delta$. Their separations 6 and 7 determine the lengths of plateaus of the class $[v_j]_3$, once the origin and endpoint convention are fixed. The first classes are 2 for 6 events, 1 for 7, and 0 for 7; the first segment of 6 is the initial restriction of a plateau. Their signatures traverse 0, -1 , $+1$. Equations (5.11) and (5.12) make explicit the link between refinement, vacancies, and regime selection. The quadratic reading of TRIT subsequently realizes these types through their respective generators; the temporal event and its geometric representation thus retain an identifiable dependence.

Inverting the monotone index supplies another useful observation:

$$\mathcal{T}(k) = \min\{t : \mathcal{C}_t = k\} = \left\lceil \frac{k}{\rho} \right\rceil - 1 \quad (k \geq 1).$$

If $\sigma = \rho^{-1} - 1$, the number of blocks needed to gain m units of capacity is

$$\mathcal{T}(k + m) - \mathcal{T}(k) = m + \lfloor (k + m)\sigma \rfloor - \lfloor k\sigma \rfloor \in \{m + \lfloor m\sigma \rfloor, m + \lceil m\sigma \rceil\}.$$

In particular, 9 units of capacity require 9 or 10 blocks. The period of a finite phase remains exact, whereas return time measured by the other index admits two values. This distinction belongs to refinement dynamics and is not an a posteriori correction of its numerical values.

Theorem 5.10 (Complexity of the phase readings). *For $m \geq 1$, the mechanical word has factor complexity $p(m) = m + 1$. Retaining the phase $n \bmod 9$ gives complexity $p_9(m) = 9(m + 1)$; retaining only its ternary class gives $p_3(m) = 3(m + 1)$. All three topological entropies are zero.*

Proof. For factors of length m , the predecessors of the two endpoints of the coding interval form $m + 1$ distinct points on the circle: the coincidence between shifted endpoints removes duplicates, and irrationality prevents further coincidences. These points determine $m + 1$ intervals with distinct constant itineraries. Every irrational orbit visits each interval, giving $p(m) = m + 1$.

Fix a phase modulo nine. The positions of that phase follow a rotation of step 9δ , also irrational. By density, every factor occurs in each phase, and the first coordinate distinguishes its nine copies. The same argument with step 3δ gives three copies after ternary reduction. Finally, $m^{-1} \log(c(m + 1)) \rightarrow 0$ for $c = 1, 3, 9$. $\square$

The phase product $C_9 \times \mathbb{T}$, translated by $(1, \delta)$, has frequency characters

$$\frac{a}{9} + k\delta \pmod{1}, \quad a \in \mathbb{Z}/9\mathbb{Z}, k \in \mathbb{Z}.$$

The spectral module is not $\log 10$: it is the additive module generated by $1/9$ and δ , modulo the integers. The factor $\log 10$ belongs to the normalization of (5.10).

An operator presentation makes memory explicit. On $\ell^2(\mathbb{Z})$, let $T|n\rangle = |n+1\rangle$, $V_9|n\rangle = \zeta_9^n|n\rangle$, $W_\delta|n\rangle = e^{2\pi i n \delta}|n\rangle$, and $D_M|n\rangle = \lfloor n/9 \rfloor |n\rangle$. On finitely supported vectors,

$$V_9 T = \zeta_9 T V_9, \quad W_\delta T = e^{2\pi i \delta} T W_\delta, \quad [D_M, T^9] = T^9. \quad (5.13)$$

The first two relations are Weyl covariances; the third expresses that one nonadic turn increases the memory degree. The base rotation is abelian, whereas the algebra of observables and translations is not. The inversion $|n\rangle \mapsto |-n\rangle$ conjugates forward motion with backward motion. This realization provides an aperiodic helical suspension with finite phase return, not a periodic word of length nine.

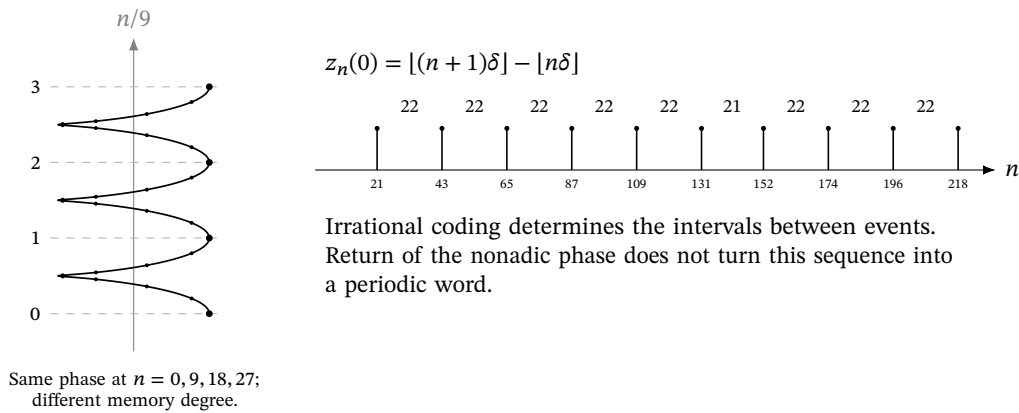


Figure 6: Complementary representations of transport. Left: projection of the helix $(\cos(2\pi n/9), \sin(2\pi n/9), n/9)$: advancing nine steps recovers the phase and increases the height. Right: the first ten events of the lower word for $\delta = \log_{10}(10/9)$, with exact separations 21 or 22. The helical representation introduces no additional physical metric.

For this count we fix the intercept $\theta = 0$ and cumulative endpoints

$$(N_0, \dots, N_4) = (0, 729, 972, 999, 1000),$$

corresponding to the cubic order $(729, 243, 27, 1)$. The lower counts are $\lfloor N_j \delta \rfloor - \lfloor N_{j-1} \delta \rfloor$, giving $(33, 11, 1, 0)$. The upper counts are $\lfloor N_j \delta \rfloor - \lfloor N_{j-1} \delta \rfloor$, with the same initial boundary zero, giving $(34, 11, 1, 0)$. These values depend on the intercept and the order of the strata; they are not asserted for an arbitrary initial phase. Their totals are forty-five and forty-six. The increment corresponds to a single boundary unit, located in the first stratum of this ordered partition. The numbers thirty-four and eleven thus appear in the same count before any units chart; identifying them with physical exponents additionally requires the corresponding dimensional derivation.

The term *aperiodic crystal* is used here for this symbolic order, with complexity, spectrum, and return specified. Zero topological entropy does not by itself imply zero physical dissipation: a thermodynamic realization requires a defined energetic dynamics and erasure operation. Likewise, this Sturmian reading does not replace all TPK operators or by itself prove periodicity or aperiodicity of the complete α output. Its function is to make explicit a structure of order that the mere return of the calendar does not describe.

5.10 Nonadic connection, holonomy, and monodromy representation

The regional structure requires distinguishing the transport producing a history, observation of its phase, and representation of its returns. APP supplies the two arithmetic evaluations with remainder and quotient; TRIT determines regime and orientation; the Topological Phase Kernel composes selection, transport, and update. The initial conditions of cardinality 104 976, their catalog of 468 emissions, and the 243 visible words in the ambient space of 729 elements belong to different stages of that construction. The chain

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

leads to a joint connection over the three regions. Its nine phases are positions of a single transport, and the regional coordinates π, φ, e , together with the dodecaphasic determination of α , arise from its compatible readings[development and scope in Sections 8 and 9.4]; [Sections 4.8, 5.10, and 9.2.1].

At prolongation index k , a state chart preserves

$$\hat{x}_k = (B_k, C_k, p_k, c_k, \Sigma_k, W_k, g_k, m_k).$$

Here B_k collects the three regional blocks, C_k is the cylinder, p_k the prefix, c_k the carry record, Σ_k the boundary signature, and W_k the incidence information. The phase is $g_k = 1 + [(k - 1)]_9$; m_k counts complete turns. The reduction $\beta_k = [m_k]_{12}$ preserves the dodecaphasic position, whereas the integer m_k distinguishes its successive turns.

Definition 5.11 (Transport through one nonadic turn). Let $\mathcal{R}_k \subseteq \hat{X}_k \times \hat{X}_{k+1}$ be the admissible extension relations of the TPK. The realization of holonomy at level k is the relational composition

$$\Gamma_{9,k} = \mathcal{R}_{k+8} \circ \cdots \circ \mathcal{R}_k.$$

For $s \geq 1$, its prolongation through s turns is

$$\Gamma_{9s,k} = \Gamma_{9,k+9(s-1)} \circ \cdots \circ \Gamma_{9,k}.$$

The relational formulation preserves admissible continuations. On an individualized history, each arrow leads to the state actually prolonged; the isolated visible block may still admit more than one continuation. This difference is precisely the function of memory and boundary conditions in determining the trajectory.

Proposition 5.12 (Iteration of return and accumulated memory). *In every admissible composition of s turns,*

$$g_{k+9s} = g_k, \quad m_{k+9s} = m_k + s, \quad \beta_{k+9s} = \beta_k + s \pmod{12}.$$

Proof. The one-turn rule preserves the phase and increments the counter by one. Applying it to the already transported state repeats exactly these two operations. Induction gives the first two equalities; reducing the second modulo 12 gives the third. $\square$

The counter thus provides an integer witness of the difference between turns. After 12 turns the pair of finite phases is restored, but the counter increases by 12. The nonsplit extension (4.5) expresses the composition of the calendar through its cocycle; the trajectory also preserves the integer lift of that datum. Preservation here means transport of prior contributions, not immobility of all coordinates.

5.10.1 Bilateral representation of the return index

The geometric realization of the cyclic graph admits an especially simple operator representation. Here we number the phases from 0 to 8, shifting the origin relative to g_k , and write $k = 9m + r$, with $0 \leq r < 9$, on the integer line covering the phase cycle. On $\mathcal{H}_{\text{ind}} = \ell^2(\mathbb{Z})$, the basis $|k\rangle = |r, m\rangle$ allows the unit shift to be expressed as

$$T|r, m\rangle = \begin{cases} |r+1, m\rangle, & r < 8, \\ |0, m+1\rangle, & r = 8. \end{cases}$$

The inverse decreases r by one when $r > 0$, and sends $|0, m\rangle$ to $|8, m-1\rangle$. Passing through the phase origin therefore incorporates the exact unit of quotient.

Proposition 5.13 (Representation of cycle monodromy). *Let $S = T^9$. Then*

$$S|r, m\rangle = |r, m+1\rangle, \quad \mathbb{Z} \longrightarrow \mathcal{U}(\mathcal{H}_{\text{ind}}), \quad s \longmapsto S^s$$

is a faithful unitary representation of the group of turns of the cycle.

Proof. Nine shifts preserve the remainder of division by 9 and increment the quotient. The powers satisfy $S^{s+t} = S^s S^t$, and $S^s|r, m\rangle = |r, m+s\rangle$ distinguishes every integer s . A permutation of an orthonormal basis is unitary. $\square$

This representation corresponds to the covering of the cycle, whose group of turns is $\mathbb{Z}$, and represents the phase with its counter. The TPK state also contains the trajectory coordinates already described. The bilateral extension admits negative indices as coordinates of the covering; a trajectory beginning at an initial condition uses its admissible half-axis. An operator inverse of the index is thus available without turning every extension relation into a bijection of the full state.

With $D_M|k\rangle = [k/9]|k\rangle$, the identity $[D_M, S] = S$ expresses the memory increment on the dense domain of finitely supported vectors. If $\mathcal{I}|k\rangle = |-k\rangle$ and P_0 projects onto indices divisible by 9, one also obtains

$$\mathcal{I}T\mathcal{I} = T^{-1}, \quad \mathcal{I}D_M\mathcal{I} = -D_M - (I - P_0). \quad (5.14)$$

Indeed, for $k = 9m + r$, $[-k/9] = -m$ if $r = 0$, and equals $-m - 1$ if $r > 0$. Inversion thus preserves the boundary term of Euclidean division.

5.11 Block clock and capacity clock

The index of a refinement map and the number of available decimal positions obey related but different laws. To express them, reserve n for the ternary-block index and define

$$\rho = \log_{1000} 729 = \log_{10} 9, \quad \delta = 1 - \rho, \quad \mathcal{C}_n = \lfloor (n+1)\rho \rfloor, \quad n \geq 0.$$

Here $\mathcal{C}_n$ denotes capacity, not the dodecaphasic record K . At the canonical phase origin, the preceding mechanical word provides

$$Z_n = \sum_{j=0}^n z_j(0) = \lfloor (n+1)\delta \rfloor.$$

Proposition 5.14 (Exact balance between capacity and vacancies). *For $n \geq 0$, and in the second identity for $n \geq 1$,*

$$\mathcal{C}_n = n - Z_n, \quad \mathcal{C}_n - \mathcal{C}_{n-1} = 1 - z_n(0).$$

Proof. Irrationality of δ implies $\lfloor (n+1)\delta \rfloor = \lfloor (n+1)\delta \rfloor + 1$. Then

$$\lfloor (n+1)(1-\delta) \rfloor = n+1 - \lfloor (n+1)\delta \rfloor = n - Z_n.$$

Subtracting two consecutive identities concludes the proof. $\square$

An event $z_n = 1$ adds a block to the history while preserving capacity $\mathcal{C}_n$; an event $z_n = 0$ increments both counters. Over any interval, capacity increments and vacancies sum exactly to its length. Reducing modulo q , for $q = 9$ or $q = 108$, gives

$$[\mathcal{C}_n]_q = [n]_q - [Z_n]_q.$$

The uniform block phase and capacity phase are related through the integrated memory of vacancies. In particular, a capacity hold and a complete turn of the connection have distinct mechanisms, although both may preserve a phase observation.

Monotone inversion of the clock gives a first-arrival time:

$$T_{\text{cap}}(a) = \min\{n : \mathcal{C}_n = a\} = \left\lceil \frac{a}{\rho} \right\rceil - 1, \quad a \geq 1.$$

With $\sigma = \rho^{-1} - 1$, the time needed to gain b units of capacity is

$$T_{\text{cap}}(a + b) - T_{\text{cap}}(a) = b + [(a + b)\sigma] - [a\sigma].$$

The difference of ceilings takes one of the two integers adjacent to $b\sigma$. This yields 9 or 10 blocks to gain 9 capacity units, and 113 or 114 to gain 108. These are inverse-clock times measured from a first arrival; the turn counter is always interpreted in the index to which it belongs.

5.12 Symbolic suspension and aperiodic time crystal

Rotation coding combines phase periodicity, local recurrence, and global aperiodicity. Let $\mathcal{X}_\delta \subseteq \{0, 1\}^{\mathbb{Z}}$ be the closure, in the product topology, of shifts of the bilateral word $z_n(\theta)$. Denote by σ_{sh} the shift of its sequences. Retaining a nonadic phase, the step is

$$F(r, z) = ([r + 1]_9, \sigma_{\text{sh}} z), \quad (r, z) \in C_9 \times \mathcal{X}_\delta.$$

Definition 5.15 (Symbolic aperiodic time crystal). The unit-roof suspension of the preceding system is the space

$$\mathcal{S}_\delta = ((C_9 \times \mathcal{X}_\delta) \times \mathbb{R}) / \sim, \quad (y, t + 1) \sim (Fy, t),$$

with flow $\Phi_s[y, t] = [y, t + s]$. This model of temporal order, together with its vacancy coding and turn-counter representation, is called a *symbolic aperiodic time crystal*.

The adjective temporal designates the action of the flow and its return section; aperiodicity characterizes its itineraries. The frequency of ones in each sequence is δ : the uniform discrepancy bound of less than one, obtained by telescoping over any window, persists under closure. A periodic sequence would have rational frequency. Therefore, $\mathcal{X}_\delta$ contains no periodic orbits. Nor does the suspension have closed orbits: a return of the flow would require an integer time and periodicity of the itinerary.

Finite words do recur. Length- m itineraries come from a partition of the circle by $m + 1$ points; every interval is visited recurrently by irrational rotation. Even when times are restricted to a fixed phase, the step 9δ remains irrational. This is why every word appears in all 9 phases and explains the already established complexities

$$p(m) = m + 1, \quad p_9(m) = 9(m + 1), \quad p_3(m) = 3(m + 1).$$

Recurrence of local configurations coexists with the absence of a global period. Its topological entropy is zero because these complexities grow linearly.

The realization through observables V_δ, W_δ and the shift T expresses the same order through the covariances (5.13). The frequencies of the uniform clock belong to $\frac{1}{9}\mathbb{Z} + \delta\mathbb{Z}$, modulo 1. The capacity clock additionally preserves the relation $[\mathcal{C}_n]_9 = [n]_9 - [Z_n]_9$; it is not replaced by the uniform phase when returns are interpreted. The term *time crystal* is thus defined through a mathematical dynamical system. Its physical realization requires specifying an energetic dynamics, its temporal observables, and its stability, data distinct from the present symbolic construction.

5.13 Specular symmetry of the regions and dual realization

Specular exchange acts on the regional block $B = (u^\pi, u^e, u^\varphi) \in (\mathbb{F}_3^6)^3$ by

$$\mathfrak{c}B = (u^e, u^\pi, u^\varphi).$$

Its fixed axis and aggregate are, respectively, u^φ and $s = u^\pi + u^e$. This fixedness concerns the involution in a fiber; both coordinates may evolve during nonadic transport. With $q = u^\pi + u^e + u^\varphi$, $a = u^\pi + u^e - u^\varphi$, and $d = u^\pi - u^e$, one recovers

$$u^\varphi = 2(q - a), \quad s = q - u^\varphi, \quad u^\pi = 2(s + d), \quad u^e = 2(s - d).$$

The equalities hold componentwise in $\mathbb{F}_3$, where $2^{-1} = 2$. The mirror preserves q, a, s and reverses d ; this antisymmetric coordinate individualizes the allocation that the aggregate preserves without distinguishing. Time-index inversion $\mathcal{J}$ and the channel involution $\mathfrak{c}$ therefore act on different data.

Phase return has an explicit witness:

$$B_1 = (012222 \mid 121221 \mid 112202),$$

$$B_{10} = (101222 \mid 112221 \mid 112002).$$

Both positions have phase 1; the block, axis, and memory have changed. Likewise, census synchronization in phases 8 and 9 annihilates their relative differences, while the diagonal component passes from (59, 59, 59) to (43, 43, 43). Equality of a relative observable does not exhaust the state determining it.

The dual realization preserves this architecture. In section 28.2, the same prefix determines nested cylinders on one side and a sequence of encoded words with their frames on the other. The first map passes to the limit by contraction of diameters; the second satisfies $r_{n,m}Q_n = Q_m p_{n,m}$ and defines Q_∞ . Figure 7 anticipates these two maps. The subsequent incidence construction produces codes, lattices, and the Moonshine realization in their own domains; the automorphism group acts on that realization, not on the digits through an implicit identification.

5.14 Temporal transport and tritic exponential composition

The Euler relations in section 5.6 describe local realizations of the three regimes. For a fixed signature τ , the generator satisfies $J_\tau^2 = -\tau I$; algebraic conjugation sends J_τ to $-J_\tau$. Consequently,

$$\exp(sJ_\tau) \exp(tJ_\tau) = \exp((s+t)J_\tau), \quad \overline{\exp(tJ_\tau)} = \exp(-tJ_\tau),$$

and the product of an evolution with its conjugate is the unit. In the elliptic type it is realized through $\cos^2 t + \sin^2 t = 1$; in the hyperbolic type, through $\cosh^2 t - \sinh^2 t = 1$; in the parabolic type, $(I + tN)(I - tN) = I$, since $N^2 = 0$. The same conservation law admits three quadratic expressions.

The elliptic realization closes a turn; the nilpotent one preserves a linear displacement; the hyperbolic one separates expansion and contraction with unit determinant. At the regional parameters already generated,

$$\exp(\pi J_e) = -I, \quad \exp(J_p) = I + J_p, \quad \exp(2 \log \varphi J_h) = F_{av}^2.$$

The evaluation $\exp(I) = eI$ recognizes the unit of the common exponential law. The number e is not assigned exclusively to the parabolic regime.

Transporting a local realization through an isomorphism U preserves these relations: if $J' = UJU^{-1}$, the exponential series gives $U \exp(tJ) = \exp(tJ')U$. The identity transports the regime and its norm. A transition between different quadratic types, by contrast, belongs to TPK sheet and regime selection; it is not a conjugation changing the value of J^2 . The temporal product therefore preserves the order of transitions and the information at their endpoints.

This articulation distinguishes the closure of a local representation from the holonomy of a history. One may have $\exp(2\pi J_e) = I$ at the same time as $S|r, m\rangle = |r, m+1\rangle$. The first equality restores the regime's orientation; the second records another turn. The discrete structure preserves both: the local return and the accumulated provenance of the state realizing it.

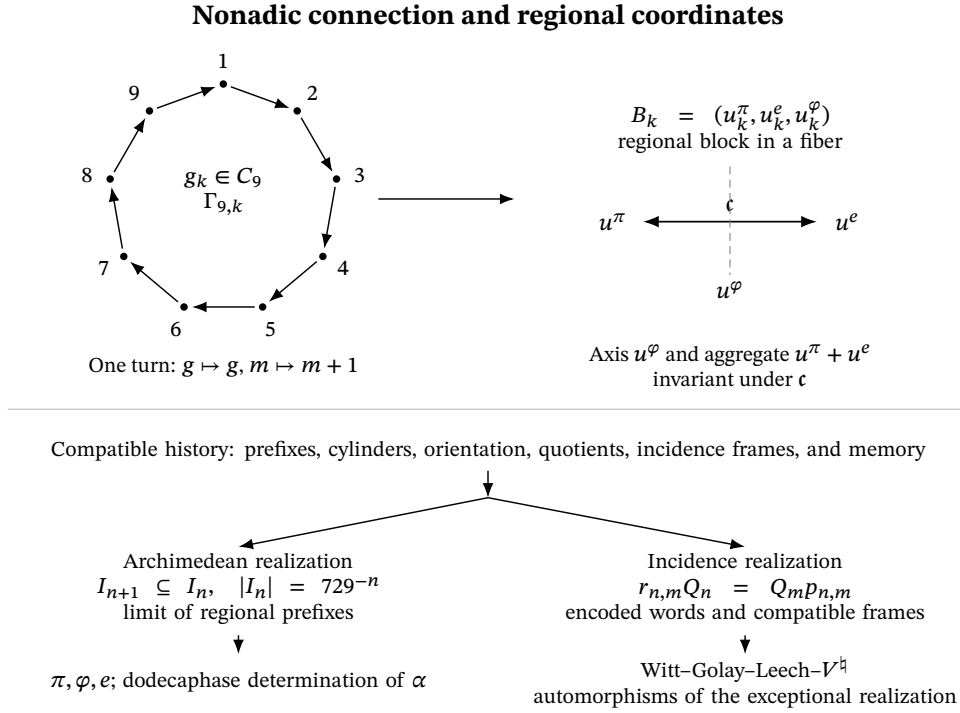


Figure 7: Nine phases of a connection and two realizations of a compatible history. The cycle represents phase; each turn increments the counter. The regional diagram represents the involution $\mathfrak{c}$, which interchanges closure and propagation and fixes the self-scaling axis in a fiber. This invariance does not require the axis to remain constant during transport. The two lower maps use the same prefix: one determines its numerical limit, while the other preserves words and incidence frames. The maps of the second realization are specified in Section 28.2; the exceptional construction is developed afterwards.

5.15 Multiplicative irreducibility and return periods

The distinction between structure, evaluation, and phase also appears in the multiplicative sector. APP's local product contains both residue and quotient; iterating it constructs normal forms and factorizations before their decimal periods are examined. This development places the expression *prime mode* in a precise domain and relates irreducibility to the operations already defined, rather than to an isolated coincidence of periods [development and scope in Sections 8 and 9.4]; [Sections 4.8, 5.10, and 9.2.1].

Let

$$\mathcal{W}_9 = \{\emptyset\} \sqcup \bigsqcup_{\ell \geq 1} \{1, \dots, 9\}^\ell.$$

Sequences are ordered from the least significant digit. The evaluation $\text{ev}_9(u) = \sum_j u_j 9^j$, with $\text{ev}_9(\emptyset) = 0$, is bijective onto the nonnegative integers: for a positive integer, division of $n - 1$ by 9 uniquely determines the first digit and continuation. This is bijective nonadic numeration; it retains the boundary representative 9.

Addition $\boxplus_\#$ is obtained by applying APP's additive sheet to the digits at each position and propagating their quotients. If there is only one digit, it is the provisional residue; a nonzero incoming quotient is normalized through a second additive operation. The sum of quotients is transported to the next position. To multiply a sequence by a digit d , the multiplicative cell produces $u_j d = 9q_j + r_j$, and the incoming quotient c_j is incorporated by

$$(w_j, c_{j+1}) = \begin{cases} (r_j, q_j), & c_j = 0, \\ (R^+(r_j, c_j), q_j + q^+(r_j, c_j)), & c_j > 0. \end{cases}$$

Start with $c_0 = 0$ and incorporate the final quotient when nonzero. During addition, $c_j \leq 2$; during multiplication by a digit, $c_j \leq 9$. Local operations thus receive arguments within their domain. Denote this second procedure by $d \odot_{\#} u$, with $d \odot_{\#} \emptyset = \emptyset$.

For $v = (v_0, \dots, v_{m-1})$, the complete product is constructed through the recurrence

$$a_m = \emptyset, \quad a_j = (9 \odot_{\#} a_{j+1}) \boxplus_{\#} (v_j \odot_{\#} u), \quad u \boxtimes_{\#} v = a_0.$$

The rule composes both local sheets. Its evaluation satisfies

$$\text{ev}_9(u \boxtimes_{\#} v) = \text{ev}_9(u) \text{ev}_9(v). \quad (5.15)$$

Indeed, each transition preserves the partial sum plus the pending quotient weighted by the power of 9. The recurrence then gives $\text{ev}_9(a_j) = 9 \text{ev}_9(a_{j+1}) + v_j \text{ev}_9(u)$, whose iteration proves (5.15). Uniqueness of normal form transports associativity and commutativity to the constructed product; its identity is (1).

Definition 5.16 (Reduced multiplicative coproduct). On the finitely supported vector space generated by nonempty normal forms, define

$$\tilde{\Delta}_{\#} e_w = \sum_{\substack{u \boxtimes_{\#} v = w \\ u, v \neq (1)}} e_u \otimes e_v.$$

A non-unit normal form is irreducible when its image under $\tilde{\Delta}_{\#}$ is zero.

Factorization fibers are finite by (5.15). In the completion $\mathcal{H}_{\#} = \ell^2(\mathcal{W}_9 \setminus \{\emptyset\})$, images of distinct forms have disjoint supports. If $d_{\#}(w)$ counts ordered non-unit factorizations, the domain of the closure is

$$\text{Dom}(\overline{\tilde{\Delta}_{\#}}) = \left\{ x \in \mathcal{H}_{\#} : \sum_w d_{\#}(w) |x_w|^2 < \infty \right\}.$$

Indeed, the squared norm of the image is this weighted sum. Hence the operator

$$A_{\#} = (\overline{\tilde{\Delta}_{\#}})^* \overline{\tilde{\Delta}_{\#}}$$

is diagonal, positive, and self-adjoint: its eigenvalue at e_w is the number of ordered non-unit factorizations of w . The projection

$$P_{\text{irr}} = (I - |e_{(1)}\rangle\langle e_{(1)}|) \mathbf{1}_{\{0\}}(A_{\#})$$

selects exactly the irreducible forms. Through multiplicative evaluation, these correspond to ordinary primes. Selection depends on factorization structure and keeps the unit state separate.

For a value $n \geq 2$ coprime to 10, decimal division induces the residue permutation $r \mapsto 10r \pmod{n}$. Return of residue 1 has period

$$\tau_n = \text{ord}_n(10).$$

If $n = \prod_p p^{a_p}$, the Chinese remainder theorem gives

$$\tau_n = \text{lcm}_{p^{a_p} \parallel n} \text{ord}_{p^{a_p}}(10).$$

Composite return synchronizes the returns of prime powers. For a prime $p \notin \{2, 3, 5\}$,

$$\tau_p \mid p - 1, \quad M_p = \frac{10^{\tau_p} - 1}{p} \in \mathbb{N}, \quad M_p \equiv 0 \pmod{9}.$$

The last congruence expresses nonadic closure of decimal repetition: $10^{\tau_p} - 1$ is a multiple of 9, and p is invertible modulo 9. The same congruence holds for every n coprime to 90. In particular, 7, 13, 77, 91, 143 have decimal period 6, although only the first two are irreducible. The coproduct and the period therefore record different properties of the same constructed value.

This differentiation specifies the relation with aperiodic refinement. The vacancy sequence determines increments of the capacity index; factorization determines multiplicative modes; decimal division determines their return times. The nonadic representative preserves phase closure, while quotient and factorization retain information that this closure does not distinguish. The connection to the treatise's spectral constructions resides in these operators and their intertwining. Proof of their subsequent analytic results retains its own development; it is not replaced by periodicity of a residue observation.

5.15.1 Vacancy discrepancy and Dirichlet readings

The connection also admits an exact analytic expression. Retain the same coding $z_n = z_n(0)$, for $n \geq 1$, and consider two subsequent observables: the series over all indices and the product over indices selected by irreducibility. For $\operatorname{Re} s > 1$,

$$Z_{\text{vac}}^+(s) = \sum_{n \geq 1} \frac{z_n}{n^s}, \quad Z_{\text{vac}}^\times(s) = \prod_p (1 - p^{-s})^{-z_p}.$$

Both converge absolutely in this half-plane. Their common factor is density δ , but their discrepancies differ.

Proposition 5.17 (Separation of vacancy density). *We have*

$$Z_{\text{vac}}^+(s) = \delta \zeta(s) + H_{\text{vac}}(s), \quad H_{\text{vac}} \in \mathcal{O}(\operatorname{Re} s > 0).$$

In $\operatorname{Re} s > 1$, with logarithms defined by absolutely convergent Euler series,

$$Z_{\text{vac}}^\times(s) = \zeta(s)^\delta G_{\text{vac}}(s), \quad \log G_{\text{vac}}(s) = \sum_p (z_p - \delta) \log(1 - p^{-s})^{-1}.$$

Proof. The centered partial sums satisfy

$$A_N := \sum_{n=1}^N (z_n - \delta) = [(N+1)\delta] - N\delta, \quad |A_N| < 1.$$

Abel summation represents the centered term by $s \int_1^\infty A_{[x]} x^{-s-1} dx$. The integral converges locally uniformly in $\operatorname{Re} s > 0$, proving holomorphy. The second identity follows by separating $z_p = \delta + (z_p - \delta)$ in the logarithm of the product, within its domain of absolute convergence. $\square$

The decomposition reveals what the reading on primes adds: it examines the same vacancy sequence on a subset determined by factorization. Controlling discrepancy over all integers gives the first holomorphic continuation; the second observable preserves its own discrepancy on primes. These are two applications of the same discrete structure, with explicit analytic domains. This articulation explains their relevance to the arithmetic kernel without transferring the proof of the treatise's terminal results to this article.

6 Integral compatibility of regional generation, cylindrical refinement, and incidence

Regional prolongation produces ordered sequences of blocks together with their reading depths. The Archimedean projection and ternary incidence are calculated on these same sequences. The connection between the two realizations can be formulated entirely through integer operations: selection by rational intervals, Euclidean division, lifting of six trits, carry reading, and the Witt transformation. This formulation makes it possible to trace each coefficient from its generating block to its subsequent use.

The construction uses the three regional readers already obtained from the APP-TRIT-TPK state. We denote them by the indices $c \in \{P, E, F\}$, corresponding, respectively, to the closure, propagation, and autoscaling regions. Their real realization will be written as v_c . The subsequent identifications $v_P = \pi_{\text{HMT}}$, $v_E = e_{\text{HMT}}$, and $v_F = \varphi_{\text{HMT}}$ preserve this causal order. The operation that emits a block inspects the rational intervals of the regional reader; the real value enters the proof of correctness of that operation.

6.1 Positional emission through rational separation of cells

For each region, two rational sequences $\ell_c(j), u_c(j)$, constructed by its reader, are available and satisfy

$$\ell_c(j) \leq v_c \leq u_c(j).$$

The cell-separation property proved for these readers states that, for each integer $s > 0$, there exists J such that, for every $j \geq J$,

$$\lfloor s\ell_c(j) \rfloor = \lfloor su_c(j) \rfloor - 1 = \lfloor sv_c \rfloor. \quad (6.1)$$

Avoidance of rational boundaries by the regional values is the hypothesis of the preceding theorems that guarantees this separation. Here we use their operational conclusion together with the readers and intervals that produce it.

Definition 6.1 (Separation index and emitted prefix). For $s > 0$, let

$$j_c(s) = \min\{j \in \mathbb{N} : \lfloor s\ell_c(j) \rfloor = \lfloor su_c(j) \rfloor - 1\}.$$

For an integer base $b \geq 2$ and a depth $n \geq 0$, define

$$P_c(b, n) = \lfloor b^n \ell_c(j_c(b^n)) \rfloor. \quad (6.2)$$

The prefix contains the integer part and the first n positions in base b ; consequently, $P_c(b, 0)$ is the regional integer part.

Theorem 6.2 (Termination and correctness of emission). *The index $j_c(s)$ exists for every $s > 0$. The emitted prefixes satisfy*

$$P_c(b, n) = \lfloor b^n v_c \rfloor, \quad P_c(b, n) \leq b^n v_c < P_c(b, n) + 1. \quad (6.3)$$

For each scale s , a single sufficiency index $J(s)$ can also be chosen that is valid simultaneously for all three regions.

Proof. The eventual equality in (6.1) makes the set defining $j_c(s)$ nonempty. Its minimum exists by the well-ordering of $\mathbb{N}$. At any index where the equality holds, the rational lower bound gives $\lfloor s\ell_c(j) \rfloor \leq \lfloor sv_c \rfloor$. The upper bound, together with avoidance of a cell boundary, gives $\lfloor sv_c \rfloor \leq \lfloor su_c(j) \rfloor - 1$. When the two integer endpoints coincide, the intervening integer coincides with them. This proves (6.3). For the joint statement, it suffices to take the maximum of the three sufficiency indices obtained at the same scale. The joint index controls all three emissions simultaneously, while each region retains its own intervals. $\square$

Proposition 6.3 (Exact compatibility of prefixes). *For $n, m \geq 0$,*

$$\left\lfloor \frac{P_c(b, n+m)}{b^m} \right\rfloor = P_c(b, n). \quad (6.4)$$

In particular, the next block $d_c(b, n) = P_c(b, n+1) \bmod b$ satisfies

$$P_c(b, n+1) = bP_c(b, n) + d_c(b, n), \quad 0 \leq d_c(b, n) < b. \quad (6.5)$$

Proof. Write $b^n v_c = N + \theta$, with $N = P_c(b, n)$ and $0 \leq \theta < 1$. Then $P_c(b, n+m) = b^m N + \lfloor b^m \theta \rfloor$, and the second summand belongs to $\{0, \dots, b^m - 1\}$. Euclidean division by b^m recovers N . The specialization $m = 1$ yields (6.5) and its residual bound. $\square$

6.2 Regional blocks of six trits at arbitrary depth

The integer equality $729 = 3^6$ determines the grouping of six ternary positions into a block. For $t \geq 0$, define

$$u_c(t) = P_c(729, t+1) \bmod 729, \quad B_{c,j}(t) = \left\lfloor \frac{u_c(t)}{3^{5-j}} \right\rfloor \bmod 3, \quad 0 \leq j < 6. \quad (6.6)$$

Thus $B_c(t)$ is an ordered word of six trits, and

$$u_c(t) = \sum_{j=0}^5 B_{c,j}(t) 3^{5-j}. \quad (6.7)$$

The word and its integer lift are mutually recoverable representations of the same emitted block.

Theorem 6.4 (Stability of extraction at any depth). *For every n ,*

$$P_c(729, n) = P_c(3, 6n).$$

If $t < n$, then

$$u_c(t) = \left\lfloor \frac{P_c(729, n)}{729^{n-t-1}} \right\rfloor \bmod 729. \quad (6.8)$$

Consequently, every extension of the prefix preserves all previously emitted blocks and all signatures calculated from them.

Proof. The first equality follows because both prefixes are separated at the same scale, $729^n = 3^{6n}$, and from Theorem 6.2. For the second, apply (6.4) at depths n and $t+1$, and then take the residue modulo 729. The ternary extraction in (6.6) is deterministic. Any explicit function of those eighteen coordinates therefore retains the same result when the prefix is extended. $\square$

The construction is quantified over $t \in \mathbb{N}$. A table materializing one hundred blocks is a finite evaluation of this construction; formula (6.8) determines its compatibility with every subsequent extension exactly.

Example 6.5 (First joint block). Evaluation of the selected regional readers gives the words

$$B_P(0) = 010211, \quad B_E(0) = 201101, \quad B_F(0) = 121200.$$

Their lifting by (6.7) produces

$$u_P(0) = 103, \quad u_E(0) = 523, \quad u_F(0) = 450.$$

For example, $201101_3 = 2 \cdot 243 + 27 + 9 + 1 = 523$. All six positions, including leading zeros, are retained in the band representation. The leading zero in 010211 has a defined position even though the decimal notation for 103 does not contain it.

6.3 Transition registers and reconstruction of linear lifts

Over $\mathbb{F}_3$, collect two consecutive emissions from each region in the matrix

$$R(t) = \begin{pmatrix} B_P(t) \\ B_P(t+1) \\ B_E(t) \\ B_E(t+1) \\ B_F(t) \\ B_F(t+1) \end{pmatrix}. \quad (6.9)$$

This definition fixes both the regional order and the temporal order. The first four registers are

$$X_0 = R(0), \quad Y_0 = R(1), \quad X_1 = R(2), \quad Y_1 = R(3).$$

Their values, obtained by extraction from the regional prefixes, are

$$\begin{aligned} X_0 &= \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 2 & 0 & 1 & 1 & 0 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 0 \\ 1 & 1 & 2 & 2 & 0 & 2 \end{pmatrix}, & Y_0 &= \begin{pmatrix} 0 & 1 & 2 & 2 & 2 & 2 \\ 0 & 1 & 0 & 2 & 1 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 1 & 1 & 2 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \end{pmatrix}, \\ X_1 &= \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 & 1 & 0 \end{pmatrix}, & Y_1 &= \begin{pmatrix} 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 1 & 0 & 2 & 2 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & 0 & 2 & 0 & 0 \end{pmatrix}. \end{aligned}$$

Theorem 6.6 (Unique reconstruction of the two initial links). *The equations $X_0 L_0 = Y_0$ and $X_1 L_1 = Y_1$ each have a unique solution in $M_6(\mathbb{F}_3)$. These solutions are*

$$\begin{aligned} L_0 &= \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, & L_1 &= \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix}. \end{aligned}$$

Both matrices are invertible.

Proof. The matrices

$$\begin{aligned} A_0 &= \begin{pmatrix} 1 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 2 & 1 & 2 & 1 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 2 \\ 2 & 1 & 1 & 0 & 2 & 2 \\ 2 & 1 & 1 & 1 & 1 & 2 \end{pmatrix}, & A_1 &= \begin{pmatrix} 1 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 & 1 & 1 \\ 1 & 1 & 0 & 2 & 0 & 0 \\ 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & 0 & 2 \end{pmatrix} \end{aligned}$$

satisfy $A_i X_i = X_i A_i = I$, by multiplication in $\mathbb{F}_3$. The solutions are therefore forced by $L_i = A_i Y_i$; multiplication produces the two displayed matrices. Their inverses are

$$L_0^{-1} = \begin{pmatrix} 1 & 2 & 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 2 & 0 & 0 \\ 2 & 1 & 2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 \\ 0 & 2 & 1 & 1 & 2 & 0 \\ 2 & 2 & 2 & 2 & 2 & 2 \end{pmatrix}, \quad L_1^{-1} = \begin{pmatrix} 2 & 0 & 2 & 1 & 2 & 1 \\ 2 & 1 & 0 & 0 & 2 & 0 \\ 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 \\ 2 & 0 & 1 & 2 & 2 & 0 \end{pmatrix}.$$

The bilateral verification $L_i L_i^{-1} = L_i^{-1} L_i = I$ proves exact recovery of each transformed word. All products use residues 0, 1, 2 with their interpretation in $\mathbb{F}_3$. $\square$

The causal reading of this result is precise: the regional emissions produce $R(t)$; the registers $R(0), \dots, R(3)$ determine the two lifts above; their inverses recover the transformed words. Continuity for every t belongs to the coinductive readers and Theorem 6.4. The matrix reconstruction proved here has the two links specified in its statement as its domain.

6.4 Simultaneous refinement in ternary and decimal bases

A ternary emission of six trits and a decimal emission of three digits are readings in different bases. To preserve their relationship, record the state

$$s = (n, k, N, D),$$

where N is a prefix at depth n in base 729, and D is a prefix at depth k in base 1000. Their Archimedean cylinders are

$$I_{729}(s) = \left[\frac{N}{729^n}, \frac{N+1}{729^n} \right), \quad I_{1000}(s) = \left[\frac{D}{1000^k}, \frac{D+1}{1000^k} \right). \quad (6.10)$$

The half-open upper endpoint assigns each value to a unique cell at a given base and depth.

Definition 6.7 (Integer compatibility defects). Define

$$A(s) = N1000^k - D729^n, \quad (6.11)$$

$$B(s) = (N+1)1000^k - D729^n = A(s) + 1000^k. \quad (6.12)$$

The state is compatible when

$$A(s) < 729^n, \quad B(s) > 0. \quad (6.13)$$

Lemma 6.8 (Integer criterion for intersection). *The cylinders in (6.10) have a nonempty intersection if and only if (6.13) holds.*

Proof. Two half-open intervals $[a, b)$ and $[c, d)$, with $a < b$ and $c < d$, intersect exactly when $a < d$ and $c < b$. Applying this condition to (6.10) and multiplying by the positive denominators gives $N1000^k < (D+1)729^n$ and $D729^n < (N+1)1000^k$. These are the two inequalities in (6.13). $\square$

Definition 6.9 (Update of depths and prefixes). For $0 \leq u < 729$ and $0 \leq d < 1000$, define

$$T_u(n, k, N, D) = (n+1, k, 729N+u, D),$$

$$T_{u,d}(n, k, N, D) = (n+1, k+1, 729N+u, 1000D+d).$$

The first operation refines only the ternary cylinder. The second refines both readings and preserves both depths in the state.

Proposition 6.10 (Exact boundary recurrences). *The updates above satisfy*

$$A(T_u s) = 729A(s) + u1000^k, \quad (6.14)$$

$$A(T_{u,d} s) = 729000A(s) + u1000^{k+1} - d729^{n+1}. \quad (6.15)$$

Proof. For the first identity,

$$(729N + u)1000^k - D729^{n+1} = 729(N1000^k - D729^n) + u1000^k.$$

For the second,

$$\begin{aligned} (729N + u)1000^{k+1} - (1000D + d)729^{n+1} \\ = 729000(N1000^k - D729^n) + u1000^{k+1} - d729^{n+1}. \end{aligned}$$

Both are integer identities preceding any real approximation. $\square$

Theorem 6.11 (Refinement of the generated regional cylinders). *Let*

$$s_c(n, k) = (n, k, P_c(729, n), P_c(1000, k)).$$

Then $s_c(n, k)$ is compatible for all n, k . Moreover,

$$\begin{aligned} T_{u_c(n)}s_c(n, k) &= s_c(n + 1, k), \\ T_{u_c(n), d_c(1000, k)}s_c(n, k) &= s_c(n + 1, k + 1). \end{aligned}$$

Euclidean division of the refined prefixes recovers the preceding prefixes exactly. For every $a, b \geq 0$,

$$\frac{P_c(729, n + a) - P_c(729, n + a) \bmod 729^a}{729^a} = P_c(729, n),$$

and the analogous identity holds in base 1000.

Proof. Theorem 6.2 places v_c simultaneously in both intervals in (6.10); the intersection criterion gives compatibility. The update identities are (6.5) in each base. Recovery follows from (6.4). Thus the defect recurrence and prefix recovery are consequences of the same update. $\square$

Example 6.12 (First pair of regional cylinders). For $n = k = 1$, evaluation of the three regions gives

c	N	D	A	B
P	2290	3141	211	1211
E	1981	2718	-422	578
F	1179	1618	-522	478

In all three rows, $A < 729$ and $B > 0$. The first prefix column decomposes as $2290 = 3 \cdot 729 + 103$, $1981 = 2 \cdot 729 + 523$, and $1179 = 729 + 450$: exactly the previously emitted six-trit blocks reappear. The different signs of A express the relative positions of the two lower endpoints without altering the regional value's membership in both intervals.

6.5 Selection of the next block by the regional reader

Compatibility of two intervals expresses a relationship between readings. Determining the next block additionally uses the rational interval produced by the region. This operation can be formulated without discarding the depth history.

Definition 6.13 (Admissibility of regional prolongation). For fixed c, n , let $s = 729^{n+1}$ and $j = j_c(s)$. A residue $u \in \{0, \dots, 728\}$ is admissible when

$$[s\ell_c(j)] \leq 729P_c(729, n) + u \leq [su_c(j)] - 1. \quad (6.16)$$

Theorem 6.14 (Uniqueness of the prolongation compatible with the reader). *Condition (6.16) is equivalent to $u = u_c(n)$. There is therefore exactly one admissible prolongation of each region at each depth.*

Proof. At the separation index, both integer endpoints in (6.16) coincide with $P_c(729, n + 1)$. The condition is therefore equivalent to $729P_c(729, n) + u = P_c(729, n + 1)$. By (6.5), the right-hand side is $729P_c(729, n) + u_c(n)$. Cancelling the common term gives equality of the residues. Conversely, this equality satisfies the condition. $\square$

The effective dependence of this selection is fully specified: selected region, rational reader, depth, separation index, preceding prefix, and next residue. The defect A summarizes a boundary relationship; the state (n, k, N, D) and the regional intervals retain the data used by the update.

6.6 Ternary signature, carry, Witt incidence, and reading clock

At a fixed depth t , write $B_{c,j} = B_{c,j}(t)$. For each column j , calculate the following integers and residues:

$$S_j = B_{P,j} + B_{E,j} + B_{F,j}, \quad q_j = S_j \bmod 3, \quad c_j = \lfloor S_j/3 \rfloor, \quad (6.17)$$

$$a_j = (B_{P,j} + B_{E,j} - B_{F,j}) \bmod 3, \quad w_j = \sum_c \mathbf{1}_{B_{c,j} \neq 0}, \quad \bar{w}_j = w_j \bmod 3. \quad (6.18)$$

The residue of the axial expression is taken in the integers. An implementation using natural numbers calculates it as $(B_{P,j} + B_{E,j} + 3 - B_{F,j}) \bmod 3$, which preserves that residue exactly.

The ternary incidence matrix used by the reader is

$$W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}. \quad (6.19)$$

Two residual charges are preserved for each region:

$$v_c^{\text{res}} = \sum_j B_{c,j} \bmod 3, \quad v_c^W = \sum_j (B_c W)_j \bmod 3. \quad (6.20)$$

The reader's complete signature is $\mathcal{S}(B) = (q, a, c, w, \bar{w}, v^{\text{res}}, v^W)$. Its structure separately preserves the column residue, carry, axial contrast, support weight, and the two charge readings.

Theorem 6.15 (Exact recovery and bounds of the signature). *For each column and each region,*

$$S_j = q_j + 3c_j, \quad 0 \leq q_j, a_j < 3, \quad 0 \leq c_j \leq 2, \quad (6.21)$$

$$B_{F,j} = 2(q_j + 3 - a_j) \bmod 3, \quad 0 \leq w_j \leq 3, \quad 0 \leq \bar{w}_j < 3. \quad (6.22)$$

Moreover,

$$v_c^W = \left(\sum_j ((B_c W)_j \bmod 3) \right) \bmod 3. \quad (6.23)$$

The stated recoveries determine each column's total and its autoscaling coordinate; the regional register additionally preserves the three ordered bands.

Proof. Identity (6.21) is Euclidean division of S_j by 3. Each summand of S_j lies between 0 and 2, so $0 \leq S_j \leq 6$ and $c_j \leq 2$. In $\mathbb{F}_3$, subtracting the two congruences defining q_j and a_j gives $q_j - a_j = 2B_{F,j}$. Since the inverse of 2 is 2, this yields (6.22); the summand 3 permits an expression using natural numbers. The bounds on w_j follow by summing three indicators. Finally, reducing an integer sum modulo 3 is equivalent to reducing each summand and then reducing their sum, which proves (6.23). $\square$

Example 6.16 (Signature of the first joint block). The three initial bands produce

$$\begin{aligned} q &= (0, 0, 2, 2, 1, 2), & a &= (1, 2, 0, 1, 1, 2), \\ c &= (1, 1, 0, 1, 0, 0), & w &= (2, 2, 2, 3, 1, 2), \\ v^{\text{res}} &= (2, 2, 0), & v^W &= (2, 1, 1). \end{aligned}$$

The fourth column has total $2 + 1 + 2 = 5$; its signature records $q_3 = 2$ and $c_3 = 1$, so $q_3 + 3c_3 = 5$. Axial recovery gives $2(2 + 3 - 1) \bmod 3 = 2$, precisely the autoscaling coordinate. The Witt images, reduced componentwise, are

$$(2, 0, 2, 1, 1, 2), \quad (0, 0, 0, 2, 0, 2), \quad (2, 1, 1, 2, 1, 0).$$

Their residual sums by row yield the three charges $(2, 1, 1)$.

Definition 6.17 (Integer clock of decimal reading). The transition index t and decimal expansion depth are related by

$$\kappa(t) = \max\{k \in \mathbb{N} : 1000^k \leq 3^{6(t+1)}\}. \quad (6.24)$$

Proposition 6.18 (Characterization and monotonicity of the clock). *The integer $\kappa(t)$ is uniquely determined by*

$$1000^{\kappa(t)} \leq 729^{t+1} < 1000^{\kappa(t)+1}. \quad (6.25)$$

The function κ is monotone and satisfies $\kappa(20) = \kappa(21) = 20$.

Proof. The powers of 1000 are strictly increasing and unbounded; there is therefore a unique interval between consecutive powers containing 729^{t+1} . Monotonicity of the latter expression proves that of κ . The final two evaluations are the integer comparisons

$$1000^{20} \leq 729^{21} < 1000^{21}, \quad 1000^{20} \leq 729^{22} < 1000^{21}.$$

These comparisons use exact integer powers. □

Equality of decimal depth at two distinct transitions explains why both indices must be retained: the reading clock can remain constant while the ternary band advances and its signature is updated. Chronological information resides in the ordered register of transitions.

6.7 Composition theorem for the block, cylinder, and signature

Theorem 6.19 (Joint compatibility of regional readings). *For each depth n , there exists a unique triple of blocks (u_P, u_E, u_F) satisfying regional admissibility (6.16) in all three regions. This triple is $(u_P(n), u_E(n), u_F(n))$. Simultaneously:*

1. *each block updates its cylinder through recurrences (6.14) and (6.15);*
2. *its six trits produce the signature $S(B_P(n), B_E(n), B_F(n))$;*
3. *the column totals are recovered as $q + 3c$, and the Witt charges are obtained from the same bands through (6.23);*
4. *every deeper prefix returns exactly these blocks, truncated cylinders, and signatures.*

Proof. Apply Theorem 6.14 to each of the three regional indices. Coordinatewise uniqueness gives joint uniqueness. Substitute those same residues into the operations $T_u, T_{u,d}$, which allows Theorem 6.11 to be applied. The deterministic application of (6.6) produces the bands used by (6.17)–(6.20). The recovery identities are those of Theorem 6.15. Finally, Theorem 6.4 and truncation in each base prove stability at greater depth. At every step the identity of the emitted block has been preserved; neither reading requires the selection of a different block. □

The conclusion is an operational compatibility between cylindrical contraction and preservation of incidence. Both use the same generated register, retain the depths, and admit integer recovery. This result is the explicit form of the connection

$$\text{emitted regional block} \longrightarrow \begin{cases} \text{cylinder and integer boundary,} \\ \text{band, signature, carry, and Witt charge.} \end{cases}$$

The subsequent development of incidence and fields uses the second reading while preserving its common provenance with the first.

6.8 Calendar census and simultaneous compatibility of the regional biographies

The preceding rows are assembled, without yet applying any decimal reader, into the three block biographies produced by the transitions:

$$\begin{aligned} B^{\text{cl}} &= 010211|012222|010211|002111|110221, \\ B^{\text{pr}} &= 201101|121221|102011|012222|102011, \\ B^{\text{au}} &= 121200|112202|121020|010210|010200. \end{aligned} \quad (6.26)$$

Indeed, the consecutive pairs of the first two segments are the rows of X_0, Y_0 , and those of the last two are the rows of X_1, Y_1 . Thus, the biographies are another representation of the record of twelve transitions, not a collection of conventional prefixes.

To verify the operative word of the calendar, let $c = c_1c_2c_3c_4 \in \{0, 1\}^4$ and, for each regime $\chi \in \{\text{cl}, \text{pr}, \text{au}\}$, define

$$u_0^\chi = b_0^\chi, \quad u_j^\chi = u_{j-1}^\chi L_{c_j} \quad (1 \leq j \leq 4).$$

To condense the census, denote by

$$N_{\text{ar}}(c) := \#\{(j, \chi) : u_j^\chi = b_j^\chi, 1 \leq j \leq 4\}, \quad N_{\text{bio}}(c) := \#\{\chi : (u_0^\chi, \dots, u_4^\chi) = B^\chi\}.$$

Exact multiplication in $\mathbb{F}_3$ for the sixteen calendars gives:

c	$N_{\text{ar}}(c)$	$N_{\text{bio}}(c)$
0000, 0001	6	0
0010	9	0
0011	12	3
0100, 0101, 0110, 0111	3	0
1000, 1001, 1010, 1011	0	0
1100, 1101, 1110, 1111	0	0

(D)

The table exhausts the $2^4 = 16$ possibilities, and each of its entries is obtained by four multiplications of a row by one of the two displayed matrices. Consequently, 0011 is the unique calendar that simultaneously reconstructs the three complete biographies. In operator notation,

$$(L_0, L_0, L_1, L_1) \quad (6.27)$$

is not an independent prescription: it is the unique binary reading compatible with the twelve edges already produced by U_t .

The calendar (6.27) produces four new six-component blocks. Their concatenations form

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30}.$$

The subscripts of w indicate cumulative length. The boundary R_{36} preserves the terminal structure and opens the prolongation relation; it is not renamed as a final periodic block that could be repeated indefinitely.

The identity $L = X^{-1}Y$ proves uniqueness once the transitions are fixed; it does not replace their production by U_t . Conversely, (D) proves within the article the uniqueness of the ordering of the four transports. The causal chain is thereby closed: the APP census produces the initial regions, TRIT fixes regime and orientation, TPK produces the transition record, and that record determines L_0, L_1 and 0011 before any Archimedean realization.

7 Regional determination of the dodecaphasic descriptor and selection of the coupling register

The determination of the coupling register combines two constructions that should be developed in their actual order. The first produces the inputs to the selector from the regional representations of the TPK: six-trit bands, column signatures, a conversion calendar, an oriented descriptor, and a functional panel. The second operates on these inputs and on the terminal eight-block repertoire, enumerates its admissible orderings, and selects a unique register through two simultaneous conditions: exceptional incidence and reading heterotypy. The value of the register appears at the end of this composition.

The present development presupposes the regional representations obtained in the preceding chapters from APP-TRIT-TPK. Its purpose is to make explicit every intermediate operation leading from those representations to the finite selector. The realization of the same register through signed incidences, Hadamard inversion, and the representation of the fine-structure constant are then incorporated. The three constructions retain their respective domains: selection of the register, recovery from its channels, and arithmetic evaluation of its representations.

7.1 Regional bands, encoding, and integer signatures

7.1.1 Domains and encoding of the regional bands

The channel order is fixed as

$$\chi \in \{\pi, e, \varphi\}, \quad B_\chi(t) \in \mathbb{F}_3^6, \quad t \in \mathbb{N}.$$

The channel symbols identify the regions already generated. Each $B_\chi(t)$ retains its place in its sequence of representations; the index t counts ternary blocks. The natural-number evaluation of a band $b = (b_0, \dots, b_5)$ is

$$\text{enc}_6(b) = 243b_0 + 81b_1 + 27b_2 + 9b_3 + 3b_4 + b_5, \quad 0 \leq \text{enc}_6(b) < 729. \quad (7.1)$$

Its fixed-width inverse is

$$\text{dig}_j(n) = \left\lfloor \frac{n}{3^{5-j}} \right\rfloor \bmod 3, \quad 0 \leq j < 6.$$

Leading zeros are thus retained: the words 002001 and 2001 could share an abbreviated integer notation, but the type of the former contains six positions. All subsequent concatenations use this width.

Within the hundred-block window, the six-hundred-trit representation of channel χ determines an integer $P_\chi^{(600)}$. Extraction is performed by

$$b_\chi(t) = \left\lfloor \frac{P_\chi^{(600)}}{729^{99-t}} \right\rfloor \bmod 729, \quad 0 \leq t < 100, \quad B_\chi(t) = \text{enc}_6^{-1}(b_\chi(t)). \quad (7.2)$$

The six-hundred-trit integer is a representation of the regional producer; the hundred-row table is the evaluation of (7.2). This separation allows a historical table to be replaced by its producer without changing the selector that consumes it.

7.1.2 Residues, quotients, occupancy, and orientation

For each column j and time t , introduce

$$T_j(t) = B_\pi(t)_j + B_e(t)_j + B_\varphi(t)_j, \quad (7.3)$$

$$q_j(t) = T_j(t) \bmod 3, \quad c_j(t) = \lfloor T_j(t)/3 \rfloor, \quad (7.4)$$

$$a_j(t) = (B_\pi(t)_j + B_e(t)_j - B_\varphi(t)_j) \bmod 3, \quad (7.5)$$

$$w_j(t) = \mathbf{1}_{B_\pi(t)_j \neq 0} + \mathbf{1}_{B_e(t)_j \neq 0} + \mathbf{1}_{B_\varphi(t)_j \neq 0}, \quad \bar{w}_j(t) = w_j(t) \bmod 3. \quad (7.6)$$

The subtraction in the third expression is performed in the integers and then reduced. In an implementation over natural numbers, the equivalent expression is $(B_\pi + B_e + 3 - B_\varphi) \bmod 3$; truncated subtraction would change the reader.

Proposition 7.1 (Reconstruction of the column sum). *For all admissible bands,*

$$T_j(t) = q_j(t) + 3c_j(t), \quad 0 \leq q_j(t) < 3, \quad 0 \leq c_j(t) \leq 2, \quad 0 \leq w_j(t) \leq 3.$$

Proof. Each column sums three representatives from $\{0, 1, 2\}$, so that $0 \leq T_j(t) \leq 6$. Euclidean division by three yields the stated residue and quotient. Occupancy counts three binary conditions; its bound follows directly from that definition. $\square$

This identity preserves the quotient of the local sum. Reconstructing the three individual bands requires the bands themselves or an additional reader that distinguishes them: a column sum has fewer coordinates than the triple that produces it. Accordingly, the temporal row used below jointly retains the three bands and the four fields $q, a, c, \bar{w}$.

7.2 Conversion clock and first zero advance

7.2.1 Ternary capacity and decimal depth

A representation of $t + 1$ bands contains $6(t + 1)$ trits. The associated conversion calendar is defined by the integer

$$d_t = \max\{k \in \mathbb{N} : 1000^k \leq 3^{6(t+1)}\} = \max\{k \in \mathbb{N} : 1000^k \leq 729^{t+1}\}. \quad (7.7)$$

The definition compares integer capacities. The counter t and the depth d_t record different quantities: a transition adds one six-trit band; the clock counts complete powers of the decimal base one thousand contained in the accumulated ternary capacity.

Lemma 7.2 (Integer characterization of the clock). *For $t, k \in \mathbb{N}$,*

$$d_t = k \iff 1000^k \leq 729^{t+1} < 1000^{k+1}.$$

Moreover, $d_t \leq d_{t+1}$.

Proof. The powers of one thousand form a strictly increasing, unbounded sequence. The power 729^{t+1} lies between two consecutive terms; the lower exponent is precisely the maximum in (7.7). The inequality $729^{t+1} < 729^{t+2}$ implies monotonicity of the maximum. $\square$

Proposition 7.3 (First zero advance). *The least time satisfying $d_{t+1} = d_t$ is*

$$t_* = 20.$$

In particular,

$$d_0 = 0, \quad d_{19} = 19, \quad d_{20} = 20, \quad d_{21} = 20, \quad d_{22} = 21.$$

Proof. The exact integer comparisons

$$1000^{20} \leq 729^{21}, \quad 729^{22} < 1000^{21}, \quad 1000^{21} \leq 729^{23} < 1000^{22}$$

give the values at times twenty, twenty-one, and twenty-two. For $0 \leq t \leq 20$, the expression $729^{t+1}/1000^t = 729(729/1000)^t$ is decreasing and its value at twenty is greater than or equal to one. Hence, $1000^t \leq 729^{t+1} < 1000^{t+1}$ and $d_t = t$. No time less than twenty can satisfy $d_{t+1} = d_t$; at twenty it does hold. All comparisons used involve integer powers. $\square$

The number twenty is therefore an output of the minimality criterion. The rule selecting the instant is “the first zero advance of the clock”; the algorithm and its proof can use this rule without receiving twenty as an external parameter.

7.2.2 Nonadic return and temporal antecedent

The length of the phase return determines

$$r = t_* - 9 = 11. \quad (7.8)$$

The descriptor queries are made at $t_* + 1$, $r + 1$, and r . This distribution preserves the relation between the pause instant, the nonadic translation, and the immediately associated phase reading. Times eleven, twelve, and twenty-one designate positions in the regional calendar; their temporal difference remains recorded even when two times produce the same decimal depth.

7.3 Oriented construction of the twenty-four-trit descriptor

7.3.1 Opposite half-band defects

Take the oriented defects of the regional chart

$$\delta_+ = (0, 0, 0, 1, 1, 1), \quad \delta_- = (0, 0, 0, 2, 2, 2) \in \mathbb{F}_3^6.$$

Their opposition is verified componentwise:

$$\delta_+ + \delta_- = 0 \quad \text{in } \mathbb{F}_3^6.$$

The representative 2 realizes -1 in the ternary chart. Concatenating the three-position tails in negative-positive order produces the word 222111. Adding the defects and concatenating their tails are different operations: the former has codomain $\mathbb{F}_3^6$, whereas the latter preserves the order of two segments of length three.

7.3.2 Dual charge and phase selection

In the regional chart, the six-coordinate transformation uses

$$A_{\text{reg}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}.$$

With row vectors, the dual charge of $b \in \mathbb{F}_3^6$ is the sum of the coordinates of bA_{reg} . The integer row sums of A_{reg} are $(5, 7, 7, 7, 7, 7)$; this gives the local expression

$$\eta^\vee(b) = 2b_0 + b_1 + b_2 + b_3 + b_4 + b_5 \pmod{3}. \quad (7.9)$$

The reading phase at instant r is $\theta_r = (r - 1) \bmod 3$. For each channel, define

$$\varepsilon_\chi(r) = \begin{cases} 1, & \eta^\vee(B_\chi(r)) = \theta_r, \\ 2, & \eta^\vee(B_\chi(r)) \neq \theta_r. \end{cases}$$

The phase-charge combination is

$$D(r) = \varepsilon_\pi(r)B_\pi(r) + \varepsilon_e(r)B_e(r) + \varepsilon_\varphi(r)B_\varphi(r) \quad \text{in } \mathbb{F}_3^6. \quad (7.10)$$

The coefficient is determined by an equality of charges before evaluating the word that will result from the combination.

7.3.3 Composition of the descriptor

Let $\parallel$ denote the concatenation of fixed-width words. The complete construction is

$$\Omega_{24} = (B_e(t_* + 1) + \delta_-) \parallel (\delta_-|_{3,4,5} \parallel \delta_+|_{3,4,5}) \parallel (B_\pi(r + 1) + B_e(r + 1)) \parallel D(r). \quad (7.11)$$

Each length-six summand is evaluated in $\mathbb{F}_3^6$; the final concatenation has length $6 + 6 + 6 + 6 = 24$. This construction distinguishes the descriptor Ω_{24} from the regional stages w_{24}^χ : the two objects share a length, but their producers and their places in the composition differ.

Proposition 7.4 (Regional evaluation of the descriptor). *The regional representations and the first-zero-advance criterion determine*

$$\Omega_{24} = 020022\ 222111\ 211201\ 021101, \quad [\Omega_{24}]_3 = 66241910521.$$

Proof. The required bands, preserving the order (π, e, φ) , are

t	$B_\pi(t)$	$B_e(t)$	$B_\varphi(t)$
11	022212	110001	100021
12	012012	202222	000120
21	221022	020100	211021

and come from extraction (7.2). At time twenty-one, $020100 + 000222 = 020022$ in $\mathbb{F}_3^6$. The defect tails provide 222111. At time twelve, $012012 + 202222 = 211201$.

At time eleven, the dual charges are 0, 1, 2, respectively; the phase is $(11 - 1) \bmod 3 = 1$. The resulting coefficients are (2, 1, 2) and

$$2(022212) + (110001) + 2(100021) = 021101 \quad \text{in } \mathbb{F}_3^6.$$

Concatenating these four results gives the stated word. Its integer evaluation is obtained by Horner's method in base three, or by three successive concatenations in base 729. $\square$

The preceding table serves a verification purpose: each of its entries is obtained from the regional producer. Construction rule (7.11) makes it possible to reproduce the descriptor without introducing the final twenty-four-trit string as an input.

7.4 Functional panel and terminal repertoire

7.4.1 Nine regional positions

The functional panel retains the first three bands of each of the three channels, in the declared regional order:

$$\mathcal{P}_9 = (b_\pi(0), b_\pi(1), b_\pi(2), b_e(0), b_e(1), b_e(2), b_\varphi(0), b_\varphi(1), b_\varphi(2)). \quad (7.12)$$

Its evaluation is

$$\mathcal{P}_9 = (103, 161, 103, 523, 457, 301, 450, 398, 438).$$

The nine positions retain their regional provenance, although the value 103 occurs twice. The panel has nine ordered incidences and eight distinct values. The subsequent enumeration uses the incidences; candidate deduplication takes place at a later stage.

7.4.2 Unordered eight-block repertoire

The terminal repertoire involved in this construction is

$$\mathcal{S}_{\text{term}} = \{002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110\} \subset \mathbb{F}_3^6. \quad (7.13)$$

Its eight elements are distinct. The integer encoding, sorted by value solely to fix a computational representation, is

$$\{28, 55, 98, 175, 437, 498, 687, 714\}.$$

The documentary provenance of this repertoire is the terminal segmentation of the seventy-two-trit word and its position-wise recovery through incidences of the phase-charge and affine readers. The selector developed below receives the repertoire without its historical ordering and tests its $8!$ orderings.

The exact formulation of the reassembled implementation maintains an explicit interface here: the regional producers materially replace the descriptor, the panel, and the temporal rows; repertoire (7.13) retains its earlier documentary producer. This distinction identifies where each input enters. Membership of eight words in a temporal reader repertoire, selection of those eight words, and determination of their terminal order are three different statements. The terminal-selection proof establishes the third for the declared repertoire.

Documentary recovery by position can be expressed exactly. Let $\mathcal{I}$ be the phase-charge incidence register and let $\mathcal{J}$ be the register of the affine extension. Each incidence retains the position ℓ in the seventy-two-trit word and the emitted word u . For $5 \leq \ell \leq 12$, form

$$\mathcal{S}_\ell = \{u : (\ell, u) \in \mathcal{I} \cup \mathcal{J}\}.$$

The recovery procedure checks that all eight positions occur and that each $\mathcal{S}_\ell$ is a singleton. If s_ℓ is its unique element, the output delivered to the selector is

$$\mathcal{S}_{\text{term}} = \{s_5, s_6, \dots, s_{12}\}.$$

The singleton proof for each position preserves the coherence of the historical witnesses; the subsequent export retains the eight words and removes only the historical ordering. The selector reconstructs the order through complete enumeration and the terminal conditions, instead of receiving it from the provenance register.

7.4.3 Temporal incidences of the terminal repertoire and recovery by position

The representation of the terminal repertoire uses the same regional blocks, the same carries, and the same calendar that produce the initial twenty-four-trit descriptor. Two operations in this composition should be made explicit separately. The first recovers eight words from recorded temporal incidences. The second presents those eight words to the dodecaphasic selector as an unordered set. The selector re-establishes the order through its incidence, cylinder, and closure conditions. The distinction preserves the provenance witnesses without turning the order of a historical segmentation into an input to the selector.

Let $B_t = (b_{t,0}, b_{t,1}, b_{t,2}) \in (\mathbb{F}_3^6)^3$ be the regional boundary already generated at time t , with $0 \leq t < 100$. In the chart used by the historical register, the incidence matrix is

$$A_{\text{hist}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}.$$

The words are regarded as row vectors. Their two charges and the chronological phase are

$$r_{t,i} = \sum_{j=1}^6 b_{t,i,j}, \quad d_{t,i} = \sum_{j=1}^6 (b_{t,i} A_{\text{hist}})_j, \quad \vartheta_t = t - 1 \pmod{3}.$$

All operations in these three expressions take place in $\mathbb{F}_3$. The historical subscript identifies a specific chart of the Witt incidence; it keeps the coordinate transformation explicit when another presentation of that same incidence is used.

For $c \in \{r_t, d_t\}$, $\delta \in \mathbb{F}_3$ and $\epsilon \in \{1, -1\}$, define

$$\eta_i(c, \delta, \epsilon) = \begin{cases} \epsilon, & c_i = \vartheta_t + \delta, \\ -\epsilon, & c_i \neq \vartheta_t + \delta, \end{cases} \quad L_{t,c,\delta,\epsilon} = \sum_{i=0}^2 \eta_i(c, \delta, \epsilon) b_{t,i}.$$

The orientation -1 is written as 2 when printing the ternary coefficients. Alongside these twelve comparative readers per boundary, the register has the affine reader

$$L_t^{\text{af}} = \sum_{i=0}^2 (d_{t,i} + \vartheta_t) b_{t,i}.$$

Adding the phase in the affine reader and comparing charges in the preceding reader are different operations, both acting on data already produced by the TPK boundary.

t	$b_{t,0}$	$b_{t,1}$	$b_{t,2}$	r_t	d_t
8	200121	212212	110110	011	202
11	022212	110001	100021	001	012
30	210112	110202	211110	100	012
34	121022	000011	112110	220	021
37	120121	011202	112221	100	201
49	110212	200212	100122	110	201
58	111121	022121	122201	122	220
67	010012	001220	011120	122	122
75	220000	020111	100002	120	021
81	122201	020202	201122	202	001
90	022112	022110	121010	202	200

Table 7: Temporal boundaries realizing the printed incidences of the terminal repertoire. Each six-digit entry preserves the order of the six coordinates of the regional boundary.

The following table displays all comparative incidences of the register on the twelve blocks of the segmentation under consideration. The letters r and d denote the visible charge and the dual charge, respectively. Column j retains the position index of the historical witness; the operation delivering the data to the selector forgets this index after forming the set of words.

The tenth block has an affine witness. At $t = 30$, $d_{30} = (0, 1, 2)$ and $\vartheta_{30} = 2$, whence

$$(d_{30,0} + \vartheta_{30}, d_{30,1} + \vartheta_{30}, d_{30,2} + \vartheta_{30}) = (2, 0, 1)$$

and, using the bands in Table 7,

$$L_{30}^{\text{af}} = 2(210112) + 0(110202) + (211110) = (001001) \quad \text{in } \mathbb{F}_3^6.$$

The equality holds coordinatewise: before reduction modulo three, the sum vector is $(6, 3, 1, 3, 3, 4)$.

Proposition 7.5 (Complete recovery of the repertoire through incidences). *Consider the comparative-incidence register in Table 8, augmented by the affine witness $(j, t, L) = (10, 30, 001001)$. For each position $5 \leq j \leq 12$, the set of output words of its incidences is a singleton. Extracting these eight words and subsequently forgetting their order produces exactly*

$$\mathcal{S}_{\text{term}} = \{002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110\}.$$

Their encoding as base-three integers is

$$\{28, 55, 98, 175, 437, 498, 687, 714\}.$$

j	t	charge	δ	ϵ	$(\eta_0\eta_1\eta_2)$	$L_{t,c,\delta,\epsilon}$
4	11	d	0	1	212	021101
5	67	d	1	-1	211	002001
5	67	d	2	1	211	002001
5	67	r	1	-1	211	002001
5	67	r	2	1	211	002001
6	81	d	0	1	222	020111
6	81	r	2	1	222	020111
7	34	r	1	-1	111	200110
7	75	d	1	-1	211	200110
7	75	r	2	-1	211	200110
8	90	r	0	1	121	121012
8	90	r	1	-1	121	121012
9	49	d	0	-1	121	010122
11	11	d	2	-1	211	221110
11	37	d	0	-1	121	221110
12	8	d	0	-1	111	222110
12	8	r	1	-1	111	222110
12	58	d	1	-1	111	222110
12	58	r	0	-1	111	222110

Table 8: Complete phase-charge incidences on the terminal segmentation within the hundred boundaries of the register. Word matches between rows retain their times, orientations, and charge types.

Proof. The incidence multiplicities at positions $j = 5, 6, 7, 8, 9, 10, 11, 12$ are, respectively,

$$(4, 2, 3, 2, 1, 1, 2, 4).$$

At each position, all rows print the same word. Each row is verified by substituting its three bands and three coefficients into the formula for L ; the tenth case has been calculated explicitly through L_{30}^{af} . For example, at $t = 67$ and with coefficients $(2, 1, 1)$, one obtains

$$2(010012) + (001220) + (011120) = (002001) \pmod{3}.$$

The eight results are distinct as words of length six. Their encoding $\sum_{k=1}^6 w_k 3^{6-k}$ gives, in position order, $(55, 175, 498, 437, 98, 28, 714, 687)$. Forgetting the order yields the stated set of integers. Thus, extraction preserves each complete word, including its zero positions, and delivers eight distinct elements to the selector. $\square$

The preceding proof is an exact recovery from identified incidences. Its input comprises the incidence register with its positions and witnesses. The subsequent orbital selection receives only $\mathcal{S}_{\text{term}}$ and the initial regional descriptor; it traverses the $8!$ orderings and determines their compatibility with the functional boundaries and the oriented incidences of the dodecaphasic wheel. Both operations are thus presented contiguously, with explicit domains: temporal provenance of the eight words and constructive determination of their order.

The comparative table contains the nineteen incidences that its formula produces on the twelve-block segmentation over the hundred declared times. This finite exhaustiveness is checked by traversing $100 \cdot 2 \cdot 3 \cdot 2 = 1200$ evaluations. The affine reader extends coverage of the terminal positions from seven to eight. This count characterizes the local reader repertoire just defined; the other TPK operations retain the domains and constructive functions established in their respective derivations.

7.5 Comparative readers, affine reader, and temporal incidence

7.5.1 Witt transformation of the selector

The selector uses the six-coordinate chart

$$A_{\text{ter}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}.$$

Define $W(b) = bA_{\text{ter}}$ in $\mathbb{F}_3^6$, together with the charges

$$\eta(b) = \sum_{j=0}^5 b_j \pmod{3}, \quad \eta_{\text{ter}}^{\vee}(b) = \eta(W(b)).$$

The descriptor chart and the selector chart are related by the permutation $\sigma = (0, 1, 2, 4, 5, 3)$, written as a list of images:

$$(A_{\text{ter}})_{ij} = (A_{\text{reg}})_{\sigma(i), \sigma(j)}. \quad (7.14)$$

Lemma 7.6 (Compatibility of the charge functional). *For every band b ,*

$$\eta_{\text{ter}}^{\vee}(b) = \eta^{\vee}(b).$$

Proof. The integer sum of each corresponding row of the two matrices agrees: it is five in the first row and seven in the other five. By distributivity,

$$\sum_j \sum_i b_i (A_{\text{ter}})_{ij} = \sum_i b_i \sum_j (A_{\text{ter}})_{ij} = \sum_i b_i \sum_j (A_{\text{reg}})_{ij}.$$

Reduction modulo three gives the equality of charges. Matrix identity (7.14) is verified by substituting its thirty-six entries and also preserves the coordinate change. $\square$

Equality of the total charge functional allows its representations to be compared. The vector transformations remain identified by their matrices and the explicit permutation; the preceding proof preserves that coordinate information.

7.5.2 Twelve comparative readers per instant

Define the temporal phase

$$\theta(t) = (t + 2) \pmod{3}.$$

This expression realizes $(t - 1) \pmod{3}$ also at $t = 0$, where the phase equals two. For each charge type $\nu \in \{\eta, \eta_{\text{ter}}^{\vee}\}$, offset $o \in \{0, 1, 2\}$ and orientation $s \in \{1, 2\}$, take

$$\epsilon_{\chi}^{\nu, o, s}(t) = s \begin{cases} 1, & \nu(B_{\chi}(t)) = \theta(t) + o \pmod{3}, \\ 2, & \nu(B_{\chi}(t)) \neq \theta(t) + o \pmod{3}. \end{cases}$$

The corresponding comparative word is

$$C_{\nu, o, s}(t) = \sum_{\chi \in \{\pi, e, \varphi\}} \epsilon_{\chi}^{\nu, o, s}(t) B_{\chi}(t) \quad \text{in } \mathbb{F}_3^6. \quad (7.15)$$

The parameters generate twelve labelled representations per row. Two labels may produce the same word; the temporal witnesses and labels remain available in the complete calendar.

7.5.3 Affine phase–charge reader

The affine reader uses coefficients

$$a_{\chi}^{\text{af}}(t) = \eta_{\text{ter}}^{\vee}(B_{\chi}(t)) + \theta(t) \pmod{3},$$

and produces

$$F(t) = \sum_{\chi \in \{\pi, e, \varphi\}} a_{\chi}^{\text{af}}(t) B_{\chi}(t) \text{ in } \mathbb{F}_3^6. \quad (7.16)$$

The comparative rule distinguishes agreement and difference of charges; the affine rule adds charge and phase before combining the bands. This operational difference is what heterotypic selection uses.

7.5.4 Word–residue incidence

Let R be the cyclic rotation of the six positions of a band. In the row at time t , form the residual repertoire

$$\mathcal{R}(t) = \{R^j v : v \in \{q(t), a(t), c(t), \bar{w}(t)\}, 0 \leq j < 6\}.$$

The temporal incidence relations are

$$\mathcal{A}_{\text{cmp}} = \{(C_{\nu, o, s}(t), u) : 0 \leq t < 100, u \in \mathcal{R}(t), \nu, o, s \text{ admissible}\}, \quad (7.17)$$

$$\mathcal{A}_{\text{af}} = \{(F(t), u) : 0 \leq t < 100, u \in \mathcal{R}(t)\}. \quad (7.18)$$

Each pair relates a read word to a residue of the same row. Time is quantified jointly across both coordinates of the pair. In the calendar used, times $0, \dots, 99$ are distinct; identifying the same row and identifying the same time are therefore equivalent operations.

Proposition 7.7 (Existential compression of the temporal repertoire). *The function*

$$M(v, u) = \mathbf{1}_{(v, u) \in \mathcal{A}_{\text{cmp}}} + 2\mathbf{1}_{(v, u) \in \mathcal{A}_{\text{af}}} \in \{0, 1, 2, 3\}$$

preserves exactly the existence of each reading type for the pair (v, u) .

Proof. The first bit equals one exactly when there exist a time, a residual field, a rotation, and a comparative label that produce the pair. The second bit expresses the corresponding condition for the affine reader. Union by bitwise disjunction preserves each existence condition as all rows are traversed. A pair may admit both types, in which case the mask equals three. $\square$

The function M is an existence reader. The witness times, their multiplicities, and the individual labels remain in (7.18) and in the calendar that produces them. The finite implementation uses M to decide heterotypy, whereas the genealogical archive retains the richer provenance data.

7.6 Enumeration of cylinders and decimal candidates

7.6.1 Concatenation of seventy-two and seventy-eight trits

For an ordering $\tau = (s_1, \dots, s_8)$ of $\mathcal{S}_{\text{term}}$, define

$$P_{72}(\tau) = [\Omega_{24} \| s_1 \| \cdots \| s_8 \|]_3.$$

Recursively, start from $p_0 = [\Omega_{24}]_3$ and compute $p_i = 729p_{i-1} + [s_i]_3$ for $1 \leq i \leq 8$. This recurrence preserves the leading zeros of each block through explicit multiplication by 729. For each position b of the functional panel,

$$P_8(\tau, b) = 729P_{72}(\tau) + b.$$

The lengths are $24 + 8 \cdot 6 = 72$ and $72 + 6 = 78$, respectively. The corresponding ternary cylinder is

$$I_{\tau, b} = \left[\frac{P_8(\tau, b)}{3^{78}}, \frac{P_8(\tau, b) + 1}{3^{78}} \right). \quad (7.19)$$

7.6.2 Exact intersection with the twelve-triad grid

The decimal reading uses twelve triads, that is, the grid $N/10^{36}$. Write $D = 10^{36}$ and $Q = 3^{78}$ for brevity.

Lemma 7.8 (Integer endpoints of the cylinder). *For an integer prefix p of length seventy-eight, the numerators of the grid points belonging to its cylinder are*

$$\left\{ N \in \mathbb{Z} : \left\lceil \frac{pD}{Q} \right\rceil \leq N < \left\lceil \frac{(p+1)D}{Q} \right\rceil \right\}. \quad (7.20)$$

Each cylinder contains at most one point of this grid.

Proof. The condition $N/D \in [p/Q, (p+1)/Q]$ is equivalent to $pD/Q \leq N < (p+1)D/Q$. For integer N , the first inequality is equivalent to $\lceil pD/Q \rceil \leq N$; the second is equivalent to $N < \lceil (p+1)D/Q \rceil$. The interval length in coordinate N is $D/Q < 1$, since $10^{36} < 3^{78}$. It can therefore contain at most one integer. $\square$

Ceilings are evaluated by integer division: $\lceil a/Q \rceil = \lfloor (a+Q-1)/Q \rfloor$ for $a \geq 0$. The entire enumeration uses exact integers; the choice of boundary points is fixed by the half-open right endpoint.

First retain the indexed list of incidences

$$\mathcal{C}_{\text{ind}} = \{(\tau, j, N) : N/D \in I_{\tau, \mathcal{P}_9(j)}\}, \quad 1 \leq j \leq 9,$$

and then obtain the set of numerators

$$\mathcal{C} = \{N : \exists \tau, j (\tau, j, N) \in \mathcal{C}_{\text{ind}}\}.$$

Passing to $\mathcal{C}$ removes repetitions of a numerator, retained in $\mathcal{C}_{\text{ind}}$ together with their provenance.

Lemma 7.9 (Recovery of the ordering from the cylinder point). *If $(\tau, j, N) \in \mathcal{C}_{\text{ind}}$, block number i of the seventy-eight-trit word is recovered by*

$$\left\lfloor \frac{N 729^i}{D} \right\rfloor \bmod 729, \quad 1 \leq i \leq 13.$$

In particular, blocks five through twelve recover τ .

Proof. Membership in the cylinder means $p \leq QN/D < p+1$, where $p = P_8(\tau, \mathcal{P}_9(j))$. Hence, $\lfloor QN/D \rfloor = p$. Extraction of the earlier blocks agrees with successive division of the prefix p by powers of 729: truncating a positional expansion before extracting a block preserves that block. The displayed formula performs exactly this extraction. The first four blocks form Ω_{24} and the next eight are the ordering τ . $\square$

Proposition 7.10 (Exact candidate count). *For Ω_{24} , $\mathcal{S}_{\text{term}}$ and $\mathcal{P}_9$ defined above,*

$$\# \text{Ord}(\mathcal{S}_{\text{term}}) = 40320, \quad \# \mathcal{C}_{\text{ind}} = 21902, \quad \# \mathcal{C} = 19446.$$

Proof. The eight blocks are distinct, so their orderings are the $8! = 40320$ permutations. For each permutation and each of the nine panel positions, compute $p = P_8(\tau, b)$, apply endpoints (7.20), and add the unique integer when the integer interval is nonempty. The sum of the cardinalities of these intervals is 21902. The union of their integers contains 19446 distinct elements. The exhaustive evaluation of these operations is certified by the theorems `indexed_cylinder_count` and `candidate_count` of the finite selector. The algorithm performs exactly the enumeration described: permutations, nine panel incidences, bounds by integer division, and duplicate removal. $\square$

The numbers 21902 and 19446 count different objects. The former counts indexed incidences containing a decimal point; the latter counts distinct numerators. The total number of cylinder queries is $40320 \cdot 9 = 362880$, including cylinders whose intersection with the grid is empty.

7.7 Exceptional incidence and heterotypy condition

7.7.1 Exceptional face produced by the panel

A band is lifted to twelve coordinates by

$$\Gamma(b) = (b, W(b)) \in \mathbb{F}_3^{12}.$$

The support of a word is the set of positions of its nonzero coordinates, with human-readable indices $1, \dots, 12$. The first regional bands of the panel produce

$$\mathcal{B} = \text{supp } \Gamma(B_\pi(0)) \cap \text{supp } \Gamma(B_e(0)) = \{4, 6, 9, 10\}. \quad (7.21)$$

The operation uses the regional words 103 and 523 and the selector's Witt matrix. The face is computed before querying the candidate coordinates.

For $N \in \mathcal{C}$, its twelve triads are

$$K_i(N) = \left\lfloor \frac{N}{1000^{12-i}} \right\rfloor \bmod 1000, \quad 1 \leq i \leq 12. \quad (7.22)$$

The crown support is

$$\mathcal{H}(N) = \{i : K_i(N) \geq 729\}, \quad \mathcal{C}_{\mathcal{B}} = \{N \in \mathcal{C} : \mathcal{H}(N) = \mathcal{B}\}.$$

The boundary $729 = 3^6$ relates the six-trit band to the decimal triad. Exhaustive evaluation of the equality of supports gives

$$\#\mathcal{C}_{\mathcal{B}} = 79. \quad (7.23)$$

The support preserves positions, while the complete register also preserves the twelve magnitudes. The next stage uses this positional information together with relations between magnitudes.

7.7.2 Ternary words and oriented differences of the candidate

The i th ternary band of the point N/D is obtained by

$$T_i(N) = \left\lfloor \frac{N 729^i}{D} \right\rfloor \bmod 729, \quad 1 \leq i \leq 13. \quad (7.24)$$

The oriented edge $a \rightarrow b$ has residue

$$R_{a,b}(N) = (K_a(N) - K_b(N)) \bmod 729. \quad (7.25)$$

Subtraction is performed in $\mathbb{Z}$ before reduction. For this edge, the pair

$$(\text{enc}_6^{-1}(T_b(N)), \text{enc}_6^{-1}(R_{a,b}(N))).$$

is queried in relations (7.18). It is therefore a joint query of the destination word and the oriented difference of the triads.

Write $\text{Cmp}(a, b; N)$ when the pair belongs to $\mathcal{A}_{\text{cmp}}$ and $\text{Af}(a, b; N)$ when it belongs to $\mathcal{A}_{\text{af}}$.

7.7.3 Determination of the transverse orbit

The twenty-four-trit descriptor determines the first three decimal triads before the terminal filtrations. Indeed, its cylinder satisfies the exact inclusion

$$\left[\frac{66241910521}{3^{24}}, \frac{66241910522}{3^{24}} \right) \subseteq \left[\frac{234543140}{10^9}, \frac{234543141}{10^9} \right).$$

The two endpoint inequalities are verified by integer cross-products. Thus, every candidate shares the three reference positions 1, 2, 3 and their decimal triads (234, 543, 140).

The step-four translation on $\mathbb{Z}/12\mathbb{Z}$ has the four orbits

$$(1, 5, 9), \quad (2, 6, 10), \quad (3, 7, 11), \quad (4, 8, 12).$$

The first three each contain one of the reference positions; the fourth is the only one disjoint from $\{1, 2, 3\}$. The oriented forest of steps three and four is fixed by the edges

$$\begin{aligned} 1 \rightarrow 4, & \quad 1 \rightarrow 5, & 2 \rightarrow 6, \\ 3 \rightarrow 7, & \quad 4 \rightarrow 8, & 5 \rightarrow 9, \\ 6 \rightarrow 10, & \quad 7 \rightarrow 11, & 8 \rightarrow 12. \end{aligned}$$

Its first edge has step three and the remaining edges have step four. The two internal edges of the distinguished orbit are precisely $4 \rightarrow 8$ and $8 \rightarrow 12$. The location of the reading condition is thus fixed by the geometry of the positions and by the initial prefix, before examining the candidates of the exceptional face.

7.7.4 Terminal heterotypy condition

The terminal condition is

$$\begin{aligned} \mathcal{J}(N) = & [\text{Cmp}(4, 8; N) \wedge \text{Af}(8, 12; N)] \\ & \vee [\text{Af}(4, 8; N) \wedge \text{Cmp}(8, 12; N)]. \end{aligned} \tag{7.26}$$

Both assignments of types to the two edges are admitted. The selection requires the existence of a realization with distinct types; the individual labels and their times remain available in the incidence witnesses.

Theorem 7.11 (Terminal selection of the coupling register). *On the candidate domain produced by Ω_{24} , $\mathcal{S}_{\text{term}}$, $\mathcal{P}_3$ and the hundred-row regional calendar, there exists a unique numerator simultaneously satisfying exceptional incidence and heterotypy:*

$$\{N \in \mathcal{C}_{\mathcal{B}} : \mathcal{J}(N)\} = \{234543140729659824621058914794146601\}.$$

Its twelve-triad register is

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \tag{7.27}$$

Proof. The preceding count produces the 79 elements of $\mathcal{C}_{\mathcal{B}}$. For each one, its twelve coordinates are extracted by (7.22); the destination bands T_8 , T_{12} and residues $R_{4,8}$, $R_{8,12}$ are computed; the two masks of the temporal repertoire are queried; and disjunction (7.26) is applied. The resulting list contains exactly the stated integer, according to the exhaustive evaluation certified by `terminal_selection`. Equality of the list with a singleton implies existence and uniqueness in the declared domain. Base-one-thousand extraction from that integer gives (7.27).

The selection procedure receives the enumerated producers and rules. The final integer appears in the statement of the verified equality after selection has been executed. This logical position allows the result to be checked without using it as a selection criterion. $\square$

7.7.5 Arithmetic witness of membership in a cylinder

One of the incidences of the selected numerator uses the ordering

$$(002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110)$$

and the boundary $438 = 121020_3$. For this incidence,

$$P_2 = 5283881584882673136514102926090957,$$

$$P_{78} = 3851949675379468716518781033120308091.$$

The integer-endpoint formula yields

$$\left\lceil \frac{P_{78} 10^{36}}{3^{78}} \right\rceil = 234543140729659824621058914794146601,$$

$$\left\lceil \frac{(P_{78} + 1) 10^{36}}{3^{78}} \right\rceil = 234543140729659824621058914794146602.$$

The unique integer in the interval is therefore N_K . The thirteen ternary bands of its decimal point are

020022	222111	211201	021101
002001	020111	200110	121012
010122	001001	221110	222110
121020			

This calculation exhibits a concrete witness of membership in the enumerated domain. Uniqueness of the register follows from the exhaustive examination of all candidates, in addition to this particular witness.

7.8 Regional composition and continuity of the realizations

7.8.1 Composition of the regional generators

Let $\text{Sel}(\Omega, \mathcal{S}, \mathcal{P}, \mathcal{N})$ be the selector defined by the enumeration and the two preceding filters. The regional producers satisfy the equalities

$$\Omega_{\text{reg}} = \Omega_{24}, \quad \mathcal{P}_{\text{reg}} = \mathcal{P}_9, \quad \mathcal{N}_{\text{reg}} = \mathcal{N}_{100},$$

where $\mathcal{N}_{\text{reg}}$ is computed by band extraction and signature reading. By congruence of functions,

$$\begin{aligned} \text{Sel}(\Omega_{\text{reg}}, \mathcal{S}_{\text{term}}, \mathcal{P}_{\text{reg}}, \mathcal{N}_{\text{reg}}) \\ = \text{Sel}(\Omega_{24}, \mathcal{S}_{\text{term}}, \mathcal{P}_9, \mathcal{N}_{100}) = \{N_K\}. \end{aligned} \tag{7.28}$$

The equality replaces three documentary interfaces with their actual regional construction. The terminal repertoire retains the provenance specified in (7.13).

The register is then formed by the twelve Euclidean divisions in (7.22). Each coordinate belongs to $\{0, \dots, 999\}$ and the length is twelve by construction. The implementation that takes the first element of the selected list is justified by equality with the singleton: the default value of a computational operation on empty lists never enters this result.

7.8.2 Relation to signed incidences and inversion

The present construction provides a prospective selection in an explicit finite domain. The subsequent development of the dodecaphasic register produces its oriented channels from the enriched terminal state and proves an integral inversion through Hadamard. That inversion recovers a register from compatible channels; here a register is selected from its regional and incidence-based inputs. Equality of their results allows the two realizations to be composed while preserving the origin of each operation.

Uniqueness of selection and uniqueness of inversion are distinct properties. The former quantifies over the candidates constructed in this chapter. The latter quantifies over the registers sharing a compatible signature. Their conjunction justifies transporting the selected register to the signed realization and recovering it from that realization, without turning a retrospective recovery into the producer of the register.

7.8.3 Relation between the terminal counts

The sequence

$$40320 \longrightarrow 21902 \longrightarrow 19446 \longrightarrow 79 \longrightarrow 1$$

records, successively, orderings, indexed incidences with the grid, distinct numerators, candidates of the exceptional face, and heterotypic survivors. The arrow notation indicates the order of the operations; the second number counts a different class of objects from the first.

The regional-catalogue count $196 \cdot 197 \cdot 196 \longrightarrow 331 \longrightarrow 2 \longrightarrow 1$, developed in its corresponding place, starts from another population: regional blocks from three catalogues and terminal conditions of Gram, occupancy, distance, and orientation. Both counts retain their domains. Their convergence to the same representation is expressed through the identification of their readers and outputs; the intermediate numbers describe different operations and remain documented separately.

7.8.4 Regional composition of the fine-structure constant

The output K subsequently enters the ordered regional composition and the base-one-thousand carry. Its later use preserves the twelve positions and the reading boundary. Coupling to the analytic root and representations at arbitrary depth uses that same output, together with the regional coordinates already generated and the rules of the analytic chart. This dependency prepares the chapter devoted to the two correlated realizations of the fine-structure constant.

The scope of this chapter is precise: production of the descriptor and its regional inputs, determination of the finite domain, unique selection on that domain, and material composition with the producers. The analytic recurrence at all orders and the compatibility of infinite carries are developed subsequently with their own proofs.

7.9 Executable certification and traceability of the operations

The formal development preserves the correspondence between the mathematical definitions and their executable realizations. The signature and clock proofs are found in `N69RegionalSignature.lean`; band extraction and reproduction of the hundred rows, in `GeneratedN69Rows.lean`; the minimality criterion and construction of Ω_{24} , in `RegionalW24.lean`. Compatibility of the two Witt matrices is assembled in `WittReaderCompatibility.lean`.

The temporal readers, rotations, oriented residues, and heterotypic condition are formalized in `TerminalReaders.lean`. Enumeration and counts are formalized in `TerminalOrbitalSelection.lean`. The regional panel is linked through `RegionalPanelBridge.lean`, and composition (7.28) is expressed in `SelectedFromRegionalInputs.lean`.

The finite enumeration checks use Lean’s native evaluation; their reports record `Lean.ofReduceBool`, in addition to the library’s customary logical axioms. The regional reconstruction of the descriptor uses kernel proofs and ordinary reduction. This distinction describes the verification mechanisms actually employed. The compilation receipts attest to their declared modules and dependencies; the mathematical proofs retain the domains, producers, and hypotheses made explicit in the body of the chapter.

7.10 Complete census of cylinders and selection of the terminal boundary

The terminal readout jointly uses two kinds of information: compatibility of the regional cylinders and an incidence profile with labelled channels and positions. This section constructs the complete catalogues, enumerates their triples, and proves the effect of each reader condition. The starting point consists of the regional outputs already produced in Section 5, not values introduced to define the APP-TRIT-TPK transport. Reduction to a matrix belongs to a subsequent readout of these outputs and preserves the order of the three channels.

The result is a census theorem with an explicit starting structure. The profile of Gram products, weights, distances, and marginals is declared as a datum of the terminal reader. The exhaustiveness of the enumeration proves what this profile selects on the specified catalogues; it does not, by itself, prove that the profile is a universal law or that it can be reconstructed from the unique triple that ultimately satisfies it. This distinction allows the complete finite proof to be incorporated without reversing its genealogy.

7.10.1 From regional outputs to ternary catalogues

Let $x_\pi, x_e, x_\varphi \in [0, 1)$ be the fractional parts of the three regional coordinates already constructed. We use their expressed cylinders of one thousand decimal places,

$$J_\chi^{(1000)} = \left[\frac{d_\chi}{10^{1000}}, \frac{d_\chi + 1}{10^{1000}} \right), \quad \chi \in \{\pi, e, \varphi\}.$$

The integers d_χ are finite output readouts. A length of one thousand is sufficient for the integers that follow, as verified from the two bounds of each interval; no infinite depth is assumed. We define

$$p = 30, \quad r = 1050, \quad L = p + r = 1080, \quad k = 171, \quad D = 3^L, \quad Q = 1000^k, \quad (7.29)$$

and

$$P_\chi = \lfloor 3^{30} x_\chi \rfloor, \quad T_\chi = \lfloor Q x_\chi \rfloor.$$

Here k counts decimal triads and does not denote the dodecaphasic register K . We have $1000^{171} \leq 3^{1080} < 1000^{172}$.

Lemma 7.12 (Stable extraction of an integer). *For $d, m, h \in \mathbb{Z}$, with $m, h > 0$, the value of $\lfloor mx \rfloor$ is constant on $[d/h, (d+1)/h)$ if and only if*

$$\left\lfloor \frac{md}{h} \right\rfloor = \left\lfloor \frac{m(d+1) - 1}{h} \right\rfloor. \quad (7.30)$$

When this holds, either side gives that value.

Proof. The minimum of mx is attained at the left endpoint. Its supremum is $m(d+1)/h$ and is not attained. The largest integer value that $\lfloor mx \rfloor$ can take is $\lceil m(d+1)/h \rceil - 1$, which equals the second side because the numerator and denominator are integers. Equality of the endpoints of the range of values proves the assertion. $\square$

The stability test holds in all three channels for $m = 3^{30}, Q, 3^{1080}$. The last choice will be used solely for a subsequent comparison, not to select the profile. The first thirty recovered positions are precisely the five six-trit stages of the regional transport:

Channel	Thirty-trit prefix	P_χ
π	01021101222201021100211110221	29152671743887
e	201101121221102011012222102011	147887858824447
φ	121200112202121020010210010200	127247717616687

The agreement preserves the prolongation already constructed; the decimal readout is not used to replace its transport matrices retrospectively.

For a fixed channel, a tail of r trits is represented by $0 \leq R < 3^r$. Its cylinder and the cylinder of the first k triads are

$$I_{\chi,R} = \left[\frac{P_\chi 3^r + R}{D}, \frac{P_\chi 3^r + R + 1}{D} \right), \quad J_{\chi,T} = \left[\frac{T_\chi}{Q}, \frac{T_\chi + 1}{Q} \right). \quad (7.31)$$

The catalogue includes all tails for which these two cylinders have a nonempty intersection.

Proposition 7.13 (Catalogue by exact intersection). *The compatible tails are the integers in the interval $[a_\chi, b_\chi]$, where*

$$a_\chi = \max\left(0, \left\lfloor \frac{T_\chi D}{Q} \right\rfloor - P_\chi 3^r\right), \quad (7.32)$$

$$b_\chi = \min\left(3^r - 1, \left\lfloor \frac{(T_\chi + 1)D - 1}{Q} \right\rfloor - P_\chi 3^r\right). \quad (7.33)$$

For the specified regional cylinders, no clipping applies and we obtain

χ	$a_\chi \bmod 729$	$b_\chi \bmod 729$	$b_\chi - a_\chi + 1$
π	67	262	196
e	584	51	197
φ	117	312	196

(7.34)

Proof. Write $N = P_\chi 3^r + R$. The two half-open intervals in (7.31) intersect exactly when

$$NQ < (T_\chi + 1)D, \quad (N + 1)Q > T_\chi D.$$

The second inequality is equivalent to $N \geq \lfloor T_\chi D/Q \rfloor$; the first, to $N \leq \lfloor ((T_\chi + 1)D - 1)/Q \rfloor$. Subtracting $P_\chi 3^r$ and retaining $0 \leq R < 3^r$ proves the formulae. Evaluation by integer division of the T_χ extracted using (7.30) gives the table. Since $r \geq 6$, $P_\chi 3^r \equiv 0 \pmod{729}$, so the residue of the first endpoint can also be obtained directly from $\lfloor T_\chi D/Q \rfloor \bmod 729$. $\square$

The last six trits of a tail form the word $\nu_6(R \bmod 729)$, where ν_6 denotes the ternary representation of length six, including its leading zeros. By the table, each catalogue contains fewer than 729 consecutive elements, so this reduction is injective on it. Thus, the three boundary catalogues are completely described, without choosing the final triples:

$$\begin{aligned} \mathcal{B}_\pi &= \{\nu_6(67 + i) : 0 \leq i < 196\}, \\ \mathcal{B}_e &= \{\nu_6((584 + j) \bmod 729) : 0 \leq j < 197\}, \\ \mathcal{B}_\varphi &= \{\nu_6(117 + \ell) : 0 \leq \ell < 196\}. \end{aligned} \quad (7.35)$$

Requiring the left endpoint of $I_{\chi,R}$ to belong to $J_{\chi,T}$ would impose a different condition: it would lose the first cylinder that intersects the boundary and would give 195, 196, 195. The census uses the intersection of the complete cylinders, in accordance with its definition.

7.10.2 Declared profile and compatibility indicators

For $u, v \in \mathbb{F}_3^6$, let $g(u, v) = \sum u_i v_i \in \mathbb{F}_3$, $n(u) = g(u, u)$, $w(u) = \#\{i : u_i \neq 0\}$, and $h(u, v) = \#\{i : u_i \neq v_i\}$. The last two functions take integer values; the first two take values in $\mathbb{F}_3$. The reader's local profile, in the order (π, e, φ) , is

$$\begin{aligned} (g_{\pi e}, g_{\pi \varphi}, g_{e \varphi}) &= (2, 2, 0), & (n_\pi, n_e, n_\varphi) &= (0, 0, 0), \\ (w_\pi, w_e, w_\varphi) &= (3, 3, 3), & (h_{\pi e}, h_{\pi \varphi}, h_{e \varphi}) &= (2, 5, 3). \end{aligned} \quad (7.36)$$

The words are ordered as in (7.35). We use $[E] \in \{0, 1\}$ for the indicator of a condition E . For each pair of channels χ, ψ , the rectangular matrix $A_{\chi\psi}$ has entry $[g(u, v) = g_{\chi\psi}]$. The matrix $A_{\chi\psi}^h$ multiplies that entry by $[h(u, v) = h_{\chi\psi}]$. Let D_χ^n be the diagonal matrix of $[n(u) = 0]$, and D_χ^{nw} that of $[n(u) = 0][w(u) = 3]$.

Theorem 7.14 (Exhaustive enumeration by finite sums). *The product of the three catalogues contains 7 567 952 triples. The successive conditions in (7.36) leave*

$$7\,567\,952 \longrightarrow 275\,794 \longrightarrow 6235 \longrightarrow 3127 \longrightarrow 331. \quad (7.37)$$

These four counts are, respectively,

$$\begin{aligned} N_g &= \text{tr}(A_{\pi e} A_{e\varphi} A_{\varphi\pi}), \\ N_{gn} &= \text{tr}(D_{\pi}^n A_{\pi e} D_e^n A_{e\varphi} D_{\varphi}^n A_{\varphi\pi}), \\ N_{gnw} &= \text{tr}(D_{\pi}^{nw} A_{\pi e} D_e^{nw} A_{e\varphi} D_{\varphi}^{nw} A_{\varphi\pi}), \\ N_{gnwh} &= \text{tr}(D_{\pi}^{nw} A_{\pi e}^h D_e^{nw} A_{e\varphi}^h D_{\varphi}^{nw} A_{\varphi\pi}^h). \end{aligned} \quad (7.38)$$

All products and traces in this formula are calculated over the integers, not over the field with three elements.

Proof. The size of the product is $196 \cdot 197 \cdot 196$. When a trace in (7.38) is expanded, its indices run through exactly one triple of words. The summand equals one if it satisfies all the represented conditions and zero otherwise. Each triple is counted once. The diagonal matrices add the individual conditions; the rectangular factors add the pairwise conditions. This proves that the traces are precisely the stated cardinalities.

To make their evaluation explicit, fix the indices i, j of the first two channels and form the sets of indices ℓ of the third that satisfy each pairwise condition. Their indicators are intersected, the relevant individual indicators are added, and the cardinality is summed. The final sum runs over $0 \leq i < 196, 0 \leq j < 197$. Substitution of the words in (7.35) and the profile (7.36) gives, in that order, 275794, 6235, 3127, 331. The supplement records the four contributions of each of the 196 indices i and the complete list of 331 triples; its program evaluates these sums with exact binary indicators and checks each survivor by scalar evaluation. No sample or tolerance is involved. $\square$

7.10.3 Labelled incidence and orientation charge

For a surviving triple, let $B \in \mathbb{F}_3^{3 \times 6}$ be its matrix of ordered rows. The second profile requires

$$\text{rank}_{\mathbb{F}_3} B = 3, \quad w_{\text{col}}(B) = (1, 1, 2, 2, 3, 0), \quad q_{\partial}(B) = (1, 2, 2, 1, 2, 0), \quad (7.39)$$

where w_{col} counts the nonzero elements in each column and q_{∂} sums its entries modulo three. This is a profile of labelled columns; permuting them without transporting the profile defines a different reader. Denote by $\mathcal{F}$ the set of 331 triples from the preceding theorem.

Proposition 7.15 (Reduction of the census to the terminal pair). *The sums of indicators over $\mathcal{F}$, successively imposing rank, column weights, and column sums, are*

$$\sum_{B \in \mathcal{F}} [\text{rank } B = 3] = 331, \quad N_{\text{rango,pesos}} = 18, \quad N_{\text{rango,pesos,sumas}} = 2. \quad (7.40)$$

With catalogue indices starting at one, the two final triples are (120, 197, 157) and (129, 197, 148), and their matrices are

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

Proof. For each triple in the already enumerated set, the rank is computed by elimination over $\mathbb{F}_3$, the nonzero elements of the six columns are counted, and their entries are summed modulo three. The products of the indicators in (7.39) give (7.40). Substituting the two final indices into (7.35) gives the displayed matrices. The complete list of survivors and the exact filters in the supplement allow the elimination to be reproduced without presupposing these two matrices as the domain. $\square$

The orientation $\sigma = (1, 1, -1) = (1, 1, 2) \in \mathbb{F}_3^3$ selects the line $\langle \sigma \rangle = \{(0, 0, 0), (1, 1, 2), (2, 2, 1)\}$. The row charges of B_0, B_1 are $(0, 2, 0)$ and $(2, 2, 1)$; only the latter belongs to this line. This recovers the last step of the selector in Section 7.15, now preceded by the construction and enumeration of its complete domain. The stable readout of $\lfloor 3^{1080} x_\chi \rfloor$, performed after the filters, gives the boundaries 021020, 001220, 100210, with the same ranks $(129, 197, 148)$. This subsequent comparison does not define the profile or enter the sums of indicators.

7.10.4 Preserved information and scope of the result

The procedure preserves the thirty-trit prefix, the compatible tail, the regional order, the position of each trit, and the orientation. Reduction modulo 729 preserves the identity of each candidate here because the interval of tails has length less than 729; this injectivity is not asserted for an arbitrary history. The census of 331 also proves that the local profile alone does not select a single matrix: the labelled marginals and the charge supply additional effective conditions.

Thus, a complete finite enumeration and its selector have been recovered on output cylinders, with an explicit readout profile. Construction of the signed register additionally uses its memory reader and its channels, as specified below; the visible matrix is not identified with that register by equality of the number of components. Nor does this count prove the universality of the profile, generate the regional coordinates anew, or decide a claim about alpha or about the complete continuum.

8 Dodecaphasic record and rational phase constant

The generation of regional coordinates does not exhaust the information in the APP-TRIT-TPK state. One Archimedean reading determines a value through compatible cylinders; another reading preserves the opposition of the sheets, the orientation of the route, and the record of the incidences traversed. The dodecaphasic record belongs to this second reading before entering the closure of alpha. It is therefore not defined by subtracting a physical constant from a combination of previously known values. It is reconstructed from a signed coordinate of the same arithmetic state with trajectory memory that produces the regional coordinates.

The distinction is necessary even when two objects have twelve coordinates. A twelve-block word from the catalog, a list of twelve oriented events, a signed integer vector, and a base-thousand dodecaphasic record are different objects. Comparing them requires the maps that relate them. Equality of length does not provide those maps, and the loss of signs is not a mere simplification of notation.

This chapter maintains the causal cut fixed in the preceding chapters: the nine APP positions on the horizontal and vertical axes produce the arithmetic sheets; TRIT types regime and orientation; TPK selects, transports, updates, and prolongs, preserving remainder, quotient, carry, sheet, route, boundary, and memory. The coordinates used here are obtained from the joint discrete structure of the continuum. The chain $w_6 \rightarrow w_{12} \rightarrow w_{18} \rightarrow w_{24} \rightarrow w_{30} \rightarrow R_{36} \rightarrow G_9$ is not replaced by a finite calendar. Nor is the order $U_t \rightarrow \Gamma_9 \rightarrow K_{\text{ph}}$ reversed: the last map is a reduced-memory reading, not the generator of the signed record.

Reversible coordinatization and incidence determination. The result organizing this chapter is the equivalence between three coordinatizations of a record previously produced by APP-TRIT-TPK. The Hadamard transformation relates the signed record to the integer record; the difference extractor relates the latter to its transverse variations and its uniform component:

$$U \xrightleftharpoons[\mathcal{H}_{12}]{\mathcal{H}_{12}/4} K \xrightleftharpoons{(D_3, D_4, Q)} (D_3 K, D_4 K, Q(K)).$$

The first equivalence is established on the integral sublattice $\mathcal{L}_H$; the second, on the image of the extractor. With base 1000, the 12 positions, and the phase origin fixed, the evaluation $K \mapsto \kappa_{\text{per}}$ adds a fourth reversible coordinatization: multiplication by $1000^{12} - 1$ followed by Euclidean division recovers the complete record. The proofs in §8.7, §8.8, and §8.10 establish these equivalences in their respective domains.

Two compositions subsequently act on this record. The positional composition combines its components with the closure, propagation, and self-scaling coordinates, and its normalization determines α_A . The incidence composition applies the cubic threshold and yields

$$K \mapsto B_K = \{j : K_j \geq 729\} = \{4, 6, 9, 10\}.$$

The complete record allows both compositions to be reconstructed. The support B_K specifically preserves membership in the class selected by the threshold. The identity $B_K = H_e \cap P_\pi$ in (28.3) links this integer extraction to the incidence of the regional codes. The negative support of their balance with K completes the flag $B_K \subset H_\alpha^-$ developed in the exceptional extension. Thus, the function of K is defined through an invertible transformation and precise subsequent maps; the additional information on depth and transport remains in the state from which the record originates.

8.1 Cyclic extension and memory grading

The observable calendar contains one hundred eight positions, organized into twelve nonadic windows. With positive indices, let

$$E_m = \{9(m-1) + 1, \dots, 9m\}, \quad 1 \leq m \leq 12.$$

The ordered support $E_1 \sqcup \cdots \sqcup E_{12}$ preserves position and window membership. The group $C_{108} = \mathbb{Z}/108\mathbb{Z}$, by contrast, preserves the law of cyclic translation. For the representative $0 \leq \tilde{\theta} < 108$, the coordinates

$$\beta(\theta) = \left[\left[\frac{\tilde{\theta}}{9} \right] \right]_{12}, \quad r(\theta) = 1 + (\tilde{\theta} \bmod 9)$$

separate the dodecaphasic sector and the interior nonadic position. Neither by itself preserves the total number of turns.

Proposition 8.1 (Calendar carry). *The maps*

$$\iota : C_{12} \longrightarrow C_{108}, \quad [b] \longmapsto [9b], \quad p : C_{108} \longrightarrow C_9, \quad [\theta] \longmapsto [\theta]_9$$

form a nonsplit exact sequence

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0.$$

For the set-theoretic section choosing the representatives $0, \dots, 8$, the defect of additivity is the carry

$$c(r, r') = \left[\left[\frac{\tilde{r} + \tilde{r}'}{9} \right] \right]_{12}.$$

Proof. The kernel of p consists of the multiples of nine and equals the image of ι . Reduction is surjective, and multiplication by nine is injective on the indicated domain. If the sequence split, the middle group would be isomorphic to $C_{12} \times C_9$, whose exponent is $\text{lcm}(12, 9) = 36$; the exponent of C_{108} is 108. This contradiction proves nonsplitting. Finally, Euclidean division of $\tilde{r} + \tilde{r}'$ by nine gives the displayed carry and verifies the section identity. $\square$

The result explains a difference that will recur in alpha: returning to the phase is not equivalent to returning to the state. If q_{mem} counts nonadic turns, the calendar retains only $\beta = [q_{\text{mem}}]_{12}$. After one hundred eight steps, β returns, but q_{mem} increases by twelve. The record also preserves the cylinder, the incidence ledger, and vertical memory. Closing a finite chart does not turn that history into a memoryless cycle.

8.2 The oriented record and the integral $1 + 5 + 6$ chart

For a history produced by TPK, let τ_m be the subroute over E_m , σ_m its orientation, κ_m the boundary carry, and Λ_m the local incidence ledger. The record is

$$\mathcal{R}_{12}(x) = ((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m))_{m=1}^{12}.$$

Its domain is the space of arithmetic histories with trajectory memory, not the set of phase labels. Projection onto the calendar forgets precisely some of the data required by the signed reader.

An initial chart of the record can be written through oriented events. The digital reader fixes the set of event codes $\{2, 4, 6, 8\}$ and the sector orientation

$$\Sigma_{12} = (+, +, +, -, -, -, +, +, +, -, -, -).$$

If d_t is the symbol of the digital record at position t , one obtains

$$e_i = \Sigma_{12}(i+1) \# \{t \in E_{i+1} : d_t \in \{2, 4, 6, 8\}\}, \quad 0 \leq i < 12.$$

The formula keeps the nonnegative event count separate from orientation. The digital symbol, cardinal direction, and TRIT sign retain their respective domains. The vector e , whose components are signed local counts, is not yet the vector U appearing below either.

Opposite positions determine, for $0 \leq i < 6$,

$$p_i = e_i + e_{i+6}, \quad a_i = e_i - e_{i+6}.$$

With

$$s_C = \sum_{i=0}^5 p_i, \quad M_i = p_i - p_5 \quad (0 \leq i < 5),$$

the integral chart is defined by

$$\mathcal{L}_{12}(e) = (s_C, M_0, \dots, M_4, a_0, \dots, a_5).$$

The 1+5+6 decomposition is not an arbitrary allocation of coordinates: one coordinate preserves charge, five compare pairs relative to a reference pair, and six preserve the opposition of the semicoronas.

Lemma 8.2 (Inversion of the integral chart). *An integer tuple $(s_C, M_0, \dots, M_4, a_0, \dots, a_5)$ belongs to the image of $\mathcal{L}_{12}$ if and only if*

$$s_C - \sum_{i=0}^4 M_i \equiv 0 \pmod{6}, \quad p_i \equiv a_i \pmod{2} \quad (0 \leq i < 6),$$

where

$$p_5 = \frac{s_C - \sum_{i=0}^4 M_i}{6}, \quad p_i = p_5 + M_i \quad (i < 5).$$

In that case its preimage is unique and is given by

$$e_i = \frac{p_i + a_i}{2}, \quad e_{i+6} = \frac{p_i - a_i}{2}.$$

Proof. The sum of the five margins is $s_C - 6p_5$, yielding division by six. Once the p_i are reconstructed, each pair of equations $p_i = e_i + e_{i+6}$, $a_i = e_i - e_{i+6}$ has the indicated solution. This solution is integral exactly under the parity congruence. All operations are inverted, so no residual freedom remains. $\square$

This reversibility shows what information is preserved; it does not authorize removing the rest of the ledger. The full record has a larger domain than this count chart. Extracting the channels feeding U uses that full record and does not follow from the list (e_i) through an identification of names.

8.3 Incidence evaluation preceding the dodecaphasic record

Recovering a record from its cyclic differences and total charge is distinct from evaluating those coordinates on the incidence ledger. The first problem is solved through the transverse extractor of Proposition 8.8. For the second, a historical presentation of the record provides three integer counts in each window. It is useful to make their composition explicit and determine exactly the information their evaluation requires.

Let $\mathcal{L}$ be a previously generated ledger of visits and oriented traces. Each visit preserves the route, APP position, sheet, instant, multiplicity, and boundary incidence. Arithmetic evaluation produces the integer and its decomposition

$$n_t = r_t + 9q_t, \quad 1 \leq r_t \leq 9.$$

The integer n_t is computed from the sum or product sheet; its remainder is not received as an independent value. The distinction between corona incidence and interior incidence remains explicit for visits with remainder 9.

In window E_m , the count presentation considers

$$A_m = \sum_{v \in E_m} w(v) \mathbf{1}_{\{1,3,5,7\}}(r(v)), \quad (8.1)$$

$$C_m = \sum_{j \geq 0} (\nu_{120,m}^-(j) - \nu_{120,m}^+(j)), \quad (8.2)$$

$$V_m = \sum_{v \in E_m} w(v) \mathbf{1}_{\{9\}}(r(v)) \epsilon_{\partial}(v). \quad (8.3)$$

Here $w(v)$ is the incidence multiplicity, and $\epsilon_\partial(v) = 1$ on the corona, -1 in the interior. The traces counted by $v^\pm$ must be enumerated through the corresponding ledger rule. The integer sum requires finite support or an alternative construction determining its meaning. The number of chronological incidences and the reduced length $120(1 + 3j)$ have different types; their relation requires the reader's length unit.

Define $[x]_{10} \in \{0, \dots, 9\}$ as the Euclidean representative and retain the quotients

$$x = [x]_{10} + 10\lfloor x/10 \rfloor.$$

The incidence block is

$$z_m = 100[A_m]_{10} + 10[C_m]_{10} + [V_m]_{10}. \quad (8.4)$$

Composition yields the map

$$\mathcal{E}_{\text{cnt}}(\mathcal{L}) = \left((z_m - z_{m+3})_{m=1}^{12}, (z_m - z_{m+4})_{m=1}^{12}, \sum_{m=1}^{12} z_m \right). \quad (8.5)$$

Indices are read cyclically modulo 12. In this subsection, $(D_j z)_m = z_m - z_{m+j}$, for $j = 3, 4$, and $Q(z) = \sum_{m=1}^{12} z_m$; thus the preceding composition is $(D_3 z, D_4 z, Q(z))$. This convention is retained in §8.8, where complete integral reconstruction is proved. This formula produces its coordinates from the counts. It does not use K , U , α , or the expected transverse coordinates as arguments. Its identification with $\mathcal{E}_{108}^{90,120}$ requires composition with the effective collection and incidences of the terminal state; it does not follow solely from invertibility of the transverse system.

Proposition 8.3 (Sufficient and necessary arithmetic information). *Let $N, N' \in \mathbb{Z}^{12 \times 3}$, ordered by the counts (A, C, V) . For the map defined by (8.4)–(8.5),*

$$\mathcal{E}_{\text{cnt}}(N) = \mathcal{E}_{\text{cnt}}(N') \iff N - N' \in 10\mathbb{Z}^{36}.$$

In particular, the 36 sector residues determine exactly the 25-coordinate output. Preservation of quotients and provenance constitutes additional information in the record.

Proof. Equality modulo 10 produces the same blocks and therefore the same transverse coordinates. Conversely, if the transverse coordinates agree, $D_3(z - z') = D_4(z - z') = 0$. Steps 3 and 4 generate the cycle of length 12, so $z - z'$ is constant. Equality of total charges annihilates that constant. Finally, the decimal representation of an integer between 0 and 999 has three unique digits. Thus the three residues of every window are recovered. The statement is an algebraic identity for all N, N' , not an inference obtained from finite tests. $\square$

8.4 Conjugation of counts and boundary terms

Let $\rho(m) = 13 - m$. Consider an incidence conjugation that reverses the window order, preserves the arithmetic and boundary classes of visits, and exchanges the trace channels. Then

$$(A_m^\dagger, C_m^\dagger, V_m^\dagger) = (A_{\rho(m)}, -C_{\rho(m)}, V_{\rho(m)}).$$

If $c_m = [C_m]_{10}$, decimal reduction produces the identity

$$z_m^\dagger = z_{\rho(m)} + 100\mathbf{1}_{\{c_{\rho(m)} \neq 0\}} - 20c_{\rho(m)}. \quad (8.6)$$

Indeed, $[-C]_{10} = 0$ when $[C]_{10} = 0$, and $[-C]_{10} = 10 - [C]_{10}$ otherwise. The formula preserves exactly this difference. Negating the channel is not equivalent to negating the entire decimal block.

Applying this identity to a concrete TPK involution requires checking its action on incidences. For an occupation reader evaluated before advancement, reversing a route $\gamma = (x_0, \dots, x_n)$ satisfies

$$\sum_{k=0}^{n-1} f(x_{n-k}) = \sum_{k=0}^{n-1} f(x_k) + f(x_n) - f(x_0).$$

Endpoint terms are incorporated before reduction modulo 10. On closed routes they vanish; in open windows they can alter the residues. This correction, already present in the bidirectional transport development, specifies what the connection between route and counts must preserve.

Proposition 8.4 (Necessity of the record's nonperiodic information). *Suppose the observable of a fixed route configuration satisfies $o(t + 54) = o(t)$, multiplicity is preserved under that return, and the same local reader acts in windows of 9 instants. Then its counts and blocks satisfy*

$$(A, C, V)_{m+6} = (A, C, V)_m, \quad z_{m+6} = z_m.$$

Proof. Translation $t \mapsto t + 54$ bijectively sends E_m to E_{m+6} , preserving each counted incidence and its multiplicity. Equality passes to the sum and decimal reduction. $\square$

Thus a presentation with 12 distinct blocks requires the reader to preserve some information that does not factor through that period-54 observable: incidence selection, effective orientation, endpoints, multiplicity, or memory. The proposition delimits an inadmissible state reduction; it does not assert that the full dynamics has period 54 or itself determine its terminal reader.

8.5 Two readings of the terminal state

Denote by x_{term} the state produced by the composition of selection, transport, and update up to the terminal cut. The operator expression is

$$x_{\text{term}} = (\mathcal{U}_{108} \circ \cdots \circ \mathcal{U}_1)(x_{\text{seed}}), \quad \mathcal{U}_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t.$$

Its two relevant readings are

$$x_{\text{term}} \xrightarrow{(\pi_{\text{term}}, R_{\text{sgn}})} (B_{\text{term}}, U).$$

The matrix B_{term} retains the visible ternary reading; the vector U preserves another coordinate of the same history. No arrow $B_{\text{term}} \rightarrow U$ is inserted between them.

The terminal selector of the reference construction [development and scope in Sections 8 and 9.4] operates on catalogs of sizes 196, 197, and 196. The local signatures reduce their $196 \cdot 197 \cdot 196 = 7\,567\,952$ triples to 331, and the column margin $q_\partial = (1, 2, 2, 1, 2, 0)$ leaves two matrices:

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

The row orientation is $\sigma = (1, 1, -1) = (1, 1, 2)$ in $\mathbb{F}_3^3$. The row charges of the candidates are $(0, 2, 0)$ and $(2, 2, 1)$. Only the second belongs to $\langle \sigma \rangle = \{(0, 0, 0), (1, 1, 2), (2, 2, 1)\}$; hence $B_{\text{term}} = B_1$. This check proves the final step of the selector from the declared census. It does not make the column margin a consequence of the row charge, nor replace the preceding enumeration with inspection of these two matrices.

The signed reader has the typed composition

$$R_{\text{sgn}} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12} : \mathcal{X}_{\text{TPK}}^{\text{term,mem}} \longrightarrow \mathbb{Z}^{12}. \quad (8.7)$$

The channel record determines

$$\mathcal{E}_{108}^{90,120}(\mathcal{R}_{12}(x_{\text{term}})) = (b^{90}, b^{120}, Q_{\text{TPK}}),$$

with coordinates

$$\begin{aligned} b^{90} &= (-495, -116, -684, 108, 601, -90, \\ &\quad -173, -88, 313, 560, -397, 461), \\ b^{120} &= (-425, -281, -481, 671, -255, 30, \\ &\quad 475, -543, 680, 251, 6, -128), \\ Q_{\text{TPK}} &= 6263. \end{aligned}$$

These quantities are used as coordinates of the preceding record, not as parameters fitted through alpha. The reconstruction proved in the following subsections starts exactly at this typed output. The preceding evaluation is developed in §8.3: visits and traces determine the counts, which produce the blocks through (8.4) and the differences and charge through (8.5). Proposition 8.3 exactly characterizes the arithmetic information in that evaluation; subsection 8.4 preserves its boundary terms under conjugation. Identification of the effective incidence collection of this terminal state has the scope specified there. The technical provenance note preserves the preceding documentary trail; the harmonic reconstruction below uses the typed output and does not replace the evaluation preceding it.

8.6 From the channel record to the signed vector

The harmonic reader Π_H can be written without using either K or alpha. To avoid confusing its orbital sums with the alpha blocks, we denote them by s_1, s_2, s_3 . Let

$$B_r = \sum_{j=0}^3 b_{r+3j}^{120}, \quad d_{r,j} = b_{r+3j}^{90}, \quad 1 \leq r \leq 3,$$

and

$$s_1 = \frac{Q_{\text{TPK}} + 2B_1 + B_2}{3}, \quad s_2 = s_1 - B_1, \quad s_3 = s_2 - B_2. \quad (8.8)$$

Division by three belongs to the three-orbit reader. Its integrality is checked on compatible record outputs; it is not assumed for an arbitrary element of $\mathbb{Z}^{25}$.

The three signed blocks are

$$U^{(r)} = (s_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}). \quad (8.9)$$

Integer substitution gives

$$(B_1, B_2, B_3) = (972, -1073, 101), \quad (s_1, s_2, s_3) = (2378, 1406, 2479),$$

and therefore

$$\begin{aligned} U^{(1)} &= (2378, -452, -668, -322), \\ U^{(2)} &= (1406, 998, -204, -28), \\ U^{(3)} &= (2479, -551, -371, -997). \end{aligned}$$

Interleaving by columns yields

$$U = (2378, 1406, 2479, -452, 998, -551, -668, -204, -371, -322, -28, -997). \quad (8.10)$$

The presence of negative values and components greater than 999 shows that U is not a base-thousand digit word. Nor is it $U_6 \parallel U_6$, a catalog concatenation of period six. The difference concerns domain and preserved information, not just values.

The constructive order so far is

$$x_{\text{term}} \longrightarrow \mathcal{R}_{12} \longrightarrow (b^{90}, b^{120}, Q_{\text{TPK}}) \longrightarrow U.$$

The inversion below produces the dodecaphasic record. The subsequent ability to recover the channels from the dodecaphasic record is a reversibility check; it does not reverse this causal order.

8.7 Hadamard transformation by orbits and integer reconstruction

On the twelve-position cycle, step three determines three orbits:

$$(1, 4, 7, 10), \quad (2, 5, 8, 11), \quad (3, 6, 9, 12).$$

The separation into four sectors of each orbit is what, once origin and orientation are fixed, admits the angular reading of ninety degrees. The transformation does not act on three consecutive four-element sets in the natural order.

Let

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

If P_{orb} reorders according to the three indicated orbits, define

$$\mathcal{H}_{12} = P_{\text{orb}}^{-1}(H_4 \oplus H_4 \oplus H_4)P_{\text{orb}}.$$

Lemma 8.5 (Hadamard inversion and integrality). *We have*

$$H_4^2 = 4I_4, \quad H_4^{-1} = \frac{1}{4}H_4, \quad \det H_4 = -16.$$

For $u \in \mathbb{Z}^4$, the preimage $H_4^{-1}u$ is integral if and only if $H_4u \equiv 0 \pmod{4}$.

Proof. The rows are orthogonal and have squared norm four; the matrix is symmetric. This gives $H_4^2 = 4I_4$ and the inverse. Integer elimination, or expansion of the determinant, gives the negative sign and value -16 . The integrality condition is exactly divisibility by four of all four coordinates of H_4u . $\square$

Definition 8.6 (Integral dodecaphasic transformation). The sublattice of signed records and its integral coordinatization are

$$\mathcal{L}_H = \{u \in \mathbb{Z}^{12} : \mathcal{H}_{12}u \equiv 0 \pmod{4}\}, \quad \mathcal{K}(u) = \frac{1}{4}\mathcal{H}_{12}u.$$

The map $\mathcal{K} : \mathcal{L}_H \rightarrow \mathbb{Z}^{12}$ is an isomorphism of abelian groups, with inverse $k \mapsto \mathcal{H}_{12}k$. The canonical record K is the image of the signed record generated by the preceding operations, with phase origin and orientation fixed.

Proof. The congruence defines exactly the integrality of $\mathcal{K}(u)$. The identity $\mathcal{H}_{12}^2 = 4I$ proves the two inverse compositions. In particular, every $k \in \mathbb{Z}^{12}$ has preimage $\mathcal{H}_{12}k \in \mathcal{L}_H$. $\square$

Proposition 8.7 (Canonical integral record). *The inversion $K^{(r)} = H_4U^{(r)}/4$ of the blocks in (8.9) produces*

$$\begin{aligned} K^{(1)} &= (234, 729, 621, 794), \\ K^{(2)} &= (543, 659, 58, 146), \\ K^{(3)} &= (140, 824, 914, 601). \end{aligned}$$

In the natural order, the dodecaphasic record is

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \quad (8.11)$$

It is the unique rational preimage of U , and belongs to $\{0, \dots, 999\}^{12}$.

Proof. The three products before division are

$$H_4 U^{(1)} = (936, 2916, 2484, 3176),$$

$$H_4 U^{(2)} = (2172, 2636, 232, 584),$$

$$H_4 U^{(3)} = (560, 3296, 3656, 2404).$$

All coordinates are divisible by four. Dividing and undoing the orbit permutation gives (8.11). Uniqueness follows from rational invertibility, not from a search among candidates close to a physical value. $\square$

Integrality can also be formulated in lattice terms. The determinantal divisors of H_4 are 1, 2, 4, 16: the entries have greatest common divisor one; the order-two minors are even, with at least one of absolute value two; the order-three minors are multiples of four, with at least one of absolute value four. Hence

$$\text{SNF}(H_4) = \text{diag}(1, 2, 2, 4),$$

and for the global transformation,

$$\text{coker } \mathcal{H}_{12} \cong (\mathbb{Z}/2\mathbb{Z})^6 \oplus (\mathbb{Z}/4\mathbb{Z})^3.$$

The index of the image is $16^3 = 4096$. Thus not every signed integer vector admits an integer preimage; the vector obtained satisfies the precise congruences required by inversion. The factor $1/4$ is forced by the matrix and does not have the status of a free coefficient.

8.8 Cyclic differences and the uniform component

With cyclic indices modulo twelve, let

$$(D_3 z)_m = z_m - z_{m+3}, \quad (D_4 z)_m = z_m - z_{m+4}, \quad Q(z) = \sum_{m=1}^{12} z_m.$$

The first operator compares sectors separated by three positions; the second compares sectors separated by four. The channel names 90 and 120 refer here to these separations in the oriented cycle, not to an identification with a physical operator from another domain.

Proposition 8.8 (Completeness of the transverse system). *The operator*

$$\mathcal{D} = (D_3, D_4, Q) : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{25}$$

has rank twelve and nonzero invariant factors

$$\text{diag}(1, \dots, 1, 12),$$

with eleven units. In particular, its outputs determine a unique dodecaphasic record.

Proof. If $D_3 z = D_4 z = 0$, the vector is constant along both steps. Since 3 and 4 generate the cycle modulo twelve, $\ker D_3 \cap \ker D_4 = \langle \mathbf{1} \rangle$. Total charge eliminates this last freedom because $Q(\mathbf{1}) = 12$.

For the integral claim, the rows of (D_3, D_4) are oriented incidences of a connected graph. A spanning tree provides eleven unit invariant factors. Every order-twelve minor must contain the row Q . Adding the first eleven columns to the last makes the latter zero in the incidence rows and twelve in the Q row. All maximal minors are divisible by twelve, and the tree produces one of absolute value exactly twelve. The last factor is therefore twelve. $\square$

Checking (8.11) gives

$$D_3 K = b^{90}, \quad D_4 K = b^{120}, \quad Q(K) = 6263.$$

Formulas (8.8) and (8.9) in turn reconstruct U from these differences. This closes a triangle of exact representations of the same object: transverse record, signed vector, and dodecaphasic record. Equality of results does not collapse their codomains into one; it proves that the declared maps recover the relevant information.

8.9 Integral compatibility and determination of the transverse record

The domain of this construction is the representation of the record of an APP-TRIT-TPK history. Evaluation of record events, compatibility of its channels, and inversion of its coordinates are distinguished. The following result gives a complete integral criterion for the second level and an explicit factorization connecting the first to it. The proof uses the record operators and establishes their relations for every element of the indicated integer domains.

8.9.1 Characterization by adjacent increments

The indices belong to $\mathbb{Z}/12\mathbb{Z}$. Let

$$(Sz)_i = z_{i+1}, \quad D_3 = I - S^3, \quad D_4 = I - S^4, \quad Q(z) = \sum_{i=0}^{11} z_i.$$

Retain the transverse extractor

$$\mathcal{D} : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{25}, \quad k \longmapsto (D_3k, D_4k, Q(k)).$$

For a triple of integer coordinates (b, c, q) , write

$$w_i = c_i - b_{i+1}, \quad P_0 = 0, \quad P_i = \sum_{j=0}^{i-1} w_j \quad (1 \leq i \leq 12), \quad T(w) = \sum_{i=0}^{11} P_i. \quad (8.12)$$

Theorem 8.9 (Integral image of the transverse extractor). *A triple $(b, c, q) \in \mathbb{Z}^{25}$ belongs to $\text{im } \mathcal{D}$ if and only if it satisfies*

$$b_i = w_i + w_{i+1} + w_{i+2} \quad (0 \leq i < 12), \quad (8.13)$$

$$\sum_{i=0}^{11} w_i = 0, \quad (8.14)$$

$$q + T(w) \equiv 0 \pmod{12}. \quad (8.15)$$

Its unique integer preimage is

$$k_i = \frac{q + T(w)}{12} - P_i, \quad 0 \leq i < 12. \quad (8.16)$$

The twelve equations of (8.13) and equation (8.14) are linearly independent over $\mathbb{Q}$.

Proof. For an output of $\mathcal{D}$, one has

$$c_i - b_{i+1} = (k_i - k_{i+4}) - (k_{i+1} - k_{i+4}) = k_i - k_{i+1}.$$

The sum of three increments gives b_i ; their full cyclic sum is zero. Moreover, $P_i = k_0 - k_i$, so $q = 12k_0 - T(w)$. All conditions and the inverse formula follow.

Conversely, (8.15) makes formula (8.16) integral; (8.14) allows its cyclic prolongation. Its adjacent differences are w_i . Thus, $D_3k = b$ and

$$(D_4k)_i = w_i + w_{i+1} + w_{i+2} + w_{i+3} = w_i + b_{i+1} = c_i.$$

Summing the inverse gives $Q(k) = q$. The same formula proves uniqueness. For independence, the invertible integer change $(b, c) \mapsto (b, w = c - Sb)$ turns the first twelve relations into $b = (I + S + S^2)w$. Each solves for a different coordinate of b . The condition $\sum w_i = 0$ is an additional independent relation. $\square$

The rational image has dimension twelve. Its integral structure adds a congruence of order twelve: this is the constructive manifestation of the final Smith factor of $\mathcal{D}$. The eleven increments $w_0, \dots, w_{10}$ and the charge q are sufficient coordinates, with $w_{11} = -\sum_{i=0}^{10} w_i$ and congruence (8.15). The original twenty-five coordinates preserve these same twelve degrees of freedom through thirteen relations.

8.9.2 Commutativity with the harmonic reading

Let $\mathcal{H}_{12}$ be the record's Hadamard transformation, with orbits $(r, r+3, r+6, r+9)$ for $r = 0, 1, 2$. Its block is

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad H_4^2 = 4I_4.$$

For compatible (b, c, q) , define

$$B_r = \sum_{j=0}^3 c_{r+3j}, \quad A_0 = \frac{q + 2B_0 + B_1}{3}, \quad A_1 = A_0 - B_0, \quad A_2 = A_1 - B_1.$$

The reader Π_H of (8.9) has blocks

$$(\Pi_H(b, c, q))^{(r)} = (A_r, b_{r+3} - b_{r+9}, b_r + b_{r+6}, b_r - b_{r+6}).$$

Proposition 8.10 (Agreement of the two reconstructions). *On $\text{im } \mathcal{D}$,*

$$\Pi_H \mathcal{D} = \mathcal{H}_{12}, \quad \mathcal{D}^{-1} = \frac{1}{4} \mathcal{H}_{12} \Pi_H.$$

The image of Π_H restricted to that domain is exactly

$$\mathcal{L}_H = \{u \in \mathbb{Z}^{12} : \mathcal{H}_{12}u \equiv 0 \pmod{4}\}.$$

Proof. For $k = \mathcal{D}^{-1}(b, c, q)$, let $a_r = \sum_{j=0}^3 k_{r+3j}$. Shifting four positions sends each orbit to the next, so $B_r = a_r - a_{r+1}$. Together with $q = a_0 + a_1 + a_2$, these identities give $A_r = a_r$. On an orbit, write (x_0, x_1, x_2, x_3) for the coordinates of k and $d_j = x_j - x_{j+1}$. Then

$$d_1 - d_3 = x_0 + x_1 - x_2 - x_3, \quad d_0 + d_2 = x_0 - x_1 + x_2 - x_3, \quad d_0 - d_2 = x_0 - x_1 - x_2 + x_3.$$

These are the remaining three rows of $H_4 k^{(r)}$. Inversion and the integral characterization follow from $\mathcal{H}_{12}^2 = 4I$. $\square$

The condition $\Pi_H(b, c, q) \in \mathcal{L}_H$ controls only the harmonic output. Membership of the twenty-five coordinates in $\text{im } \mathcal{D}$ additionally requires (8.13)–(8.15). For example, adding the vector $e_0 - e_3$ to c preserves the three orbital sums and leaves Π_H unchanged. Starting from $(0, 0, 0)$, one thus obtains a triple with $\Pi_H = 0$ that violates (8.13). This is an exact negative control of channel compatibility, independent of any terminal evaluation.

8.9.3 Determination and transport on the record's orbits

Theorem 8.9 determines k once its channels and charge have been produced. As a condition on outputs not yet individualized, the same image is a group isomorphic to $\mathbb{Z}^{12}$. Fixing only a charge q , the solutions form an affine class of

$$L_0 = \{v \in \mathbb{Z}^{12} : \sum v_i = 0\} = \bigoplus_{i=0}^{10} \mathbb{Z}(e_i - e_{11}).$$

For example, $1001 + e_0$ and $1001 + e_1$ are two different records of charge 1201, with components in $\{0, \dots, 999\}$. Each satisfies every image condition and every Hadamard congruence through its own channels. These witnesses compare compatibility equations; they are not attributed the status of terminal HMT histories.

Proposition 8.11 (Cyclic covariance and stabilizer of the record). *Let ρ be window translation on a record domain $\mathcal{R}$, and S the preceding translation on $\mathbb{Z}^{12}$. On the orbit of $R \in \mathcal{R}$, a covariant reader is determined by a vector*

$$k \in \text{Fix}(S^p),$$

where $p \mid 12$ is the least period of R under ρ . The charge condition $Q(k) = q$ admits integer solutions if and only if $12/p$ divides q ; when they exist, they form an affine class of rank $p - 1$. In particular, on a free orbit, covariance and charge leave eleven degrees of freedom.

Proof. The rule $F(\rho^j R) = S^j k$ is well defined exactly when $S^p k = k$. Such vectors repeat a word of length p exactly $12/p$ times. Their charge is $\frac{12}{p} \sum_{i=0}^{p-1} k_i$. Divisibility and rank follow from this formula. Channel covariance follows from $D_a S = S D_a$ and $Q S = Q$. On $\mathcal{L}_H$, the corresponding action is $\hat{S}_H = \mathcal{H}_{12} S \mathcal{H}_{12}/4$; by Proposition 8.10, both reconstructions intertwine these actions. $\square$

If the domain carries a marked origin, the action transporting that mark as well must be used. The proposition concerns this declared cyclic action; it does not assume periodicity of the enriched history. Naturality under truncations after the 108 cutoff is obtained, for any already defined reader F_{108} , through $F_N = F_{108} \circ \tau_{N,108}$. This naturality preserves the initial evaluation rule; it does not by itself choose one of the possible rules on the first one hundred eight events.

8.9.4 Event evaluation and composition of contributions

The preceding characterization provides a factorization of extraction through two effective readings of the record:

$$\omega : \mathcal{R} \longrightarrow \{w \in \mathbb{Z}^{12} : \sum w_i = 0\}, \quad q_{\text{reg}} : \mathcal{R} \longrightarrow \mathbb{Z}, \quad q_{\text{reg}}(R) + T(\omega(R)) \equiv 0 \pmod{12}. \quad (8.17)$$

These maps must be evaluated on the events, orientation, incidence, and memory produced by APP-TRIT-TPK. Once their actions on these data are given, the composition

$$R \longmapsto (w, q) \longmapsto ((I + S + S^2)w, (I + S + S^2 + S^3)w, q) \xrightarrow{\Pi_H} u \xrightarrow{\mathcal{H}_{12}/4} k$$

is determined and satisfies all inverse checks. The formula $w = c - Sb$ also allows connection to a producer already delivering both channels: it is a check on its output, prior to any comparison with the terminal witness.

Proposition 8.12 (Accumulation of contributions and prospective uniqueness). *Suppose the initial state and each event a_t of the record produce, through previously fixed rules, contributions $(\delta w_t, \delta q_t)$ with*

$$\sum_i \delta w_{t,i} = 0, \quad \delta q_t + T(\delta w_t) \equiv 0 \pmod{12}.$$

With compatible initial conditions $(w^{(0)}, q^{(0)})$, the update

$$w^{(t+1)} = w^{(t)} + \delta w_t, \quad q^{(t+1)} = q^{(t)} + \delta q_t$$

produces a unique record at each stage. Its increment is

$$\delta k_{t,i} = \frac{\delta q_t + T(\delta w_t)}{12} - P_i(\delta w_t).$$

The construction respects concatenation of records, and the reading of a prefix remains unchanged under subsequent prolongations.

Proof. The compatibility conditions are preserved under addition. Formula (8.16) is additive in (w, q) and produces the indicated increment. Induction fixes each state successively. The sum over two segments is the concatenated sum, and a prefix receives exactly the same summands before and after prolonging the history. $\square$

The proposition provides a sufficient criterion, not a requirement of linearity for every TPK reader. If events are transported by nontrivial actions, their contributions can be expressed in the terminal chart through the corresponding affine cocycle. The cocycle equation then preserves the same independence from the partition of the trajectory. In both cases, event rules and the initial state constitute the constructive data that must come from the ledger: choosing them to reproduce an expected vector would replace that provenance with another task.

The result of this development is the exact determination condition (8.17), its proved inversion, and its transport identities. Its scope is the transverse record; the remaining HMT constructions retain their own domains and proofs.

8.10 Two rational readings: κ_{36} and κ_{per}

Once K has been constructed, the base-thousand chart permits two distinct operations. The first reads a single turn; the second prolongs the dodecaphasic record periodically. Let $B = 1000$, and define its concatenation integer

$$N_K = \sum_{m=1}^{12} K_m B^{12-m} = 234543140729659824621058914794146601. \quad (8.18)$$

The leading zeros of a block, such as that of 058, are part of its three-digit representation. They do not change its integer value, but allow the twelve blocks to be concatenated unambiguously.

Definition 8.13 (Finite and periodic readings of the dodecaphasic record). The one-turn reading is

$$\kappa_{36} = \sum_{m=1}^{12} K_m B^{-m} = \frac{N_K}{B^{12}}.$$

The periodic prolongation is

$$K_j^{\text{per}} = K_{1+((j-1) \bmod 12)}, \quad \kappa_{\text{per}} = \sum_{j \geq 1} K_j^{\text{per}} B^{-j}.$$

Proposition 8.14 (Exact value and period of the prolonged dodecaphasic record). *We have*

$$\kappa_{\text{per}} = \frac{N_K}{B^{12} - 1}, \quad \kappa_{\text{per}} - \kappa_{36} = \frac{N_K}{B^{12}(B^{12} - 1)} = B^{-12} \kappa_{\text{per}}. \quad (8.19)$$

The infinite word of the dodecaphasic record has least period twelve in base thousand and least period thirty-six in base ten.

Proof. The sum of the first turn is $N_K B^{-12}$; each new turn multiplies its contribution by B^{-12} . The geometric series yields $N_K/(B^{12} - 1)$, and subtraction gives the second identity.

The twelve components of K are distinct. A proper period of the cycle would identify two of them; the least block period is therefore twelve. The decimal expansion is canonical, since the dodecaphasic record has no tail of 999 blocks. If its least decimal period were d , it would divide 36. Grouping digits in threes would produce a block period $d/\gcd(d, 3)$. No proper divisor of 36 gives twelve through this operation. Hence the least decimal period is 36. $\square$

Both readings are rational, but they are not the same rational number. The first terminates after twelve blocks; the second repeats the turn. Moreover, $0 < \kappa_{\text{per}} - \kappa_{36} < B^{-12}$. Their closeness does not permit substituting one for the other in an exact identity. In particular, the finite alpha closure uses κ_{36} , whereas the sum of the prolonged section uses κ_{per} .

8.11 Reversibility of the rational constant and phase change

The rational constant κ_{per} determines the entire record K when the base, length, and reading origin are retained. Its numerical and structural functions are related by an exact inversion.

Proposition 8.15 (Integral recovery and cyclic transformation). *Let $I(K) = \sum_{j=1}^{12} K_j 1000^{12-j}$. Then*

$$I(K) = (1000^{12} - 1)\kappa_{\text{per}}.$$

The integer expansion of $I(K)$ into twelve base-1000 blocks recovers every K_j , including the leading zeros in each block. If $\rho(K) = (K_2, \dots, K_{12}, K_1)$, then

$$\kappa(\rho K) = 1000\kappa(K) - K_1. \quad (8.20)$$

Proof. The first identity inverts the definition of periodic evaluation. Iterated Euclidean division by 1000, with twelve fixed positions, recovers the vector. Furthermore,

$$I(\rho K) = 1000I(K) - K_1(1000^{12} - 1).$$

Dividing by $1000^{12} - 1$ yields (8.20). □

Recovery therefore composes $\kappa \mapsto K \mapsto U$ and the already constructed transformations D_3K, D_4K, Q . Rational evaluation preserves the signed record and the cyclic differences that depend on it. The total depth and transport operators of a history remain in their own coordinates: the domain of the preceding inversion is the integral dodecaphasic record.

Write $\kappa_j = \kappa(\rho^j K)$, with indices modulo 12. The same identity gives

$$K_{j+1} = 1000\kappa_j - \kappa_{j+1}, \quad \sum_{j=0}^{11} \kappa_j = \frac{\sum_{j=1}^{12} K_j}{999} = \frac{6263}{999}.$$

Phase acts exactly on the constant: its value is fixed in the canonical reading, while its rotations follow an affine law. The orbit sum is invariant under the choice of origin.

The subsequent incidence also has an expression in these coordinates. The cubic threshold determines

$$B_K = \{j + 1 : 1000\kappa_j - \kappa_{j+1} \geq 729\} = \{4, 6, 9, 10\}.$$

The tetrad is recovered from the rational orbit because the latter first recovers the integer components. The signed support associated with α additionally uses its regional records and boundary normalization. The relationship between coordinate and incidence is thus expressed through identifiable operations.

This is an operational definition of the function of K : it provides a reversible integral coordinatization of the oriented record, whose periodic evaluation and incidence functions participate in subsequent constructions. The value, transformation, and domain are preserved together.

8.12 Periodicity and cokernel of the transport operator

The periodicity just proved belongs to the dodecaphasic record. It does not imply periodicity of the regional coordinates, the closure carries, or alpha. It is useful to clarify this separation through a second, purely integral quotient.

Let S be the cyclic shift of twelve coordinates, with fixed orientation, and let $b \geq 2$ be an integer base. The periodic carry operator is $\partial_b = S - bI$. If a periodic identity is written

$$A = P + E - \Phi - K + \partial_b C,$$

then the transformations

$$K \mapsto K + \partial_b h, \quad C \mapsto C + h$$

preserve its right-hand side. This invariance compares representatives of a carry class; it does not authorize changing the dodecaphasic record already selected by the record.

Proposition 8.16 (Quotient of periodic carry). *With the preceding orientation,*

$$\text{coker}(S - bI) \cong \mathbb{Z}/(b^{12} - 1)\mathbb{Z}.$$

Proof. The cyclic module is presented as $\mathbb{Z}[t]/(t^{12} - 1)$. Imposing $S = bI$ is equivalent to imposing $t = b$ and leaves the relation $b^{12} - 1 = 0$. Evaluating the coefficients at b , with the chosen concatenation order, represents the resulting class. $\square$

The denominator $B^{12} - 1$ of κ_{per} and this quotient arise from the same cyclic length, but their objects remain different: one is a rational evaluation and the other a quotient of a carry module. The finite alpha closure will be an oriented chain with a right boundary. Its prolongation will transport a compatible remote boundary, not a carry identified by decree between the first and thirteenth positions.

The next chapter is thus prepared. The dodecaphasic record has been fixed before solving the alpha recurrence; its Hadamard reconstruction is integral and unique; its transverse differences recover the record; and its two rational evaluations have separate domains and uses. None of these facts uses alpha as an input, and none yet determines the arithmetic type of the output of the complete closure.

9 Derivation of the fine-structure constant

In this development, the fine-structure constant receives an arithmetic determination that brings together the three regional coordinates and the integral dodecaphase record. The operation combines their blocks with fixed orientation and transports the quotients of positional normalization. The resulting parameter expresses compatibility of that oriented difference with cylinder prolongation. Its value and the incidence data used in its construction will remain related throughout the exceptional development.

The internal order is important. The coordinates π , e , and φ , already constructed in the preceding section, are prior coordinates of the same arithmetic state with trajectory memory and may legitimately act as inputs to a subsequent reader. They are not conventional constants injected to select alpha. Correlated generation of the four coordinates shares origin and coinductive compatibility without requiring that all four components be evaluated in a single step.

The two routes express different mathematical roles of the same parameter. The first realizes an integer closure by Euclidean division in base 1000. The second studies the analytic equilibrium of vacancies through the resolvent $E_{21/22}$ and its graded prolongations. One route determines blocks and quotients; the other determines a root through an equation and a uniqueness domain. Their identification is formulated afterward in the common cylinders at every depth. The exposition distinguishes these three results: positional construction, analytic determination, and compatibility of the two.

9.1 Positional determination through integer dodecaphase closure

9.1.1 Domain and orientation of integer closure

Let $P_j, E_j, \Phi_j \in \{0, \dots, 999\}$ be the canonical blocks of the three regional coordinates, and prolong the dodecaphase record by

$$K_j = K_{1+((j-1) \bmod 12)}.$$

The channel orientation is $(1, 1, -1)$. The contribution of the dodecaphase record is subtracted according to the already typed dodecaphase closure. For $B = 1000$, the local operation is

$$s_j = P_j + E_j - \Phi_j - K_j + c_{j+1}, \quad A_j = s_j - B \left\lfloor \frac{s_j}{B} \right\rfloor, \quad c_j = \left\lfloor \frac{s_j}{B} \right\rfloor. \quad (9.1)$$

Thus $A_j \in \{0, \dots, B-1\}$, even when s_j is negative. The carry entering from the right belongs to the domain of the operation; it cannot be omitted because the table is printed from left to right.

Definition 9.1 (Finite-depth closure). For a depth N and an integer boundary t , the map $\mathcal{C}_N$ receives $(P_{1:N}, E_{1:N}, \Phi_{1:N}, K_{1:N}, c_{N+1} = t)$ and applies (9.1) in the order $N, N-1, \dots, 1$. Its output is the pair of words

$$(A_{1:N}, c_{1:N+1}).$$

Euclidean division has no residual sign choice: the remainder is fixed in $\{0, \dots, 999\}$, and the quotient is determined. For example, $-431 = 569 - 1000$ and $-1200 = 800 - 2 \cdot 1000$. The negative carries of closure are not anomalies; they are the integer information needed to maintain the oriented equality.

9.1.2 Dodecaphase window and carry conditions

The first twelve triads obtained are

$$\begin{aligned} P &= (141, 592, 653, 589, 793, 238, 462, 643, 383, 279, 502, 884), \\ E &= (718, 281, 828, 459, 045, 235, 360, 287, 471, 352, 662, 497), \\ \Phi &= (618, 033, 988, 749, 894, 848, 204, 586, 834, 365, 638, 117). \end{aligned}$$

In the closure channel, the five regions determining coordinate P preserve their distinct antecedents. Sharing this reading does not identify them as regions or histories.

With boundary $c_{13} = 0$, the recurrence produces the following table. Each input, incoming carry, total before normalization, obtained block, and outgoing carry is shown. The table is a reproducible integer computation, not a list of blocks selected for their resemblance to a target.

j	R_j	E_j	Φ_j	K_j	c_{j+1}	s_j	A_j	c_j
1	141	718	618	234	0	7	007	0
2	592	281	033	543	0	297	297	0
3	653	828	988	140	-1	352	352	0
4	589	459	749	729	-1	-431	569	-1
5	793	045	894	659	-2	-717	283	-1
6	238	235	848	824	-1	-1200	800	-2
7	462	360	204	621	0	-3	997	-1
8	643	287	586	058	-1	285	285	0
9	383	471	834	914	-1	-895	105	-1
10	279	352	365	794	0	-528	472	-1
11	502	662	638	146	0	380	380	0
12	884	497	117	601	0	663	663	0

Table 9: Initial closure with zero right boundary. The dependency propagates from the last row to the first.

The carry word is

$$(c_1, \dots, c_{13}) = (0, 0, 0, -1, -1, -2, -1, 0, -1, -1, 0, 0, 0),$$

and the window reading is the rational number

$$\alpha_{12}^{(0)} = \frac{7297352569283800997285105472380663}{10^{36}} = 0.007297352569283800997285105472380663. \quad (9.2)$$

The superscript recalls the chosen boundary. This window is exact in its finite domain; it is not simply identified with the truncation of the infinite section. The difference between the two readings will become visible when the carry of the remote tail is transported.

9.1.3 Telescoping and reversibility in a boundary fiber

The local form of the equality is

$$R_j + E_j - \Phi_j - K_j - A_j = Bc_j - c_{j+1}. \quad (9.3)$$

The carry term is a link between adjacent positions. Eliminating it coordinate by coordinate gives a false equality. Only an evaluation preserving its endpoints allows it to telescope.

Proposition 9.2 (Integer identity at arbitrary depth). *For $\iota_N(z) = \sum_{j=1}^N z_j B^{N-j}$, every execution of $\mathcal{C}_N$ satisfies*

$$\iota_N(P) + \iota_N(E) - \iota_N(\Phi) - \iota_N(K) - \iota_N(A) = B^N c_1 - c_{N+1}. \quad (9.4)$$

Proof. Multiplying (9.3) by B^{N-j} and summing produces on the right-hand side

$$\sum_{j=1}^N (Bc_j - c_{j+1}) B^{N-j} = B^N c_1 - c_{N+1}.$$

Each interior carry appears twice with opposite signs; only the two endpoints survive. $\square$

For Table 9, both endpoints are zero. If p_{12}, e_{12}, f_{12} are the three truncated fractional sums,

$$\alpha_{12}^{(0)} = p_{12} + e_{12} - f_{12} - \kappa_{36}.$$

The dodecaphase-record constant appearing here is κ_{36} , not κ_{per} . The domain of this identity contains truncations and a finite boundary; its form does not authorize erasing both conditions when passing to the complete section.

Local inversion is exact as well:

$$K_j = R_j + E_j - \Phi_j + c_{j+1} - A_j - Bc_j.$$

Given P, E, Φ , blocks A , and the complete carries, it recovers the dodecaphase record. Concatenation additionally recovers

$$\iota_N(K) = \iota_N(P) + \iota_N(E) - \iota_N(\Phi) - \iota_N(A) - B^N c_1 + c_{N+1}.$$

This reversibility is a property of the fiber with fixed data and boundaries. It must not be turned into a new causal recipe starting from alpha and manufacturing the record afterward. The direction of construction remains signed record, dodecaphase record, closure, and output.

9.1.4 Prolongation and canonical normalization of the complete section

Prolongation preserves the periodic dodecaphase record and receives the complete regional sequences. Let

$$p = \sum_{j \geq 1} R_j B^{-j}, \quad e_0 = \sum_{j \geq 1} E_j B^{-j}, \quad f = \sum_{j \geq 1} \Phi_j B^{-j}.$$

Each series is a realization of the corresponding internal character. The integer parts already obtained are 3, 2, and 1, so $p = \pi - 3$, $e_0 = e - 2$, and $f = \varphi - 1$.

Proposition 9.3 (Existence and uniqueness of complete normalization). *The series*

$$y = p + e_0 - f - \kappa_{\text{per}} = \sum_{j \geq 1} (R_j + E_j - \Phi_j - K_j) B^{-j} \quad (9.5)$$

converges absolutely. On the domain of the obtained blocks, $0 < y < 10^{-2}$. There is a unique canonical expansion (A_j) of y , with no eventually constant 999 tail, and a unique bounded integer sequence (c_j) satisfying (9.3) with $c_1 = 0$. These carries belong to the invariant set $\mathcal{C} = \{-2, -1, 0, 1, 2\}$.

Proof. The coefficients of the series are bounded in absolute value by $2(B - 1)$, so a geometric series dominates it. If $Z_j = R_j + E_j - \Phi_j - K_j$ and $R_j(Z) = \sum_{r \geq 0} Z_{j+r} B^{-r-1}$, each tail is the difference of two sums of two canonical tails and belongs to $(-2, 2)$. The first coefficient is $Z_1 = 141 + 718 - 618 - 234 = 7$; hence $y = (7 + R_2(Z))/1000 \in (0.005, 0.009)$.

Let (A_j) be the canonical expansion of y . Define

$$c_j = R_j(Z) - R_j(A).$$

Multiplying equality of the two series by B^{j-1} and separating their first $j - 1$ terms gives

$$c_j = \sum_{r=1}^{j-1} (A_r - Z_r) B^{j-1-r} \in \mathbb{Z},$$

and $c_1 = 0$. Furthermore, $R_j(A) \in [0, 1)$, so $-3 < c_j < 2$, placing all carries in $\mathcal{C}$. The tail-shift identities give $Bc_j = Z_j - A_j + c_{j+1}$, which is the required recurrence.

Conversely, any bounded solution with canonical output and $c_1 = 0$ telescopes in the limit to the same series y . The canonical expansion is unique, and once it is fixed, the tail formula determines all carries. Finally, if a finite execution is applied with $c_{j+1} \in \mathcal{C}$, then $-2000 \leq s_j \leq 2000$, so its Euclidean quotient again belongs to $\mathcal{C}$. $\square$

The proposition makes explicit the normalization of the subsequent reader: it does not replace generation of the three sequences with a conventional series introduced as data. Once those sequences have been produced by the coinductive state, their combination and the normalization boundary are determined without consulting alpha.

The finite procedure preserved in the corpus propagates the five boundaries of $\mathcal{C}$ and retains only the prefix common to all of them. If two depths have coalesced before a boundary, the deterministic right-to-left recurrence gives the same blocks to its left. This is compatibility with truncation. A boundary is not chosen because it gives favorable digits, nor is all-depth compatibility inferred from observing a long run.

The initial comparison shows why this precaution matters. Stabilized prolongation gives $c_{13}^\infty = -1$, whereas the isolated window imposed $c_{13} = 0$. Thus,

$$884 + 497 - 117 - 601 - 1 = 662$$

in the last block of the first turn. The first eleven triads do not change, but the prolongation begins

$$(007, 297, 352, 569, 283, 800, 997, 285, 105, 472, 380, 662, 999, \dots).$$

An occurrence of block 999 does not violate the canonical convention: what is excluded is an eventually constant 999 tail. The zero-boundary window ends in 663; the truncation of the complete section ends in 662. Both statements describe different and exact operations. They must not be merged into a single projection notation.

9.1.5 Operative definition and complete additive equation

Definition 9.4 (Alpha coordinate of closure). The integer route determines the coordinate

$$\alpha_A = \sum_{j \geq 1} A_j B^{-j} = p + e_0 - f - \kappa_{\text{per}}, \quad (9.6)$$

where (A_j, c_j) is the compatible canonical normalization of (9.1). In the scalar realization of the HMT coordinates,

$$\boxed{\alpha_A = \pi + e - \varphi - 4 - \kappa_{\text{per}}.} \quad (9.7)$$

The integer four is not calibrated: it is $3 + 2 - 1$, the combination of the integer parts of the three coordinates. Base thousand arises from the cubic chart, and period twelve from the dodecaphase record. The signs are those of the oriented channel and closure, and the remote boundary is that imposed by the compatible section. These are the effective origins of the coefficients of the equation.

At this level, alpha can be described as the scalar coordinate of the closure defect between the three regional realizations and the rational dodecaphase return, computed with compatible carry. The term *defect* denotes a determinate algebraic difference, not an arbitrary loss or an imperfection of the system. This description does not suppress the genealogy: without the regions, signed record, orientation, and carry law, the scalar equality would not be a sufficient exposition of the construction.

Three objects that alpha notation may accompany in subsequent readers must also be kept separate: A is a word of blocks; the negative pre-carry support is obtained from the signs of the totals before normalization; and an exceptional radial vector belongs to an incidence codomain. None is interchangeable with the real value α_A .

Signed balance and sum with incoming quotient. It is useful to distinguish the two consecutive stages of positional division by means of a common notation. The balance before quotient transport is

$$Z_j := R_j + E_j - \Phi_j - K_j^{\text{per}}.$$

The variable s_j in (9.1) denotes the sum that already includes the incoming contribution, so that

$$s_j = Z_j + c_{j+1}, \quad A_j = Z_j + c_{j+1} - 1000c_j.$$

In (28.2), the letter s_j locally denotes the balance before that contribution, namely Z_j in the present notation. The two formulas therefore express the same Euclidean division with differently named stages. The incidence configuration H_α^- uses the signs of the balance Z_j in the initial window; the determination of the digits also uses the linking quotients c_{j+1} and c_j . This distinction jointly preserves the signed support and the normalization of the expansion.

Exact dependence on the truncation boundary. The regional coordinates and the dodecaphasic record originate in the common APP–TRIT–TPK generation. Once produced, their positional composition has an explicit dependence on reading depth. Let $B = 1000$,

$$Z_j = P_j + E_j - \Phi_j - K_j^{\text{per}}, \quad y_N = \sum_{j=1}^N Z_j B^{-j}, \quad R_{N+1}(Z) = \sum_{r \geq 0} Z_{N+1+r} B^{-r-1}.$$

The complete series in (9.5) satisfies exactly

$$\alpha_A = y_N + B^{-N} R_{N+1}(Z), \quad -2 < R_{N+1}(Z) < 2.$$

This identity combines the contribution of the window with that of its continuation, both expressed in the same coordinates.

Proposition 9.5 (Boundary transport and canonical cylinder). *Let $A_j^{(N,t)}$ be the output of finite positional division with $c_{N+1} = t \in \mathcal{C} = \{-2, -1, 0, 1, 2\}$, and let $a_N^{(t)} = \sum_{j=1}^N A_j^{(N,t)} B^{-j}$. For the records in the domain under consideration,*

$$\alpha_A - a_N^{(t)} = B^{-N}(R_{N+1}(Z) - t), \quad |\alpha_A - a_N^{(t)}| < 4B^{-N}. \quad (9.8)$$

The quotient of the complete continuation is $c_{N+1}^\infty = \lfloor R_{N+1}(Z) \rfloor$. With this boundary,

$$0 \leq \alpha_A - a_N^{(c_{N+1}^\infty)} < B^{-N},$$

so that the resulting window is the canonical prefix of order N .

Proof. Telescoping (9.3) gives $a_N^{(t)} = y_N + tB^{-N} - c_1$. Since $Z_1 = 7$, each incoming state belongs to $\mathcal{C}$, and the first dividend lies between 5 and 9, we have $c_1 = 0$. Substituting this equality into the series decomposition proves (9.8). The bound follows from $R_{N+1}(Z) \in (-2, 2)$ and $t \in \mathcal{C}$.

For complete canonical normalization, $c_{N+1}^\infty = R_{N+1}(Z) - R_{N+1}(A)$ and $R_{N+1}(A) \in [0, 1)$. The integrality of the quotient implies $c_{N+1}^\infty = \lfloor R_{N+1}(Z) \rfloor$. The error formula then becomes $\alpha_A - a_N^{(c_{N+1}^\infty)} = B^{-N} R_{N+1}(A)$, proving membership in the canonical cylinder. $\square$

The proposition identifies the meaning of the right boundary: it is the integer part of the signed tail expressed at the scale of the next position. Computation over the five boundaries preserves this information as a finite collection of possibilities until the blocks on the left stabilize. Quantitative contraction follows from depth and tail transport; its object is the normalization of the coordinates already generated.

Graded domain of the analytic resolvent. The analytic composition exhibited in §9.2.2 acts on $\text{span}\{e_7, e_8, e_9\}$ and satisfies $Ne_9 = 0$. Its evaluation yields exactly $T_{7,9}$; degree 9 is the final component of this domain. The identity $N^3 = 0$ explains why the resolvent is expressed as three summands. To describe a continuation to higher degrees, one must additionally specify the transport between successive graded domains and its action on the initial component of each domain. Local nilpotence and global compatibility thus perform different mathematical functions.

9.1.6 Constructive determination and the function of the dodecaphasic record

The preceding definition makes it possible to specify what it means to determine alpha from the structure. The domain of the operation contains four records of known provenance: the three regional sequences and the integral phase record. Normalization additionally incorporates a boundary condition compatible with continuation. Once these data are fixed, Euclidean division determines each remainder and its quotient. Proposition 9.3 extends this determination to the complete system. Independence from a target physical value is therefore a property of the construction order: the value compared at the end does not participate in selecting any argument of this map.

This causal independence must be distinguished from algebraic independence between numbers. The former is established by identifying the domain, operations, and provenance of the coefficients; the latter would be a statement about polynomial relations. The result used here is the former. The construction determines a coordinate through records produced by APP-TRIT-TPK and subsequently permits verification of the relations it satisfies. This order makes it possible to use an already generated coordinate in a later operation without turning it into an external parameter.

The function of the record K is more precise than that of a numerical correction added to a formula. Its twelve positions originate in the reversible transformation of the signed record. The equation $U = \mathcal{H}_{12}K$ preserves the possibility of returning to that record, while the transverse differences and total sum allow another reconstruction. In periodic evaluation, the order of the positions determines the positional weight of each component. The same set of components, arranged in a different order, generally produces a different rational coordinate. The phase origin and orientation therefore form part of the definition of the number occurring in (9.7).

Alpha normalization composes this information with the regions. Its local identity distinguishes two successive operations. First, the oriented balance of blocks is formed; then that balance is represented by a canonical remainder and a quotient linking adjacent positions. The first operation preserves the sign of the balance and allows a support to be extracted subsequently. The second allows prefixes of the real coordinate to be formed. Both readings remain related because the state preserves the data before and after normalization. The signed support and the positional expansion are thus different realizations of an explicit arithmetic composition.

The telescoping identity expresses the global content of this composition. The interior quotients cancel in pairs when a prefix is evaluated, while the endpoint terms remain in the equality. This cancellation is a property of the weighted sum; it neither removes the quotients from the state nor allows them to be discarded before the sum is performed. The right boundary of a window receives the contribution of the continuation. For this reason, evaluating an isolated window and restricting a continued section are different operations. The change of the final block from 663 to 662, proved above, provides an explicit realization of this difference without altering the already stabilized blocks to its left.

In this sense, bidirectionality has a concrete arithmetic formulation. Generation provides compatible continuations; normalization transmits the integer information at their boundary from positions of lower weight towards those of higher weight. The equation preserves the contribution of both directions through its indices and boundary conditions. Reversibility in the fixed fibre is verified by reconstructing K from the other records and the complete quotients. This inversion certifies the consistency of the original operation; the genealogical direction remains the one that first produces the record and then the alpha coordinate.

The second determination has complementary content. In it, the immediate object is a vacancy balance, and the parameter appears as its root in an isolating domain. The positional route provides a canonical normalization of records; the analytic route studies the vanishing of an operator. Presenting both requires precisely this difference in function to be retained. Their relationships with the common state must be expressed through the refinement maps and the comparison of their complete realizations, as developed in the corresponding subsection. Agreement at a finite approximation constitutes a localized check; the identification of the complete constructions has the domain and scope of its own statement.

Finally, this organization explains alpha's place in the article. Its equation appears after regional generation and the construction of K , because it uses both results. The electronic section subsequently uses this coordinate as a factor in its composition. The exceptional extension receives the records and their supports, in which incidence information is preserved. These two continuations return to a common provenance and develop different mathematical functions: evaluating a central section and constructing a boundary configuration. Developing each makes it possible to follow the structure through the equation, rather than restricting the comparison to its scalar value.

9.2 Analytic determination through the vacancy structure

9.2.1 Transport invariants and graded resolvent

The analytic route receives invariants of the oriented state, not the output α_A . Its sixth-order truncation is

$$P_6(x) = 2\pi x - \frac{7}{4}x^2 + \frac{x^3}{2\pi} + \frac{x^4}{20} - \frac{2x^5}{21} - \frac{x^6}{46}, \quad (9.9)$$

and the vacancy condition uses

$$\delta = \log_{10} \frac{10}{9}.$$

The ratio 10/9 compares the decimal chart with the nine APP positions of the nonadic construction. Its logarithm belongs to this subsequent compactification reading; it is not a target value of alpha.

The genealogy of the coefficients must be preserved in full, but with its exact logical strength. Recent developments bring together a marked signature of the state. The synchronization matrix and associated census are

$$D_{\text{syn}} = \begin{pmatrix} 85 & 112 \\ 41 & 52 \end{pmatrix}, \quad N_9 = 43.$$

From its determinant one extracts

$$a = -\frac{\det D_{\text{syn}}}{N_9} = \frac{172}{43} = 4.$$

The clock induced on the short gaps has secondary spectrum $\{6, 7\}$; its upper endpoint gives $b = 7$. The tritic cardinality $m = 3$ determines

$$t_* = mb = 21, \quad k_* = mb - 1 = 20.$$

The vacancy reading in the thousand chart gives

$$v_- = \lfloor 1000\delta \rfloor = 45, \quad v_+ = \lceil 1000\delta \rceil = 46.$$

These integers are typed antecedents of the truncation's evaluation. They are not obtained from its root or from a metrological approximation to it.

The lower branch additionally coincides with the determinant of the APP block

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}, \quad \det B_c = 45.$$

The transverse rank of the preceding chapter's record gives $r_{\text{dod}} = 11$, and the phase semicrown provides $f_{\text{ph}} = 8\binom{6}{2} = 120$. In particular, $45r_{\text{dod}} = 495$.

With $\omega = 2\pi$, the signature

$$\mathcal{I}_9 = (\omega, m, a, b, k_*, t_*, v_-, v_+, f_{\text{ph}}, r_{\text{dod}})$$

feeds the graded reader

$$\begin{aligned} \mathfrak{G}_9(\mathcal{I}_9)(x) = & \omega x - \frac{b}{a}x^2 + \omega^{-1}x^3 + \frac{x^4}{k_*} - \frac{2x^5}{t_*} - \frac{x^6}{v_+} \\ & - \frac{x^7}{f_{\text{ph}}} + \frac{x^8}{v_-} + \frac{2x^9}{v_-r_{\text{dod}}}. \end{aligned} \quad (9.10)$$

Substitution recovers the coefficients of P_6 and its ninth-order prolongation. The pair 45/46 crosses the separation between the two: the upper branch occupies degree six, and the lower reappears in degrees eight and nine.

The factorization of (9.10) is formulated for a state preserving origin, channel order, and orientation. Naturality with respect to morphisms preserving those data is not equivalent to uniqueness of every imaginable reader after they are forgotten. The technical note preserves the evolution of this formulation in the sources [Sections 4.8, 5.10, and 9.2.1], without downgrading the recovered factorization to the reservation of an earlier version or promoting it to a universal uniqueness not being proved here.

9.2.2 Nilpotent resolvent in degrees 7, 8, and 9

The first prolongation admits an especially explicit finite operator construction. On the space with basis e_7, e_8, e_9 , let

$$Ne_7 = -\frac{8}{3}e_8, \quad Ne_8 = \frac{2}{11}e_9, \quad Ne_9 = 0, \quad v_7 = -\frac{1}{120}e_7.$$

The symbol N here denotes a nilpotent operator, not a truncation depth. Its domain and grading are fixed before the polynomial is evaluated at a number.

Lemma 9.6 (Exact nilpotent prolongation). *One has $N^3 = 0$ and*

$$(I - N)^{-1}v_7 = -\frac{e_7}{120} + \frac{e_8}{45} + \frac{2e_9}{495}.$$

Evaluation $e_n \mapsto x^n$ produces

$$T_{7:9}(x) = -\frac{x^7}{120} + \frac{x^8}{45} + \frac{2x^9}{495}. \quad (9.11)$$

Proof. The operator raises degree and vanishes on the last basis vector, so its cube is zero. The inverse is the finite resolvent $I + N + N^2$. Moreover,

$$Nv_7 = \frac{1}{45}e_8, \quad N^2v_7 = \frac{2}{495}e_9.$$

Summing the three terms and evaluating gives the formula. $\square$

Here the denominators 120, 45, and 495 are not decimal fits: they arise from the semicrown, APP determinant, and its product with the dodecaphase rank. The transfers $-8/3$ and $2/11$, together with the sign and degree of the seed, belong to the typed reader. Preserving only cycle symmetry and replacing this operator by another symmetric operator does not necessarily preserve the coefficients.

9.2.3 Multiplicity-one root of the ninth-order truncation

Define

$$R_9 = P_6 + T_{7:9}, \quad E_{21/22}^{(9)}(x) = R_9(x) - \delta, \quad I_\alpha = (0, 10^{-2}).$$

This finite operator has its own existence and uniqueness result.

As a subsequent check of the characters already constructed, the rational isolation can be reproduced without a decimal table. Let

$$A_n(t) = \sum_{j=0}^n \frac{(-1)^j t^{2j+1}}{2j+1}, \quad L_n(t) = 2 \sum_{j=0}^n \frac{t^{2j+1}}{2j+1} \quad (0 < t < 1).$$

The alternating-series criterion and the geometric tail sum give

$$A_{2n+1}(t) < \arctan t < A_{2n}(t), \quad (9.12)$$

$$L_n(t) < \log \frac{1+t}{1-t} < L_n(t) + \frac{2t^{2n+3}}{(2n+3)(1-t^2)}. \quad (9.13)$$

Apply (9.12) to Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$, and (9.13) to

$$\log(10/9) = \log \frac{1+1/19}{1-1/19}, \quad \log 10 = \log(10/9) + 2 \log \frac{1+1/2}{1-1/2}.$$

Write

$$\delta := \frac{\log(10/9)}{\log 10}.$$

Truncation orders 80, 20, 40, 120, applied respectively to the four preceding series, give rational intervals for π and δ , both of width less than 10^{-70} . These bounds check a subsequent realization; they neither select the APP-TRIT-TPK state nor receive a value of alpha.

Proposition 9.7 (Multiplicity-one root of the truncation). *The operator $E_{21/22}^{(9)}$ has a unique root ξ_9 in I_α , and that root is simple. It also has the rational isolation*

$$\frac{729735256928380099}{10^{20}} < \xi_9 < \frac{729735256928380100}{10^{20}}. \quad (9.14)$$

Proof. The internal-coordinate bounds $3.14 < \pi < 3.15$ and $0.045 < \delta < 0.046$ suffice for existence. At zero, $E_{21/22}^{(9)}(0) = -\delta < 0$. At 10^{-2} , the linear contribution minus the negative terms exceeds 0.0626, so the operator value is positive.

For $0 \leq x \leq 10^{-2}$, discarding the positive terms of the derivative gives

$$(E_{21/22}^{(9)})'(x) \geq 2\pi - \frac{7}{2} \cdot 10^{-2} - \frac{10}{21} \cdot 10^{-8} - \frac{6}{46} \cdot 10^{-10} - \frac{7}{120} \cdot 10^{-12} > 6.24.$$

Continuity gives existence; the derivative bound gives uniqueness and simplicity. For the left endpoint l and the right endpoint u of (9.14), rational substitution of the bounds (9.12)–(9.13) in the polynomial gives, with directed rounding,

$$-4.6 \cdot 10^{-20} < E_{21/22}^{(9)}(l) < -4.5 \cdot 10^{-20} < 0, \quad 0 < 1.6 \cdot 10^{-20} < E_{21/22}^{(9)}(u) < 1.8 \cdot 10^{-20}.$$

The monotonicity already proved places the unique root between these endpoints. The interval is checked after constructing the operator and does not enter the selection of its coefficients. $\square$

The result concerns ξ_9 . It does not authorize writing $\xi_9 = \alpha_A$ as an exact equality of complete numbers. A truncation root may share a prefix with the root of the prolonged operator and differ at a later position. Simplicity allows control of root movement under perturbations; it does not annul those perturbations.

9.3 Identification criterion for compatible readouts

Identification of two complete readouts requires their cylinders and restriction maps to be preserved. For a compatible graded collection,

$$E_{21/22}^{(6)} \longleftarrow E_{21/22}^{(9)} \longleftarrow E_{21/22}^{(12)} \longleftarrow \cdots, \quad E_{21/22} = \varprojlim_m E_{21/22}^{(6+3m)},$$

the limit denotes the readout together with its transports, not a freely selected tail. The following criterion separates the consequence of compatibility from the effective construction establishing it below.

Proposition 9.8 (Identification of compatible limits). *If the values of the integer and analytic routes belong to the same cylinders C_N^α for every N , and $\text{diam } C_N^\alpha \rightarrow 0$, then*

$$\alpha_A = \alpha_B = \alpha.$$

Proof. For every N , the distance between the two values is bounded by $\text{diam } C_N^\alpha$. Letting N tend to infinity gives zero distance. Existence of compatible sections ensures that this is not the empty intersection of an approximation system. $\square$

Cylinder contraction establishes equality of the limits; the complete analytic recurrence and its compatibility squares are developed below. The ninth-order truncation remains a finite object distinct from the complete function.

The difference can be expressed algebraically. Let $F = \mathbb{Q}(\pi, \delta)$, and let $j_9 : F[[x]] \rightarrow F[x]/(x^{10})$ be truncation. Then

$$\ker j_9 = x^{10}F[[x]].$$

All operators

$$E_h(x) = E_{21/22}^{(9)}(x) + x^{10}h(x)$$

have the same ninth-order truncation. However, taking $h = 0$ and $h = 1/729$,

$$E_{1/729}(\xi_9) = \frac{\xi_9^{10}}{729} > 0,$$

so they do not share that root. The observation does not permit choosing a tail foreign to HMT transport; it demonstrates precisely why the generated tail and its compatibility must be preserved. Nor does an arbitrary formal series suffice for speaking of analytic roots without its convergence domain or the realization of the cylinder system.

9.4 Compatibility between the dodecaphasic and analytic representations of alpha

Compatibility is established at every depth. The reversible dodecaphasic representation and the analytic representation of the alpha coordinate $E_{21/22}$ are two internal representations with distinct codomains, produced from the same enriched APP-TRIT-TPK state. The jet $E_{21/22}^{(9)}$ is an exact finite witness, but does not by itself determine the tail. The completion constructed below is not deduced from the bare jet: it receives the shared coordinate of the state and constructs its analytic image through an explicit recurrence. It therefore does not constitute an independent causal selection; it constitutes a typed analytic representation of the same generated point.

This separation preserves the causal order

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow \mathcal{X}_\alpha^{\text{enr}} \longrightarrow (P, E, \Phi, U_{12}^{\text{sgn}}, K) \longrightarrow \alpha_{\text{HMT}}.$$

The coordinates P, E, Φ and the register K are earlier outputs of the same enriched state. Their use in the closure of alpha is therefore a subsequent internal composition, not an injection of conventional constants.

For $r \in 6\mathbb{N}$, $r \geq 36$, let $\widehat{\mathcal{X}}_r^{\text{enr}, \alpha}$ be the sector of truncated states that extend the internal alpha seed fixed at R_{36} . Write

$$x_r^\alpha = (\tilde{x}_r, \chi_\alpha, \varepsilon_r^\alpha, N_r^\alpha),$$

where χ_α is the internal character of the sector, not the expressed scalar value. Each state preserves remainder, quotient, carry, sheet, orientation, phase, route, cylinder, boundary, signature, register, memory, survival, provenance and depth. If

$$d_r^\alpha = \text{Dig}_{729}(\varepsilon_r^\alpha), \quad \varepsilon_{r+6}^\alpha = 729\varepsilon_r^\alpha - d_r^\alpha, \quad N_{r+6}^\alpha = 729N_r^\alpha + d_r^\alpha,$$

the effective transition is

$$\mathcal{U}_r^\alpha = \text{Upd}_r^\alpha \circ \text{Tra}_r^\alpha \circ \text{Sel}_r^\alpha, \quad \text{Ext}_{r+6, r}^{\alpha, \rightarrow} = \text{graph}(\mathcal{U}_r^\alpha).$$

The single-valuedness of these extensions, with the seed R_{36} fixed, produces a unique compatible section: the phase returns while memory and depth advance. The global domain of both determinations is

$$\mathcal{X}_\alpha^{\text{enr}} := \widehat{\Omega}_\alpha := \varprojlim_{r \geq 36} (\widehat{\mathcal{X}}_r^{\text{enr}, \alpha}, \text{Ext}_{r+6, r}^{\alpha, \rightarrow}). \quad (9.15)$$

The notation κ_{per} retains the periodic readout of the register already constructed in the base-one-thousand chart fixed in (9.7).

9.4.1 The dodecaphasic closure as a complete section

For $N \geq 1$, set

$$\mathcal{W}_N := \{0, \dots, 999\}^N, \quad \mathsf{T}_{\alpha, N}(\mathbf{s}) := (A_1, \dots, A_N), \quad \mathbf{s} \in \mathcal{X}_\alpha^{\text{enr}},$$

where the blocks A_j are obtained through (9.1) with the compatible boundary $c_{N+1}^\infty = [R_{N+1}(Z)]$. If $N' \geq N$, let $\rho_{N', N} : \mathcal{W}_{N'} \rightarrow \mathcal{W}_N$ be the restriction to the first N blocks.

Theorem 9.9 (Complete dodecaphasic output and projective compatibility). *The collection $(T_{\alpha,N})_{N \geq 1}$ is compatible:*

$$\rho_{N',N} \circ T_{\alpha,N'} = T_{\alpha,N} \quad (N' \geq N). \quad (9.16)$$

Its canonical cylinders

$$C_N^A(s) := \left[\sum_{j=1}^N A_j 1000^{-j}, \sum_{j=1}^N A_j 1000^{-j} + 1000^{-N} \right) \cap I_\alpha \quad (9.17)$$

are nested, have diameter at most 1000^{-N} and satisfy

$$\bigcap_{N \geq 1} C_N^A(s) = \{\alpha_A(s)\}, \quad \alpha_A = \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4 - \kappa_{\text{per}}. \quad (9.18)$$

Consequently, the first route determines a unique infinite section, which we denote by $\alpha_{\text{HMT}} := \alpha_A$.

Proof. Proposition 9.3 provides the unique canonical expansion and the unique bounded sequence of carries. The local identity (9.3) and its telescoping (9.4) preserve the remote boundary; Proposition 9.5 proves that restricting an extended execution recovers exactly the preceding prefix. This gives (9.16). Membership of the full sum in each cylinder follows from the tail identity, and $\text{diam } C_N^A \leq 1000^{-N} \rightarrow 0$; the intersection contains a single point. Finally, (9.7) identifies that point with the expression in (9.18). All its coefficients and signs are fixed before α is evaluated. $\square$

The window $\alpha_{12}^{(0)}$, closed with $c_{13} = 0$, remains an exact finite calculation; it must not be confused with the restriction of the complete section, whose compatible boundary satisfies $c_{13}^\infty = -1$. In particular,

$$\rho_{36,12}(\alpha_{A,36}) = \alpha_{A,12} \quad (9.19)$$

is an equality of compatible words, not an equality between words of different lengths or an identification of the isolated window with the limit.

9.4.2 Analytic representation of the alpha coordinate

The analytic extension is specified through all its coefficients. Let $s \in \mathcal{X}_\alpha^{\text{enr}}$. Before decimal normalization, the same state already has the pre-coordinate

$$a(s) := p(s) + e_0(s) - f(s) - \kappa_{\text{per}}(s) \in I_\alpha, \quad (9.20)$$

with $p = \pi_{\text{HMT}} - 3$, $e_0 = e_{\text{HMT}} - 2$ and $f = \varphi_{\text{HMT}} - 1$. Equivalently, $a = \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4 - \kappa_{\text{per}}$. No conventional α is introduced here: p, e_0, f are the three regional characters already generated, and κ_{per} is the rational reading of the terminal register. The integer closure applies to (9.20) the normalization in (9.1); the chart below expresses a compatible analytic representation on the same pre-coordinate. Its dependency graph receives $(p, e_0, f, \kappa_{\text{per}})$ directly: it does not contain an edge $\alpha_A \rightarrow \lambda$, although the equality proved below allows both representation names to be identified subsequently.

Write $q = 1/729$, let F_s be the ordered field generated by the coordinates of s , and set $E_{9,s}(X) = E_{21/22}^{(9)}(s; X)$. We define the linking coefficient

$$\lambda(s) := -\frac{E_{9,s}(a(s))(1 - q a(s)^3)}{a(s)^{10}}. \quad (9.21)$$

All its arguments belong to the state before metrological recognition. In particular, λ is not obtained by fitting a target decimal string. Equation (9.21) must be read with its exact type: once the first representation of the state has produced $a(s)$, it fixes the unique extension within the geometric class defined below. It is not a second independent selection of $a(s)$, but a correlated analytic representation of the same enriched state.

Indeed, for $\mu \in F_s$ set

$$E_{s,\mu}(X) := E_{9,s}(X) + \mu \frac{X^{10}}{1 - qX^3}.$$

Since $0 < a(s) < 10^{-2}$ and $qa(s)^3 < 1$, we have

$$E_{s,\mu}(a(s)) = 0 \iff \mu = \lambda(s). \quad (9.22)$$

Thus, the coefficient seed contains no freedom once the state and the class of q -geometric extensions have been fixed. This uniqueness is relative to the declared class; it does not assert that the ninth-order jet by itself selects a tail among all possible analytic series.

The local action on blocks of three coefficients is

$$\mathcal{C}_\alpha : F_s^3 \longrightarrow F_s^3, \quad \mathcal{C}_\alpha(u, v, w) = q(u, v, w), \quad c^{(0)} = (\lambda(s), 0, 0). \quad (9.23)$$

Therefore, for every $n \geq 0$,

$$b_{10+3n} = q^n \lambda(s), \quad b_{11+3n} = b_{12+3n} = 0. \quad (9.24)$$

This is the rule that computes each coefficient beyond degree nine. No free parameter remains at each extension.

For $M \geq 0$, let

$$E_{s,M}(X) := E_{9,s}(X) + \lambda(s) \sum_{n=0}^M q^n X^{10+3n}. \quad (9.25)$$

The complete function is

$$E_{s,\infty}(X) := E_{9,s}(X) + \lambda(s) \frac{X^{10}}{1 - X^3/729}. \quad (9.26)$$

To formulate the projective system without implicit types, we fix $d_M = 12 + 3M$ and define the jet fibration

$$\mathfrak{J}_M := \coprod_{s \in \mathcal{X}_\alpha^{\text{enr}}} \{s\} \times F_s[X]/(X^{d_M+1}). \quad (9.27)$$

If $M' \geq M$, the restriction is

$$\tau_{M',M}(s, [P]_{d_{M'}}) := (s, [P]_{d_M}). \quad (9.28)$$

Proposition 9.10 (Naturality of the analytic recurrence). *Let*

$$\Gamma_M(s) = (s, [E_{s,M}]_{d_M}).$$

If $M' \geq M$, the truncation $\tau_{M',M}$ satisfies

$$\tau_{M',M} \circ \Gamma_{M'} = \Gamma_M, \quad \Gamma_{E21}(s) = \varprojlim_M \Gamma_M(s). \quad (9.29)$$

The projection onto the three new degrees is precisely $\mathcal{C}_\alpha^{M+1}(c^{(0)})$.

Proof. Equation (9.24) fixes the block of degrees $10 + 3n, 11 + 3n, 12 + 3n$. Passing from M to $M + 1$ adds only the next block; truncating it recovers the preceding polynomial literally. Compatibility for $M' \geq M$ is obtained by composition. Thus, the inverse limit is not a section chosen from the bare jet, but the unique orbit of $c^{(0)}$ under $\mathcal{C}_\alpha$. $\square$

9.4.3 Uniform convergence, simple root and rational intervals

The regional cylinders and the periodic register provide bounds through rational interval arithmetic. For each coordinate

$$c \in \{\pi_{\text{HMT}}, e_{\text{HMT}}, \varphi_{\text{HMT}}\},$$

let P_c be its certified thousand-digit nonadic prefix. The datum used is not the rational number P_c treated as though it were the complete value, but the half-open cylinder and its outer closure

$$\mathcal{C}_c^{(1000)} = [P_c, P_c + \varepsilon), \quad \overline{\mathcal{C}}_c^{(1000)} = [P_c, P_c + \varepsilon], \quad \varepsilon = 10^{-1000}.$$

Directed arithmetic operates conservatively with these closures. If P_π, P_e, P_φ denote the three prefixes, we obtain

$$\begin{aligned} a_- &:= P_\pi + P_e - (P_\varphi + \varepsilon) - 4 - \kappa_{\text{per}}, \\ a_+ &:= (P_\pi + \varepsilon) + (P_e + \varepsilon) - P_\varphi - 4 - \kappa_{\text{per}}, \\ A_{1000} &:= [a_-, a_+], \quad a(s) \in [a_-, a_+] = A_{1000}, \quad \text{diam } A_{1000} = 3\varepsilon. \end{aligned} \quad (9.30)$$

The natural interval extension of (9.21) then transports A_{1000} , the cylinder of π_{HMT} and the directed rational bound for δ to a closed interval $\Lambda_{1000} \ni \lambda(s)$; none of the three tails is frozen. In particular,

$$\frac{7297352569283800997285105472380662}{10^{36}} < a(s) < \frac{7297352569283800997285105472380663}{10^{36}}, \quad (9.31)$$

$$-8.711 \times 10^{-6} < \lambda(s) < -8.709 \times 10^{-6}.$$

These inequalities use certified prefixes with their rational tail bound; they do not receive a table of alpha values. The package verifier reproduces them from the three nonadic outputs and from the terminal chart.

Let

$$r := \frac{10^{-6}}{729} < 1.372 \times 10^{-9}.$$

For $0 \leq x \leq 10^{-2}$, the tail beyond M satisfies

$$|E_{s,\infty}(x) - E_{s,M}(x)| \leq 8.711 \times 10^{-26} \frac{r^{M+1}}{1-r}. \quad (9.32)$$

Moreover,

$$\left| \frac{d}{dx} \left(\lambda \frac{x^{10}}{1-x^3/729} \right) \right| \leq 8.711 \times 10^{-6} \left(\frac{10 \cdot 10^{-18}}{1-r} + \frac{3 \cdot 10^{-24}/729}{(1-r)^2} \right) < 8.712 \times 10^{-23}. \quad (9.33)$$

The tail of the derivatives beyond block M also admits the depth-dependent estimate

$$\sup_{0 \leq x \leq 10^{-2}} |E'_{s,\infty}(x) - E'_{s,M}(x)| \leq 8.711 \times 10^{-24} r^{M+1} \left(\frac{13+3M}{1-r} + \frac{3r}{(1-r)^2} \right) \xrightarrow{M \rightarrow \infty} 0. \quad (9.34)$$

Theorem 9.11 (Limit function and unique simple root of the correlated chart). *The sequence $E_{s,M}$ converges uniformly, together with its derivatives, to $E_{s,\infty}$ on $\bar{I}_\alpha = [0, 10^{-2}]$. We have*

$$E_{s,\infty}(a(s)) = 0, \quad \inf_{x \in \bar{I}_\alpha} E'_{s,\infty}(x) > 6.239. \quad (9.35)$$

Consequently, $a(s)$ is the unique simple root of the complete function in I_α .

Proof. The ratio of two consecutive terms of the tail is bounded by $r < 1$, which gives (9.32); term-by-term differentiation and summation of the two geometric series give (9.33) and (9.34). The proof for the jet establishes $E'_{9,s} > 6.24$ on $\bar{I}_\alpha$; hence, the complete perturbation leaves the derivative above 6.239. Finally, substituting (9.21) into (9.26), the two summands evaluated at $a(s)$ cancel exactly. The function is strictly increasing; its unique root is therefore simple and coincides with that pre-coordinate. $\square$

To make the identification effective at every depth, directed endpoint evaluations first certify

$$E_{s,\infty}(0) < 0 < E_{s,\infty}(10^{-2}), \quad E'_{s,\infty} \geq \mu := 6239/1000 > c := 6 \quad \text{on } [0, 10^{-2}].$$

By continuity there is a root in the interval and, by the second inequality, it is unique. Only after this existence, monotonicity and the invariant that the bracket contains the root have been established is the mean-value-theorem fallback applied. We start from $D_0 = [0, 10^{-2}]$. If $D_j = [r_j, s_j]$, let $m_j = (r_j + s_j)/2$, and let $[F_j^-, F_j^+]$ be a directed rational enclosure of $E_{s,\infty}(m_j)$, obtained by propagating the cylinders of $\pi_{\text{HMT}}, e_{\text{HMT}}, \varphi_{\text{HMT}}, \delta$ and λ , with

$$0 \leq F_j^+ - F_j^- \leq \frac{c}{4} (s_j - r_j). \quad (9.36)$$

When $F_j^+ < 0$, the right half is retained; when $F_j^- > 0$, the left half is retained. If $F_j^- \leq 0 \leq F_j^+$, no attempt is made to decide whether the midpoint is exactly the root. Let $w_j = s_j - r_j$. The uniform bound $E'_{s,\infty} \geq c = 6$ and the mean value theorem imply that any root contained in D_j belongs to

$$D_{j+1} = D_j \cap \left[m_j - \frac{w_j}{4}, m_j + \frac{w_j}{4} \right]. \quad (9.37)$$

Indeed, let ξ_j be the root and let $y_j = E_{s,\infty}(m_j) \in [F_j^-, F_j^+]$. Since the enclosure contains zero and satisfies (9.36), we have $|y_j| \leq cw_j/4$. The mean value theorem gives $|m_j - \xi_j| \leq |y_j|/c \leq w_j/4$, proving (9.37). The argument includes the case $y_j = 0$: the root is then at the midpoint, but the algorithm does not need to recognize this equality in order to retain it.

If the available enclosure does not satisfy (9.36), the source cylinders are first refined and the evaluation is repeated; no sign is declared and no insufficient contraction is accepted. In each of the three branches the new interval retains the root and has diameter at most $w_j/2$. For each finite target depth, directed refinement allows the width bound to be met without an uncertified comparison of a real number with zero. Monotonicity proves inductively that

$$a(s) \in D_{j+1} \subseteq D_j, \quad \text{diam } D_{j+1} \leq \frac{1}{2} \text{diam } D_j, \quad \text{diam } D_j \longrightarrow 0, \quad (9.38)$$

where the inequality is a contraction bound, not an imposed equality. The executable certificate applies this rule to twenty-four base-one-thousand levels and checks that the complete outer closure A_{1000} , not only its lower endpoint, lies inside each expressed cylinder. Its negative tests reject an excessively wide enclosure, an incorrectly ordered enclosure, a point other than the midpoint, a domain without a sign change, and a degenerate bracket. Coinductive extension allows one thousand to be replaced by any required depth.

Specifically, for $S_N = 1000^N$ the verifier computes

$$j_N^- := \lfloor S_N a_- \rfloor, \quad j_N^+ := \lfloor S_N a_+ \rfloor \quad (9.39)$$

and certifies the level only if $j_N^- = j_N^+$. This equality, which includes the upper endpoint of the outer closure, proves $A_{1000} \subset [j_N^-/S_N, (j_N^- + 1)/S_N)$ and avoids assigning to the left cell an endpoint lying exactly on its right boundary.

Let $C_N^A(s)$ be the integer cylinder in (9.17). We recursively choose $m(N) > m(N-1)$ so that $\text{diam } D_{m(N)} < 1000^{-N}/4$, and define

$$I_{B,N} := D_{m(N)} \cap C_N^A(s). \quad (9.40)$$

Define $\ell_N := \inf I_{B,N}$ and $u_N := \sup I_{B,N}$. Both are rational; the interval is nonempty because both factors contain $a(s)$. Since the cylinder is half-open, u_N may not belong to $I_{B,N}$. Evaluation at that endpoint is understood through the continuous extension of $E_{s,\infty}$ to the outer closure. Thus,

$$I_{B,N+1} \subseteq I_{B,N} \subseteq C_N^A(s), \quad \text{diam } I_{B,N} \longrightarrow 0, \quad E_{s,\infty}(\ell_N) \leq 0 \leq E_{s,\infty}(u_N). \quad (9.41)$$

The bound (9.32) allows a truncation to be chosen that preserves the sign at each endpoint where the complete function does not vanish. If an endpoint coincides with the root, the identity of the

complete function and the membership already proved are used; that same sign is not attributed to its truncations. Indeed, the exact remainder is

$$E_{s,\infty}(x) - E_{s,M}(x) = \lambda(s)x^{10} \frac{(qx^3)^{M+1}}{1 - qx^3}.$$

In the positive domain under consideration, where $\lambda(s) < 0$, we obtain at the root

$$E_{s,M}(a(s)) = -\lambda(s)a(s)^{10} \frac{(qa(s)^3)^{M+1}}{1 - qa(s)^3} > 0 \quad (M < \infty).$$

Therefore, the certification of the weak signs in (9.41) refers to $E_{s,\infty}$. Its evaluation uses the complete function or the truncation accompanied by the signed remainder. This covers both the included infimum and the supremum that may be excluded from the cylinder.

Proposition 9.12 (Compatibility squares and common cylinders). *The restrictions of words and of analytic representations commute:*

$$\begin{array}{ccc} \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{T_{\alpha,N'}} & \mathcal{W}_{N'} \\ \parallel & & \downarrow \rho_{N',N} \\ \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{T_{\alpha,N}} & \mathcal{W}_N \end{array} \quad \begin{array}{ccc} \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{\Gamma_{M'}} & \mathfrak{F}_{M'} \\ \parallel & & \downarrow \tau_{M',M} \\ \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{\Gamma_M} & \mathfrak{F}_M. \end{array}$$

The collection $I_{B,N}$ provides the effective connection between the two squares and satisfies concretely $I_{B,N} \subseteq C_N^A$ for every N .

Proof. The first square is the compatibility of integer normalization; the second is Proposition 9.10. The inclusion follows from the definition (9.40), and nonemptiness and the signs follow from Theorem 9.11. No step identifies the root of a truncation with that of an arbitrary tail. $\square$

We now define, without leaving the prefix reader implicit,

$$\alpha_B(s) := \arg \text{zero}_{x \in I_\alpha} E_{s,\infty}(x), \quad \text{pref}_N(x) := \frac{\lfloor 1000^N x \rfloor}{1000^N}, \quad \alpha_{B,N} := \text{pref}_N(\alpha_B). \quad (9.42)$$

Theorem 9.13 (Projective equality of the two correlated representations). *For every N , the integer and analytic routes yield the same prefix:*

$$\boxed{\alpha_{A,N} = \text{pref}_N(\alpha_A) = \text{pref}_N(\alpha_B) = \alpha_{B,N}.} \quad (9.43)$$

In the limit,

$$\boxed{\alpha_A = \arg \text{zero}_{x \in I_\alpha} E_{s,\infty}(x) = \alpha_B = \alpha_{\text{HMT}}.} \quad (9.44)$$

Proof. Integer normalization represents exactly $a(s)$. Theorem 9.11 proves that the analytic chart represents that same point as its unique simple root. The half-open cylinders C_N^A fix a canonical prefix, and $I_{B,N} \subseteq C_N^A$ contains $\alpha_B = \alpha_A$. Therefore, their prefixes coincide for each N ; nesting and diameters tending to zero give equality of the limits. Metrological recognition occurs afterwards. $\square$

The proved identity connects two representations of the same state through the coefficient recurrence and compatible cylinders. The analytic function is a correlated representation of the already generated precoordinate, not a second causally independent selection. The jet, geometric class, and state coordinates retain their respective roles; conventional recognition comes afterwards.

10 Dodecaphasic update and observable memory

Let d_t be the integer direction code preserved by a TPK trace. This code and the cursor's cardinal direction are distinct coordinates. The twelve windows $E_m = \{9m + 1, \dots, 9m + 9\}$, $0 \leq m < 12$, divide the record's 108 steps. The orientation is

$$\Sigma_{12} = (+, +, +, -, -, -, +, +, +, -, -, -).$$

The event of a window is the output

$$a_m = \Sigma_{12}(m) \sum_{t \in E_m} \mathbf{1}_{\{2,4,6,8\}}(d_t). \quad (10.1)$$

The notation a_m avoids confusing this event with Euler's number. The source record also preserves TRIT orientation, signature, carry, and the corresponding history segment [Sections 8.2 and 11].

Proposition 10.1 (Increment produced by transport). *Let $\mathbf{e}_m$ be the basis of $\mathbb{R}^{12}$ and $\mathbf{S}\mathbf{e}_m = \mathbf{e}_{m+1 \pmod{12}}$. If $\mathbf{z}_{m+1} = \mathbf{z}_m + a_m \mathbf{e}_m$ and $y_m = S^{-m} \mathbf{z}_m$, then*

$$y_{m+1} = S^{-1} y_m + \eta_m, \quad \eta_m = a_m S^{-1} \mathbf{e}_0. \quad (10.2)$$

After one complete turn,

$$y_{12} = y_0 + \sum_{m=0}^{11} a_m \mathbf{e}_m. \quad (10.3)$$

Proof. Multiplying the update of $\mathbf{z}_m$ by $S^{-(m+1)}$ gives

$$y_{m+1} = S^{-1} y_m + a_m S^{-(m+1)} \mathbf{e}_m = S^{-1} y_m + a_m S^{-1} \mathbf{e}_0.$$

Upon iteration, the coefficient of event a_m is $S^{-(12-m)} \mathbf{e}_0 = \mathbf{e}_m$. Since $S^{12} = I$, one obtains (10.3). $\square$

Boldface here identifies the accumulated vector $\mathbf{z}_m \in \mathbb{R}^{12}$, whereas the incidence block z_m of (8.4) is an integer between 0 and 999. The moving frame y_m transports the first object; the decimal coding of the second retains its own definition.

Thus η_m is not a free force chosen to impose a conservation law: it is the vector expression of the TPK event. Equality of phase after one turn does not cancel accumulated events. This reader preserves counts, while the full record also preserves their order and provenance. The vector increment η_m is distinguished from η_{ret} , the scalar action mantissa, and η_{el} , the shape parameter to be introduced in section 15.

10.1 Quadratic balance

Define

$$\nabla_3 = I - S^3, \quad \nabla_4 = I - S^4, \quad B = \nabla_3^T \nabla_3 + \nabla_4^T \nabla_4, \quad \mathcal{M}(y) = \frac{1}{2} y^T B y.$$

The oriented readers of the signed record satisfy $(D_j z)_m = z_m - z_{m+j}$; with the convention $\mathbf{S}\mathbf{e}_m = \mathbf{e}_{m+1}$, one has $D_j = I - S^{-j} = \nabla_j^T$. Both directions produce the same quadratic form B , since powers of S commute. We have $S^T B S = B$ and $B_{00} = 4$. Substituting (10.2) yields exactly

$$\mathcal{M}(y_{m+1}) - \mathcal{M}(y_m) = a_m (B y_m)_0 + 2a_m^2. \quad (10.4)$$

Likewise, the charge $q(y) = \sum_j y_j$ satisfies $q(y_{m+1}) - q(y_m) = a_m$. The proof is expansion of the quadratic polynomial after removing the permutation S^{-1} . No metrological value enters that calculation. The form $\mathcal{M}$ measures this record; its interpretation as a specific physical energy would require the corresponding dimensional reader.

10.2 Transport resolvents and preservation of memory

The prolonged history determines the windows $I_m = \{9m + 1, \dots, 9m + 9\}$, for $m \geq 0$. With the numbering of this section, $I_m = E_m$ during the first cycle. Let $\sigma_m \in \{-1, 1\}$ be the orientation preserved by that history; for $0 \leq m < 12$, it coincides with $\Sigma_{12}(m)$. Define

$$a_m = \sigma_m \sum_{t \in I_m} \mathbf{1}_{\{2,4,6,8\}}(d_t), \quad |a_m| \leq 9.$$

The record and the moving frame are prolonged by

$$\mathbf{z}_{m+1} = \mathbf{z}_m + a_m \mathbf{e}_{m \bmod 12}, \quad y_m = S^{-m} \mathbf{z}_m.$$

The initial state $\mathbf{z}_0$ preserves the preceding history; it is zero for an initially empty prefix. In this subsection we write $R = S^{-1}$ for frame transport. The update (10.2) is retained in every window. Subsequent orientation and events are obtained from the prolonged history: frame return does not presuppose periodicity of the event sequence.

Proposition 10.2 (Truncation identity with terminal state). *For $N \geq 1$, let*

$$Y_N(u) = \sum_{m=0}^N y_m u^m, \quad H_{\eta,N}(u) = \sum_{m=0}^{N-1} \eta_m u^m.$$

The following exact polynomial identity holds:

$$(I - uR)Y_N(u) = y_0 + uH_{\eta,N}(u) - u^{N+1}Ry_N. \quad (10.5)$$

Proof. In degrees $1 \leq m \leq N$, the coefficient on the left-hand side is $y_m - Ry_{m-1} = \eta_{m-1}$, by (10.2). Its coefficients in degrees zero and $N + 1$ are y_0 and $-Ry_N$, respectively. Comparing coefficients proves the equality. $\square$

The terminal term preserves the transported memory when a history is cut. The identity allows a finite evaluation to be compared with its prolongation while keeping explicit the state left at the boundary.

10.2.1 Correspondence with common-core transport

The window numbering of Section 2.4 and that used here describe the same partition of the history. The correspondence is

$$\underbrace{I_m = E_{m+1}}_{\text{common-core index}} \longleftrightarrow \underbrace{I_m = E_m}_{\text{index in this section}}, \quad 0 \leq m < 12.$$

The subscript changes the name of the window; its nine transitions, its orientation $\sigma_m = \Sigma_{12}(m)$, its event a_m , and its position in the record remain the same. For every prolongation, $R = S^{-1}$, $y_m = S^{-m} \mathbf{z}_m$, and update (10.2) coincide with (2.24). Orientation and the event sequence retain their dependence on the history, also after frame return.

This identification allows Proposition 2.18 to be applied directly: series (2.28), bound (2.29), and uniform convergence of its derivatives correspond to this same record and the same window clock. The finite identity (10.5) also determines the terminal term when that series is truncated. Its hypotheses retain the initial history and the bounded event $|a_m| \leq 9$.

When the record is reused, Proposition 2.19 jointly preserves the operator, source, and reader through (2.30) and (2.31). Example (2.32) specifies the effect of the excursions omitted by a compression. The affine composition (2.33), the intermediate blocks (2.34), and identity (2.35) ensure consistency when segments are concatenated or several layers eliminated. Their complete proofs reside in Section 2.4, with the domains, sources, and reading maps specified there.

Balance (10.4) preserves the correspondence between the oriented readers of the signed record and the common quadratic form. The distinction between the accumulated vector $\mathbf{z}_m$ and the incidence decimal block $\mathbf{z}_m$ remains that established in this section. The transport identities apply to memory thus typed; they do not by themselves determine additional analytic coefficients.

11 Dodecaphasic seal and reversible rational reading

The terminal record has a visible coordinate and a signed coordinate. The seal is obtained from the latter through the Hadamard transformation by orbits; a subsequent scalar reading allows the seal to be recovered when its representation frame is preserved. This property specifies what a structural definition of a constant means here: the production and recovery maps are made explicit, along with its value.

Let $U \in \mathbb{Z}^{12}$ be the signed coordinate produced by the record, H_4 the Hadamard matrix developed in section 13, and P_{orb} the permutation grouping

$$(1, 4, 7, 10), \quad (2, 5, 8, 11), \quad (3, 6, 9, 12).$$

Define

$$\mathcal{H}_{12} = P_{\text{orb}}^{-1}(H_4 \oplus H_4 \oplus H_4)P_{\text{orb}}.$$

We have $\mathcal{H}_{12}^2 = 4I$. When $\mathcal{H}_{12}U \equiv 0 \pmod{4}$, the integral seal is

$$K = \frac{1}{4}\mathcal{H}_{12}U, \quad U = \mathcal{H}_{12}K.$$

The transverse differences D_3K, D_4K and charge $Q(K)$ recover the channels specified by the record [Sections 8.2 and 11]. The seal is thus a reconstructible coordinate of that oriented record; depth and the full history remain distinct fields.

Proposition 11.1 (Block recovery and origin transport). *Fix an integer base $b \geq 2$ and twelve ordered positions, and let $K_j \in \{0, \dots, b-1\}$. Define*

$$I(K) = \sum_{j=1}^{12} K_j b^{12-j}, \quad \kappa(K) = \frac{I(K)}{b^{12}-1}.$$

The exact reading κ uniquely determines the twelve blocks. For the rotation $\rho K = (K_2, \dots, K_{12}, K_1)$,

$$\boxed{\kappa(\rho K) = b\kappa(K) - K_1.}$$

Proof. The integer $I = (b^{12}-1)\kappa$ decomposes uniquely into twelve base- b positions, filling the necessary initial positions with zeros. Moreover,

$$I(\rho K) = bI(K) - K_1(b^{12}-1).$$

Dividing by $b^{12}-1$ gives the transformation law. For the constant block $K_j = b-1$, the reading is $\kappa = 1$; the fixed periodic representation type likewise preserves its recovery. $\square$

In the record's base 1000, the three-digit blocks remain recoverable, including leading zeros. The periodic rational reading preserves more than a truncated decimal approximation. Changing the phase origin generally changes its value according to the preceding identity; after twelve rotations it returns. Memory advance in the TPK state is an additional transformation: the reader's periodicity does not identify accumulated histories.

Remark: reversible reading and information. Exact block recovery is a property of the rational reader in its ordered frame. Word complexity, the entropy of a symbolic dynamics, reduced-state entropy, and thermodynamic production are distinct objects. A relation between them requires the distribution or state and the corresponding transport map to be specified; reversibility of the seal alone does not identify these functionals.

12 Cyclic decomposition and ratios of determinants

The cyclic projectors

$$P_3 = \frac{1}{4}(I + S^3 + S^6 + S^9), \quad P_4 = \frac{1}{3}(I + S^4 + S^8)$$

have images of dimensions three and four. If $E = P_3 - P_4$, then

$$\frac{\text{Tr } E}{12} = -\frac{1}{12}. \quad (12.1)$$

The scalar comes from a normalized rank difference, not an ordinary sum of positive integers.

Theorem 12.1 (Representation of the channel reader as a ratio of determinants). *Let $\mathcal{H}_3 = \text{Fix}(S^3)$ and $\mathcal{H}_4 = \text{Fix}(S^4)$. For $0 < s < 1$,*

$$F(s) = \frac{\det_{\mathcal{H}_3}(I - sS)}{\det_{\mathcal{H}_4}(I - sS)} = \frac{1 - s^3}{1 - s^4} = \frac{1 + s + s^2}{1 + s + s^2 + s^3}. \quad (12.2)$$

Its continuous extension at $s = 1$ equals $3/4$.

Proof. The restriction of S to each image is a cycle of three or four positions, respectively. Their characteristic polynomials give $\det(I - sS) = 1 - s^r$ for $r = 3, 4$. Canceling the common factor $1 - s$ leaves the last expression, regular at $s = 1$. $\square$

Substitution $s = q^{30}$ gives $F(q^{30}) = (1 - q^{90})/(1 - q^{120})$. The same pair of spaces producing (12.1) produces the channel quotient. This equality does not identify a twelve-coordinate matrix with the entire TPK state.

12.1 Oriented information and the complex sector

In the antipodal sector $V_- = \text{Ran}((I - S^6)/2)$, the operator $J = S^3$ satisfies $J^2 = -I$ and $J^\top = -J$. With $Q = P_4(I - S^6)/2$, one has

$$\text{rank } Q = 2, \quad E|_{V_-} = -Q, \quad B|_{V_-} = 5I - 3Q.$$

On $\text{Ran } Q$, $S^2 = -I$ and the determinant is $1 + s^2$. Thus,

$$F(s) = \frac{1 + s + s^2}{1 + s} \frac{1}{1 + s^2}.$$

Orientation reversal changes the sign of the form $\omega(u, v) = \langle Ju, v \rangle$, but preserves B and F . Scalar evaluation does not distinguish all oriented transports.

The transported trace also produces

$$a_n^{\text{tr}} = \text{Tr}(S^n E) = 3\mathbf{1}_{3|n} - 4\mathbf{1}_{4|n},$$

and, for $|s| < 1$,

$$-\log F(s) = \sum_{n \geq 1} \frac{a_n^{\text{tr}}}{n} s^n.$$

This sequence of moments links the cyclic decomposition to the spectral representation of the reader. The arithmetic nature of particular values, such as Catalan's constant, additionally requires the corresponding arithmetic analysis.

13 Planck constants and construction of the action scale

13.1 Hadamard quotient, multiplicative norm, and decimal order

The absolute scale and the relative corrector originate in distinct objects. The former uses the local block of the step-three dodecaphase record,

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad G_H = \mathbb{Z}^4 / H_4 \mathbb{Z}^4. \quad (13.1)$$

The operation uses a local quotient. The three blocks of the full record are not replaced by a tensorization that would multiply the number of sheets of this reader once again.

Lemma 13.1 (Number of sheets of the local quotient). *The Smith normal form of H_4 is $\text{diag}(1, 2, 2, 4)$. Consequently,*

$$G_H \simeq \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}, \quad |G_H| = 16.$$

Proof. The greatest common divisor of the entries is one. The minors of order two are even, and one has value -2 , so their greatest common divisor is two. Since $H_4 H_4^T = 4I$ and $\det H_4 = -16$, the cofactors are ± 4 ; the greatest common divisor of the minors of order three is four. The determinant has absolute value 16. The successive quotients of the divisors 1, 2, 4, 16 are 1, 2, 2, 4. Their product gives the order of the quotient. $\square$

Definition 13.2 (Action reader defined by the multiplicative norm). The multiplicative norm of the scalar section α , constant over the sheets of G_H , and one application of the self-scale φ produce

$$\Sigma_H = \varphi N_{G_H}(\alpha), \quad N_{G_H}(\alpha) = \prod_{g \in G_H} \alpha = \alpha^{16}. \quad (13.2)$$

The associated decimal chart determines $\text{ord}_{10}(\Sigma_H) = \lfloor \log_{10} \Sigma_H \rfloor$.

The sixteenth power thus arises from the number of sheets of the quotient, once this local reader has been fixed. The factor φ acts once on the basis of the action line. The Smith computation alone would not select any possible functional on the quotient: the multiplicative norm and the self-scale form part of the declared operation.

Theorem 13.3 (Decade of the action section). *Reader (13.2), applied to the already generated HMT coordinates, satisfies*

$$10^{-34} < \Sigma_H < 10^{-33}, \quad \text{ord}_{10}(\Sigma_H) = -34. \quad (13.3)$$

Proof. The following rational intervals suffice; they are much coarser than the available cylinders:

$$\frac{1618}{1000} < \varphi < \frac{1619}{1000}, \quad \frac{7297}{10^6} < \alpha < \frac{7298}{10^6}.$$

The function $(x, a) \mapsto xa^{16}$ is increasing in both positive arguments. The two integer inequalities

$$1618 \cdot 7297^{16} > 10^{65}, \quad 1619 \cdot 7298^{16} < 10^{66}$$

imply

$$10^{-34} < \frac{1618}{1000} \left(\frac{7297}{10^6} \right)^{16} < \Sigma_H < \frac{1619}{1000} \left(\frac{7298}{10^6} \right)^{16} < 10^{-33}.$$

The definition of decimal order completes the proof. No SI unit or reference mass enters these inequalities. $\square$

The normalized section begins with $\Sigma_H = 1.0462438381 \dots \times 10^{-34}$. Its mantissa is not identified with the action coordinate before or after return. The norm-and-self-scale reader fixes the decimal order; the two return coordinates are obtained through the following readers.

13.2 Regional persistence and uniqueness of the action selector

The closure microfibre in Section 5 supplies the regional antecedent of the action reader. For the oriented ternary reduction $\Psi : \mathcal{U}_6 \rightarrow \mathbb{F}_3^6$ already constructed, let $\mathcal{F}_\pi = \Psi^{-1}(010211)$. Its five emissions, separated into initial and terminal blocks, are

$$\begin{aligned} &(501, 614, 498 \mid 169, 272, 272), \\ &(810, 923, 870 \mid 169, 272, 272), \\ &(810, 983, 810 \mid 169, 272, 272), \\ &(870, 923, 810 \mid 169, 272, 272), \\ &(870, 923, 810 \mid 418, 674, 521). \end{aligned}$$

The return projection is

$$p_{\text{ret}} : \mathbb{Z}^6 \longrightarrow \mathbb{Z}^3, \quad p_{\text{ret}}(u_1, \dots, u_6) = (u_4, u_5, u_6).$$

Each emission is counted once at this step: persistence is selected among the five regions, while the multiplicities of the seeds producing those emissions remain a separate record.

Proposition 13.4 (Persistent terminal block). *The multiset $p_{\text{ret}}(\mathcal{F}_\pi)$ has a unique mode,*

$$b_\pi = (169, 272, 272), \quad \text{mult}(b_\pi) = 4.$$

The remaining block, $b_\pi^- = (418, 674, 521)$, has multiplicity one.

Proof. The first four emissions project to b_π ; the fifth projects to b_π^- . The two blocks differ. The maximum multiplicity is therefore four and is attained by exactly one block. $\square$

The selector is defined on multisets of terminal blocks with a unique mode of type (r, s, s) , $r \neq s$. The group $\mathfrak{S}_3$ acts by relabelling the positions of these blocks; this equivariance compares their representations without adding a dynamical symmetry to the fixed fibre $\Psi^{-1}(010211)$. The stabilizer of b_π exchanges the positions occupied by 272 and fixes the position occupied by 169. For a block of type (r, s, s) , with $r \neq s$, the value at this fixed position is

$$\text{sing}(b) = \sum_{j=1}^3 b_j - 2 \text{moda}(b).$$

Selecting the regional mode and then applying this operation defines the residual selector ρ_π .

Theorem 13.5 (Equivariant uniqueness of the return). *Among selectors invariant under permutations of the five regions, equivariant under permutations of terminal positions, selecting the block of maximum multiplicity and retaining a position fixed by its stabilizer, the return value is unique:*

$$\rho_\pi = \text{sing}(b_\pi) = 169.$$

Proof. Invariance with respect to the regions makes the selection depend on the multiset rather than its enumeration. Modal persistence selects b_π by Proposition 13.4. Its stabilizer has two position orbits: a pair and a fixed point. The fixed-position requirement selects the latter, whose value is 169. Under a permutation of positions, equivariance transports this position while retaining its value. The construction satisfies all four conditions, which also proves existence. $\square$

The oriented difference between the two returns retains further data:

$$\omega_\pi = b_\pi^- - b_\pi = (249, 402, 249), \quad \langle (1, 1, 1), \omega_\pi \rangle = 900.$$

The selector 169 is terminal data of degree zero. The difference ω_π assigns degree-one data to the oriented edge between the two blocks; treating it as a cocycle requires specifying the complex and

checking the corresponding coboundary. This distinction preserves both the return and its variation as their respective objects.

The selection is regional: in the full catalogue of 468 emissions, 169 occurs 84 times, 14 in each position. Its role in the reader follows jointly from the microfibre, modal persistence and the stabilizer; its global frequency remains another count of the same catalogue.

13.3 Word length and the determinant-based return character

Let $\mathcal{P}_{\text{pal}} = \bigoplus_{n \geq 0} \mathcal{P}_{\text{pal},n}$ be the free module on finite words over the emission alphabet, graded by length, with the induced filtration. This domain contains regional emissions and their concatenations; concatenation adds degrees. The formal length character

$$\chi_z([w]) = z^{|w|}, \quad \chi_z([uv]) = \chi_z([u])\chi_z([v])$$

preserves this operation. Cylindrical compatibility of the reader sends $\mathcal{P}_{\text{pal},n}$ to the homogeneous component $z^n \mathbb{R}$ of $\mathbb{R}[z]$; evaluation at $z = \alpha$ follows afterwards. Since every $w \in \mathcal{F}_\pi$ has length six, the residual impulse is independent of the chosen region and has degree six:

$$\rho_\pi \chi_z([w]) = 169z^6, \quad \rho_\pi \chi_z([w])|_{z=\alpha} = 169\alpha^6.$$

The selection records one common impulse; it does not sum the five emissions again after extracting their persistent return. Emission length and path length in the incidence graph grade different objects. In particular, closed paths through the four regional anchors have minimum length eight; the degree used here counts the six positions of an emission rather than the edges of such a path.

Dodecaphase closure and completion of the three nonadic pairs supply the common coordinate $A = 10^3\alpha$. In radians,

$$x = \frac{\pi A}{180} = \frac{50\pi}{9}\alpha.$$

Let R_h be a real two-dimensional involution with zero trace, $R_h^2 = I_2$, $\text{tr } R_h = 0$. The two-sheet transport is

$$\mathcal{T}(x, y) = \exp(-xI_2 - yR_h).$$

The oriented coordinate y remains unevaluated. Since the polynomial $X^2 - 1$ has distinct roots and R_h has zero trace, its eigenvalues are $1, -1$. In an eigenbasis, the transport factors are e^{-x-y} and e^{-x+y} . Hence

$$\det \mathcal{T} = e^{-x-y}e^{-x+y} = e^{-2x} = \exp\left(-\frac{100\pi\alpha}{9}\right) = D_A.$$

On the determinant line, $\bigwedge^2 \mathcal{T}$ acts as D_A times the identity. Their product eliminates the oriented coordinate and preserves the common contraction. The determinant is invariant under conjugation and sheet exchange $R_h \mapsto -R_h$; thus D_A is constructed before the counter-angle is evaluated.

Proposition 13.6 (Multiplicative return character). *The multiplicative character $\delta_A : (\mathbb{N}_0, +) \rightarrow (\mathbb{R}_{>0}, \cdot)$ whose value on one elementary extension is D_A is determined by $\delta_A(k) = D_A^k$.*

Proof. A character preserves the identity, so $\delta_A(0) = 1$. Concatenating k extensions gives $\delta_A(k) = \delta_A(1)^k = D_A^k$. These powers satisfy multiplicativity and the prescribed elementary value, proving existence and uniqueness. $\square$

Definition 13.7 (Rules of regional continuation). The regional determinant-based reader preserves five operations:

1. cylindrical length determines analytic degree;
2. the return impulse and strict continuation are distinct additive terms in a renewal;
3. after recording ρ_π once, continuation starts from the unit of the determinant line;
4. each extension acts through $\bigwedge^2 \mathcal{T}$, with scalar factor D_A ;
5. in the cardinal orientation of APP, regional return occupies the compensating sector and is subtracted from the fifth-degree action phase.

Unit normalization determines the amplitude of continuation as well as its ratio. Indeed, all series

$$F_\lambda(z) = 169z^6 + \lambda \frac{D_A z^7}{1 - D_A z} \quad (\lambda \in \mathbb{R})$$

preserve the finite impulse and the same multiplicative recurrence for tail coefficients. The third rule fixes $\lambda = 1$ before evaluation at $z = \alpha$. The residue is recorded once; extension counts the continuing line with its unit. The renewal equation (13.12) and its uniqueness proof, developed below, complete the chain from region and degree through transport to the action section.

13.4 Action sections and dimensionless return quotient

Let $\mathcal{L}_S$ be an oriented action line, and let $\mathcal{U}_S$ be its positive basis. The fifth-order character and the regional return term have the expressions

$$H_5 = \frac{1}{2} \sqrt{\frac{1000\alpha}{\varphi}} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5, \quad (13.4)$$

$$D_A = \exp\left(-\frac{100\pi\alpha}{9}\right), \quad (13.5)$$

$$\mathcal{C}_\pi(\alpha) = 169\alpha^6 + \frac{D_A \alpha^7}{1 - D_A \alpha}, \quad (13.6)$$

$$\eta_{\text{ret}} = H_5 - \frac{90}{\pi} \mathcal{C}_\pi(\alpha). \quad (13.7)$$

These are subsequent readings of the dodecaphase closure and its regional incidence. The integer 169 is retained as the regional selector already constructed, and the coefficients of H_5 as those of its local character; they are not replaced by an interpolation of the Planck constant. The formulas make explicit exactly which readings are needed for the electronic stage. It is not necessary to introduce other functionals of mass theory here.

For $0 < \alpha < 1$, one has $0 < D_A < 1$, so $D_A \alpha < 1$. The rational term in (13.6) preserves the complete sum of the geometric tail

$$\frac{D_A \alpha^7}{1 - D_A \alpha} = \sum_{n=0}^{\infty} D_A^{n+1} \alpha^{n+7}. \quad (13.8)$$

It is not a truncation that may be silently modified when the precision changes. After the term of index N , the remaining tail is $D_A^{N+2} \alpha^{N+8} / (1 - D_A \alpha)$, allowing evaluation to be controlled without recourse to a target quantity.

Definition 13.8 (Action sections and return reader). The pre-return and post-return sections are

$$\hbar_{\text{pre}} = H_5 10^{-34} \mathcal{U}_S, \quad \hbar_{\text{ret}} = \eta_{\text{ret}} 10^{-34} \mathcal{U}_S. \quad (13.9)$$

Whenever $\eta_{\text{ret}} > 0$, define

$$\delta_{\text{ret}} = H_5 - \eta_{\text{ret}}, \quad R_{\text{act}} = \frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}} = \frac{H_5}{\eta_{\text{ret}}}. \quad (13.10)$$

Proposition 13.9 (Positivity and invariance of the ratio). *In the intervals used in the proof of (13.3), $0 < \eta_{\text{ret}} < H_5$, and therefore $R_{\text{act}} > 1$. The ratio does not change when the basis $\mathcal{U}_S$ is replaced by any other common positive basis.*

Proof. The terms of $\mathcal{C}_\pi$ are positive. For an elementary bound, it suffices to use $\alpha < 0.008$, $\varphi < 1.619$, $\alpha > 0.007$, $D_A < 1$, and $\pi > 3$. The root in (13.4) is greater than one, whereas the sum of the negative terms is less than 0.008001; in particular, $H_5 > 0.99$. Likewise,

$$0 < \frac{90}{\pi} \mathcal{C}_\pi < 30 \left(169(0.008)^6 + \frac{(0.008)^7}{1 - 0.008} \right) < 10^{-6}.$$

Thus $\eta_{\text{ret}} > 0.989999 > 0$ and $\eta_{\text{ret}} < H_5$. Finally, the basis change divides both action coordinates by the same positive factor, which cancels in their quotient. $\square$

Subsequent evaluation gives

$$\begin{aligned} H_5 &= 1.0545718176461544696 \dots, \\ \eta_{\text{ret}} &= 1.0545718169150390611 \dots, \\ R_{\text{act}} &= 1.0000000006932817630 \dots. \end{aligned}$$

The difference and the quotient are two readings of the same sections, one additive and the other multiplicative. They do not represent two independent Planck constants. The passage from angular action to action per complete turn is expressed, in each section, by $h = 2\pi\hbar$. The subsequent chart $\mathcal{U}_S \mapsto \mathcal{J}_S$ assigns a unit but does not act retroactively on the decimal order already proved.

Regional return can also be written in an angular chart. This reparametrization is not an additional indispensable dependency: (13.4)–(13.7) directly supply H_5 , η_{ret} , and R_{act} . If angle and counter-angle are introduced, both must be transported in the same chart; degrees and radians are not mixed within a difference.

13.4.1 Action character and provenance of its coefficients

The reduced Planck constant, $\hbar$, is the physical object that motivates realization of the angular-action section. Its position in the electronic dependency must be specified at two levels. The multiplicative norm, composed with the self-scale in Σ_H , determines the decade of the scale. The character H_5 and the regional return then determine the coordinates of the action sections. This separation allows the result needed for the electron to be presented without anticipating the full developments of other sectors of dimensional mechanics.

The provenance of the fifth-order character can be made explicit through invariants already fixed in the support. In the central chart, consider

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}, \quad \lambda_+ = 9, \quad \lambda_- = 5.$$

The calendar has length 12, the step-3 orbit has length 4, and the relevant census quotient is $104976/1944 = 54$. The character coefficients satisfy

$$\frac{9}{16} = \frac{\lambda_+}{4^2}, \quad \frac{5}{9} = \frac{\lambda_-}{\lambda_+}, \quad \frac{7}{48} = \frac{\text{tr}(B_c)/2}{4 \cdot 12}, \quad \frac{1}{54} = \frac{1944}{104976}. \quad (13.11)$$

The assignment of these invariants to degrees 2, 3, 4, 5, with the alternating signs of (13.4), belongs to the definition of the analytic character used. Making it explicit is essential: the collection of cardinalities, considered without the character and without the ordering of its degrees, would admit other expressions. The constructive content lies in the composition of the support, orientation, and declared analytic rule.

13.4.2 Regional continuation and renewal equation

The selector 169 determines the degree-six impulse. Subsequent prolongation is organized by the transfer D_A and a unit normalization of the determinant line. These rules admit a formulation in formal series before substituting $z = \alpha$, with D_A already constructed. Write

$$\mathcal{C}(z) = 169z^6 + R(z), \quad R(z) \in z^7\mathbb{R}[[z]].$$

Concatenating a prolongation increases the degree by one and multiplies its weight by D_A . The first strict prolongation has weight D_A . Its renewal equation is therefore

$$R(z) = D_A z^7 + D_A z R(z). \quad (13.12)$$

Proposition 13.10 (Formal uniqueness of the normalized continuation). *Equation (13.12) determines a unique series, and*

$$\mathcal{C}(z) = 169z^6 + \frac{D_A z^7}{1 - D_A z} = 169z^6 + \sum_{n=7}^{\infty} D_A^{n-6} z^n.$$

The realization is analytic for $|D_A z| < 1$.

Proof. The element $1 - D_A z$ has constant term one and is invertible in the ring of formal series. Solving the equation produces the displayed series. Equivalently, the degree-seven coefficient is D_A , and each subsequent coefficient is D_A times the preceding one. The geometric series converges absolutely in the indicated disk. $\square$

Normalization of the first prolongation is an indispensable operative datum. If only the degree-six impulse and the concatenation ratio were retained, the series

$$169z^6 + \lambda \frac{D_A z^7}{1 - D_A z}$$

would still share both data for every λ . The unit rule fixes $\lambda = 1$ before evaluation is performed. This locates the contribution of each rule to the uniqueness of the reader, and the rational expression preserves the full continuation.

13.4.3 Reduced Planck constant and return of the action section

In the dimensional chart of (13.9), the pre-return section is the model construction intended for comparison with the reduced Planck constant:

$$\hbar_{\text{pre}} = H_5 10^{-34} \mathcal{U}_S.$$

The post-return section additionally preserves the regional compensation $90\mathcal{C}(\alpha)/\pi$. Both belong to the same action line, and their quotient R_{act} measures the multiplicative effect of return. The passage to h through 2π uses the relation between angular parametrization and a complete turn.

This construction enters effectively into $R_{\text{act}}^{80-54\alpha+6\Delta_4}$. The electron preserves its central coordinate and its trajectory; the chronological scale does not replace that structure with an undifferentiated quantum. Nor is a mass already expressed in MeV multiplied again by 10^{-34} : the decade belongs to the action section, cancels in its dimensionless quotient, and reappears only where the corresponding dimensional realization acts.

The numerical comparison of H_5 , the returned section, and $h/(2\pi)$ is presented in Section A.9. This distinguishes the name of the physical object, the equation assigned to it by the model, and the precision with which that equation reproduces it.

13.5 Structural definition and role of the Planck constants

Definition 13.11 (Structural determination of angular action). On the oriented line $\mathcal{L}_S$, the HMT determination of angular action is the pair of sections $\hbar_{\text{pre}}, \hbar_{\text{ret}}$ in (13.9), together with their identification by return. Its structure combines the valuation of the local norm $\varphi N_{G_H}(\alpha)$, the character H_5 , and the regional continuation $\mathcal{C}_\pi$. The action per complete turn is $h_a = 2\pi\hbar_a$ in each orientation a .

This definition preserves three successive determinations. The local quotient in (13.1) fixes the sixteen sheets of the norm; the inequality (13.3) fixes its decimal order. The character and the normalized renewal in (13.12) then determine the pre-return and post-return coordinates. Uniqueness of the tail uses its first weight and its concatenation rule; the distribution of the coefficients of H_5 preserves the character declared in Section 13.4.1. On the positive branch, the multiplicative information of the return is the invariant

$$\frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}} = R_{\text{act}} = \frac{H_5}{\eta_{\text{ret}}} > 1. \quad (13.13)$$

Its physical role is expressed through the contraction between action and frequency: a section $\hbar_a$ assigns energy to an angular frequency, whereas h_a expresses the action of one turn. The quotient (13.13) preserves the structural modification between the two readings even when the common basis of units is changed. Identification with the physical constant and its precision are examined in the chart of Section A.9, with the two return coordinates distinguished.

13.6 Angular coordinates, normalization of the turn, and units of action

The quantity of action and the angular coordinate belong to distinct spaces. The section $\hbar_{\text{ret}}$ occupies the action line $\mathcal{L}_S$; η_{ret} is its coefficient in the declared basis. The map to angular coordinates acts on the dimensionless pair $(\alpha, \eta_{\text{ret}})$, after its construction:

$$\mathcal{P}(\alpha, \eta_{\text{ret}}) = ([1000\alpha]_{360}, [2(\eta_{\text{ret}} + \alpha)]_{360}). \quad (13.14)$$

Nonadic completion determines the factor 1000; bidirectional closure determines the factor two. The circular chart of 360 units makes the partitions $12 \cdot 30 = 4 \cdot 90 = 3 \cdot 120$ visible. On the positive branch of the article, the representatives are

$$A_{\text{deg}} = 1000\alpha, \quad C_{\text{deg}}^* = 2(\eta_{\text{ret}} + \alpha).$$

Proposition 13.12 (Shift of the base-thousand expansion). *Let $\alpha = \sum_{n \geq 1} b_n 1000^{-n}$, with $b_n \in \{0, \dots, 999\}$, be the compatible expansion already generated. The lifted coordinate $A_{\text{deg}} = 1000\alpha$ satisfies*

$$A_{\text{deg}} = b_1 + \sum_{n \geq 1} b_{n+1} 1000^{-n}.$$

Multiplication by 1000 preserves the block sequence from the second block onward; the initial block becomes the integer part.

Proof. The series converges absolutely. Multiplying each term by 1000 and separating the initial term gives the identity. The operation acts on the constructed expansion, with its blocks and memory; the subsequent circular projection takes its class modulo 360. $\square$

Definition 13.13 (Charts of the angular coordinate). The degree chart expresses a complete turn in 360 units. The radian chart expresses it as 2π . The coordinate changes and their inverses are

$$x = \frac{\pi}{180} A_{\text{deg}}, \quad y = \frac{\pi}{180} C_{\text{deg}}^*, \quad A_{\text{deg}} = \frac{180}{\pi} x, \quad C_{\text{deg}}^* = \frac{180}{\pi} y.$$

On a path with preserved winding number, the lifted coordinates are used; the circular quotient represents its class modulo the corresponding turn.

The definition characterizes degrees and radians through their exact relation to the turn. The record also retains orientation, turns, and transport memory. Projection onto its angular class preserves phase, whereas the lift distinguishes paths with the same final phase. The physical reading of action requires specifying which of these coordinates enters the integral.

Proposition 13.14 (Covariance of the action reader). *For $h_a = 2\pi\hbar_a$, the action of a lifted coordinate satisfies*

$$S_a(\theta) = \hbar_a \theta_{\text{rad}} = h_a \frac{\theta_{\text{deg}}}{360}. \quad (13.15)$$

The action of a positive turn is h_a .

Proof. Substituting $\theta_{\text{rad}} = \pi\theta_{\text{deg}}/180$ gives $\hbar_a\pi\theta_{\text{deg}}/180 = (2\pi\hbar_a)\theta_{\text{deg}}/360$. In one turn $\theta_{\text{deg}} = 360$, yielding h_a . $\square$

The periodicity of a representation is declared separately. If the transported representation satisfies $U(2\pi) = -I$, its operator restoration occurs at 4π . This property of the lift coexists with the per-turn normalization of (13.15). Each preserves its own object: integrated quantity, circular phase, and represented state.

13.7 Covariance of the channels and the incidence functional

The channels are written equivalently as

$$q_{\pm} = \exp[-(x \pm y)] = \exp\left[-\frac{\pi}{180}(A_{\text{deg}} \pm C_{\text{deg}}^*)\right]. \quad (13.16)$$

Proposition 13.15 (Recovery of the even and odd coordinates). *In the lifted real chart, the positive channels uniquely determine the two coordinates:*

$$A_{\text{deg}} = -\frac{90}{\pi} \log(q_+ q_-), \quad C_{\text{deg}}^* = \frac{90}{\pi} \log \frac{q_-}{q_+}.$$

The exchange $(q_+, q_-) \mapsto (q_-, q_+)$ preserves A_{deg} and reverses C_{deg}^ .*

Proof. The sum and difference of the logarithms of (13.16) are, respectively, $-\pi A_{\text{deg}}/90$ and $-\pi C_{\text{deg}}^*/90$. The real logarithm is single-valued on the positive channels. The product is symmetric, and the quotient is inverted when they are exchanged. $\square$

In the twelve-position record, the antipodal involution $r \mapsto r + 6 \pmod{12}$ forms six pairs. The uniform distribution of the odd coordinate over these pairs is $C_{\text{deg}}^*/6$: summing their six contributions recovers C_{deg}^* . This sectorial coordinate is distinguished from the Planck constant h , which belongs to the action line.

The conversion is applied only once to each argument. The quantity α remains dimensionless, and the quantity $\hbar$ continues to belong to the action line; their preceding coordinates enter the map $\mathcal{P}$.

In addition to the channels, the Barbero–Immirzi functional contains a component bilinear in the angular representatives. Its transport is

$$\frac{\pi}{180} A_{\text{deg}} C_{\text{deg}}^* = \frac{180}{\pi} xy. \quad (13.17)$$

The identity follows by substituting both degree representatives with their expressions in radians. The complete formula in analytic coordinates is therefore

$$\gamma_{\text{HMT}} = \frac{180}{\pi} xy + 12[S_{90}(e^{-x+y}) - S_{90}(e^{-x-y})] - [S_{120}(e^{-x+y}) - S_{120}(e^{-x-y})].$$

Equivalence of the two charts transports the bilinear component and the exponential functions simultaneously. This transformation makes explicit the precise role of the degree in the construction and allows the evaluation to be reproduced without ambiguity in angular units.

14 Hexadic–octadic incidence and the Barbero–Immirzi functional

Let $H \hookrightarrow O$ be a marked inclusion, $|H| = 6$, $|O| = 8$. The incidence objects

$$\mathcal{F}_6 = H \times \binom{H}{2}, \quad \mathcal{F}_8 = O \times \binom{H}{2}$$

have cardinalities 90 and 120. Their normalized difference gives

$$\frac{|\mathcal{F}_6| - |\mathcal{F}_8|}{24\binom{6}{2}} = -\frac{1}{12}.$$

This is a combinatorial realization of the same scalar. A numerical equality with (12.1) does not erase the distinction between the spaces on which it was computed.

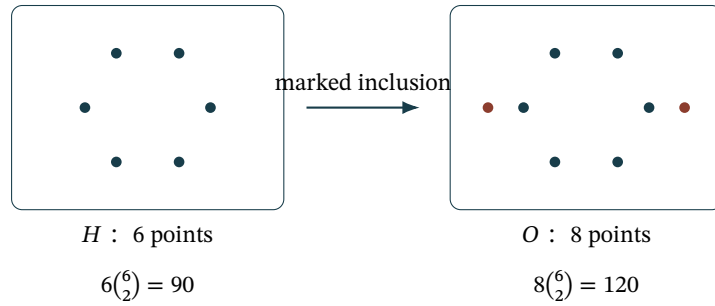


Figure 8: Marked incidence from six to eight points. The drawing represents the sets and the inclusion; it does not purport to turn the faces or vertices of a cube into Steiner blocks without their incidence map.

14.1 The weights precede the pole coefficient

In the free module of oriented events, the source fixes the fundamental chain of the turn [Section 14, equation (14.1)]

$$c_{12}^+ = \sum_{j \in \mathbb{Z}/12\mathbb{Z}} [j, \mathcal{F}_6] - [\partial_{\text{ret}}, \mathcal{F}_8].$$

Multiplicity twelve corresponds to the sectors, and multiplicity one to the oriented seam. The involution changes the orientation of the complete chain. Let

$$S_m(q) = \frac{q^m}{1 - q^{3m}}, \quad 0 < q < 1.$$

The linear reader assigns S_{90} to each sector and S_{120} to the seam; consequently,

$$\mathcal{R}(c_{12}^+) = 12S_{90} - S_{120}.$$

Theorem 14.1 (Pole coefficient and conjugate evaluation). *The image of the fundamental chain has coefficient $1/24$ with respect to $(1 - q)^{-1}$ and residue $-1/24$ with respect to $(q - 1)^{-1}$. Its evaluation in the previously generated channels determines*

$$\gamma_{\text{HMT}} = \frac{\pi AC^*}{180} + 12[S_{90}(q_-) - S_{90}(q_+)] - [S_{120}(q_-) - S_{120}(q_+)]. \quad (14.1)$$

Proof. The factorization $1 - q^{3m} = (1 - q) \sum_{j=0}^{3m-1} q^j$ implies $\lim_{q \uparrow 1} (1 - q)S_m(q) = 1/(3m)$. By linearity,

$$\frac{12}{270} - \frac{1}{360} = \frac{1}{24}.$$

The coordinate $q - 1$ reverses the sign of the residue. Conjugate evaluation and the angular component of the source produce (14.1). $\square$

The Diophantine certificate $4a - 3b = 45$ admits the pairs $(12 + 3t, 1 + 4t)$, $t \geq 0$; it certifies minimality of $(12, 1)$, but does not replace the origin of the multiplicities in the chain. The evaluation of (14.1) is

$$\gamma_{\text{HMT}} = 0.27400767269445927474725526077398041940 \dots$$

This number is an output of the source functional; it is not an external value used to select A , C^* , or their coefficients.

14.2 Parity and structural sensitivity

Reversing C^* exchanges q_+ and q_- and changes the sign of the angular component. Hence

$$\gamma_{\text{HMT}}(A, -C^*) = -\gamma_{\text{HMT}}(A, C^*).$$

The functional preserves orientation, whereas other readers, such as q_+q_- , are even. This difference precedes their physical interpretation.

14.3 Structural dependence of the functional and area normalization

The preceding construction places the Barbero–Immirzi parameter at the meeting of three determinations: marked incidence, dodecaphasic orientation, and angular evaluation. Incidence supplies the event spaces; orientation fixes the chain on those spaces; evaluation transports the chain to a dimensionless quantity. This decomposition preserves the origin of each coefficient when the functional is subsequently used in area geometry.

The inclusion $\mathcal{F}_6 \hookrightarrow \mathcal{F}_8$ retains the marked pair of the hexad. Its complement has cardinality $2\binom{6}{2} = 30$. This increment reappears as the determinant of the map reconstructing edge occupations from area and the weighted observable. Equality of the two integers has a concrete explanation here: the sector weights differ precisely by the number of events added by the marked inclusion.

Proposition 14.2 (Incidence and recoverability of occupations). *Let n_6, n_8 be the aggregate occupations of two sectors, and let $r_6 = |\mathcal{F}_6|$, $r_8 = |\mathcal{F}_8|$. The map*

$$(n_6, n_8) \longmapsto (n_6 + n_8, r_6 n_6 + r_8 n_8)$$

is invertible on its image when the inclusion adds at least one event. For the constructed inclusion, its determinant is 30.

Proof. The matrix of the map is $\begin{pmatrix} 1 & 1 \\ r_6 & r_8 \end{pmatrix}$. Its determinant is $r_8 - r_6 = |\mathcal{F}_8 \setminus \mathcal{F}_6|$. The inclusion considered adds $2\binom{6}{2} = 30$ events. Linear inversion recovers both occupations and preserves the integer restrictions of its image. $\square$

The result recovers aggregate occupations. Information about each link and the order of events remains in the full state and its additional readers. This hierarchy gives a precise formulation of the difference between area value and boundary genealogy: area preserves a sum; the pair of observables preserves the decomposition of that sum by sectors; the record additionally preserves its distribution and history.

The parameter γ_{HMT} transports that incidence to the area scale. Its structural interpretation is an oriented coefficient of the hexadic–octadic return functional. The gravitational realization subsequently recognizes it in the Holst action. Its intrinsic definition, value, and function in the physical equation are thus distinguished.

14.4 Structural definition and role of the Barbero–Immirzi parameter

Definition 14.3 (Oriented evaluation of hexadic–octadic incidence). The parameter γ_{HMT} is the conjugate evaluation of the fundamental chain c_{12}^+ , completed by its angular component. On the domain $0 < q_{\pm} < 1$, its definition (14.1) takes the form

$$\gamma_{\text{HMT}} = \frac{\pi AC^*}{180} + [\mathcal{R}(c_{12}^+)](q_-) - [\mathcal{R}(c_{12}^+)](q_+). \quad (14.2)$$

The marked incidence supplies the event spaces; the oriented chain fixes their multiplicities; the channels in (13.16) perform the evaluation.

The determination jointly uses the cardinalities 90, 120, the twelve sector contributions, and the return seam. The pole coefficient $1/24$ and Diophantine minimality verify the compatibility of these weights with the normalization. Their provenance is preserved in c_{12}^+ , before the channels are evaluated. Angular inversion exchanges q_+ and q_- and produces $\gamma_{\text{HMT}}(A, -C^*) = -\gamma_{\text{HMT}}(A, C^*)$: the dimensionless value transports an orientation of the incidence.

In the geometric realization, this same coefficient enters (21.2) and the area increment $\gamma_{\text{HMT}} a_0 \ell_P^2$. The Holst action uses $\gamma_{\text{HMT}} \neq 0$; the positive areal magnitude uses $|\gamma_{\text{HMT}}|$. Proposition 21.10 makes the coordinated change of orientation explicit. Thus, the structure of the parameter explains what it weights in the incidence, what it preserves under transport, and how it participates in the torsional response and in the geometric resolution of information.

15 Action ellipse and modular reciprocity

With the outputs $A, C^*, \hbar > 0$ fixed in the chart $|C^*/A| < 1$, we define

$$\eta_{\text{el}} = \text{artanh}(C^*/A), \quad a_S = \sqrt{2\hbar} e^{\eta_{\text{el}}/2}, \quad b_S = \sqrt{2\hbar} e^{-\eta_{\text{el}}/2}.$$

The balanced canonical coordinates have units of the square root of action; position and momentum are not added here without a dimensional chart. The following hold:

$$a_S b_S = 2\hbar, \quad \tau_{\text{el}} = i \frac{b_S}{a_S}, \quad R_{\text{el}} = \sqrt{\frac{b_S}{a_S}} = \left(\frac{A - C^*}{A + C^*} \right)^{1/4}.$$

The positive root determines the composition $\tau_{\text{el}} = iR_{\text{el}}^2$, and makes it possible to recover the angular ratio:

$$\frac{C^*}{A} = \frac{1 - R_{\text{el}}^4}{1 + R_{\text{el}}^4}.$$

Shape inversion sends R_{el} to R_{el}^{-1} and τ_{el} to $-1/\tau_{\text{el}}$ [Section 15 and Proposition 29.1].

Proposition 15.1 (Coincident energy, distinguishable action). *Let*

$$D(\eta) = \begin{pmatrix} e^\eta & 0 \\ 0 & e^{-\eta} \end{pmatrix}, \quad S_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Both maps satisfy $T^\top D(-\eta)T = D(\eta)$. However, for $\omega = dQ \wedge dP$,

$$S_2^* \omega = -\omega, \quad J_2^* \omega = \omega.$$

If $\mathcal{I}(\gamma) = \oint_\gamma P dQ$ on a closed curve,

$$\mathcal{I}(S_2 \gamma) = -\mathcal{I}(\gamma), \quad \mathcal{I}(J_2 \gamma) = \mathcal{I}(\gamma).$$

Proof. Conjugation exchanges the diagonal entries of D . The determinants of S_2, J_2 are $-1, 1$, respectively, which proves the transformation of ω . For $\lambda = P dQ$,

$$S_2^* \lambda = d(PQ) - \lambda, \quad J_2^* \lambda = \lambda - d(PQ).$$

Integrating the exact differential over the closed curve completes the proof. $\square$

The quadratic energy and the modulus do not suffice to distinguish the two transports. The orientation of action does distinguish them, even in the circular shape. The geometric marking is thus part of the recoverable reading, not an embellishment added to the value.

15.1 Semiaxes, foci, and interior radii

The figure is obtained from the angular and action outputs of the preceding sections. APP preserves the sheets and their residues and quotients; TRIT fixes orientation; TPK transports boundary, carry, and memory through to the representations of the same enriched state and the joint discrete structure of the continuum. The pair (A, C^*) and the positive section $\hbar$ are received here after that construction. The geometry makes explicit what information their readers preserve, without using an observed length to select the source state [Section 15 and Proposition 29.1]; [Section 29].

Let $\xi = C^*/A \in (-1, 1)$, with $A \neq 0$ and both angles in the same unit. Write $\eta = \text{artanh } \xi$ and

$$a_S = \sqrt{2\hbar} e^{\eta/2}, \quad b_S = \sqrt{2\hbar} e^{-\eta/2}, \quad \gamma(\theta) = (a_S \cos \theta, b_S \sin \theta).$$

The semiaxes are labeled by their coordinates: if $\eta < 0$, the vertical axis is the larger. Both belong to the square-root-of-action line of the balanced chart, not to the line of spatial lengths.

Proposition 15.2 (Focal geometry and radial interior). *Let $a = \max(a_S, b_S)$, $b = \min(a_S, b_S)$. The focal half-distance and eccentricity are*

$$f_S = \sqrt{a^2 - b^2} = \sqrt{4\hbar \sinh |\eta|}, \quad e_f = \frac{f_S}{a} = \sqrt{1 - e^{-2|\eta|}}. \quad (15.1)$$

For $\eta > 0$, the foci are $(\pm f_S, 0)$; for $\eta < 0$, they are $(0, \pm f_S)$. At $\eta = 0$, they coincide with the center. The boundary radius in polar direction ϕ is

$$\rho_\partial(\phi) = \left(\frac{\cos^2 \phi}{a_S^2} + \frac{\sin^2 \phi}{b_S^2} \right)^{-1/2}. \quad (15.2)$$

The interior segment in that direction consists of $r(\cos \phi, \sin \phi)$, with $0 \leq r < \rho_\partial(\phi)$. In particular, $b \leq \rho_\partial(\phi) \leq a$.

Proof. The squares of the major and minor semiaxes are $2\hbar e^{|\eta|}$ and $2\hbar e^{-|\eta|}$. Their difference gives (15.1). To verify the focus, suppose $\eta > 0$. At $X = \gamma(\theta)$, the distances to $F_+ = (f_S, 0)$ and $F_- = (-f_S, 0)$ satisfy

$$|X - F_+|^2 = (a_S - f_S \cos \theta)^2, \quad |X - F_-|^2 = (a_S + f_S \cos \theta)^2.$$

Since $a_S > f_S$, their positive roots sum to $2a_S$. For $\eta < 0$, exchanging coordinates gives the same proof. In the circular case, $f_S = 0$. Finally, substituting $(Q, P) = r(\cos \phi, \sin \phi)$ into $Q^2/a_S^2 + P^2/b_S^2 = 1$ gives (15.2). The coefficient of r^2 lies between a^{-2} and b^{-2} , proving the bounds and the characterization of the interior. $\square$

15.2 Parametric phase, polar angle, and action area

Proposition 15.3 (Areal law and two angular coordinates). *The phase θ of γ and its continuous polar angle ϕ , lifted from zero, satisfy*

$$\phi(\theta) = \arg(a_S \cos \theta + ib_S \sin \theta), \quad \frac{d\phi}{d\theta} = \frac{a_S b_S}{a_S^2 \cos^2 \theta + b_S^2 \sin^2 \theta} > 0. \quad (15.3)$$

In a chart where both tangents are defined, $\tan \phi = (b_S/a_S) \tan \theta$. The oriented swept area is

$$\mathcal{A}(\theta) = \frac{1}{2} \int_0^\theta (QP' - PQ') dt = \hbar \theta, \quad \mathcal{A}(1) = \hbar, \quad \mathcal{A}(2\pi) = h. \quad (15.4)$$

Proof. The derivative of the argument of a nonzero vector is $(QP' - PQ')/(Q^2 + P^2)$. Here the numerator is $a_S b_S = 2\hbar$, proving the derivative and monotonicity. The quotient P/Q gives the tangent relation; the continuous lift fixes the quadrants and the number of turns. The areal integral is $a_S b_S \theta/2 = \hbar \theta$, and $h = 2\pi \hbar$ gives the final identity. $\square$

Parametric phase does not generally coincide with polar angle. One radian of phase sweeps $\hbar$, but is not thereby identified with a circular arc or a sector of unit polar angle. In polar coordinates, the same law is written $d\mathcal{A} = \rho_\partial(\phi)^2 d\phi/2$. With the orientation of γ used here, $\oint P dQ = -h$: positive area is h , whereas the integral of that one-form preserves its sign. This agrees with the distinction between symplectic and antisymplectic transports in the preceding section.

For the control case of the figure, $\xi = 1/3$, the normalized coordinates are exactly

$$\eta = \frac{1}{2} \log 2, \quad \frac{a_S}{\sqrt{2\hbar}} = 2^{1/4} = 1.1892071150 \dots, \quad \frac{b_S}{\sqrt{2\hbar}} = \frac{f_S}{\sqrt{2\hbar}} = R_{\text{el}} = 2^{-1/4} = 0.8408964153 \dots$$

These are regression numbers for the drawing, not values assigned to the angular generator. The local certificate also checks the circular shape and the orientations $\xi = \pm 1/3, \pm 3/5$.

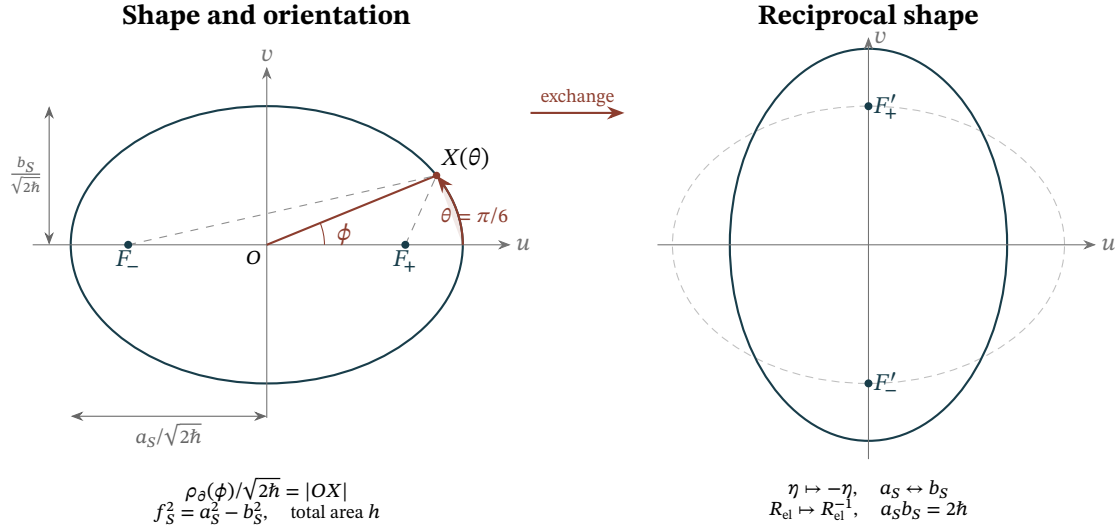


Figure 9: Semiaxes, foci, and reciprocity of the action ellipse in the normalized coordinates $u = Q/\sqrt{2\hbar}$, $v = P/\sqrt{2\hbar}$. The marked sector has parametric phase $\theta = \pi/6$, different from its polar angle ϕ ; its action area is $\hbar\pi/6$. The figure uses the algebraic case $\xi = 1/3$ to make the relations legible: it does not assign that value to the HMT angular output. The foci are computed from the semiaxes. On the right, exchanging axes preserves total area h and inverts the dimensionless modular radius.

15.3 Shape reciprocity and interior distance

The representation in rectangular coordinates recovers

$$R_{el} = \sqrt{b_S/a_S} = e^{-\eta/2}, \quad R_{el}^4 = \frac{1-\xi}{1+\xi}, \quad \xi = \frac{1-R_{el}^4}{1+R_{el}^4}. \quad (15.5)$$

The proof consists in substituting $2\eta = \log((1+\xi)/(1-\xi))$. Exchanging axes sends η to $-\eta$, R_{el} to R_{el}^{-1} , and $\tau = iR_{el}^2$ to $-1/\tau$. The product $a_S b_S$ is preserved. The radius R_{el} is dimensionless; the polar radius ρ_∂ has square-root-of-action units.

The interior also admits the Hilbert projective distance. If z_- , x , y , z_+ are ordered on a chord of the ellipse, define

$$d_H(x, y) = \frac{1}{2} \log \frac{|y - z_-| |x - z_+|}{|x - z_-| |y - z_+|}.$$

It is a dimensionless distance: the units cancel in the cross-ratio. The normalization $(Q, P) \mapsto (Q/a_S, P/b_S)$ takes the ellipse to the unit disk and preserves that ratio, because lengths on a single line are multiplied by a common factor. It thus transports the metric to the Beltrami–Klein model. Neither this interior distance nor the distance between foci is identified with an atomic length.

15.4 Bohr radius: a consequence of the action scale

In the electrodynamic realization, the Bohr scale is expressed as $a_B = \hbar/(m_e c \alpha)$ [6]. In the chain of the present article, this reading follows the constructions of action, vacuum, fine structure, and electronic mass. Here $\hbar, m_e, c, \alpha > 0$ are received in the same dimensional realization.

Proposition 15.4 (Areal expression of the Bohr scale). *In that realization,*

$$a_B = \frac{a_S b_S}{2m_e c \alpha} = \frac{\mathcal{A}(2\pi)}{2\pi m_e c \alpha} = \frac{\bar{\lambda}_C(m_e)}{\alpha}, \quad \bar{\lambda}_C(m_e) = \frac{\hbar}{m_e c}. \quad (15.6)$$

Proof. The identities $a_S b_S = 2\hbar$ and $\mathcal{A}(2\pi) = 2\pi\hbar$, substituted into the Bohr expression, give the first two members. The definition of $\bar{\lambda}_C(m_e)$ gives the third. $\square$

The numerator has action dimension, and $m_e c$ has momentum dimension; the quotient is a length. This composition uses the product of the semiaxes and the other three quantities: it does not assert that Bohr is a semiaxis, a focus, a Hilbert distance, or the modular radius. Nor does it equate electronic mass with the circular quantum of order 108. The notation a_B avoids confusing this length with the normalization a_0 of the area operator.

15.5 Inductive–capacitive realization and constitutive continuity

The electrodynamic bridge of the ellipse preserves an additional chart: a clock $\omega > 0$ and a reference impedance $Z_* > 0$. The construction recovered from the corpus defines

$$L_\eta = \frac{Z_*}{\omega} e^{-\eta}, \quad C_\eta = \frac{1}{Z_* \omega} e^\eta, \quad Z_\eta = \sqrt{L_\eta / C_\eta}. \quad (15.7)$$

Here C_η is capacitance, distinct from the counter-angle C^* .

Proposition 15.5 (Dynamical product and constitutive quotient). *In the preceding chart,*

$$L_\eta C_\eta = \omega^{-2}, \quad \frac{Z_\eta}{Z_*} = e^{-\eta} = \frac{b_S}{a_S} = R_{\text{el}}^2. \quad (15.8)$$

With $Q = \sqrt{Z_*} q$, $P = \Phi / \sqrt{Z_*}$, the energy $H_{LC} = q^2 / (2C_\eta) + \Phi^2 / (2L_\eta)$ takes the form

$$H_{LC} = \frac{\omega}{2} (e^{-\eta} Q^2 + e^\eta P^2).$$

On $\gamma(\theta)$, its electric and magnetic components are $U_E = \hbar \omega \cos^2 \theta$ and $U_B = \hbar \omega \sin^2 \theta$; their sum is $\hbar \omega$.

Proof. The product in (15.7) cancels the exponents and Z_* ; the quotient is $Z_*^2 e^{-2\eta}$. Taking the positive root and using (15.5) gives (15.8). Substituting $q = Q / \sqrt{Z_*}$ and $\Phi = \sqrt{Z_*} P$ into H_{LC} gives its balanced form. Finally, the semiaxes of γ cancel the factors e^η , and $\cos^2 \theta + \sin^2 \theta = 1$. $\square$

This bridge distinguishes shape, clock, and transducer. The choice of dynamical direction must agree with the symplectic convention: for the equations $\dot{Q} = \partial_P H$, $\dot{P} = -\partial_Q H$, the preceding parametrization satisfies $\dot{\theta} = -\omega$. The energy identities do not depend on reversing that path.

The vacuum resolvent reader acts on the already produced channels $q_\pm = e^{-(A_{\text{rad}} \pm C_{\text{rad}}^*)}$ and incidence 90/120. Its responses $r_\pm = (1 - q_\pm^{90}) / (1 - q_\pm^{120})$ give $\hat{Z} = r_- / r_+$ and $\hat{c} = (r_+ r_-)^{-1}$. These are spectral responses, not geometric radii. Identity (15.8) fixes the LC realization here; identifying it with the constitutive vacuum impedance preserves the map between the two readers and their charts. Their values are not equated merely because they share the angular antecedent.

15.6 Canonical covariance and reciprocity of the action ellipse

The preceding construction determines two complementary invariants: the product of the semiaxes fixes the action area, and their quotient preserves the anisotropy of the angular pair. We now specify the canonical realization in which those semiaxes are two standard deviations. The starting structure is the ellipse already obtained from the angular and action representations of APP–TRIT–TPK; the operator realization is established after that construction.

In this subsection we write $\eta = \eta_{\text{el}} = \text{artanh}(C^*/A)$, with $|C^*/A| < 1$ and $\hbar > 0$. The angular quotient uses the same unit for its two components. In the balanced coordinates (Q, P) , both having the dimension of the square root of action, define

$$V_\eta := \frac{1}{4} \begin{pmatrix} a_S^2 & 0 \\ 0 & b_S^2 \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} e^\eta & 0 \\ 0 & e^{-\eta} \end{pmatrix}. \quad (15.9)$$

This positive matrix has entries of action dimension. Its interpretation as quantum covariance is specified by the following realization.

Proposition 15.6 (Gaussian realization of action covariance). *Consider the canonical representation on $L^2(\mathbb{R}, dQ)$, with $\hat{Q}f(Q) = Qf(Q)$ and $\hat{P}f(Q) = -i\hbar f'(Q)$, initially on the Schwartz space $\mathcal{S}(\mathbb{R})$. The normalized state*

$$\psi_\eta(Q) = (\pi\hbar e^\eta)^{-1/4} \exp\left(-\frac{Q^2}{2\hbar e^\eta}\right) \quad (15.10)$$

has zero means and covariance matrix V_η . In particular,

$$(\Delta Q)^2 = \frac{\hbar e^\eta}{2}, \quad (\Delta P)^2 = \frac{\hbar e^{-\eta}}{2}, \quad \text{Cov}(Q, P) = 0, \quad \det V_\eta = \frac{\hbar^2}{4}. \quad (15.11)$$

This realization saturates the Robertson–Schrödinger inequality.

Proof. The preceding operators are essentially self-adjoint on $\mathcal{S}(\mathbb{R})$, which is a common invariant core; on this space, $[\hat{Q}, \hat{P}] = i\hbar I$. For $\hat{Y} = (\hat{Q}, \hat{P})$, the symmetric covariance is

$$V_{jk} = \frac{1}{2} \langle (\hat{Y}_j - \langle \hat{Y}_j \rangle)(\hat{Y}_k - \langle \hat{Y}_k \rangle) + (\hat{Y}_k - \langle \hat{Y}_k \rangle)(\hat{Y}_j - \langle \hat{Y}_j \rangle) \rangle.$$

With $s = \hbar e^\eta > 0$, let $I_s = \int_{\mathbb{R}} e^{-Q^2/s} dQ$. The product of two equal integrals, evaluated in polar coordinates, gives $I_s^2 = 2\pi \int_0^\infty r e^{-r^2/s} dr = \pi s$. Furthermore, the integral of $\frac{d}{dQ}(Qe^{-Q^2/s})$ is zero, by decay at both ends; hence $\int_{\mathbb{R}} Q^2 e^{-Q^2/s} dQ = sI_s/2$. These identities and parity prove normalization, zero mean, and $\langle \hat{Q}^2 \rangle = s/2$. Moreover, $\hat{P}\psi_\eta = ie^{-\eta}Q\psi_\eta$. Thus $\langle \hat{P} \rangle = 0$, $\|\hat{P}\psi_\eta\|^2 = \hbar e^{-\eta}/2$, and $\text{Re}\langle \hat{Q}\psi_\eta, \hat{P}\psi_\eta \rangle = 0$.

For any normalized state in that domain, Cauchy–Schwarz applied to $(\hat{Q} - \langle \hat{Q} \rangle)\psi$ and $(\hat{P} - \langle \hat{P} \rangle)\psi$ gives

$$(\Delta Q)^2(\Delta P)^2 \geq \text{Cov}(Q, P)^2 + \frac{1}{4} |[\hat{Q}, \hat{P}]|^2 = \text{Cov}(Q, P)^2 + \frac{\hbar^2}{4}.$$

The computed variances attain equality. Equivalently, the operator

$$a_\eta = \frac{e^{-\eta/2}\hat{Q} + ie^{\eta/2}\hat{P}}{\sqrt{2\hbar}}$$

satisfies $[a_\eta, a_\eta^\dagger] = I$ and $a_\eta\psi_\eta = 0$, by direct substitution. $\square$

Proposition 15.7 (Area, dispersion, and reciprocal transport). *The action ellipse is exactly the covariance sublevel set*

$$\mathcal{E}_\eta = \{y = (Q, P)^\top : y^\top V_\eta^{-1} y \leq 4\} = \left\{ \frac{Q^2}{a_S^2} + \frac{P^2}{b_S^2} \leq 1 \right\}. \quad (15.12)$$

Its area is $h = 2\pi\hbar$, its semiaxes are $a_S = 2\Delta Q$, $b_S = 2\Delta P$, and

$$\frac{\Delta P}{\Delta Q} = e^{-\eta} = \frac{b_S}{a_S} = R_{\text{el}}^2. \quad (15.13)$$

The canonical rotation J_2 of Section 15 transports V_η to $V_{-\eta}$ and preserves the symplectic form.

Proof. Inverting the diagonal matrix (15.9) gives (15.12). Its semiaxes are the positive roots of four times its variances, and

$$\text{Area}(\mathcal{E}_\eta) = 4\pi \sqrt{\det V_\eta} = 2\pi\hbar = h,$$

in agreement with the areal law (15.4). The ratio of the positive roots proves (15.13). If $M_\eta = \text{diag}(e^{\eta/2}, e^{-\eta/2})$, then

$$V_\eta = M_\eta V_0 M_\eta^\top, \quad M_\eta^\top J_2 M_\eta = J_2, \quad J_2 V_\eta J_2^\top = V_{-\eta}.$$

These identities follow by multiplication. In operator terms, $(\hat{Q}', \hat{P}') = (-\hat{P}, \hat{Q})$ preserves $[\hat{Q}', \hat{P}'] = i\hbar I$. The permutation $S_2 : (Q, P) \mapsto (P, Q)$ also exchanges the variances but reverses the sign of the commutator and of the symplectic form. Thus covariance transport respects the distinction between canonical reciprocity and orientation reversal already established in Section 15. $\square$

Position, momentum, and dimensional transduction. Let $\lambda_{xp} > 0$ be a factor of the dimensional chart, with dimension momentum per unit length. The transformation

$$Q = \sqrt{\lambda_{xp}} x, \quad P = \frac{p_x}{\sqrt{\lambda_{xp}}}$$

realizes the balanced pair as a position x and its conjugate momentum p_x ; it preserves $dQ \wedge dP = dx \wedge dp_x$ and $[\hat{x}, \hat{p}_x] = i\hbar I$. By transformation of covariance,

$$V_\eta^{(x,p_x)} = \frac{\hbar}{2} \begin{pmatrix} e^\eta/\lambda_{xp} & 0 \\ 0 & \lambda_{xp}e^{-\eta} \end{pmatrix}, \quad \Delta x \Delta p_x = \frac{\hbar}{2}, \quad \frac{\Delta p_x}{\Delta x} = \lambda_{xp}e^{-\eta}. \quad (15.14)$$

The dispersion quotient here acquires the dimension momentum per length. By contrast, (15.13) belongs to the balanced chart. For the inductive–capacitive realization already defined, (15.8) supplies the dimensionless identity $Z_\eta/Z_* = e^{-\eta} = \Delta P/\Delta Q$. Each reading uses its declared transduction and preserves the same action determinant.

Contour energy and mean energy of the state. The inductive–capacitive realization has Hamiltonian

$$H_\eta(Q, P) = \frac{\omega}{2} (e^{-\eta} Q^2 + e^\eta P^2), \quad \omega > 0.$$

On $\partial\mathcal{E}_\eta$, its value is $H_\eta = \hbar\omega$, by the semiaxis equations. In the quantum realization of Proposition 15.6,

$$\langle \hat{H}_\eta \rangle_{\psi_\eta} = \frac{\omega}{2} \left(e^{-\eta} \frac{\hbar e^\eta}{2} + e^\eta \frac{\hbar e^{-\eta}}{2} \right) = \frac{\hbar\omega}{2}. \quad (15.15)$$

The two-standard-deviation contour therefore has twice the mean energy of this Gaussian state. The areal law fixes action normalization; the angular pair fixes its distribution between the two coordinates; the canonical representation and explicit state realize that distribution as covariance. Saturation of uncertainty is thus established in the realization specified by its operators and its state.

16 Symplectic transport of the action ellipse

The ellipse constructed above distinguishes the areal measure of action, angular anisotropy, and the rate of evolution. This distinction determines an exact transport between elliptical realizations and fixes the connection term in its Hamiltonian expression. Its antecedent is the angular pair expressed by APP–TRIT–TPK after regional generation and the joint selection of K ; the following operations act on those outputs and retain their orientation labels.

16.1 Quadratic form and inductive–capacitive realization

Write $\zeta = \eta_{\text{el}} = \text{artanh}(C^*/A)$, with both angular coordinates in the same unit and $|C^*/A| < 1$. The symbol η_{ret} is reserved for the action coefficient. For the positive section $\hbar = \eta_{\text{ret}} s_0$, $s_0 = 10^{-34} \mathcal{U}_S$, the balanced coordinates, with the dimension of the square root of action, are

$$Q = \sqrt{2\hbar} e^{\zeta/2} \cos \theta, \quad P = \sqrt{2\hbar} e^{-\zeta/2} \sin \theta.$$

The product of the semiaxes is $2\hbar$, the total area is $2\pi\hbar = h$, and the accumulated parametric area is $\hbar\theta$. The phase θ is the parameter of this elliptical representation. Retaining the transducer $Z_* > 0$ and the frequency $\omega > 0$ of the preceding realization gives

$$L_\zeta = \frac{Z_*}{\omega} e^{-\zeta}, \quad C_\zeta = \frac{e^\zeta}{Z_* \omega}, \quad Q = \sqrt{Z_*} q, \quad P = \frac{\Phi}{\sqrt{Z_*}}.$$

Proposition 16.1 (Energy composition of the ellipse).

$$H_0 = \frac{\omega}{2} (e^{-\zeta} Q^2 + e^\zeta P^2), \quad E_{\text{el}} = \hbar \omega \cos^2 \theta, \quad E_{\text{mag}} = \hbar \omega \sin^2 \theta.$$

On the boundary, $H_0 = E_{\text{el}} + E_{\text{mag}} = \hbar \omega$.

Proof. Substitution of $q = Q/\sqrt{Z_*}$, $\Phi = \sqrt{Z_*} P$ into $q^2/(2C_\zeta) + \Phi^2/(2L_\zeta)$ yields H_0 . Substituting the elliptical parametrization then gives the two contributions and their sum. Action and shape precede the circuit realizing them. $\square$

The impedance of this realization satisfies $\sqrt{L_\zeta/C_\zeta}/Z_* = e^{-\zeta}$. The constitutive impedance of the vacuum uses the ratio of its 90/120 readers; each impedance retains its definition and transduction.

16.2 Transport invariant and Hamiltonian connection

Let

$$S_\zeta = \text{diag}(e^{\zeta/2}, e^{-\zeta/2}), \quad g_\zeta = S_\zeta^{-\top} S_\zeta^{-1}, \quad J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad R(-\theta) = e^{-\theta J_0}.$$

Theorem 16.2 (Exact conservation of quadratic action). *For a sequence of shapes ζ_n and phases θ_n , the transport*

$$X_{n+1} = T_n X_n, \quad T_n = S_{\zeta_{n+1}} R(-\theta_n) S_{\zeta_n}^{-1}$$

has determinant one and preserves

$$I_n = \frac{1}{2} X_n^\top g_{\zeta_n} X_n, \quad T_n^\top g_{\zeta_{n+1}} T_n = g_{\zeta_n}.$$

Proof. The determinants of S_ζ and R are one. In the quadratic product, the factors $S_{\zeta_{n+1}}$ cancel their inverses; $R^\top R = I$ leaves $S_{\zeta_n}^{-\top} S_{\zeta_n}^{-1}$. This identity proves conservation at every step and, by composition, at any length. A sequence expressed by a TPK route is transported together with that route and its labels. $\square$

Theorem 16.3 (Differentiable generator and mixed term). *If ζ is C^1 , $\omega > 0$ is continuous, and $X = S_\zeta Y$, $\dot{Y} = -\omega J_0 Y$, then*

$$\dot{Q} = \frac{1}{2}\dot{\zeta}Q + \omega e^\zeta P, \quad \dot{P} = -\omega e^{-\zeta}Q - \frac{1}{2}\dot{\zeta}P.$$

With the convention $\dot{Q} = \partial_P H$, $\dot{P} = -\partial_Q H$, its Hamiltonian, up to an additive function of time, is

$$H_{\text{tot}} = \frac{\omega}{2}(e^{-\zeta}Q^2 + e^\zeta P^2) + \frac{\dot{\zeta}}{2}QP. \quad (16.1)$$

The invariant $I = (e^{-\zeta}Q^2 + e^\zeta P^2)/2$ is conserved exactly.

Proof. Differentiating $X = S_\zeta Y$ yields $\dot{S}_\zeta S_\zeta^{-1} = \text{diag}(\dot{\zeta}/2, -\dot{\zeta}/2)$ and the conjugated rotational term. The partial derivatives of (16.1) reproduce both equations. In $\dot{I}$, the explicit contributions $-\dot{\zeta}e^{-\zeta}Q^2/2 + \dot{\zeta}e^\zeta P^2/2$ cancel those from the diagonal of the generator; the rotational cross terms cancel each other. $\square$

In the elliptical parametrization, $QP = I \sin 2\theta$ and $\dot{\theta} = -\omega$. Thus $H_{\text{tot}} = I\omega + (I\dot{\zeta}/2) \sin 2\theta$. The quadratic matrix has determinant $\omega^2 - \dot{\zeta}^2/4$: it is positive definite precisely when $|\dot{\zeta}| < 2\omega$, and positive semidefinite at equality. Conservation of I and time dependence of H_{tot} express different properties. As an algebraic control, omitting the mixed term gives $\dot{I} = \dot{\zeta}(e^\zeta P^2 - e^{-\zeta}Q^2)/2$. In the corresponding quantum representation, the symmetric mixed term is $\dot{\zeta}(\hat{Q}\hat{P} + \hat{P}\hat{Q})/4$.

16.3 Genealogical compatibility and conformally symplectic transport

On the generated section,

$$\zeta = \text{artanh} \frac{2(\eta_{\text{ret}} + \alpha)}{1000\alpha}.$$

Fixing α and η_{ret} fixes the shape. The variable version of the theorem describes transport between admissible shapes; its dynamical realization retains the update determining those shapes. For positive sections $\hbar_n$ in the same dimensional chart, let $\rho_n = \hbar_{n+1}/\hbar_n$. Then

$$\tilde{T}_n = \sqrt{\rho_n} T_n, \quad \det \tilde{T}_n = \rho_n, \quad \tilde{T}_n^\top g_{\zeta_{n+1}} \tilde{T}_n = \rho_n g_{\zeta_n}, \quad \frac{I_{n+1}}{\hbar_{n+1}} = \frac{I_n}{\hbar_n}.$$

These equalities follow by multiplying the preceding transport by $\sqrt{\rho_n}$. The symplectic form and normalized action are transported together. The differentiable limit has trace $\hbar/\hbar$; for a fixed symplectic form, a linear Hamiltonian generator has zero trace. A variable section therefore requires retaining the transported form or the degrees of freedom exchanging action. If the variation is a change of units, the dimensional bases are transported first.

The operational distinction is precise. Under a passive change $X = S_\zeta Y$, the reader also satisfies $L_X(S_\zeta Y) = L_Y(Y)$; the mixed term represents the coordinate connection. A physical deformation is specified by relative evolution between system and reference, obtained from their updates and read with the same device. An energy attribution includes the field, the record, and the controller. This separates the proved conservation of quadratic action, its Hamiltonian realization, and the conditions for a testable physical response.

17 CKM mixing angles, CP phase, and leptonic transport

17.1 Common antecedent and separation of the two transports

The article's common foundation produces the enriched state through the two APP evaluations, the oriented selection of TRIT, and the effective composition of selection, transport, carry, memory, and return in TPK. Angular representations do not replace that state: they are readers of its record. In particular, the prolongation $w_6 \rightarrow w_{12} \rightarrow w_{18} \rightarrow w_{24} \rightarrow w_{30} \rightarrow R_{36} \rightarrow G_9$ preserves the provenance of the numerical readings, and the holonomy Γ_9 acts before the reduced representation K_{ph} . Phase return does not restart the record.

Two classes of objects are received here from that same antecedent. The first contains the direct angle, the counter-angle, and the torsional defect Δ_4 , already constructed through their own readers. The second contains the surviving extensions, their observable routes, the dodecaphasic record, and the operators of the material fibers. The extension–route–record–signature–character–towers chain preserves the atlas 81/56/13/324 and distinguishes the null chart from the positive branch. The mixing angles do not retrospectively choose those routes or generate the mass eigenvalues.

Quark transport uses a rational angular chart and its complex unitary realization. Leptonic transport uses the spectral frames of the charged and neutral fibers. Both come from the common discrete structure, but their domains, readers, and changes of basis are distinct. The Barbero–Immirzi functional and mixing are not defined by one another either: they receive typed representations of the angular and incidence antecedent. Their relationship is expressed through those maps, not through a retrospective formula using a Barbero value as a mixing selector.

Remark 17.1 (Sectorial rules and evaluation of the reader). The three sectorial rules and their evaluation are developed below in Section 17.2. Theorem 17.5 obtains the columns as images of the angular basis. Subsequent proofs establish inversion of the chart, sheet transport, the metric, and the complex realization. The choice of rules and evaluation of their coefficients are two distinct mathematical questions: the starting system is declared in Definition 17.4, and the theorem's proof uses that complete system. The preceding composition that selects those rules from incidence requires its own proof.

17.2 Sectorial incidence and construction of the angular reader

The angular construction uses the action and closure representations previously obtained from APP–TRIT–TPK. The direct angle, counter-angle, and torsional defect preserve their own domains. The quark reader combines these three coordinates through sectorial incidence; its evaluation precedes complex amplitudes and metrological comparison. Hexadic–octadic inclusion enters the third sector through the difference of its cardinalities.

17.2.1 Marked incidence and bidirectional normalization

The reader invariants preserve the nature of the object from which they arise. The marked intersection $B_K = H_e \cap P_\pi$ has cardinality $b = 4$; the integer $q = 7$ is the coordinate of the pivot p_φ in the oriented numbering. The determination order of the Steiner design is $p = 5$. The hexadic and octadic supports have cardinalities $h_6 = 6$ and $o_8 = 8$, and the tritic orientation alphabet has cardinality $t = 3$. A support cardinality and a pivot coordinate are used through their respective maps, even when both are expressed as integers.

Normalization of the pentagonal self-crossing admits the following explicit realization. Let P be the five-position support of the closure, and let $X = P \times P$ be its space of ordered pairs. Bidirectional inversion on this space is

$$J : X \longrightarrow X, \quad J(x, y) = (y, x), \quad J^2 = I.$$

The stabilizer information of each orbit is preserved. For an orbit $[z]$, define its weight as $1/|\text{Stab}_{C_2}(z)|$. The sum of these weights is the cardinality of the action groupoid $X//C_2$.

Proposition 17.2 (Orbital measure of bidirectional self-crossing). *For a finite support of cardinality p , inversion of ordered pairs determines*

$$|(P \times P) // C_2| = \sum_{[z] \in (P \times P) / C_2} \frac{1}{|\text{Stab}_{C_2}(z)|} = \frac{p^2}{2}.$$

For $p = 5$, the self-crossing has ten free orbits and five fixed orbits, and its orbital measure is $10 + 5/2 = 25/2$.

Proof. Each diagonal pair (x, x) forms a fixed orbit, whose stabilizer has order two; there are p such orbits. The remaining $p(p - 1)$ pairs are grouped in twos, with trivial stabilizer. Their contribution is therefore $p(p - 1)/2$, and that of the diagonal is $p/2$. The sum is $p^2/2$. Equivalently, the orbit–stabilizer identity gives $1/|\text{Stab}(z)| = |[z]|/2$; summing over all orbits produces $|P \times P|/2$. $\square$

The measure explicitly preserves the contribution of fixed points. The ordinary orbit set has cardinality $p(p + 1)/2$, equal to fifteen in the pentagonal closure. The orbital measure and that cardinality are two determinate readings of the same inversion. The reader’s bidirectional normalization uses the former: the ten free orbits have unit weight, and the five diagonals have half-unit weight.

This realization specifies the factor $p^2/2$ used in the following sectorial system. The assignment of that contribution to sector 23, its negative orientation, and the other two incidence rules remain specified by the complete sectorial system. The proof establishes the normalization of self-crossing and preserves the selection of the angular functionals as a distinct matter.

17.2.2 Carry transport and integer holonomy of the arithmetic crossing

The modular reading of the crossing admits an integral realization retaining the information accumulated while crossing the edge. This realization is developed here through transport quotients. Its domain is a lifted chart of the APP support; the subsequent assignment of a region to an angular sector retains its own incidence map.

Nonadic chart and lifted transport. We use the exact decomposition

$$\rho_9(n) = 1 + ((n - 1) \bmod 9), \quad q_9(n) = \left\lfloor \frac{n - 1}{9} \right\rfloor, \quad n = \rho_9(n) + 9q_9(n).$$

A lifted position is $z = (x, y, q_x, q_y) \in D_9^2 \times \mathbb{Z}^2$, where $X = x + 9q_x$ and $Y = y + 9q_y$. The integer increment u of the first coordinate acts by

$$T_u^x(x, y, q_x, q_y) = (\rho_9(x + u), y, q_x + c_9(x; u), q_y), \quad c_9(x; u) = \left\lfloor \frac{x + u - 1}{9} \right\rfloor.$$

The second coordinate is transported in the same way. Cardinal advances correspond to $u = \pm 1$; the zero increment preserves the state. The cocycle satisfies

$$c_9(x; u + v) = c_9(x; u) + c_9(\rho_9(x + u); v).$$

Indeed, both sides are the quotient of the unique decomposition of $x + u + v$ with remainder in D_9 . Consequently, $T_v^x T_u^x = T_{u+v}^x$ and $T_{-u}^x = (T_u^x)^{-1}$; the transports of the two coordinates commute. The state preserves the number of edge crossings even when the visible position returns.

Lifted sheets and update of records. The integral realization is fixed by prolonging the APP arithmetic operations to the transported coordinates:

$$\begin{aligned} \tilde{S}(z) &= X + Y = x + y + 9(q_x + q_y), \\ \tilde{P}(z) &= XY = xy + 9(q_x y + q_y x) + 81q_x q_y. \end{aligned}$$

At each position the record of both sheets is preserved:

$$(\rho_9(\tilde{S}), q_9(\tilde{S}), \rho_9(\tilde{P}), q_9(\tilde{P})).$$

If $e : z \rightarrow z'$ is a link and T the selected sheet, its full increment is obtained from the record before and after transport:

$$\delta_e \tilde{T} = \rho_9(\tilde{T}(z')) - \rho_9(\tilde{T}(z)) + 9(q_9(\tilde{T}(z')) - q_9(\tilde{T}(z))). \quad (17.1)$$

This identity makes preservation explicit: the residue jump and quotient increment together reconstruct the variation of the sheet. Sheet selection, cardinal transport, and record update are distinct operations. In this chart, the ordered sheet selection (T_x, T_y) is first composed with T_d and the update of its quotients; the formula does not postulate that the entire TPK is reduced to this local representation.

Proposition 17.3 (Integral curvature and boundary transport). *Define $\tilde{A}_x = \delta_x \tilde{T}_x$ and $\tilde{A}_y = \delta_y \tilde{T}_y$ through (17.1), and let*

$$\tilde{F}_{xy}(X, Y) = \tilde{A}_x(X, Y) + \tilde{A}_y(X + 1, Y) - \tilde{A}_x(X, Y + 1) - \tilde{A}_y(X, Y).$$

Homogeneous choices have zero curvature. The mixed choices $(\tilde{S}, \tilde{P})$ and $(\tilde{P}, \tilde{S})$ have curvature +1 and -1, respectively, as integers. For the positive boundary of a rectangle $R_{a,b}$ with integer sides $a, b > 0$,

$$\tilde{W}_{SP}(\partial R_{a,b}) = ab, \quad \tilde{W}_{PS}(\partial R_{a,b}) = -ab.$$

The result is independent of the base point and the number of crossings of nonadic seams.

Proof. Exact reconstruction of the two sheets gives

$$\delta_x \tilde{S} = \delta_y \tilde{S} = 1, \quad \delta_x \tilde{P} = Y, \quad \delta_y \tilde{P} = X.$$

Thus the SP connection is $(1, X)$, with curvature $1 + (X + 1) - 1 - X = 1$; the PS connection is $(Y, 1)$, with curvature $Y + 1 - (Y + 1) - 1 = -1$. For a single sheet, mixed differences cancel. Summing the curvatures of the ab plaquettes cancels each interior link with its inverse and preserves the boundary links. This yields $\pm ab$ in $\mathbb{Z}$. The transported quotients ensure the same identity holds across any seam. $\square$

The boundary accumulator can in turn be retained in nonadic form: $W = r_W + 9q_W$. For an integer contribution w_e , it is updated by

$$r'_W = \rho_9(r_W + w_e), \quad q'_W = q_W + \left\lfloor \frac{r_W + w_e - 1}{9} \right\rfloor.$$

Composition by links provides the same sum as direct integer evaluation. In this chart, integer zero has coordinates $(r_W, q_W) = (9, -1)$; this is an arithmetic representation of the accumulator, without further identification with the distinct oriented zeros of TRIT.

Three operations on orientation. Let γ be a closed cardinal route. Its time reversal γ^{-1} reverses the order of links and the orientation of each; by additivity,

$$\tilde{W}_{SP}(\gamma^{-1}) = -\tilde{W}_{SP}(\gamma).$$

Exchange of sheets likewise satisfies

$$\tilde{W}_{PS}(\gamma) = -\tilde{W}_{SP}(\gamma).$$

To check this on any route, note that the sum of both connections is the exact increment of $\tilde{S} + \tilde{P} = X + Y + XY$; its integral over a loop is zero.

The simultaneous half-turn of both directions, on the other hand, acts through $J_c(X, Y) = (2c_x - X, 2c_y - Y)$, with $2c \in \mathbb{Z}^2$, so that it preserves the lattice chart. For a loop,

$$\tilde{W}_{SP}(J_c\gamma) = \tilde{W}_{SP}(\gamma).$$

Indeed, the closed horizontal contribution is zero, and the vertical one transforms as $\sum (2c_x - X)(-\delta Y) = \sum X\delta Y$, since $\sum \delta Y = 0$. A spatial half-turn therefore preserves the area orientation of this chart. Its combination with time reversal or sheet exchange is computed by composing the corresponding operations.

The difference matters for the electronic record: simultaneous reversal of both displacements every 27 advances produces, in windows of 9, the sequence

$$\tau_j = (-1)^{\lfloor j/3 \rfloor}, \quad 0 \leq j < 12, \quad (+, +, +, -, -, -, +, +, +, -, -, -).$$

This sequence records advance orientations. Holonomy additionally requires the links traversed, selected sheets, and accumulated quotients. Transport of electronic orientation is thus preserved without identifying a half-turn of both axes with time reversal of a loop.

Application to the marked crossing and sector scope. For the rectangular crossing with sides 4 and 7, this realization evaluates exactly

$$\tilde{W}_{SP} = 28 = 1 + 9 \cdot 3, \quad \tilde{W}_{SP}(\gamma^{-1}) = -28 = 8 + 9(-4).$$

The integer preserves the oriented area; the isolated remainder represents only its nonadic class. For example, from (7, 1) the periodic representatives of the links sum to -35 , whereas the carry correction adds 63 and recovers 28.

In the marked incidence of the angular reader, 4 is $|H_e \cap P_\pi|$ and 7 is the marked pivot p_φ . The rectangular reading supplies an integer realization of the product $|H_e \cap P_\pi| p_\varphi$ once that crossing has been declared. The map identifying that region with sector 12 must preserve the mark and orientation of the sector system in Definition 17.4. Sector assignments 12, 23, 13 and their combinations of A, h, Δ° are an additional specification beyond the rectangular-holonomy law. The present result makes the area lift, carry, and their involutions explicit, with these starting data declared.

Definition 17.4 (Sectorial system of angular incidence). The sectorial system consists of the ordered sextuple of invariants

$$(b, p, q, h_6, o_8, t) = (4, 5, 7, 6, 8, 3)$$

and the linear maps, on angular coordinates (a, u, d) ,

$$\mathcal{A}_{12}^{\text{ang}}(a, u, d) = 2a - pu + bq d, \quad (17.2)$$

$$\mathcal{A}_{23}^{\text{ang}}(a, u, d) = qu - \frac{p^2}{2} d, \quad (17.3)$$

$$\mathcal{A}_{13}^{\text{ang}}(a, u, d) = a - pbu - \frac{o_8 - h_6}{t} d. \quad (17.4)$$

The symbols h_6, o_8 preserve the cardinalities of the marked incidence; $t = 3$ is the declared tritic normalization of that sector. The sextuple and the three rules jointly form the starting system of this construction.

Theorem 17.5 (Exact evaluation of the sectorial system). *The joint map $(\mathcal{A}_{12}^{\text{ang}}, \mathcal{A}_{23}^{\text{ang}}, \mathcal{A}_{13}^{\text{ang}})^\top$ of Definition 17.4 has the unique matrix*

$$M_{\text{CKM}} = \begin{pmatrix} 2 & -5 & 28 \\ 0 & 7 & -25/2 \\ 1 & -20 & -2/3 \end{pmatrix}.$$

The three columns are the images of the coordinate vectors (1, 0, 0), (0, 1, 0), and (0, 0, 1) under these maps.

Proof. The sectorial operations give

$$bq = 4 \cdot 7 = 28, \quad p^2/2 = 5^2/2 = 25/2, \quad pb = 5 \cdot 4 = 20, \quad (o_8 - h_6)/t = (8 - 6)/3 = 2/3.$$

Thus the successive images of the coordinate basis are

$$c_A = (2, 0, 1)^T, \quad c_h = (-5, 7, -20)^T, \quad c_\Delta = (28, -25/2, -2/3)^T.$$

A linear map is determined by its images on a basis. The matrix with these images as its columns is the one indicated. The proof uses the complete sectorial system of the definition. $\square$

On the previously produced coordinates, take

$$(a, u, d) = \left(A_{\text{deg}}, \frac{C_{\text{deg}}^*}{6}, \frac{180}{\pi} \Delta_4 \right).$$

The direct, conjugate, and torsional contributions to each angle are thereby individualized. In particular, the difference $o_8 - h_6 = 2$ is preserved in the coefficient of the torsional defect in sector 13. This concrete presence makes it possible to follow incidence through to the angular reader, keeping the sectorial system, its evaluation, and the unitary parametrization separate.

17.3 Rational CKM chart and angular units

We write A_{deg} and C_{deg}^* for the coordinates in degrees of the HMT angles already generated. We use the abbreviations

$$h_\theta = \frac{C_{\text{deg}}^*}{6}, \quad \Delta_{\text{deg}} = \frac{180}{\pi_{\text{HMT}}} \Delta_4, \quad x_\theta = (A_{\text{deg}}, h_\theta, \Delta_{\text{deg}})^T. \quad (17.5)$$

Here, h_θ is an angular coordinate, distinct from the Planck constant h . Conversion to degrees is a subsequent change of coordinates; it does not introduce a conventional value of π into the generator.

Definition 17.6 (Quark angular reader). The inherited reader $M_{\text{CKM}} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ evaluates the coordinates

$$\theta_{\text{deg}} = M_{\text{CKM}} x_\theta, \quad M_{\text{CKM}} = \begin{pmatrix} 2 & -5 & 28 \\ 0 & 7 & -25/2 \\ 1 & -20 & -2/3 \end{pmatrix}, \quad \delta_{\text{deg}} = 9A_{\text{deg}}. \quad (17.6)$$

The component order is (12, 23, 13). Subsequent trigonometric functions are evaluated at $\vartheta_{ab} = (\pi_{\text{HMT}}/180)\theta_{ab,\text{deg}}$ and $\delta = (\pi_{\text{HMT}}/180)\delta_{\text{deg}}$.

Proposition 17.7 (Rational reconstruction of the coordinates). *The reader has determinant $-3857/6$ and inverse*

$$M_{\text{CKM}}^{-1} = \begin{pmatrix} 1528/3857 & 3380/3857 & 801/3857 \\ 75/3857 & 176/3857 & -150/3857 \\ 6/551 & -30/551 & -12/551 \end{pmatrix}. \quad (17.7)$$

In particular,

$$\begin{aligned} A_{\text{deg}} &= \frac{1528\theta_{12,\text{deg}} + 3380\theta_{23,\text{deg}} + 801\theta_{13,\text{deg}}}{3857}, \\ C_{\text{deg}}^* &= \frac{6(75\theta_{12,\text{deg}} + 176\theta_{23,\text{deg}} - 150\theta_{13,\text{deg}})}{3857}, \\ \Delta_{\text{deg}} &= \frac{6\theta_{12,\text{deg}} - 30\theta_{23,\text{deg}} - 12\theta_{13,\text{deg}}}{551}. \end{aligned} \quad (17.8)$$

The phase satisfies the exact linear check

$$3857\delta_{\text{deg}} = 9(1528\theta_{12,\text{deg}} + 3380\theta_{23,\text{deg}} + 801\theta_{13,\text{deg}}). \quad (17.9)$$

Proof. Expansion of the determinant in (17.6) gives $-3857/6$. Multiplying the matrix given in (17.7) by M_{CKM} , in either order, gives I_3 . Its rows recover the three components of x_θ ; the second is multiplied by six to recover C_{deg}^* . Finally, we use $\delta_{\text{deg}} = 9A_{\text{deg}}$. $\square$

Inversion reconstructs the reader's coordinates once the reader is fixed. It does not turn observed angles into antecedents of A , C^* , or Δ_4 .

17.4 Geometric decomposition under sheet inversion

In the angular chart, sheet inversion preserves a, d and reverses u . The expressed vector admits the decomposition

$$\theta_+ = ac_A + uc_h + dc_\Delta, \quad \theta_- = ac_A - uc_h + dc_\Delta.$$

Proposition 17.8 (Even and odd components of angular transport). *The two vectors satisfy*

$$\frac{\theta_+ + \theta_-}{2} = ac_A + dc_\Delta, \quad \frac{\theta_+ - \theta_-}{2} = uc_h.$$

The contribution that changes sign occupies the line $\text{span}(c_h)$; the preserved component occupies $\text{span}(c_A, c_\Delta)$.

Proof. The sum cancels the terms uc_h , and the difference cancels the other two. The columns are linearly independent, since $\det M_{\text{CKM}} = -3857/6$. The two components therefore form a direct decomposition. $\square$

The same inversion acts on the Barbero–Immirzi evaluation: it exchanges q_+ and q_- , and reverses the component bilinear in A and C^* . Consequently, the incidence functional is odd. An observable symmetric in both channels preserves its value. This difference in parity expresses how the same operation on the structure produces different responses depending on the reader: areal orientation, symmetric information, and the mixing vector preserve different information from the same antecedent.

17.5 Transport of sheet inversion

Conjugation of the angular charts preserves A_{deg} and Δ_{deg} , and reverses h_θ . We define

$$D_\theta = \text{diag}(1, -1, 1), \quad c_h = (-5, 7, -20)^\top, \quad \ell_h = \frac{1}{3857}(75, 176, -150). \quad (17.10)$$

If $c_A = (2, 0, 1)^\top$ and $c_\Delta = (28, -25/2, -2/3)^\top$, then $\ell_h c_A = \ell_h c_\Delta = 0$ and $\ell_h c_h = 1$.

Theorem 17.9 (Rational reflection and transported metric). *The operator*

$$\mathcal{R}_{\text{CKM}} = M_{\text{CKM}} D_\theta M_{\text{CKM}}^{-1} = I_3 - 2c_h \ell_h \quad (17.11)$$

satisfies

$$\mathcal{R}_{\text{CKM}}^2 = I_3, \quad \det \mathcal{R}_{\text{CKM}} = -1, \quad \mathcal{R}_{\text{CKM}} M_{\text{CKM}} = M_{\text{CKM}} D_\theta. \quad (17.12)$$

It is an orthogonal reflection for the positive metric

$$G_\theta = (M_{\text{CKM}}^{-1})^\top M_{\text{CKM}}^{-1}, \quad \mathcal{R}_{\text{CKM}}^\top G_\theta \mathcal{R}_{\text{CKM}} = G_\theta. \quad (17.13)$$

Proof. The operator $P_h = c_h \ell_h$ satisfies $P_h^2 = P_h$, has image $\mathbb{R}c_h$, and kernel $\text{span}(c_A, c_\Delta)$. Thus $(I_3 - 2P_h)^2 = I_3$, with eigenvalues $1, 1, -1$. Its action on those three columns is that of D_θ in the original basis, proving the conjugation and intertwining. For $v \neq 0$, $v^\top G_\theta v = \|M_{\text{CKM}}^{-1}v\|^2 > 0$. The last identity follows from $D_\theta^\top D_\theta = I_3$. $\square$

Its complete expression, preserved from the preceding supporting material, is

$$\mathcal{R}_{\text{CKM}} = \begin{pmatrix} 4607/3857 & 1760/3857 & -1500/3857 \\ -150/551 & 199/551 & 300/551 \\ 3000/3857 & 7040/3857 & -2143/3857 \end{pmatrix}. \quad (17.14)$$

The phase $\delta_{\text{deg}} = 9A_{\text{deg}}$ remains fixed under this reflection. Consequently, sheet transport must not be identified with CP conjugation of amplitudes, which changes the sign of the complex phase. The combinatorial parity of a route, its sheet conjugation, and its physical representation retain the distinct domains in [equations (17.15) and (17.16)]; [Section 17, equations (17.18) and (17.26)].

17.6 Record parity and CP realization

To preserve the discrete support of this distinction, let ρ be a recorded route of length L , with directions $d_t \in \{N, E, S, W\}$ and sheets $\ell_t \in \{\Sigma, \Pi\}$. The intermediate states are those produced by TPK transport. Reflection P acts by $(i, j) \mapsto (i, -j) \pmod{9}$ and permutes E, W ; C_{hoja} permutes Σ, Π ; T_{reg} reverses the order of the record and replaces each direction by its opposite. Inversion is performed on the enriched record, not on a projection that has forgotten its predecessor.

Proposition 17.10 (Invariance of the combinatorial route action). *For $\lambda \geq 0$, the dimensionless functional*

$$\mathcal{A}_\lambda(\rho) = \sum_{t=2}^L [1 - \delta_{d_t, d_{t-1}} + \lambda(1 - \delta_{\ell_t, \ell_{t-1}})] \quad (17.15)$$

is invariant under P , C_{hoja} , T_{reg} , and their compositions.

Proof. A global permutation of the alphabet preserves equality between neighbors. Reversing the order preserves the number of consecutive changes. Thus each operation separately preserves the changes of direction and sheet summed in (17.15). $\square$

λ is the declared parameter of this collection of functionals; $\mathcal{A}_\lambda$ is not identified with a dimensional Planck section. Invariance of the counter also does not imply that every oriented reader is even. In a mixing realization $\mathcal{V}$, CP compatibility takes the precise form

$$\mathcal{V}(CP_{\text{HMT}}\rho) = D_u \overline{\mathcal{V}(\rho)} D_d^\dagger, \quad (17.16)$$

with the diagonal phase changes and representation spaces declared. The route counter, rational sheet transport, and complex conjugation are three distinct maps of the preserved support.

Remark 17.11 (CP compatibility of the physical reader). Equation (17.16) is the CP compatibility condition for the complete physical reader. The proofs in this section establish invariance of the record and the identities of the complex reader, but do not automatically turn $\mathcal{R}_{\text{CKM}}$ into CP_{HMT} . This qualification concerns that representation square, not the existence of the internal route transport already constructed.

17.7 Unitary amplitudes, CP phase, and the Jarlskog invariant

We write $c_{ab} = \cos \vartheta_{ab}$ and $s_{ab} = \sin \vartheta_{ab}$. The complex realization of the reader is

$$V_{\text{CKM}} = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13} \end{pmatrix}. \quad (17.17)$$

The complex structure used is the realization of the oriented structure already constructed; the unitary parametrization represents the angular result, rather than determining its coefficients.

Proposition 17.12 (Unitarity and phase invariant). *The matrix (17.17) is unitary. The quartet*

$$J_{\text{CKM}} = \text{Im}(V_{11} V_{22} \overline{V_{12}} \overline{V_{21}}) = c_{12} c_{23} c_{13}^2 s_{12} s_{23} s_{13} \sin \delta \quad (17.18)$$

is invariant under $V \mapsto D_u V D_d$, with D_u, D_d diagonal unitary matrices, and changes sign under $V \mapsto \overline{V}$.

Proof. The matrix is the product $R_{23} U_{13}(\delta) R_{12}$, where

$$\begin{aligned} R_{12} &= \begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix}, \\ U_{13}(\delta) &= \begin{pmatrix} c_{13} & 0 & s_{13} e^{-i\delta} \\ 0 & 1 & 0 \\ -s_{13} e^{i\delta} & 0 & c_{13} \end{pmatrix}, \\ R_{23} &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix}. \end{aligned} \quad (17.19)$$

Each factor is unitary because $c_{ab}^2 + s_{ab}^2 = 1$. Substituting the four entries of the quartet, its imaginary part is $c_{12} s_{12} c_{13}^2 c_{23} s_{23} s_{13} (c_{12}^2 + s_{12}^2) \sin \delta$, giving (17.18). The phase factors of each row and each column cancel with their conjugates. Complex conjugation reverses the imaginary part. $\square$

A nonzero J_{CKM} obstructs making the entire matrix real through those phase changes. Its physical interpretation also retains the corresponding sector and representation; it is not a scalar correction added to masses. If the spectra have degeneracies, their change-of-basis freedoms must be preserved before formulating a CP conclusion.

17.8 Conjugation of amplitudes and oriented invariant

The complex realization preserves the three mixing angles and the interference phase as explicit coordinates. Write $V(\vartheta, \delta) = R_{23} U_{13}(\delta) R_{12}$, with the rotations and phase defined in the unitary development. The arguments of the trigonometric functions are their representatives in radians.

Proposition 17.13 (Conjugation in the mixing realization). *For the three declared real angles and the declared real phase,*

$$V(\vartheta, -\delta) = \overline{V(\vartheta, \delta)}, \quad J(\vartheta, -\delta) = -J(\vartheta, \delta).$$

The invariant J also preserves its value under diagonal phase changes of the charged bases.

Proof. The matrices R_{12} and R_{23} have real entries. In U_{13} , the change $\delta \mapsto -\delta$ conjugates the exponential factors. Conjugating the product gives the first identity. In the quartet $V_{11} V_{22} \overline{V_{12}} \overline{V_{21}}$, the diagonal factors of each row and column cancel. Conjugation reverses the sign of its imaginary part. Equivalently, the result follows by substitution into

$$J = c_{12} c_{23} c_{13}^2 s_{12} s_{23} s_{13} \sin \delta.$$

$\square$

In the three-generation sector, the nonzero internal value of J expresses the obstruction to realizing all amplitudes by a real matrix through diagonal phase changes. This is the charge–parity violation reading of the constructed matrix. The route reading, in turn, preserves the discrete operation of conjugation and parity and its representation. Their full compatibility is expressed by the square

$$\mathcal{V}(CP_{\text{HMT}} \rho) = D_u \overline{\mathcal{V}(\rho)} D_d^\dagger,$$

with the representation spaces and phase changes declared. The conjugation identity proved here fixes exactly the action that this square must transport.

Proposition 17.14 (Mixing and spectrum). *Let $B_u = \text{diag}(a_1, a_2, a_3)$ and $B_d = \text{diag}(b_1, b_2, b_3)$ be the Hermitian mass readers already produced in their fibers. The transport*

$$B_d^{(u)} = V_{\text{CKM}} B_d V_{\text{CKM}}^\dagger \quad (17.20)$$

preserves the spectrum of B_d , and satisfies

$$|\det[B_u, B_d^{(u)}]| = 2|J_{\text{CKM}}| \prod_{i < j} |a_i - a_j| \prod_{r < s} |b_r - b_s|. \quad (17.21)$$

Proof. Unitary conjugation preserves the characteristic polynomial. If $B = B_d^{(u)}$, the entries of the commutator are $[B_u, B]_{ij} = (a_i - a_j)B_{ij}$, with zero diagonal. Its determinant is

$$2i(a_1 - a_2)(a_2 - a_3)(a_3 - a_1) \text{Im}(B_{12}B_{23}B_{31}).$$

Substituting $B_{ij} = \sum_r b_r V_{ir} \overline{V_{jr}}$, the imaginary part is an alternating polynomial of degree three in the b_r . It is divisible by their three differences: if two b_r coincide, $B = bI + (b' - b)vv^\dagger$, and the off-diagonal product is real. Expanding a coefficient, using orthogonality of the columns, gives the quartet in (17.18), with the sign of the chosen order. Taking absolute values removes that convention and yields (17.21), including its vanishing at degeneracies. $\square$

Remark 17.15 (Evaluation and comparison). The decimal CKM coordinates in Table 12 are subsequent evaluations of the preceding reader. The quantitative comparison section also presents the angles and moduli of the amplitudes. Those evaluations are not used to choose its columns or to prove (17.12) or (17.18). This block assigns no new metrological significance to those numbers: a joint comparison requires its external edition, definition of observables, and covariance. Marginal deviations do not replace it.

17.9 Leptonic record, operators, and spectral frames

The leptonic sector preserves another domain. The spectral construction of (17.22) brings together four neutral depths of dimensions 2, 4, 6, 8, before the three-dimensional compression:

$$\mathcal{H}_\nu^{\text{reg}} = \bigoplus_{j=1}^4 \mathcal{H}_{\nu, G_j}, \quad \dim \mathcal{H}_\nu^{\text{reg}} = 20, \quad M_\nu = (P_\nu \mathcal{M}_\nu P_\nu)|_{\text{Im } P_\nu}, \quad \text{rank } P_\nu = 3. \quad (17.22)$$

The four depths are not four flavors or four masses. P_ν is the neutral projector inherited from the record reader, and $M_\ell = (P_\ell \mathcal{M}_\ell P_\ell)|_{\text{Im } P_\ell}$, with $\text{rank } P_\ell = 3$, is the compressed charged operator. The scale of these operators comes from the action–clock–mass branch; no PMNS angle determines it.

Definition 17.16 (Transport of leptonic frames). In the declared common three-dimensional realization, let $F_\ell, F_\nu : \mathbb{C}^3 \rightarrow \mathcal{H}_3$ be complete orthonormal spectral frames of the charged and neutral operators. Identification with $\mathcal{H}_3$ is fixed before taking their product. The reader is

$$U_{\text{PMNS}} = F_\ell^\dagger F_\nu : \mathbb{C}_\nu^3 \longrightarrow \mathbb{C}_\ell^3. \quad (17.23)$$

The frames and identification come from the representation and the record; they are not chosen from an experimental matrix one wishes to reproduce.

Theorem 17.17 (Unitarity, degeneracies, and naturality). *Under the preceding identification, U_{PMNS} is unitary. If the spectra are simple, their frame changes are diagonal phases and permutations; once spectral order is fixed, the phases remain. If there are multiplicities $m_{\ell, \lambda}$ and $m_{\nu, \eta}$, the basis-independent object is the double coset*

$$\left(\prod_{\lambda} U(m_{\ell, \lambda}) \right) \backslash U(3) / \left(\prod_{\eta} U(m_{\nu, \eta}) \right). \quad (17.24)$$

A common unitary refinement W intertwining both operators preserves transport when the frames are prolonged as $F'_\ell = W F_\ell$, $F'_\nu = W F_\nu$.

Proof. Completeness in the same space gives $F_\ell F_\ell^\dagger = F_\nu F_\nu^\dagger = I_{\mathcal{H}_3}$. Therefore, $U_{\text{PMNS}}^\dagger U_{\text{PMNS}} = F_\nu^\dagger F_\ell F_\ell^\dagger F_\nu = I_3$, and analogously in the other order. Basis changes preserving a Hermitian operator act in blocks within its eigenspaces. Substituting $F_\ell D_\ell, F_\nu D_\nu$ yields $U \mapsto D_\ell^\dagger U D_\nu$, which determines the double coset. Finally, $(W F_\ell)^\dagger W F_\nu = F_\ell^\dagger F_\nu$. $\square$

The common-space condition is substantive. For two isometric frames in a larger ambient space, separate orthonormality yields only a compression; it does not authorize canceling $F_\ell F_\ell^\dagger$ as though it were the ambient identity.

17.10 Record kernel and rank sufficiency

The readers Ξ_r preserve the carry, sheet, orientation, vacancy, and holonomy components of the record. On the finite set of admitted routes, one forms

$$K_{\text{reg}}(c, d) = \sum_{\substack{r \in \{\text{car}, \text{hoj}, \text{ori}, \text{vac}, \text{hol}\} \\ d_r > 0}} \frac{\langle \Xi_r(c), \Xi_r(d) \rangle}{d_r}, \quad d_r = \sum_x \|\Xi_r(x)\|^2. \quad (17.25)$$

Normalization comes from the norms of the record itself. Zero-norm components are omitted; they are not replaced by terms taken from a target matrix. Each Ξ_r and each aggregation class must preserve its route residence and compatibility with refinement.

Proposition 17.18 (Positivity and rank of the record). *The matrix K_{reg} is positive semidefinite. If its feature vectors are aggregated into three sectors, the resulting Gram matrix G_ν is invertible exactly when those three aggregated vectors are linearly independent.*

Proof. Define $v_c = \bigoplus_{r: d_r > 0} d_r^{-1/2} \Xi_r(c)$. Then $K_{\text{reg}}(c, d) = \langle v_c, v_d \rangle$, and for any coefficients z_c , $\sum_{c,d} \bar{z}_c K_{\text{reg}}(c, d) z_d = \|\sum_c z_c v_c\|^2 \geq 0$. The same identity holds for the three aggregated vectors. The kernel of their Gram matrix consists exactly of their linear relations, yielding the invertibility criterion. $\square$

That single-sector Gram matrix is distinguished from the *cross* matrix between the charged classes L_a and neutral classes N_b , expressed in [Section 17, equations (17.18) and (17.26)]:

$$G_{ab} = \frac{1}{\sqrt{|L_a||N_b|}} \sum_{c \in L_a} \sum_{d \in N_b} K_{\text{reg}}(c, d). \quad (17.26)$$

The nonempty classes come from the internal classifier. The cross matrix need not be Hermitian, unlike a Gram matrix constructed with the same three vectors on both sides.

Proposition 17.19 (Polar factor and its scope). *If G is invertible,*

$$U_G = G(G^\dagger G)^{-1/2} \quad (17.27)$$

is unitary. If G is also Hermitian positive definite, then $U_G = I_3$. Consequently, the rank of a positive Gram matrix certifies the sufficiency of the observables, but does not by itself determine nontrivial mixing between the charged and neutral frames.

Proof. $G^\dagger G$ is positive definite; its positive square root is invertible, and

$$U_G^\dagger U_G = (G^\dagger G)^{-1/2} (G^\dagger G) (G^\dagger G)^{-1/2} = I_3.$$

By finite dimensionality, also $U_G U_G^\dagger = I_3$. If $G > 0$, the positive square root of $G^\dagger G = G^2$ is G , and $U_G = I_3$. $\square$

Remark 17.20 (Polar factorization and spectral frames). Equations (17.25) and (17.26) preserve, respectively, the record's rank criterion and the cross matrix of the charged and neutral sectors. This exposition separates their functions to avoid turning a positive Gram matrix into an artificial angle selector. Identifying U_G with (17.23) requires compatibility of the cross kernel with the declared spectral frames. The identity of a polar factor or mere unitarity does not replace that identification. No new numerical PMNS matrix is specified here.

17.11 Check of the rank-one phase

Let b_K be the source's normalized incidence witness and $Q_K = |b_K\rangle\langle b_K|$. The operator

$$\Phi_K = I + (e^{i\pi_{\text{HMT}}/6} - 1)Q_K \quad (17.28)$$

is unitary in its ambient space. For isometric frames, consider $U_\Phi = F_\ell^\dagger \Phi_K F_\nu$.

Proposition 17.21 (Exact criterion for a unitary compression). *The operator U_Φ is a contraction. It is unitary exactly when*

$$\Phi_K(\text{Im } F_\nu) = \text{Im } F_\ell. \quad (17.29)$$

Proof. Let $P_\ell = F_\ell F_\ell^\dagger$ and $P_\nu = F_\nu F_\nu^\dagger$. Then $U_\Phi^\dagger U_\Phi = F_\nu^\dagger \Phi_K^\dagger P_\ell \Phi_K F_\nu \leq I_3$. Equality is equivalent to $(I - P_\ell)\Phi_K F_\nu = 0$, that is, to inclusion of the left-hand image in (17.29) in the right-hand one. Both have dimension three, so the inclusion is equality. $\square$

The source [Section 17, equations (17.18) and (17.26)] preserves this minimal ansatz as a negative check on physical identification. Its ambient unitarity, and even satisfaction of (17.29), do not replace the complete record or the frames of (17.23). The check is not removed when angular propagation is assembled in article II.

17.12 Leptonic phase and the Dirac–Majorana criterion

Phase changes of bases must be distinguished from the self-conjugation data of the neutral sector. For any unitary leptonic transport, one can form the quartet

$$J_\ell = \text{Im}(U_{e1} U_{\mu 2} \overline{U_{e2}} \overline{U_{\mu 1}}). \quad (17.30)$$

The phase cancellation proved for (17.18) applies equally here. The quartet does not fix all the data of the neutral realization or determine by itself whether it is of Dirac or Majorana type.

The Majorana realization is specified by the antiunitary self-conjugation $\mathcal{C}_\nu$, compatible with mass and dynamics:

$$\mathcal{C}_\nu^2 = I, \quad \mathcal{C}_\nu M_\nu = M_\nu \mathcal{C}_\nu, \quad \mathcal{C}_\nu H_\nu = H_\nu \mathcal{C}_\nu, \quad (17.31)$$

with domains preserved. The representation is realized in the corresponding fixed real form. The Dirac alternative also preserves a charge Q distinguishing the mode from its conjugate:

$$\mathcal{C}_\nu Q \mathcal{C}_\nu^{-1} = -Q, \quad Q\psi \neq 0. \quad (17.32)$$

These statements preserve their representational data. Neutrality, the involution of dodecaphasic words, and a topological Ising sector do not separately decide the alternative.

Remark 17.22 (Basis freedom and Majorana phases). The double coset (17.24) describes the basis freedom of the Hermitian pair. If a Majorana real form is additionally fixed, admissible phase changes must preserve that additional datum; not all the freedoms of a Dirac representation can be transferred without qualification. The present block preserves the self-conjugation criterion and assigns no values to Majorana phases or neutrino masses. Such values are not obtained from neutrality or from a rank-one polarization.

17.13 Closure of angular transport

The quark construction expresses the angular coordinates first, then the complex amplitudes, and finally the phase invariants and operator transport. The leptonic construction expresses record operators and spectral frames before the change of basis. Both routes preserve their residence in the common enriched state and in the discrete structure of the continuum; neither introduces an experimental table as a selector of coefficients or states.

The checks in this section are local and explicit: rational CKM inversion and intertwining; spectral preservation by conjugation; completeness and common space of the PMNS frames; rank of the record; compatibility of the cross matrix; and preservation of the self-conjugation data. Angular propagation is thus incorporated into the theory of action, area, and information without making mixing a second mass law or turning the Barbero reader into its cause.

18 Equivariant realization of the angular reader in Witt incidence

The angular chart and exceptional incidence preserve different information from the same APP-TRIT-TPK development. The former expresses rotation coordinates; the latter preserves relations between positions, supports, and orientations. Comparing them requires a transport that preserves these relations and allows the complete coordinates to be recovered. Here we construct that transport and prove its compatibility with the oriented permutations of the regional triplet.

The antecedents of the composition are the dodecaphase seal K , its marked face B_K , the sector of the three-dimensional isotypic projector P_3^{ico} , the compatible branches of the nonadic filtration, and the sectorial system of Definition 17.4. Each enters with its structure, not merely through its scalar value. The real vector spaces in this section are subsequent realizations of those already expressed sectors. Generation of the seal, transport of its coordinates, and evaluation of the reader correspond to distinct stages of the causal composition.

18.1 Regional triplet and orientation line

Let V be the counting space of the three positive regions of π_{HMT} , with ordered orthonormal basis

$$(v_1, v_2, v_3) = (933555, 393555, 339555).$$

The notation identifies each basis vector with the region it represents. In the prolongation window of index seven, the eleven admitted branches are distributed into orbits of sizes $3 + 3 + 1 + 1 + 3$ under simultaneous permutation of the final three positions of the first two words. The sector retaining continuation to horizon two consists of

branch	word triple	positive region
0	200112 212221 110110	339555
1	200121 212212 110110	393555
2	200211 212122 110110	933555.

The central branch additionally extends to horizon three. These horizons belong to the same filtration of compatible prolongations; the successive support cardinalities are 11, 3, and 1.

The map from the tail of the first word to the positive region substitutes $1 \mapsto 3$ and $2 \mapsto 9$ coordinate-wise and adds the regional tail 555. It therefore preserves the position of the exceptional symbol and commutes with every permutation of the three positions. The branch order is (339555, 393555, 933555); the change from the order of V is the permutation $R = (1\ 3)$.

The orientation line of V is $o(V) = \det V$. The map

$$\Psi : \Lambda^2 V \longrightarrow \text{Hom}(V, \det V), \quad \Psi(u \wedge v)(w) = u \wedge v \wedge w \quad (18.1)$$

is identified through the counting metric with an isometry $\Lambda^2 V \rightarrow o(V) \otimes V$. For $\omega = v_1 \wedge v_2 \wedge v_3$, its values on the exterior basis are

$$v_1 \wedge v_2 \mapsto \omega \otimes v_3, \quad v_2 \wedge v_3 \mapsto \omega \otimes v_1, \quad v_1 \wedge v_3 \mapsto -\omega \otimes v_2.$$

In angular order (12, 23, 13), joint transport of basis and orientation has matrix

$$D = \det(R)P_R \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}. \quad (18.2)$$

Thus $D(\theta_{12}, \theta_{23}, \theta_{13})^\top = (-\theta_{12}, \theta_{13}, -\theta_{23})^\top$. Throughout this section, $P_p e_j = e_{p(j)}$ defines the permutation-matrix convention, and the asterisk denotes the adjoint for the declared counting metrics.

Proposition 18.1 (Naturality of the oriented identification). *For every $p \in S_3$,*

$$D\Lambda^2 P_p = \det(p)P_{RpR^{-1}}D, \quad D^*D = I_3.$$

Proof. The identity $(P_p u) \wedge (P_p v) \wedge (P_p w) = \det(p)(u \wedge v \wedge w)$ proves naturality of (18.1). Simultaneous change of regional order introduces conjugation by R . Finally, the columns of D are signed coordinate vectors, whence $D^* D = I_3$. $\square$

The orientation line explains the difference between the character $(3, 1, 0)$ of the position triplet and the character $(3, -1, 0)$ of the oriented triplet, on the identity, transposition, and order-three cycle classes. The three transpositions provide an explicit control: omitting the determinant factor changes the sign of the preceding identity. A global change of the oriented anchor is instead transported as a joint coordinate change.

18.2 Dodecaphase sector and variational selection of orientation

The chart of the previously expressed seal is

$$K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601). \quad (18.3)$$

To make its incidence realization explicit, write the ternary Paley matrix in the gauge used:

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}, \quad C(a) = (a, aA_W), \quad a \in \mathbb{F}_3^6.$$

Support is taken on the twelve coordinates of $C(a)$. The marked intersection has the two compatible expressions

$$B_K = \text{supp } C(010211) \cap \text{supp } C(201101) = \{p : K_p \geq 729\} = \{4, 6, 9, 10\}. \quad (18.4)$$

The equality is evaluated by ternary multiplications of the displayed matrix. In particular, the intersection is obtained as a set of positions before its cardinality four is used.

To fix the representation of the twelve positions completely, consider A_5 acting by left multiplication on the cosets $r_p C_5$, where $C_5 = \langle (1\ 2\ 3\ 4\ 5) \rangle$. The representatives, written as image lists on $\{1, \dots, 5\}$, are

p	r_p	p	r_p
1	(4, 3, 5, 2, 1)	7	(5, 4, 3, 2, 1)
2	(4, 2, 1, 3, 5)	8	(1, 3, 4, 2, 5)
3	(1, 2, 4, 5, 3)	9	(4, 2, 3, 5, 1)
4	(2, 5, 3, 4, 1)	10	(3, 2, 5, 4, 1)
5	(4, 3, 1, 5, 2)	11	(4, 5, 2, 3, 1)
6	(5, 1, 2, 3, 4)	12	(5, 3, 2, 4, 1).

Denote this permutation representation by ρ , and put $V_{11} = \mathbf{1}^\perp \subset \mathbb{R}^{12}$. The icosahedral three-dimensional isotypic projector is determined by

$$P_3^{\text{ico}} = \frac{3}{60} \sum_{a \in A_5} \chi_3(a^{-1}) \rho(a), \quad (18.5)$$

where the values of χ_3 on classes of types $1, 2^2, 3, 5_a, 5_b$ are

$$\left(3, -1, 0, \frac{1 + \sqrt{5}}{2}, \frac{1 - \sqrt{5}}{2} \right).$$

Class 5_a contains $(1\ 2\ 3\ 4\ 5)$, and class 5_b contains its square. This convention distinguishes the two three-dimensional summands. The superscript ico identifies this component of the A_5 representation;

the cyclic projector P_3 entering (12.2) acts on twelve-position transport with its own definition. Finite evaluation of (18.5) satisfies $P_3^{\text{ico}} = (P_3^{\text{ico}})^* = (P_3^{\text{ico}})^2$, $P_3^{\text{ico}} \mathbf{1} = 0$, and $\text{Tr } P_3^{\text{ico}} = 3$.

The two marked vectors of the sector are

$$b = P_3^{\text{ico}} \left(\mathbf{1}_{B_K} - \frac{1}{3} \mathbf{1} \right), \quad u = P_3^{\text{ico}} (K - \bar{K} \mathbf{1}), \quad \bar{K} = \frac{1}{12} \sum_{p=1}^{12} K_p, \quad \|b\|^2 = 1.$$

For each of the ten order-three subgroups $C_3 \subset A_5$, let $H = N_{A_5}(C_3) \cong S_3$. The character sgn_H takes value +1 on C_3 and -1 on $H \setminus C_3$. It is the character of the quotient H/C_3 , whereas the ambient sign of elements of A_5 is always positive. Define

$$e_{\text{sgn}, H} = \frac{1}{6} \sum_{a \in H} \text{sgn}_H(a) \rho(a), \quad \mathcal{E}_B(H) = \|e_{\text{sgn}, H} b\|^2. \quad (18.6)$$

Proposition 18.2 (Normalizer and orientation determined by the seal). *The maximization rule of (18.6) selects a unique normalizer H_* . Its energy and the second distinct energy are*

$$\frac{2}{3} + \frac{2\sqrt{5}}{15}, \quad \frac{1}{3} + \frac{2\sqrt{5}}{15},$$

respectively; the gap is 1/3. Projection of u selects the same normalizer. Within H_* , maximizing $s(a) = \langle b, \rho(a)u \rangle$, separately among order-three elements and involutions, determines

$$c_* = (1, 2, 4, 5, 3), \quad r_* = (2, 1, 5, 4, 3)$$

in image notation on five points. The identity $r_* c_* r_* = c_*^{-1}$ holds.

Proof. Evaluation of the preceding finite sums in $\mathbb{Q}(\sqrt{5})$ gives the following distribution of the ten energies:

energy	multiplicity	energy	multiplicity
$2/3 + 2\sqrt{5}/15$	1	$2/3 - 2\sqrt{5}/15$	1
$1/3 + 2\sqrt{5}/15$	2	$1/3 - 2\sqrt{5}/15$	2
$1/6 + \sqrt{5}/30$	2	$1/6 - \sqrt{5}/30$	2.

The greatest belongs to the subgroup generated by c_* . The next is obtained by subtracting 1/3. The same evaluation with u has a unique maximum at that subgroup, of value $2526869/12 + 2813609\sqrt{5}/30$. The scores of the two cycle orientations are

$$s(c_*) = \frac{1341}{4} + \frac{2409\sqrt{5}}{20}, \quad s(c_*^{-1}) = \frac{357}{2} + \frac{1853\sqrt{5}}{10}.$$

Their difference is $627/4 - 1297\sqrt{5}/20 > 0$, as is checked by comparing positive squares. The scores of the three involutions are

image list	$s(a)$
(2, 1, 3, 5, 4)	$-1341/4 - 2409\sqrt{5}/20$
(2, 1, 4, 3, 5)	$-1659/4 - 2667\sqrt{5}/20$
(2, 1, 5, 4, 3)	$-357/2 - 1853\sqrt{5}/10$.

The unique maximum corresponds to r_* . Composition of the two image lists verifies the dihedral relation. All these operations use K, B_K, P_3^{ico} , and the finite actions already specified. $\square$

Write $\vartheta : S_3 \rightarrow H_*$ for the isomorphism determined by $\vartheta((123)) = c_*$ and $\vartheta((13)) = r_*$. The position reflection fixes the central branch reaching horizon three. This orientation completes regional identification before any angle evaluation.

18.3 Reynolds intertwiner and polar normalization

In the branch basis, let $\rho_o(g) = \det(g)P_g$ and $w = (0, 1, 0)^\top$. The Reynolds operator defined by summation over the group is

$$A_R = \sum_{g \in S_3} \rho_o(g) |w\rangle\langle b| \rho(\vartheta(g))^{-1}. \quad (18.7)$$

The symbol A_R distinguishes this operator from the angle $A_{\deg}$. Each summand has unit scale; metric normalization is performed through its polar factor.

Proposition 18.3 (Polar isometry of the oriented sector). *The following hold:*

$$A_R = A_R P_3^{\text{ico}}, \quad \rho_o(g)A_R = A_R \rho(\vartheta(g)), \quad G = A_R A_R^* > 0, \quad A_R^* G^{-1} A_R = P_3^{\text{ico}}.$$

With $E = \mathbf{1}\mathbf{1}^\top/3$,

$$G = \lambda_s E + \lambda_t (I_3 - E), \quad \lambda_s = 8 + \frac{8\sqrt{5}}{5}, \quad \lambda_t = \frac{1}{2} - \frac{\sqrt{5}}{5} > 0. \quad (18.8)$$

Consequently,

$$U_{GK} = G^{-1/2} A_R : P_3^{\text{ico}} V_{11} \longrightarrow o(V) \otimes V$$

is a surjective isometry: $U_{GK}^* U_{GK} = P_3^{\text{ico}}$ and $U_{GK} U_{GK}^* = I_3$.

Proof. The projector P_3^{ico} commutes with ρ , and $b = P_3^{\text{ico}} b$; hence $A_R = A_R P_3^{\text{ico}}$. Reindexing the Reynolds sum proves equivariance. Multiplication of the finite sum gives diagonal entries $3 + 2\sqrt{5}/5$ and off-diagonal entries $5/2 + 3\sqrt{5}/5$ in G . The projections E and $I_3 - E$ give (18.8). Both eigenvalues are positive; for the smaller, $25 > 20$ suffices. Thus A_R has rank three. Its row space is contained in the rank-three sector of P_3^{ico} , and equals it. The orthogonal projector onto that row space is $A_R^* (A_R A_R^*)^{-1} A_R$, giving the final identity. The two polar-factor equations follow by substitution. $\square$

The positive root preserves equivariance because both spectral projectors commute with the oriented representation. Uniqueness of the polar factor pertains to the Reynolds operator and orientation determined by the preceding rules.

18.4 Point-hexad incidence and joint gauge change

Let $\mathcal{W}$ be the design $S(5, 6, 12)$ of 132 hexads obtained from the weight-six supports of the ternary code $C(a)$. The incidence matrix is

$$I_{H,p} = \mathbf{1}_{p \in H}, \quad (I\mathcal{Z})_H = \sum_{p \in H} z_p.$$

The blocks are reconstructed by enumerating $a \in \mathbb{F}_3^6$, retaining weight six, and collecting equal supports. The finite verification that each five-point subset belongs to a unique hexad fixes the design property used below.

Lemma 18.4 (Metric and spectral incidence identities). *Let J_{12} and $J_{132,12}$ be the all-ones matrices indicated by their dimensions, and let*

$$(G_{\mathcal{W}})_{H,H'} = \binom{|H \cap H'|}{4}.$$

Then

$$I^* I = 36I_{12} + 30J_{12}, \quad G_{\mathcal{W}} I = 30I + 15J_{132,12}. \quad (18.9)$$

In particular, $U_{\mathcal{W}} = I/6$ is an isometry on $V_{11} = \mathbf{1}^\perp$, with image contained in $E_{30} = \ker(G_{\mathcal{W}} - 30I_{132})$.

Proof. The design property gives 66 hexads per point and 30 per pair of points:

$$\frac{\binom{11}{4}}{\binom{5}{4}} = 66, \quad \frac{\binom{10}{3}}{\binom{4}{3}} = 30.$$

These are the diagonal and off-diagonal entries of I^*I . For the second identity, count pairs (T, H') , where $T \subset H \cap H'$, $|T| = 4$, and $p \in H'$. If $p \in H$, ten tetrads $T \subset H$ contain p , and each belongs to four hexads; the other five tetrads, adjoined to p , determine one hexad each. The result is $10 \cdot 4 + 5 = 45$. If $p \notin H$, each of the fifteen tetrads determines a hexad together with p , and the result is 15. This yields $30I_{H,p} + 15$. The all-ones matrices vanish on V_{11} , proving the isometry and spectral membership. $\square$

The gauge change of the twelve positions is

$$\sigma = (1, 5, 2, 6, 7, 4, 3, 10, 11, 9, 8, 12), \quad (18.10)$$

written as an image list with indices starting at one. If ρ_{perm} is the point action underlying ρ , the six permutations

$$\tau_g = \sigma \rho_{\text{perm}}(\vartheta(g)) \sigma^{-1}, \quad g \in S_3,$$

preserve $\mathcal{W}$. Their two generators have image lists

$$\begin{aligned} \tau_{(123)} &= (3, 6, 5, 2, 1, 4, 9, 12, 11, 8, 7, 10), \\ \tau_{(13)} &= (2, 1, 4, 3, 6, 5, 8, 7, 10, 9, 12, 11). \end{aligned}$$

The assertion is checked on the finite design: these two generating permutations are applied to the weight-six supports of $C(a)$, and their images are verified to run through the same 132 supports. Products of these generators give the other four permutations. The induced action Q_{τ_g} on hexads satisfies

$$IP_{\tau_g} = Q_{\tau_g} I, \quad P_{\sigma} \rho(\vartheta(g)) = P_{\tau_g} P_{\sigma}. \quad (18.11)$$

The first identity expresses exactly the equivalence $p \in H \iff \tau_g(p) \in \tau_g(H)$.

This change jointly transports positions, code, incidence, origin, chamber, and orientation of the seal. In the initial gauge, five elements of this order-six subgroup do not preserve the fixed design. Conjugation (18.10) provides the compatible gauge and maintains that control over the initial labeling. The procedure exhibits a gauge change; its uniqueness does not enter the proof.

18.5 Isometric composition and coordinate recovery

Theorem 18.5 (Dodecaphase–Witt angular transport). *The composition*

$$F = \frac{1}{6} IP_{\sigma} A_R^* G^{-1/2} D : \mathbb{R}_{\text{ang}}^3 \longrightarrow E_{30} \quad (18.12)$$

is an isometry. Its coordinate action and equivariance are

$$(F\theta)_H = \frac{1}{6} \sum_{p \in H} (P_{\sigma} A_R^* G^{-1/2} D \theta)_p, \quad (18.13)$$

$$FX_g = Q_{\tau_g} F, \quad X_g = \Lambda^2 P_{R^{-1}gR}. \quad (18.14)$$

Proof. The columns of A_R^* belong to $P_3^{\text{ico}} V_{11}$, and P_{σ} preserves V_{11} . Lemma 18.4 therefore proves membership of the image of F in E_{30} . Moreover,

$$\begin{aligned} F^* F &= D^* G^{-1/2} A_R P_{\sigma}^* \frac{I^* I}{36} P_{\sigma} A_R^* G^{-1/2} D \\ &= D^* G^{-1/2} A_R A_R^* G^{-1/2} D = D^* D = I_3. \end{aligned}$$

The coordinate formula expands multiplication by incidence. For equivariance, compose in order the naturality of D , equivariance of A_R^* , commutation of $G^{-1/2}$ with ρ_o , and the two identities of (18.11). This produces (18.14) for each $g \in S_3$. $\square$

Proposition 18.6 (Covariance of the complete gauge). *Let S, T, V_o be orthogonal coordinate changes in the dodecaphase space, Witt point space, and oriented sector, respectively. Let Q_T be the corresponding coordinate change in the incidence space; for a design relabeling, it is the induced hexad permutation. If one jointly transports*

$$\begin{aligned} A'_R &= V_o A_R S^*, & G' &= V_o G V_o^*, & P'_\sigma &= T P_\sigma S^*, \\ D' &= V_o D, & U'_W &= Q_T U_W T^*, \end{aligned}$$

the composition satisfies $F' = Q_T F$.

Proof. Uniqueness of the positive root gives $(G')^{-1/2} = V_o G^{-1/2} V_o^*$. Substituting the five transformations into $F' = U'_W P'_\sigma (A'_R)^* (G')^{-1/2} D'$, the factors $T^* T$, $S^* S$, and $V_o^* V_o$ cancel in pairs, leaving $Q_T U_W P_\sigma A_R^* G^{-1/2} D = Q_T F$. $\square$

The preceding covariance describes a coordinate change of all marked data. Its application to an active cardinal transformation requires the corresponding identity between actions on the state; the two notions thus preserve their own domains.

18.6 Faithful realization of the angular sectorial system

Use chart x_θ of (17.5),

$$x_\theta = \begin{pmatrix} A_{\text{deg}} \\ C_{\text{deg}}^*/6 \\ (180/\pi_{\text{HMT}})\Delta_4 \end{pmatrix}.$$

The component $h_\theta = C_{\text{deg}}^*/6$ distributes the odd component of the record among six antipodal pairs. Definition 17.4 fixes the three rules

$$R_{12} = (2, -p, bq), \quad R_{23} = (0, q, -p^2/2), \quad R_{13} = (1, -pb, -(o_8 - h_6)/t)$$

for $(b, p, q, h_6, o_8, t) = (4, 5, 7, 6, 8, 3)$. The face of cardinality four, pentadic order, pivot coordinate, hexadic–octadic difference, and tritic alphabet retain distinct types. The bidirectional orbital normalization of 17.2 preserves the term $p^2/2$, including the contribution of fixed pairs.

Its evaluation, already proved in Theorem 17.5, gives

$$M_{\text{CKM}} = \begin{pmatrix} 2 & -5 & 28 \\ 0 & 7 & -25/2 \\ 1 & -20 & -2/3 \end{pmatrix}, \quad \det M_{\text{CKM}} = -\frac{3857}{6}.$$

Transport F acts on the coordinates expressed by those rules; the preceding choice of functionals retains its place in the sectorial construction.

Corollary 18.7 (Angular reader in the incidence module). *The map*

$$\mathcal{C} = F M_{\text{CKM}} \tag{18.15}$$

satisfies

$$\mathcal{C}^* \mathcal{C} = M_{\text{CKM}}^* M_{\text{CKM}}, \quad M_{\text{CKM}}^{-1} F^* \mathcal{C} = I_3.$$

It is an isometry from the space of x_θ , equipped with metric $M_{\text{CKM}}^ M_{\text{CKM}}$, onto its image.*

Proof. Substitute $F^* F = I_3$ into both products. The nonzero determinant of M_{CKM} permits use of its inverse, displayed in (17.7). Positivity of the metric follows from $x^* M_{\text{CKM}}^* M_{\text{CKM}} x = \|M_{\text{CKM}} x\|^2 > 0$ for $x \neq 0$. $\square$

The isometry F preserves the counting metric of $\theta = M_{\text{CKM}} x_\theta$; its transport to coordinates x_θ gives the metric of the corollary. This distinction avoids identifying a basis of angular generators with an orthonormal basis of their parameters.

18.7 Exact certification and finite unitary composition

The positive roots of the polar factor can be checked through their projectors while preserving exact arithmetic. Define

$$B = \frac{1}{6}IP_{\sigma}A_R^*D, \quad H = D^*GD = B^*B, \quad L = H^{-1}B^*. \quad (18.16)$$

The identities

$$LB = I_3, \quad L(BM_{\text{CKM}}) = M_{\text{CKM}}, \quad M_{\text{CKM}}^{-1}L(BM_{\text{CKM}}) = I_3$$

are evaluated in $\mathbb{Q}(\sqrt{5})$. If $E_D = D^*ED$,

$$H^{-1/2} = \lambda_s^{-1/2}E_D + \lambda_t^{-1/2}(I_3 - E_D), \quad F = BH^{-1/2}.$$

The products $E_D^2 = E_D$, $E_D(I_3 - E_D) = 0$, and the decomposition of H prove this formula without approximating its radicals. The 132 rows of B are determined by (18.13) and by the hexad construction rule. The executable certificate of this delivery preserves those rows, those of BM_{CKM} , the left inverse L , the projectors, and the labels of each hexad. Its 906 exact checks verify this composition and its orientation, incidence, and recovery identities.

Conversion to radians is performed before the trigonometric realization:

$$\vartheta = M_{\text{CKM}} \begin{pmatrix} (\pi_{\text{HMT}}/180)A_{\text{deg}} \\ (\pi_{\text{HMT}}/180)C_{\text{deg}}^*/6 \\ \Delta_4 \end{pmatrix}, \quad \delta = \frac{\pi_{\text{HMT}}}{180} 9A_{\text{deg}}. \quad (18.17)$$

The finite realization preserves the order $V_{\text{CKM}} = R_{23}U_{13}(\delta)R_{12}$ of (17.19). To specify the additional information of the complex phase, let E_{ab} be the elementary matrix and define

$$J_{13} = E_{13} - E_{31}, \quad S_{13} = E_{13} + E_{31}, \quad X_{13}(\delta) = \cos \delta J_{13} - i \sin \delta S_{13}.$$

Then $X_{13}(\delta)^* = -X_{13}(\delta)$ and $U_{13}(\delta) = \exp(\vartheta_{13}X_{13}(\delta))$. Indeed, $X_{13}(\delta)^2 = -(E_{11} + E_{33})$, and its exponential reproduces the unitary 13 block of (17.19). Exterior transport preserves the oriented generators; the symmetric imaginary component preserves phase information. The ordered product of the three finite factors maintains these two contributions and their order of composition.

18.8 Structural result and scope of the composition

The construction provides an equivariant isometric transport from the three angular coordinates to Witt incidence, together with an explicit reconstruction of A_{deg} , $C_{\text{deg}}^*/6$, and Δ_{deg} . It preserves exterior orientation, the dodecaphase sector, the marked face of the seal, and the joint gauge change. In particular, the incidence realization represents the same angular content on another support and allows its three coordinates to be recovered.

Prior generation of K , selection of the sectorial functionals, and physical identification of the amplitudes retain their own proofs. The recovery identities proved here establish faithfulness of the composition; its subsequent use keeps those antecedents explicit. The complementary development of quartic memory planes remains another construction of the state and does not interpose itself as an additional requirement for the alignment realized in (18.12).

19 Area operator and reconstruction of sectorial information

Regrouping six trits constructs a bijection between $(\mathbb{Z}/9\mathbb{Z})^3$ and $(\mathbb{Z}/27\mathbb{Z})^2$, both of cardinality 729. The bijection is not presented as an additive isomorphism. On a 9×9 boundary block, the source decomposes 36 links into two sets of 18, labeled by channels 90 and 120.

19.1 Tritic regrouping and boundary incidence

The correspondence preserves an ordered word, in addition to its cardinality. Write each coordinate $x_j \in \mathbb{Z}/9\mathbb{Z}$, for $j = 1, 2, 3$, through its unique positional representatives $x_j = r_{2j-1} + 3r_{2j}$, with $r_k \in \{0, 1, 2\}$. The map is

$$\beta_{729}(x_1, x_2, x_3) = (r_1 + 3r_2 + 9r_3, r_4 + 3r_5 + 9r_6) \in (\mathbb{Z}/27\mathbb{Z})^2. \quad (19.1)$$

Proposition 19.1 (Positional correspondence between interior and boundary). *The map β_{729} is bijective and preserves the order of the six trits. Its inverse extracts the two triples of digits and regroups them into three pairs.*

Proof. The positional expansion of the representatives $0, \dots, 8$ and $0, \dots, 26$ is unique. Extracting the six digits, regrouping them, and extracting them again returns the same word. The inverse operation thus restores each of the three initial coordinates. $\square$

Carry preserves its position in the enriched word. The additive operations of the two packagings have different boundaries: for example, $\beta_{729}(0, 2, 0) = (18, 0)$ and $\beta_{729}(0, 1, 0) = (9, 0)$, whose sum in the order-twenty-seven torus is $(0, 0)$, whereas $\beta_{729}(0, 3, 0) = (0, 1)$. This distinction allows the positional correspondence to be reused while keeping the operations typed.

In the torus $(\mathbb{Z}/27\mathbb{Z})^2$, the block $B_{\partial} = \{0, \dots, 8\}^2$ has the oriented link sets

$$\partial B_{90} = \{((-1, j), (0, j)), ((8, j), (9, j)) : 0 \leq j \leq 8\}, \quad (19.2)$$

$$\partial B_{120} = \{((i, -1), (i, 0)), ((i, 8), (i, 9)) : 0 \leq i \leq 8\}. \quad (19.3)$$

All coordinates are read modulo twenty-seven. Each collection contains eighteen links, and the two are disjoint. Their labels preserve the incidence channels; the number of links is $18 + 18$.

Theorem 19.2 (Minimum capacity of the cubic cut). *In the cubic graph with vertices $\{0, \dots, N-1\}^3$, with $N \geq 2$, links between neighbors, and unit capacity per link, the minimum cut between two opposite faces has capacity N^2 . For $N = 9$, a maximally mixed product state on the trits of a minimum layer has information*

$$\text{cap}_{\min} = 81, \quad s_{\text{corte}} = 81 \log 3. \quad (19.4)$$

Proof. The N^2 lines parallel to the normal of the faces form edge-disjoint paths. Every separating cut intercepts each path, so it contains at least N^2 links. The links crossing a transverse layer constitute a cut of that cardinality. Each uniform trit contributes $\log 3$ nats; entropy is additive for the product state of the eighty-one trits of the layer. $\square$

The capacity of the cubic cut and the state of the thirty-six links in (19.3) belong to observables defined on different cuts. The genealogy preserves both domains and their measures.

19.2 Transport of the collective incidence modes

The spaces $\mathcal{F}_6$ and $\mathcal{F}_8$ of Section 14 provide the uniform vectors

$$u_{90} = \frac{1}{\sqrt{90}} \sum_{f \in \mathcal{F}_6} (\delta_f, 0), \quad u_{120} = \frac{1}{\sqrt{120}} \sum_{f \in \mathcal{F}_8} (0, \delta_f)$$

in $\ell^2(\mathcal{F}_6) \oplus \ell^2(\mathcal{F}_8)$. In $\ell^2(\partial B_{90}) \oplus \ell^2(\partial B_{120})$, construct

$$v_{90} = \frac{1}{\sqrt{18}} \sum_{e \in \partial B_{90}} (\delta_e, 0), \quad v_{120} = \frac{1}{\sqrt{18}} \sum_{e \in \partial B_{120}} (0, \delta_e).$$

The two pairs are orthonormal. Denote their respective spanned planes by $\mathcal{H}_{\text{inc}}^{\text{coll}}$ and $\mathcal{H}_{\partial}^{\text{coll}}$.

Theorem 19.3 (Collective intertwining of the sectorial labels). *There is a unique surjective linear isometry between these planes such that*

$$\mathcal{J}_{6 \rightarrow 8} u_m = v_m \quad (m = 90, 120). \quad (19.5)$$

For the operators

$$\begin{aligned} D_{\text{inc}} &= 90|u_{90}\rangle\langle u_{90}| + 120|u_{120}\rangle\langle u_{120}|, \\ D_{\partial} &= 90|v_{90}\rangle\langle v_{90}| + 120|v_{120}\rangle\langle v_{120}|, \end{aligned}$$

the following identity holds:

$$\mathcal{J}_{6 \rightarrow 8} D_{\text{inc}} = D_{\partial} \mathcal{J}_{6 \rightarrow 8}. \quad (19.6)$$

The sectorial projectors and their oriented combination with coefficients 12 : -1 are also transported.

Proof. The image of a basis fixes the unique linear map. Norms and inner products are preserved because both bases are orthonormal. Identity (19.6) holds on each vector u_m , since its two sides are mu_m . The projectors are expressed as $(120I - D)/30$ and $(D - 90I)/30$; their linear functional calculus and any combination of the two preserve intertwining. $\square$

This transport acts on the two uniform modes and their spectral labels. Individual incidences and fluctuations within each sector remain in their complete spaces. The link preserves exactly the collective datum that the areal weighting will use.

19.3 Angular character and area operator

Let $N_e = \text{diag}(0, 1, 2)$ on each link, and let N_{90}, N_{120} be the sector sums. With the scale

$$\theta_{\text{clk}} = \frac{C^*}{6} \frac{\pi}{180}, \quad a_0 = \frac{1 + 2 \cos(10^\circ)}{3} \theta_{\text{clk}},$$

fixed, the angular factor is the normalized real character of the triplet

$$\chi_{10} = \frac{1}{3} \text{Tr} \text{diag}(1, e^{i\pi/18}, e^{-i\pi/18}) = \frac{1 + 2 \cos(10^\circ)}{3}, \quad a_0 = \chi_{10} \theta_{\text{clk}}. \quad (19.7)$$

The sum of the conjugate phases proves the equality. On the positive branch considered, $\theta_{\text{clk}} > 0$, $\chi_{10} > 0$, and $\gamma_{\text{HMT}} > 0$; hence a_0 and the areal scale are positive. The angular triplet and its normalization precede the entropic distribution on the links. With these quantities, the normalized area operator is

$$\hat{\mathcal{A}}/\ell_P^2 = \gamma_{\text{HMT}} a_0 (N_{90} + N_{120}).$$

Its spectrum has values $n\gamma_{\text{HMT}}a_0$, $0 \leq n \leq 72$, with multiplicities given by $(1 + z + z^2)^{36}$. The proof consists in summing the eigenvalues 0, 1, 2 of the 36 tensor factors [Section 19, equation (19.8)].

Theorem 19.4 (Recovery of the boundary sectors). *Let*

$$N = N_{90} + N_{120}, \quad D = 90N_{90} + 120N_{120}.$$

Then

$$N_{90} = 4N - \frac{D}{30}, \quad N_{120} = \frac{D}{30} - 3N. \quad (19.8)$$

Thus the total and the weighted reader jointly recover the two occupations.

Proof. The matrix $\begin{pmatrix} 1 & 1 \\ 90 & 120 \end{pmatrix}$ has determinant 30. Its inversion gives (19.8). On the domain of integer occupations, the image satisfies $D \in 30\mathbb{Z}$ and $90N \leq D \leq 120N$. $\square$

The modular Hamiltonian in the source is $K_A = -\log \rho_A = cI + \Delta_{\text{mod}} D$. For $\Delta_{\text{mod}} \neq 0$ and known normalization, $D = (K_A - cI)/\Delta_{\text{mod}}$ recovers the second observable. This operator describes the spectral structure of the reduced state; it is not identified by notation with the generator of temporal evolution. Area preserves total occupation, and the pair adds provenance. For example, $(N_{90}, N_{120}) = (1, 0)$ and $(0, 1)$ share $N = 1$, but have $D = 90$ and $D = 120$.

19.4 State normalization and the meaning of the modular Hamiltonian

The second observable is made explicit through the state that defines it. The preceding tensor decomposition is retained: eighteen qutrits per sector, each with number operator $\text{diag}(0, 1, 2)$; N_m is the sum of these operators, and the sectors act on distinct factors. Let

$$z_m = 1 + e^{-m\Delta_{\text{mod}}} + e^{-2m\Delta_{\text{mod}}}, \quad m \in \{90, 120\}.$$

The state collection in the source has the product realization

$$\rho_A = Z^{-1} e^{-\Delta_{\text{mod}}(90N_{90} + 120N_{120})}, \quad Z = z_{90}^{18} z_{120}^{18}.$$

Commutativity of the number operators and tensor factorization prove the expression for Z . The state has full rank for real, finite Δ_{mod} , and its spectral logarithm gives

$$K_A = -\log \rho_A = (\log Z)I + \Delta_{\text{mod}} D.$$

For $\Delta_{\text{mod}} \neq 0$, K_A weights the excitations of the two sectors differently. The joint reading with N recovers the two occupations through equation (19.8).

The need for the joint reading is seen in two complementary examples. The occupation states $(1, 0)$ and $(0, 1)$ have the same total $N = 1$ and different modular weights. Conversely, $(4, 0)$ and $(0, 3)$ have the same $D = 360$ and different totals. The pair (N, D) separates both cases. This recovery concerns sectorial occupations, not each microscopic vector within a degeneracy of those occupations.

When $\Delta_{\text{mod}} = 0$, the state is proportional to the identity and K_A is scalar: that reading no longer distinguishes the sectors. For $\Delta_{\text{mod}} \neq 0$, the known normalization makes it possible to recover D and compose it with area. This defines the role of the modular Hamiltonian: it characterizes the reduced state and its information distribution. Its inclusion in the article is motivated by that mathematical role, whereas the evolution Hamiltonian belongs to the dynamical development of action.

19.5 Reduced state, Schmidt decomposition, and entanglement entropy

If $\rho_A = \sum_{\nu} p_{\nu} |\nu\rangle\langle\nu|$, the purification

$$|\Psi\rangle = \sum_{\nu} \sqrt{p_{\nu}} |\nu\rangle_A |\nu\rangle_B$$

satisfies $\text{Tr}_B |\Psi\rangle\langle\Psi| = \rho_A$. Thus $-\text{Tr}(\rho_A \log \rho_A)$ is the entanglement entropy of this particular pure bipartition. The enlarged state declares which system carries that interpretation.

For a two-sheet involution with one eigenvector of each sign,

$$\rho_y = \frac{e^{-yR}}{2 \cosh y}, \quad R^2 = I,$$

diagonalization gives

$$S(\rho_y) = \log(2 \cosh y) - y \tanh y, \quad D(\rho_y \| \rho_{-y}) = 2y \tanh y.$$

Entropy is even in y ; the distance explicitly compares the two orientations. This two-level entropy is not identified with the capacity $81 \log 3$ of the triadic cut.

19.6 Genealogical memory and informational content of a reading

The area operator combines boundary occupations. The weighted observable adds their sector provenance. This difference expresses a physical principle of the manuscript: accessible information depends on the map extracting the observable coordinate from the state. In APP, remainder and quotient jointly preserve the integer operation; in TPK, the visible phase coexists with orientation and the memory increment. The physical realization transports this same distinction between state, observable, and recovery record [Sections 19.6 and 19.8].

Let Ω be a finite set of states reached by the update under consideration. A reader $q : \Omega \rightarrow Y$ assigns the observable coordinate, and $M : \Omega \rightarrow \mathcal{M}$ preserves fiber memory: the relevant sheet, orientation, quotient, carry, and route. For a distribution p on those states, denote by H_p the dimensionless entropy, using the natural logarithm.

Proposition 19.5 (Informational resolution of the fiber). *If (q, M) is injective on the support of p , then*

$$H_p(X) = H_p(q(X)) + H_p(X | q(X)), \quad (19.9)$$

$$H_p(X | q(X), M(X)) = 0. \quad (19.10)$$

Proof. Factoring the joint distribution into the visible distribution and the conditional distribution on the fiber gives the first equality. Injectivity provides an inverse on the image of (q, M) ; each expressed pair determines a unique state in the support, and its conditional distribution is concentrated at that state. $\square$

Equation (19.8) realizes this mechanism for the two aggregate occupations: N gives the total and D resolves its sector distribution. The states $(N_{90}, N_{120}) = (1, 0)$ and $(0, 1)$ share the aggregate area and are distinguished by the second observable. Under an equiprobable distribution on that pair, reading N alone leaves $\log 2$ nats of uncertainty; the pair (N, D) resolves it exactly. The statement concerns these occupations. Microhistories within each sector retain their additional record.

The interpretation of edge memory is thus determined by a concrete recovery. A reduction can hide information that the joint state continues to transport. For a bijective update U , the equality $H_p(X | U(X)) = 0$ expresses logical reversibility. A physical memory-discarding or reset operation, for its part, has its own information and energy balance. Entanglement is evaluated with respect to the bipartition and reduced state specified in the preceding section.

19.7 Areal and volumetric sheets: incidence, growth, and measure

The areal–volumetric distinction has a dynamic realization in the TPK calendar. The block $(+1, -1, 0)$, applied to the modes Σ, Π through

$$+1 : m \mapsto \Sigma, \quad -1 : m \mapsto \Pi, \quad 0 : m \mapsto m,$$

produces the cyclic orbit (Σ, Π, Π) . The geometric reader realizes the correspondences $\Sigma \mapsto \mathcal{H}_{\text{ar}}$ and $\Pi \mapsto \mathcal{H}_{\text{vol}}$. Its consecutive transitions fix the incidence

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad N_9 = 3F_{\text{av}}, \quad N_{27} = 9F_{\text{av}}, \quad N_{108} = 36F_{\text{av}}. \quad (19.11)$$

Indeed, the cycle contains one transition of each type $\Sigma \rightarrow \Pi, \Pi \rightarrow \Pi, \Pi \rightarrow \Sigma$. The larger calendars repeat this block with the indicated multiplicities.

The associated symbolic envelope allows every concatenation admitted by F_{av} . Its number of paths depends on powers of that matrix. With $F_0 = 0, F_1 = 1$, induction gives

$$F_{\text{av}}^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}, \quad h_{\text{top}} = \log \varphi. \quad (19.12)$$

The matrix identity uses the Fibonacci recurrence. Exponential growth is fixed by the Perron root φ of the incidence already constructed.

Proposition 19.6 (Stationary distribution of the sheets). *The maximal-entropy measure of the symbolic envelope has weights*

$$\mu_{\text{ar}} = \frac{1}{1 + \varphi^2}, \quad \mu_{\text{vol}} = \frac{\varphi^2}{1 + \varphi^2}, \quad \frac{\mu_{\text{ar}}}{\mu_{\text{vol}}} = \varphi^{-2}. \quad (19.13)$$

Proof. The positive vector $r = (1, \varphi)$ satisfies $F_{\text{av}} r = \varphi r$. The transition probabilities $P_{ij} = (F_{\text{av}})_{ij} r_j / (\varphi r_i)$ sum to one. Symmetry of F_{av} gives stationary weights proportional to r_i^2 . Their normalization produces (19.13). In the stationary mean, $-\log P_{ij} = \log \varphi + \log r_i - \log r_j$ on the allowed edges. The final two terms cancel, and the entropy per transition is $\log \varphi$, equal to the topological entropy in (19.12). $\square$

The periodic calendar and its symbolic envelope are two explicit objects: the former has chronological frequencies $(1/3, 2/3)$; the latter admits the incidence concatenations and has measure (19.13). This separation allows each quantity to be assigned its physical-mathematical meaning. The ratio φ^{-2} characterizes the distribution of sheets under that measure; $\log \varphi$ characterizes the growth of their allowed histories. The angular boundary reading subsequently incorporates the corresponding regional coordinate. The construction distinguishes volumetric growth, areal representation, and transport memory.

19.8 Admissible continuations and survival information

For a fixed TPK state x , let $\Omega_n(x)$ be the finite set of its length- n continuations admitted by the prolongation rule. Restrictions between levels preserve prefixes. A compatible probability collection p_n , obtained from the statistical reader to be declared, defines continuation uncertainty

$$s_n(x) = - \sum_{\omega \in \Omega_n(x)} p_n(\omega) \log p_n(\omega). \quad (19.14)$$

Compatibility requires the probability of a prefix to be the sum of the probabilities of its prolongations. In the preceding envelope, the stationary measure provides a collection of this type.

Proposition 19.7 (Informational balance of refinement). *For the preceding compatible collection,*

$$0 \leq s_n(x) \leq \log |\text{supp } p_n|, \quad (19.15)$$

$$s_{n+1}(x) - s_n(x) = H(X_{n+1} | X_n) \geq 0. \quad (19.16)$$

The upper bound is attained by the uniform distribution on the support. The increment is zero when every prefix of positive probability determines its prolongation with probability one.

Proof. The elementary entropy inequality for a finite distribution gives the first line. Since X_n is the restriction of X_{n+1} , factoring probabilities over each prolongation fiber gives $H(X_{n+1}) = H(X_n) + H(X_{n+1} | X_n)$. Each conditional entropy is nonnegative and vanishes exactly for a point distribution. $\square$

Continuations remaining admissible thus have precise informational content. Their multiplicity and weights describe the residual uncertainty of the reading at finite depth. A realized history also preserves its orientation and memory. Informational entropy, reduced-state entropy, and thermodynamic entropy are linked through their respective readers; each retains the domain information necessary for its interpretation.

19.9 Cyclic correlation and preservation of path order

Let $\tau = (\tau_1, \dots, \tau_{12}) \in \{-1, 0, 1\}^{12}$ be an ordered word, with indices taken modulo 12. Define

$$\begin{aligned} G_a &= \{a, a + 4, a + 8\} \quad (1 \leq a \leq 4), \\ v_{120}(\tau) &= \sum_{a=1}^4 \text{sgn} \left(\sum_{m \in G_a} \tau_m \right), \\ v_{270}(\tau) &= \sum_{m=1}^{12} \tau_m \tau_{m+3}. \end{aligned}$$

The marginal histogram is $p_j(\tau) = \#\{m : \tau_m = j\}/12$ for $j \in \{-1, 0, 1\}$, and its entropy is $H(\tau) = -\sum_j p_j(\tau) \log p_j(\tau)$, with $0 \log 0 = 0$.

Proposition 19.8 (Order information beyond the marginal distribution). *There exist words with identical histograms and distinct correlations ν_{270} , even when ν_{120} has the same value.*

Proof. Consider

$$\tau^{(a)} = (1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0), \quad \tau^{(b)} = (1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0).$$

Both have $p_1 = 1/6$, $p_0 = 5/6$ and $p_{-1} = 0$, so their marginal entropies coincide. In each case, two of the sets G_a contain a unit and the other two contain only zeros: $\nu_{120} = 2$. In $\tau^{(a)}$, no pair of units has cyclic separation three, whereas in $\tau^{(b)}$ precisely the term $m = 1$ contributes. Thus $\nu_{270}(\tau^{(a)}) = 0$ and $\nu_{270}(\tau^{(b)}) = 1$. $\square$

This separation identifies the information retained by the ordered record: incidences between positions in addition to symbol frequencies. A reduced state retaining only the histogram assigns the same output to both words and loses this correlation. Marginal entropy, uncertainty between bases and entanglement entropy are defined on different objects; relating them requires retaining the distribution, the change of basis, and the state together with its partition, respectively.

19.10 Equation of state for sectorial information

The incidence already constructed provides eighteen tritic factors in each sector. This multiplicity allows the relation between area, the occupation distribution, and the entropy of the preceding purification to be written explicitly. We retain the reduced state defined there and abbreviate $\Delta = \Delta_{\text{mod}}$. For $m \in \{90, 120\}$, put

$$x_m = e^{-m\Delta}, \quad z_m = 1 + x_m + x_m^2, \quad f_m = \frac{x_m + 2x_m^2}{z_m}, \quad v_m = \frac{x_m + 4x_m^2}{z_m} - f_m^2. \quad (19.17)$$

Here f_m and v_m are, respectively, the mean and variance of the occupation of one factor in sector m . The parameter Δ preserves the status and normalization of the previously specified reduced state.

19.10.1 Explicit purification of the two sectors

The preceding Schmidt decomposition is made concrete on each link. For a finite real weight w , define

$$z(w) = 1 + e^{-w} + e^{-2w}, \quad \rho(w) = \frac{1}{z(w)} \sum_{n=0}^2 e^{-wn} |n\rangle\langle n|.$$

The sectorial weights are $w_m = m\Delta$, with $m = 90, 120$; hence $z(w_m) = z_m$.

Proposition 19.9 (Thermofield-double purification of sectorial incidence). *The normalized vector*

$$|\text{TFD}(w)\rangle = \frac{1}{\sqrt{z(w)}} \sum_{n=0}^2 e^{-wn/2} |n\rangle_A \otimes |n\rangle_{\bar{A}} \quad (19.18)$$

purifies $\rho(w)$. The product

$$|\Psi_\Delta\rangle = |\text{TFD}(90\Delta)\rangle^{\otimes 18} \otimes |\text{TFD}(120\Delta)\rangle^{\otimes 18} \quad (19.19)$$

has reduction $\rho_A(\Delta)$ on the thirty-six original links. Its entanglement entropy for the bipartition $A | \bar{A}$ is $s(\Delta)$.

Proof. The squared norm is $z(w)^{-1} \sum_{n=0}^2 e^{-wn} = 1$. In the partial trace, orthogonality of the second factor eliminates terms with distinct indices; the remaining terms form $\rho(w)$. The partial trace commutes with the tensor product. One thus obtains $\rho(90\Delta)^{\otimes 18} \otimes \rho(120\Delta)^{\otimes 18}$, which coincides with the normalized state defined previously. Equality between the entropy of a reduction and the entanglement of this purification is read directly from its Schmidt coefficients. $\square$

The two factors of the bipartition are determined within the construction. The physical realization of the second factor preserves that subsystem identification and its algebra of observables.

19.10.2 Occupation moments and information content

Proposition 19.10 (Mean area and entropy of the sectorial purification). *In this finite collection of states,*

$$\langle N \rangle_\Delta = 18(f_{90} + f_{120}), \quad (19.20)$$

$$\langle D \rangle_\Delta = 18(90f_{90} + 120f_{120}), \quad (19.21)$$

$$\frac{\langle \hat{\mathcal{A}} \rangle_\Delta}{\ell_P^2} = 18\gamma_{\text{HMT}}a_0(f_{90} + f_{120}), \quad (19.22)$$

$$s(\Delta) = 18 \log z_{90} + 18 \log z_{120} + \Delta \langle D \rangle_\Delta. \quad (19.23)$$

The dimensional entropy is $S(\Delta) = k_B s(\Delta)$, with the thermal transducer of Section 22.

Proof. In each factor, the normalized weights are $(1, x_m, x_m^2)/z_m$. Their first moment gives f_m , and their second moment gives the first fraction in v_m . Tensor factorization and additivity of the number operators produce the first three identities. Finally, $\log \rho_A = -(\log Z)I - \Delta D$, with $Z = z_{90}^{18}z_{120}^{18}$. Taking the trace against $-\rho_A$ gives (19.23). The purification specified previously has this same entanglement entropy. $\square$

Loss of uniformity has its own quantitative reading. In dimension $d_\partial = 3^{36}$, let $\tau_\partial = I/d_\partial$. Define

$$I_{\text{or}}(\Delta) = 36 \log 3 - s(\Delta) = D(\rho_A(\Delta) \| \tau_\partial) \geq 0. \quad (19.24)$$

The equality follows by substituting $\log \tau_\partial = -36 \log 3 I$. If p_ν are the eigenvalues of ρ_A , convexity of $t \log t$ gives $\sum_\nu p_\nu \log(d_\partial p_\nu) \geq 0$, with equality exactly for the uniform state. For this collection, that occurs at $\Delta = 0$. The deficit is dimensionless information; the areal quantum $\gamma_{\text{HMT}}a_0$ is another reading. Each expression preserves its definition even when their evaluations may be close.

Proposition 19.11 (Information response and boundary fluctuations). *For every finite real Δ ,*

$$\frac{d\langle D \rangle_\Delta}{d\Delta} = -\text{Var}_\Delta(D) = -18(90^2 v_{90} + 120^2 v_{120}), \quad (19.25)$$

$$\frac{ds}{d\Delta} = -\Delta \text{Var}_\Delta(D), \quad (19.26)$$

$$\frac{d\langle \hat{\mathcal{A}} \rangle_\Delta}{d\Delta} = -18\gamma_{\text{HMT}}a_0\ell_P^2(90v_{90} + 120v_{120}). \quad (19.27)$$

Proof. The sum defining Z is finite. Differentiating it gives $\partial_\Delta \log Z = -\langle D \rangle_\Delta$ and $\partial_\Delta^2 \log Z = \text{Var}_\Delta(D)$. The sectors and their factors are independent in the product state; their variances therefore add. On differentiating (19.23), the terms $-\langle D \rangle_\Delta$ and $\langle D \rangle_\Delta$ cancel. For area, use $\partial_\Delta f_m = -mv_m$ in (19.22). $\square$

These equations relate three levels of the same construction. Incidence fixes multiplicities and sectorial weights; the reduced state fixes their probabilities; fluctuations of those occupations determine the response of area and entropy. The scale $\gamma_{\text{HMT}}a_0\ell_P^2$ converts occupation into area, whereas k_B converts information into dimensional entropy. In the gravitational section, $\ell_P^2 = G\hbar/c^3$ makes explicit the dependence of that areal unit on action and gravitation. Comparison with a horizon additionally preserves the geometric map and the state specified there.

19.11 Occupation inversion and entropic invariants

Let $\mathcal{R}$ be the unitary involution that exchanges $|0\rangle$ and $|2\rangle$ and fixes $|1\rangle$ in each of the thirty-six tritic factors. Its action on occupations preserves the entire microscopic configuration through a bijective change of its labels.

Theorem 19.12 (Preserved entropy and complementary area). *The preceding involution satisfies*

$$\mathcal{R}N\mathcal{R}^* = 72I - N, \quad \mathcal{R}D\mathcal{R}^* = 7560I - D, \quad (19.28)$$

$$\rho_A(-\Delta) = \mathcal{R}\rho_A(\Delta)\mathcal{R}^*, \quad s(-\Delta) = s(\Delta), \quad (19.29)$$

$$\langle \hat{\mathcal{A}} \rangle_{-\Delta} + \langle \hat{\mathcal{A}} \rangle_{\Delta} = 72\gamma_{\text{HMT}}a_0\ell_P^2. \quad (19.30)$$

The relative entropy between the two orientations is

$$D(\rho_A(\Delta)\|\rho_A(-\Delta)) = \Delta(7560 - 2\langle D \rangle_{\Delta}). \quad (19.31)$$

Proof. Each local occupation n transforms into $2-n$. Thus, $N_m \mapsto 36I - N_m$ and $D \mapsto 36(90+120)I - D = 7560I - D$. Furthermore, $z_m(-\Delta) = e^{2m\Delta}z_m(\Delta)$, whence $Z(-\Delta) = e^{7560\Delta}Z(\Delta)$. Substituting both identities into the normalized state proves unitary conjugation. Spectral invariance proves equality of entropies; transformation of N proves the area sum. Finally, subtracting the two state logarithms and taking their expectation in $\rho_A(\Delta)$ yields (19.31). $\square$

At $\Delta = 0$, the reduced state is uniform and $s(0) = 36 \log 3$. In the limit $\Delta \rightarrow +\infty$, the zero-occupation configuration survives; entropy tends to zero. Inversion takes that limit to the maximum-occupation configuration. At both extremes, entropy is the same, and area distinguishes the configurations. For $\Delta > 0$, response (19.26) is strictly negative because the state has full rank and D has more than one eigenvalue. The law thus preserves the difference between reversibility of the complete configuration, equality of entropies, and change in areal reading. These properties belong to the specified sectorial collection; physical transport of the bipartition is treated in Section 26.

19.12 First modular law and finite area variations

The derivatives in (19.26) vary the parameter Δ of a collection of states. The first modular law, by contrast, fixes a reference state and considers variations of the observed state around it. Take $\rho_0 = \rho_A(\Delta_0)$, with finite real Δ_0 , and keep $\Delta_0, \gamma_{\text{HMT}}, a_0, \ell_P$ fixed. The branch $\gamma_{\text{HMT}} > 0, a_0 > 0$ is retained. The normalized sectorial areas are

$$\mathcal{A}_m^n = \gamma_{\text{HMT}}a_0N_m, \quad \hat{\mathcal{A}}_m = \ell_P^2\mathcal{A}_m^n, \quad m \in \{90, 120\}. \quad (19.32)$$

The logarithm of the reference state takes the form

$$K_0 = -\log \rho_0 = c_0I + \beta_{90}\mathcal{A}_{90}^n + \beta_{120}\mathcal{A}_{120}^n, \quad \beta_m = \frac{m\Delta_0}{\gamma_{\text{HMT}}a_0}, \quad c_0 = \log Z(\Delta_0). \quad (19.33)$$

In particular, $\beta_{120} = \frac{4}{3}\beta_{90}$; for $\Delta_0 \neq 0$, their quotient is 4/3. The coefficients arise from expressing the same modular operator in areal units.

Theorem 19.13 (First modular law in areal coordinates). *Let $\sigma(\varepsilon)$ be a differentiable collection of normalized states on an interval containing zero, with $\sigma(0) = \rho_0$. Then*

$$\left. \frac{ds(\sigma(\varepsilon))}{d\varepsilon} \right|_0 = \sum_{m=90,120} \beta_m \left. \frac{d\langle \mathcal{A}_m^n \rangle_{\sigma(\varepsilon)}}{d\varepsilon} \right|_0. \quad (19.34)$$

Proof. The state ρ_0 has full rank. In its neighborhood, the derivative of the trace of $-\sigma \log \sigma$ is $-\text{Tr}[\dot{\sigma}(\log \rho_0 + I)]$; this follows by diagonalizing ρ_0 in the formula for the spectral derivative of the trace. Normalization implies $\text{Tr} \dot{\sigma} = 0$, so the derivative is $\text{Tr}(\dot{\sigma}K_0)$. Substituting (19.33) also cancels c_0I and produces the identity. $\square$

Theorem 19.14 (Finite area identity and relative-entropy deficit). *For any normalized state σ on the same finite space, define $\Delta_\sigma\langle B \rangle = \text{Tr}[(\sigma - \rho_0)B]$. The identity*

$$s(\sigma) - s(\rho_0) = \sum_{m=90,120} \beta_m \Delta_\sigma\langle \mathcal{A}_m^n \rangle - D(\sigma\|\rho_0). \quad (19.35)$$

holds. Here $D(\sigma\|\rho_0) = \text{Tr}[\sigma(\log \sigma - \log \rho_0)]$ is finite, nonnegative, and interpreted with $0 \log 0 = 0$.

Proof. By definition, $D(\sigma\|\rho_0) = -s(\sigma) + \text{Tr}(\sigma K_0)$, whereas $s(\rho_0) = \text{Tr}(\rho_0 K_0)$. Subtraction gives $s(\sigma) - s(\rho_0) = \Delta_\sigma\langle K_0 \rangle - D(\sigma\|\rho_0)$. Substituting (19.33) proves the formula because the traces of both states are one.

To see positivity, diagonalize σ with eigenvalues p_i and vectors u_i , and ρ_0 with eigenvalues $q_j > 0$ and vectors v_j . Put $r_i = \langle u_i, \rho_0 u_i \rangle$. Concavity of the logarithm gives $\sum_j |\langle u_i, v_j \rangle|^2 \log q_j \leq \log r_i$. Hence $D(\sigma\|\rho_0) \geq \sum_i p_i \log(p_i/r_i) \geq 0$; the last inequality is convexity of $t \log t$ applied with weights r_i , which sum to one. Strict positivity of the q_j ensures that all reference logarithms are finite. $\square$

The finite identity preserves information distance from the reference state; its tangent term is (19.34). For physical areas, the corresponding coefficients are β_m/ℓ_P^2 . Multiplying the equation by the common transducer k_B expresses dimensional entropy. Thus the relation between geometry and information remains exact also for finite variations, with its explicit correction and without modifying the reference-state parameters.

20 Simultaneous rigidity of the longitudinal reader and radial coupling

The construction of Triadic Modular Holography transmits to the radial realization an enriched state together with its coordinates, orientation, carry, memory and incidences. The sequence

$$\begin{aligned} & \text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow \text{enriched state} \\ & \longrightarrow \text{joint discrete structure of the continuum} \end{aligned}$$

preserves the Archimedean projection and the incidence projection, together with the five constitutive constructions of the continuum. The prolongation

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

is coordinated with the operator order $U_t \longrightarrow \Gamma_9 \longrightarrow K_{\text{ph}}$: the phase returns and the memory record advances. The regional coordinates, the selected record K , the coordinate α and the return action are received with this common provenance and their previously constructed readers.

This section classifies realizations that simultaneously preserve the generated data and the longitudinal, metric and temporal rules specified below. Each positive character determines a radial response; the coefficient relating that response to mass is common to all admissible characters. The gravitational interpretation is formulated after both readers have been constructed on the same space.

20.1 Incidence record and constitutive restrictions

Let V_{in} and V_{out} be complex Hilbert spaces of dimension N , equipped with fixed inner products. The marked cardinal specified in R1 is positive; in particular, both fibres are nonzero. Adjoint operators are taken with respect to these products. The symbol U will denote the normalized archive transport, while U_t retains its meaning as the evolution of the enriched state.

R1. Marked record. The incidence set is

$$\Omega = \left\{ (j, k, q, u, v) : \begin{array}{l} j, k \in \mathbb{Z}/12\mathbb{Z}, \quad k - j \in \{0, 3, 6, 9\}, \\ 1 \leq q \leq 8, \quad u < v, \quad u, v \in \{1, 2, 4, 5, 7, 8\} \end{array} \right\}. \quad (20.1)$$

The twelve phases, four relative positions, eight positions of the index q and fifteen pairs from the final set give

$$N = |\Omega| = 12 \cdot 4 \cdot 8 \binom{6}{2} = 12 \cdot 4 \cdot 120 = 5760. \quad (20.2)$$

Each incidence retains its individual mark. In the corresponding basis $(e_i)_{i \in \Omega}$ of V_{out} , let $P_i = e_i e_i^*$. The common mode u is normalized by $|u_i|^2 = 1/N$, and $P = uu^*$. The marked algebra is $\mathcal{A}_\Omega = \text{span}\{P_i : i \in \Omega\}$.

R2. Normalization and complete archive. The generated inverse map $B : V_{\text{in}} \rightarrow V_{\text{out}}$ satisfies

$$B^* B = \frac{1}{4} I_{\text{in}}, \quad B B^* = \frac{1}{4} I_{\text{out}}. \quad (20.3)$$

Both components are preserved:

$$C_0 = (I - P)B, \quad M_0 = PB, \quad C_0 + M_0 = B, \quad U = 2B. \quad (20.4)$$

The first records variation relative to the common mode; the second preserves that mode. Their images are orthogonal, and their sum recovers the complete transport. By (20.3), U is unitary.

R3. Individual inscription. The map $\mathcal{E} : \text{End}(V_{\text{out}}) \rightarrow \mathcal{A}_\Omega$ is a linear conditional expectation that fixes each P_i and satisfies

$$\mathcal{E}(A_1 T A_2) = A_1 \mathcal{E}(T) A_2, \quad A_1, A_2 \in \mathcal{A}_\Omega. \quad (20.5)$$

The inscription of the result retains the complete construction data in a complementary record. The expectation determines the inscribed observable; preservation of the archive determines its subsequent reconstruction.

R4. Longitudinal composition. Fix the generated coordinate $\alpha > 0$ and the longitudinal reference section $L_* > 0$. Define the longitudinal transport by

$$D = L_* \alpha^{16} \mathcal{E}(C_0 C_0^*) U : V_{\text{in}} \longrightarrow V_{\text{out}}. \quad (20.6)$$

This rule applies the inscribed coefficient linearly to the longitudinal section. Its type fixes the position of the power α^{16} and the return coefficient within the composition. Reading the square root of an intensity is a different operation with its own transformation rule.

R5. Positive radial metric. Let $X = X^* > 0$ be the complete character on V_{in} . The positive radial reader R_X on V_{out} is defined by

$$\|R_X v\|^2 = \|XD^* v\|^2 \quad (v \in V_{\text{out}}). \quad (20.7)$$

This identity fixes the metric on every vector of the fibre. In polar notation, R_X is the arrival modulus $|(DX)^*| = \sqrt{DX^2 D^*}$.

R6. Action and clock. The generated action section $\hbar_{\text{ret}} > 0$ and the real phase lift satisfy

$$S(\theta) = \hbar_{\text{ret}} \theta. \quad (20.8)$$

The constitutive velocity $c > 0$ is preserved. Once the intensive length L has been obtained from (20.6), the conjugate clock satisfies $\omega L = c$. For $\theta(t) = \omega t$, the action rate determines the energy scale

$$E_L = \frac{dS}{dt} = \hbar_{\text{ret}} \omega = \frac{\hbar_{\text{ret}} c}{L}. \quad (20.9)$$

The lift and its refinements remain associated with the path memory.

R7. Readers of the same character. On $Y = UXU^* > 0$, define

$$H_X = E_L Y, \quad M_X = \frac{H_X}{c^2}, \quad \Lambda_X = LY^{-1}. \quad (20.10)$$

Energy, mass and conjugate length preserve the same character and the same transport. The inverse acts on the positive sector; the null sector retains its separate treatment.

R8. Reduced radius and dimensional sections. The physical identification of this realization is

$$R_X = \frac{G}{c^2} M_X. \quad (20.11)$$

Thus R_X represents the reduced gravitational radius. The corresponding Schwarzschild radius is $2R_X$. The reference L_* , the action section and the constitutive velocity section are preserved jointly; a change of units transforms their coordinates according to their dimensions.

Provenance of the restrictions. Hadamard transport, archive preservation, inscription, phase action and reciprocity have antecedents in the corpus. The enlarged record (20.1) and the compositions (20.6)–(20.7) are explicitly incorporated into the assembled construction. R4 and R5 are constitutive rules of this realization; the following proof establishes their joint consequence with the remaining antecedents. R8 specifies the physical identification that fixes the meaning of the coefficient.

20.2 Simultaneous determination theorem

Theorem. For the same generated data $B, (P_i), P, \alpha, L_*, \hbar_{\text{ret}}, c$ and the same character X , restrictions R1–R8 uniquely determine the inscription, longitudinal transport and radial response. The coefficient G is common to all admissible positive characters. One has

$$\mathcal{E} = \Delta_\Omega, \quad \Delta_\Omega(T) = \sum_{i \in \Omega} R_i T R_i, \quad (20.12)$$

$$r_N = \frac{N-1}{4N} = \frac{5759}{23040}, \quad L = L_* \alpha^{16} r_N, \quad (20.13)$$

$$D = LU, \quad R_X = LUXU^*, \quad (20.14)$$

$$R_X \Lambda_X = L^2 I, \quad H_X \Lambda_X = \hbar_{\text{ret}} c I. \quad (20.15)$$

The coupling is determined by

$$G = \frac{c^3 L^2}{\hbar_{\text{ret}}} = \frac{c^3 L_*^2}{\hbar_{\text{ret}}} \left(\frac{5759}{23040} \right)^2 \alpha^{32}. \quad (20.16)$$

Proof: uniqueness of inscription. Let $E_{ij} = e_i e_j^*$ be the matrix units. Bimodularity gives

$$\mathcal{E}(E_{ij}) = R_i \mathcal{E}(E_{ij}) R_j.$$

For $i \neq j$, membership of $\mathcal{E}(E_{ij})$ in the diagonal algebra implies $R_i \mathcal{E}(E_{ij}) R_j = 0$. For $i = j$, fixation of R_i gives $\mathcal{E}(E_{ii}) = R_i$. Linearity therefore determines (20.12) on all matrices. The marked inscription is thereby fixed by its algebraic properties.

Proof: return coefficient and archive. The normalization of B , together with $P^2 = P = P^*$, yields

$$C_0 C_0^* = \frac{1}{4}(I - P), \quad M_0 M_0^* = \frac{1}{4}P.$$

Since $R_i P R_i = |u_i|^2 R_i = R_i/N$, it follows that

$$\Delta_\Omega(C_0 C_0^*) = \frac{N-1}{4N} I, \quad \Delta_\Omega(M_0 M_0^*) = \frac{1}{4N} I. \quad (20.17)$$

The two readers sum to $I/4$. The first is scalar, so every normalized distribution (p_i) on the incidences, with $p_i \geq 0$ and $\sum_i p_i = 1$, gives $\sum_i p_i r_N = r_N$. Hence an arbitrary diagonal weighting preserves the same coefficient.

Proof: length and metric. Substitution of (20.17) into R4 gives $D = LU$, with L as in (20.13). Unitarity of U implies $DD^* = L^2 I$. By R5, the quadratic forms of R_X^2 and $DX^2 D^*$ coincide. The polarization identity determines

$$R_X^2 = DX^2 D^* = L^2 UX^2 U^*.$$

The operator $LUXU^*$ is positive and its square is the operator on the right-hand side. Uniqueness of the positive square root proves (20.14). In particular, $R_I = LI$.

Proof: action, reciprocity and coupling. R6 and R7 give

$$H_X = \frac{\hbar_{\text{ret}} c}{L} Y, \quad M_X = \frac{\hbar_{\text{ret}}}{cL} Y, \quad \Lambda_X = LY^{-1}.$$

Multiplication gives both products in (20.15). R8 then becomes

$$LY = \frac{G \hbar_{\text{ret}}}{c^3 L} Y.$$

Invertibility of Y determines (20.16). The same substitution verifies the identity for every positive character. If G_1 and G_2 satisfy the identity with the same data, then $(G_1 - G_2)Y = 0$, hence $G_1 = G_2$. This conclusion distinguishes the state-dependent response R_X from the common coefficient G . $\square$

20.3 Reciprocity and phase rigidity

The power-logarithmic antecedent uses, for $x > 0$,

$$\ell_p(x) = \frac{x^p - 1}{p} \quad (p \neq 0), \quad \ell_0(x) = \log x.$$

Direct substitution proves $\ell_{-p}(x^{-1}) = -\ell_p(x)$. On the positive domain of their inverses, this identity is equivalent to

$$\exp_p(s) \exp_{-p}(-s) = 1.$$

Spectral composition on a positive character transmits this product to the pair LY, LY^{-1} . In the present realization, once R1–R4 and R7 have been fixed, R5 can equivalently be written as $R_X \Lambda_X = L^2 I$: multiplication by Λ_X^{-1} determines $R_X = LUXU^*$, and this expression satisfies R5.

The two products in (20.15) also give a direct form of rigidity:

$$R_X = \frac{L^2}{\hbar_{\text{ret}} c} H_X. \quad (20.18)$$

For a positive rescaling s , the pair $(sR_X, \Lambda_X/s)$ preserves the radial product; the product $H_X(\Lambda_X/s) = \hbar_{\text{ret}} c I/s$, with fixed energy and action, requires $s = 1$. Joint preservation determines the scale. Likewise, every positive radial operator with the same metric R5 coincides with R_X by uniqueness of the positive square root.

The lifted action is also determined exactly. For $\theta \neq 0$, the identity $S/\hbar_{\text{ret}} = \theta$ fixes S . Even when one preserves the rotations of all subdivisions $\theta/9^n$, a fixed real rescaling a satisfying

$$\exp(-ia\theta/9^n) = \exp(-i\theta/9^n) \quad (n \geq 0)$$

must obey $(a - 1)\theta/9^n \in 2\pi\mathbb{Z}$. This sequence tends to zero; for sufficiently large n , its only possible value is zero. Since $\theta \neq 0$, it follows that $a = 1$.

20.4 Planck scales and the unit-character specialization

We write $L_\alpha = L$; in the return section, $E_{P,\text{ret}} = E_L$.

The constructed length $L_\alpha > 0$, the action section $\hbar_a > 0$, and the speed $c > 0$ determine the scales

$$t_P = \frac{L_\alpha}{c}, \quad m_{P,a} = \frac{\hbar_a}{cL_\alpha}, \quad E_{P,a} = \frac{\hbar_a c}{L_\alpha}.$$

Indeed, substituting $G_a = c^3 L_\alpha^2 / \hbar_a$ into $\sqrt{\hbar_a G_a / c^5}$, $\sqrt{\hbar_a c / G_a}$, and $\sqrt{\hbar_a c^5 / G_a}$ yields these three identities, respectively. Positivity fixes each square root. The length precedes the recognition of these scales; their conventional expressions are identities of the constructed realization.

For a mode with character $y > 0$, the mass and longitudinal readings are

$$m(y) = m_{P,a} y, \quad r_g(y) = L_\alpha y, \quad \bar{\lambda}(y) = \frac{L_\alpha}{y}.$$

Thus $r_g(y) \bar{\lambda}(y) = L_\alpha^2$ for every positive mode. Equality of the two lengths characterizes the mode $y = 1$, since $y = y^{-1}$ and $y > 0$ imply $y = 1$. In this mode, the mass is $m_{P,a}$ and both lengths equal L_α . This distinguishes general reciprocity from its unit-character specialization.

The chronological step and the Planck time satisfy $t_0 = (\pi_{\text{HMT}}/54)t_P$. The complete 108-step phase period is $108t_0 = 2\pi_{\text{HMT}}t_P$, with angular frequency $\omega_{108} = 1/t_P = c/L_\alpha$. A horizon of radius $R_h = \zeta_h L_\alpha$, where $\zeta_h > 0$, retains its own geometric variable. The Schwarzschild radius associated with a mass is $2r_g$.

When the two action sections are compared at fixed L_α and c , the ratio $\rho = \hbar_b / \hbar_a$ gives $G_b = G_a / \rho$, $m_{P,b} = \rho m_{P,a}$, and $E_{P,b} = \rho E_{P,a}$; L_α and t_P remain unchanged. This comparison between sections remains distinct from temporal evolution of the constants.

20.5 Archive covariance and change of units

Let $T : V_{\text{in}} \rightarrow V'_{\text{in}}$ and $S : V_{\text{out}} \rightarrow V'_{\text{out}}$ be unitary. Joint transport of the data is given by

$$B' = SBT^*, \quad P' = SPS^*, \quad P'_i = SP_iS^*, \quad X' = TXT^*.$$

The transported expectation $\mathcal{E}'(A') = S\mathcal{E}(S^*A'S)S^*$ fixes P'_i and is bimodular over the transformed marked algebra. Substitution gives

$$\begin{aligned} U' &= SUT^*, & D' &= SDT^*, \\ R'_{X'} &= SR_XS^*, & H'_{X'} &= SH_XS^*, & \Lambda'_{X'} &= S\Lambda_XS^*. \end{aligned}$$

Conjugation preserves the identities R1–R8 and the same coefficient G . Marks, orientations and records are transported together with the bases. Permutations of incidences are a particular case.

Adjoining an auxiliary memory space $\mathcal{F}$ transports P to $P \otimes I_{\mathcal{F}}$, U to $U \otimes I_{\mathcal{F}}$, and Δ_{Ω} to $\Delta_{\Omega} \otimes \text{id}$. The inscription of $C_0C_0^* \otimes I_{\mathcal{F}}$ is $r_N I \otimes I_{\mathcal{F}}$; the coefficient retains the original marked cardinality.

To specify dimensional covariance, let ℓ , τ and m be the positive factors of the new length, time and mass units relative to the previous ones. The numerical coordinates satisfy

$$L' = L/\ell, \quad c' = c\tau/\ell, \quad \hbar'_{\text{ret}} = \hbar_{\text{ret}}\tau/(m\ell^2).$$

Consequently,

$$\frac{(c')^3(L')^2}{\hbar'_{\text{ret}}} = G \frac{m\tau^2}{\ell^3},$$

which is the transformation law for a quantity of dimension $\text{length}^3 \text{mass}^{-1} \text{time}^{-2}$. The section L_* transforms as L , while α and r_N remain dimensionless.

20.6 Exact elimination of coupled memory

Let $\mathcal{K}$ and $\mathcal{H}$ be finite-dimensional Hilbert spaces, let $W = W^* > 0$ be a global weight, and let $\mathcal{C} : \mathcal{K} \rightarrow \mathcal{H}$ be surjective. The action of residuals r , with boundary defect z , is

$$\mathcal{S}(r) = \alpha r^* W r, \quad \mathcal{C}r = z, \quad \alpha > 0.$$

The global weight retains its cross-blocks between incidences. Define $\mathcal{R} = \mathcal{C}W^{-1}\mathcal{C}^*$. For $v \neq 0$, injectivity of $\mathcal{C}^*$ gives

$$v^* \mathcal{R} v = \|W^{-1/2}\mathcal{C}^*v\|^2 > 0.$$

Thus $\mathcal{R}$ is invertible. The minimum-action residual is

$$r_*(z) = W^{-1}\mathcal{C}^*\mathcal{R}^{-1}z.$$

It satisfies $\mathcal{C}r_* = z$. Every admissible residual can be written as $r = r_* + k$, where $\mathcal{C}k = 0$, and $k^* W r_* = k^* \mathcal{C}^* \mathcal{R}^{-1} z = 0$. Therefore,

$$\mathcal{S}(r) = \alpha z^* \mathcal{R}^{-1} z + \alpha k^* W k. \quad (20.19)$$

Positivity of W proves uniqueness of the minimum. The pair (z, k) preserves and reconstructs the complete residual.

For a path with transports U_j , the residuals $r_j = f_j - U_j f_{j-1}$ reconstruct the fields through $f_j = U_j f_{j-1} + r_j$. The map $\mathcal{C}$ collects the subsequent transports of each residual to the endpoint. If its blocks are $\mathcal{C}_j$, then

$$\mathcal{R} = \sum_{j,k} \mathcal{C}_j (W^{-1})_{jk} \mathcal{C}_k^*.$$

This expression preserves correlations between segments of the path. For an invertible longitudinal transport D , the boundary $y = Dz$ has response

$$\mathcal{S}_{\text{ef},L}(y) = \alpha y^* (D \mathcal{R} D^*)^{-1} y = \alpha z^* \mathcal{R}^{-1} z.$$

The same action is obtained by eliminating memory first or transporting the boundary first. The prefactor retains the normalization of the source action; when that action has $\alpha = \hbar_{\text{ret}}$, reduction transmits the same section.

20.7 Joint reduction of radius, energy and conjugate length

Write the same positive character in a boundary–memory decomposition:

$$Y = \begin{pmatrix} A & F \\ F^* & C \end{pmatrix}, \quad C > 0, \quad Y_{\text{ef}} = A - FC^{-1}F^*.$$

The identity

$$\begin{pmatrix} b \\ y \end{pmatrix}^* Y \begin{pmatrix} b \\ y \end{pmatrix} = b^* Y_{\text{ef}} b + \eta^* C \eta, \quad \eta = y + C^{-1}F^*b,$$

proves $Y_{\text{ef}} > 0$, characterizes the minimum-energy extension and preserves the reconstruction $y = \eta - C^{-1}F^*b$. For $s > 0$, substitution into the Schur complement gives $\text{Schur}(sY) = sY_{\text{ef}}$. Consequently,

$$R_{\text{ef}} = LY_{\text{ef}}, \quad H_{\text{ef}} = E_L Y_{\text{ef}}.$$

To compute the compressed conjugate length, solve $Y(x, y) = (f, 0)$. The second equation gives $y = -C^{-1}F^*x$, and the first gives $x = Y_{\text{ef}}^{-1}f$. The boundary block of Y^{-1} is therefore Y_{ef}^{-1} . Thus,

$$\Lambda_b = LY_{\text{ef}}^{-1}, \quad R_{\text{ef}}\Lambda_b = L^2I, \quad R_{\text{ef}} = \frac{L}{E_L}H_{\text{ef}}. \quad (20.20)$$

Reduction acts on the same blocks in both readers and retains the residual η . The effective character may depend on memory through a matrix-valued expression, while L/E_L remains fixed.

20.8 Dynamic memory and induced temporal norm

Dynamic elimination retains an additional spectral dependence. In a stationary representation, let

$$H = \begin{pmatrix} H_{\text{bb}} & V \\ V^* & H_{\text{ii}} \end{pmatrix}, \quad H_{\text{ii}} > 0.$$

For z in the relevant resolvent domain, eliminating the interior component in $(H - zI)(b, y) = (f, 0)$ yields

$$\mathcal{K}_H(z) = H_{\text{bb}} - zI - V(H_{\text{ii}} - zI)^{-1}V^*. \quad (20.21)$$

When both inverses exist, the boundary block of $(H - zI)^{-1}$ is $\mathcal{K}_H(z)^{-1}$. For $|z| < \lambda_{\min}(H_{\text{ii}})$, the convergent resolvent series gives

$$\mathcal{K}_H(z) = H_{\text{est}} - zZ - z^2VH_{\text{ii}}^{-3}V^* - \dots, \quad (20.22)$$

$$H_{\text{est}} = H_{\text{bb}} - VH_{\text{ii}}^{-1}V^*, \quad Z = I + VH_{\text{ii}}^{-2}V^* \geq I. \quad (20.23)$$

Positivity follows from $Z - I = (H_{\text{ii}}^{-1}V^*)^*(H_{\text{ii}}^{-1}V^*)$. The static extension $(b, -H_{\text{ii}}^{-1}V^*b)$ has squared norm b^*Zb ; energy and temporal norm arise from the same extension. The physical parameter is $z = \hbar_{\text{ret}}\omega$.

At first temporal order, $T = Z^{1/2}$ and $\psi = Tb$ transform the pair (H_{est}, Z) into $(T^{-*}H_{\text{est}}T^{-1}, I)$. The action section $\hbar_{\text{ret}}$ remains fixed. If T depends on time, the transformation also retains the term $\dot{T}b$ in $\dot{\psi} = \dot{T}b + T\dot{b}$.

Radial proportionality is preserved in the complete resolvent. Writing $\chi_{\text{rad}} = L/E_L$ and $R = \chi_{\text{rad}}H$, substitution in each block proves

$$\mathcal{K}_R(\chi_{\text{rad}}z) = \chi_{\text{rad}}\mathcal{K}_H(z). \quad (20.24)$$

This identity transports the operator and the spectral parameter together, preserving every order of dynamic memory. For the effective forms in (20.20), canonical normalization requires the dual transport of inverse length:

$$H_{\text{can}} = T^{-*}H_{\text{ef}}T^{-1}, \quad R_{\text{can}} = T^{-*}R_{\text{ef}}T^{-1}, \quad \Lambda_{\text{can}} = T\Lambda_bT^*.$$

Multiplication gives

$$R_{\text{can}} = \frac{L}{E_L}H_{\text{can}}, \quad R_{\text{can}}\Lambda_{\text{can}} = L^2I.$$

The proof allows noncommuting T , R_{ef} and Λ_b . The induced temporal norm and the dual length jointly preserve the same G .

20.9 Refinement, spectral domains and scope

Let J and K be isometric inscriptions between two levels satisfying $X'J = JX$ and $U'J = KU$, with the same intensive section L . Then

$$R'K = LU'X'U'^*K = KLUXU^* = KR,$$

and the same calculation proves $H'K = KH$. The intertwining of inverses is formulated on the corresponding domains. When the compatible system has a positive injective self-adjoint limit X , spectral calculus defines $Y = UXU^*$, $R = LY$, $H = E_L Y$ and $\Lambda = LY^{-1}$. The proportionality $R = (L/E_L)H$ holds on $\text{Dom}(Y)$, while $R\Lambda = L^2I$ and $H\Lambda = \hbar_{\text{ret}}cI$ hold on $\text{Dom}(Y^{-1})$, with bounded extensions of the products. The spectral cutoffs $\mathbf{1}_{[\varepsilon, M]}(Y)$ verify the identities before passing to the limit on these domains. A refined edge length $L/9$ carries its own scale factor; the intensive section is preserved according to the declared intertwining.

The classification distinguishes transformations preserving the realization from those modifying a constitutive rule. Diagonal weightings, unitary equivalences, the auxiliary archive and joint reduction preserve G . Changing the marked resolution modifies R1; replacing r_N by $\sqrt{r_N}$, α^{16} by its square, or the inscribed component by the total norm modifies R4. An isolated radial rescaling changes R5, and an isolated action rescaling changes R6. R8 maintains the distinction between reduced radius, Schwarzschild radius and a change of units.

This establishes the mathematical uniqueness of realization R1–R8: the generated data and the character fix the response, and their common composition fixes G . The longitudinal and metric rules appear explicitly in this conclusion as incorporated compositions; the gravitational identification retains its physical status. The proof covers the declared class with its constitutive definitions and compatible transformations. The rational checks of the development verify finite instances of the displayed identities; the general proof is contained in the preceding equations and arguments.

21 Action, gravity, and thermal information

21.1 Dodecaphasic period and circular realization of action

The length in Section 20 fixes, within the same realization, the compatible clock

$$t_0 = \frac{\pi_{\text{HMT}} L}{54 c_{\text{int}}}.$$

Its circular representation then satisfies $R_{108} = L$. This substitution expresses the constructed length chronologically and retains the longitudinal reference, action and velocity of that realization. The calendar already constructed determines $T_{108} = 108 t_0$. In its circular realization,

$$\omega_{108} = \frac{2\pi}{108 t_0}, \quad R_{108} = \frac{c_{\text{int}}}{\omega_{108}} = \frac{54}{\pi} \ell_0, \quad \ell_0 = c_{\text{int}} t_0.$$

The unit ℓ_0 and the time and velocity charts remain explicit; the SI chart does not select the cycle. For each oriented action section $a \in \{\text{pre}, \text{ret}\}$,

$$E_{108,a} = \hbar_a \omega_{108}, \quad m_{108,a} = \frac{E_{108,a}}{c_{\text{int}}^2}.$$

The reduced and full lengths are, respectively,

$$\frac{\hbar_a}{m_{108,a} c_{\text{int}}} = R_{108}, \quad \frac{h_a}{m_{108,a} c_{\text{int}}} = 2\pi R_{108}.$$

The same action and duration enter both readings [Section 21].

Proposition 21.1 (Radius-coincidence selector). *In the dimensional collection*

$$G_a(\vartheta) = \vartheta \frac{c_{\text{int}}^5 t_0^2}{\hbar_a}, \quad \vartheta > 0,$$

the circular-realization condition $G_a(\vartheta) m_{108,a} / c_{\text{int}}^2 = R_{108}$ has the unique solution

$$\vartheta_{108}^\circ = (54/\pi)^2.$$

Proof. The first radius is $r_g(\vartheta) = \vartheta(\pi/54)\ell_0$. Comparing it with $(54/\pi)\ell_0$, with $\ell_0 > 0$, reduces the equality to a strictly increasing linear equation. $\square$

The coincidence condition is an explicit part of this realization; the theorem solves for its coefficient without using a metrological value of G . The radius used here is Gm/c^2 , so the factor two from the convention $2Gm/c^2$ is not inadvertently transferred.

Proposition 21.2 (Reciprocal action and gravity, preserved area). *In the two preceding charts,*

$$\frac{G_{\text{pre}}}{G_{\text{ret}}} = R_{\text{act}}^{-1}, \quad \hbar_a G_a = \vartheta_{108}^\circ c_{\text{int}}^5 t_0^2, \quad \ell_P^2 = \frac{\hbar_a G_a}{c_{\text{int}}^3} = \vartheta_{108}^\circ \ell_0^2.$$

Proof. The definition of G_a contains $\hbar_a^{-1}$, and the remaining factors are common to both charts. Taking the quotient and product gives the three equalities. $\square$

This reciprocity explains why orientation information may vary while area remains unchanged. A precise relation is thus available between the action ellipse, Planck scale, and area operator of section 19.

21.2 Thermal normalization and energy per cycle

Fix the same oriented section a for energy and temperature. The corpus's thermal relation gives

$$k_B \Theta_{\text{clk},a} \log 3 = \frac{h_a}{108 t_0} = \frac{\hbar_a}{t_P} = E_{P,a}, \quad t_P = \frac{54}{\pi} t_0. \quad (21.1)$$

The factor $\log 3$ measures the information in the equiprobable triple; clock energy incorporates the action already generated. The identity preserves the product $k_B \Theta_{\text{clk}}$ as an energy section and makes the thermal chart used explicit [equation (21.1) and Section 22].

At this scale, the radiative realization provides

$$n_\gamma \ell_P^3 = \frac{2\zeta(3)}{\pi^2 \log^3 3}, \quad \frac{u_\gamma \ell_P^3}{E_P} = \frac{\pi^2}{15 \log^4 3}, \quad \frac{s_\gamma \ell_P^3}{k_B} = \frac{4\pi^2}{45 \log^3 3}.$$

Two distinct invariants occur here: $\log 3$ preserves TRIT information and $\zeta(3)$ the bosonic occupation moment. Their occurrence in the same expression relates the thermal constants to the spectral moments of bosonic occupation, without identifying these quantities with one another.

Section 22.2 develops the Boltzmann transducer, its action on tritic information, and the change of thermal coordinates. Equation (21.1) fixes the energy–temperature product used by that construction. The selection of an individual coordinate is distinguished there from the identity for the product. Within this same development, the preceding radiative formulas preserve the occupation and entropy content of the bosonic realization.

21.3 Barbero–Immirzi in the Cartan–Holst realization

Once γ_{HMT} has been constructed, the source realizes it as a coefficient of the Holst action. In the declared geometric chart, let e^I be a nondegenerate co-tetrad, ω^{IJ} a connection, $\Sigma^{IJ} = e^I \wedge e^J$, and $F^{IJ} = d\omega^{IJ} + \omega^I_K \wedge \omega^{KJ}$. With $\kappa = 8\pi G/c^4$, the specialized action is

$$S[e, \omega, \Psi] = \frac{1}{2\kappa c} \int \Sigma_{IJ} \wedge (\star F^{IJ} + \gamma_{\text{HMT}}^{-1} F^{IJ}) - \frac{\Lambda}{\kappa c} \int \text{vol}_e + S_m[e, \omega, \Psi]. \quad (21.2)$$

The current convention is $\delta_\omega S_m = -\frac{1}{2} \int \delta\omega^{IJ} \wedge \sigma_{IJ}$. For this variation, the connection is fixed at the boundary, $\delta\omega^{IJ}|_{\partial M} = 0$, or variations compactly supported in the interior are used. These sufficient conditions annihilate the boundary term resulting from integrating $\Sigma_{IJ} \wedge (\star + \gamma_{\text{HMT}}^{-1} I) D_\omega \delta\omega^{IJ}$. Variation of (21.2) under that condition gives

$$D_\omega[(\star + \gamma_{\text{HMT}}^{-1} I) \Sigma^{IJ}] = \kappa c \sigma^{IJ}.$$

This follows by using $\delta F = D_\omega \delta\omega$, integrating by parts, and setting the coefficient of the arbitrary variation to zero [action (21.2) and adjacent variation]. Boundary orientation and current normalization are part of this formula.

In Lorentzian signature, $\star^2 = -I$ on bivectors. The identity

$$(\star + \gamma^{-1} I)^{-1} = \frac{\gamma^2}{1 + \gamma^2} (-\star + \gamma^{-1} I)$$

is checked by multiplication: the cross terms cancel, leaving $(1 + \gamma^{-2})I$ before normalization. Thus the parameter already constructed enters the torsional response, not merely a table value. The identities $D_\omega T^I = F^I_J \wedge e^J$ and $D_\omega F^{IJ} = 0$ keep torsion, curvature, and balance connected.

The geometric realization preserves route memory and boundary soldering. The cosmological constant Λ in (21.2) is the coefficient declared in the action; it is not identified with the vector representation $\Lambda(R)$. The following identities retain the holonomy, smoothness, and nondegeneracy conditions of the realization.

21.4 Phase return and representation of accumulated memory

The chronology of enriched routes distinguishes positional return, orientational return, and phase return. The observable projection identifies certain arrival states, while the cumulative record preserves the sequence traversed. The geometric representation must transport both data jointly.

21.4.1 Inversion cocycle and change of sheet

Let F_{obs} be the chronological representation of TPK transport. Its canonical blocks have the hierarchy

27	positional return with reversal of orientation,
54	positional and orientational return,
108	complete return of the observable phase.

The record counts reversals of orientation $g(a)$ and effective changes of sheet $s(a)$. For a route,

$$c(\gamma) = \sum_{a \in \gamma} (g(a), s(a)), \quad c(\gamma_2 \gamma_1) = c(\gamma_2) + c(\gamma_1). \quad (21.3)$$

The canonical computation of the block of length 27 yields $c_{27} = (1, 18)$. Composition preserves those increments in the four blocks completing the observable period. On forward-oriented routes the record is cumulative; its extension to the route groupoid uses $\mathbb{Z}^2$ and $c(\bar{\gamma}) = -c(\gamma)$.

Proposition 21.3 (Memory of the complete observable circuit). *Writing $\mathcal{K}_{27} = F_{\text{obs}}^{27}$, the lift to the record, for $x \in \mathcal{Q}_{\text{obs}}$ and $z \in \mathbb{Z}^2$, satisfies*

$$\tilde{\mathcal{K}}_{27}(x, z) = (\mathcal{K}_{27}x, z + (1, 18)), \quad \tilde{\mathcal{K}}_{27}^4(x, z) = (x, z + (4, 72)).$$

After N complete circuits, the increment is $N(4, 72)$.

Proof. The observable projection of $\mathcal{K}_{27}$ has order four. The preceding canonical computation fixes $c_{27} = (1, 18)$ constantly on that observable orbit. Additivity of (21.3) sums the four increments and gives $c_{108} = 4(1, 18) = (4, 72)$. Repetition applies the same concatenation law and produces Nc_{108} . Return of the first coordinate preserves the increment of the second. $\square$

21.4.2 Affine representation of the cocycle

The discrete realization uses an element R_4 of order four in $\text{Spin}^+(1, 3)$, a unit timelike vector u fixed by its vector action, and a length unit $\ell_0 > 0$. The scale ℓ_0 preserves its provenance in the dimensional section. We define

$$\rho_C^{\text{disc}}(\gamma) = (R_4^{c_g(\gamma)}, \ell_0 c_s(\gamma)u) \in \text{Spin}^+(1, 3) \ltimes \mathbb{R}^{1,3}. \quad (21.4)$$

The power acts on the vector through the associated vector representation of the spin group.

Theorem 21.4 (Composition and translational amplitude of return). *The map ρ_C^{disc} preserves the identity, concatenation, and inversion. At complete returns,*

$$\rho_C^{\text{disc}}(\gamma_{108}) = (I, 72\ell_0 u), \quad \rho_C^{\text{disc}}(\gamma_{108}^N) = (I, 72N\ell_0 u). \quad (21.5)$$

Inversion reverses the sign of the translation.

Proof. The semidirect product is $(R, a)(S, b) = (RS, a + Rb)$. Since $R_4 u = u$,

$$\begin{aligned} \rho_C^{\text{disc}}(\gamma_2) \rho_C^{\text{disc}}(\gamma_1) &= (R_4^{c_g(\gamma_2) + c_g(\gamma_1)}, \ell_0 (c_s(\gamma_2) + c_s(\gamma_1))u) \\ &= \rho_C^{\text{disc}}(\gamma_2 \gamma_1). \end{aligned}$$

The cocycle of the identity is zero and that of the inverse is its negative. For the complete circuit, substitute $c_{108} = (4, 72)$ and $R_4^4 = I$. Concatenation of N circuits keeps the rotational part equal to the identity and sums their translations. $\square$

The amplitude $72\ell_0$ is a length associated with a finite route. A realization through a cotetrad and connection transports it to its affine holonomy. The local contribution to the gravitational equations is then obtained through the curvature of that connection, variation of the material action, and normalization of its density. The representation thus preserves the exact cumulative content that must enter that realization.

21.5 Cubic incidence and boundary soldering

The cubic realization of transport preserves six oriented incidences. Its geometric quotient, Lorentzian form, and energetic response are obtained from the same incidence matrix. Transport originates in the enriched APP-TRIT-TPK route: the face, sign, carry, and memory accompany each edge. This section develops the linear operation acting on that boundary representation.

21.5.1 Cubic incidence and the rank-four quotient

Let $H_F = \mathbb{R}^F$, with $F = \{(i, \epsilon) : 1 \leq i \leq 3, \epsilon \in \{-1, 1\}\}$, its Euclidean inner product, and its basis $e_{i,\epsilon}$. Face opposition is the orthogonal involution $Re_{i,\epsilon} = e_{i,-\epsilon}$. Define

$$u = \sum_{i,\epsilon} e_{i,\epsilon}, \quad P_u = \frac{uu^\top}{6}, \quad P_- = \frac{I - R}{2}, \quad P_0 = \frac{I + R}{2} - P_u, \quad P_\square = P_u + P_-.$$

The vertices are $V = \{-1, 1\}^3$. The incidence matrix $M : \mathbb{R}^V \rightarrow H_F$ is determined by

$$M_{(i,\epsilon),v} = \mathbf{1}_{\{v_i=\epsilon\}}.$$

Proposition 21.5 (Incidence decomposition). *One has*

$$MM^\top = 12P_u + 4P_-, \quad \ker M^\top = P_0H_F.$$

Consequently, the quotient $Q_\square = Q = H_F/P_0H_F$, orthogonally identified with $P_\square H_F$, has dimension four.

Proof. A face contains four vertices. Opposite faces have no common vertices, and two faces on different axes have two common vertices. Writing $\mathbf{J} = uu^\top$ gives

$$MM^\top = 4I + 2(\mathbf{J} - I - R) = 2(I - R) + 2\mathbf{J} = 4P_- + 12P_u.$$

The three projectors P_u, P_-, P_0 are orthogonal and sum to I ; their ranks are one, three, and two, respectively. The identity $\|M^\top x\|^2 = \langle x, MM^\top x \rangle$ identifies the stated kernel. The complementary rank is four. $\square$

On Q , the form

$$B = P_- - P_u$$

has signature $(3, 1)$. Time orientation is fixed by u . With

$$t = \frac{u}{\sqrt{6}}, \quad d_i = \frac{e_{i,+} - e_{i,-}}{\sqrt{2}}, \quad W = \sqrt{6}P_u + \sqrt{2}P_-,$$

one has $We_{i,\epsilon} = t + \epsilon d_i$. These six vectors are null for B , whereas

$$3t + \nu_1 d_1 + \nu_2 d_2 + \nu_3 d_3$$

has squared norm -6 . The same incidence satisfies $MM^\top = 2W^*W$ on Q . The proper rotations of the cube commute with the projectors, preserve both forms, and fix t . These identities determine the metric of the cubic representation.

21.5.2 Cellular soldering and memory transport

For each oriented edge a , the reading of the route provides the face $\lambda(\underline{a})$, the sign $\sigma(a)$, the source and target fibers, and the transport $U(a) : Q_{s(a)} \longrightarrow Q_{t(a)}$. Reversal satisfies $U(\bar{a}) = U(a)^{-1}$. Boundary half-edges preserve their formal reversal and their incidence on the boundary. With $\ell_m = \ell_0 9^{-m} > 0$, soldering is

$$\beta_m(a) = \ell_m \sigma_m(a) n_{\lambda(\underline{a})}^{s(a)}, \quad n_{i,\epsilon} = t + \epsilon d_i. \quad (21.6)$$

The face and sign are transported together with the route memory. The reversal convention requires

$$\beta_m(\bar{a}) = -U_m(a) \beta_m(a). \quad (21.7)$$

Applying this relation twice recovers $\beta_m(a)$; the carry record separately preserves the ordered composition.

Proposition 21.6 (Naturality of soldering under refinement). *Let $a = a_1 \cdots a_9$ be a compatible refinement of an edge. Suppose that the nine successor incidences, transported to the predecessor fiber, preserve the face and sign, and that $\ell_{m+1} = \ell_m/9$. Let $\mathcal{U}_j : Q_{m,s(a)} \longrightarrow Q_{m+1,s(a_j)}$ be the accumulated transport, including the refinement identification and the path to the source of successor j . Then*

$$\beta_m(a) = \sum_{j=1}^9 \mathcal{U}_j^{-1} \beta_{m+1}(a_j).$$

The equality is compatible with reversal and with concatenation of the memory record.

Proof. Each summand, expressed in the source fiber of the predecessor, is $(\ell_m/9) \sigma_m(a) n_{\lambda(\underline{a})}$. The sum of nine terms reproduces the predecessor soldering. Reversal reverses the order of the transports and replaces each by its inverse; relation (21.7) supplies the overall sign. Memory is concatenated in the same order, so refinement preserves both the vector representation and its enriched record. $\square$

21.6 Geometric realization of transport and soldering

The discrete representation of return and screen soldering determine two aspects of transport: accumulation of memory along routes and comparison of their fibres. Their geometric realization preserves both operations. This section fixes the target space, the dimensional normalization of the connection, and the identities that its variation will use.

21.6.1 Screen, cotetrad, and connection

The screen quotient $Q_{\square}$, its Lorentzian form, and the soldering β_m come from 21.5. In each cell, a screen realization $J_{\text{scr},m}$ transports the form and soldering to the corresponding geometric fibre. Its cellular image is

$$e_m = J_{\text{scr},m} \beta_m.$$

The scale ℓ_m remains within the soldering. Frame changes act jointly on $J_{\text{scr},m}$, the connection, and cell transport.

A smooth realization of these data on an oriented manifold $\mathcal{U}$ of dimension four consists of a nondegenerate cotetrad $e = (e^I)$, a Lorentz connection $\omega = (\omega^I_J)$, and a route realization $\mathfrak{R}_C$ satisfying

$$\text{Hol}_{\mathcal{A}_{\ell_0}}(\mathfrak{R}_C(\gamma)) = \rho_C(\text{Hol}_{\text{TPK}}(\gamma)). \quad (21.8)$$

In this matrix equality, the representation is normalized by

$$N_{\ell_0}(R, a) = (\Lambda(R), a/\ell_0), \quad \rho_C = N_{\ell_0} \circ \rho_C^{\text{disc}},$$

where $\Lambda : \text{Spin}^+(3, 1) \rightarrow \text{SO}^+(3, 1)$ is the vector representation. The map N_{ℓ_0} preserves the semidirect product: dividing the translation $a + \Lambda(R)b$ by ℓ_0 is equivalent to composing the two normalized

translations. The complete circuit therefore yields $72u$ in the normalized connection and $72\ell_0 u$ in the dimensional chart. The spin lift preserves, separately, $R_4^{c_g(\gamma)}$; the vector representation has central kernel $\{1, -1\}$, so that lift is retained when spinorial fields are used. Identity, concatenation, inversion, and subdivision are maintained in the corresponding realization. The screen pairing and holonomy condition specify the data of the realization used in the field equations.

With $\eta = \text{diag}(-1, 1, 1, 1)$, define $g = \eta_{IJ} e^I \otimes e^J$. For a positive elementary scale ℓ_0 , constant in the chart, the affine connection is

$$\mathcal{A}_{\ell_0} = \begin{pmatrix} \omega & \ell_0^{-1} e \\ 0 & 0 \end{pmatrix}. \quad (21.9)$$

The cotetrad has dimensions of length; the factor ℓ_0^{-1} makes its translational component dimensionally homogeneous.

Proposition 21.7 (Curvature and torsion of the realized connection). *The curvature of (21.9) is*

$$\mathcal{F}_{\ell_0} = d\mathcal{A}_{\ell_0} + \mathcal{A}_{\ell_0} \wedge \mathcal{A}_{\ell_0} = \begin{pmatrix} F_\omega & \ell_0^{-1} T \\ 0 & 0 \end{pmatrix}, \quad F_\omega = d\omega + \omega \wedge \omega, \quad T = de + \omega \wedge e. \quad (21.10)$$

Proof. The product of matrices of forms produces $\omega \wedge \omega$ in the diagonal block and $\ell_0^{-1} \omega \wedge e$ in the translational block. The exterior derivative contributes $d\omega$ and $\ell_0^{-1} de$, since ℓ_0 is constant. Their sum proves the equality. $\square$

Curvature describes the transport defect of frames and torsion that of the cotetrad. Both belong to the same realized connection; their components preserve distinct tensor domains.

Proposition 21.8 (Covariance and Bianchi identities). *Under a frame change $g_L : \mathcal{U} \rightarrow SO^+(3, 1)$,*

$$e' = g_L e, \quad \omega' = g_L \omega g_L^{-1} - dg_L g_L^{-1},$$

one has $T' = g_L T$ and $F_{\omega'} = g_L F_\omega g_L^{-1}$. Moreover,

$$D_\omega T = F_\omega \wedge e, \quad D_\omega F_\omega = 0. \quad (21.11)$$

Proof. In $d(g_L e) + \omega' \wedge g_L e$, the terms containing $dg_L \wedge e$ cancel. The same substitution into the curvature gives its transformation by conjugation. For the first Bianchi identity, expand

$$D_\omega (de + \omega \wedge e) = (d\omega + \omega \wedge \omega) \wedge e.$$

In $dF_\omega + \omega \wedge F_\omega - F_\omega \wedge \omega$, the terms containing $d\omega$ and the cubic products cancel in pairs. This proves the second identity. $\square$

21.6.2 The three quadratic realizations of curvature

The TRIT sheet $\tau \in \{+1, 0, -1\}$ is represented in the Cartan algebra by generators $J_{ab} = -J_{ba}$ and $P_a^{(\tau)}$, with relations

$$\begin{aligned} [J_{ab}, J_{cd}] &= \eta_{bc} J_{ad} - \eta_{ac} J_{bd} - \eta_{bd} J_{ac} + \eta_{ad} J_{bc}, \\ [J_{ab}, P_c^{(\tau)}] &= \eta_{bc} P_a^{(\tau)} - \eta_{ac} P_b^{(\tau)}, \\ [P_a^{(\tau)}, P_b^{(\tau)}] &= -\tau J_{ab}. \end{aligned}$$

For constant $\ell > 0$, write

$$\mathcal{A}_\tau^{\text{Cart}} = \frac{1}{2} \omega^{ab} J_{ab} + \ell^{-1} e^a P_a^{(\tau)}.$$

Proposition 21.9 (Curvature of the trit-typed realization). *With the preceding relations,*

$$\mathcal{F}_\tau^{\text{Cart}} = \frac{1}{2} (F_\omega^{ab} - \tau \ell^{-2} e^a \wedge e^b) J_{ab} + \ell^{-1} T^a P_a^{(\tau)}. \quad (21.12)$$

Proof. Expand $d\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}]$. The bracket of the Lorentz sector produces F_ω , the mixed bracket produces $D_\omega e$, and the translational bracket contributes $-\frac{1}{2}\tau\ell^{-2}e^a \wedge e^b J_{ab}$. $\square$

The central sheet recovers the affine connection; the outer sheets preserve the two signs of the constant-curvature term. The representation thus maintains the route's tritic regime before applying the variational equations. The holonomy equality (21.8) corresponds to the central sheet. Each outer sheet uses the representation $\rho_{C,\tau}$ in its Cartan group and its respective holonomy condition. The current and metric tensor use the fields and representation of the same realization.

21.7 Oriented transport of the connection and area

The parity of the incidence functional allows the charts γ and $-\gamma$ to be compared. In the canonical geometric realization, let E_i^a be a densitized triad, $\Gamma_a^i(E)$ its intrinsic connection, and K_a^i the extrinsic curvature in the chosen normal convention. The Ashtekar–Barbero connection is $\mathcal{A}_a^i = \Gamma_a^i(E) + \gamma K_a^i$.

Proposition 21.10 (Involution between oriented charts). *For $\gamma \in \mathbb{R} \setminus \{0\}$, the map*

$$\mathfrak{b} : (\gamma, K, E) \longmapsto (-\gamma, -K, E)$$

is involutive. It preserves $\mathcal{A}$ and reverses the form

$$\Omega_{KE} = \frac{1}{\kappa} \int_{\Sigma} \delta K_a^i \wedge \delta E_i^a,$$

with κ fixed and the same boundary conditions.

Proof. Applying $\mathfrak{b}$ twice restores all three coordinates. Since E remains fixed, so does $\Gamma(E)$. The product of the two odd coordinates satisfies $(-\gamma)(-K) = \gamma K$, preserving the connection. In the space of variations, $\delta(-K) = -\delta K$ and δE remains fixed: the pullback of the form is $\mathfrak{b}^* \Omega_{KE} = -\Omega_{KE}$. $\square$

The oriented area $\mathcal{A}_{\text{or}} = \gamma a_0 N$ and its physical magnitude $\mathcal{A}_{\text{fis}} = |\gamma| a_0 N$, for $a_0 > 0$ and $N \geq 0$, agree in the positive chart. Upon prolongation to the opposite chart, the former is odd and the latter remains invariant. This distinction preserves both the orientation of incidence and positivity of the geometric observable. The proposition establishes algebraic transport of these charts; its realization as reversal of the spatial normal additionally uses transport of matter and of the boundary data of action (21.2).

21.8 Icosahedral incidence and five-component gravitational coupling

Scalar normalization is received from (20.16): $G_{\text{ret}} = c_{\text{int}}^3 L^2 / \hbar_{\text{ret}}$. Transport between the two action sections preserves L^2 and yields $G_a = c_{\text{int}}^3 L^2 / \hbar_a$. Tensorial normalization is determined below by the icosahedral frame. Each has its own proof: rules R1–R8 fix the coefficient; incidence and the intertwiner fix the space and metric of its components.

The discrete structure of the continuum preserves an incidence reading alongside the arithmetic reading. The five-component gravitational carrier is constructed through that incidence and its tensor representation. The coupling section $G_a(\mathfrak{g}_{108}^\circ)$, with $a \in \{\text{pre}, \text{ret}\}$, is that obtained in Section 21; the space on which it acts, its five components, and its transverse projection are made explicit below.

Let $X_{\text{TPK}}^{\text{exc}}$ be the subspace of the enriched TPK state that preserves, in addition to arithmetic output, orientation, route, carry, and exceptional incidence. To make the target geometric space explicit, let $\varphi_g = (1 + \sqrt{5})/2$, $\nu = \sqrt{1 + \varphi_g^2}$, and define

$$\Omega_{12}^{\text{ico}} = \frac{1}{\nu} (\{0\} \times \{\pm 1\} \times \{\pm \varphi_g\} \cup \{\pm 1\} \times \{\pm \varphi_g\} \times \{0\} \cup \{\pm \varphi_g\} \times \{0\} \times \{\pm 1\}) \subset S^2. \quad (21.13)$$

Its twelve elements are the oriented vertex classes of an icosahedron. Incidence is not left verbal: it is fixed by

$$\mathcal{J}_{12}^{\text{ico}} = \{(v, w) \in \Omega_{12}^{\text{ico}} \times \Omega_{12}^{\text{ico}} : v \neq w, \langle v, w \rangle = 1/\sqrt{5}\}. \quad (21.14)$$

Each vertex has five neighbors in this relation. The reading bridge is typed as a map

$$\pi_{\text{ico}} : X_{\text{TPK}}^{\text{exc}} \longrightarrow \Omega_{12}^{\text{ico}}. \quad (21.15)$$

In addition to the twelve elements of the image, the incidence realization requires the following simultaneous properties of the map:

- (i) to be surjective and constant exactly on the fibers that discard the arithmetic chart but preserve the exceptional route;
- (ii) to intertwine the action of the oriented automorphisms of the antecedent with a transitive action on Ω_{12}^{ico} ;
- (iii) to transport the declared TPK incidence to $\mathcal{J}_{12}^{\text{ico}}$, not merely its cardinalities;
- (iv) to preserve sheet involution and nonadic return in the stabilizer of the corresponding fiber.

These conditions fix the construction order: first the map π_{ico} , then identification of the induced action, and finally its linear representation. Orientation, route, and carry belong to the enriched antecedent of that descent.

21.8.1 Calendar projection and representation of its twelve positions

The calendar of Section 2.6 provides an observable coordinate of the complete state. In its notation, write $r_0(\theta) = r(\theta) - 1 \in \{0, \dots, 8\}$ and $b = \beta(\theta) \in \mathbb{Z}/12\mathbb{Z}$. The clock projection is

$$Q_{\text{clk}}(x) = (\beta(\theta(x)), r_0(\theta(x))).$$

For an admissible route of n updates, the induced transport is

$$k(r_0, n) = \left\lfloor \frac{r_0 + n}{9} \right\rfloor, \quad T_n(b, r_0) = (b + k(r_0, n) \bmod 12, (r_0 + n) \bmod 9). \quad (21.16)$$

Here $n \in \mathbb{N}_0$ counts effective updates; the integer quotient k records the nonadic returns crossed.

Proposition 21.11 (Compatibility of the calendar with route composition). *The projection Q_{clk} , together with the length of each route, defines a functor from the category of admissible TPK routes to the action category of calendar (21.16). The identities $T_{m+n} = T_m T_n$, $T_0 = I$, $T_9(b, r_0) = (b + 1, r_0)$, and $T_{108}(b, r_0) = (b, r_0)$ hold.*

Proof. Putting $r'_0 = (r_0 + n) \bmod 9$, Euclidean division gives

$$k(r_0, n + m) = k(r_0, n) + k(r'_0, m).$$

This identity simultaneously preserves residue and quotient when the routes are concatenated. The update of θ established in Section 2.6 induces exactly (21.16); the remaining identities follow by substitution. The target category has pairs (b, r_0) as objects and admissible lengths as arrows, with addition as composition. $\square$

The index $p = b + 1 \in \{1, \dots, 12\}$, taking the representative $0 \leq b < 12$, indicates the dodecaphase position. One turn of 108 updates traverses its twelve positions. The proved periodicity belongs to this projection: route, accumulated carry, and memory remain in the source state and continue to be updated.

Incidence realization from the angular projector. The representatives $r_p C_5$ and representation ρ of Section 18 fix the chart of the twelve positions. The projector already defined by (18.5) produces

$$\begin{aligned} \mathcal{H}_3 &= P_3^{\text{ico}} \mathbb{R}^{12}, & v_p &= 2P_3^{\text{ico}} e_p, \\ (P_3^{\text{ico}})_{pq} &= \frac{1}{20} \sum_{\substack{g \in A_5 \\ gr_q C_5 = r_p C_5}} \chi_3(g^{-1}). \end{aligned} \quad (21.17)$$

The last sum contains five terms per entry and uses the representatives and the two classes of order-five cycles fixed there.

Proposition 21.12 (Icosahedral realization of the dodecaphase positions). *The vectors v_p are twelve distinct unit vectors in $\mathcal{H}_3$. Their incidence is $p \sim q \iff \langle v_p, v_q \rangle = 1/\sqrt{5}$. The map $p \mapsto v_p$ is A_5 -equivariant, and its image is isometric to Ω_{12}^{ico} .*

Proof. Transitivity and $\text{Tr } P_3^{\text{ico}} = 3$ give $(P_3^{\text{ico}})_{pp} = 1/4$; by idempotence, $\langle v_p, v_q \rangle = 4(P_3^{\text{ico}})_{pq}$ and $\|v_p\| = 1$. Evaluation of the five terms in (21.17) gives the antipodal pairs

$$(1, 4), (2, 3), (5, 8), (6, 7), (9, 12), (10, 11).$$

For the representatives $(1, 2, 5, 6, 9, 10)$, their Gram matrix is

$$I_6 + \frac{D}{\sqrt{5}}, \quad D = \begin{pmatrix} 0 & -1 & -1 & 1 & -1 & 1 \\ -1 & 0 & 1 & 1 & -1 & -1 \\ -1 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & -1 & 1 \\ -1 & -1 & 1 & -1 & 0 & 1 \\ 1 & -1 & 1 & 1 & 1 & 0 \end{pmatrix}, \quad D^2 = 5I_6. \quad (21.18)$$

The remaining inner products are obtained by antipodality. Thus, apart from a vertex and its opposite, the products are $\pm 1/\sqrt{5}$, and each vertex has five neighbors. The following table makes explicit the comparison with the signed six-position chart $\mathcal{U} = (\infty, 0, 1, 2, 3, 4)$:

p	$u(p)$	$\sigma(p)$	p	$u(p)$	$\sigma(p)$
1	∞	+1	7	3	-1
2	0	-1	8	1	+1
3	0	+1	9	2	-1
4	∞	-1	10	4	+1
5	1	-1	11	4	-1
6	3	+1	12	2	+1

(21.19)

Indeed, for $C = \text{bal}(A_W)$, with A_W given in (28.8), the entries of D are $D_{ab} = \sigma(p_a)\sigma(p_b)C_{u(p_a), u(p_b)}$. Consequently, the table and (21.18) isometrically identify v_p with $\sigma(p)\sqrt{2}E_C e_{u(p)}$, where $E_C = (I_6 + C/\sqrt{5})/2$. Both systems span three-dimensional spaces and have identical Gram matrices, proving that the induced linear map is well defined and isometric. The explicit realization in (21.24), proved below, identifies it with Ω_{12}^{ico} and its incidence. Finally, $P_3^{\text{ico}}\rho(g) = \rho(g)P_3^{\text{ico}}$ implies $v_{g \cdot p} = \rho(g)v_p$. The bijection of the twelve positions gives $\text{Stab}_{A_5}(v_p) = r_p C_5 r_p^{-1}$. $\square$

One thus obtains the representational composition

$$x \mapsto Q_{\text{clk}}(x) \mapsto p = \beta(\theta(x)) + 1 \mapsto v_p. \quad (21.20)$$

The signed expression of this composition is $r_{\pm}^{\text{rep}}(x) = (u(p), \sigma(p))$, determined by Table (21.19). The table is an explicit choice of alignment between charts; the actions of A_5 provide its equivalent alignments. The complete record is preserved in the domain of (21.20).

Link with the five-component tensor carrier. In $\mathcal{H}_3$, form $Q_p = v_p v_p^* - I_{\mathcal{H}_3}/3$. Let $J = \mathbf{11}^*$, and let A be the matrix of the preceding antipodal permutation. The Frobenius product and (21.18) give

$$\begin{aligned} \text{Gram}(Q_p) &= (4P_3^{\text{ico}}) \circ (4P_3^{\text{ico}}) - \frac{1}{3}J = \frac{4}{5}(I + A) - \frac{2}{15}J, \\ P_5^{\text{ico}} &= \frac{5}{8} \text{Gram}(Q_p) = \frac{1}{2}(I + A) - \frac{1}{12}J. \end{aligned} \quad (21.21)$$

The symbol $\circ$ denotes the entrywise product. The last expression is the orthogonal projector onto coordinates even under antipodality and orthogonal to $\mathbf{1}$, of dimension five. Its character is $(5, 1, -1, 0, 0)$: on the six antipodal pairs, the fixed-point counts are $(6, 2, 0, 1, 1)$, and the constant component is removed. It therefore coincides with the central projector of (21.35). This identity links the three-dimensional construction with Γ_0 and Γ_5 , defined in (21.40) and (21.42), through the explicit quadratic operation $v_p \mapsto Q_p$.

21.8.2 Transport of the representational frame

Let $s = (1\ 2\ \cdots\ 12)$, and let $S_c e_p = e_{s(p)}$. A nonadic calendar return takes p to $s(p)$. The joint transport of frame, representation, and projector is expressed by

$$\begin{aligned} P_{3,k}^{\text{ico}} &= S_c^k P_3^{\text{ico}} S_c^{-k}, & \rho^{(k)}(g) &= S_c^k \rho(g) S_c^{-k}, \\ v_{s^k(p)}^{(k)} &= 2P_{3,k}^{\text{ico}} e_{s^k(p)} = S_c^k v_p. \end{aligned} \quad (21.22)$$

The last equality follows by substitution; orthogonality of S_c preserves all inner products and transports incidence. For a route of length n , take $k = k(r_0, n)$. The carry identity in Proposition 21.11 proves compatibility of these transports with concatenation. Although $S_c^{12} = I$, the twelve turns update the memory of the source route.

Equality (21.22) establishes covariance between frames. Active action on a fixed incidence has additional content: it requires identifying transport of the complete state and its action in that chart. In particular, s has order twelve, whereas the orders of elements of A_5 are 1, 2, 3, 5. Likewise, the stabilizer of a vertex has order five, and that of its antipodal pair has order ten; the involution of the signed chart exchanges the two vertices. These distinctions preserve the meaning of orientation, sheet, and return in properties (i)–(iv): the proved representational realization and the descent of the complete exceptional route are related through those properties, with their domains and fibers declared.

Construction of the carrier from six-position incidence. The matrix A_W of (28.8), which previously enters angular transport in section 18, provides an explicit antecedent of the icosahedral realization. Let $C = \text{bal}(A_W)$, where $\text{bal}(0) = 0$, $\text{bal}(1) = 1$, and $\text{bal}(2) = -1$. Its coordinate order is $\mathcal{U} = (\infty, 0, 1, 2, 3, 4)$. The integer chart satisfies $C = C^\top$, $C^2 = 5I_6$, and $\text{Tr } C = 0$. Define its signed covering by

$$\mathcal{D}_C = \mathcal{U} \times \{+1, -1\}, \quad (u, \sigma) \sim_C (v, \tau) \iff u \neq v \text{ and } \sigma\tau = C_{uv}. \quad (21.23)$$

Orientation doubles each coordinate; the matrix decides the incidences between the two corresponding fibers. The involution of this covering is $J_C(u, \sigma) = (u, -\sigma)$.

Proposition 21.13 (Isometric realization of signed incidence). *The operator and vectors*

$$E_C = \frac{1}{2} \left(I_6 + \frac{C}{\sqrt{5}} \right), \quad \mathcal{H}_C = \text{im } E_C, \quad v_{u,\sigma} = \sigma \sqrt{2} E_C e_u \quad (21.24)$$

realize $\mathcal{D}_C$ as twelve distinct unit vectors in a three-dimensional Euclidean space. There is an explicit isometry $T_C : \mathcal{H}_C \rightarrow \mathbb{R}^3$ for which $\iota_C(u, \sigma) = T_C v_{u,\sigma}$ is a bijection onto Ω_{12}^{ico} , preserves incidence, and satisfies $\iota_C J_C = -\iota_C$.

Proof. From $C^2 = 5I_6$, one obtains $E_C^2 = E_C = E_C^*$; its trace is three. The metric identities are

$$\langle v_{u,\sigma}, v_{v,\tau} \rangle = \begin{cases} \sigma\tau, & u = v, \\ \sigma\tau C_{uv}/\sqrt{5}, & u \neq v. \end{cases}$$

Thus the twelve vectors are distinct, and their inner-product $1/\sqrt{5}$ incidence is exactly (21.23). To make the isometry explicit, let the columns of B , in order $\mathcal{U}$, be

$$B = \frac{1}{\nu} \begin{pmatrix} 0 & -1 & -\varphi_g & 0 & \varphi_g & 1 \\ -1 & -\varphi_g & 0 & 1 & 0 & -\varphi_g \\ -\varphi_g & 0 & -1 & -\varphi_g & -1 & 0 \end{pmatrix}.$$

Substitution of $\varphi_g^2 = \varphi_g + 1$ gives $B^* B = I_6 + C/\sqrt{5} = 2E_C$. Hence $T_C = B/\sqrt{2}|_{\mathcal{H}_C}$ is an isometry, and $T_C v_{u,\sigma} = \sigma B e_u$. The six columns of B and their opposites are precisely the twelve vertices of (21.13). The last identity also proves compatibility of the involutions. $\square$

The preceding dodecaphasic realization constructs the twelve vectors directly from the projector P_3^{ico} and the calendar positions. The signed covering provides another coordinate expression of that carrier. For the reader π_{ico} in the descent theorem, the relation between the two expressions is written

$$X_{\text{TPK}}^{\text{exc}} \xrightarrow{r_{\pm}} \mathcal{D}_C \xrightarrow[\simeq]{\iota_C} \Omega_{12}^{\text{ico}}, \quad \pi_{\text{ico}}(x) = \sigma(x)Be_{u(x)}, \quad r_{\pm}(x) = (u(x), \sigma(x)). \quad (21.25)$$

Since ι_C is a bijection, this factorization determines $r_{\pm} = \iota_C^{-1}\pi_{\text{ico}}$. Thus $r_{\pm}$ expresses the reader being used in the signed chart: its introduction does not add an independent hypothesis for constructing the dodecaphasic carrier. The preceding correspondence table gives this expression for its twelve positions. Properties (i)–(iv) retain their own scope when the realization is identified with the quotient of the full exceptional route. In particular, joint transport of frame, projector, and incidence proves covariance between charts; preservation of a fixed incidence by an active transformation is a different identity. Orientation and memory remain in the source state during both readings. A signed permutation transporting C jointly transports E_C , its vectors, and incidence; this covariance allows the charts to be compared while preserving each fiber's orientation.

Theorem 21.14 (Incidence descent to the icosahedral quotient). *Suppose that π_{ico} satisfies (i)–(iv) and that the induced oriented action on Ω_{12}^{ico} has group $G_{\text{ico}} \simeq A_5$. Then, for every class $v_0 \in \Omega_{12}^{\text{ico}}$,*

$$H_{v_0} = \text{Stab}_{G_{\text{ico}}}(v_0) \simeq C_5, \quad \Omega_{12}^{\text{ico}} \simeq G_{\text{ico}}/H_{v_0} \simeq A_5/C_5. \quad (21.26)$$

The map

$$\kappa_{v_0} : G_{\text{ico}}/H_{v_0} \longrightarrow \Omega_{12}^{\text{ico}}, \quad gH_{v_0} \longmapsto g \cdot v_0, \quad (21.27)$$

is an equivariant bijection and preserves incidence if the quotient is equipped with

$$gH_{v_0} \sim g'H_{v_0} \iff (g \cdot v_0, g' \cdot v_0) \in \mathcal{I}_{12}^{\text{ico}}. \quad (21.28)$$

Thus the composite $\kappa_{v_0}^{-1}\pi_{\text{ico}}$ is the typed map from the TPK antecedent to the quotient A_5/C_5 .

Proof. The list in (21.13) has twelve elements. By transitivity, $|G_{\text{ico}} \cdot v_0| = 12$, and the orbit–stabilizer theorem gives $|H_{v_0}| = 60/12 = 5$. Every group of prime order five is cyclic; hence $H_{v_0} \simeq C_5$. The formula for κ_{v_0} is independent of the representative: if $gH_{v_0} = g'H_{v_0}$, then $g^{-1}g' \in H_{v_0}$ and $g'v_0 = gv_0$. Surjectivity follows from transitivity, and equality of cardinalities gives injectivity. Equivariance is immediate: $\kappa_{v_0}(agH_{v_0}) = a\kappa_{v_0}(gH_{v_0})$. Finally, (21.28) is precisely transport of structure by that bijection; hypothesis (iii) makes this transport commute with π_{ico} . $\square$

The theorem establishes descent once the four properties of π_{ico} have been verified. Its domain is the enriched state of the exceptional reading, which preserves the incidence transported by APP–TRIT–TPK. In this construction, the explicit antecedent is map (21.15) with its four properties; the segment from A_5/C_5 to the transverse traceless reduction is proved below with that antecedent and the induced action of A_5 .

The provenance chain is expressed through all its morphisms:

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow X_{\text{TPK}}^{\text{exc}} \xrightarrow{\pi_{\text{ico}}} \Omega_{12}^{\text{ico}} \xleftarrow[\simeq]{\kappa_{v_0}} A_5/C_5. \quad (21.29)$$

Composition (21.29) preserves as its antecedent the existence of π_{ico} with properties (i)–(iv). On that domain, descent to the quotient is Theorem 21.14; from it are constructed the projector P_5^{ico} , intertwiner Γ_5 , and projection $\Lambda(n)$. Realization of the carrier as a gravitational field additionally uses the dynamics and physical constraints specified at the end of this development.

Let $\mathcal{G} = A_5$, let $H \simeq C_5$ be the stabilizer of an oriented vertex of the icosahedron, and consider

$$V_{12} = \mathbb{R}[\mathcal{G}/H], \quad |\mathcal{G}/H| = \frac{60}{5} = 12. \quad (21.30)$$

Characters of icosahedral incidence. The following calculation is performed on the icosahedral action obtained in Theorem 21.14, with its incidence and orientation hypotheses. It determines the decomposition of the vertex representation and the irreducibility of the tensor space through their traces, preserving the antecedent π_{ico} declared in that theorem.

Proposition 21.15 (Vertex decomposition and tensor character). *Let $\tau = (1 + \sqrt{5})/2$ and $\bar{\tau} = (1 - \sqrt{5})/2$. In the class order 1, 2A, 3A, 5A, 5B, the characters of the rotational representation V_3 , its algebraic conjugate $V_{3'}$, the twelve-vertex representation V_{12} , and $\text{STF}_3 = \text{Sym}_0^2(V_3)$ are*

class	1	2A	3A	5A	5B
cardinality	1	15	20	12	12
class of g^2	1	1	3A	5B	5A
χ_3	3	-1	0	τ	$\bar{\tau}$
$\chi_{3'}$	3	-1	0	$\bar{\tau}$	τ
χ_{12}	12	0	0	2	2
χ_{STF}	5	1	-1	0	0

(21.31)

The representations V_3 , $V_{3'}$ and STF_3 are absolutely irreducible and distinct. Moreover,

$$V_{12} \simeq \mathbf{1} \oplus V_3 \oplus V_{3'} \oplus \text{STF}_3, \quad \langle \chi_{12}, \chi_{\text{STF}} \rangle = 1. \quad (21.32)$$

In particular, the five-dimensional summand occurs exactly once.

Proof. In A_5 , there are $5 \cdot 3 = 15$ double transpositions and $\binom{5}{3} \cdot 2 = 20$ three-cycles. The $4! = 24$ five-cycles split into two classes of twelve: their centralizer is the cyclic subgroup of order five and is contained in A_5 . Identifying the positions of a cycle with $\mathbb{Z}/5\mathbb{Z}$, the permutation inducing $r \mapsto 2r$ is a four-cycle on the nonzero elements, with negative sign. Squaring therefore exchanges the two classes. Inversion $r \mapsto -r$ is a product of two transpositions and preserves each class. This determines the first two rows of (21.31).

The representation V_3 is the rotation action on the vertex coordinates. A rotation through angle θ has trace $1 + 2 \cos \theta$. The classes of orders two and three give -1 and 0 , respectively. The class of angles $\pm 2\pi/5$ is called 5A, and that of $\pm 4\pi/5$ is called 5B; their traces are τ and $\bar{\tau}$. These expressions follow from $u^2 + u - 1 = 0$, for $u = 2 \cos(2\pi/5)$, and the double-angle formula. The equation for u follows by dividing $1 + z + z^2 + z^3 + z^4 = 0$ by z^2 , with $z = \exp(2\pi i/5)$. The character of an already geometrically defined representation has thus been calculated.

To construct the other representation, use the vertices without the common normalization v^{-1} . Their coordinates belong to $K = \mathbb{Q}(\sqrt{5})$. If B is the matrix of three linearly independent vertices and B_g contains their images, the rotation matrix is $R_g = B_g B^{-1} \in M_3(K)$. The conjugation $\sigma : \sqrt{5} \mapsto -\sqrt{5}$ then defines $R'_g = \sigma(R_g)$, satisfying

$$R'_{gh} = R'_g R'_h, \quad (R'_g)^T R'_g = I, \quad \det R'_g = 1.$$

Thus $V_{3'}$ is an effectively constructed real representation. Its character is $\sigma(\chi_3)$, exchanging τ and $\bar{\tau}$.

The permutation character counts fixed vertices. The identity fixes twelve; nontrivial rotations of order five fix the two endpoints of their axis. Those of order two or three fix no vertices, since the stabilizer of a vertex is C_5 . Hence $\chi_{12} = (12, 0, 0, 2, 2)$.

For the tensor space, if $\lambda_1, \lambda_2, \lambda_3$ are the eigenvalues of R_g , the sum $\sum_{i \leq j} \lambda_i \lambda_j$ is $(\sum_i \lambda_i)^2 + \sum_i \lambda_i^2 / 2$. The metric form generates an invariant line in $\text{Sym}^2(V_3)$, whose trace-free complement has character

$$\chi_{\text{STF}}(g) = \frac{\chi_3(g)^2 + \chi_3(g^2)}{2} - 1. \quad (21.33)$$

Using the squaring map in the table and $\tau + \bar{\tau} = 1$, $\tau \bar{\tau} = -1$, gives $\chi_{\text{STF}} = (5, 1, -1, 0, 0)$.

The real-character inner product is $\langle \chi, \psi \rangle = \frac{1}{60} \sum_C |\chi(C)\psi(C)|$. In particular,

$$\begin{aligned}\langle \chi_3, \chi_3 \rangle &= \langle \chi_{3'}, \chi_{3'} \rangle = \frac{9 + 15 + 12(\tau^2 + \bar{\tau}^2)}{60} = 1, \\ \langle \chi_3, \chi_{3'} \rangle &= \frac{9 + 15 + 24\tau\bar{\tau}}{60} = 0, \\ \langle \chi_{\text{STF}}, \chi_{\text{STF}} \rangle &= \frac{25 + 15 + 20}{60} = 1.\end{aligned}$$

Norm one proves irreducibility of the complexifications; consequently, it also proves absolute irreducibility of the constructed real representations. Their dimensions and orthogonality distinguish the three summands. Finally, the equality of class functions

$$\chi_{12} = 1 + \chi_3 + \chi_{3'} + \chi_{\text{STF}}$$

is verified in all five columns. Semisimplicity of finite-group representations in characteristic zero gives decomposition (21.32). The multiplicity of the tensor summand can be checked directly:

$$\langle \chi_{12}, \chi_{\text{STF}} \rangle = \frac{12 \cdot 5}{60} = 1.$$

□

Thus the central idempotent associated with $\chi_{\text{STF}} = \chi_5$ has rank $5 \cdot 1 = 5$ in V_{12} . The irreducibility entering normalization of the tensor intertwiner and the multiplicity entering gravitational coupling are established by this same calculation.

The permutation representation ρ_{12} decomposes as

$$V_{12} \simeq \mathbf{1} \oplus V_3 \oplus V_{3'} \oplus V_5. \quad (21.34)$$

The central idempotent of the summand with character $\chi_5 = (5, 1, -1, 0, 0)$ is

$$P_5^{\text{ico}} = \frac{5}{60} \sum_{g \in A_5} \chi_5(g^{-1}) \rho_{12}(g) = \frac{1}{12} \left(5I + \sum_{g \in 2A} \rho_{12}(g) - \sum_{g \in 3A} \rho_{12}(g) \right). \quad (21.35)$$

Theorem 21.16 (Rank-five projector and gravitational coupling). *The operator P_5^{ico} satisfies*

$$(P_5^{\text{ico}})^2 = P_5^{\text{ico}} = (P_5^{\text{ico}})^*, \quad \text{Tr } P_5^{\text{ico}} = \text{rank } P_5^{\text{ico}} = 5. \quad (21.36)$$

For

$$\mathbb{G}_{5,a} := G_a(\mathfrak{g}_{108}^\circ) P_5^{\text{ico}} \quad (21.37)$$

one has

$$\text{Spec}(\mathbb{G}_{5,a}) = \{G_a(\mathfrak{g}_{108}^\circ) \text{ with multiplicity } 5, 0 \text{ with multiplicity } 7\}. \quad (21.38)$$

Proof. Character orthogonality turns (21.35) into the central idempotent acting as the identity on the unique copy of V_5 and as zero on the other three summands of (21.34). Reality of the character and orthogonality of ρ_{12} give self-adjointness. Its trace and rank are five. Multiplying a rank-five projector by the positive scalar G_a immediately produces the indicated spectrum. □

The gravitational section G_a is common to the five components of the selected tensor summand. Its spectral multiplicity expresses the rank of the carrier; the other seven modes of the permutation module belong to the kernel of $\mathbb{G}_{5,a}$. The symbol P_5^{ico} here identifies this central rank-five projector on V_{12} ; averaging the five positions of an arithmetic penta-fiber has a different domain and rank one.

21.9 Intertwiner with the space of symmetric traceless tensors

Let $\mathcal{V} = \Omega_{12}^{\text{ico}} \subset S^2$ be the set of twelve unit vertices already constructed. For each $v \in \mathcal{V}$, define

$$Q_v = vv^T - \frac{1}{3}I_3 \in \text{STF}_3 := \text{Sym}_0^2(\mathbb{R}^3). \quad (21.39)$$

The frame map

$$\Gamma_0 \left(\sum_{v \in \mathcal{V}} x_v e_v \right) = \sum_{v \in \mathcal{V}} x_v Q_v \quad (21.40)$$

is A_5 -equivariant and satisfies

$$\Gamma_0 \Gamma_0^* = \frac{8}{5} I_{\text{STF}_3}, \quad \Gamma_0 = \Gamma_0 P_5^{\text{ico}}. \quad (21.41)$$

Theorem 21.17 (Normalized tensor intertwiner). *The map*

$$\Gamma_5 := \sqrt{\frac{5}{8}} \Gamma_0|_{V_5} : V_5 \xrightarrow{\simeq} \text{STF}_3 \quad (21.42)$$

is an A_5 -equivariant isometry. Its adjoint is

$$\Gamma_5^* Q = \sqrt{\frac{5}{8}} \sum_{v \in \mathcal{V}} (v^T Q v) e_v. \quad (21.43)$$

Proof. Equivariance follows from $Q_{gv} = gQ_v g^T$. The operator $\Gamma_0 \Gamma_0^*$ commutes with the irreducible representation of A_5 on STF_3 , so it is scalar. Its trace is

$$\sum_{v \in \mathcal{V}} \|Q_v\|_F^2 = 12 \cdot \frac{2}{3} = 8;$$

dividing by dimension five gives $8/5$. Hence, $\Gamma_5 \Gamma_5^* = I$. Since domain and codomain have dimension five, $\Gamma_5^* \Gamma_5 = I$ as well. The adjoint formula follows from $\langle Q, Q_v \rangle_F = v^T Q v$. $\square$

Normalization through the icosahedral frame constructs the isometry transporting the finite summand to the real spin-two tensor space.

The coupling already determined satisfies

$$\Gamma_5(\mathbb{G}_{5,a}|_{V_5})\Gamma_5^* = \frac{c_{\text{int}}^3 L^2}{\hbar_a} I_{\text{STF}_3}. \quad (21.44)$$

The restriction of the projector to V_5 is the identity and $\Gamma_5 \Gamma_5^* = I$, proving the equality. The factor $\sqrt{5/8}$ normalizes the intertwiner; the factor $c_{\text{int}}^3 L^2 / \hbar_a$ retains the longitudinal, action, and radial normalization of the preceding construction.

21.10 Transverse traceless projection

For $n \in S^2$, let $\Pi(n) = I - nn^T$. Define

$$\Lambda(n)Q = \Pi Q \Pi - \frac{1}{2} \text{Tr}(\Pi Q \Pi) \Pi. \quad (21.45)$$

Theorem 21.18 (Reduction from five components to two). *The restriction of $\Lambda(n)$ to STF_3 is the orthogonal projector onto*

$$\mathcal{H}_{\text{TT}}(n) = \{Q \in \text{STF}_3 : Qn = 0\}. \quad (21.46)$$

In particular,

$$\Lambda(n)^2 = \Lambda(n) = \Lambda(n)^*, \quad \text{rank } \Lambda(n) = 2. \quad (21.47)$$

For $n = e_3$, the image is spanned by

$$E_+ = \frac{1}{\sqrt{2}} \text{diag}(1, -1, 0), \quad E_\times = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (21.48)$$

Proof. Since $\Pi n = 0$, the output is transverse. Its trace is

$$\text{Tr}(\Pi Q \Pi) - \frac{1}{2} \text{Tr}(\Pi) \text{Tr}(\Pi Q \Pi) = 0,$$

because $\text{Tr} \Pi = 2$. If $Qn = 0$ and $\text{Tr} Q = 0$, then $\Pi Q \Pi = Q$ and $\text{Tr}(\Pi Q \Pi) = 0$, so $\Lambda(n)Q = Q$. The formula is self-adjoint for the Frobenius product. In the frame $n = e_3$, its matrix in an adapted STF basis is $\text{diag}(1, 1, 0, 0, 0)$, proving the rank. $\square$

The complete chain is therefore

$$V_{12} \xrightarrow{P_5^{\text{ico}}} V_5 \xrightarrow{\Gamma_5} \text{STF}_3 \xrightarrow{\Lambda(n)} \mathcal{H}_{\text{TT}}(n), \quad 12 \longrightarrow 5 \longrightarrow 5 \longrightarrow 2. \quad (21.49)$$

The reduced coupling

$$\mathbb{G}_{\text{TT},a}(n) = G_a(\vartheta_{108}^\circ) \Lambda(n) \Gamma_5 P_5^{\text{ico}} \quad (21.50)$$

has rank two.

21.11 Decomposition into 5 polarization modes

Dimension five acquires tensor content upon choosing a unit propagation direction n and an oriented orthonormal frame (e_1, e_2, n) . Define the five real tensors

$$E_+ = \frac{1}{\sqrt{2}}(e_1 e_1^T - e_2 e_2^T), \quad E_\times = \frac{1}{\sqrt{2}}(e_1 e_2^T + e_2 e_1^T), \quad (21.51)$$

$$E_1 = \frac{1}{\sqrt{2}}(e_1 n^T + n e_1^T), \quad E_2 = \frac{1}{\sqrt{2}}(e_2 n^T + n e_2^T), \quad (21.52)$$

$$E_0 = \frac{1}{\sqrt{6}}(2nn^T - e_1 e_1^T - e_2 e_2^T). \quad (21.53)$$

All are symmetric and traceless. The first two are transverse; the next two mix the transverse plane with the propagation direction, and the last is the traceless longitudinal scalar mode.

Theorem 21.19 (Complete helicity decomposition of the carrier). *The tensors in (21.51), (21.52), (21.53) form an orthonormal basis of STF_3 . In the complexification, the combinations*

$$E_{\pm 2} = \frac{1}{\sqrt{2}}(E_+ \mp iE_\times), \quad E_{\pm 1} = \frac{1}{\sqrt{2}}(E_1 \mp iE_2), \quad E_0 = E_0 \quad (21.54)$$

are vectors of weights $\pm 2, \pm 1, 0$, respectively, under rotations around n . Moreover,

$$\Lambda(n)E_{\pm 2} = E_{\pm 2}, \quad \Lambda(n)E_{\pm 1} = 0, \quad \Lambda(n)E_0 = 0. \quad (21.55)$$

Proof. In the frame (e_1, e_2, n) , each tensor has Frobenius norm one. Their off-diagonal entry patterns are disjoint, and the three diagonal entries of E_+, E_0 have zero inner product; thus the five tensors are orthonormal. Since $\dim \text{STF}_3 = 6 - 1 = 5$, they form a basis.

A rotation through angle ψ around n sends (e_1, e_2) to $(\cos \psi e_1 + \sin \psi e_2, -\sin \psi e_1 + \cos \psi e_2)$. Substituting into the formulas gives a rotation through angle 2ψ on $\text{span}\{E_+, E_\times\}$, one through angle ψ on $\text{span}\{E_1, E_2\}$, and fixes E_0 . This gives the weights in (21.54). Finally, $\Pi E_+ \Pi = E_+$ and $\Pi E_\times \Pi = E_\times$, whereas $\Pi E_1 \Pi = \Pi E_2 \Pi = 0$. For E_0 , $\Pi E_0 \Pi = -(e_1 e_1^T + e_2 e_2^T)/\sqrt{6}$, which is pure trace in the plane, and subtraction in (21.45) annihilates it. $\square$

The theorem specifies the meaning of “five polarizations” without assigning them all the same dynamical status. Before field equations are imposed, the spin-two carrier has five irreducible coordinates. In a massive Fierz–Pauli realization, the constraints leave precisely the five weights of (21.54); in a massless realization with gauge invariance, sectors ± 1 and 0 are eliminated, leaving ± 2 . HMT here provides the carrier, basis, and projector; the choice of massive or massless dynamics belongs to the

physical interface. Thus five means not “five observed waves,” but five tensor degrees available before reduction.

Orientation also remains visible: changing (e_1, e_2, n) to $(e_1, -e_2, -n)$ preserves E_+, E_0 and exchanges the phases of the helicity pairs. That information is not in the scalar G_a ; it resides in the carrier on which $\mathbb{G}_{5,a}$ acts.

21.12 Physical interpretation of the representational chain

Dimension five appears in distinct objects, and only (21.49) authorizes transport between them:

1. V_5 is the irreducible summand of the representation on the twelve vertices. It is a finite result of representation theory.
2. STF_3 is the five-dimensional real carrier of the spin-two weights $m = -2, -1, 0, 1, 2$.
3. A massive Fierz–Pauli theory can realize those five weights as five physical states, after action and constraints are fixed.
4. Massless general relativity preserves two radiative helicities, ± 2 , represented by E_+ and E_- .
5. The $E(2)$ classes of metric waves, such as III_5 , classify possible amplitudes under other hypotheses; they are not automatically V_5 , Fierz–Pauli, or general relativity.

The structural content is thus determined: under the incidence-descent conditions, HMT constructs a five-component tensor carrier before the constraints. The massless realization preserves the two radiative helicities through the rank-two TT projection. Orientation information is preserved in the tensors and their transport; the scalar value G_a determines the common coupling.

21.13 Structural definition and function of the gravitational constant

The common realization first receives the length $L = (5759/23040)\alpha_{\text{HMT}}^{16}L_*$ from (20.13). Its elementary duration is $t_0 = \pi_{\text{HMT}}L/(54c_{\text{int}})$; hence $R_{108} = 54c_{\text{int}}t_0/\pi_{\text{HMT}} = L$. Equality with the Planck length of this realization, $\ell_P = L$, follows after the coupling has been determined. The positive reference L_* remains explicit; evaluation in metres uses the declared section $L_* = 1 \text{ m}$.

Definition 21.20 (Gravitational section contragredient to action). For a positive section $\hbar_a$, a velocity $c_{\text{int}} > 0$, and the chronological radius R_{108} of section 21, the circular gravitational section is

$$G_a = \frac{c_{\text{int}}^3 R_{108}^2}{\hbar_a}, \quad R_{108} = \frac{54}{\pi} \ell_0, \quad \ell_0 = c_{\text{int}} t_0. \quad (21.56)$$

It is the action-reciprocal section realizing coincidence between the cycle’s reduced gravitational radius and its chronological radius, in the convention $r_g = Gm/c_{\text{int}}^2$.

The radius-coincidence proposition of section 21 expresses its uniqueness in the collection $G_a(\vartheta) = \vartheta c_{\text{int}}^5 t_0^2 / \hbar_a$. The circular condition fixes $\vartheta = (54/\pi)^2$ and leads to (21.56). That condition and the positive dimensional sections are part of the realization. The operator proof in 20 extends that expression to the complete positive character: R4 defines the longitudinal composition and R5 its radial metric; R8 identifies the resulting operator with the reduced gravitational radius. With those data, the same coefficient $G_{\text{ret}} = c_{\text{int}}^3 L^2 / \hbar_{\text{ret}}$ acts in every admissible positive mode. The circular formula is its specialization to the identity character and the clock conjugate to the length. Keeping them common to both orientations yields

$$\frac{G_{\text{pre}}}{G_{\text{ret}}} = R_{\text{act}}^{-1}, \quad \frac{G_a \hbar_a}{c_{\text{int}}^3} = R_{108}^2 = L^2 = \ell_P^2. \quad (21.57)$$

The proof is direct substitution of (21.56); the action change is compensated by the reciprocal change in G_a .

This structure assigns gravitation a precise function: transporting the action section to a common geometric unit. In (21.2), G fixes coupling normalization; in the area operator, $G\hbar/c^3$ fixes the area

unit. Reciprocity preserves that unit while the oriented action reading changes. The reduced state and its occupations subsequently supply the information and entanglement associated with the boundary.

The tensorial content of the coupling is preserved in $\mathbb{G}_{5,a} = G_a P_5^{\text{ico}}$, constructed in 21.8. Section G_a determines its common magnitude; the projector selects the five-component carrier, and Γ_5 transports its orientation to the space of symmetric trace-free tensors. The basis in (21.51)–(21.53) makes the five components explicit. Projection $\Lambda(n)$ then defines the transverse reduction. Magnitude, carrier, and reduction are related operations with different domains within this realization.

22 Boltzmann constant, information, and thermal transduction

22.1 Dimensional structure of transduction

The structural definition of the Boltzmann constant uses the temperature and energy lines. The positive transduction

$$k_B : \mathcal{L}_\Theta \longrightarrow \mathcal{L}_E$$

assigns a temperature its energy scale. In the tritic normalization, the cycle's temperature section satisfies

$$k_B \Theta_{\text{clk}} = \frac{E_P}{\log 3}.$$

The right-hand side is fixed by the previously constructed action and period. The factor $\log 3$ comes from the informational content of a uniform distribution on the three local states.

Let u_E, u_Θ be positive bases, and write $E_P = \epsilon_P u_E$, $\Theta_{\text{clk}} = \theta_{\text{clk}} u_\Theta$, and $k_B(u_\Theta) = b u_E$. Then

$$b \theta_{\text{clk}} = \frac{\epsilon_P}{\log 3}.$$

For $u'_E = s_E u_E$ and $u'_\Theta = s_\Theta u_\Theta$, the coordinates transform by

$$b' = \frac{s_\Theta}{s_E} b, \quad \theta'_{\text{clk}} = \frac{\theta_{\text{clk}}}{s_\Theta}, \quad \epsilon'_P = \frac{\epsilon_P}{s_E}.$$

The energy identity remains covariant. This formulation combines the individual structure of the transducer with the normalization of the section on which it acts.

Remark 22.1 (Thermal coordinate and choice of bases). The transducer and the preceding energy identity are covariant. An individual numerical value of k_B is referred to the positive bases u_E, u_Θ ; its transformation is $b' = (s_\Theta/s_E)b$. Internal selection of a numerical coordinate therefore requires, in addition, a construction fixing those bases. The energy identity does not by itself select that choice. The metrological coordinate does not define action or period.

The distinction helps interpret entropy. The functional $s(\rho) = -\text{Tr}(\rho \log \rho)$ is dimensionless. Dimensional entropy is $S(\rho) = k_B s(\rho)$, understood on the corresponding line. The quantity ΘS belongs to the energy line. This preserves the mathematical meaning of the equalities combining information, temperature, and action.

In particular, continuation information (19.14) admits the dimensional expression $S_n^{\text{gen}}(x) = k_B s_n(x)$. Admissible branches and their measure determine s_n ; the transducer fixes the entropy scale. For a uniform distribution on three alternatives, the value is $k_B \log 3$. When prolongations have different weights, the weighted expression preserves that structural difference. The Boltzmann constant can be defined here by its function: it is the dimensional coefficient transporting information from the admitted distribution to the entropy line and, under a declared temperature, linking that information to energy.

Action enters before thermal expression. The section $\hbar_a$ converts evolution frequency into energy; the section h_a expresses action accumulated over one turn. The equality $k_B \Theta_{\text{clk},a} \log 3 = h_a/T_{108}$ combines both functions: energy of the period and local information of the tritic dynamics. A thermodynamic identification additionally preserves the statistical state, temperature, and the physical exchange or erasure operation under study.

22.2 Genealogical memory, information reduction, and thermal transduction

The thermal construction uses two representations from the same APP-TRIT-TPK development. The first preserves states, admissible routes, and record information; the second provides oriented action and the duration of return. Their composition determines an energy per tritic decision. Temperature and energy coordinates are introduced afterward through explicit dimensional maps. This sequence makes it possible to identify what information is preserved, what information a reader extracts, and what quantity converts that information into dimensional entropy.

22.2.1 Reader, memory, and state recovery

Fix a finite level of the development, and denote by Ω the set of admissible enriched states. The record preserves sheet, orientation, residue, quotient, carry, route, phase, and memory in the domains where each coordinate enters. Let

$$q : \Omega \longrightarrow Y, \quad M : \Omega \longrightarrow \mathcal{M}$$

be the observable map and retained memory, respectively. For a distribution p on Ω , write $H(p) = -\sum_x p_x \log p_x$, with $0 \log 0 = 0$. The logarithm is applied to dimensionless probabilities.

Definition 22.2 (Genealogical recovery from observation and its memory). The pair $\hat{q} = (q, M)$ is a recoverable reader on support Ω_p when there exists

$$d : \hat{q}(\Omega_p) \longrightarrow \Omega_p, \quad d \circ \hat{q} = \text{id}_{\Omega_p}.$$

The fiber information of q is $H_{\text{fib}}(q; p) = H(X | q(X))$, where X has law p .

Proposition 22.3 (Recovery identity and cost of reduction). *For a recoverable reader,*

$$H(X | q(X), M(X)) = 0, \quad (22.1)$$

$$H(X) = H(q(X)) + H(X | q(X)), \quad (22.2)$$

$$H(X | q(X)) = H(M(X) | q(X)). \quad (22.3)$$

Proof. The map d recovers X from the pair $(q(X), M(X))$, so the first conditional entropy is zero. Since q is a function of X , the chain rule gives the second equality. For the third, expand $H(X, M(X) | q(X))$ in two ways. One gives $H(X | q(X))$, because M is a function of X ; the other gives $H(M(X) | q(X)) + H(X | q(X), M(X))$. $\square$

Euclidean division of the APP sheets displays the elementary arithmetic mechanism. With $i, j \in \{1, \dots, 9\}$, one preserves

$$i + j = 9q_{ij}^+ + R_{ij}^+, \quad ij = 9q_{ij}^\times + R_{ij}^\times, \quad 1 \leq R_{ij}^\sigma \leq 9 \quad (\sigma \in \{+, \times\}).$$

The residue–quotient pair recovers the integer value of the operation. Summing the differences over the eighty-one positions gives

$$2025 - 810 = 54 + 9 \cdot 129.$$

Quotient memory restores the contribution not retained by residual reduction. Recoverability of the complete state additionally requires preserving the relevant position and route coordinates; the residue–quotient pair here identifies the result of the operation.

22.2.2 Reversible update, tritic decision, and erasure

Let $U : \Omega \rightarrow \Omega$ be the TPK update on a domain where enriched transport has an inverse. The transported law is $p'_y = p_{U^{-1}y}$. The bijection preserves the multiset of probabilities, and hence

$$H(U(X)) = H(X), \quad H(X | U(X)) = 0. \quad (22.4)$$

If the observed output is $q(U(X))$, the unobserved information is quantified by $H(X | q(U(X)))$. The complete state, its observation, and the reset of a record are thus three distinct maps.

For the three oriented TRIT states, let $p = (p_{-1}, p_0, p_{+1})$ be a normalized distribution. Its information is

$$I_{\text{TRIT}}(p) = -\sum_{a=-1}^1 p_a \log p_a, \quad 0 \leq I_{\text{TRIT}}(p) \leq \log 3. \quad (22.5)$$

The upper bound corresponds to $p_a = 1/3$. Indeed,

$$D(p \| (1/3, 1/3, 1/3)) = \sum_a p_a \log(3p_a) = \log 3 - I_{\text{TRIT}}(p) \geq 0;$$

the inequality follows from convexity of $t \log t$, and equality characterizes the uniform distribution. The lower bound corresponds to a distribution concentrated on one state.

The ideal Landauer realization uses a memory with energetically degenerate logical states, an isothermal environment at temperature Θ , and a balance that preserves the logical map's output and accounts for auxiliary records. In that model, the decrease in logical entropy is $k_B[H(X) - H(F(X))] = k_B H(X | F(X))$ for a deterministic map F . Entropy balance requires transferring at least that amount to the environment; its energy expression is $Q \geq k_B \Theta H(X | F(X))$. The distinction between reversible logical evolution and physical erasure corresponds to the Landauer–Bennett realization [1]. The invertible update (22.4) has zero logical contribution; resetting a uniform triple requires transferring $\log 3$ nats to the environment and satisfies $Q \geq k_B \Theta \log 3$. Control energy and dissipation in a particular physical realization belong to its additional thermodynamic balance.

22.2.3 Action, frequency, and decision energy

Let $\mathcal{L}_{\text{act}}$ and $\mathcal{L}_T$ be the dimensional lines of action and duration; the energy line is $\mathcal{L}_E = \mathcal{L}_{\text{act}} \otimes \mathcal{L}_T^*$. The preceding development supplies the positive sections $\hbar_a \in \mathcal{L}_{\text{act}}$, with $a \in \{\text{pre}, \text{ret}\}$, and the elementary duration $t_0 \in \mathcal{L}_T$. The complete calendar provides

$$T_{108} = 108t_0, \quad \omega_{108} = \frac{2\pi}{T_{108}}, \quad h_a = 2\pi\hbar_a.$$

The dimensional contraction between action and frequency defines

$$\epsilon_a := \hbar_a \omega_{108} = \frac{h_a}{108t_0} = \frac{\hbar_a}{t_P}, \quad t_P = \frac{54}{\pi}t_0. \quad (22.6)$$

The last member uses the same reduced circular period as the gravitational section. The three expressions correspond to one energy section, with an orientation a maintained in all factors. The normalized information of a uniform decision produces the energy-per-nat section

$$\epsilon_a := \frac{\epsilon_a}{\log 3} \in \mathcal{L}_E^+. \quad (22.7)$$

The previously constructed action determines the energy scale once duration is fixed; the cardinality of TRIT alternatives determines its information normalization. The value $\log 3$ enters through the count and measure declared in (22.5).

22.2.4 Dimensional definition of Boltzmann and coordinate selection

Let $\mathcal{L}_\Theta$ be the oriented temperature line. The dimensional entropy space is $\mathcal{L}_{\text{ent}} = \mathcal{L}_E \otimes \mathcal{L}_\Theta^*$. A positive linear map $k_B : \mathcal{L}_\Theta \rightarrow \mathcal{L}_E$ is, equivalently, a positive element of $\mathcal{L}_{\text{ent}}$. The same object can therefore act on a temperature or multiply dimensionless information.

Definition 22.4 (Thermal normalization of the tritic decision). A thermal pair compatible with oriented section a consists of a positive transducer k_B and a positive section $\Theta_{\text{clk},a} \in \mathcal{L}_\Theta$ such that

$$k_B(\Theta_{\text{clk},a}) = \epsilon_a, \quad k_B \Theta_{\text{clk},a} \log 3 = \epsilon_a. \quad (22.8)$$

The Boltzmann constant serves as the energy–temperature transducer of tritic information in this realization.

Theorem 22.5 (Uniqueness relative to the thermal section and dimensional covariance). *With a positive section $\Theta_{\text{clk},a}$ fixed, there is a unique positive linear map satisfying (22.8). For positive bases u_E, u_Θ , if*

$$\epsilon_a = \epsilon_a^{\text{num}} u_E, \quad \Theta_{\text{clk},a} = \theta_a u_\Theta, \quad k_B(u_\Theta) = b u_E,$$

its coordinate is

$$b = \frac{\epsilon_a^{\text{num}}}{\theta_a \log 3}. \quad (22.9)$$

Under $u'_E = s_E u_E$, $u'_\Theta = s_\Theta u_\Theta$, with $s_E, s_\Theta > 0$, one has

$$b' = \frac{s_\Theta}{s_E} b, \quad \theta'_a = \frac{\theta_a}{s_\Theta}, \quad (\epsilon_a^{\text{num}})' = \frac{\epsilon_a^{\text{num}}}{s_E}.$$

Proof. A nonzero section spans a line. Prescribing its image determines the unique linear map; positivity of both sections proves its positivity. Substituting the coordinates into (22.8) gives $b\theta_a \log 3 = \epsilon_a^{\text{num}}$. The three transformation laws follow by expressing the same sections and the same map in the new bases. $\square$

The same Boltzmann constant must act in both orientations. Since t_P is common, simultaneous compatibility is equivalent to

$$\frac{\Theta_{\text{clk,pre}}}{\Theta_{\text{clk,ret}}} = \frac{\epsilon_{\text{pre}}}{\epsilon_{\text{ret}}} = \frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}}. \quad (22.10)$$

Indeed, an isomorphism of lines preserves quotients of sections. Conversely, if the preceding equality holds, the isomorphism determined in one orientation also satisfies the normalization of the other. Thus return modifies the thermal section of the cycle in the same proportion as energy, while the transducer remains common to both readings.

The preceding equation makes explicit the choice required to fix an individual value. If the thermal section has not yet been selected by an independent reader, then for every $\lambda > 0$ the pair

$$(k_B, \Theta_{\text{clk},a}) \mapsto (\lambda k_B, \lambda^{-1} \Theta_{\text{clk},a})$$

preserves ϵ_a . Consequently, presenting an internally selected individual coordinate requires providing the reader of $\Theta_{\text{clk},a}$, the bases, and their dimensional transport. The action and information equations remain determined before that specification. Subsequent identification of the bases with joules and kelvins belongs to the metrological expression of the result.

22.2.5 Dimensional entropy and information of the reduced state

For a distribution p of admitted routes, or a reduced state with density matrix ρ , let

$$s(p) = - \sum_x p_x \log p_x, \quad s(\rho) = - \text{Tr}(\rho \log \rho).$$

Their dimensional expressions are

$$S(p) = k_B s(p), \quad S(\rho) = k_B s(\rho) \quad \text{in } \mathcal{L}_{\text{ent}}. \quad (22.11)$$

Evaluation at Θ produces an energy $\Theta S = k_B \Theta s$. For the uniform triple at the cycle temperature, (22.8) gives exactly

$$S_{\text{TRIT}} = k_B \log 3, \quad \Theta_{\text{clk},a} S_{\text{TRIT}} = \epsilon_a.$$

If ρ_A is the reduction of the bipartite purification constructed in the areal section, $s(\rho_A)$ measures the entanglement of that bipartition. Unit transport preserves its eigenvalues, orientation, and sectorial occupations. The areal scale $\gamma a_0 \ell_P^2$ converts occupations into area; the transducer k_B converts information into dimensional entropy. Both readings use the same record through different operators.

22.2.6 Spectral normalization of thermal histories

Let $\mathcal{G}$ be an already constructed finite graph of admissible extensions, with adjacency matrix A . Each edge e preserves its route coordinates and receives a dimensionless action increment $a_*(e) > 0$ from the corresponding reader. With energy section ϵ_* chosen, put $t = \beta\epsilon_*$, where $\beta = (k_B\Theta)^{-1}$. The thermal operator is

$$(L_t f)(y) = \sum_{e: x \rightarrow y} \exp[-ta_*(e)]f(x). \quad (22.12)$$

All its exponents are dimensionless. The condition $\rho(L_t) = 1$ normalizes the weighted growth of histories.

Proposition 22.6 (Existence and uniqueness of normalized pressure). *Suppose that A is irreducible, $\rho(A) > 1$, and $0 < m \leq a_*(e) \leq M$ for all edges. There is a unique $t_* > 0$ such that $\rho(L_{t_*}) = 1$, and*

$$\frac{\log \rho(A)}{M} \leq t_* \leq \frac{\log \rho(A)}{m}. \quad (22.13)$$

Proof. Entrywise comparison gives $e^{-tM}A \leq L_t \leq e^{-tm}A$ for $t \geq 0$. Monotonicity of the spectral radius of nonnegative matrices implies the corresponding bounds for $\rho(L_t)$. The spectral radius is continuous, equals $\rho(A) > 1$ at zero, and tends to zero at infinity. For $s > 0$, moreover, $L_{t+s} \leq e^{-sm}L_t$, so $\rho(L_{t+s}) \leq e^{-sm}\rho(L_t) < \rho(L_t)$. Continuity and strict decrease prove existence and uniqueness. Substituting the value one into the initial bounds proves (22.13). $\square$

The local realization of three equally weighted alternatives, represented by a vertex with three loops and $a_*(e) = 1$, satisfies $\rho(L_t) = 3e^{-t}$. Its normalization gives $t_* = \log 3$. With $\epsilon_* = \epsilon_a$, one obtains $\beta_{\text{clk}} = \log 3/\epsilon_a$, exactly relation (22.8). This computation identifies the local tritic confluence. For a history graph with additional constraints, confluence with the same clock requires checking

$$\rho(L_{(\log 3/\epsilon_a)\epsilon_*}) = 1$$

with its effective increments. Local normalization and the pressure of the complete system thus preserve their respective domains.

The relation obtained brings together structure, value, and meaning. The record determines recoverable information; probabilities on continuations determine its entropic reading; action and period determine ϵ_a ; the energy–temperature transducer expresses dimensional entropy. Area and entanglement are related through the reduced state and its boundary operators, keeping visible which datum produces each quantity.

23 Spectral capacity, energy-bearing memory and thermal transport

The following construction combines extension capacity, renewal of the two sheets and their spectral realizations. Its purpose is to determine what a thermal reading preserves when phase returns while the record continues to grow. Its operations originate in the enriched APP-TRIT-TPK state: the signature catalogue, calendar, orientation and memory precede the traces to be evaluated. The linear spaces, compressions and preparations added to those antecedents are defined explicitly. Each result thus retains its construction conditions and its role in the argument.

23.1 Capacity, vacancy and joint renewal

The comparison between the 729 six-trit words and the 1000 positions of a decimal block defines, with the origin fixed in 5.9.1,

$$C_t = \lfloor (t+1)\rho \rfloor, \quad \rho = \log_{1000} 729 = \log_{10} 9, \quad z_t = 1 - (C_t - C_{t-1}). \quad (23.1)$$

Since $0 < \rho < 1$, one has $z_t \in \{0, 1\}$. Capacity counts blocks in this comparison; its exact evaluation uses $10^{C_t} \leq 9^{t+1} < 10^{C_{t+1}}$. For $1 \leq t \leq 20$, $C_t = t$, whereas $C_{21} = 20$. Indeed, for $t \leq 20$ the lower inequality is equivalent to $9(9/10)^t \geq 1$, and checking $t = 20$ suffices; at $t = 21$ it reverses. The remaining comparisons place 9^{22} between 10^{20} and 10^{21} . The first positive vacancy is therefore $p_1 = 21$, with

$$C_{20} = C_{21} = 20, \quad C_{22} = 21, \quad C_{23} = 22, \quad C_{24} = 23.$$

The twenty capacity increments provide twenty effective labels. The cut index 24, the 25 positions from 0 to 24, and capacity 23 are records of different types.

The joint emission catalogue is $\Omega = \mathcal{S}_+ \times \mathcal{S}_\times$, with cardinalities 18 and 26. On $\ell^2(\Omega)$ with uniform counting measure, renewal of the active coordinate is represented by

$$E_+ = \frac{\mathbf{J}_{18}}{18} \otimes I_{26}, \quad E_\times = I_{18} \otimes \frac{\mathbf{J}_{26}}{26}.$$

These are commuting orthogonal projections. Their product is the constant projection Π , all of whose matrix entries equal $1/468$. The reduced phase reading uses the departure regime:

$$K_{\text{ph}}((u, r), (v, r+1)) = P_{\tau(r)}(u, v), \quad (P_+, P_-, P_0) = (E_+, E_\times, I), \quad \tau(r) = M_{\text{ph}}(1+r).$$

This operation follows the enriched dynamics and its holonomy, in the order $U_t \longrightarrow \Gamma_9 \longrightarrow K_{\text{ph}}$.

Proposition 23.1 (Two-sheet renewal on the capacity clock). *Evaluate the preceding reader through $\mathcal{R}_t = K_{\text{ph}}^{a_t}$, where $a_t = C_{t+1} - C_t$. Starting at the first vacancy, the earliest instant at which its matrix output is independent of the initial emission is 24. The successive ranks are 468, 26, 1; in the conjugate orientation they are 468, 18, 1.*

Proof. The departure capacities 20, 21, 22 correspond to regimes 0, +1, -1. The transitions $21 \rightarrow 22 \rightarrow 23 \rightarrow 24$ therefore apply $I, E_+, E_\times$, with accumulated products I, E_+, Π . Tensor-product ranks multiply: rank $E_+ = 26$, rank $E_\times = 18$, and rank $\Pi = 1$. The rows of I and E_+ depend on the initial emission; the rows of Π coincide. This establishes minimality. The conjugate sequence applies $I, E_\times, E_+$ and has the same final product. Composition through capacity preserves chronological time:

$$(t, C_t, v_t) \mapsto (t+1, C_t + a_t, v_t + 1 - a_t), \quad v_t = t - C_t.$$

The identity $t = C_t + v_t$ is maintained also when $a_t = 0$. □

Reduction from phase modulo nine to regime modulo three is exact for this reader. With $Q(u, r) = (u, [r]_3)$,

$$\sum_{s: [s]_3 = b} K_{\text{ph}}((u, r), (v, s)) = \mathbf{1}_{b=[r+1]_3} P_{\tau(r)}(u, v).$$

The sum depends only on $Q(u, r)$, proving strong lumpability. The lift, however, retains $C = 9m + 3b + a$. The transition $20 \rightarrow 23$ sends $(m, b, a) = (2, 0, 2)$ to $(2, 1, 2)$: a returns and b advances. The positive phase modulo nine changes from 3 to 6. Uniform renewal is a rank-one matrix output; the earlier microstate and path remain in the enriched archive rather than in that matrix output alone. The information of a uniform regime is $\log 3$; that of renewed emissions is $\log 468$; the entropy rate of a deterministic rotation is zero. These readings refer to different ensembles and operations.

23.2 Capacity returns and cylinder valuation

Define strict regime return by

$$R_3(s) = \min\{t > s : C_t > C_s, C_t \equiv C_s \pmod{3}\}.$$

The sequence C_t is unbounded and has increments zero or one. It therefore reaches $C_s + 3$ before any larger positive value congruent to C_s , so $C_{R_3(s)} = C_s + 3$. Starting at $s_0 = 24$, the corrected localization is

$$s_j = \min\{t \geq 24 : C_t = 23 + 3j\}, \quad C_{s_j} = 23 + 3j. \quad (23.2)$$

The first times are 24, 27, 30. Later intervals may contain vacancies; the capacity increase nevertheless remains exactly three.

To explain the valuation, write each block as $d_r = 100a_{r,1} + 10a_{r,2} + a_{r,3}$, with $0 \leq a_{r,k} \leq 9$. A prefix of c blocks is transported to the three integers $X_k = \sum_{r=1}^c a_{r,k} 10^{c-r}$. Euclidean division gives a bijection; its inverse regroups the digits at each position. Truncating one block truncates one digit in each coordinate. Addition with carry is transported through this complete bijection, retaining its quotients.

The scalar cylinder of length 1000^{-c} corresponds to the half-open cube

$$\prod_{k=1}^3 [X_k 10^{-c}, (X_k + 1) 10^{-c}), \quad J_c = 10^{-c} I_3, \quad \det J_c = 1000^{-c}.$$

The equality expresses a correspondence of cylinder measures. Between two capacities, $J_{c',c} = 10^{-(c'-c)} I_3$ satisfies $J_{c'',c'} J_{c',c} = J_{c'',c}$. The positive isotropic similarity is determined by its determinant, because $s^3 = 1000^{-(c'-c)}$ has one positive root. At the marked returns,

$$J_{C_{s_j}} = 10^{-23} 1000^{-j} I_3. \quad (23.3)$$

The factor 10^{-23} is the scale of one cube coordinate; the scalar cylinder measure at that capacity is 10^{-69} . This distinction preserves the geometric degree of the reader. An extractor fixed at s_0 , followed by compatible truncation, satisfies $E_n \pi_{m,n} = E_m$ for $m \geq n \geq s_0$. Extension thus preserves the marked record without recomputing its ranks from a later cut.

23.3 Relation carriers and their interchange realization

Let U_m be the space of capacity labels and P_p the space of prefix positions. The diagonal expectation on $\text{End}(U_m)$ is $E_{\text{diag}}(X) = \sum_i E_{ii} X E_{ii}$. Its orthogonal complement has basis E_{ij} , $i \neq j$, and dimension $m(m-1)$. The second record space is $\text{End}(P_p) \oplus A_{p-1}$, where A_{p-1} retains adjacent transitions as typed observations. Its dimension is $p^2 + p - 1$. For $m = 20$, $p = 25$, the ranks are 380 and 649. This choice of spaces and cut defines a realization of the reader; the identities below prove its operations without attributing physical selection of that domain to the count alone.

Based reversal acts by $E_{ij} \mapsto E_{m-1-i, m-1-j}$ and, on the second space, by $E_{ij} \mapsto E_{p-1-i, p-1-j}$ and $a_j \mapsto a_{p-2-j}$. For $m = 20$ there are 190 two-element orbits. For $p = 25$ there is exactly one fixed matrix element, $E_{12,12}$; the remaining elements form 312 matrix orbits and 12 transition orbits. Thus there is one fixed line and 324 regular representations of C_2 , with eigenspace dimensions 325 and 324. This is the antecedent of $649 = 1 + 2 \cdot 324$; it preserves record reversal rather than specifying an energy grading.

A second realization uses the same interchange J of two orientations and retains mixed tensor terms. Let W, V have dimensions m, n . In the exterior sector of $W_+ \oplus W_-$ and the symmetric oscillators of $V_+ \oplus V_-$, the fixed characters are

$$\chi_F(q) = \frac{1}{2}((1+q)^{2m} + (1-q^2)^m), \quad (23.4)$$

$$\chi_B(q) = \frac{1}{2}(P_n(q)^2 + P_n(q^2)), \quad P_n(q) = \prod_{r \geq 1} (1 - q^r)^{-n}. \quad (23.5)$$

Proof. On one particle, the eigenspaces $J = \pm 1$ each have dimension m . The inserted trace on the exterior algebra is $(1+q)^m(1-q)^m = (1-q^2)^m$. The projection $(I+J)/2$ averages this trace with the ordinary character. Its degree-two coefficient is $((\binom{2m}{2} - m)/2 = m(m-1)$. At each bosonic frequency, both signs have multiplicity n , giving inserted trace $\prod_r [(1-q^r)(1+q^r)]^{-n} = P_n(q^2)$. The ordinary coefficients in degrees zero, one and two are $1, 2n, 2n^2 + 3n$, whereas the inserted coefficients are $1, 0, n$. Their averages are $1, n, n(n+2)$. Every formal coefficient depends on finitely many factors, so no further analytic convergence assumption is required. $\square$

The additional n in energy two comes from frequency-two creation operators. Indeed, $M(V)_2 = V(-2) \oplus \text{Sym}^2(V(-1))$; at rank 24 its dimension is $24 + 300 = 324$. For the two interchanged orientations, the bosonic fixed space accumulated through energy two has dimension $1 + n + n(n+2) = (n+1)^2 + n$, hence 649 for $n = 24$. Linear identification with the earlier matrix-and-transition space requires transport of its bases; equal dimensions do not automatically identify Hamiltonians. The homogeneous exterior cut and accumulated bosonic cut remain declared components of this reader.

23.4 Multiplicative seeds and exact fibre compensation

The multiplicative seed support is $X_\times = (\mathbb{Z}/9\mathbb{Z})^2 \times \{N, E, S, O\}$, with cardinality 324. The six-window emitter $e : X_\times \rightarrow \mathcal{S}_\times$ of the formal kernel produces 26 signatures. Its fibres have size 8 for 24 signatures, 24 for 555555, and 108 for 000000, summing to 324. The temporal set $I = \{1, \dots, 24\}$ retains a type distinct from the fibre of 555555, despite their equal cardinalities.

Construct the microscopic space

$$\tilde{T} = \{\Omega\} \sqcup \{a_i : i \in I\} \sqcup \{b_{i,y} : i \in I, y \in X_\times\}, \quad |\tilde{T}| = 7801.$$

The reading $p(b_{i,y}) = b_{i,e(y)}$, fixing the other records, has image T of size $1 + 24 + 24 \cdot 26 = 649$. Put $m_e = |e^{-1}(e)|$. Assign mass 1 to Ω, a_i and $1/m_{e(y)}$ to $b_{i,y}$. This defines a measure ν whose fibres under p all have mass one. For $r > 0$, the rates are r in both directions between Ω and a_i , and $r/m_{e(y)}$ from a_i to $b_{i,y}$, with reverse rate r .

Proposition 23.2 (Compensated transport of the signature tree). *The microscopic generator is reversible for ν . The map p induces the symmetric tree on T , and the lift $J_p g = g \circ p$ is an isometry intertwining the two generators.*

Proof. Each terminal edge satisfies $\nu(a_i)r/m_{e(y)} = \nu(b_{i,y})r$. Moreover, $\sum_{y:e(y)=e} r/m_e = r$, while the reverse rate has the same value from every representative. This proves strong lumpability. The identity $\sum_{x:p(x)=z} \nu(x) = 1$ proves $J_p^* J_p = I$; the rate sums prove $\tilde{L} J_p = J_p L_T$. The tree is connected and has 648 edges, hence its normalized stationary measure is uniform. $\square$

The cursor half-turn induces a signature involution preserving m_e . Together with $i \mapsto 25 - i$, it preserves ν and the rates; the intertwiner also transports this reversal. Seeds remain recorded in $\tilde{T}$ when T evaluates only their signatures. Selection of a mark and compensated signature renewal are the operations of this reader. Their domain and rates remain explicit: cursor paths and the full enriched update retain additional operations. The tree connects the internal bosonic levels; communication between all three grades is constructed in 23.9.

23.5 Double grading and recovery from its marginals

The carrier selected for comparison is

$$\mathcal{D} = \mathbb{C}\Omega_{\text{ext}} \oplus F_2^J \oplus B_{\leq 2}^J.$$

The operator L retains Fock degree; N assigns external grades 0, 1, 2 to the three summands. For $m = 20, n = 24$, their joint decomposition is

$$(N, L; d) = (0, 0; 1), (1, 2; 380), (2, 0; 1), (2, 1; 24), (2, 2; 624). \quad (23.6)$$

The two vacua are orthogonal and retain their provenance. The bivariate function and its specializations are

$$\mathcal{Z}(u, v) = 1 + 380uv^2 + u^2(1 + 24v + 624v^2), \quad (23.7)$$

$$Z_N(q) = \mathcal{Z}(q, 1) = 1 + 380q + 649q^2, \quad (23.8)$$

$$Z_L(q) = \mathcal{Z}(1, q) = 2 + 24q + 1004q^2. \quad (23.9)$$

Each monomial follows by applying $u^N v^L$ on a cell of (23.6) and taking its trace. At $q = 1/1000$ the values are respectively 1.380649 and 2.025004. The latter contains two vacua, 24 energy-one states and 1004 energy-two states.

The complete marginals n_i, l_j of a distribution on these five cells recover the cell masses:

$$p_{00} = n_0, \quad p_{12} = n_1, \quad p_{21} = l_1, \quad p_{20} = l_0 - n_0, \quad p_{22} = l_2 - n_1.$$

Summation along rows and columns proves recovery when these differences are nonnegative and the totals agree. The support is a bipartite tree; recovery concerns its five masses, while coherences and internal indices remain in the full state. Two isolated partition values do not replace those marginals. As a control, $Z_N - Z_L = (q - 1)(1 - 355q)$: at $q = 1/355$ the traces coincide, but the internal-vacuum probabilities are $1/261574$ and $126025/261574$. Equal scalars coexist with distinct states.

The based reversal of the prefix reader also determines the orientation information omitted by its quotient. There are 190 pairs in degree one and 324 in degree two; each equiprobable pair contributes $\log 2$ times its mass. Hence

$$I_{\text{or}}(q) = \frac{380q + 648q^2}{Z_N(q)} \log 2.$$

Restoring the orientation label recovers that information. This quotient and its entropy reading remain distinct from a physical erasure operation.

23.6 Positive energy dilation of memory

Since N, L commute, $Q_c = N - L$ takes values 0, $-1, 2, 1, 0$ on the preceding cells. Let $J_m = \text{diag}(0, 1, 2, 3)$ and $g > 0$. Define the isometry

$$V|n, l, i\rangle = |n, l, i\rangle \otimes |n - l + 1\rangle, \quad H_F = gL, \quad H_m = gJ_m. \quad (23.10)$$

The auxiliary levels are 1, 0, 3, 2, 1. The first coordinate retains the complete basis, so $V^*V = I$. On each vector,

$$(H_F \otimes I + I \otimes H_m)V = Vg(N + I), \quad (23.11)$$

because $l + (n - l + 1) = n + 1$. Auxiliary energy is nonnegative and total energies are $g, 2g, 3g, 3g, 3g$. The image is reducing for this diagonal Hamiltonian. For $q = e^{-\beta g}$, functional calculus gives

$$V^* e^{-\beta H_{\text{tot}}} V = q^{N+I}, \quad \frac{V^* e^{-\beta H_{\text{tot}}} V}{\text{Tr}(V^* e^{-\beta H_{\text{tot}}} V)} = \frac{q^N}{Z_N(q)}.$$

The filter $M_q = q^{(Q_c+I)/2}$ satisfies $V^*(I \otimes q^{J_m})V = M_q^* M_q$ and $q^L M_q^* M_q = q^{N+I}$. For $0 < q < 1$, it is contractive and conditionally transforms $\rho_L = q^L/Z_L$ into $\rho_N = q^N/Z_N$ with probability qZ_N/Z_L .

Recording its outcome also retains the complementary outcome. The isometry V preserves coherences; partial trace over memory is a different operation, removing precisely the coherences between different values of Q_c . The shift $+1$ is the least integer making every auxiliary level nonnegative. A larger common shift preserves the normalized state and changes the recorded energy origin.

23.7 Memory loop and decimal spectrum

The Gaussian realization receives the collective $8 : 1$ decomposition of nine copies and the unit memory loop. Prepare both modes in the centred pure Gaussian reference of joint covariance $I \oplus I$, in the same symplectic coordinates used for the following matrices:

$$Q_8 = \frac{1}{3} \begin{pmatrix} \sqrt{8}I & -I \\ I & \sqrt{8}I \end{pmatrix}, \quad H_1 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad U_{H_1} = Q_8^T \text{diag}(I, H_1) Q_8.$$

These matrices inherit their preparation and loop from system–memory transport; the thermal calculation follows. The visible block of $U_{H_1} U_{H_1}^T$ is

$$W = \frac{8I + H_1 H_1^T}{9} = \frac{1}{9} \begin{pmatrix} 13 & 3 \\ 3 & 10 \end{pmatrix}, \quad \det W = \frac{121}{81}, \quad \nu = \frac{11}{9}.$$

In dimension two, $S = \sqrt{\nu} W^{-1/2}$ has determinant one and is symplectic, with $SW S^T = \nu I$. Its metaplectic realization reduces the Gaussian state to that isotropic covariance. The geometric state $(1-r) \sum_{j \geq 0} r^j |j\rangle\langle j|$ has mean occupation $r/(1-r)$ and covariance $(1+r)/(1-r)I$. Uniqueness of a centred Gaussian state with given covariance implies

$$r = \frac{\nu - 1}{\nu + 1} = \frac{1}{10}, \quad \rho_r = (1-r) \sum_{j \geq 0} r^j |j\rangle\langle j|. \quad (23.12)$$

The decimal ratio follows from the stated loop and preparation. For example, $H_0 = I$ would give $\nu = 1$ and $r = 0$, displaying the result's effective dependence on the memory operator.

Euclidean division $j = 3w + b$, $0 \leq b < 3$, defines a unitary $U_3 |j\rangle = |w, b\rangle$, with inverse $|w, b\rangle \mapsto |3w + b\rangle$. The factorization $1 - r^3 = (1-r)(1+r+r^2)$ proves, term by term,

$$U_3 \rho_r U_3^* = \rho_q \otimes \sigma_r, \quad q = r^3 = \frac{1}{1000}, \quad \sigma_r = \frac{\text{diag}(100, 10, 1)}{111}. \quad (23.13)$$

The unilateral shift is transported to

$$T = I \otimes (|1\rangle\langle 0| + |2\rangle\langle 1|) + S_w \otimes |0\rangle\langle 2|, \quad T^3 = S_w \otimes I_3.$$

Applying this formula to each basis vector proves the identity and exhibits the carry from remainder to quotient. One has $T^* T = I$ and $T T^* = I - |0, 0\rangle\langle 0, 0|$; remainder return retains the occupation advance. Acting as $U_3 \otimes I$, the unitary also transports every purification of the enlarged system.

For the energy section $H_W = \epsilon_a(j + 1/2)$, transport gives

$$U_3 H_W U_3^* = \epsilon_a(3N_w + B + \frac{1}{2}I), \quad B = \text{diag}(0, 1, 2).$$

The diagonal identity preserves the self-adjoint domain defined by square summability of energy. At $\beta_W \epsilon_a = \log 10$, the quotient has spacing $3\epsilon_a$ and ratio q , with the same thermometer. Its means satisfy $\langle N_w \rangle = 1/999$, $\langle B \rangle = 4/37$ and $3\langle N_w \rangle + \langle B \rangle = 1/9$. Moreover, $s(\rho_r) = s(\rho_q) + s(\sigma_r)$ by exact factorization.

This fixes $g = 3\epsilon_a$ when composing (23.10) with the Williamson quotient. Preparing $\rho_L \otimes \rho_q$ and recording projection onto the image of V gives

$$V^*(\rho_L \otimes \rho_q)V = \frac{1-q}{Z_L} q^{N+I}.$$

Its probability is $(1-q)qZ_N/Z_L$, and its conditioned state is ρ_N . The factor σ_r is retained through $V \otimes I_3$; the transported total energy is $3\epsilon_a(N+1) + \epsilon_a(B+1/2)$.

23.8 Marked spectral section and complete intensity

Composition with capacity uses the declared map $C_{s_n} \mapsto j = C_{s_n}$. By (23.2), the levels are $23 + 3n = 3(7 + n) + 2$. This section requires both remainder 2 and quotient at least 7; remainder alone would also admit 2, 5, 8, Its projection and probability are

$$P_{23} = \sum_{n \geq 0} |23 + 3n\rangle\langle 23 + 3n|, \quad p_{23} = \frac{(1-r)r^{23}}{1-r^3}.$$

Conditioning and sending $|23 + 3n\rangle$ to $|n\rangle$ produces ρ_q . The weight p_{23} records the threshold that disappears from the normalized density.

To incorporate multiplicities $d = (1, 380, 649)$, define $|n, a\rangle = |23 + 3n\rangle \otimes |a\rangle$ in a memory of dimension 649, with $1 \leq a \leq d_n$. This is an isometry. Compressing $\rho_r \otimes I_{649}/649$ to its image gives mass

$$p_{\mathcal{D}} = \frac{1-r}{649} r^{23} Z_N(q)$$

and conditioned state $\rho_N(q)$. If $\lambda_j = (1-r)r^j$, the intensity relative to the reference population λ_0 is

$$B_* := \frac{\lambda_{23} + 380\lambda_{26} + 649\lambda_{29}}{\lambda_0} = 10^{-23} \left(1 + \frac{380}{1000} + \frac{649}{1000^2} \right) = 1.380649 \times 10^{-23}. \quad (23.14)$$

This formula retains spectrum, threshold, multiplicities and reference. The historical formula was known during construction of this reader; the operator development specifies its realization and conditions rather than retrospectively attributing an independent prediction to it. Interpretation as a map between energy and temperature lines requires composition of the dimensional readings, treated in the following development.

23.9 Boundary currents, compression and offset rigidity

The area incidence developed in 19.1 provides $18 + 18$ tritic links. Put $u = N_{90}$, $v = N_{120}$, $a = u + v$, and

$$F = 3u + 4v, \quad H_R = \epsilon_0 F, \quad \epsilon_0 = 30\Delta\tau_R > 0, \quad \tau_R = \frac{\hbar c}{2\pi R}.$$

Physical area is $\delta a a$, with δa fixed by incidence and Barbero. Provenance remains in u, v : equal areas may have different energies. The currents below use this existing structure while retaining its Hamiltonian.

On one link, $T^+ = |1\rangle\langle 0| + |2\rangle\langle 1|$ and $T^- = (T^+)^*$ satisfy $[n, T^\pm] = \pm T^\pm$. Define

$$J_{ef} = T_{120,f}^+ T_{90,e}^-, \quad B_c = \left(\sum_e T_{90,e}^+ \right)^3 \left(\sum_f T_{120,f}^- \right)^2.$$

The rule $[F, AB] = [F, A]B + A[F, B]$ proves

$$[F, J_{ef}] = J_{ef}, \quad [a, J_{ef}] = 0, \quad [F, B_c] = B_c, \quad [a, B_c] = B_c.$$

The occupation sequence $(0, 2) \rightarrow (3, 0) \rightarrow (2, 1)$ has primitive energies 8, 9, 10 and areas 2, 3, 3. The same energy increment retains two distinguishable area actions.

The coefficients $c_j = [z^j](1 + z + z^2)^{18}$ count occupations in each channel: $c_0 = 1, c_1 = 18, c_2 = 171, c_3 = 1122$. The boundary character

$$P(x) = (1 + x^3 + x^6)^{18} (1 + x^4 + x^8)^{18}$$

gives multiplicities $g_7 = 324, g_8 = 171, g_9 = 1122, g_{10} = 3078$. For ranks $d = (1, 380, 649)$, the first admissible triple of consecutive energies starts at 8. Earlier starts are excluded using $g_0 = 1, g_1 = g_2 = g_5 = 0, g_3 = g_4 = 18, g_6 = 171, g_7 = 324$ and $g_8 = 171$: each earlier triple fails at least one required rank.

A connected compression is constructed by selecting one $(0, 2)$ state, 380 states $(3, 0)$ and 649 distinct neighbours under the J_{ef} . There are sufficiently many neighbours by a counting proof: every occupation of sum three has at least two predecessors of sum two, and each predecessor has at most 18 successors. The 380 states therefore yield at least $\lceil 760/18 \rceil = 43$ distinct predecessors, and $18 \cdot 43 = 774$ neighbours $(2, 1)$, from which 649 may be selected. The current B_c connects the initial state to all 380 intermediate states, and every final neighbour has a selected predecessor. The graph is connected. Its labels and subspace selection are retained. The based isometry ι_N satisfies

$$H_R \iota_N = \iota_N \epsilon_0 (8I + N). \quad (23.15)$$

Evaluation on each basis vector proves this identity.

Let $P_F = \mathbf{1}_{N=1}$ and let $B_N : \mathcal{D}_0 \rightarrow \mathcal{D}_1$, $J_N : \mathcal{D}_1 \rightarrow \mathcal{D}_2$ be the nonzero compressed currents. For $H_\delta = \epsilon_0(8I + N) + \delta P_F$,

$$[H_\delta, B_N] = (\epsilon_0 + \delta)B_N, \quad [H_\delta, J_N] = (\epsilon_0 - \delta)J_N.$$

Preserving both increments, or requiring their equality, forces $\delta = 0$. More generally, a diagonal correction preserves each increment exactly when its values agree at the two ends of every edge. Connectivity makes it constant. This rigidity classifies diagonal corrections; its proof does not extend that classification to all non-diagonal operators.

For $x = e^{-\epsilon_0/\tau}$ and $\tau > 0$, Gibbs compression gives $\iota_N^* e^{-H_R/\tau} \iota_N = x^8 x^N$. Its selection probability from the boundary is $x^8 Z_N(x)/P(x)$. Recording selection and complement through $P_N \otimes |s\rangle + (I - P_N) \otimes |f\rangle$ is isometric and preserves energy for degenerate flags. On each upward edge, rates $x r_e$ and reverse rate $r_e > 0$ satisfy detailed balance for weights x^N . Connectivity proves uniqueness of the stationary state; grade masses are $d_n x^n / Z_N(x)$. This specifies a thermal preparation with a bath rather than an isolated unitary evolution.

23.10 Thermal equivalence with transported reference and scale

Theorem 23.3 (Transport of the centred thermal exponent). *Let H_1, H_2 be finite self-adjoint operators, $\tau_i > 0$, and T an isometry whose image reduces H_2 . Traces in the second space are taken on that image. Let σ_{ref} be a normalized state on the first space, and define the transported reference energies by*

$$e_1 = \text{Tr}(\sigma_{\text{ref}} H_1), \quad e_2 = \text{Tr}(T \sigma_{\text{ref}} T^* H_2).$$

If

$$\frac{H_2 - e_2 I}{\tau_2} T = T \frac{H_1 - e_1 I}{\tau_1},$$

then the centred partition function, Gibbs state, entropy and probabilities of all transported observables, including the marker, are preserved. Conversely, for faithful states, equality of the transported states implies this identity with the references defined above.

Proof. Functional calculus intertwines exponentials of both exponents. Trace invariance under an isometry onto its image gives equality of centred partitions; normalization yields $\rho_2 = T \rho_1 T^*$. Trace cyclicity transports observables, and spectral calculus transports entropy. Conversely, logarithms of faithful states give equality of H_i/τ_i up to a scalar, the difference of their log partition functions. Taking its expectation in σ_{ref} identifies that scalar with $e_2/\tau_2 - e_1/\tau_1$; subtracting it gives the stated centred identity. $\square$

For the capacity reader, $C = 23I + 3N$ and

$$H_{\text{cap}} = \frac{\epsilon_a \log 10}{\log 3} C, \quad \tau_{\text{clk}} = \frac{\epsilon_a}{\log 3}, \quad H_{W,\text{rel}} = \epsilon_a C, \quad \tau_W = \frac{\epsilon_a}{\log 10}. \quad (23.16)$$

Both exponents equal $C \log 10$ although energies and temperatures have different scales. On the boundary compression, preparation at $\tau_* = \epsilon_0 / \log 1000$ satisfies

$$\frac{H_R - 8\epsilon_0 I}{\tau_*} \iota_N = \iota_N \frac{H_{\text{cap}} - 23\tau_{\text{clk}} \log 10 I}{\tau_{\text{clk}}} = \iota_N N \log 1000.$$

The native horizon activity remains $e^{-30\Delta}$; this preparation has ratio $\tau_*/\tau_R = 30\Delta / \log 1000$. Varying R changes ϵ_0 and τ_R together, preserving their quotient. Before centring, boundary and capacity exponents differ by $\log 10 I$, because $8 \log 1000 - 23 \log 10 = \log 10$. Their traces are $10^{-24} Z_N$ and $10^{-23} Z_N$. The factor $1/10$ also occurs in their references and must be retained when comparing absolute intensities.

If ι_1, ι_2 realize the same ranks, grading and marker, $U = \iota_2 \iota_1^*$ is unitary between their images and intertwines Hamiltonian and marker. The preceding theorem implies the same reader, all its spectral moments and its thermal derivatives for any admissible basis choice. A unique microscopic representative is unnecessary for that conclusion. Comparing area, currents or relaxation times also requires transport of those additional observables.

Finally, for $P_0 = |\Omega_{\text{ext}}\rangle\langle\Omega_{\text{ext}}|$ and $\rho_N = q^N / Z_N$, one has

$$p_0 = \text{Tr}(P_0 \rho_N) = Z_N^{-1}, \quad Z_N = 1 + \frac{\text{Var}_{\rho_N}(P_0)}{p_0^2}.$$

The second identity uses $P_0^2 = P_0$, hence variance $p_0(1 - p_0)$. Normalized state and marker recover the relative trace; the recorded threshold recovers the factor in (23.14). The result thereby preserves the structure, preparation, reference and memory required for composition with the action–temperature reading.

23.11 Thermal response of the marker and preparation cost

The marker distinguishes the reference state within the complete carrier. Its probability, its response to a perturbation and the cost of preparing that sector are three calculable readings of the same distribution, with memory retained. They recover the thermal meaning of the normalizer in addition to its numerical evaluation.

Consider the carrier with degrees and multiplicities

$$(N, L; d) = (0, 0; 1), (1, 2; 380), (2, 0; 1), (2, 1; 24), (2, 2; 624).$$

For an energy gap $a > 0$, let $H = aL$, $Q = N - L$ and $\mu = -a$. The Gibbs generator is therefore

$$K = H - \mu Q = aN, \quad q = e^{-\beta a}, \quad Z_N = \text{Tr}(q^N) = 1 + 380q + 649q^2, \quad \rho_N = \frac{q^N}{Z_N}.$$

Here $\beta = (k_B T)^{-1}$, and P projects onto the marked reference of degree $N = 0$. This reference is distinct from the state of degree $L = 0$ in the sector $N = 2$. Its probability is $p = \text{Tr}(\rho_N P) = Z_N^{-1}$.

Proposition 23.4 (Susceptibility of the marked sector). *For the perturbed generator $K_u = K - uP$, with the carrier fixed, the response $\chi_P = \partial_u \text{Tr}(\rho_u P)|_{u=0}$ satisfies*

$$\chi_P = \beta p(1 - p), \quad Z_N = 1 + \frac{\chi_P}{\beta p^2}.$$

Proof. Since P commutes with K , the perturbation multiplies the marked-sector weight by $e^{\beta u}$ and leaves the other weights unchanged. Thus

$$p(u) = \frac{p e^{\beta u}}{1 - p + p e^{\beta u}}.$$

Differentiation at $u = 0$ gives $\chi_P = \beta p(1 - p)$. Substitution of $p = Z_N^{-1}$ proves the second identity. Equivalently, $P^2 = P$ yields $\chi_P = \beta \text{Var}_{\rho_N}(P)$. $\square$

Proposition 23.5 (Relative entropy of preparation). *Let $\rho_P = P \rho_N P / p$. With $s(\sigma) = -\text{Tr}(\sigma \ln \sigma)$ and $\mathcal{G}(\sigma) = \text{Tr}(K \sigma) - \beta^{-1} s(\sigma)$,*

$$D(\rho_P \| \rho_N) = \ln Z_N, \quad \mathcal{G}(\rho_P) - \mathcal{G}(\rho_N) = \beta^{-1} \ln Z_N.$$

Proof. On the support of P , the relation $\rho_P = \rho_N / p$ gives $\ln \rho_P - \ln \rho_N = -\ln p$. Its trace against ρ_P is $-\ln p = \ln Z_N$. Moreover, $\ln \rho_N = -\beta K - \ln Z_N$ implies, for any state σ ,

$$D(\sigma \| \rho_N) = \beta [\mathcal{G}(\sigma) - \mathcal{G}(\rho_N)].$$

Taking $\sigma = \rho_P$ proves the energy identity. $\square$

The potential difference expresses reversible preparation work when the corresponding control and thermal and charge reservoirs are available. The identity describes that preparation; it does not assign a fixed cost to every observation. For an independent memory η , replacing (ρ_N, P) by $(\rho_N \otimes \eta, P \otimes I)$ preserves p , the susceptibility and the relative entropy; the state η remains intact.

23.12 Entropy and heat capacity of the graded carrier

In the spectral realization $H_{\text{rel}} = 3\epsilon N$, the condition $3\beta\epsilon = \ln 1000$ fixes $q = 10^{-3}$. The normalizer and the first two degree moments are

$$Z_N = 1.380649, \quad \langle N \rangle = \frac{381298}{1380649}, \quad \text{Var}(N) = \frac{382842620000}{1906191661201}.$$

Indeed, the probabilities of degrees 0, 1, 2 are $1/Z_N$, $380q/Z_N$ and $649q^2/Z_N$. Hence $\langle N \rangle = (380q + 1298q^2)/Z_N$ and $\langle N^2 \rangle = (380q + 2596q^2)/Z_N$; subtracting the squared mean gives the stated variance.

Holding the Hamiltonian and the carrier fixed while varying temperature, the entropy $S = k_B s(\rho_N)$ and heat capacity C_V are

$$\begin{aligned}\frac{S}{k_B} &= \ln Z_N + \ln 1000 \langle N \rangle = 2.230289295827403 \dots, \\ \frac{C_V}{k_B} &= (\ln 1000)^2 \text{Var}(N) = 9.58357621855691 \dots.\end{aligned}$$

The first formula follows from $\ln \rho_N = N \ln q - \ln Z_N$; the second uses $C_V = k_B \beta^2 \text{Var}(H_{\text{rel}})$. These are observables of this thermal realization. The spectral-origin shift recorded by 23 changes preparation energy and intensity relative to the reference; when held fixed, it changes neither the conditional probabilities nor the degree variance. The normalized distribution and the origin record thus retain distinct functions within the same state.

24 Composition of action, area, and marked thermal reading

The preceding construction determines a graded carrier, its energy-bearing memory, the currents between sectors, and a marked spectral reference. This section establishes how these data act on the article's energy and information readers. The argument has two demonstrative levels: composition of the previously constructed action, area, and horizon sections, and an explicit thermometric realization through two functionals of the same enlarged record. The second operation fixes a thermal coefficient within the reading rule stated and proved here.

The antecedents retain the APP–TRIT–TPK sequence: the sum and product sheets retain remainder and quotient; TRIT preserves regime and orientation; TPK transports path, boundary, carry, and memory. The nonadic connection returns the phase while prolonging the enriched state. The coordinates $\pi_{\text{HMT}}, \varphi_{\text{HMT}}, e_{\text{HMT}}, \alpha_{\text{HMT}}$, oriented action, and hexadic–octadic incidence are preceding outputs of that same joint discrete structure of the continuum. In the equations below, π, α and the other scalar readers retain this provenance.

24.1 Return energy and areal resolution

Fix an orientation a and the article's radial realization. Write

$$L_\alpha = \frac{5759}{23040} \alpha^{16} L_*, \quad T_{108} = 108 t_0 = \frac{2\pi L_\alpha}{c}, \quad \epsilon_a = \frac{h_a}{T_{108}} = \frac{\hbar_a c}{L_\alpha}. \quad (24.1)$$

The symbol L_α denotes a length; the energy degree L of the preceding carrier is a different, dimensionless operator. The previously constructed oriented functional and angular character are

$$\gamma = \frac{\pi A C^*}{180} + 12[S_{90}(q_-) - S_{90}(q_+)] - [S_{120}(q_-) - S_{120}(q_+)],$$

$$S_m(q) = \frac{q^m}{1 - q^{3m}}, \quad a_0 = \frac{1 + 2 \cos(10^\circ)}{3} \frac{C^*}{6} \frac{\pi}{180}. \quad (24.2)$$

In their positive sector, the area quantum is $q_A = \gamma a_0 L_\alpha^2$. Define energy per unit area by $\mathcal{F}_{I/A} = \epsilon_a / q_A$.

Proposition 24.1 (Areal, constitutive, and gravitational composition). *With the radial coefficient $G_a \hbar_a = c^3 L_\alpha^2$, one has*

$$\mathcal{F}_{I/A} = \frac{c^4}{\gamma a_0 G_a L_\alpha}, \quad \frac{\mathcal{F}_{I/A} q_A}{\ln 3} = \frac{h_a}{108 t_0 \ln 3}, \quad (24.3)$$

$$G_a k_B \Theta_{\text{clk},a} \ln 3 = c^4 L_\alpha. \quad (24.4)$$

If ϵ_a denotes charge in the constitutive section $\epsilon_a^2 = 2\alpha \hbar_a / Z_{\text{vac}}$, the same energy per nat is

$$\frac{\epsilon_a}{\ln 3} = \frac{Z_{\text{vac}} \epsilon_a^2}{216 \alpha t_0 \ln 3}. \quad (24.5)$$

Proof. Substituting q_A and (24.1) gives $\mathcal{F}_{I/A} = \hbar_a c / (\gamma a_0 L_\alpha^3)$. The radial identity transforms this into the first equation in (24.3); multiplication by q_A proves the second. The established thermal relation is $k_B \Theta_{\text{clk},a} \ln 3 = \epsilon_a$. Multiplying it by G_a cancels $\hbar_a$ and proves (24.4). Finally, $h_a = Z_{\text{vac}} \epsilon_a^2 / (2\alpha)$ in (24.1) yields (24.5). $\square$

Barbero determines the areal resolution entering the tension. The product of the two readings recovers return energy. This cancellation explains compatibility between the geometric, constitutive, and thermal formulations: all three transport the same action section.

24.2 Provenance information and horizon response

On the thirty-six tritic boundary factors, retain $N = N_{90} + N_{120}$ and $D = 90N_{90} + 120N_{120}$. Area is $\hat{\mathcal{A}} = q_A N$, whereas the sector state and its modular Hamiltonian are

$$\rho_A = Z_A^{-1} e^{-\Delta D}, \quad K_A = -\ln \rho_A = (\ln Z_A)I + \Delta D. \quad (24.6)$$

Here Δ is finite and Z_A is the known normalization. Incidence allows recovery of

$$N_{90} = 4\hat{\mathcal{A}}/q_A - D/30, \quad N_{120} = D/30 - 3\hat{\mathcal{A}}/q_A.$$

Recovering D from K_A additionally requires $\Delta \neq 0$. Configurations $(1, 0)$ and $(0, 1)$ have the same area q_A but modular costs 90Δ and 120Δ . Their difference comes from the sector incidence retained by the state in addition to its total occupation.

The purification in 19.9 preserves the complete state and its bipartition. In a horizon realization of radius $R > 0$, the energy scale and response operator are

$$\tau_R = \frac{\hbar_a c}{2\pi R}, \quad H_R = \tau_R \Delta D + e_R I, \quad (24.7)$$

with registered origin e_R . If $\delta E_R = \text{Tr}[(\sigma - \rho_A)H_R]$ and $\delta s = s(\sigma) - s(\rho_A)$, the origin cancels in the increment.

Theorem 24.2 (Finite balance and conversion between thermal sections). *For every state σ on the sector carrier,*

$$\delta E_R - \tau_R \delta s = \tau_R D(\sigma \| \rho_A) \geq 0. \quad (24.8)$$

If the same positive transducer B satisfies $B(T_H(R)) = \tau_R$ and $B(\Theta_{\text{clk},a}) = \epsilon_a / \ln 3$, then

$$\frac{T_H(R)}{\Theta_{\text{clk},a}} = \frac{L_\alpha}{R} \frac{\ln 3}{2\pi}. \quad (24.9)$$

Proof. Subtracting expectations in (24.6) gives $\delta s = \Delta \text{Tr}[(\sigma - \rho_A)D] - D(\sigma \| \rho_A)$. Multiplication by τ_R proves (24.8); positivity and its conditions were proved in 19.14. A positive linear map between lines preserves positive ratios. Hence the temperature ratio is $\tau_R/(\epsilon_a / \ln 3)$, evaluated by (24.1). $\square$

At $R = L_\alpha$, the conversion factor is $\ln 3/(2\pi)$. The two sections correspond to different energy scales and are compared by exact transport. Barbero's factors convert occupation into area; ΔD retains provenance cost; τ_R expresses it as energy. The finite identity additionally preserves the informational cost of the trial state relative to its reference.

24.3 Spectral reference and preserved multiplicities

The preceding bilateral loop produces the spectral ratio $r = 1/10$, and tritic grouping preserves $q = r^3 = 1/1000$ together with each index remainder. The first strict return to the capacity regime after the first vacancy fixes $C_{24} = 23$. The carrier retains its three sectors of dimensions $d_0 = 1$, $d_1 = 380$, $d_2 = 649$; their incidence, double grading, and memory constructions were established in the preceding section. Retain the declared assignment of these sectors to degrees $23 + 3n$. This assignment is an explicit datum of the joint realization, distinct from the calculation producing the three counts.

On $\mathcal{D} = \bigoplus_{n=0}^2 \mathbb{C}^{d_n}$, let

$$\eta = \bigoplus_{n=0}^2 r^{23+3n} I_{d_n}, \quad b_* := \text{Tr } \eta = 10^{-23} \left(1 + \frac{380}{1000} + \frac{649}{1000^2} \right). \quad (24.10)$$

Exact evaluation gives $b_* = 1.380649 \times 10^{-23}$ and $0 < b_* < 1$. The operator retains every carrier label; its trace aggregates their weights. Throughout the following thermodynamic variations, η remains fixed: it is the constructed reference, whereas the trial-system state varies. The operations, degrees, and normalization are thereby fixed together with the reference entering the variation.

24.4 Entropy increment of the tensor record

Let $\mathcal{H}$ be finite-dimensional and let ρ be a density matrix on it. Write $s(\rho) = -\text{Tr}(\rho \ln \rho)$ and, for any positive finite-trace operator X , set $\mathcal{H}(X) = -\text{Tr}(X \ln X)$, with $0 \ln 0 = 0$. The latter function admits references whose trace differs from one. The trial record $\eta \otimes \rho$ retains reference and system separately.

Theorem 24.3 (Marked entropy of the product record). *The exact identity is*

$$\mathcal{H}(\eta \otimes \rho) - \mathcal{H}(\eta) = b_* s(\rho). \quad (24.11)$$

Proof. Let a_i and p_j be the respective eigenvalues. The product eigenvalues are $a_i p_j$, so

$$-\sum_{i,j} a_i p_j \ln(a_i p_j) = -\sum_i a_i \ln a_i \sum_j p_j - \sum_i a_i \sum_j p_j \ln p_j.$$

The identities $\sum_j p_j = 1$ and $\sum_i a_i = b_*$ prove the formula. The convention $0 \ln 0 = 0$ includes rank-deficient states by continuity. $\square$

The difference separates the reference's informational contribution in a recorded balance. The tensor operation specifies independence of the trial from that reference; a correlated state requires its corresponding conditional information. The coefficient b_* appears through trace factorization of the constructed record.

24.5 Normalized realization and conditioned energy reading

The reference admits a normalized enlargement on $\hat{\mathcal{D}} = \mathcal{D} \oplus \mathbb{C}f$:

$$\hat{\eta} = \eta \oplus (1 - b_*)|f\rangle\langle f|, \quad \Omega_\rho = \hat{\eta} \otimes \rho. \quad (24.12)$$

Let P project onto the summand $\mathcal{D}$, extended by the identity on $\mathcal{H}$, and let $H = H^*$ be the system energy operator with registered origin. The flag f completes the trace of this auxiliary realization. The received TPK archive, including paths and memory, retains its own factor; it is transported with the record and is not replaced by f .

Proposition 24.4 (Two functionals of a normalized state). *With logarithms restricted to the support, the readings*

$$\mathcal{S}_P(\rho) = -\text{Tr}[P\Omega_\rho(I \otimes \ln \rho)] = b_* s(\rho), \quad (24.13)$$

$$\mathcal{E}_P(\rho) = \frac{\text{Tr}[P\Omega_\rho(I \otimes H)]}{\text{Tr}(P\Omega_\rho)} = \text{Tr}(\rho H) \quad (24.14)$$

retain, respectively, information relative to the marker and energy per state conditioned on it. The denominator is $\text{Tr}(P\Omega_\rho) = b_$.*

Proof. On the marked summand, $P\Omega_\rho = \eta \otimes \rho$. Trace factorization gives $b_* s(\rho)$ and $b_* \text{Tr}(\rho H)$ in the two numerators. Dividing the latter by $b_* > 0$ proves the energy reading. Both η and ρ are recovered from the enlarged state and marker: the former through its marked reduction, the latter through the trace over $\hat{\mathcal{D}}$. $\square$

The complementary block also contains ρ . Total information satisfies $s(\Omega_\rho) - s(\hat{\eta}) = s(\rho)$, and total mean energy is $\text{Tr}(\rho H)$. Choosing marked information and conditioned energy thus specifies two observations of a state retaining both blocks. Observable selection and physical deletion of information are different operations.

24.6 Transducer fixed by the joint reading rule

Work in transported positive bases of the energy, temperature, and entropy lines, with $u_S = u_E/u_\Theta$. In coordinate formulas, H expresses energy in u_E and S_P expresses entropy in u_S . The realization rule comprises four operations: receive the fixed reference (24.10); evaluate (24.13); evaluate (24.14); and define temperature as conjugate to these two functionals. The bases are transported with the operators, without a target temperature input.

Theorem 24.5 (Thermal selection within the marked realization). *Let H be nonscalar and let $\rho_\beta = e^{-\beta H}/Z(\beta)$ for $\beta > 0$. The preceding rule produces*

$$\frac{dS_P}{dE_P} = b_*\beta, \quad \Theta_P = \frac{1}{b_*\beta}, \quad B_*(\Theta_P u_\Theta) = \beta^{-1} u_E, \quad (24.15)$$

where $B_*(u_\Theta) = b_*u_E$ is a unique positive linear map.

Proof. For $E = \text{Tr}(\rho_\beta H)$, differentiation of the finite spectral sum gives $E' = -\text{Var}_{\rho_\beta}(H) < 0$ and $(\ln Z)' = -E$. The identity $s(\rho_\beta) = \beta E + \ln Z$ yields $s' = \beta E'$. The reference remains fixed, hence $S'_P = b_*s'$ and $E'_P = E'$. Their ratio and its inverse give the first two identities. The image of a positive basis determines a unique linear map; substitution of Θ_P proves the last. $\square$

The factor depends on the specific pair of readings. If marked energy and entropy are both used, (b_*E, b_*s) , or both are conditioned, (E, s) , the entropy derivative with respect to energy is β . For the pair (24.13)–(24.14), it is $b_*\beta$. All three identities follow from the same differentiation and explain which normalization determines the coefficient. Once the marked rule has been fixed, multiplying only its entropy channel by another factor changes the rule and ceases to preserve its hypotheses.

Corollary 24.6 (Marked variational principle). *For $\Theta > 0$, the functional*

$$\Phi_\Theta(\rho) = \mathcal{E}_P(\rho) - \Theta S_P(\rho) = \text{Tr}(\rho H) - b_*\Theta s(\rho)$$

has a unique minimum over density matrices:

$$\rho_\Theta^* = \frac{\exp[-H/(b_*\Theta)]}{\text{Tr} \exp[-H/(b_*\Theta)]}. \quad (24.16)$$

Proof. Substitution of $\ln \rho_\Theta^* = -H/(b_*\Theta) - \ln Z$ gives

$$\Phi_\Theta(\rho) - \Phi_\Theta(\rho_\Theta^*) = b_*\Theta D(\rho \| \rho_\Theta^*) \geq 0.$$

The reference has full rank. Equality in relative-entropy positivity occurs exactly for $\rho = \rho_\Theta^*$, proving existence and uniqueness of the minimum. $\square$

The fixed reference and pair of functionals (24.13)–(24.14) are the constitutive conditions of this realization. Under these conditions, trace b_* determines the thermometric ratio and equilibrium state; the relative-entropy identity proves existence and uniqueness of the minimum.

24.7 Composition with the chronological section and its transports

Consider the chronological capacity realization on $\mathcal{D} = \bigoplus_{n=0}^2 \mathbb{C}^{d_n}$, with $(d_0, d_1, d_2) = (1, 380, 649)$ and $N|_{\mathbb{C}^{d_n}} = nI$. Retain the degree assignment $C = 23I + 3N$ of Section 23.10. The operator in (23.16) is

$$H_{\text{clk},a} := H_{\text{cap}} = \epsilon_a \frac{\ln 10}{\ln 3} (23I + 3N), \quad \beta_{\text{clk},a} = \frac{\ln 3}{\epsilon_a}. \quad (24.17)$$

It is self-adjoint and positive on this finite-dimensional space. Its three distinct eigenvalues, proportional to 23, 26, 29, prove that it is nonscalar. Multiplication and spectral calculus give exactly

$$\begin{aligned}\beta_{\text{clk},a} H_{\text{clk},a} &= (23I + 3N) \ln 10, \\ e^{-\beta_{\text{clk},a} H_{\text{clk},a}} &= 10^{-23} q^N = \eta, \quad q = 10^{-3}, \\ \rho_{\text{clk},a} &= \frac{q^N}{Z_N(q)} = \frac{\eta}{b_*}, \quad Z_N(q) = 1 + 380q + 649q^2.\end{aligned}\quad (24.18)$$

The origin factor 10^{-23} remains in trace b_* and cancels when the state is normalized. To apply Theorem 24.5, the orientation and ϵ_a , hence H_{cap} , are fixed; differentiation is with respect to β and is then evaluated at $\beta_{\text{clk},a}$. Reference η , independent of ϵ_a , stays fixed in the first factor, while ρ_β varies on a second copy of $\mathcal{D}$ with fixed Hamiltonian (24.17). Evaluation at $\beta = \beta_{\text{clk},a}$ gives state (24.18) and energy per nat $\tau_{\text{clk},a} = \epsilon_a / \ln 3$.

The relative Williamson section retains $H_{W,\text{rel}} = \epsilon_a C$ and $\tau_W = \epsilon_a / \ln 10$. Thus $H_{\text{clk},a} / \tau_{\text{clk},a} = H_{W,\text{rel}} / \tau_W = C \ln 10$: the exponents and state coincide, whereas the scales are related by $\tau_{\text{clk},a} / \tau_W = \ln 10 / \ln 3$. The marked rule, with its two fixed functionals, produces

$$\Theta_{\text{clk},P,a} = \frac{\epsilon_a}{b_* \ln 3}, \quad B_*(\Theta_{\text{clk},P,a} u_\Theta) = \frac{h_a}{108 t_0 \ln 3} u_E. \quad (24.19)$$

Temperature is obtained here from the proved entropy derivative, rather than by imposing the numerical result as a fitting condition.

Let J_E be the effective energy transport to these coordinates and J_Θ the transport between the thermometric sections of the same reference state. In particular, J_Θ maps the preceding clock section to the section in (24.19), according to their two temperature readers. Uniqueness between lines yields the square

$$B_* J_\Theta = J_E B_{\text{clk}}. \quad (24.20)$$

Indeed, the two maps agree on the positive clock section by (24.19), and that section generates the line. The transport declares the two realizations being compared: the typed equality preserves their bases and sections, without identifying by notation alone two differently fixed temperature coordinates.

For the horizon, $\Theta_{H,P}(R) = \tau_R / b_*$. Substitution into (24.9) preserves the exact ratio of the two temperatures. Moreover, with $S_P = b_* s$, identity (24.8) becomes

$$\delta E_R - \Theta_{H,P}(R) \delta S_P = \tau_R D(\sigma \| \rho_A).$$

This composition incorporates the new reader into an existing balance while preserving incidence, action, area, and the relative-entropy term.

24.8 Energy origin, reciprocity, and the electronic section

The energy reference also enters radial applications. For a transported positive character Y , retain

$$H_a = \epsilon_a Y, \quad M_a = H_a / c^2, \quad R_a = L_\alpha Y, \quad \Lambda_a = L_\alpha Y^{-1}.$$

Thus $H_a \Lambda_a = \hbar_a c I$, $R_a \Lambda_a = L_\alpha^2 I$, and $R_a = (G_a / c^2) M_a$. A shift $Y \mapsto Y + \delta I$, retaining positivity, changes mass and radius by $\epsilon_a \delta / c^2$ and $L_\alpha \delta$, respectively, although the scalar Hamiltonian shift cancels from its normalized thermal state. The marked provenance of energy distinguishes these realizations.

In the capacity record, the received origin is $Y_{\text{cap}} - Y_{\text{rel}} = 23 \ln 10 / \ln 3$. The energy difference is therefore $\epsilon_a (23 \ln 10 / \ln 3) I$; its radial contribution remains in the preceding formulas. When reconstructing a distribution from $-\ln \rho$, $\ln Z$ must also be restored before reading absolute energy. The marker allows both data to be retained.

The previously constructed electronic corrector satisfies

$$\begin{aligned}\kappa_e &= 80 - 54\alpha + 6\Delta_4, & \Delta_4 &= \frac{\pi - e \ln \pi}{270} \left(1 + \frac{\pi}{729}\right), \\ \frac{\mathcal{E}_e^\beta}{\mathcal{E}_e^0} &= R_{\text{act}}^{\kappa_e}, & R_{\text{act}} &= \frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}}.\end{aligned}\quad (24.21)$$

The electronic energies use the same unit, and $\kappa_e > 0$ in the constructed electronic section. The positive ratio and its logarithm determine the real power. With common L_α , c , and transducer, the preceding identities imply

$$\left(\frac{\mathcal{E}_e^\beta}{\mathcal{E}_e^0}\right)^{1/\kappa_e} = R_{\text{act}} = \frac{\Theta_{\text{clk,pre}}}{\Theta_{\text{clk,ret}}} = \frac{G_{\text{ret}}}{G_{\text{pre}}}. \quad (24.22)$$

In the return branch, let $\mathcal{E}_e = m_e c^2$ and $y_e = \mathcal{E}_e / \epsilon_{\text{ret}}$. Direct substitution gives

$$\Lambda_e = \frac{\hbar_{\text{ret}} c}{\mathcal{E}_e}, \quad \frac{G_{\text{ret}} m_e^2}{\hbar_{\text{ret}} c} = y_e^2, \quad \beta_{\text{clk,ret}} \mathcal{E}_e = (\ln 3) y_e. \quad (24.23)$$

Indeed, $\epsilon_{\text{ret}} = \hbar_{\text{ret}} c / L_\alpha$ and $G_{\text{ret}} = c^3 L_\alpha^2 / \hbar_{\text{ret}}$ cancel the scales in the second ratio, while the third uses the definition of $\beta_{\text{clk,ret}}$. These relations compose the same return, with the electronic dimensional conversion retained.

The degree assignment $C = 23I + 3N$, fixed spectral reference η , and marked-conditioned pair of functionals jointly determine the thermal realization studied here. Its tensor and variational proofs establish the coefficient and equilibrium; chronological, areal, and radial composition preserves the state, scales, and energy origin required by their respective readings.

25 Additive selection and the thermal reading instrument

The spectral reference and action constructed in the preceding sections make it possible to specify what is counted when information and energy are measured. The thermal coefficient appears as a relative intensity between two events in the same record. Additivity characterizes its informational role, and both events admit an explicit apparatus preserving the complete state. This develops the marked rule of 24.5, including its realization, its selection within a class of readers, and its transport conditions.

The composition retains its APP-TRIT-TPK antecedent: additive and multiplicative sheets, regime, orientation, remainder, quotient, boundary, route, and memory. The prolongation $w_6 \rightarrow w_{12} \rightarrow w_{18} \rightarrow w_{24} \rightarrow w_{30} \rightarrow R_{36} \rightarrow G_9$ and the order $U_t \rightarrow \Gamma_9 \rightarrow K_{\text{ph}}$ precede this spectral reading. Phase return prolongs the record; the observed marks replace neither that record nor the two projections of the joint discrete structure of the continuum. The ratio $r = 1/10$, the ranks $(1, 380, 649)$, and their assignment to degrees $23 + 3n$ are received with the conditions of 24.3. This assignment remains an explicit part of the realization, together with the Hamiltonian and its energy origin.

25.1 Spectral events and relative measure

Let $\mathcal{K} = \ell^2(\mathbb{N}_0) \otimes \mathbb{C}^M$, with $M = 649$, and consider the normalized density

$$\Sigma = \rho_r \otimes I_M / M, \quad \rho_r = (1 - r) \sum_{j \geq 0} r^j |j\rangle\langle j|. \quad (25.1)$$

For $n = 0, 1, 2$, let $j_n = 23 + 3n$ and let Q_n be memory projectors of rank d_n , where $(d_0, d_1, d_2) = (1, 380, 649)$. Fix a unit vector χ_B . The letters A, B, C here denote reading events, not geometric area:

$$\begin{aligned} P_A &= \sum_{n=0}^2 |j_n\rangle\langle j_n| \otimes Q_n, & P_B &= |0\rangle\langle 0| \otimes |\chi_B\rangle\langle \chi_B|, \\ P_C &= I - P_A - P_B, & \mu(P) &= \text{Tr}(\Sigma P). \end{aligned} \quad (25.2)$$

The three projectors are mutually orthogonal, sum to the identity, and commute with Σ . Event B fixes a rank-one reference; C retains the entire complement of the marked sectors.

Proposition 25.1 (Relative intensity and compression). *Writing $p_A = \mu(P_A)$ and $p_B = \mu(P_B)$ gives*

$$p_B = \frac{1-r}{M}, \quad p_A = \frac{1-r}{M} \sum_{n=0}^2 d_n r^{23+3n}, \quad \frac{p_A}{p_B} = b_* = \frac{1380649}{10^{29}} = 1.380649 \times 10^{-23}. \quad (25.3)$$

If $\iota : \mathcal{D} \rightarrow \mathcal{K}$ embeds each carrier sector in its degree j_n and in $\text{ran } Q_n$, then

$$p_B^{-1} \iota^* \Sigma \iota = \bigoplus_{n=0}^2 r^{23+3n} I_{d_n} = \eta. \quad (25.4)$$

Proof. At degree j , the density acts as $(1-r)r^j/M$ times the memory identity. Taking its trace over a subspace of rank d multiplies that scalar by d . Applying this to degree zero and the three marked degrees gives the two probabilities; their ratio cancels the common factor. Since $r = 10^{-1}$, $\sum_n d_n r^{23+3n} = 10^{-23}(1 + 380 \cdot 10^{-3} + 649 \cdot 10^{-6})$, which yields the displayed fraction. The same scalar action, restricted by ι , proves (25.4). $\square$

The intensity b_* is a relative measure, not the raw probability p_A . Using the whole vacuum $|0\rangle\langle 0| \otimes I_M$ as reference would give b_*/M : this changes the reference event. By contrast, adjoining an independent auxiliary factor and extending both projectors by its identity preserves the two probabilities and their ratio.

25.2 Unitary apparatus, recovery, and energy domain

On a three-dimensional pointer with basis $|A\rangle, |B\rangle, |C\rangle$, define

$$V\psi = \sum_{y=A,B,C} P_y \psi \otimes |y\rangle. \quad (25.5)$$

For a unitary realization, let $R_C = I$, let R_A exchange $|C\rangle$ and $|A\rangle$, and let R_B exchange $|C\rangle$ and $|B\rangle$. The coupling is

$$\mathbb{U} = \sum_{y=A,B,C} P_y \otimes R_y, \quad \mathbb{U}(\psi \otimes |C\rangle) = V\psi. \quad (25.6)$$

Proposition 25.2 (Conservative recording of the three alternatives). *The identities $\mathbb{U}^* \mathbb{U} = \mathbb{U} \mathbb{U}^* = I$ and $V^* V = I$ hold. For any density ξ , the joint state $\hat{\xi} = V \xi V^*$ admits exact recovery $V^* \hat{\xi} V = \xi$. Reading the pointer defines the instrument*

$$\mathcal{I}_y(\xi) = P_y \xi P_y, \quad p_y = \text{Tr}(\xi P_y), \quad \xi_y = \mathcal{I}_y(\xi)/p_y \quad (p_y > 0). \quad (25.7)$$

Its nonselective channel preserves trace and leaves Σ invariant.

Proof. Cross-products of distinct P_y vanish. Thus $\mathbb{U}^* \mathbb{U} = \sum_y P_y \otimes R_y^* R_y = I$, and likewise for the other product. Pointer orthonormality gives $V^* V = \sum_y P_y = I$; applying this to both sides of $\hat{\xi}$ proves recovery. Compressing the pointer to $|y\rangle$ gives (25.7). Moreover, $\sum_y \text{Tr}(P_y \xi P_y) = \text{Tr } \xi$. Since every P_y commutes with Σ , we also have $\sum_y P_y \Sigma P_y = \Sigma$. $\square$

For a state with coherences between marks, forgetting the pointer may change them. The recovery above uses the joint state, which retains them. Reading an outcome and retaining the complete apparatus are distinct operations.

Let $J|j\rangle = j|j\rangle$ and $H_{\text{sp}} = f(J) \otimes I_M$, with $f : \mathbb{N}_0 \rightarrow \mathbb{R}$. Its self-adjoint domain is

$$\mathcal{D}(H_{\text{sp}}) = \left\{ \psi : \sum_{j,a} |f(j)|^2 |\psi_{j,a}|^2 < \infty \right\}.$$

The mark projectors reduce this domain and commute with the spectral projections of H_{sp} . On that domain,

$$(H_{\text{sp}} \otimes I_3)V = V H_{\text{sp}}, \quad V^*(H_{\text{sp}} \otimes I_3)V = H_{\text{sp}}. \quad (25.8)$$

Indeed, applying H_{sp} to each summand of (25.5) allows it to be commuted with P_y ; the squared norm of the sum equals the sum of the squared sector norms. The second identity follows from $V^*V = I$. Energy expectations are therefore preserved whenever they exist, including the realization $f(j) = \epsilon_a(j + 1/2)$. A pointer Hamiltonian requires separate specification: if its three levels are degenerate, the coupling also preserves total energy. This identity assigns no zero cost to physically preparing or resetting the apparatus.

25.3 Tritic transport with remainder and carry

Division $j = 3w + b$, $0 \leq b < 3$, defines the unitary $U_3|j\rangle = |w, b\rangle$. Let $W = U_3 \otimes I_M$. The transported marks are

$$\begin{aligned} P'_A &= \sum_{n=0}^2 |7+n, 2\rangle\langle 7+n, 2| \otimes Q_n, \\ P'_B &= |0, 0\rangle\langle 0, 0| \otimes |\chi_B\rangle\langle \chi_B|, \quad P'_C = I - P'_A - P'_B. \end{aligned} \quad (25.9)$$

Proposition 25.3 (Naturality of the apparatus). *If V' is the apparatus for the transported marks, then*

$$V'W = (W \otimes I_3)V, \quad \mathcal{J}'_y(W\xi W^*) = W\mathcal{J}_y(\xi)W^*. \quad (25.10)$$

Probabilities and relative states are transported without loss.

Proof. Euclidean division gives $P'_yW = WP_y$ for each y . Substituting these three identities in the sum defining V' proves the first square; multiplication by the adjoints proves the second. Unitary invariance of the trace proves the probability statement, and division by the same positive probability proves the statement for relative states. $\square$

The transported density retains quotient and remainder separately:

$$W\Sigma W^* = \rho_{r^3} \otimes \sigma_r \otimes I_M/M, \quad \sigma_r = \frac{\text{diag}(1, r, r^2)}{1 + r + r^2}. \quad (25.11)$$

Indeed, $(1 - r)r^{3w+b} = (1 - r^3)(r^3)^w r^b / (1 + r + r^2)$. For the unilateral shift $S|j\rangle = |j + 1\rangle$,

$$U_3 S U_3^* = I \otimes (|1\rangle\langle 0| + |2\rangle\langle 1|) + S_w \otimes |0\rangle\langle 2|. \quad (25.12)$$

The cases $b = 0, 1$ increment the remainder; $b = 2$ resets the remainder to zero and increments w . This proof retains the carry. The shift may change the mark: (25.10) expresses independence of the order of observation and reversible representation change, not immobility under the dynamics.

25.4 Equivalent embeddings and memory enlargement

Let $\tilde{Q}_n$ be other projectors of the same ranks and $\tilde{\chi}_B$ another unit vector. At each degree j_n , a unitary R_{j_n} takes $\text{ran } Q_n$ to $\text{ran } \tilde{Q}_n$: send orthonormal bases of those subspaces and their complements to one another. Choose likewise $R_0\chi_B = \tilde{\chi}_B$ and complete with unitaries at the remaining degrees. Then $R = \bigoplus_{j \geq 0} R_j$ is unitary, preserves Σ , and commutes with H_{sp} . The identities

$$\tilde{P}_y R = R P_y, \quad \tilde{V} R = (R \otimes I_3)V \quad (25.13)$$

hold block by block. Conjugation and cyclicity of the trace show that they preserve probabilities, spectral energy, and b_* . A single degree-independent memory unitary need not realize every embedding simultaneously. Any additional operator is transported by the same R ; this conjugation does not assert that its matrix remains unchanged.

For an enlargement, let $L : \mathbb{C}^M \rightarrow \mathbb{C}^{M'}$ be an isometry and let $J_L = I \otimes L$. Define $P_A^+ = J_L P_A J_L^*$, $P_B^+ = J_L P_B J_L^*$, and $P_C^+ = I - P_A^+ - P_B^+$. Since $J_L^* J_L = I$, all three marks satisfy $P_y^+ J_L = J_L P_y$. Consequently,

$$V^+ J_L = (J_L \otimes I_3)V, \quad \Sigma_L = J_L \Sigma J_L^* = \rho_r \otimes L L^* / M. \quad (25.14)$$

Isometric transport preserves the original probabilities. If the uniform density of the whole enlarged ambient space is also prepared, $\Sigma^+ = \rho_r \otimes I_{M'}/M'$, the preparation changes and

$$p_A^+ = \frac{M}{M'} p_A, \quad p_B^+ = \frac{M}{M'} p_B, \quad p_A^+/p_B^+ = b_*. \quad (25.15)$$

The proof again takes the trace of each degree's scalar over the same ranks. On A and B , normalization cancels M/M' and gives the isometric images of the original relative states; the complement receives the added weight. Preservation of the apparatus, equality of the intensity, and equality of the complete state are thereby distinguished.

25.5 Complete preparation, entropy, and conditioning

Let ρ be a state of a finite trial system, $H = H^*$ its inherited Hamiltonian, and ρ_0 a fixed pure preparation. With $P = P_A$, define

$$\Omega_\rho = (P\Sigma P) \otimes \rho + ((I - P)\Sigma(I - P)) \otimes \rho_0, \quad \Omega_0 = \Sigma \otimes \rho_0. \quad (25.16)$$

This preparation retains the entire spectral complement. It differs from the auxiliary product in (24.12), although it realizes the same two functionals on the trial system.

Theorem 25.4 (Information per reference and conditional energy). *The trace of Ω_ρ is one, and its partial trace over the trial system recovers Σ . Moreover,*

$$s_B(\rho) := \frac{s(\Omega_\rho) - s(\Omega_0)}{p_B} = b_* s(\rho), \quad (25.17)$$

$$\rho_A := \frac{\text{Tr}_{\mathcal{K}}[(P \otimes I)\Omega_\rho(P \otimes I)]}{p_A} = \rho, \quad \mathcal{E}_A(\rho) = \text{Tr}(\rho_A H) = \text{Tr}(\rho H). \quad (25.18)$$

Proof. The positive summands have traces p_A and $1 - p_A$. Since $[P, \Sigma] = 0$, partial trace over the trial system gives $P\Sigma P + (I - P)\Sigma(I - P) = \Sigma$. Let λ_i be the eigenvalues of the marked block and p_j those of ρ . Their products replace λ_i ; the entropy increment of that block is

$$- \sum_{i,j} \lambda_i p_j \ln(\lambda_i p_j) + \sum_i \lambda_i \ln \lambda_i = p_A s(\rho).$$

The complement is unchanged. The sums are legitimate because $s(\Sigma) = -\ln(1 - r) - r \ln r/(1 - r) + \ln M < \infty$ and the trial system is finite. Division by p_B proves (25.17). Compression to $P \otimes I$ and partial trace give $p_A \rho$; division by $p_A > 0$ proves (25.18) and its energy expectation. $\square$

Conditioning preserves the observational operation of the corpus. If U_a couples the state to the initial record m_0 , the output and its fibre are

$$\text{Out}_a(x) = q_a(\text{pr}_M U_a(x, m_0)), \quad \Omega_{a,y} = \{x : \text{Out}_a(x) = y\}.$$

For $\mu_a(\Omega_{a,y}) > 0$, the measure is restricted to that fibre:

$$\mu_{a,y}(D) = \frac{\mu_a(D \cap \Omega_{a,y})}{\mu_a(\Omega_{a,y})}. \quad (25.19)$$

When these events belong to a common projection-valued measure and $\mu_a(D) = \text{Tr}(\omega P_D)$, we have $P_D P_y = P_{D \cap \Omega_{a,y}}$. Cyclicity of the trace proves

$$\text{Tr}\left(\frac{P_y \omega P_y}{\text{Tr}(\omega P_y)} P_D\right) = \mu_{a,y}(D).$$

The equality extends to bounded observables of that algebra by spectral sums and uniform approximation. Applied to event A in (25.16), it gives exactly the preceding conditional mean energy. It requires no dynamical ergodicity assumption. Any additional purification is retained in its own factor; the entropy computed here belongs to the specified observable algebra.

25.6 Characterization by additivity and refinement

The preceding evaluation admits a characterization fixing its weight without postulating b_* as a coefficient. For a fixed trial state ρ , consider a contribution $F_\rho(x)$ for an alternative of received weight $x \in [0, 1]$. In the relative reading, this weight is measured against reference B ; only alternatives within the stated domain are considered. Require dependence on the weight rather than the alternative's name; additivity $F_\rho(x + y) = F_\rho(x) + F_\rho(y)$ for $x + y \leq 1$; positivity; and normalization $F_\rho(1) = s(\rho)$. The domain must realize these partitions or the ordered refinements described below.

Theorem 25.5 (Uniqueness of the additive informational weight). *On the complete scalar domain, the four conditions above imply $F_\rho(x) = xs(\rho)$. The same conclusion holds for a realization of alternatives with equiponderant refinements of weights 3^{-n} and, for each mark P , approximations $P_n^- \leq P \leq P_n^+$ whose weights converge to that of P and whose contributions belong to those refinements.*

Proof. Additivity gives $F_\rho(0) = 0$. If $x \leq y$, then $F_\rho(y) - F_\rho(x) = F_\rho(y - x) \geq 0$, so the reader is monotone. Partitioning the unit into 3^n equal alternatives gives $F_\rho(3^{-n}) = 3^{-n}s(\rho)$, and joining k of them gives $F_\rho(k3^{-n}) = k3^{-n}s(\rho)$. For $0 \leq x < 1$, take

$$\ell_n = 3^{-n} \lfloor 3^n x \rfloor, \quad u_n = \ell_n + 3^{-n}.$$

They bound x within $[0, 1]$. Monotonicity implies $\ell_n s(\rho) \leq F_\rho(x) \leq u_n s(\rho)$; their difference $3^{-n}s(\rho)$ tends to zero. This proves the formula, including $s(\rho) = 0$; $x = 1$ is the normalization. For ordered marks, repeat the argument with P_n^- and P_n^+ , since positive additivity implies isotonicity. No differentiable continuity of the reader is assumed. $\square$

In particular, the contribution of A is $b_* s(\rho)$ when its relative measure is retained. The characterization also applies to the normalized mark of (24.12). There, the functional $-\text{Tr}[(P \otimes I)(\dot{\eta} \otimes \rho)(I \otimes \ln \rho)]$ satisfies the conditions by linearity of the trace and assigns $s(\rho)$ to the certain mark.

Refinement requires an effective realization: the rank-one projector P_B contains no three nonzero equiponderant subprojectors. An auxiliary enlargement can realize refinements, but must transport the measure and the reading. Likewise, three descendants of a cylinder do not imply equal weights merely from their number. For a rational weight, an equiponderant partition with the corresponding denominator suffices if it exists in the realization; the ternary proof does not assume that particular denominator.

The assumptions isolate precisely the freedoms excluded. The reader $f(x) = x^2$ respects products but gives $f(x + y) - f(x) - f(y) = 2xy$; for $x = 1/3$, $y = 1/5$, the defect is $2/15$. Without normalization, $f(x) = cx$ preserves positivity and additivity. Without identifying the received measure, $(1/3, 2/3)$ and $(1/2, 1/2)$ are distinct positive, additive, normalized measures on two alternatives. The fixed entropy of the label record remains separate: the theorem does not remove the mixing term from a joint entropy.

25.7 Realization by visits between returns

Relative intensity also has a precise dynamical interpretation. Let (X, μ, T) be an invertible, ergodic, measure-preserving probability system with events A, B of measures $p_A, p_B > 0$. For $x \in B$, let

$$R_B(x) = \min\{k \geq 1 : T^k x \in B\}, \quad C_A(x) = \sum_{k=0}^{R_B(x)-1} \mathbf{1}_A(T^k x).$$

The excursion retains its complete sequence; C_A is only one observable of that sequence.

Theorem 25.6 (Reward per return to a reference). *With $\mu_B = \mu|_B/p_B$, for every $f \in L^1(\mu)$,*

$$\mathbb{E}_{\mu_B} \left[\sum_{k=0}^{R_B-1} f \circ T^k \right] = \frac{1}{p_B} \int_X f d\mu. \quad (25.20)$$

In particular, $\mathbb{E}_{\mu_B} C_A = p_A/p_B = b_*$.

Proof. Let $B_k = \{x \in B : R_B(x) > k\}$. Recurrence, invertibility, and ergodicity ensure that the levels $T^k B_k$, $k \geq 0$, partition X up to a null set: every point has a most recent visit to B in its past. For $f \geq 0$, measure preservation and monotone convergence give

$$\int_B \sum_{k=0}^{R_B-1} f(T^k x) d\mu = \sum_{k \geq 0} \int_{T^k B_k} f d\mu = \int_X f d\mu.$$

For $f \in L^1$, applying the identity first to $|f|$ establishes integrability; the positive and negative parts can then be subtracted. Division by p_B proves the formula; $f = \mathbf{1}_A$ gives its consequence. $\square$

For a finite irreducible chain with matrix P and stationary distribution $\pi > 0$, there is also an algebraic proof. Starting at the reference state B , let v_i be the expected number of visits to i before the first positive return. That return has finite expectation; summing transitions up to it shows that arrivals and departures have equal expectations, hence $vP = v$. Since $v_B = 1$, uniqueness of the stationary distribution gives $v_i = \pi_i/\pi_B$. Summing over A reproduces (25.20). Grouping the spectral distribution into B , the three sectors of A , and the complement provides a finite realization with the exact ratio (25.3).

If every visit to A carries a trial ρ , independent of the others conditional on the excursion record, the trials form $\rho^{\otimes C_A}$. Tensor additivity gives

$$\begin{aligned} \mathbb{E}_{\mu_B} s(\rho^{\otimes C_A}) &= b_* s(\rho), \\ \frac{\mathbb{E}_{\mu_B} [C_A \operatorname{Tr}(\rho H)]}{\mathbb{E}_{\mu_B} C_A} &= \operatorname{Tr}(\rho H). \end{aligned} \quad (25.21)$$

The second expression is the ratio of total energy to total visits, not an average of ratios per excursion, which would be undefined when $C_A = 0$. Accumulated energy per return equals $b_* \operatorname{Tr}(\rho H)$.

Information per return and energy per visit thus realize the two denominators of (25.17)–(25.18). Correlated trials require their joint conditional entropy. A noninvertible process admits this formulation on its stationary bilateral extension; without ergodicity, the domain must be restricted to the support swept out by the return section. These dynamical assumptions are unnecessary for the operator theorem 25.4. Neither a stationary test chain is identified by itself with the TPK update, nor R_B with the material period $108t_0$.

25.8 Common reference and correlations between systems

One reference and m independent systems produce the record $\hat{\eta} \otimes \rho_1 \otimes \cdots \otimes \rho_m$, with a single mark $P \otimes I$. Its weight remains b_* . For $H_{\text{tot}} = \sum_j H_j$, with the appropriate tensor extensions,

$$\mathcal{S}_{P,\text{tot}} = b_* \sum_j s(\rho_j), \quad \mathcal{E}_{P,\text{tot}} = \sum_j \operatorname{Tr}(\rho_j H_j). \quad (25.22)$$

The proof uses trace factorization and $\ln(\rho_1 \otimes \rho_2) = \ln \rho_1 \otimes I + I \otimes \ln \rho_2$ on the support, iterated over the factors. For a correlated state $\rho_{1\dots m}$, the same mark gives $b_* s(\rho_{1\dots m})$. For two systems,

$$\mathcal{S}_{P,1} + \mathcal{S}_{P,2} - \mathcal{S}_{P,12} = b_* I(1 : 2)_\rho, \quad (25.23)$$

by the definition of mutual information. An interacting Hamiltonian also retains a single reference even when its Gibbs state does not factorize.

Two independent references with joint mark $P_1 \otimes P_2$ have weight $b_1 b_2$; for correlated references the weight is $\operatorname{Tr}[\omega_{R_1 R_2}(P_1 \otimes P_2)]$ and need not factorize. Copying a label does not create independence. Indeed, the isometry

$$V_P v = P v \otimes |1\rangle + (I - P) v \otimes |0\rangle$$

satisfies $V_P^*(P \otimes |1\rangle\langle 1|)V_P = P$; the trace in the transported state preserves b_* , not b_*^2 . More generally, for an isometry L , $\operatorname{Tr}[(L\omega L^*)(LPL^*)] = \operatorname{Tr}(\omega P)$ because $L^*L = I$. Returning while retaining the same record does not automatically raise its weight to a power. A new survival selection uses its own mark.

25.9 Thermometric selection and composition with action and horizon

Fix the reference and a finite nonscalar Hamiltonian H . For $\rho_\beta = e^{-\beta H}/Z(\beta)$, $\beta > 0$, the additive theorem and conditional energy produce the pair $\mathcal{S}_B = b_* s(\rho_\beta)$, $\mathcal{E}_A = \text{Tr}(\rho_\beta H)$. Writing $E = \text{Tr}(\rho_\beta H)$, the finite spectral sum gives $E' = -\text{Var}_{\rho_\beta}(H) < 0$ and $(\ln Z)' = -E$. Differentiating $s = \beta E + \ln Z$ gives $s' = \beta E'$, hence

$$\frac{d\mathcal{S}_B}{d\mathcal{E}_A} = b_* \beta, \quad \Theta_{B,A} = \frac{1}{b_* \beta}, \quad B_*(u_\Theta) = b_* u_E. \quad (25.24)$$

The positive bases satisfy $u_S = u_E/u_\Theta$. The last equation determines the unique positive linear map sending $\Theta_{B,A} u_\Theta$ to $\beta^{-1} u_E$. The balance

$$\text{Tr}(\rho H) - b_* \Theta s(\rho) - [\text{Tr}(\rho_\Theta^* H) - b_* \Theta s(\rho_\Theta^*)] = b_* \Theta D(\rho \| \rho_\Theta^*)$$

with $\rho_\Theta^* \propto e^{-H/(b_* \Theta)}$ retains the unique minimum proved in 24.6. Additive selection now characterizes the weight entering that balance.

With $\epsilon_a = \hbar_a/(108t_0) = \hbar_a c/L_\alpha$, $q_A = \gamma a_0 L_\alpha^2$, and $\mathcal{T}_{I/A} = \epsilon_a/q_A$, the thermal sections are

$$\begin{aligned} b_* \Theta_{\text{clk},P,a} &= \frac{\hbar_a}{108t_0 \ln 3} = \frac{\mathcal{T}_{I/A} q_A}{\ln 3}, \\ b_* \Theta_{H,P}(R) &= \frac{\hbar_a c}{2\pi R}, \quad \frac{\Theta_{H,P}(R)}{\Theta_{\text{clk},P,a}} = \frac{L_\alpha}{R} \frac{\ln 3}{2\pi}. \end{aligned} \quad (25.25)$$

These are coordinate equations for B_* in the transported bases. The first uses the section $\beta_{\text{clk},a} = \ln 3/\epsilon_a$; the second uses the scale $\tau_R = \hbar_a c/(2\pi R)$. Their quotient proves the temperature ratio. The radial relation $G_a \hbar_a = c^3 L_\alpha^2$ implies

$$G_a b_* \Theta_{\text{clk},P,a} \ln 3 = c^4 L_\alpha. \quad (25.26)$$

At the horizon, $\mathcal{S}_B = b_* s$ and $\Theta_{H,P} = \tau_R/b_*$ convert the previous balance into

$$\delta E_R - \Theta_{H,P} \delta \mathcal{S}_B = \tau_R D(\sigma \| \rho_A).$$

The composition retains Barbero's area resolution, sector incidence, action, and the energy origin: cancellation of scales preserves the same energy rather than adding a metrological parameter to the constructor.

The selection has a specified domain. Conditioning both channels gives (E, s) ; counting both per reference gives $(b_* E, b_* s)$. In both cases the derivative is β . The pair characterized here is $(E, b_* s)$, using energy per visit and information per reference. Preservation of the record allows all three observations but does not identify them. Nor do energy products alone select the weight: for $p > 0$, replacing b_* by b_*^p and dividing temperatures by the same relative factor preserves those products and serial composition; the quadratic reader, however, violates the additivity proved above.

The mark instrument, its recovery and transport, the additive informational contribution, conditional energy, and this realization's transducer are thus determined. Identifying it with a thermometric instrument of the complete constructor requires transporting its events, measure, and pair of normalizations to A and B while retaining refinement and memory. Existence of the apparatus does not replace that particular identification. If the reference varies with β , its derivatives also enter; if the projection does not commute with Σ , preparation (25.16) preserves its block diagonal rather than necessarily the original density. These conditions specify the scope of the identities without altering the preceding results on action, area, capacity, and transport.

25.10 Structural definition and function of the Boltzmann constant

Definition 25.7 (Thermal transducer of energy per decision). Fix a positive thermal section $\Theta_{\text{clk},a} \in \mathcal{L}_\Theta$. The Boltzmann transducer is the positive linear map $k_B : \mathcal{L}_\Theta \rightarrow \mathcal{L}_E$ realizing

$$k_B(\Theta_{\text{clk},a}) = \frac{\epsilon_a}{\log 3}, \quad \epsilon_a = \frac{h_a}{108t_0}. \quad (25.27)$$

Action and duration determine ϵ_a ; the uniform tritic decision determines its informational normalization.

Uniqueness, proved in section 22.2, follows by prescribing the image of a nonzero section of a line. Equations (22.9) and (22.10) specify, respectively, its coordinate in declared bases and compatibility of the same transducer with both orientations. Determination of the product $k_B\Theta_{\text{clk},a}$ precedes individualization: the latter retains as explicit data the thermal section, the bases, and, when internal selection is claimed, its constructive reader. The pressure in (22.13), under the hypotheses enumerated there, normalizes the energy parameter $\beta = (k_B\Theta)^{-1}$.

The physical function of k_B is to express the information of a distribution in dimensional form: $S = k_B s$. Evaluating that entropy at a temperature produces the energy ΘS . A uniform triple contributes $\log 3$ nats; a distribution of continuations contributes its weighted entropy; a quantum reduction contributes the entropy of the specified bipartition. The transducer acts on these contents while preserving their domains. Genealogical memory determines what information each reading retains; the thermodynamic balance declares which operation exchanges or erases it. This separation explains the relation between reversibility, entanglement, and decision energy.

26 Confluence of action, area, information, and gravitation

The area scale in this section is the same one determined by the longitudinal and radial composition in 20:

$$\ell_P^2 = L^2 = \frac{G_a \hbar_a}{c_{\text{int}}^3}, \quad E_{P,a} = \frac{\hbar_a c_{\text{int}}}{L} = k_B \Theta_{\text{clk},a} \log 3. \quad (26.1)$$

In the return section, $E_{P,\text{ret}} = E_L$. The variable horizon radius R retains its own meaning; condition $R = L$ specifies the particular cell studied below. The increment $\delta\mathcal{A} = \gamma_{\text{HMT}} a_0 L^2$ transmits that same area unit to incidence, retaining action, the thermal transducer, and the occupations of the reduced state.

26.1 Energy and entropy of the half-action cell

Fix the same oriented action section and its corresponding gravitational and thermal sections. The preceding reciprocity allows the same cell to be expressed through action, duration, and length. With $\ell_P^2 = \hbar G/c^3$, $t_P = \ell_P/c$, and $E_P = \hbar/t_P$, one has

$$\frac{c^4 \ell_P}{G} = \frac{\hbar}{t_P} = E_P.$$

The equality follows by substituting the area scale already constructed. In the horizon realization defined by (26.2), the quantities are normalized as

$$E_\Lambda(R) = \frac{c^4 R}{2G}, \quad S_H(R) = \frac{k_B \pi R^2}{\ell_P^2}, \quad T_H(R) = \frac{\hbar c}{2\pi k_B R}. \quad (26.2)$$

The temperature corresponds to the cosmological-horizon normalization of this realization. The factor 2π and the definition of the radius are part of the set of equations used.

Theorem 26.1 (Identity of the gravitational half-action cell). *In realization (26.2),*

$$T_H(R) S_H(R) = E_\Lambda(R), \quad (26.3)$$

$$T_H(\ell_P) S_H(\ell_P) = \frac{E_P}{2} = \frac{k_B \Theta_{\text{clk}} \log 3}{2}, \quad (26.4)$$

$$E_\Lambda(\ell_P) t_P = \frac{\hbar}{2}, \quad E_\Lambda(\ell_P) T_{108} = \frac{h}{2}. \quad (26.5)$$

Proof. Multiplying temperature by entropy cancels the thermal normalization and gives $\hbar c R / (2\ell_P^2) = c^4 R / (2G)$. At $R = \ell_P$, the expression is $\hbar c / (2\ell_P) = \hbar / (2t_P) = E_P/2$. The thermal identity of the cycle gives the second member. Finally, $T_{108} = 2\pi t_P$ and $h = 2\pi \hbar$ produce the two action relations. $\square$

The theorem brings together quantities that previously belonged to different descriptions. The cell energy, the temperature–entropy product, tritic information, and action integrated over its duration are compatible readings under the declared normalization. The gravitational constant enters the area scale; action fixes the energy of the period; Boltzmann transports dimensionless information to the entropy line. The proof shows how these functions are joined in a single identity.

26.2 Conversion between decision energy and area increment

Energy per tritic decision and the elementary area increment are obtained through different readers in the preceding development:

$$E_P = k_B \Theta_{\text{clk}} \log 3 = \frac{\hbar}{t_P}, \quad \delta\mathcal{A} = \gamma_{\text{HMT}} a_0 \ell_P^2. \quad (26.6)$$

The same oriented section and the positive branch $\hbar, G, \gamma_{\text{HMT}}, a_0, t_P > 0$ are maintained. The first quantity belongs to the energy line and the second to the area line. Their quotient defines the conversion coefficient

$$\mathcal{J}_{I/A} := \frac{E_P}{\delta\mathcal{A}} = \frac{k_B \Theta_{\text{clk}} \log 3}{\gamma_{\text{HMT}} a_0 \ell_P^2}, \quad [\mathcal{J}_{I/A}] = [\text{energy}]/[\text{area}]. \quad (26.7)$$

This definition makes explicit the normalization required to transport a decision energy to a geometric resolution. All its factors are specified before the quotient is formed.

Proposition 26.2 (Gravitational expression of the conversion coefficient). *In the same dimensional realization,*

$$\mathcal{J}_{I/A} = \frac{c^3}{\gamma_{\text{HMT}} a_0 G t_P} = \frac{c^4}{\gamma_{\text{HMT}} a_0 G \ell_P}. \quad (26.8)$$

Proof. Substituting $E_P = \hbar/t_P$ and $\ell_P^2 = \hbar G/c^3$ into (26.7) cancels the common action section and gives the first member. The identity $\ell_P = c t_P$ produces the second. The dependence of t_P and ℓ_P on the previously constructed realization is preserved. $\square$

In the minimal cut of (19.4), the realization of one uniform decision per link gives

$$S_{\text{corte}}^{\text{fis}} = 81 k_B \log 3, \quad E_{\text{corte}}^{(1)} = 81 E_P = \Theta_{\text{clk}} S_{\text{corte}}^{\text{fis}}. \quad (26.9)$$

The first equality uses the proved capacity and the uniform product state; the second assigns the decision energy of the same cycle to each of its eighty-one links. Multiplying the thermal identity by eighty-one proves the equation of state of this realization. The thirty-six-link sector state preserves its own partition function, entropy, and area variations in (19.35).

The composition identifies the complementary functions of the constants: action fixes energy per duration; the angular character and Barbero–Immirzi fix the sector resolution of area; Boltzmann dimensionally realizes information; the product $G\hbar$ determines the area unit. The coefficient $\mathcal{J}_{I/A}$ preserves these dependencies by expressing their relation in units of energy per area.

26.3 Physical meaning of area confluence

For a horizon of radius R , the area is $A_H = 4\pi R^2$. The same entropy relation admits two expressions

$$\frac{S_H}{k_B} = \frac{A_H}{4\ell_P^2} = \frac{\pi R^2 c^3}{G\hbar}.$$

The first quotient compares the macroscopic area with the construction's area unit. The second makes explicit the composition of that unit through action and gravitation. Transporting the sections so that $G_a \hbar_a$ remains fixed makes the geometric reading preserve this normalization. The dimensionless entropy of the same horizon remains invariant; its dimensional expressions preserve the bases they use.

The Barbero–Immirzi parameter acts at a different resolution of the same development: oriented incidence organizes the elementary area spectrum, and the reduced state describes the information accessible at the cut. The areal and volumetric sheets provide the geometric transport and history measure described in section 19.7. Comparing a microscopic spectrum with a horizon entropy retains its realization map, state, and scale as data. These specifications allow the physical relation to be interpreted at each level and avoid replacing a correspondence of observables with an indiscriminate identification of their spaces.

The theoretical unity of the argument resides in complementary functions:

The area unit combines the product $G\hbar$; oriented incidence organizes its elementary resolution through $\gamma_{\text{HMT}} a_0$; the state and bipartition determine the information accessible to the subsystem. Response equations (19.26)–(19.27) make this articulation explicit in the sector collection. The inversion (19.29) preserves entropy and transforms area according to (19.30): the two readings distinguish different properties of the same configuration.

In this articulation, the dynamics produces and preserves history; the reader determines what information is accessible; action relates evolution and energy; Boltzmann realizes its thermal expression; gravitation enters the geometric scale by which area is measured. The identity $T_H S_H = E_\Lambda$ shows an exact confluence of these functions in the declared realization. Physical theory is thus formulated from mathematical objects and their effective transformations.

26.4 Spectral entropy and preservation under transport

The entropy of a reduced state depends on its spectrum and on the bipartition defining the reduction. To specify continuation toward duality, let $\mathcal{H}_A, \mathcal{H}_B$ be the two factors and ρ_{AB} a state with finite entropies. A factorized transport $U_A \otimes U_B$ produces

$$\rho'_A = U_A \rho_A U_A^*, \quad S(\rho'_A) = S(\rho_A).$$

The first identity follows from the definition of partial trace, and the second from spectrum invariance. For a general unitary transport, the equivalent formulation also transports the inclusion of the subsystem's observable algebra. This qualification identifies the geometric datum that must accompany a claim of entanglement preservation.

The reciprocity of the ellipse and momentum–winding exchange preserve the quantities indicated in their respective theorems. Transport of the bipartition determines how to use these identities in an entropy reading. Modular continuation thus maintains the article's informational content and keeps distinct the energy spectrum, action orientation, and algebraic reduction defining entanglement.

26.5 Effective multiplicity and hyperbolic realization

In the preceding cell, $S_H(\ell_P)/k_B = \pi$. The effective multiplicity associated with that entropy is

$$\Omega_{\text{eff}} = \exp(S_H/k_B) = e^\pi.$$

If H is the hyperbolic involution of TRIT, with eigenvalues $+1, -1$, the exponential $\exp(\pi H)$ has spectrum $\{e^\pi, e^{-\pi}\}$. Agreement of the expansive eigenvalue with the effective multiplicity locates a specific spectral relation. Its microscopic meaning retains as additional data the map between the horizon space and the hyperbolic representation. This localization allows the correspondence to be developed without confusing the spectral value with its entire realization space.

27 Evaluation of the constants and quantitative comparison

The preceding constructions allow three operations to be separated: producing a quantity, expressing its coordinate in a chart, and comparing that coordinate with a reference. The APP-TRIT-TPK state preserves sheets, orientation, quotient, carry, and memory before these readings. Its nonadic prolongation determines the correlated coordinates π, φ, e, α ; the multiplicative norm over the local quotient and regional return produce the action sections; the marked 90/120 incidence and conjugate channels produce the Barbero functional; angular transport subsequently determines the CKM coordinates. No entry from the external tables enters the choice of these operations.

Here we collect the evaluations of the preceding constructions and their comparison references. The digits of the internal outputs express computational precision, not experimental uncertainty. For a value x and a nonzero reference x_0 , both in the same chart, we use

$$\Delta x = x - x_0, \quad r_x = \frac{x - x_0}{x_0}.$$

When the reference has asymmetric marginal uncertainties u_-, u_+ , we also define the descriptive residual

$$z_x = \begin{cases} \Delta x/u_+, & \Delta x \geq 0, \\ \Delta x/u_-, & \Delta x < 0. \end{cases} \quad (27.1)$$

This normalization does not assume independence between rows and does not by itself constitute a joint statistical test.

27.1 Reduced Planck constant and action per turn: two orientations

In the chart $\mathcal{U}_S \mapsto \text{J s}$, the sections already constructed are

$$\hbar_{\text{pre}} = H_5 10^{-34} \text{ J s}, \quad H_5 = 1.0545718176461544696 \dots, \quad (27.2)$$

$$\hbar_{\text{ret}} = \eta_{\text{ret}} 10^{-34} \text{ J s}, \quad \eta_{\text{ret}} = 1.0545718169150390611 \dots, \quad (27.3)$$

$$h_a = 2\pi\hbar_a, \quad a \in \{\text{pre}, \text{ret}\}. \quad (27.4)$$

The decade comes from $\Sigma_H = \varphi\alpha^{16}$; the difference between the two sections comes from regional return. Multiplication by 2π changes action per radian to action per turn, without introducing another orientation or altering the relative residual.

The SI reference is

$$h_{\text{SI}} = 6.62607015 \times 10^{-34} \text{ J s}, \quad \hbar_{\text{SI}} = \frac{h_{\text{SI}}}{2\pi} = 1.0545718176461563912 \dots \times 10^{-34} \text{ J s}.$$

Both expressions are exact. The infinite expansion of $\hbar_{\text{SI}}$ is not a measurement uncertainty. Fixing the numerical value of h belongs to the definition of the SI in effect since 2019 [10]; comparison takes place after the HMT construction.

Quantity	Coordinate in 10^{-34} J s	r_x relative to SI
$\hbar_{\text{pre}}$	1.0545718176461544696	$-1.822221 \times 10^{-15}$
$\hbar_{\text{ret}}$	1.0545718169150390611	$-6.932836 \times 10^{-10}$
h_{pre}	6.62607014999998793	$-1.822221 \times 10^{-15}$
h_{ret}	6.62607014540625433	$-6.932836 \times 10^{-10}$

Table 10: Action before and after return. The final digits are evaluations of the resulting coordinates; the residuals are not experimental errors in h or $\hbar$.

Proposition 27.1 (Invariance of the comparison under a complete turn). *In a common chart,*

$$\frac{h_a - h_{\text{SI}}}{h_{\text{SI}}} = \frac{\hbar_a - \hbar_{\text{SI}}}{\hbar_{\text{SI}}}.$$

Proof. The two terms of the difference in the numerator on the left contain the common factor 2π , which cancels against that in the denominator. $\square$

The pre/ret distinction is quantitatively relevant: replacing one with the other changes the comparison by several orders of magnitude. The regional corrector must remain visible and must not be confused with numerical rounding.

27.2 Barbero–Immirzi: internal value and comparison between constructions

The functional evaluation is reconstructed from its three contributions:

$$\begin{aligned}\Delta S_{90} &= 2.95169439297457621139847987861 \times 10^{-4}, \\ \Delta S_{120} &= 1.96835112502951720093738706217 \times 10^{-5}, \\ \pi AC^*/180 &= 0.270485322934140078465586458790 \dots, \\ \gamma_{\text{HMT}} &= \pi AC^*/180 + 12\Delta S_{90} - \Delta S_{120} \\ &= 0.27400767269445927474725526077398041940 \dots\end{aligned}$$

This composition preserves the oriented weight $12 : -1$, prior to its decimal evaluation. The stated summands reconstruct the final coordinate with an absolute residual below 3×10^{-31} .

The Ghosh–Mitra counting is considered as a bibliographic comparison, not a measurement of the parameter. Their equation (7), written with $\lambda = 2\pi\gamma_{\text{GM}}$, is

$$\sum_{j \in \{1/2, 1, 3/2, \dots\}} (2j+1)e^{-2\pi\gamma_{\text{GM}}\sqrt{j(j+1)}} = 1. \quad (27.5)$$

The reference is A. Ghosh and P. Mitra, *An improved estimate of black hole entropy in the quantum geometry approach*, *Physics Letters B* **616** (2005), 114–117, arXiv:gr-qc/0411035, equations (7)–(8). The root is evaluated here for that specific equation; puncture counting is not identified with the HMT generator.

Construction	Dimensionless value	Nature
Functional $\Gamma_{6 \rightarrow 8}$	0.2740076726944593	HMT incidence and return
Root of (27.5)	0.2740668582328300	Declared bibliographic counting
$\gamma_{\text{HMT}} - \gamma_{\text{GM}}$	$-5.918553837 \times 10^{-5}$	Difference between constructions
$\gamma_{\text{HMT}}/\gamma_{\text{GM}} - 1$	$-2.159529202 \times 10^{-4}$	Relative difference

Table 11: Barbero–Immirzi comparison in the positive orientation. No experimental uncertainty or statistical significance is assigned to this table.

The external root evaluation is reproducible without arbitrarily truncating its tail. With $n = 2j$, $N = 128$, and $r = e^{-0.27\pi}$, for $\gamma \geq 0.27$,

$$\sum_{n>N} (n+1)e^{-\pi\gamma\sqrt{n(n+2)}} \leq \frac{r^{N+1}[(N+2) - (N+1)r]}{(1-r)^2} < 10^{-43}.$$

Every summand decreases strictly with γ . The endpoints 0.27 and 0.28 enclose the root; bisection and the tail bound control the decimals in the table. This is a check of the bibliographic comparator, not a route for selecting γ_{HMT} .

27.3 CKM angles, amplitudes, and CP violation

The angular reader already constructed determines, in degrees,

$$(\theta_{12}, \theta_{23}, \theta_{13}, \delta) = (13.003313589509, 2.398056157332, 0.213977382244, 65.676173123554)^\circ. \quad (27.6)$$

These values arise from $M_{\text{CKM}}(A, C^*/6, \Delta)^T$ and $\delta = 9A$. Conversion to radians is performed once before evaluating sines and cosines. The unitary parameterization of the angular section gives

$$|V_{\text{CKM}}| \simeq \begin{pmatrix} 0.974350259 & 0.225005836 & 0.003734601 \\ 0.224873110 & 0.973489279 & 0.041841465 \\ 0.008582400 & 0.041121745 & 0.999117283 \end{pmatrix},$$

and the invariant quartet preserves the phase:

$$J = c_{12}c_{23}c_{13}^2 s_{12}s_{23}s_{13} \sin \delta = 3.1189723326632974 \times 10^{-5} \quad \text{in the angular reader evaluation.}$$

This evaluation uses (17.18) with the coordinates of (27.6). The nine printed moduli are rounded; unitarity belongs to the complex matrix before rounding, not to a matrix formed solely from moduli.

The comparator is the three-generation fit in the Particle Data Group, 2025 update, *CKM Quark-Mixing Matrix*, equations (12.27)–(12.28) and the adjacent value of J . To avoid altering its uncertainties through a nonlinear transformation, the table retains the observables expressed by PDG: $s_{ij} = \sin \theta_{ij}$, δ in radians, and J .

Observable	HMT evaluation	PDG 2025	z_x
s_{12}	0.225007404757	0.22501 ± 0.00068	−0.003817
s_{23}	0.041841757010	$0.04183^{+0.00079}_{-0.00069}$	0.014882
s_{13}	0.003734601164	$0.003732^{+0.000090}_{-0.000085}$	0.028902
δ (rad)	1.146265461116	1.147 ± 0.026	−0.028251
$10^5 J$	3.118972333	$3.12^{+0.13}_{-0.12}$	−0.008564

Table 12: Marginal comparison in the declared parameterization and edition. The factor 10^5 jointly multiplies the value and uncertainty in the last row; it does not change its normalized residual.

Residuals are computed with the coordinates before table rounding. The PDG fit combines data and unitarity constraints; its rows are not independent measurements. Therefore $\sum z_x^2$ is not summed, nor is a number of degrees of freedom assigned to this table. The result is a reproducible marginal comparison with that edition, not a probability measure of the model. The historical marginal deviations from preceding calculations are not transferred to this table: their documentation does not identify the edition and covariance of that comparison.

The PMNS sector retains the transport $U = F_\ell^\dagger F_\nu$, the spectral domains, and phase data developed in the angular section. A numerical row requires specifying the frames of the record used and the realization compared. This table does not attribute CKM numbers to the leptonic reader or determine a new PMNS matrix.

27.4 Boltzmann and universal gravitation: quantity, chart, and reference

The thermal construction already recovered determines the positive transducer $k_B : \mathcal{L}_\Theta \rightarrow \mathcal{L}_E$ and the relation

$$k_B \Theta_{\text{clk},a} \log 3 = \frac{h_a}{108 t_0} = \frac{\hbar_a}{t_P}, \quad t_P = \frac{54}{\pi} t_0. \quad (27.7)$$

Local information, action, and period appear before the kelvin–joule chart. For positive bases u_E, u_Θ , if $k_B(u_\Theta) = b u_E$, $\Theta_{\text{clk},a} = \theta_a u_\Theta$, and $E_{P,a} = \epsilon_a u_E$, the equation is equivalent to $b \theta_a = \epsilon_a / \log 3$. A change $u'_E = s_E u_E$, $u'_\Theta = s_\Theta u_\Theta$ produces

$$b' = \frac{s_\Theta}{s_E} b, \quad \theta'_a = \frac{\theta_a}{s_\Theta}, \quad \epsilon'_a = \frac{\epsilon_a}{s_E}.$$

The marked realization of 24.5 further determines the coefficient through the spectral reference and the pair information per reference–conditional energy:

$$b_* = 10^{-23} (1 + 380 \cdot 10^{-3} + 649 \cdot 10^{-6}) = \frac{1380649}{10^{29}} = 1.380649 \times 10^{-23}. \quad (27.8)$$

The degrees, their multiplicities and the capacity origin are retained in 23; the additive characterization and the conservative instrument are developed in 25. With H and the reference fixed, $dS/d\mathcal{E} = b_*\beta$ yields $\Theta_P = (b_*\beta)^{-1}$ and $B_*(u_\Theta) = b_*u_E$. The table therefore displays the coefficient actually evaluated, alongside the SI reference

$$k_B^{\text{SI}} = 1.380649 \times 10^{-23} \text{ J K}^{-1},$$

The numeral agrees exactly in this realization and in the specified transported bases. Its thermometric proof includes the spectral measure and both functionals: changing only their normalization changes the reader. Identifying u_E/u_Θ with J/K is the dimensional transport enabling comparison. The SI value is definitional and is therefore assigned no experimental uncertainty.

In gravity, the circular realization fixes

$$G_a = \vartheta_{108}^\circ \frac{c_{\text{int}}^5 t_0^2}{\hbar_a}, \quad \vartheta_{108}^\circ = (54/\pi)^2 = 295.4525715010569 \dots \quad (27.9)$$

The selector and reciprocity $G_a \hbar_a = \vartheta_{108}^\circ c_{\text{int}}^5 t_0^2$ are internal quantities of the declared realization. A dimensional evaluation of G_a must preserve, in addition to orientation a , the time, velocity, and action charts. The longitudinal and radial composition in Section 20 first determines

$$L = \frac{5759}{23040} \alpha_{\text{HMT}}^{16} L_*, \quad \hbar_{\text{ret}} = \eta_{\text{ret}} S_*, \quad t_0 = \frac{\pi_{\text{HMT}} L}{54 c_{\text{int}}}.$$

With the same constitutive velocity section, substitution into (27.9) yields

$$G_{\text{ret}} = \frac{c_{\text{int}}^3 L_*^2}{S_*} \left(\frac{5759}{23040} \right)^2 \frac{\alpha_{\text{HMT}}^{32}}{\eta_{\text{ret}}}.$$

Dimensional speed retains the constitutive normalization of the angular channels. With the already generated coordinates A_{deg} and C_{deg}^* ,

$$q_\pm = \exp \left[-\frac{\pi_{\text{HMT}}}{180} (A_{\text{deg}} \pm C_{\text{deg}}^*) \right], \quad (27.10)$$

$$r_\pm = \frac{1 - q_\pm^{90}}{1 - q_\pm^{120}}, \quad \hat{c} = \frac{1}{r_+ r_-}, \quad c_{\text{int}} = c_* \hat{c}.$$

In compatible positive bases, $\varepsilon = \varepsilon_* r_+^2$, $\mu = \mu_* r_-^2$, and $c_* = (\mu_* \varepsilon_*)^{-1/2}$; multiplication gives $c_{\text{int}} = (\mu \varepsilon)^{-1/2}$. Hence

$$G_{\text{ret}} = \frac{c_*^3 L_*^2}{S_*} r_N^2 \alpha_{\text{HMT}}^{32} \frac{\hat{c}^3}{\eta_{\text{ret}}} = \frac{L^2}{\hbar_{\text{ret}} (\mu \varepsilon)^{3/2}}. \quad (27.11)$$

The evaluator uses the dimensional section c_{int} declared below and recovers $c_* = c_{\text{int}}/\hat{c}$. Substitution verifies equivalence of these expressions in the same representation: $\hat{c}$ enters once in passing from c_* to c_{int} [16].

In the declared representation $L_* = 1 \text{ m}$, $S_* = 10^{-34} \text{ J s}$, and $c_{\text{int}} = 299792458 \text{ m s}^{-1}$, the result is

$$L = 1.616254982625126330168262501334562 \dots \times 10^{-35} \text{ m},$$

$$G_{\text{ret}} = 6.674299659414022706367776189 \dots \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}.$$

The comparison value $6.67430(15) \times 10^{-11}$ is retained separately. The difference is $-3.40585977 \dots \times 10^{-18} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, approximately -0.00227057 standard uncertainties of that reference. Stable computed digits describe the evaluation of the composition with its explicit readers and bases; experimental uncertainty belongs to the measured reference.

The external sources for this last table are the BIPM defining constants and the NIST CODATA 2022 table [6]. The experimental uncertainty in G is not transferred to h , $\hbar$, or k_B . Nor does an equality of digits in a chart eliminate the need to identify the reader producing the quantity.

Object	Internal evaluation	Dimensional reference
B_*	$1.380649 \times 10^{-23} u_E/u_\Theta$, from (27.8) and the marked thermometric rule.	$1.380649 \times 10^{-23} \text{ J/K}$, exact in SI.
G_{ret}	$6.6742996594140227 \dots \times 10^{-11} \text{ m}^3/(\text{kg s}^2)$, from $c_{\text{int}}^3 L^2/\hbar_{\text{ret}}$.	$6.67430(15) \times 10^{-11} \text{ m}^3/(\text{kg s}^2)$, CODATA 2022.

Table 13: Evaluations and references in the specified realizations. The thermal row retains the identification of bases explained in the text. The standard relative uncertainty of the G reference is approximately 2.24743×10^{-5} ; k_B has no definitional uncertainty. The gravitational evaluation and its difference from the reference use the longitudinal and radial composition declared in the text.

27.5 Reproduction of the evaluations

The program `verificar_comparaciones_II.py`, included with the sources, receives the stored internal coordinates in files separate from the external data. It checks the composition of the Barbero functional, the conversion between h and $\hbar$, the bibliographic root and its tail, the CKM moduli, the quartet J , and the residuals of those comparisons. It uses 75-digit decimal arithmetic and elementary functions evaluated after freezing the internal outputs. Execution instructions and program versions are included in the reproducible package. Working precision does not increase the digits available from the original calculation: only digits stable under its rounding are presented. The check records the files used and their hashes, so that a change of evaluation or external edition remains visible.

The added gravitational evaluation is reproduced by `verificar_reproduccion.py`, in the source directory `reproduccion_gravitatoria_20260926`. The focal set retains the tests of polar transport, metric composition, simultaneous rigidity, phase action, and joint composition, together with structural and angular substitution. The substitution verifiers read the archived regional expansions and register K , reconstruct the subsequent expressions, and compare two working precisions; the gravitational evaluations retain 71 stable significant digits. The general identities are proved in the body; the rational checks examine finite cases of those identities. Inputs, copy provenance, hashes, and the scope of each program are documented in the same directory. This execution reuses previously generated outputs and keeps focal reproduction, complete TPK execution, and Lean formalization distinct.

28 Exceptional prolongation: from dodecaphasic incidence to Moonshine

The numerical expression does not exhaust the state that generated it. In the preceding construction, APP preserves the two sheets and division with remainder and quotient; the TRIT determines the local regime and its orientation; the TPK composes selection, transport, carry, memory, and return. The Archimedean reading contracts compatible cylinders. The reading now under study instead preserves the incidence of positions and their transport between frames. Both start from the same arithmetic state with trajectory memory. An exceptional geometry is not reconstructed from the final decimals of π , φ , e , or α .

This point clarifies the meaning of the prolongation $\alpha/K \rightarrow$ exceptional incidence. The arrow does not assert that an isolated real number determines a code, a lattice, or a group. It transports the dodecaphasic record and the signed data that entered the determination of α , together with the words and frames arising from APP–TRIT–TPK. In particular, the ordered record K can be recovered from its rational constant κ under the previously fixed base, length, and origin conventions. Projection onto a support, by contrast, preserves only the selected incidences. For its part, K_{ph} denotes a different, reduced-memory reading, subsequent to the holonomy Γ_9 .

The exposition separates three mathematical operations: constructing the finite objects and their maps; prolonging them to a lattice and a vertex operator algebra; and subsequently recognizing the resulting objects through the relevant classical theorems. The names Paley, Witt, Mathieu, Golay, Leech, and Monster designate those recognitions and the explicit realizations that make them verifiable. They are not retrospective selectors of the seeds or coefficients of the generation.

28.1 Structural function of K and α in exceptional incidence

The link between numerical determination and exceptional realization admits an operational description. In the arithmetic reading, K acts in the normalization of regional records. In the incidence reading, its components select the support

$$B_K = \{j : K_j \geq 729\} = \{4, 6, 9, 10\}.$$

The same ordered record therefore participates in two specified functions: a positional composition and an incidence selection. Its rational realization preserves the record when base, length, and origin are fixed, while passage to the support retains the information needed for the combinatorial flag.

The signed record preceding normalization of α supplies the hexad H_{α}^- . The intersections of this hexad with the regional supports and APP incidence determine the marked point. That point individualizes the norm-54 vector used in the lattice neighbor. Thus the orientation entering arithmetic normalization acts again in geometric selection. The relation does not require deriving a lattice from a decimal approximation: it uses the signed data preceding its evaluation.

The exceptional realization successively preserves different types of relation. The code preserves linear combinations; the design preserves incidences between supports; gluing preserves discriminant classes; the lattice preserves its bilinear form; the vertex algebra preserves grading and the corresponding products. Their automorphism groups are defined with respect to these structures. The appearance of the Monster at the end of the construction expresses a property of the algebra $V^{\natural}$ obtained through the cited procedure, and Moonshine relates the graded traces of that action to modular functions.

This hierarchy gives a concrete mathematical meaning to the linking function attributed to K and α . The former is a reversible integral record with positional and incidence realizations; the latter is the arithmetic-compatibility coordinate whose oriented record also enters the marked incidence. The definitions acquire content through these operations and the identities presented in this chapter. The incidence development is a substantive part of the article's argument: it shows which relations persist when the construction is no longer evaluated as a real coordinate and is instead realized as an algebraic structure.

Articulation of regional coordinates and closure incidence. The role of this prolongation is understood through a distinction between three operations: preserving a record, evaluating a coordinate, and extracting an incidence relation. All three act on information produced by APP-TRIT-TPK. Their articulation makes explicit the article's structural thesis: a numerical realization has an operator provenance, and certain data from that provenance additionally admit a combinatorial and lattice realization. The equations allow both paths to be followed with their domains and the information remaining in each.

The starting point is the terminal state with its compatible prolongations. It preserves the residues and quotients of the two APP sheets, the TRIT regime and orientation, and TPK transport with its memory. The closure, propagation, and self-scale regions are determined in that domain before their readers establish the coordinates π, e, φ . At the same time, the oriented record of the twelve windows determines the signed vector U . The subsequent appearance of K arises from the integral transformation proved in §8.7:

$$U = \mathcal{H}_{12}K, \quad K = \frac{1}{4}\mathcal{H}_{12}U, \quad \mathcal{H}_{12}^2 = 4I_{12}. \quad (28.1)$$

The congruence defining $\mathcal{L}_H$ ensures integrality of the inverse. Thus K is a reversible coordinatization of the signed record in the relevant sublattice. Its positions preserve phase order; its differences D_3K, D_4K preserve transverse relations; the sum $Q(K)$ determines the uniform component. That structure explains why the record enters both numerical composition and the selection of a boundary configuration.

The periodic evaluation

$$K \mapsto \kappa_{\text{per}} = \frac{\sum_{m=1}^{12} K_m 1000^{12-m}}{1000^{12} - 1}$$

keeps the twelve blocks recoverable when base, length, and phase origin are fixed. Extracting a support performs a different operation: it preserves membership of each position in a class defined by a threshold and groups records with the same support. The distinction between these two maps is decisive. The scalar precision of κ_{per} and the incidence of B_K are two specific mathematical contents, related through the complete record from which they arise.

Alpha normalization and its signed configuration. The coordinate of α is determined in the positional route through composition of the regional blocks and the dodecaphase record. Before positional division, the oriented combination has components $s_j = P_j + E_j - \Phi_j - K_j^{\text{per}}$. Normalization satisfies

$$A_j = s_j + c_{j+1} - 1000c_j, \quad 0 \leq A_j < 1000. \quad (28.2)$$

This identity simultaneously preserves the normalized block and the contribution of the transported quotient. The limit of compatible prefixes determines the scalar coordinate of closure. The signed configuration, in turn, preserves which positions of the combination before normalization have a negative defect. In the initial window, that configuration is the vector s_0 of (28.12), and its negative support is H_{α}^- .

Two contents of the same operation are thereby defined. Positional normalization determines a real coordinate through its compatible remainders and quotients; the negative support determines a configuration of positions in the twelve-component cycle. The sign refers to the balance of records before normalization. Its incidence reading remains available because the state preserves that balance together with the normalized result. This articulation corresponds to the proved positional route; comparison with the analytic vacancy operators preserves the precise scope established in the chapter on α .

From ternary coding to the marked flag. The regions also supply the codes $w_{\pi}, w_e, w_{\varphi}$, and w_{APP} in the inherited frame. The map

$$\gamma_{A_W} : w \mapsto (w, wA_W)$$

preserves the six-component word as its first coordinate and adds its transformation by A_W . The relation $A_W^2 = -I_6$ fixes the quadratic compatibility of this realization. The next step, $c \mapsto \text{supp}(c)$, preserves the nonzero positions. For weight-six words, the support identifies exactly the pair $\{c, -c\}$, as proved by the minimum distance of the code. Thus, the sign quotient produces the hexads, whereas the complete code remains available for operations requiring its symbols.

In this chart, the two constructions of the tetrad coincide:

$$\{i : K_i \geq 729\} = H_e \cap P_\pi = B_K = \{4, 6, 9, 10\}. \quad (28.3)$$

The left-hand side arises from an integer comparison of the components of K ; the right-hand side, from the intersection of the coded supports of propagation and closure. The threshold belongs to the declared block chart and remains explicit in the reader's definition. The complete record preserves the values on which it acts, even when different thresholds produce the same subset. The equality therefore expresses compatibility between specified maps.

The support H_α^- completes that tetrad to a hexad. The condition of membership in support P_π , together with the intersection sizes $(4, 3, 2)$ relative to $H_e, H_\varphi, H_{\text{APP}}$, selects a unique hexad. The following proofs bring together the two determinations: that obtained through the sign of s_0 and that obtained through incidence. The result is organized as

$$\tilde{J}_* = (B_K \subset H_\alpha^-; P_\pi, H_e, H_\varphi, H_{\text{APP}}). \quad (28.4)$$

The inclusion is the flag; the four additional supports constitute its marked frame. Simultaneous permutations transport the frame and preserve its inclusions and intersections. The relevant information resides in this transported configuration and its isomorphism class.

The five regions of π recover their internal differentiation here. They share w_π as a visible ternary code but preserve distinct emissions, signatures, and multiplicities. In the binary realization on a marked dodecad, the transverse region corresponds to the octad whose intersection with the dodecad is the tetrad; the other four regions correspond to the lifts of the four hexads of its star. The negative region is distinguished by H_α^- , and chart order individualizes the three positive regions. This correspondence preserves the regional roles and distribution $432 + 4 \cdot 144 = 1008$, while also realizing the incidence partition $1 + 4$.

From the incidence origin to lattice modification. The marked flag determines a position through an equivariant set-theoretic operation:

$$o_* = (H_e \cap H_\varphi) \setminus (H_\alpha^- \cup H_{\text{APP}}) = \{1\}.$$

Its role is to distinguish one component among the twelve entering the lattice realization. The complete ternary code acts as a glue code in the discriminant module $(A_2^*/A_2)^{12}$. Its preimage defines $N(C_W)$, and the self-duality, weights, and distance of the code translate, respectively, into integrality and determinant properties, parity, and norm bounds for the glue classes. Here the symbolic information of the code performs a role that the isolated support no longer performs.

The A_2 metric and positive radial chamber are fixed in the lattice development. Within that chamber, the congruence needed for an even index-three neighbor has twelve realizations of minimum norm 54, permuted by changing the distinguished component. The origin o_* selects

$$v_\alpha = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \quad v_\alpha^2 = 2(4^2 + 11) = 54.$$

The coefficient 4 belongs to that solution of the congruence in the declared chamber. The subscript α records the incidence provenance of the selection: its content is the frame that entered closure and now individualizes a lattice direction.

Pairing with v_α determines the index-three sublattice $N_{\alpha,0}$, and adjoining the class $v_\alpha/3$ produces the neighbor N_α of (28.17). This modification preserves rank and recovers unimodularity and evenness; congruence selection eliminates the preceding roots, and local bounds exclude roots in the new classes.

Identification with Leech appears after those verifications. The binary realization of the five regions and this lattice construction are two prolongations of the common frame: the latter directly receives the ternary code as glue data.

Grading, unit, and automorphisms. The Frenkel–Lepowsky–Meurman construction is applied to the lattice N_α once its properties have been verified. It incorporates the oscillator space, central extension of the lattice, vertex product, and sector twisted by the lift of negation. The result is the algebra $V_{\text{HMT}}^\natural$ of (28.18), whose automorphism group is isomorphic to the Monster. Its action domain is thus fixed by an algebraic structure with grading and product. Borchers’ Moonshine theorem subsequently determines the graded traces with insertions of those automorphisms [7]; [8].

At this level, the unit acquires a precise realization: the weight-zero component is the vacuum line and contributes the coefficient 1 of q^{-1} in the graded character. At the cylinder level, refinement maps preserve the constant function through precomposition. Every conservation statement therefore has its object and its map. The development of the proofs makes this continuity of construction visible, from arithmetic operations and memory to incidence, metric, and vertex products.

This articulation places K and the closure of α in the main argument. K reversibly preserves an integral coordinate of the history; normalization of α composes the regional records with that coordinate; the signed configuration determines a flag; and the marked flag selects the lattice modification on which Moonshine is realized. The following proofs develop these maps, distinguishing reversible equivalences, information quotients, and realizations that add metric or algebraic structure. A provable relation between structure, equation, and value is thus expressed, preserving the attribution of each construction and the scope of its theorems.

28.2 Compatibility of the Archimedean and incidence realizations

Let X_n^{cal} be the space of admissible prefixes of length n of the arithmetic state with trajectory memory, with truncations $p_{n,m} : X_n^{\text{cal}} \rightarrow X_m^{\text{cal}}$ for $m \leq n$. The superscript indicates that the declared initial frame and its transport are preserved, not that a parameter is fitted against an experimental value. We write (w_j, f_j) for the visible six-trit word and the frame at step j . The frames satisfy a causal law

$$f_{j+1} = g_j(x)f_j,$$

where $g_j(x)$ depends on the prefix already constructed. Consequently, a prolongation does not alter earlier frames.

The chain

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

is preserved in this reading. The return of the phase resets neither the frame nor the memory of the state. Likewise, the census of 468 decimal emissions, the 243 visible six-trit words, and the ambient space $\mathbb{F}_3^6$, with 729 words, remain distinct. The ambient space will be used to construct a linear code; it does not replace the admissible domain of emissions.

For a six-trit block b , the Archimedean reading takes its integer address $\nu(b) \in \{0, \dots, 728\}$. A prefix determines

$$q_n = m + \sum_{j=0}^{n-1} \frac{\nu(b_j)}{729^{j+1}}, \quad I_n = [q_n, q_n + 729^{-n}].$$

Prefix compatibility gives nested cylinders whose diameter tends to zero. Their Cauchy class belongs first to the internal completion $\mathbb{R}_{\text{HMT}}$; its subsequent identification with the usual real line does not enter the choice of blocks.

For the incidence reading we introduce, on the same prefix, the graph map

$$\gamma_A : \mathbb{F}_3^6 \longrightarrow \mathbb{F}_3^{12}, \quad w \longmapsto (w, wA). \quad (28.5)$$

Words are written as row vectors. If $A^f = fAf^{-1}$, the change of frame satisfies the concrete identity

$$\gamma_{A^f}(wf^{-1}) = \gamma_A(w)(f^{-1} \oplus f^{-1}). \quad (28.6)$$

The calibrated codomain retains f together with the encoded word. We therefore define

$$Q_n(x) = ((f_j, \gamma_{f_j A_W f_j^{-1}}(w_j)))_{0 \leq j < n}. \quad (28.7)$$

Proposition 28.1 (Compatibility and prolongation of the incidence reading). *If $r_{n,m}$ truncates the encoded sequences, then*

$$r_{n,m} Q_n = Q_m p_{n,m}.$$

Thus the Q_n determine a unique map Q_∞ between the respective inverse limits. In each frame, the map preserves the visible word exactly, since $\text{pr}_1 \gamma_A = \text{id}$.

Proof. Two compatible prefixes have the same words and the same transport operators through index $m - 1$. By induction, they have the same frames at those indices. The first m components of (28.7) agree, proving the square. The universal property of the inverse limit gives Q_∞ . The recovery claim is the projection onto the first six coordinates of each graph. $\square$

This local injectivity has a precise domain. It does not prove that Q_∞ recovers all the deep memory of the TPK state if only the sequence of visible words and frames has been retained. Still less does it prove injectivity after passing from words to supports. Moreover, a general change of basis in $\text{GL}_6(\mathbb{F}_3)$ does not by itself preserve ordinary Hamming weight. In a general chart, the incidence and metric of the initial chart are also transported. The support calculations below are performed in the declared Paley chart; monomial relabelings directly preserve their coordinate interpretation.

28.3 The finite realization and the ternary code

The elliptic operator of the TRIT admits the following finite realization in the six-position chart. These positions coordinatize the six units $\{1, 2, 4, 5, 7, 8\}$ modulo nine, preserved by modular transport. On $\mathbb{P}^1(\mathbb{F}_5)$, with the order $\infty, 0, 1, 2, 3, 4$ declared, let C be the matrix with zero diagonal, $C_{\infty,a} = C_{a,\infty} = 1$, and $C_{a,b} = \chi(b - a)$ at the five finite positions. Here $\chi(1) = \chi(4) = 1$, $\chi(2) = \chi(3) = -1$, and $\chi(0) = 0$. This is a coordinate realization of the quadratic structure already transported; identifying the six positions with the projective line is chart data, not a new numerical input into the generator.

The quadratic-sum identities give $C^2 = 5I_6$. Reducing modulo three yields

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}, \quad A_W^2 = 2I_6 = -I_6. \quad (28.8)$$

The quadratic equality realizes the elliptic regime; it does not identify the field $\mathbb{F}_3$ with an Archimedean realization of the imaginary unit. They are distinct realizations of a common operator relation, with their corresponding domains.

Definition 28.2 (Code of the dodecaphasic chart). The ternary code associated with (28.8) is

$$C_W = \{(w, wA_W) : w \in \mathbb{F}_3^6\} \subset \mathbb{F}_3^{12}.$$

Its set of hexads is

$$\mathcal{H} = \{\text{supp}(c) : c \in C_W, \text{wt}(c) = 6\}.$$

Proposition 28.3 (Self-duality, distance, and the Witt design). *The code C_W is self-dual with parameters $[12, 6, 6]_3$. Its weight enumerator is*

$$\sum_{c \in C_W} z^{\text{wt}(c)} = 1 + 264z^6 + 440z^9 + 24z^{12}. \quad (28.9)$$

The set $\mathcal{H}$ has 132 elements and constitutes an $S(5, 6, 12)$ system: every set of five positions is contained in a unique hexad.

Proof. The generator matrix $[I_6 \mid A_W]$ has rank six and, by the symmetry of A_W , satisfies

$$[I_6 \mid A_W][I_6 \mid A_W]^\top = I_6 + A_W^2 = 0.$$

The code is self-orthogonal with dimension half its length and is therefore self-dual. Finite evaluation of the 3^6 words gives (28.9); this is an enumeration over the explicit matrix, without real values or external targets. In particular, the minimum distance is six.

If two weight-six words have the same support, one of the signs 1, -1 occurs at least three times among the six ratios of their nonzero coordinates. Subtracting the corresponding multiple would produce a nonzero word of weight at most three, which is impossible. Hence a support corresponds to exactly the pair $\{c, -c\}$, and there are $264/2 = 132$ supports. If two distinct supports shared five positions, the same argument would cancel at least three coordinates in a union of size seven, producing weight at most four. Thus a five-element set belongs to at most one hexad. Finally,

$$132 \binom{6}{5} = 792 = \binom{12}{5},$$

so it belongs to one and only one. □

The subsequent recognition is the extended ternary Golay code and the small Witt design [9]. The preceding proof explains which object is recognized. It does not replace the APP-TRIT-TPK genealogy with a search among known codes.

28.4 Dodecaphasic flag associated with K and α

In the frame inherited from the dodecaphasic record, the data are

$$\begin{aligned} K &= (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601), & (28.10) \\ w_\pi &= 010211, & w_e &= 201101, \\ w_\varphi &= 121200, & w_{\text{APP}} &= 022020. \end{aligned}$$

Applying γ_{A_W} yields the following incidences:

$$\begin{aligned} P_\pi &= \{2, 4, 5, 6, 7, 8, 9, 10, 11\}, \\ H_e &= \{1, 3, 4, 6, 9, 10\}, \\ H_\varphi &= \{1, 2, 3, 4, 7, 9\}, \\ H_{\text{APP}} &= \{2, 3, 5, 10, 11, 12\}. \end{aligned}$$

The first is the support of a weight-nine word; the other three are hexads. Through its high-block reading, the record supplies

$$B_K = \{i : K_i \geq 729\} = \{4, 6, 9, 10\} = H_e \cap P_\pi. \quad (28.11)$$

Here the integer 729 marks a boundary of the declared block format. It is not identified merely by cardinality with the ternary ambient space used in (28.2). In this particular record, all integer thresholds between 660 and 729 produce the same support: the support recovers neither the threshold nor the twelve blocks.

Before carry, the arithmetic route to α preserves the signed vector

$$s_0 = (7, 297, 353, -430, -715, -1199, -3, 286, -894, -528, 380, 663). \quad (28.12)$$

Its negative support is

$$H_{\alpha}^{-} = \{4, 5, 6, 7, 9, 10\}. \quad (28.13)$$

These signs are not extracted from the decimal expansion of the scalar α ; they are retained in the state preceding its numerical expression.

Proposition 28.4 (Agreement of the two flag selectors). *Within $\mathcal{H}$, the support (28.13) is the unique hexad H satisfying*

$$B_K \subset H \subset P_{\pi}, \quad (|H \cap H_e|, |H \cap H_{\varphi}|, |H \cap H_{\text{APP}}|) = (4, 3, 2).$$

Thus the signed reading and the incidence reading select the same flag $B_K \subset H_{\alpha}^{-}$.

Proof. The four hexads containing B_K are obtained by adjoining, respectively, the pairs

$$\{1, 3\}, \quad \{2, 11\}, \quad \{5, 7\}, \quad \{8, 12\}.$$

Only the second and third pairs are contained in P_{π} . For the second, the intersection with H_{φ} has size three, and that with H_{APP} has size three; it does not meet the last condition. For the third, the three intersections have sizes 4, 3, 2. This last pair produces precisely the negative support of (28.12). $\square$

It is useful to retain both the flag and its marked frame:

$$\mathcal{J}_* = (B_K \subset H_{\alpha}^{-}), \quad \tilde{\mathcal{J}}_* = (\mathcal{J}_*; P_{\pi}, H_e, H_{\varphi}, H_{\text{APP}}).$$

Union, intersection, and difference are equivariant under a simultaneous relabeling of the design. The marked class, not the position numbers taken in isolation, is the transported object.

Lemma 28.5 (Origin selected by the marked incidence). *For the preceding frame,*

$$o_* = (H_e \cap H_{\varphi}) \setminus (H_{\alpha}^{-} \cup H_{\text{APP}}) = \{1\}.$$

The rule is equivariant under every automorphism applied simultaneously to the data of $\tilde{\mathcal{J}}_$.*

Proof. The first intersection is $\{1, 3, 4, 9\}$. The subtracted union contains 3, 4, 9 and does not contain 1. Equivariance follows because a bijection commutes with set operations. $\square$

This origin will select a direction of the lattice neighbor once the A_2 metric and the positive chamber are declared. Incidence alone does not contain that metric. The lattice step will explicitly preserve both data.

28.5 Isometric intertwining of the incidence representations

Incidence defines a linear map whose action can be checked position by position:

$$I : \mathbb{R}^{12} \longrightarrow \mathbb{R}^{\mathcal{H}}, \quad (Iv)(H) = \sum_{p \in H} v_p. \quad (28.14)$$

Let $V_{11} = \mathbf{1}^{\perp} \subset \mathbb{R}^{12}$. The two Euclidean realizations are subsequent languages for the same finite design here.

Proposition 28.6 (Isometry of the dodecaphasic screen). *We have*

$$I^{\top} I = 36I_{12} + 30J_{12},$$

where J_{12} is the all-ones matrix. Consequently,

$$U_W = \frac{1}{6} I|_{V_{11}} : V_{11} \xrightarrow{\sim} E_{\text{inc}} := I(V_{11})$$

is an equivariant isometry for the permutation actions of the automorphisms of the design.

Proof. Each point belongs to $r = \binom{11}{4}/\binom{5}{4} = 66$ hexads, and each pair to $\lambda_2 = \binom{10}{3}/\binom{4}{3} = 30$. These are, respectively, the diagonal and off-diagonal entries of $I^\top I$. On $\mathbf{1}^\perp$, the J_{12} term vanishes, leaving $36I$. Finally, if g is an automorphism, $p \in H$ is equivalent to $g(p) \in g(H)$; hence $I\rho_{12}(g) = \rho_{\mathcal{H}}(g)I$. $\square$

The incidence information also has a concrete spectral reading. Define

$$G_{H,H'} = \binom{|H \cap H'|}{4}.$$

Lemma 28.7 (Action of the intersection kernel). *If $J_{132,12}$ is the rectangular all-ones matrix,*

$$GI = 30I + 15J_{132,12}.$$

Thus G acts by the scalar 30 on E_{inc} .

Proof. The (H, p) entry of GI counts pairs (T, H') with $T \subset H \cap H'$, $|T| = 4$, and $p \in H'$. If $p \in H$, there are ten four-element sets T containing p , each contained in four hexads, and five not containing it, each with a unique extension of the five-element set $T \cup \{p\}$. The total is $10 \cdot 4 + 5 = 45$. If $p \notin H$, the fifteen four-element subsets of H each give one extension: the total is 15. This yields the identity. The rectangular term vanishes on V_{11} . $\square$

There is indeed a material intertwiner here: its domain, codomain, entries, and normalization are determined. For example, any orthogonal projector P already constructed on V_{11} is transported to $Q = U_W P U_W^*$ on E_{inc} , with $U_W P = Q U_W$. This statement does not identify all the operators acting on these spaces solely by their trace.

28.6 The Witt star and the generation of M_{12}

Definition 28.8 (Star on a four-element set). For $B \in \binom{[12]}{4}$, the Witt star is the collection of the four hexads containing B . Its petals are the pairs $H \setminus B$. The permutation σ_B fixes B and exchanges the two points of each petal.

The existence of four hexads follows from

$$\lambda_4 = \frac{\binom{8}{1}}{\binom{2}{1}} = 4.$$

Two petals cannot share a point, since two hexads would then contain the same five-element set. They are therefore four disjoint pairs partitioning $[12] \setminus B$. This description is neither a five-pointed star nor a geometric metaphor: it is a four-to-one incidence of the design.

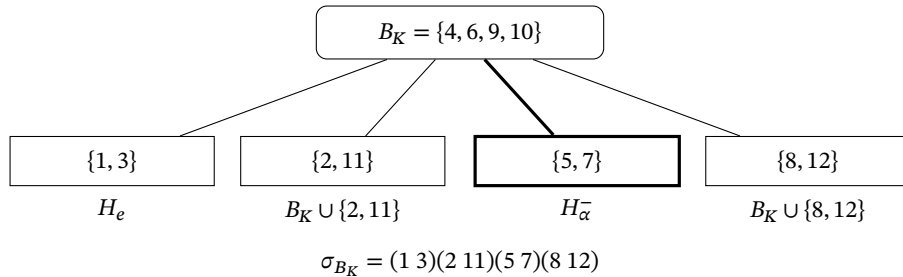


Figure 10: The concrete star of the dodecaphasic flag. Each line indicates the union of the central four-element set with the pair at its endpoint; the thick line marks the hexad also selected by the signs preceding the carry of α . The four pairs exhaust the eight outer positions.

Proposition 28.9 (The collection of stars). *The $\binom{12}{4} = 495$ permutations σ_B preserve $\mathcal{H}$. The group they generate has order 95040 and is the Mathieu group M_{12} in its action on the small Witt design.*

Proof. The preservation claim is a finite check on the 132 hexads of (28.2): for each four-element set the four petals are constructed and one verifies $\{\sigma_B(H) : H \in \mathcal{H}\} = \mathcal{H}$. Closure under composition of these permutations produces 95040 elements. In the lexicographic enumeration, it suffices to add successively six stars not yet contained in the group; the intermediate orders are 2, 6, 18, 360, 7920, 95040, and all 495 stars belong to the final group. The procedure and the six generators are made explicit in the technical provenance note. Finally, one applies the recognition of the automorphism group of $S(5, 6, 12)$ as M_{12} , whose order is $12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 = 95040$. $\square$

An individual star generates a cyclic group of order two. It does not by itself generate M_{12} , still less the Monster. Nor does the existence of 495 generators of a permutation group prove that each is the transport of a specific TPK loop. That assertion requires the corresponding loop map; the result established here is the finite incidence action.

Remark 28.10 (Equality of traces does not identify operators). On V_{11} , a star has spectrum $+1$ with multiplicity seven and -1 with multiplicity four. Its normalized trace is $3/11$. A rank-three projector on the same space also has normalized trace $3/11$, but spectrum $1^3, 0^8$. They are not conjugate: their minimal polynomials and spectra differ. The intertwiner (28.14) transports defined operators; it does not turn this scalar coincidence into an equivalence of operators.

28.7 The binary residue and the five regions of π

The next recognition uses the extended binary Golay code G_{24} , with parameters $[24, 12, 8]_2$. Its weights are 0, 8, 12, 16, 24, and the weight-eight supports form the design $S(5, 8, 24)$. Fix a dodecad D , the support of a weight-twelve word. This datum and the incidence isomorphism to be declared below belong to the realization chart; they are not used to generate or select the decimals of π .

Proposition 28.11 (Residue of a dodecad and lifting of hexads). *The set*

$$\mathcal{H}(D) = \{O \cap D : O \text{ is an octad, } |O \cap D| = 6\}$$

is an $S(5, 6, 12)$ system on D . Each of its hexads has a unique lift to an octad. For any four-element set $B \subset D$, the five octads containing it split into four residual lifts and a unique octad $O_B^\perp$ with $O_B^\perp \cap D = B$.

Proof. A five-element set $A \subset D$ belongs to a unique octad O . If $j = |O \cap D|$, the binary word given by the sum of O and D has weight $20 - 2j$. Among the permitted weights, with $0 \leq j \leq 8$, only $j = 2, 4, 6$ are possible. Since $A \subset O \cap D$, we have $j \geq 5$ and therefore $j = 6$. This gives existence and uniqueness of the residual hexad, as well as uniqueness of its lift by selecting five of its points. Moreover,

$$\lambda_4(S(5, 8, 24)) = \frac{\binom{20}{1}}{\binom{4}{1}} = 5, \quad \lambda_4(S(5, 6, 12)) = 4.$$

The four residual hexads on B give four distinct octads. The fifth cannot meet D in six points, since it would be another residual hexad; because it contains B , its intersection has size four and equals B . $\square$

A hexad-preserving chart $\ell : [12] \rightarrow D$ identifies $\mathcal{H}$ with $\mathcal{H}(D)$. Its existence is the recognition of the design already constructed as the small Witt design; a concrete choice of ℓ fixes the relabeling. The lift

$$\iota_D : \mathcal{H} \longrightarrow \{\text{octads of } G_{24}\}, \quad H \longmapsto O \text{ such that } O \cap D = \ell(H),$$

is injective because intersection with D recovers $\ell(H)$. The domain of this injectivity is the hexads, not the record K .

The relation to the five regions of π preserves richer data than the number five. The emissions and their signatures are:

Region	Multiplicity	Emission U_6	Signature
$\perp$	432	(501, 614, 498, 169, 272, 272)	555555
$++_1$	144	(810, 923, 870, 169, 272, 272)	339555
$++_2$	144	(810, 983, 810, 169, 272, 272)	393555
$++_3$	144	(870, 923, 810, 169, 272, 272)	933555
$--$	144	(870, 923, 810, 418, 674, 521)	933717

In each three-digit decimal block, the signature uses the modular difference of the first two digits. The five emissions have the same visible tritic word $w_\pi = 010211$, but not the same regional genealogy. The three positive regions preserve the tail 169|272|272; the negative region changes it to 418|674|521; the transverse region has a constant signature and greater multiplicity.

The marked extension of the regional chart sends the transverse region to $O_{\ell(B_K)}^\perp$, the negative region to $\iota_D(H_\alpha^-)$, and the three positive regions to the other three lifts of the star. This realizes the $4 + 1$ partition of the five octads on $\ell(B_K)$. The individual correspondence between the three positive names and the three remaining petals requires a chart order; incidence does not determine it without that datum. The precise claim is therefore a marked correspondence of regions, compatible with their roles and with the flag, not a canonical bijection inferred from the cardinality five.

The identity $432 + 4 \cdot 144 = 1008$ preserves the regional multiplicities. Also, $1008 = 36 \cdot 28$, and an octad has $\binom{8}{2} = 28$ pairs. These equalities are arithmetic catalog checks, not substitutes for the map ι_D or the regional chart.

28.8 From the ternary code to the Niemeier lattice

The passage to dimension twenty-four need not first go through the binary code. It is performed directly from C_W by discriminant gluing. Two compatible prolongations thus remain distinct: one preserves the residue $W_{12} \subset W_{24}$; the other constructs the lattice on which the vertex operator algebra will be realized.

Let $A_2 = \mathbb{Z}a_1 \oplus \mathbb{Z}a_2$ with Gram matrix

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad \rho = a_1 + a_2, \quad \rho^2 = 2.$$

Represent the three classes of A_2^*/A_2 by

$$\varpi_0 = 0, \quad \varpi_1 = \frac{2a_1 + a_2}{3}, \quad \varpi_2 = \frac{a_1 + 2a_2}{3}.$$

The map $j \mapsto \varpi_j + A_2$ additively identifies $\mathbb{F}_3$ with this discriminant. For $c \in C_W$, write $\phi(c) = (\varpi_{c_1}, \dots, \varpi_{c_{12}})$ and define

$$N(C_W) = \bigcup_{c \in C_W} (A_2^{12} + \phi(c)). \quad (28.15)$$

Theorem 28.12 (Ternary gluing). *The object (28.15) is a positive, even, unimodular lattice of rank 24. Its root system is exactly A_2^{12} ; it is therefore recognized as the Niemeier lattice with that root system.*

Proof. Linearity of the code makes the union of classes stable under addition. Self-orthogonality annihilates the discriminant pairing: the latter is a multiple of $c \cdot d/3$ modulo $\mathbb{Z}$. Thus the lattice is integral. The norm of a glue class is $2 \text{ wt}(c)/3$ modulo $2\mathbb{Z}$. The weights in (28.9) are multiples of three, so the lattice is even.

The index of A_2^{12} in $N(C_W)$ is $|C_W| = 3^6$. Consequently,

$$\det N(C_W) = \frac{3^{12}}{(36)^2} = 1.$$

Positivity follows from the orthogonal sum of the twelve A_2 metrics. In a nonzero class of A_2^*/A_2 , the minimum norm is $2/3$; every nonzero word has at least six nonzero positions. Hence a vector in a nontrivial glue class has norm at least $6 \cdot 2/3 = 4$. No norm-two roots occur outside A_2^{12} . Within it, the roots are precisely those of its twelve components. $\square$

28.9 Incidence selection of the rootless unimodular neighbor

The label $o_* = 1$ of the flag is now realized as one of the twelve A_2 components. This operation declares the preceding metric and the positive radial chamber

$$v(a) = \sum_{i=1}^{12} a_i \rho_i, \quad a_i = 1 + 3b_i, \quad b_i \in \mathbb{Z}_{\geq 0}.$$

The evenness requirement for the index-three neighbor is $v(a)^2 \equiv 0 \pmod{18}$. Since

$$v(a)^2 = 2 \sum_i (1 + 3b_i)^2,$$

this congruence is equivalent to $\sum_i b_i \equiv 1 \pmod{3}$. The least norm in the chamber is obtained with a single $b_i = 1$ and all others zero; it is 54. The marked origin selects

$$v_\alpha = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \quad v_\alpha^2 = 54. \quad (28.16)$$

The word “least” refers to this positive chamber. It does not assert minimality throughout the residual class: the vector $(a_1 - 2a_2, \rho, \dots, \rho)$ represents the same class modulo three and has norm $14 + 11 \cdot 2 = 36$. The chamber restriction is part of the construction data, not an omissible conclusion.

Let $N = N(C_W)$. Define

$$N_{\alpha,0} = \{x \in N : \langle x, v_\alpha \rangle \equiv 0 \pmod{3}\}, \quad N_\alpha = N_{\alpha,0} + \mathbb{Z} \frac{v_\alpha}{3}. \quad (28.17)$$

Theorem 28.13 (Leech neighbor determined by the marked flag). *With the declared metric and chamber, N_α is a positive, even, unimodular lattice of rank 24, with no vectors of norm two. By the corresponding classification theorem, it is isometric to the Leech lattice Λ_{24} .*

Proof. The character $x \mapsto \langle x, v_\alpha \rangle \pmod{3}$ is nonzero: a simple root of a component pairs with $a_i \rho_i$ to give $a_i \equiv 1 \pmod{3}$. Hence $[N : N_{\alpha,0}] = 3$ and $\det N_{\alpha,0} = 9$. Moreover, $v_\alpha/3 \in N_{\alpha,0}^*$ and $v_\alpha \in N_{\alpha,0}$. If $v_\alpha/3$ belonged to N , the integrality of N would make all pairings with v_α divisible by three, contradicting the nonzero character. The class of $v_\alpha/3$ therefore has order three. It follows that

$$[N_\alpha : N_{\alpha,0}] = 3, \quad \det N_\alpha = 9/3^2 = 1.$$

The extension is integral, and for $x \in N_{\alpha,0}$,

$$\left(x + m \frac{v_\alpha}{3}\right)^2 = x^2 + 2m \frac{\langle x, v_\alpha \rangle}{3} + 6m^2 \in 2\mathbb{Z}.$$

Hence it is even.

It remains to exclude roots, the decisive step. The roots of N belong to A_2^{12} . Their pairings with v_α are congruent to ± 1 or ± 2 modulo three; none belongs to $N_{\alpha,0}$. For the other two classes of the neighbor, each component belongs to one of the local classes

$$A_2 + \varpi_j \pm \rho/3, \quad j = 0, 1, 2.$$

Since $4\rho/3 \equiv \rho/3 \pmod{A_2}$, this description also holds in the distinguished component. Writing a local vector as $(ua_1 + va_2)/3$, its norm is $2(u^2 - uv + v^2)/9$. The six preceding classes have nonzero residues and minimum $2/9$: representatives minimizing the quadratic form are $(\pm 1, 0)$, $(0, \pm 1)$, $(1, 1)$, $(-1, -1)$ in the appropriate residues. A vector in a new class therefore has norm at least $12 \cdot 2/9 = 8/3 > 2$. There are no roots in any of the three classes. The classification of positive even unimodular lattices of rank twenty-four identifies the unique rootless case with Λ_{24} . $\square$

The argument does not recognize Leech from a coincidence of dimension. It constructs the gluing, proves integrality, evenness, and the determinant, and eliminates roots through a local bound. Only then does it apply the classification theorem. The dependence on α is preserved in the flag selecting o_* and in the vector (28.16); the scalar value of the constant is not converted into a length of the lattice.

28.10 The vertex operator algebra and the Monster

Let $\Lambda = N_\alpha$, already recognized as Leech. The next step uses its integral lattice, not a decimal expansion. The lattice vertex operator algebra construction is written

$$V_\Lambda = M(1) \otimes \mathbb{C}_\varepsilon[\Lambda].$$

Here $M(1)$ is the rank-twenty-four Heisenberg oscillator space; $\mathbb{C}_\varepsilon[\Lambda]$ is the group algebra twisted by the cocycle of the central extension of the lattice. The cocycle and the lift of the isometries are part of the construction: a bare vector-space sum does not determine the vertex products.

Negation $\lambda \mapsto -\lambda$ admits the standard lift θ . With its twisted module and the compatible parity choice, the Frenkel–Lepowsky–Meurman construction produces

$$V_{\text{HMT}}^{\natural} = V_\Lambda^+ \oplus (V_\Lambda^T)^+. \quad (28.18)$$

The sign $+$ denotes the even part of the lifted action. The expression includes the orbifold vertex product; it is not merely a sum of graded spaces.

Theorem 28.14 (Prolongation to the Moonshine module). *The algebra (28.18), constructed on the neighbor (28.17), is a realization of the Moonshine algebra $V^{\natural}$. It has central charge 24, zero weight-one component, and automorphism group isomorphic to the Monster $\mathbb{M}$. Its graded character is*

$$\text{Tr}_{V_{\text{HMT}}^{\natural}} q^{L_0-1} = J(\tau) = j(\tau) - 744 = q^{-1} + 196884q + 21493760q^2 + \cdots, \quad q = e^{2\pi i\tau}. \quad (28.19)$$

Proof. Theorem 28.13 verifies the lattice hypotheses: positivity, evenness, unimodularity, rank twenty-four, and absence of roots. The Frenkel–Lepowsky–Meurman construction and theorem for the Leech lattice then apply. They identify the orbifold with $V^{\natural}$ and establish its character and its automorphism group. Composing the flag, the neighbor, and the VOA construction functor gives the realization stated here. The group is not inferred from the coefficient 196884. $\square$

The theorem used is a subsequent classical recognition with verified hypotheses, not a new proof of the FLM construction in this article [7]. Moreover, the notation $q = e^{2\pi i\tau}$ is the conventional Fourier variable used to recognize the character. It does not introduce π or e as inputs into the coinductive generator that has already determined them.

Part of the trace mechanism can be seen before the orbifold. Only the zero lattice vector is fixed by negation; each oscillator contributes an alternating sign. Hence

$$\text{Tr}_{V_\Lambda} \theta q^{L_0-1} = t_2(\tau) := q^{-1} \prod_{n \geq 1} (1 + q^n)^{-24}.$$

This trace in the lattice algebra must not be confused with the character of $V^{\natural}$ without insertion. Nor does it by itself identify a Monster conjugacy class: the space, insertion, and lift must be specified before functions are compared.

28.11 Graded Moonshine traces and representation of the Monster

For $g \in \text{Aut}(V_{\text{HMT}}^{\natural})$, the trace with insertion is

$$T_g(\tau) = \text{Tr}_{V_{\text{HMT}}^{\natural}} g q^{L_0-1}.$$

The monstrous Moonshine theorem identifies these series as Hauptmoduln of the corresponding genus-zero groups [8, Theorem 1.1]. The character J , the group $\mathbb{M}$, and the collection $g \mapsto T_g$ are realizations of the algebra already constructed. They do not form a causal sequence in which one first guesses J and then infers the Monster.

The unit has a precise algebraic location here. The weight-zero subspace is the vacuum line, $V_0^{\natural} = \mathbb{C}\mathbf{1}$, and the coefficient of q^{-1} in J is therefore 1. In weight one the dimension is zero, accounting for the

absence of the constant term in $J = j - 744$. A morphism of vertex algebras preserves the vacuum vector. This property provides the technical meaning of preservation of the unit at this level; the unit of cylindrical observables is preserved by the precomposition proved earlier. The relation between the two levels must follow the construction maps with their units.

In weight two, the equality

$$196884 = 1 + 196883 = 196560 + 324 = 54(5 \cdot 729 + 1).$$

appears. The first decomposition corresponds to the trivial conformal line and the irreducible component of dimension 196883 in the Monster representation. The other expressions have lattice-theoretic or arithmetic readings. Their equality does not automatically identify their summands with TPK regions, words, or routes. Doing so would require presenting the maps between those spaces and proving that they respect the relevant actions.

In particular, an isometry $N_\alpha \simeq \Lambda_{24}$ induces a realization of (28.18), and a choice of isomorphism with $V^\natural$ transports its automorphism group. This does not construct an action of the Monster on the digits of π , on the blocks of K , or on all TPK histories. Such an additional assertion would require a representation ρ_{TPK} and a map F with an explicit square

$$F\rho_{\text{TPK}}(g) = \rho_{V^\natural}(g)F,$$

including domain, codomain, and preserved information. The absence of that step from the present claim does not remove the incidence intertwiner U_W , the flag, or the neighbor already constructed; it only delimits a different promotion, namely an action on the source state.

The chain established in this section can be summarized, keeping its transport data visible, as

$$\begin{aligned} &\text{APP-TRIT-TPK and arithmetic state with trajectory memory} \longrightarrow (w_j, f_j; K, s_0) \\ &\longrightarrow (C_W, \mathcal{H}; B_K \subset H_\alpha^-, P_\pi, H_e, H_\varphi, H_{\text{APP}}) \\ &\longrightarrow (N(C_W), o_*, \text{metric and chamber}) \longrightarrow N_\alpha \longrightarrow V_{\text{HMT}}^\natural. \end{aligned}$$

At the incidence level, the stars generate M_{12} , and the residual chart realizes the regional $4 + 1$ structure in W_{24} . At the lattice and VOA level, the construction gives Leech and the Moonshine module. These are prolongations of the same marked source, with different objects and maps. Causal continuity does not require identifying their codomains or erasing the explicit choices of frame, chamber, and lift.

The result of this section is therefore an assembled derivation: the finite and lattice proofs make the passage from dodecaphasic incidence to the hypotheses of the recognition theorems effective. The conclusion is algebraic: it preserves incidence, lattice structure, and the resulting module under the declared choices of frame, chamber, and lift. Mathematical recognition is distinct from the experimental comparison of constants.

29 From the action ellipse to T-duality

The reciprocity of the ellipse provides a shape coordinate for the exchange between visible value and boundary memory. The antecedent is the same enriched APP-TRIT-TPK state: APP preserves the sum and product sheets with their remainders and quotients; TRIT preserves their orientation; TPK transports sheet, carry, route, and memory. The angular and action outputs already constructed on the joint discrete structure of the continuum then determine the ellipse. This section composes its subsequent readers: elliptic shape, lattice exchange, momentum-winding realization, and modular evaluation [Section 15 and Proposition 29.1]; [Section 29].

The cutoff thus lies after the generation of $A, C^*, \hbar$. No prescribed radius or string constant selects those outputs. Boundary memory is first preserved in the remainder-quotient pair and then transported through the maps proved below.

29.1 The action shape determines the normalized radius

Return to the chart of section 15. Let $A \neq 0, \hbar > 0$, and $|C^*/A| < 1$, with both angles expressed in the same unit. The semiaxes remain labeled:

$$\eta_{\text{el}} = \text{artanh}(C^*/A), \quad a_S = \sqrt{2\hbar} e^{\eta_{\text{el}}/2}, \quad b_S = \sqrt{2\hbar} e^{-\eta_{\text{el}}/2}. \quad (29.1)$$

The balanced phase coordinates and their semiaxes have units of the square root of action. The radius below is dimensionless.

Proposition 29.1 (Shape radius and labeled reciprocity). *The rectangular chart of the ellipse uniquely determines*

$$R_0 = \sqrt{\frac{b_S}{a_S}} = e^{-\eta_{\text{el}}/2} = \left(\frac{A - C^*}{A + C^*} \right)^{1/4} > 0, \quad \tau_{\text{el}} = iR_0^2. \quad (29.2)$$

It preserves the signed angular ratio through

$$\frac{C^*}{A} = \frac{1 - R_0^4}{1 + R_0^4}. \quad (29.3)$$

Exchanging the labeled semiaxes transforms $(\eta_{\text{el}}, R_0, \tau_{\text{el}})$ into $(-\eta_{\text{el}}, R_0^{-1}, -1/\tau_{\text{el}})$, leaving the scale $a_S b_S = 2\hbar$ fixed.

Proof. Dividing the semiaxes of (29.1) gives $b_S/a_S = e^{-\eta_{\text{el}}}$. The map $R \mapsto R^2$ is bijective on $\mathbb{R}_{>0}$, so its positive root is unique. For $x = C^*/A \in (-1, 1)$, $2 \text{artanh } x = \log((1+x)/(1-x))$; hence $R_0^4 = (1-x)/(1+x)$. Solving for x proves (29.3). Exchanging a_S, b_S inverts their ratio and changes the sign of η_{el} . Finally, $-1/(iR_0^2) = iR_0^{-2}$, while the product of the semiaxes remains $2\hbar$. $\square$

The angular ratio is recovered; the two absolute angles and their generating history continue to reside in the enriched state. The point $R_0 = 1$ corresponds to the circular shape $C^*/A = 0$, not to an identification of all orientations or all histories.

29.2 Lattice exchange and equivalence of operators

Definition 29.2 (Stable domain of the dual reading). On $\mathcal{D}_0 = \mathbb{Z} \times \mathbb{R}_{>0}$, define

$$E_{R_0}(m, w) = \frac{m^2}{R_0^2} + w^2 R_0^2, \quad T_0(m, w, R_0) = (w, m, R_0^{-1}). \quad (29.4)$$

The first coordinate prolongs the visible reading, and the second the quotient reading. The full integer domain permits their exchange without requiring the quotient to belong to a section of nine representatives.

For the radius of (29.2), the matrix of this form is

$$D(\eta_{\text{el}}) = \begin{pmatrix} e^{\eta_{\text{el}}} & 0 \\ 0 & e^{-\eta_{\text{el}}} \end{pmatrix} = \frac{1}{2\hbar} \begin{pmatrix} a_S^2 & 0 \\ 0 & b_S^2 \end{pmatrix}. \quad (29.5)$$

Indeed, $R_0^{-2} = e^{\eta_{\text{el}}}$ and $R_0^2 = e^{-\eta_{\text{el}}}$. The ellipse equation $Q^2/a_S^2 + P^2/b_S^2 = 1$, multiplied by $2\hbar$, uses the inverse matrix $D(-\eta_{\text{el}})$. This distinguishes the squared-semiaxis matrix from the matrix of the elliptic equation.

Theorem 29.3 (Lattice duality and unitary conjugation). *The transformation T_0 is involutive and satisfies*

$$E_{R_0^{-1}}(w, m) = E_{R_0}(m, w). \quad (29.6)$$

On $\mathcal{H}_{\text{lat}} = \ell^2(\mathbb{Z}^2)$, let $H(R_0)$ be the diagonal operator with values $E_{R_0}(m, w)$, with domain

$$\mathcal{D}(H(R_0)) = \left\{ c \in \ell^2(\mathbb{Z}^2) : \sum_{m,w \in \mathbb{Z}} E_{R_0}(m, w)^2 |c_{m,w}|^2 < \infty \right\}.$$

Then $H(R_0)$ is self-adjoint and nonnegative. The exchange $U_T|m, w\rangle = |w, m\rangle$ is a unitary involution that transports the domains and satisfies

$$U_T H(R_0) U_T^{-1} = H(R_0^{-1}). \quad (29.7)$$

Proof. Exchanging the integers twice and inverting the radius twice returns the initial point. Moreover,

$$E_{R_0^{-1}}(w, m) = w^2 R_0^2 + m^2 R_0^{-2} = E_{R_0}(m, w).$$

The diagonal domain contains the finitely supported sequences and is dense. Multiplication by a real sequence is symmetric on that domain. If c belongs to the adjoint domain, evaluation against each basis vector forces the adjoint to have coordinates $E_{R_0}(m, w)c_{m,w}$. Their membership in ℓ^2 is equivalent exactly to the condition defining $\mathcal{D}(H(R_0))$. Hence the operator and its adjoint agree. Its nonnegative diagonal values prove positivity.

The exchange bijectively permutes the orthonormal basis, so it is unitary and its own inverse. For every sequence c ,

$$\sum_{m,w} E_{R_0^{-1}}(m, w)^2 |(U_T c)_{m,w}|^2 = \sum_{m,w} E_{R_0}(m, w)^2 |c_{m,w}|^2.$$

This equality proves exact transport of domains. Finally, $(U_T H(R_0) U_T^{-1} c)_{m,w} = E_{R_0}(w, m) c_{m,w} = E_{R_0^{-1}}(m, w) c_{m,w}$; this proves (29.7) on the entire domain, not only on the basis. $\square$

Orientation adds information that energy does not distinguish. In the phase plane, the exchange $S_2(Q, P) = (P, Q)$ and rotation $J_2(Q, P) = (-P, Q)$ transport the same quadratic form, but $S_2^*(dQ \wedge dP) = -dQ \wedge dP$ and $J_2^*(dQ \wedge dP) = dQ \wedge dP$. With $\lambda = P dQ$, one obtains directly

$$S_2^* \lambda = d(PQ) - \lambda, \quad J_2^* \lambda = \lambda - d(PQ).$$

Integration over a closed curve gives opposite actions under S_2 and equal actions under J_2 . Thus the index exchange in (29.4) and the symplectic transport of a trajectory are different maps, even when they yield the same energy.

29.3 Nonadic sections and carry transport

For $n \in \mathbb{Z}$, the positive section writes

$$r_9^+(n) = 1 + ((n-1) \bmod 9), \quad Q_9^+(n) = \frac{n - r_9^+(n)}{9}, \quad n = r_9^+(n) + 9Q_9^+(n).$$

The remainder of $n - 1$ is taken in $\{0, \dots, 8\}$. The change to the remainder section $0, \dots, 8$ is

$$C(r, Q) = (r - 9\varepsilon, Q + \varepsilon), \quad \varepsilon = \begin{cases} 1, & r = 9, \\ 0, & r \neq 9. \end{cases} \quad (29.8)$$

Thus $r + 9Q$ remains fixed. For $n = 9$, the two coordinates are $(9, 0)$ and $(0, 1)$: their uncorrected energies are $81/R_0^2$ and R_0^2 . Equality of the integer does not imply equality of those energy insertions. The following covariant form resolves the change of section without removing carry.

Definition 29.4 (Two quotient charts of the nonadic reading). Let $L_9 = \mathbb{Z}(9, -1)$, $S(m, w) = (w, m)$, and $L_9^\vee = SL_9 = \mathbb{Z}(-1, 9)$. For $L \in \{L_9, L_9^\vee\}$, define

$$\mathcal{E}_{R_0}^L([x]_L) = \min_{\ell \in L} E_{R_0}(x + \ell).$$

Transport between charts is

$$T_{\text{cov}} : (\mathbb{Z}^2/L_9) \times \mathbb{R}_{>0} \longrightarrow (\mathbb{Z}^2/L_9^\vee) \times \mathbb{R}_{>0}, \quad ([x]_{L_9}, R_0) \longmapsto ([Sx]_{L_9^\vee}, R_0^{-1}).$$

Theorem 29.5 (Covariance with crossing memory). *The preceding energies and transport are well defined. The change of section (29.8) preserves the class in $\mathbb{Z}^2/L_9$, and*

$$\mathcal{E}_{R_0^{-1}}^{L_9^\vee}([Sx]_{L_9^\vee}) = \mathcal{E}_{R_0}^{L_9}([x]_{L_9}). \quad (29.9)$$

The return exchange recovers the initial class and radius. The integer coordinate maps $N([x]) = x_1 + 9x_2$ and $N^\vee([y]) = 9y_1 + y_2$ are isomorphisms with $\mathbb{Z}$ and satisfy $N^\vee([Sx]) = N([x])$.

Proof. Replacing x with $x + \ell_0$, where $\ell_0 \in L$, permutes the candidates for the minimum. This minimum exists: for $L = L_9$,

$$E_{R_0}(x + k(9, -1)) = \left(\frac{81}{R_0^2} + R_0^2 \right) k^2 + 2 \left(\frac{9x_1}{R_0^2} - x_2 R_0^2 \right) k + E_{R_0}(x),$$

which tends to infinity with $|k|$; for $L_9^\vee$, the leading coefficient is $R_0^{-2} + 81R_0^2 > 0$. The change $C(x) - x = -\varepsilon(9, -1)$ belongs to L_9 . If $x' - x \in L_9$, then $Sx' - Sx \in L_9^\vee$, proving that transport is well defined. By (29.6),

$$\min_{\ell \in L_9} E_{R_0^{-1}}(Sx + S\ell) = \min_{\ell \in L_9} E_{R_0}(x + \ell).$$

This proves (29.9); $S^2 = I$ proves the return between the two charts.

Finally, the homomorphism $x \mapsto x_1 + 9x_2$ is surjective and its kernel is exactly L_9 , since $x_1 = -9x_2$. Similarly, the kernel of $y \mapsto 9y_1 + y_2$ is $L_9^\vee$. Their induced maps on the quotients are therefore isomorphisms. Substituting $y = Sx$ proves the final identity. $\square$

The recovered integer determines its remainders and quotients once the section is fixed. The energy-minimizing representative need not be that canonical representative. The quotient is a covariant representation of the nonadic reading; the source state also preserves sheet, route, orientation, boundary, memory, and survival. Nor is $L_9^\vee$ replaced by L_9 : the exchange is between two transported charts.

29.4 Momentum, winding, and realization scale

The circular realization introduces an explicit dimensional chart. Denote its radius by $R > 0$ and let $\alpha' > 0$ be a parameter with dimension length squared. This symbol differs from the dimensionless output α_{HMT} . Fixing the chart does not alter the angular generator or the shape radius. The normalization and its inverse are

$$\mathcal{N}_{\alpha'}(m, w, R) = \left(m, w, \frac{R}{\sqrt{\alpha'}} \right), \quad R = \sqrt{\alpha'} R_0.$$

The indices are now read as momentum number and winding number; the zero-mode components, expressed in units of inverse length, are

$$p_L = \frac{m}{R} + \frac{wR}{\alpha'}, \quad p_R = \frac{m}{R} - \frac{wR}{\alpha'}, \quad E_{\text{str}} = \frac{p_L^2 + p_R^2}{2} = \frac{m^2}{R^2} + \frac{w^2 R^2}{\alpha'^2}. \quad (29.10)$$

Theorem 29.6 (Dimensional conjugation of T-duality). *The transformation*

$$\mathsf{T}_{\alpha'}(m, w, R) = \left(w, m, \frac{\alpha'}{R} \right)$$

is an involution and satisfies

$$\mathcal{N}_{\alpha'} \mathsf{T}_{\alpha'} = \mathsf{T}_0 \mathcal{N}_{\alpha'}, \quad E_{\text{str}} = \alpha'^{-1} E_0 \circ \mathcal{N}_{\alpha'}, \quad (29.11)$$

where $E_0(m, w, R_0) = E_{R_0}(m, w)$. It preserves E_{str} , fixes p_L , and reverses the sign of p_R . For fixed oscillator data N_L, N_R and intercepts a_L, a_R , it also preserves the condition

$$\frac{\alpha'}{4}(p_L^2 - p_R^2) + N_L - N_R - a_L + a_R = 0, \quad \text{that is} \quad N_L - N_R = a_L - a_R - mw. \quad (29.12)$$

Proof. Two applications exchange the indices twice and send R to $\alpha' / (\alpha' / R) = R$. Also,

$$\mathcal{N}_{\alpha'} \mathsf{T}_{\alpha'}(m, w, R) = \left(w, m, \frac{\sqrt{\alpha'}}{R} \right) = \mathsf{T}_0 \mathcal{N}_{\alpha'}(m, w, R).$$

Substitution into the normalized form gives

$$E_0 \left(m, w, \frac{R}{\sqrt{\alpha'}} \right) = \frac{\alpha' m^2}{R^2} + \frac{w^2 R^2}{\alpha'} = \alpha' E_{\text{str}}(m, w, R).$$

For $R' = \alpha' / R$, the transformed modes are

$$p'_L = \frac{wR}{\alpha'} + \frac{m}{R} = p_L, \quad p'_R = \frac{wR}{\alpha'} - \frac{m}{R} = -p_R.$$

Thus both their sum and difference of squares are preserved. Direct expansion gives $p_L^2 - p_R^2 = 4mw/\alpha'$, transforming the first equation of (29.12) into the second. The exchange preserves mw ; with the other data fixed, it preserves the full condition. $\square$

With these orientations, the abbreviated formula $N_L - N_R = nw$ requires $a_L = a_R$ and $n = -m$. Retaining the sign in (29.12) preserves the dictionary between the two readings. The proved result covers conjugation of radii, zero modes, energy, and the declared level compatibility. The dimensional interpretation identifies coordinates through its chart; it does not turn R_0 alone into a length or infer a complete field theory from an equality of energies.

29.5 Modular evaluation and the Moonshine character

The exceptional construction already developed preserves the incidence of the same antecedent up to $V^{\natural}$. Its graded character has the expression $J(\tau) = j(\tau) - 744$, with the modular invariance used in Moonshine [7]; [8]. This J is a function on the upper half-plane $\mathbb{H}$; it is not the rotation J_2 , the exchange S , or the nonadic holonomy Γ_9 .

Proposition 29.7 (Modular semiconjugacy of the radius). *Let $\iota : \mathbb{R}_{>0} \rightarrow \mathbb{H}$, $\iota(R_0) = iR_0^2$, and $S_{\text{mod}}(\tau) = -1/\tau$. Then*

$$S_{\text{mod}} \circ \iota(R_0) = \iota(R_0^{-1}), \quad J(iR_0^2) = J(iR_0^{-2}). \quad (29.13)$$

In particular, the two labeled shapes of the ellipse have the same value under J .

Proof. The first identity is $-1/(iR_0^2) = iR_0^{-2}$. The matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ belongs to $\text{SL}_2(\mathbb{Z})$ and acts on $\mathbb{H}$ as S_{mod} . The modular invariance of J , established in its exceptional construction, gives $J(S_{\text{mod}}\tau) = J(\tau)$. Applying it to $\tau = \iota(R_0)$ yields the second identity. Proposition 29.1 identifies these two arguments with the moduli of the ellipses of opposite shapes. $\square$

The composition thus preserves an explicit chain:

$$(A, C^*, \hbar) \longmapsto (a_S, b_S) \longmapsto R_0 \longmapsto iR_0^2 \longmapsto J(iR_0^2).$$

Shape inversion is transported to radius inversion and to S_{mod} ; oriented action and change of section retain their own maps. The function J assigns a value to a modular class, while the marking of the axes distinguishes its two presentations and the TPK state preserves its genealogy. This is the proved relation between the action ellipse, T-duality, and Moonshine: a composition with domains and transport identities, whose modular value replaces neither the lattice operator nor the construction of $V^{\natural}$.

30 Conclusions

The construction obtains the Planck action sections and their return ratio, evaluates the Barbero–Immirzi parameter from oriented incidence, and produces the CKM angles with their phase and unitary amplitudes. It also determines the Boltzmann transducer for a positive thermal section and the common gravitational coefficient of realization R1–R8, whose circular specialization retains radius coincidence. Inscription determines $r_N = 5759/23040$; longitudinal composition gives $L = r_N \alpha_{\text{HMT}}^{16} L_*$; the radial metric and action fix $G_{\text{ret}} = c_{\text{int}}^3 L^2 / \hbar_{\text{ret}}$ under the explicit reduced-radius identification. The resulting identities connect the constants to their functions: energy per frequency, elementary area resolution, mixing information, thermal conversion, and geometric invariance.

The central relation is preservation of the geometric scale. The product $G_a \hbar_a$ preserves the area scale while action and gravitation transform reciprocally; incidence weighted by γ_{HMT} determines the resolution of that area; and $k_B \Theta_{\text{clk},a} \log 3 = \hbar_a / T_{108}$ relates the same action to the energy of a tritic decision. Joint observation of area and the modular Hamiltonian, with $\Delta_{\text{mod}} \neq 0$ and known normalization, recovers the sector occupations that a single aggregate reading does not distinguish. Direct inversion of (N, D) remains valid without that condition. Each identity retains the conditions of its dimensional, geometric, or spectral realization.

30.1 Planck constants, Barbero–Immirzi, and CKM mixing

The action scale contains two distinct determinations. The local quotient of the record has sixteen sheets; its multiplicative norm and autoscaling fix $\text{ord}_{10}(\varphi \alpha^{16}) = -34$. The regional characters fix the oriented sections and return quotient. Conversion between angular action and action per turn maintains the identity $h = 2\pi \hbar$. The angular chart transports representatives in degrees to radians while preserving both channel arguments and functional coefficients. This precision allows the dependence of action to be followed without a retrospective inversion of its coordinates.

The hexadic–octadic transition supplies the specific content of the Barbero–Immirzi functional. The incidence spaces fix the cardinalities; the oriented turn fixes the multiplicities; evaluation of the channels determines the scalar. Its subsequent use in the area operator and gravitational action makes visible the meaning of the structure producing the value. The parameter thus serves a demonstrative function in the relation between incidence, orientation, and geometry.

The CKM construction determines mixing through an explicit reader. The rational reader transports the preceding angular coordinates to three mixing angles and fixes their phase; its inverse reconstructs the initial coordinates. Unitary factorization makes the amplitudes explicit, and the Jarlskog invariant preserves CP-violation information under admissible phase changes. Its quantitative comparison uses the declared angular convention and physical reference. The leptonic construction is presented in its own spectral frames, preserving the distinction between the two species transports.

The numerical mixing results are

$$(\theta_{12}, \theta_{23}, \theta_{13}) = (13.003313589509, 2.398056157332, 0.213977382244)^\circ, \\ \delta_{\text{CP}} = 65.676173123554^\circ, \quad J_{\text{CKM}} = 3.1189723326632974 \times 10^{-5}.$$

Table 12 compares their sines, phase and invariant with PDG 2025 in the same parametrization: all five normalized marginal deviations satisfy $|z| < 0.029$. The result connects the generated coordinates with unitary amplitudes and with a phase measure that remains invariant under reparametrization.

The angular realization in Witt incidence establishes an explicit, polar-normalized isometric transport with inverse on its image. The metric equality in Corollary 18.7 specifies its function: composition with the angular reader preserves the induced metric, distinguishes different angular coordinates, and permits reconstruction of the three coordinates feeding the sector rules. Comparison between the record and incidence is thus made through a faithful map. The determinant representation preserves orientation; gauge changes transform the data jointly, according to 18.6; the complex realization preserves phase and the order of the unitary mixing factors. The selection of sector operations, their

evaluation, and transport of their result remain distinct, without identifying subsequent isometric transport with selection of the sector rules it evaluates.

30.2 The Boltzmann constant and energy per tritic decision

Action and period yield $\epsilon_a = h_a/T_{108} = \hbar_a/t_P$; the information of the uniform ternary distribution fixes the energy per nat $\epsilon_a/\log 3$. For a fixed positive thermal section $\Theta_{\text{clk},a}$, there is a unique positive linear transducer $k_B : \mathcal{L}_\Theta \rightarrow \mathcal{L}_E$ mapping it to that energy. Its expression in positive bases retains the transformation law $b' = (s_\Theta/s_E)b$. This determines both the dimensional function of Boltzmann and the conditions that individualize its coordinate: the thermal section, the bases, and, when internal selection is claimed, the reader that fixes them.

The same transducer acts on both orientations when the cycle temperatures have the ratio of the action sections, as in (22.10). It assigns the corresponding entropy quantity to information through $S = k_B s$, without identifying a distribution of continuations with a quantum bipartition or a reversible update with physical erasure. Energy, information, and entropy thus retain their domains when related to area and to the gravitational cell.

The capacity construction in 23 retains the calendar with its vacancies, the two carrier gradings, and the positive energy realization of memory. The decimal spectrum, capacity origin, and multiplicities determine intensity $b_* = 1.380649 \times 10^{-23}$ under the stated degree assignment. Currents fix the offset in their connected realization; centered transport preserves the reader and its reference.

Theorem 24.5 adds a precise thermometric consequence. Fixed reference and trial state produce an entropy increment $b_* s$; its joint reading with conditioned energy determines $\Theta_P = 1/(b_* \beta)$ and a unique positive transducer of coefficient b_* . Corollary 24.6 also proves the unique free-energy minimum. The fixed reference and pair of functionals are the conditions of that realization. On the graded carrier, the chronological capacity Hamiltonian gives exactly $\rho = q^N/Z_N(q)$; its relation to area and horizon preserves the different scales and finite relative-entropy identity.

Section 25 supplies the additive characterization of this weight and the conservative realization of its marks. Disjoint alternatives, their received measure and the reference unit fix the contribution $b_* s$; conditioning recovers the mean energy of the same trial. The instrument preserves the spectral complement, system energy and tritic transport with quotient and carry. A shared reference retains a single factor b_* when systems are composed and preserves their correlations. These proofs specify the operations of the thermal realization as well as evaluating its coefficient.

30.3 Gravitation and preservation of the area scale

The gravitational construction also preserves tensorial information not exhausted by the scalar coupling section. The incidence descent of 21.8, under its explicit conditions, produces the five-component carrier. The intertwiner Γ_5 realizes that carrier as symmetric trace-free tensors; its basis distinguishes the weights $\pm 2, \pm 1, 0$. Transverse reduction preserves two of them through a rank-two projector. This sequence simultaneously transmits the common coupling G_a , the orientation of its five components, and the reduced observation, according to (21.49). The physical interpretation preserves the dynamics and constraints specified in that development.

Inscription, the complete archive, and longitudinal and metric compositions first determine the radial reader. Energy and radius act on the same positive character; cancellation proves that the coupling coefficient is common to every mode in the declared domain. Rules R4 and R5 remain incorporated compositions, and R8 retains the physical identification of the reduced radius. The clock $t_0 = \pi_{\text{HMT}} L / (54c_{\text{int}})$ subsequently expresses the same length in the circular calendar.

Unitary transport preserves the coefficient and the records. Joint elimination of boundary and memory preserves radial–energy proportionality, the radial product, and the recoverable residual; in the complete resolvent the spectral parameter is transported as well, according to (20.24). This conservation includes static and dynamic memory within their domains. Reciprocity of the action sections preserves the product of gravitation and action and, with it, the area scale. The thermal

realization links the energy of the elementary duration to tritic information. In the specified horizon cell, energy, temperature, entropy, and half-action satisfy the proved identities. The article thus places the relation between information and gravity within an explicit mathematical composition.

30.4 Boundary information and entanglement entropy

The joint reading of area and sector information provides a concrete expression of boundary genealogy. Area aggregates occupations, while the weighted observable preserves their distribution between the two sectors. Their recovery is proved by inverting the incidence map. Entanglement entropy adds the reduced state and bipartition that give it meaning. Transporting these data allows spectral preservation, orientation preservation, and preservation of the information accessible in a reduction to be distinguished.

The sector formulation also obtains the mean area and mean modular information from the finite partition function. Their derivatives are expressed through occupation variances and covariances. Unitary inversion of the occupations preserves entropy and transforms area into its complement. This specifies a central phenomenon of the genealogical reading: a conserved quantity can coexist with a structural transformation detected by another observable. The Boltzmann transducer preserves its dimensional function in this relation between information, energy, and entropy.

Transport of the collective modes makes explicit how hexadic–octadic incidence reaches the area labels. At a fixed reference state, the modular first law relates variations of entropy and weighted occupation. For finite changes, the identity includes the relative entropy between states; retaining that term specifies the informational content distinguishing an infinitesimal variation from a finite transformation. Joint recovery of the occupations through area and incidence places this relation directly on the two boundary observables.

30.5 Reciprocity and exceptional continuation

The action ellipse and T-duality prolong the argument through transport invariants. Exchanging semi-axes preserves the determinant of the form; symplectic orientation distinguishes realizations with the same energy; momentum–winding exchange preserves the spectral operator under the corresponding domain identification. The exceptional realization maintains incidence up to Moonshine and allows modular reciprocity to be formulated in its own space.

The covariance realization in section 15.6 specifies the operator content of this reciprocity. In the two-standard-deviation normalization, the area h fixes $\det V_\eta = \hbar^2/4$, while angular anisotropy determines $\Delta P/\Delta Q = e^{-\eta}$. The constructed Gaussian state and its canonical operators realize exactly these two relations; the symplectic rotation exchanges their dispersions while preserving the commutator. The action determinant, ellipse geometry, and uncertainty saturation are thereby related in an explicit representation. The distinction between contour energy, $\hbar\omega$, and mean state energy, $\hbar\omega/2$, preserves the normalization of both readings.

The extension to M-theory requires a specification of compactification, world sheets, and the corresponding physical representations. The particle–antiparticle relation through orientation, advance, and delay constitutes a proposed extension, distinct from the covariance and reciprocity identities proved here.

Section 16 proves exact conservation of quadratic action under transport between ellipses, derives its Hamiltonian connection term, and establishes the positivity condition of the generator. The conformally symplectic version preserves normalized action and specifies the transported form.

30.6 Evaluation of the record and preservation of observation

The incidence evaluation in (8.5) makes explicit the operation converting a ledger of visits and oriented traces into twenty-five coordinates. Each triple of counts is represented through its decimal residues, while its quotients and provenance remain in the ledger. Proposition 8.3 determines when two collections of counts have the same image. Its content makes it possible to specify what information the resulting value uses and what additional information is preserved by the history that produced it. The incidence collection and traces feeding the evaluator retain their construction conditions; the subsequent invertibility of the record has a complementary function: exact recovery of that representation.

Preservation of observation acquires a dynamic formulation in 10.2. The finite identity retains the terminal state of the history; its prolongation controls the contribution of subsequent increments through a tail bound. When an auxiliary sector is eliminated, identities (2.30)–(2.31) also transport the source and the reading map. The content of a trajectory can thus be expressed in fewer components while preserving the effects of its excursions and returns. The transitivity of (2.35) ensures coherence of this operation when elimination proceeds in several stages. This is a precise consequence of treating memory as an operative part of the state.

30.7 Structural definitions and quantitative results

Definition through structure preserves the object prior to the scalar reading. For the action constants, that object is an oriented section with a multiplicative norm over the local quotient and a regional character. For Barbero–Immirzi, it is the evaluation of an oriented incidence. For mixing, it is a transport between coordinates and amplitudes. For Boltzmann, it is a map between lines of quantities. For gravitation, it is a reciprocal action section whose magnitude follows from the inscribed length and radial reading specified in R1–R8. These definitions determine different operations within the same genealogy.

For each quantity, the following table fixes the object that defines it, the operation that determines its reading, and the physical role developed in the body. The reconstruction and transport proofs establish how that information is preserved when the result is reused. In particular, 8.3 characterizes the representation of the record, 2.19 preserves observation under reduction, and 18.7 preserves the angular coordinates in the incidence realization. These three operations have their own domains and distinct demonstrative roles within the same sequence.

Table 14: Structural object, produced reading, and physical meaning of the article’s results.

Quantity	Structural definition and result	Physical reading and internal development
Planck	Section of the action line: $\hbar_{\text{pre}} = H_5 10^{-34} \mathcal{U}_S$, $\hbar_{\text{ret}} = \eta_{\text{ret}} 10^{-34} \mathcal{U}_S$. The local quotient has sixteen sheets, and $\text{ord}_{10}(\varphi \alpha^{16}) = -34$.	The section relates frequency and energy; the complete turn determines $h_a = 2\pi \hbar_a$. Development in 13; coordinates and comparison in 10.
Barbero–Immirzi	Oriented evaluation of the channels and incidence: $\gamma_{\text{HMT}} = \pi AC^*/180 + 12\Delta S_{90} - \Delta S_{120}$. The printed evaluation is 0.2740076726944592747	The coefficient enters elementary area resolution and the Cartan–Holst realization. The definitions in 14 and 19 preserve these two uses; (21.2) specifies the second.
CKM mixing	Image of the angular vector under the three sectorial maps; phase $\delta_{\text{deg}} = 9A_{\text{deg}}$. The angles and amplitudes are evaluated after the reader is constructed.	The Jarlskog quartet expresses phase information invariant under diagonal basis changes. The rules, their evaluation, and the unitary realization are developed in 17.

Quantity	Structural definition and result	Physical reading and internal development
Boltzmann	Positive transduction $k_B : \mathcal{L}_\Theta \rightarrow \mathcal{L}_E$ with $k_B \Theta_{\text{clk},a} \log 3 = h_a/T_{108}$. The state and measure of the continuations determine the information content on which it acts.	$S = k_B s$ dimensionally realizes the entropy of the reduced state. Energy and temperature bases and their transformations are preserved in 22; the additive characterization and instrument are proved in 25; the comparison is in 13.
Marked thermal realization	Fixed reference η , with multiplicities $(d_0, d_1, d_2) = (1, 380, 649)$ at degrees 23, 26, 29 and trace $b_* = 1.380649 \times 10^{-23}$. The readings $\mathcal{S}_P = b_* s$ and $\mathcal{E}_P = \text{Tr}(\rho H)$ determine $d\mathcal{S}_P/d\mathcal{E}_P = b_* \beta$ and $\Theta_P = (b_* \beta)^{-1}$.	With fixed reference, transported positive bases and a nonscalar Hamiltonian, 24.5 determines the transducer. Corollary 24.6 proves the unique minimum. Composition preserves $q^N/Z_N(q)$, the energy origin and the clock-horizon transports of (24.20).
Gravitation	Reciprocal section $G_a = c_{\text{int}}^3 L^2/\hbar_a$, with $L = (5759/23040)\alpha_{\text{HMT}}^{16} L_*$. The clock $t_0 = \pi_{\text{HMT}} L/(54c_{\text{int}})$ preserves the circular expression $G_a = \vartheta_{108}^\circ c_{\text{int}}^5 t_0^2/\hbar_a$, with $\vartheta_{108}^\circ = (54/\pi_{\text{HMT}})^2$.	The product $G_a \hbar_a$ preserves areal normalization. The longitudinal and radial construction, its rules, and uniqueness of the coefficient are proved in 20; the circular realization is retained in 21 and thermal confluence in 26.

Quantitative comparison preserves the particular meaning of each reference. For Planck, Table 10 compares dimensional coordinates with an SI defining constant; the relative residuals of the pre-return and post-return orientations are, respectively, approximately -1.82×10^{-15} and -6.93×10^{-10} . For Barbero–Immirzi, Table 11 compares two constructions: the internal evaluation and the root of the Ghosh–Mitra equation, whose approximate value is 0.2740668582328300. The relative difference is approximately -2.16×10^{-4} ; the object compared is a theoretical coefficient.

For CKM, Table 12 brings together the sines of the three mixing angles, the phase in radians, and the Jarlskog invariant. The angles and the matrix of moduli are made explicit in the adjacent development. The normalized marginal deviations defined there satisfy $|z| < 0.029$ relative to the PDG rows used. Each residual uses the indicated marginal uncertainty. Its reading is individual; a joint evaluation additionally requires the covariance matrix of those observables. For Boltzmann and gravitation, Table 13 presents the transducer, longitudinal and radial composition, dimensional charts, and external reference separately. This organization makes it possible to identify exactly which quantity each construction produces and which quantity each comparison evaluates.

Table 15 brings together the results of these comparisons alongside the preceding structural definitions.

30.8 Conservation of the areal scale and recovery of boundary information

The relation between information and gravitation is concentrated in two complementary maps. The first preserves the areal scale through $G_a \hbar_a$ and organizes its occupations through $\gamma_{\text{HMT}} a_0$. The second preserves sectorial resolution through the joint observation of area and weighted incidence. Inversion (19.8) recovers the occupations from area and the weighted observable; equations (19.26) and (19.27) link their variations to fluctuations of the state.

This gives the article’s central physical reading: the same configuration admits a geometric observation and an information observation related by a recoverable structure. An orientation change can preserve the entropy of the reduced state and transform area, as established by (19.29) and (19.30). Information conservation concerns the state and its transport; entanglement entropy additionally

Quantity	Result in the specified realization	Reference and comparison
Planck, before return	$\hbar_{\text{pre}} = 1.0545718176461544696 \dots \times 10^{-34} \text{ J s.}$	SI: $\hbar_{\text{SI}} = \hbar_{\text{SI}}/(2\pi)$, exact. Its coordinate is $1.0545718176461563912 \dots \times 10^{-34} \text{ J s.}$ Relative residual $-1.822221 \times 10^{-15}$.
Planck, after return	$\hbar_{\text{ret}} = 1.0545718169150390611 \dots \times 10^{-34} \text{ J s.}$	Same SI reference. Relative residual $-6.932836 \times 10^{-10}$. For $\hbar_a = 2\pi\hbar_a$, the residuals are identical.
Barbero–Immirzi	$\gamma_{\text{HMT}} = 0.2740076726944592747 \dots$	Ghosh–Mitra: approximately 0.2740668582328300 . Relative difference $-2.159529202 \times 10^{-4}$ between theoretical constructions.
CKM mixing	$\delta_{\text{CP}} = 65.676173123554^\circ$, $J = 3.1189723326632974 \times 10^{-5}$. The three angles in (27.6) are $\theta_{12} = 13.003313589509^\circ$, $\theta_{23} = 2.398056157332^\circ$, and $\theta_{13} = 0.213977382244^\circ$.	PDG 2025: all five comparisons of $s_{12}, s_{23}, s_{13}, \delta, J$ satisfy $ z_x < 0.029$ with the marginal uncertainties in Table 12.
Boltzmann	$B_* = 1.380649 \times 10^{-23} u_E/u_\Theta$, with the bases and thermometric rule of (27.8).	SI: $1.380649 \times 10^{-23} \text{ J/K}$, exact. The numerals coincide under the declared dimensional transport.
Length L	$L = 1.6162549826251263301 \dots \times 10^{-35} \text{ m}$, for $L_* = 1 \text{ m}$.	Longitudinal and radial composition: $G_{\text{ret}} = c_{\text{int}}^3 L^2 / \hbar_{\text{ret}}$, with the same velocity and action bases.
Gravitation	$G_{\text{ret}} = 6.6742996594140227 \dots \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$.	CODATA 2022: $6.67430(15) \times 10^{-11}$ in the same units. Difference of approximately -0.00227057 standard uncertainties.

Table 15: Summary of the quantitative comparison. Evaluations, bases, and references are those in Tables 10–13. The relative residuals of the action constants, the theoretical comparison of Barbero–Immirzi, and the normalized residuals of CKM and gravitation retain their respective meanings.

Observable	HMT evaluation	PDG 2025	z_x
s_{12}	0.225007404757	0.22501 ± 0.00068	-0.003817
s_{23}	0.041841757010	$0.04183^{+0.00079}_{-0.00069}$	0.014882
s_{13}	0.003734601164	$0.003732^{+0.000090}_{-0.000085}$	0.028902
$\delta \text{ (rad)}$	1.146265461116	1.147 ± 0.026	-0.028251
$10^5 J$	3.118972333	$3.12^{+0.13}_{-0.12}$	-0.008564

Table 16: CKM values and marginal comparison collected in the conclusions. The three angles in Table 15 are compared through $s_{ij} = \sin \theta_{ij}$, together with the phase in radians and the invariant J . The values, uncertainties, and residuals from Table 12 are retained, corresponding to the PDG 2025 three-generation fit. The last row expresses both the value and its uncertainty in units of 10^{-5} . The residuals are marginal comparisons of correlated observables.

concerns the bipartition and reduction. The horizon identity in 26 uses these roles in the specified geometric and thermal realization.

The memory identities also allow the scope of variable reduction to be specified. Equality of observations in (2.31) preserves contributions of the eliminated sector through the transformed effective operator, source, and reader. The entropy of the reduced state, by contrast, is evaluated with the bipartition and measure specified in 19.10. The two constructions explain different roles of memory: the first preserves a reading of transport; the second quantifies the information accessible to the subsystem. Their articulation keeps explicit which structure is transported and which reduction defines the physical observable.

The explanatory advance is therefore expressed through a precise order: generated structure, reading map, resulting value, and physical role. Equality of two values is studied together with equality or difference of their structures; the recovery maps determine which of those differences retains observable meaning.

The quantitative comparisons retain the distinction between a unit-defining constant, a theoretical coefficient, and mixing observables with uncertainties. They examine the coordinates produced in the declared charts, while the structural identities determine what is preserved under transport: the action return ratio, the orientation of the Barbero–Immirzi functional, the metric and phase of mixing, thermal transduction, and the gravitational product.

The article's contribution is this interconnected determination of constants and their physical relations. The APP–TRIT–TPK composition constructs numerical regions and dodecaphase memory before their evaluations; subsequent readers produce the action sections, angular coordinates, and boundary observables. In the specified horizon realization, $T_H S_H = E_\Lambda$, and the cell of radius ℓ_P combines decision energy and half-action. Action fixes energy per duration; Barbero–Immirzi resolves areal incidence; Boltzmann relates energy and information; and gravitation preserves the geometric unit $G_a \hbar_a / c_{\text{int}}^3$. These relations express the meaning of the constants within a common structure and retain the precise conditions under which each is determined.

A Electronic antecedent and dimensional realization

The electronic construction receives the arithmetic state with trajectory memory produced by APP-TRIT-TPK and its numerical coordinates. At this point in the development, π , φ , e and α designate the coordinates already obtained from the same genealogy, in the constructive order set out above. They are not parameters taken from a physical table. The electronic section is a subsequent realization of that joint discrete structure of the continuum: it preserves the cell of origin, the two arithmetic sheets, orientation, carry, and return memory before producing a scalar.

Three objects should be distinguished from the outset. The first is the central section and its transport history; the second is the dimensionless coordinate obtained from it; the third is its realization on an energy or mass line. Equality between two numerical coordinates does not identify these three levels. In particular, Euler's number e , the electronic coordinate ε_e , and a rest energy E_e will have distinct notations.

The aim of this section is to establish the composition

$$\varepsilon_e^\beta = \frac{\sqrt{3}}{4} \left[\exp\left(\frac{\varphi}{\pi^2}\right) - 22\alpha^3 \right] (1 + 15\Delta_4) R_{\text{act}}^{80-54\alpha+6\Delta_4}, \quad (\text{A.1})$$

explaining separately the origin of the prefactor, the two integers of the basal stage, the normalization of Δ_4 , the action ratio, and the return triple. The dimensional chart is introduced only after this composition. The result does not consist in making a dimensionless number acquire units merely through its resemblance to a tabulated quantity.

Identity of the section and composition of its observables. The structural identity of the electron is introduced through the central section and its transport operations. Inversion of the APP chart selects the cell (5, 5) by its fixed-point property. The two arithmetic sheets allow the deformations preserving the sum and the product residue to be studied. The result distinguishes the two orientations (8, 2) and (2, 8) and preserves a difference of one quotient unit. This is the first level of information: a cell, an involution, a preserved fibre, and a minimal integer defect. No mass evaluation is required to state or prove it.

The passage to arithmetic projections introduces a second level. The sum and product fibres determine conditional expectations on the same finite space. Their commutator measures dependence on the order of these two projections. The norm $\sqrt{3}/4$ follows from the spectrum of their pairing; the block attaining this norm also preserves the quadratic relations proved below. Thus, the scalar prefactor and the local algebra have a common provenance, but contain different information. The norm retains the magnitude of the commutator, while the matrices preserve its orientation and its composition law.

Areal-volumetric transfer incorporates the regional coordinates at a third level. The mode calendar determines the incidence matrix and its quadratic action. The closure mark produces the ratio π/φ^2 ; interchanging the two arguments provides the orientation φ/π^2 used in the electronic exponential character. The two expressions are related by an explicit involution. This relationship explains why a regional coordinate may appear in the electronic equation with a power or in a position different from that in its first reading: the operation applied is part of the mathematical datum that is preserved.

The cubic alpha term belongs to the regional complement of a marked return. Counting its positions yields $27 - 5 = 22$, while the three sheets of each of the five regions yield $3 \cdot 5 = 15$. The proofs preserve the domain of these counts and the orientation determining their signs. The basal formula therefore composes an incompatibility norm, a transfer character, a position deficit, and a torsional memory. Writing it on a single line expresses the composition of these operations; explaining it requires reconstructing them in that order.

The transition from the basal coordinate to the complete coordinate preserves the same central section. The return factor uses the history records, and the two stated readers recover the triple (80, 54, 6) along that trajectory. Multiplication by $R_{\text{act}}^{80-54\alpha+6\Delta_4}$ modifies the evaluation, but keeps its origin and antecedents identified. The basal value constitutes an intermediate stage of the composition;

the complete value also includes this return contribution. The physical comparison must use the stage specified by the formula being evaluated.

The action scale intervenes subsequently with a different function. A ratio of two action sections is dimensionless and cancels their common scale. The absolute realization instead uses an action basis, a temporal reader, and the map to energy or mass. This separation allows the construction of the scale and the identity of the electronic section to be preserved simultaneously. The chronological quantum provides a scale reference; the central section determines the structural factor expressed relative to it. The dimensional equations of this chapter keep that composition explicit.

The electron's relevance to the overall argument arises from this convergence of operations. The same arithmetic that produced the regions now determines a central cell, its deformations, and its projections. The fine-structure constant, constructed earlier, acts in this composition as a closure coordinate. Return information determines the complete evaluation, and dimensional realization allows subsequent comparison. Explaining the electron from its structure consists in preserving this succession of objects, maps, and values at every step of the proof.

A.1 Center, orientation, and vacancy on a preserved fiber

The cellular support under consideration is $\{1, \dots, 9\}^2$, with both APP evaluations, $r + c$ and rc . The central inversion is

$$\rho(r, c) = (10 - r, 10 - c). \quad (\text{A.2})$$

The center is not chosen by a mass: it is characterized by a property of the support that precedes its physical reading.

Proposition A.1 (Uniqueness of the center). *The inversion ρ has a unique fixed point, $(5, 5)$.*

Proof. The equations $r = 10 - r$ and $c = 10 - c$ are equivalent to $2r = 2c = 10$. Their unique solution in the nonadic chart is $(5, 5)$. $\square$

The preceding property identifies a cell, not a unique deep history. A route adds the initial choice of orientation and its compatible prolongations. In the chart used here, the central oriented section begins at $(5, 5)$ with both orientations positive; it will be denoted by Γ_e . The return of its cellular projection does not erase the transported orientations or the carry record of the section with its trajectory memory.

Vacancy is now considered, after the closure of α , as a local operation on this section. Its arithmetic can be isolated in a finite lemma; this isolation does not make it a substitute for the preceding generator of constants.

Definition A.2 (Central fiber and quotient defect). The central additive fiber is

$$\mathcal{F}_{10} = \{(5 + t, 5 - t) : t \in \mathbb{Z}, |t| \leq 4\}.$$

In the chart $n = \rho_9(n) + 9q_9(n)$, with $\rho_9(n) \in \{1, \dots, 9\}$, the quotient defect relative to the center is defined as

$$v(t) = q_9(25) - q_9(25 - t^2),$$

when both products have the same nonadic residue.

Lemma A.3 (Minimal central vacancy). *The only points of $\mathcal{F}_{10}$ that preserve the multiplicative residue of the center are*

$$(5, 5), \quad (8, 2), \quad (2, 8).$$

At the two noncentral points, the quotient defect is exactly one.

Proof. The sum is constantly ten, whereas

$$(5 + t)(5 - t) = 25 - t^2.$$

Preservation of the residue is equivalent to $t^2 \equiv 0 \pmod{9}$, that is, to $3 \mid t$. In the integer interval $[-4, 4]$, the solutions are $t = -3, 0, 3$. For the two nonzero solutions,

$$25 = 7 + 9 \cdot 2, \quad 16 = 7 + 9 \cdot 1.$$

The residue seven is preserved and one unit of quotient is lost. There is no other admissible noncentral displacement in that domain, so the positive defect is minimal. The interchange $t \mapsto -t$ permutes the two outputs and preserves the defect. $\square$

Vacancy here is not a numerical zero: it is a difference between two states with the same residual reading and different quotients. If only the residue were preserved, that difference would disappear. Orientation also distinguishes the two output points. Neither the quotient defect nor that orientation is automatically identified with an electric charge or a mass; their physical realization requires the corresponding readers.

A.2 Commutator norm of the arithmetic projections

The electronic prefactor is obtained without using α , torsion, or a dimensional unit. The domain is the third realization of APP, $\mathcal{A}_3 = \{1, \dots, 9\}^3$, equipped with the uniform measure. We define

$$\Sigma(x) = \rho_9(x_1 + x_2 + x_3), \quad \Pi(x) = \rho_9(x_1 x_2 x_3).$$

The conditional expectations R_Σ and R_Π are the orthogonal projections onto functions constant on each additive and multiplicative fiber, respectively. Thus, for a function f ,

$$(R_\Sigma f)(x) = \frac{1}{|\Sigma^{-1}(\Sigma(x))|} \sum_{y: \Sigma(y)=\Sigma(x)} f(y),$$

and analogously for R_Π . The two projections act on the same space; they do not represent two independent arithmetic supports.

Lemma A.4 (Two projections and principal angles). *If $s \in [0, 1]$ is a singular value of the pairing between orthonormal bases of the ranges of two projections P, Q , its nontrivial block contributes $s\sqrt{1-s^2}$ to $\|[P, Q]\|$.*

Proof. On the corresponding principal plane, one can write

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} s^2 & s\sqrt{1-s^2} \\ s\sqrt{1-s^2} & 1-s^2 \end{pmatrix}.$$

Their commutator is

$$[P, Q] = s\sqrt{1-s^2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The final matrix is orthogonal, so its norm is one. Common or orthogonal blocks have zero commutator. The orthogonal decomposition into principal planes gives the claim and reduces the total norm to the maximum of these contributions. $\square$

Proposition A.5 (Exact incompatibility norm). *On $\ell^2(\mathcal{A}_3)$,*

$$\|[R_\Sigma, R_\Pi]\| = \frac{\sqrt{3}}{4}. \quad (\text{A.3})$$

Proof. Let $N_{ab} = |\Sigma^{-1}(a) \cap \Pi^{-1}(b)|$. Integer enumeration of the three symbols yields

$$N = 9 \begin{pmatrix} 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 1 & 0 & 2 & 0 & 1 & 2 & 0 & 0 & 3 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 0 & 0 & 2 & 1 & 0 & 2 & 0 & 1 & 3 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 0 & 1 & 2 & 0 & 0 & 2 & 1 & 0 & 3 \end{pmatrix}.$$

Each row sums to 81; the column sums are

$$(m_1, \dots, m_9) = (36, 36, 108, 36, 36, 108, 36, 36, 297).$$

The normalized indicator functions of the fibers have pairing matrix $C_{ab} = N_{ab}/\sqrt{81m_b}$. Consequently,

$$M = CC^T, \quad M_{ac} = \sum_{b=1}^9 \frac{N_{ab}N_{cb}}{81m_b}$$

is a rational matrix whose spectrum consists of the squares of the required singular values.

Exact multiplication shows that $(1, \dots, 1)$, $(1, -1, 0)^3$, and $(1, 1, -2)^3$ are eigenvectors with eigenvalues 1, 1/4, and 7/132, respectively. The subspace of vectors supported on $\{3, 6, 9\}$ whose sum is zero has dimension two and eigenvalue 1/18. The zero-sum vectors supported on $\{1, 4, 7\}$ or $\{2, 5, 8\}$ form a four-dimensional kernel. These subspaces are independent and their dimensions sum to nine. The complete spectrum is therefore established:

$$\text{spec}(M) = \left\{1, \frac{1}{4}, \frac{7}{132}, \frac{1}{18}, \frac{1}{18}, 0, 0, 0, 0\right\}.$$

The maximum of $\lambda(1 - \lambda)$ on this set is 3/16, attained at $\lambda = 1/4$. The preceding lemma gives $\sqrt{3/16} = \sqrt{3}/4$. $\square$

The mode $(1, -1, 0)^3$ is precisely the distribution of values of the tritic phase selector in the nonadic chart. The scalar norm retains a measure of incompatibility, but does not by itself preserve the orientation of that mode or reconstruct the two projectors. That memory remains in the route state accompanying the scalar reading.

Proposition A.6 (Algebra of the local electronic block). *On the principal plane attaining (A.3), let*

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 1/4 & \sqrt{3}/4 \\ \sqrt{3}/4 & 3/4 \end{pmatrix}, \quad S = 2P - I, \quad J = \frac{4}{\sqrt{3}}[P, Q].$$

Then $S^2 = I$, $J^2 = -I$, $SJ = -JS$, and the real algebra generated by S, J is $M_2(\mathbb{R})$, a realization of $\text{Cl}_{1,1}$. Moreover,

$$n_{\pm} = \frac{S \pm J}{2}, \quad n_{\pm}^2 = 0, \quad n_+ n_- + n_- n_+ = I.$$

Proof. We have $S = \text{diag}(1, -1)$ and $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Multiplication verifies the three relations. The matrices I, S, J, SJ are linearly independent and form a basis of $M_2(\mathbb{R})$, proving the claim about the algebra. Finally, $(S \pm J)^2 = S^2 + J^2 \pm (SJ + JS) = 0$, and the sum of the cross products is $(S^2 - J^2)/2 = I$. $\square$

In this block, an elliptic direction, a hyperbolic direction, and two parabolic null directions coexist. The result describes a local matrix representation of the arithmetic of the two sheets. It does not identify all of APP with $M_2(\mathbb{R})$, or with a quantum Latin square; such claims would require other domains and other relations. Nor does the exponential belong exclusively to the parabolic regime: it can be applied to any of the generators of this block.

A.3 Transfer between sheets and coefficients of the basal stage

The primitive tritic calendar visits the modes (Σ, Π, Π) . Counting consecutive pairs, including the closure of the block, yields one transition $\Sigma \rightarrow \Pi$, one $\Pi \rightarrow \Pi$, and one $\Pi \rightarrow \Sigma$. Their primitive incidence is

$$F_{av} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

This matrix realization is obtained after the calendar. It is not used to replace the coinductive reader that has already generated φ . Its characteristic polynomial is $X^2 - X - 1$, and recognition of the two roots as φ and $-\varphi^{-1}$ describes the action of that output in modal transfer. The quadratic action has eigenvalues $\varphi^2, -1, \varphi^{-2}$; the last is the positive stable mode. The regional mark π , applied once, gives π/φ^2 .

Lemma A.7 (Interchange of the two orientations). *Let $\mathcal{R}(x, y) = x/y^2$ and $J(x, y) = (y, x)$, for $x, y > 0$. Then*

$$J^2 = I, \quad \mathcal{R}(\pi, \varphi) = \frac{\pi}{\varphi^2}, \quad (\mathcal{R} \circ J)(\pi, \varphi) = \frac{\varphi}{\pi^2} = \frac{1}{\varphi^3(\pi/\varphi^2)^2}. \quad (\text{A.4})$$

Proof. Applying the interchange twice gives the identity. Moreover, $\varphi^3(\pi/\varphi^2)^2 = \pi^2/\varphi$; taking its inverse proves the last equality. $\square$

The electronic reader uses the orientation φ/π^2 , followed by the exponential character. The preceding identity explains the interchange, but does not replace the rule that chooses that character or by itself produce the integers of the correction. These have a different provenance.

The phase-compatible central return has minimal length twenty-seven. With the phase and sheet record, its finite space is

$$\mathcal{K}_e = \mathbb{Z}/9\mathbb{Z} \times \mathbb{F}_3.$$

The five regions of π are distinct genealogical identities, not five interchangeable copies of a visible expansion. In the regional gauge fixed by the construction, their injection into the return occupies the positions

$$(4, 0), \quad (5, 0), \quad (6, 0), \quad (6, 1), \quad (6, 2). \quad (\text{A.5})$$

The transverse orientation, mutated return, and three positive regions remain separated by that record. Admissible changes of gauge transport the labels jointly; they do not reselect the positions from an electronic value.

Proposition A.8 (Deficit and retained memory). *Given the central return and the preceding regional injection, the oriented boundary deficit and torsional memory are*

$$n_e = -|\mathcal{K}_e \setminus \iota(\mathcal{R}_\pi)| = -22, \quad m_e^{\text{tor}} = |\mathcal{R}_\pi \times \mathbb{F}_3| = 15. \quad (\text{A.6})$$

Proof. The five positions in (A.5) are distinct. Thus, $|\mathcal{K}_e| = 9 \cdot 3 = 27$, and the complement of the image has cardinality $27 - 5 = 22$. The orientation convention records the deficit removed from the boundary as negative. The retained memory has three sheets per region, so its positive cardinality is $5 \cdot 3 = 15$. $\square$

The same pair has the integer check supplied by the collection's APP chart:

$$396 = 18 + 72 + 108 + 198, \quad \frac{396}{18} = 22, \quad \frac{270}{18} = 15 = 3 \cdot 5 = \binom{6}{2}. \quad (\text{A.7})$$

The weight 18 belongs to the central 2×2 kernel, and 270 to the declared global channel. These equalities check the normalization of the count; the return argument also fixes the signs. The quotient of the alternating rings, $396/(18 + 108) = 22/7$, is rational and is not identified with π . Nor does $\binom{6}{2}$ mean $6/2$: it counts the unordered pairs of a six-element set.

Evaluation of the deficit in the third symmetric realization uses α^3 . This power is the product of three copies of the closure scalar already obtained; it does not assert a universal rule associating any dimension with the same power of α . The specified domain and reader are essential to the composition.

A.4 Global torsion and the basal coordinate

Definition A.9 (Electronic torsional normalization). The reduced defect d_4 and the complete reader Δ_4 are distinguished:

$$d_4 = \frac{\pi - e \log \pi}{270}, \quad \Delta_4 = d_4 \left(1 + \frac{\pi}{729}\right). \quad (\text{A.8})$$

The denominator 270 corresponds to the census channel, whereas the factor $1 + \pi/729$ retains the coupling with the cubic cell of 729 positions. The electronic reading uses Δ_4 both in its basal stage and in the return exponent.

This definition acts on previously generated π and e . The logarithm is a subsequent operation in their real realization, not a retrospective input of conventional values. Torsion is global: it is not determined independently for each section by an observed mass.

Lemma A.10 (The two normalizations are not interchangeable). *We have*

$$\Delta_4 - d_4 = \frac{\pi}{729} d_4. \quad (\text{A.9})$$

In particular, $d_4 > 0$ and $\Delta_4 > d_4$ for the coordinates under consideration.

Proof. The identity follows by distributivity. For positivity, the function $f(x) = x - e \log x$, on $x > 0$, has derivative $1 - e/x$ and second derivative $e/x^2 > 0$. Its unique minimum is at $x = e$, where it is zero. Since $\pi \neq e$, we have $f(\pi) > 0$. Both factors in (A.8) are positive and $1 + \pi/729 > 1$. $\square$

Remark A.11 (Preservation of the normalization). Removing the second factor in (A.8) changes the reader; it is not an algebraic simplification. If d_4 is introduced as an auxiliary coordinate, the same object must be written as $\Delta_4 = (1 + \pi/729)d_4$ in every occurrence. Keeping the integers 15 and 6 and replacing only Δ_4 by d_4 produces a different formula.

Definition A.12 (Basal electronic coordinate). The composition of incompatibility, modal transfer, deficit, and retained memory is

$$\varepsilon_e^{(0)} = \frac{\sqrt{3}}{4} \left[\exp\left(\frac{\varphi}{\pi^2}\right) - 22\alpha^3 \right] (1 + 15\Delta_4). \quad (\text{A.10})$$

The equation combines three operations of different kinds. The norm $\sqrt{3}/4$ quantifies the incompatibility of the sheet projections; its origin is a principal angle of the APP fibers. The exponential acts on the orientation φ/π^2 of the transfer reader. The cubic subtraction records the 22 complementary positions of the marked return, and the torsional factor preserves the 15 components corresponding to three sheets per region. The composition preserves these domains even though its final evaluation is scalar.

The presence of π , φ , and e thus has operational meanings. Euler's number enters through the exponential and the logarithmic normalization of Δ_4 ; φ and π enter through oriented transfer. The constant α acts after its own construction through the cubic closure term. The chronological return must still act on this basal coordinate. Separating the stages makes it possible to explain what changes when an orientation, normalization, or record is modified, while preserving the central structure that identifies the electronic section.

The parentheses in this equation are part of its content. The term $-22\alpha^3$ is subtracted *outside* the exponential; the memory $1 + 15\Delta_4$ multiplies the entire bracket. Placing the deficit in the exponent or applying a different torsion to it would alter the composition.

Proposition A.13 (Basal positivity). *For $0 < \alpha < 10^{-2}$, $\varphi > 0$, and the normalization (A.8), we have $\varepsilon_e^{(0)} > 0$.*

Proof. The exponential is greater than one, and $22\alpha^3 < 22 \cdot 10^{-6} < 1$. The bracket is therefore positive. Also, $1 + 15\Delta_4 > 1$ and $\sqrt{3}/4 > 0$. $\square$

The subsequent evaluation of (A.10) begins with

$$\varepsilon_e^{(0)} = 0.5109989224880660725 \dots$$

This number is still a dimensionless coordinate and is not the final stage: the return memory of the route itself remains to be applied.

A.5 Two readers of the electronic return

Electronic memory is represented by a tritic word of length 108 and a signed twelve-phase word. To make the exponent calculation reproducible, we specify the chronological projection used. Initially, let $(r_0, c_0) = (5, 5)$ and $(h_0, v_0) = (1, 1)$. At each step, the selector repeats $(+1, -1, 0)$. With orientation fixed within each twenty-seven-step block, the visible rule is

mode	integer read	cellular transport
+1	$r + c$	$c \mapsto 1 + (c - 1 + h) \bmod 9$
-1	rc	$r \mapsto 1 + (r - 1 + v) \bmod 9$
0	9	no displacement

The emitted symbol is $\rho_9(\text{integer read}) \bmod 3$. After steps 27, 54, 81, h and v are simultaneously reversed. This rule computes the observable projection; the state update additionally preserves residue, quotient, carry, sheet, boundary, and memory. The record of 108 symbols is not identified with the complete state.

In each complete three-step block, each coordinate advances once; nine such blocks are needed to return to the center in the same phase. The minimal length of this recorded return is therefore 27. If phase is forgotten, the cell is already reached after step 26, before the neutral threshold completing the block; that is not the typed return counted by $\mathcal{K}_e$. The explicit reading of the first two return blocks gives

$$a = 100000220, \quad b = 120200000, \quad \gamma_e = (a^3 b^3)^2. \quad (\text{A.11})$$

Here, powers denote word concatenation. In particular, $\gamma_{e,t+54} = \gamma_{e,t}$. In the record of nine steps per phase, the central orientations give

$$\tau_e = (1, 1, 1, -1, -1, -1, 1, 1, 1, -1, -1, -1). \quad (\text{A.12})$$

The periodicity of these representations does not amount to restarting the history that produces them.

Definition A.14 (Cyclic discrepancy reader). For a word $\gamma \in \mathbb{F}_3^{108}$, define

$$D_k(\gamma) = \#\{t \in \mathbb{Z}/108\mathbb{Z} : \gamma_t \neq \gamma_{t+k}\}.$$

On the domain of words whose period divides 54, the integer reader is

$$K_D(\gamma) = \left(D_{27} + D_{36}, D_1, \frac{D_4 - D_3}{2} \right). \quad (\text{A.13})$$

Lemma A.15 (Integrality and central evaluation). *The D_k on the preceding domain are even. For (A.11),*

k	1	3	4	27	36
D_k	54	56	68	48	32

and $K_D(\gamma_e) = (80, 54, 6)$.

Proof. The map $t \mapsto t + 54$ pairs discrepancy indices without fixed points and preserves their condition; each cardinality is even. For the explicit word $a^3 b^3$ of length 54, the discrepancy count for each displacement is, in the same order, 27, 28, 34, 24, 16. Doubling the word doubles these five cardinalities. Finally,

$$(D_{27} + D_{36}, D_1, (D_4 - D_3)/2) = (48 + 32, 54, (68 - 56)/2) = (80, 54, 6).$$

□

The displacements 27 and 36 are the chronological lifts of the 90- and 120-degree channels in a chart where nine steps correspond to thirty degrees. The angular reading explains that representation, but the integers are computed on the word, without introducing a mass or a target real value.

Definition A.16 (Dodecaphasic window reader). For $\tau \in \{-1, 0, 1\}^{12}$, with cyclic indices, let

$$\begin{aligned} C_k(\tau) &= \sum_{r=0}^{11} \tau_r \tau_{r+k}, \\ A_\ell(\tau) &= \sum_{r=0}^{11} \left(\sum_{j=0}^{\ell-1} \tau_{r+j} \right)^2, \\ P_6(\tau) &= \sum_{r=0}^5 \tau_r \tau_{r+6}. \end{aligned}$$

Define

$$\Omega(\tau) = 4A_3(\tau) - 3A_4(\tau), \quad K_\Omega(\tau) = (\Omega(\tau), 54, P_6(\tau)). \quad (\text{A.14})$$

The 54 in this reader records the declared chronological half-turn. It is not confused with the local APP jump of value four.

Lemma A.17 (Primitive contrast and correlation form). *Among integer linear contrasts $uA_3 + vA_4$ that vanish whenever $A_3 = 3a$ and $A_4 = 4a$, the primitive pair, with orientation $90 - 120$, is $(u, v) = (4, -3)$. Moreover,*

$$\Omega = -2C_1 - 4C_2 - 6C_3. \quad (\text{A.15})$$

Proof. Vanishing for every a requires $3u + 4v = 0$. By coprimality, the solutions are $(u, v) = n(4, -3)$, and orientation fixes the sign of the primitive generator. Expanding the squares gives

$$A_\ell = \ell C_0 + 2 \sum_{k=1}^{\ell-1} (\ell - k) C_k.$$

Therefore, $A_3 = 3C_0 + 4C_1 + 2C_2$ and $A_4 = 4C_0 + 6C_1 + 4C_2 + 2C_3$. Their combination $4A_3 - 3A_4$ cancels C_0 and yields (A.15). $\square$

The uniqueness proved belongs to that class of integer linear contrasts; it does not by itself exclude every nonlinear functional of the windows. Specifying the functional class avoids turning an identity of two channels into a claim of universal selection.

Theorem A.18 (Return triple of the electronic section). *The two readers applied to the records of Γ_e coincide and give*

$$K_D(\gamma_e) = K_\Omega(\tau_e) = (80, 54, 6). \quad (\text{A.16})$$

Their evaluation in the already generated coordinates is

$$\kappa_e = 80 - 54\alpha + 6\Delta_4. \quad (\text{A.17})$$

Proof. The first reader was computed above. For τ_e , the antipodal product is always one, and hence $P_6 = 6$. The cyclic count gives $C_1 = 4$, $C_2 = -4$, $C_3 = -12$; thus

$$\Omega = -2 \cdot 4 - 4(-4) - 6(-12) = 80.$$

Equivalently, $A_3 = 44$, $A_4 = 32$, and $4 \cdot 44 - 3 \cdot 32 = 80$. Both triples are therefore $(80, 54, 6)$. Evaluating the functional $(A, B, C) \mapsto A - \alpha B + \Delta_4 C$ proves the last formula. $\square$

This coincidence concerns the central section. It does not identify the two readers on every route domain: one compares tritic symbols and the other compares windows of a signed record. Nor is 80 obtained from the period of a word assigned to Euler's number. Its origin in this composition is the triple of the electronic route.

A.6 Complete electronic composition and normalization

Definition A.19 (Multiplicative return). The action of return memory on the basal coordinate is

$$\varepsilon_e^\beta = \varepsilon_e^{(0)} \exp(\kappa_e \log R_{\text{act}}). \quad (\text{A.18})$$

Theorem A.20 (Positive electronic factorization). *With the declared readers, orientation, and normalization, (A.18) coincides with (A.1). All its factors are dimensionless and the resulting coordinate is positive.*

Proof. The incompatibility norm is (A.3); modal transfer, deficit, and retained memory compose (A.10). The triple (A.16) produces (A.17), and the ratio of the two action sections is (13.10). Substituting those three identities into (A.18) yields exactly (A.1). The basal stage is positive, as is the exponential of a real number. The common factor with the dimension of action has already canceled in forming R_{act} . $\square$

The evaluation of this composition begins with

$$\varepsilon_e^\beta = 0.5109989506900008086082571648 \dots \quad (\text{A.19})$$

The digits are a subsequent evaluation of the formula, not data used to select 22, 15, or (80, 54, 6). The numerical representation of the basal coordinate by a finite prefix introduces only its truncation error; it does not modify the exact formula defining it.

The effect of a different torsional normalization can be located without comparison to a measurement. Write $B = \frac{\sqrt{3}}{4}[\exp(\varphi/\pi^2) - 22\alpha^3]$, $R = R_{\text{act}}$, and

$$\mathcal{E}(d) = B(1 + 15d)R^{80-54\alpha+6d}.$$

For the two defects in (A.8),

$$\frac{\mathcal{E}(\Delta_4)}{\mathcal{E}(d_4)} = \frac{1 + 15\Delta_4}{1 + 15d_4} R^{6(\Delta_4-d_4)} > 1. \quad (\text{A.20})$$

The first torsional inequality and $R > 1$ prove that they are not the same coordinate. If only the defect in the exponent is changed, keeping the complete basal coordinate, the ratio of the two outputs is $R^{6(\Delta_4-d_4)}$, again different from one. A very small difference is still a difference of formula; it cannot be attributed to a mere conversion of units.

A.7 Dimensional realization: energy, mass, and chronological scale

The electronic coordinate finally acts on a dimensional line. Let $\mathcal{L}_E$ be an energy line with positive basis $\mathcal{U}_E$. The realization is

$$E_e^{(0)} = \varepsilon_e^{(0)} \mathcal{U}_E, \quad E_e^\beta = \varepsilon_e^\beta \mathcal{U}_E, \quad m_e^\beta = \frac{E_e^\beta}{c_{\text{int}}^2}. \quad (\text{A.21})$$

The last expression belongs to the mass line when c_{int} has the dimension of velocity in the declared realization. In the comparative chart $\mathcal{U}_E = 1 \text{ MeV}$, (A.19) expresses an energy in MeV; the corresponding mass is expressed in MeV/c^2 . It is common to abbreviate both quantities by taking $c = 1$, but that abbreviation must not conceal the dimensional map.

The action scale also admits an action-clock realization. If t_0 is a positive temporal section, the 108-step clock defines

$$\omega_0 = \frac{2\pi}{108t_0}, \quad E_{\text{clk}} = \hbar_{\text{ret}}\omega_0, \quad m_{\text{clk}} = E_{\text{clk}}c_{\text{int}}^{-2}. \quad (\text{A.22})$$

Action times frequency has the dimension of energy. This chronological quantum is a common dimensional section; it is not, by definition, the electronic section. The route word and its factors are not obtained from the uniformity of the clock.

Proposition A.21 (Cancellation of the common scale). *The factor $10^{-34}\mathcal{U}_S$ in (13.9) is not part of κ_e . On the same energy line,*

$$\frac{E_e^\beta}{E_e^{(0)}} = R_{\text{act}}^{\kappa_e}. \quad (\text{A.23})$$

The result is independent of the choice of the common dimensional basis.

Proof. In forming $\hbar_{\text{pre}}/\hbar_{\text{ret}}$, the factor $10^{-34}\mathcal{U}_S$ cancels exactly. Dividing the two energy expressions in (A.21) also cancels $\mathcal{U}_E$, and (A.18) gives the stated ratio. $\square$

Thus, a coordinate already expressed in MeV is not multiplied again by 10^{-34} . If E_{clk} is chosen as the basis of $\mathcal{L}_E$, the coordinates must be transported by the change-of-basis factor; retaining the number from the one-MeV chart unchanged would identify two bases without proof. This is the precise point at which the action scale enters the absolute realization and ceases to enter the relative correction.

There is also a linear realization that makes the separation between clock and genealogy visible. On $\ell^2(\mathbb{Z}/108\mathbb{Z}; \mathbb{R})$, the cyclic shift S has fixed space

$$\ker(S - I) = \mathbb{R}n_0, \quad n_0 = \frac{1}{\sqrt{108}} \sum_{t=0}^{107} \delta_t.$$

If g_{Γ_e} is the route symbol in the free linearization of genealogies and $\mathcal{U}_M$ is a basis of the mass line, the rank-one map is defined by

$$\mathfrak{C}_e : \mathbb{R}n_0 \otimes \mathbb{R}g_{\Gamma_e} \longrightarrow \mathcal{L}_M, \quad \mathfrak{C}_e(n_0 \otimes g_{\Gamma_e}) = \varepsilon_e^\beta \mathcal{U}_M. \quad (\text{A.24})$$

The map assembles the preceding factorization once its bases have been declared. The uniform vector n_0 does not select ε_e^β : that coordinate comes from APP, the oriented return, and its readers. An algebraic realization of the null mode is thus distinguished from an identification of the electron with the clock quantum.

The internal conclusion is the positive factorization of a central section, with its operations, memory, and normalizations made explicit. Assignment to a physical electron adds the charge representation, spin, coupling, and corresponding unit chart; it does not follow exclusively from a decimal coincidence. Metrological comparison is presented after the exceptional prolongation and does not alter the inputs of any of the constructions performed here.

A.7.1 Structural dependence of the electronic coordinate

The electronic equation expresses a hierarchical composition. Its first factor, $\sqrt{3}/4$, belongs to the central matrix structure and fixes a geometric normalization. The exponent φ/π^2 uses the already generated regional coordinates in the indicated orientation of the areal–volumetric reader. The term $22\alpha^3$ incorporates the dodecaphasic coupling into the cubic correction. The factor $1 + 15\Delta_4$ preserves the global torsional normalization. Finally, the quotient of the action sections acts with the exponent determined by the return readers. Each operation thus has a different antecedent within the same construction.

The articulation becomes more precise when its dependencies are considered. The central norm can be calculated before the regional coordinates are evaluated. The exponential and cubic correction require those coordinates and the closure of α . Multiplicative memory additionally requires the action sections and the 108-position trajectory. Knowing the basal expression therefore does not amount to having evaluated the complete coordinate. The succession of operations does not add an external mass to the point (5, 5); it develops the readers assigned to its persistent section.

The distinction between scale and structure also provides an interpretation of the physical comparison. Mass is the dimensional realization of the complete coordinate, while the centre, involution, and vacancy describe the relations that produce it. A common change of unit modifies the dimensional coordinates, but leaves the dimensionless ratios and the proved transport identities unchanged.

This is the sense in which the electron preserves its structural character when expressed relative to a chronological scale.

A.8 Lorentzian transport and electromagnetic representation

The electronic section provides a structure and a coordinate before its application to an interaction. The Lorentzian realization of cubic incidence determines, under the stated temporal convention, the space of forms in which that interaction can be expressed. To formulate it, an oriented local realization $(\mathcal{M}, g)$ of dimension four is fixed, with tangent fibres identified with the model $Q_{\square}$. A connection ∇ with $\nabla g = 0$, a potential one-form $A \in \Omega^1(\mathcal{M})$, its curvature $F = dA$, and a charge representation of the section are also specified. These data are the arguments of the physical representation that follows.

Hodge duality satisfies $\star^2 = -I$ on two-forms. The electric-magnetic separation corresponds to a temporal choice in the $1 + 3$ decomposition; F and $\star F$ preserve their relationship within the same space of two-forms. Neutrality of a charge representation means $q = 0$, whereas the section may retain orientation, phase, and internal transport. The absence of electric coupling in that representation has a different content from the absence of phase information.

In this dimensional realization, let $m_e > 0$, let q be the charge, let $\gamma(s)$ be a parametrized trajectory, and let $u = \dot{\gamma}$ be its four-velocity. The Lorentz force is expressed by

$$m_e \nabla_u u^\mu = q F^\mu{}_\nu u^\nu. \quad (\text{A.25})$$

The antisymmetry of F implies $F_{\mu\nu} u^\mu u^\nu = 0$. Consequently,

$$\frac{d}{ds} g(u, u) = 2g(\nabla_u u, u) = 0.$$

The normalization of the trajectory is preserved provided that ∇ is metric and (A.25) holds. The mass used is the dimensional realization of the preceding electronic section; the coupling parameter is identified in the adopted electromagnetic convention. The concrete deduction of the potential and of a field configuration constitutes an additional problem with its own boundary conditions. The result presented here determines the domain, the representation data, and the conservation implied by its equation.

TPK route inversion and the orientation of an electronic line then offer a precise comparison: each has its involution, transport, and representation. The work of Wheeler and Feynman places the question historically; the equivalence of two realizations is decided through their maps and compatibility conditions. This arrangement keeps the electron's arithmetic origin connected to the formulation of its interaction, making the function of each stage explicit.

A.9 Comparison of the reduced Planck constant

The International System in force since 20 May 2019 fixes $h = 6.62607015 \times 10^{-34}$ J s exactly [10]. Its reduced constant is therefore

$$\hbar_{\text{SI}} = \frac{6.62607015}{2\pi} 10^{-34} \text{ J s} = 1.0545718176461563912 \dots \times 10^{-34} \text{ J s}.$$

The ellipsis denotes the expansion of an exact expression, not an experimental uncertainty in h in the current SI. This reference value is used here after evaluation of the HMT readers; its function is comparative.

The article's two return coordinates are

$$\begin{aligned} H_5 &= 1.0545718176461544696 \dots, \\ \eta_{\text{ret}} &= 1.0545718169150390611 \dots \end{aligned}$$

Under the identification of $\mathcal{U}_S$ with J s, the relative difference between the preceding section and $\hbar_{\text{SI}}$ is approximately -1.82×10^{-15} . For the subsequent section, it is approximately -6.93×10^{-10} . The second difference contains the return effect used by the electronic corrector; it does not equal the evaluation error of the first section.

The precision of these comparisons allows the quantitative proximity of the construction of H_5 to be recognized while distinguishing proximity from exact identity. In the present manuscript, the displayed expressions do not provide an exact equality with $h/(2\pi)$. The result proved for Σ_H is its decade -34 ; the analytic result for H_5 is the evaluation of the stated character. The physical identification is examined in this chart and with the explicit differences just given. The SI value does not participate in the definition of the norm-and-self-scale reader or in the regional recurrence.

B Conditions of the realizations

The action, mixing, area, and gravitational realizations retain the domains and normalizations of the results they compose.

B.1 Alpha coordinate and mixing frames

The dodecaphasic closure and the analytic representation of alpha express the same coordinate at every depth (Theorem 9.13). The analytic recurrence is fixed from the enriched state and the declared geometric class; it does not independently select that coordinate. In the angular realization, complete spectral frames enter (17.23). Unitarity of a polar factor does not replace its identification with that change of frames. CP compatibility retains condition (17.16), and a Majorana real form restricts the admissible phases.

B.2 Thermal bases and gravitational normalization

Section 22 defines the thermal transducer and its transformation under a change of bases. The product $k_B \Theta_{\text{clk}} \log 3$ is fixed by the cycle energy; the individual coordinate of k_B requires the positive bases of the energy and temperature lines to be specified. This covariance does not alter the preceding action or period.

Section 21 fixes the gravitational chart through the reduced radius Gm/c^2 and its coincidence condition with R_{108} . In the present composition, length precedes that circular expression: R4 and R5 determine L and the positive radius of the character, and the clock $t_0 = \pi_{\text{HMT}} L / (54c_{\text{int}})$ gives $R_{108} = L$. R8 identifies this radius with the reduced gravitational reading. The reciprocal action sections preserve the product $\hbar G$ and hence the corresponding area scale. The Cartan–Holst equation is formulated for a nondegenerate cotetrad, a connection, and the declared spin current, with the boundary conditions of its variation.

B.3 Provenance of the longitudinal and radial composition

Archive conservation, Hadamard transport, phase action, and reciprocity of the sections originate in preceding developments of the corpus. The revision of 25–26 September 2026 assembles the marked incidence register and makes longitudinal composition R4 and metric composition R5 explicit. The theorem in 20 proves their joint consequence with R1–R3 and R6–R8: uniqueness of inscription, the radial reader, and the common coefficient in the declared realization [14]. This provenance distinguishes the retained antecedents from the definitions and proofs incorporated in the revision.

Transports with a global metric, including its cross-blocks, preserve action and the recoverable residual. Restitution of the counterangle and constitutive speed provides equivalent expressions of the same evaluation [15]; [16]. The reference longitudinal section, action, and dimensional speed remain explicit. The choice $L_* = 1 \text{ m}$ is the section used for the SI digits; declaring it enables their reproduction with the same dimensional meaning. Physical identification R8 retains its place in the proof.

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